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Ioannis D. Platis

Geometries

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To my mother,
Anastasia Charitou-Plati.

Preface

The purpose of the book in hand is to introduce the reader to the modern viewpoint of geometry which is based on the brilliant idea of Klein: a *geometry* is a pair comprising a nonempty set and a group of transformations of this set. In this way, the importance of the interconnection of modern day geometry and group theory becomes immediately transparent. The book is mostly addressed to undergraduate students of mathematics but also of sciences in general (middle to upper level). Here, most of the so-called Classical Geometries modelled as Kleinian Geometries are presented in due detail.

The text of this book resulted from the evolution and complete rewriting of the teaching notes that were the companion notes of the corresponding course I taught in the Department of Mathematics and Applied Mathematics at the University of Crete in consecutive times from 2019 to 2021. An effort was made so that the text is on the one hand self-contained: the reader does not need in principle to refer to other sources for the knowledge of calculus, linear algebra and group theory that are necessary for understanding the text. Furthermore, through the sufficient number of examples and exercises listed at the end of each chapter, the text should be as friendly as possible to the reader, but without discounting anything that concerns mathematical rigour.

The text covers the material which can be taught in two academic semesters, during which, e.g. in the first semester, Chaps. 1–5 can be taught, followed by Chaps. 6, 7, 8, and 9 in the second semester. The teacher who has to condense the course into one teaching semester will find it comfortable to teach the first four chapters and then two of his/her preference.

There are certainly a lot of good Geometry textbooks; many of them are listed in the bibliography at the end of the book. So, the question is perfectly natural: what is the need for *this* book? It is clear that the answer (and the justification) given below contains a strong subjective component, but I think that in the end something like this is rather inevitable. Having taught the material in this book over many years (not necessarily in the form presented here), I found that the majority of my audience members came to class at least prejudiced (if not afraid) of the subject of geometry. There are many reasons why this happened (and why it still happens), but

this is not the place to analyse them. However, when writing those first notes that formed the basis of this book, I had the following in mind:

(a) To present the material in an understandable way, but not *easy to digest*, considering the huge difference between simplicity and simplification at the expense of rigour. In contrast to many textbooks that I have in mind, here I dealt with Classical Geometries in the arbitrary dimension and not in the two or three dimensions, which is the norm for such textbooks—this is perhaps also the main innovation of this book. In the manner of presentation in many dimensions, the reader perhaps understands a little more the usefulness of the many variables on the one hand, while on the other hand, he gains a complete understanding of the subject so that he/she is well prepared when he/she meets geometry again in a differentiable manifolds course, for example. In addition, because it is pedagogically correct to focus a little more to the examples of low dimensions, these are presented in the book either separately or in the form of examples.
(b) To write, as I mentioned above, a text with as much autonomy as possible, but also uniformity in presentation, with the least prerequisites as possible. The great Polish mathematician A. Zygmund, when asked by a member of his audience about the prerequisites of the complex analysis course he was going to teach, answered: the knowledge of the number i. In this spirit, the only requirement I have from the reader is *good mood*—although some basic knowledge of calculus and linear algebra, topology and especially group theory, whose summaries are listed here anyway, would facilitate the situation quite a bit!
(c) Finally, to highlight the beauty of Classical Geometries as such, but also of the method by which they are studied in this book. All that is described in the text that follows resulted from the intellectual efforts of dozens of generations of great mathematicians—we may for instance just think that Greek Geometry made such enormous achievements without the existence of a handy numerical system and with the complete absence of algebra. So I thought it right to expand a little on the historical data not only about geometry in general, but also about every branch of it that we study here. And this, because I think that the reader should find motivation to understand more broadly the reasons that lead to the study of geometry.

At this point, I would like to give a special mention to Georgia Plati, architect B.Sc., Dipl., Glasgow School of Art, for the designs that adorn this book.

As Euclid said, *there is no royal path to Geometry.* I hope that the readers of his book will forge their relationship with geometry through the effort to master its first secrets, which are contained and analysed in the text that follows. This will be their reward, and, in my opinion, it is worth it.

Heraklion, Crete, Greece Ioannis D. Platis
September 2024

Declarations

Competing Interests The authors has no competing interests to declare that are relevant to the content of this manuscript.

Introduction

Geometry (from Greek: $\Gamma\tilde{\eta}+\mu\varepsilon\tau\rho\tilde{\omega}$, measuring the Earth) is the branch of mathematics that deals with the *shapes* of various objects, the relationships between them and possibly the properties of the surrounding space. It is commonly accepted that geometry initially arose from purely practical reason. For example, the first civilisations of Mesopotamia built monumental buildings in the ancient Sumerian, Akkadian and Babylonian cities in a particular pyramid shape, the *ziggurat*. During the time of the Pharaohs, the empirical knowledge of basic geometry was necessary for Egyptians to understand the impact of the flooding of Nile for their land, and for the construction of the pyramids and their gigantic religious buildings. Similarly, for Indians and other East Asian civilisations, geometry, like mathematics, lacked theoretical foundations. To illustrate this, a good example is what one can read in one of the remaining Sumerian clay tablets: in order to solve such and such a problem, you will take the following steps; and the author proceeds to the detailed explanation of the steps of the method without any justification and/or proof of the method they used to solve the problem they tried to solve.

The rise of the sciences and the first explosion of mathematics and geometry occurs with the dawn of the ancient Greek city-state. The ancient Greeks enjoyed liberal political systems, broadly construed, that provided the opportunity for the development of the arts and the sciences. The trade and travelling (but also colonising) that city- states pursued in the Mediterranean Sea brought them into contact with all the neighbouring cultures. This was a decisive factor which marked the shift from Mythos to Logos, that is, the explanation of physical phenomena based on a logical proof and not on a divine intervention. The move from the mythical narrative to logical explanation of physical phenomena marks the birth to Philosophy, and theoria (theory) in Mathematics. During this time, we begin to see thinkers who were actively involved in geometry. For example, Thales, the first philosopher, Pythagoras of Samos (c.570–c.495 BC) to whom and his students we owe the proof of the Pythagorean Theorem, as well as many other mathematical/geometric achievements. Also, it is worth mentioning, Hippocrates of Chios (c. 470–c. 421 BC), who, in addition to squaring the menisci (lunes) that bear

his name, wrote the first *Elements*, that is, the first attempt at a scientific/axiomatic foundation of geometry.

Euclid, however, *the Elementor*, was the first to use rigorous methods, obviously influenced by the teachings of Plato, and presented a complete system of geometry. Euclid was the librarian of the Library of Alexandria in the time of Ptolemy I (first half of the fourth century BC), and was able to have access to all the known texts of the time. In his opus magnum, the Elements, comprising 13 books, Euclid presented geometry in a rigorous way, providing justificatory proofs for a multitude of theorems of geometry and arithmetic. All his proofs were based on clearly defined axioms (postulates) and common notions that are considered true from the preamble and that are listed at the beginning of each book of the Elements. This colossal work was the source of mathematics for generations upon generations, at least until the nineteenth century AD.

One should note here, geometry, mathematics and sciences in general made relatively little progress until the end of the ancient Greek world, with Archimedes of Syracuse, c. 287–c. 212 BC, being a notable exception. Moreover, during early Christian times we would note the mathematician, astronomer, astrologist and geographer Claudius Ptolemy of Alexandria (c. 100–160/170 AD) and Pappus. During the Middle Ages, the Arabs invented algebra based on the ideas of Diophantus of Alexandria (c. 210–290 AD) while at the same time they developed, analysed in detail and took geometry a step further. However, the second great explosion for geometry will occur as late as in the beginning of the nineteenth century. After the revolutionary discovery of calculus by the two great polymaths, Newton (Isaac Newton, 1643–1727) and Leibniz (Gottfried Wilhelm Leibniz, 1646–1716), as well as that of coordinates by the French philosopher and mathematician Descartes (René Descartes, 1596–1650), Hungarian mathematician Bolyai (János Bolyai, 1802–1860) and Russian mathematician Lobachevsky (Nikolay Ivanovich Lobachevsky, 1792–1856), independently of one after the other, introduced the first *non-Euclidean Geometry*, by simply changing Euclid's Fifth Postulate. For over 20 centuries, generations of mathematicians believed that the Fifth Postulate, which essentially says that there is a unique parallel to a given straight line that can be drawn from a point not in the straight line,could be proven as a theorem. Bolyai and Lobachevsky simply replaced it with another one which says that from a point outside the straight line more than one parallel is drawn to the straight line (and thus, *infinite* parallels). This new Geometry, the Hyperbolic Geometry, was the precursor to the introduction of many new Geometries at the behest of the great German mathematician Riemann (Georg Friedrich Bernhardt Riemann, 1826–1866), the father of Riemann Geometry.

All this, along with the appearance of set theory, created the need to re-examine Euclid from a new perspective. Many great mathematicians of the nineteenth century tried to establish geometry in a way freed from any inconsistencies of the Elements. It was Hilbert who made the first comprehensive such attempt, introducing a rigorous axiomatic foundation of geometry in his small but thoroughly ground-breaking book *Grundlagen der Geometrie*.

On the other hand, Klein (Felix Klein, 1849–1925), with the Erlangen Programme, proposed a consideration of geometry from the point of view of group theory: a geometry is nothing more than a pair consisting of a non-empty set and a group of self-transformations of the set. Klein's genius idea evolved during the twentieth century by Cartan (Élie Joseph Cartan, 1869–1951). Perhaps, the most innovative thing in Klein's view of Geometry is the absence of axioms: everything can be proven as theorems.

The development of geometry from the nineteenth century onward was and still is staggering. A multitude of new branches of 0 have been created, which interact with other branches of mathematics. As for the applications of geometry, they are found in many branches of science: to list only a few, architecture (especially in the environmental design and the design of the internal spaces, medical sciences, computer graphics, robotics, the arts in general, etc.).

This book explores geometry from the Kleinian viewpoint: a *geometry* is a pair (X, G) where X is a non-empty set and G is a group comprising transformations of X. Without using Cartan's definition for Kleinian geometries which involves Lie groups, considering that Lie groups are a subject beyond the scope of his book, we follow in pretty much solid lines the following pattern for the (X, G) geometries we study:

- We examine the space X and its properties.
- We define the group G of the transformations of X and examine its constituent elements as well as its action on X.
- We formulate and prove the Fundamental Theorems of each of the Geometries we study. Usually, these theorems have to do with special forms of the action of G on X.
- In the case that there is a metric for our geometry, we find its geodesics, that is, the shortest paths for this geometry.

Explicitly, in this book:

In the first, introductory chapter, we present in brief those episodes of the History of Mathematics that we believe set the necessary background for the reader to understand how Geometry came to life, and its evolution over the centuries and its enormous impact on human civilisation.

In the second chapter, after briefly listing those elements of Group Theory that are necessary for a better understanding of the text, we proceed to the general definition of Geometry according to Klein and mention the most important properties. As an introductory example, we cite the case of Euclidean two-dimensional Geometry, seen from the point of view we have throughout the book.

The third chapter focuses on Euclidean Geometry. For the sake of completeness, we mention in a rather analytical way all the elements and properties of the underlying real space, before proceeding to the group of transformations, which is precisely the group of Euclidean isometries. For this, we define in detail the reflections in hyperplanes, the group of orthogonal transformations and the group of translations, before formulating and proving the Isometric Theorem and the Cartan-

Dieudonné Reflections Theorem. In between, we prove that the geodesics of the Euclidean metric are straight lines.

The fourth chapter is a supplement to the previous one, as we deal there with the Similarity Geometry, which is inextricably intertwined with Euclidean Geometry (to be precise, Similarity Geometry contains Euclidean Geometry in a natural way since the similarity group of any metric space contains the corresponding group of isometries as a normal subgroup). We present the group of dilations, we show that every Euclidean similarity is a composition of a Euclidean isometry and a dilation, and we proceed to study the action of the group of similarities, after proving the corresponding Fundamental Theorem. Defined in antiquity and already studied exhaustively by Euclid within the Elements, the metric ratio of three points is presented, which we all essentially used in school when we were taught similar shapes and especially similar triangles.

Affine Geometry follows in the fifth chapter, which we owe to Euler. Affine Geometry can roughly be defined as Euclidean Geometry stripped of the property of the invariance of lengths (as well as of angles and areas). Here, the only invariant is parallelism. Before proceeding to the study of the Affine Geometry of real space (which, as we have already hinted, contains the previous Geometries) we list, for the sake of completeness and consistency, certain elements of abstract affine spaces. Then, we define the affine group; we study its action and prove the Fundamental Theorem of Affine Geometry as well as the Parallel Projections Theorem. Finally, because many Euclidean theorems are in essence theorems of Affine Geometry, we close the chapter by quoting an affine proof of the Theorems of Medians, Menelaus and Ceva.

Spherical Geometry, which we deal with in the sixth chapter, is the first (metric) Geometry that we encounter here whose underlying space (in this particular case, the unit sphere) is compact. After describing the unit sphere in detail, emphasising its intersections with affine planes of various dimensions, we focus on the great circles which are also the geodesics of the spherical metric. The group of isometries is the orthogonal group, whose action on the sphere we also study. We dedicate a section to spherical trigonometry, first demonstrating the fact that spherical triangles have an angle sum greater than two right angles. As in Euclidean Trigonometry, there is a Law of Cosines and Sines, but additionally, and in contrast to the Euclidean case, the existence of a Second Law of Cosines identifies spherical similarities with spherical isometries. Finally, we give the elegant Legendre's proof of Euler's formula for vertices, sides and the number of triangles of any triangulation of the sphere. With the help of this formula, we precisely determine the Platonic Solids.

Projective Geometry, the topic of the seventh chapter, most probably finds its roots in Euclid's *Optics*. Even if one of its famous theorems bears the name of the Greek mathematician of the late antiquity Pappus of Alexandria (c. 290–c. 350 AD), Projective Geometry was not developed as a distinguished branch of mathematics until the Renaissance; at that time, for example, Italian painters stopped painting in the static ancient way and introduced *perspective* into their works. After stating some generalities about abstract projective spaces, we focus on the study of the real projective space and the projective group (the group of

homographies.) We analyse the behaviour of projective lines, prove the Fundamental Theorem, present the Duality principle, a phenomenon appearing only in Projective Geometry, and we introduce the cross-ratio of four collinear points, which is also the only projective invariant. Projective Geometry contains Affine Geometry, which in turn contains Euclidean Geometry, but none of the invariants of all these Geometries is a projective invariant-parallelism in (compact) projective space does not exist. The chapter closes with two sections, the first of which is dedicated to perspectivities and the second to the classical Theorems of Pappus and Desargues.

In the eighth chapter we deal with the Inversive Geometry. Its beginnings may be found in Euclid and the geometric inversion, as well as in Claudius Ptolemy, who is considered to have known stereographic projection. The latter is the tool we use to introduce the underlying space of Inversive Geometry: that is, the extended real space, which is nothing more than the one point compactification of the real space which is done by adding a point at infinity. On the other hand, the inversions on spheres and on hyperplanes are the constituent elements of the respective group of transformations, that is, of the General Möbius group. We study this group in detail, focusing on its action on the extended real space. After proving the Fundamental Theorem, we show the connection of the Möbius group with the projective group in low dimensions.

Hyperbolic Geometry, which occupies the ninth and last chapter of the book, arises as a natural continuation of Inversive Geometry, in the sense that hyperbolic transformations are Poincaré extensions of Möbius transformations that preserve the upper half-space at $(n + 1)$- dimensions. These transformations are exactly the isometries of the hyperbolic metric, whose geodesics are either semicircles or half straight lines, both orthogonal to the boundary. In this environment, from a point outside the hyperbolic straight line there are infinite parallels to the given straight line. We study the action of hyperbolic transformations, emphasising the two-dimensional case, and dedicate a section to hyperbolic trigonometry. We see here that the sum of the angles of a hyperbolic triangle is smaller than two right angles, and we prove the corresponding Laws of Cosines and Sines. As in Spherical Trigonometry, the existence of the Second Law of Cosines assures us that hyperbolic similarities are necessarily hyperbolic isometries. We close the chapter by presenting various models of Hyperbolic Geometry in the two-dimensional case, leaving it to be implied how the corresponding treatment is done in any dimension.

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Chapter 1
Historical Background



In this chapter we list various historical facts in relation to the beginnings of mathematics and geometry in particular. Starting from prehistory and crossing the first Mesopotamian civilisations, the Indians and the Chinese, the Greeks and the Arabs, we end up in the nineteenth century with the strict foundation of geometry by Hilbert and the revolutionary idea of Klein as it was described in the Erlangen Programme. There is an immense bibliography related to the History of Mathematics. For example, we refer the reader to Katz's detailed book [30], as well as to [14], [20], [52].

1.1 From Prehistoric Times to the Greeks

1.1.1 Prehistoric Times

It is thought that our prehistoric ancestors had a general sense of numbers and certainly knew the difference between one and two. But it took a great mental leap to invent a symbol for the meaning of "two". In the absence of settled agriculture and trade, there was little need for a formal number system. Early man regularly tracked events such as the phases of the Moon and the seasons. Some of the earliest evidence that humanity thought about numbers is from bone carvings in Africa dating from 35,000 to 20,000 years ago. But this is really just counting rather than mathematics per se.

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1.1.2 Mesopotamia and Egypt

Mathematics began to develop widely due to the bureaucratic needs of the first civilisations formed by the settlement of Sumerians and Egyptians in large cities and the development of agriculture and animal husbandry. Land measurement, individual taxation, etc., first appeared in the Sumerian and Egyptian cultures. Sumer (a place in Mesopotamia, roughly modern southern Iraq), is considered the birthplace of writing, the wheel, agriculture, the bow, the plough, irrigation and many other innovations and is called the *Cradle of Civilisation*. The Sumerians built the first great cities, such as Ur, Uruk and Umma, whose administration and bureaucratic needs brought world-historical discoveries to the fore. The Sumerians developed the earliest form of writing, the so-called *cuneiform* writing. Cuneiform characters were inscribed on baked clay tablets, for various purposes, commercial, didactic, and others. In the case of the Sumerians and Babylonians, who were their successors, we even have school exercises in arithmetic and geometric problems. The Sumerians were perhaps the first people to assign symbols to groups of objects in an attempt to make it easier to describe larger numbers. Their number system was sexagesimal (60 is the basis in this system) upon which the mathematics of the Sumerians and Babylonians were built. This probably arose from their astronomical observations and the lunar cycle. It is speculated that their progress in Mathematics was probably facilitated by the fact that 60 has many divisors; in fact, 60 is the smallest positive integer divisible by all positive integers from 1 to 6. The continuous use to this day of 60 seconds in a minute, 60 minutes in an hour, and 360 ($60 \cdot 6$) degrees in a circle, traces its origins to the ancient Sumerian/Babylonian system. For similar reasons 12 has historically been the basis for such a popular system (e.g. 12 months, 12 inches, 12 pennies, $2 \cdot 12$ hours, etc.). Babylonian tiles dating from about 1800 to 1600 BC., cover topics as varied as fractions, algebra, methods of solving linear, quadratic and even some cubic equations, and calculating regular reciprocals (pairs of numbers that are multiplied together to give 60). The geometry of the Sumerians and the Babylonians extended to calculating the areas of rectangles, triangles, and trapezoids, as well as calculating the volumes of simple shapes such as bricks and cylindres. Although the Sumerian pyramids (ziggurats) were re-deposition of parallelepiped volumes, they are considered the first forms of building pyramids used for both political (residence of the emperor) and religious purposes. The myth of the Tower of Babel clearly draws its origins from the great Sumerian ziggurats. Finally, the famous (and controversial) Plimpton 322 clay tablet, believed to date from 1800 BC, suggests that the Babylonians may have been practicing the Pythagorean Theorem many centuries before Pythagoras.

The first Egyptians settled along the fertile Nile Valley in about 6000 BC, and began to record the patterns of the lunar phases and seasons, both for agricultural and religious reasons. The surveyors of the Pharaohs (according to Herodotus, the *harpedonaptes*) were the ones responsible for measuring the Nile floods) used measurements based on body parts. The Egyptians are believed to have introduced

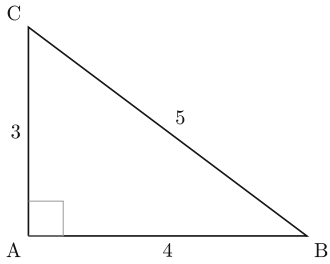


Fig. 1.1 Egyptian triangle

the earliest fully developed decimal (base 10) numerical system at least as early as 2700 BC.

The Rhind Papyrus, dating from around 1650 BC, is a kind of arithmetic and geometry instruction manual and gives us clear indications of how multiplication and division were carried out at that time. It also contains elements of other mathematical knowledge, such as unit fractions (fractions with a unit numerator), complex and prime numbers, arithmetic, geometric and harmonic means, how to solve first-order linear equations, and arithmetic and geometric series. The Berlin Papyrus, dating from around 1300 BC, shows that the ancient Egyptians could solve quadratic equations. The Egyptians approximated the area of the circle using figures whose area they knew. They observed that the area of a circle with a diameter of 9 units, for example, was approximately equal to the area of a square with sides of 8 units. Thus, the areas of circles of other diameters could be obtained by multiplying the diameter by $8/9$ and then squaring that number. This gives an efficient approximation of π with an error of less than 1%. The pyramids themselves are another indication of the complexity of Egyptian Mathematics. There is definite proof that the Egyptians knew the formula for the volume of a pyramid: $1/3$ times the height times the length by the width, plus that for the volume of the truncated pyramid. They also knew, long before Pythagoras, the rule that a triangle with sides of 3, 4, and 5 units yields a perfect right angle, and Egyptian builders used ropes tied at intervals of 3, 4, and 5 units to ensure the correct angles for their masonry (in fact, the 3-4-5 right triangle is often called *Egyptian triangle*) (Fig. 1.1).

1.1.3 India and China

Mathematics appeared in India around 1200 BC. Ancient Indian Mathematics (1200 BC–400 AD) as well as those of the middle period (400 AD–1200 AD) were all written in Sanskrit. They usually consisted of a section of *sutras* where a set of problems or rules were set out in verse for easy memorisation by students, followed by a second section containing prose commentary. In these comments, the solutions to the problems were explained. All mathematical works were spread orally until about 500 BC. From then on they were spread either orally or in writing through manuscripts. The oldest surviving Indian mathematical text is a text

written on birch bark dating from the seventh century AD. It is worth noting that to the Indian Mathematicians of post-Christian times we owe important discoveries, among others that of the zero and the decimal numbering system; we refer the interested reader to [40].

On the other hand, China has a huge history of contributions to human culture and the sciences in particular. Its geographical position and its isolation were the reasons why these contributions were slow to become known to the peoples of the West. From long ago, as early as the second millennium BC, the imperial polity and the pyramidal system of administration of the vast country brought about the need to use Mathematics in the vast bureaucratic machine of the state. We assume that the oldest mathematical text is the *Zhou Bi Suan Jing*: Numerology of the Gnomon and the Circular Paths of Heaven, dating from around 500–200 BC. Several texts survive from around this time-in one of them *Jiŭzhāng Suànshù*: A Treatise on the Nine Chapters of the Mathematical Art, there is an approximation of π and a version of the Pythagorean Theorem. Chinese mathematics used the decimal number system and was written on bamboo sticks. Their heyday continued in post-Christian times, and among the many scholars of this period that we could mention, we single out the imperial general and mathematician Sun-Tzu (544–496 BC) who, in addition to the book he wrote on the Art of War, is also known for the Remainder Theorem in Number Theory. For details on Chinese mathematics we refer, among others, to [35].

1.2 Greeks

Once the Greeks emerged from the Dark Age (1200–800 BC), they began to expand their sphere of influence in Asia Minor through colonies, as well as in Mesopotamia and the shores of the Mediterranean through trade. The Greeks adopted and adapted useful elements from the societies they occupied/interacted with, the first perhaps being the Phoenician alphabet, which, after first adapting it to fit their language, spread it to others, among of which were the Latins. And for their Mathematics, they adopted a multitude of elements from both the Babylonians and the Egyptians. Soon, they began to make significant contributions and, for the first time, we can record *individual* contributions. Until the Hellenistic period and until about 400 AD, the Greeks were the protagonists in one of the most dramatic and important revolutions in mathematical thought of all time. The earliest Greek numeral system, known as Attic or Herodian, was fully developed by 450 BC, and was in regular use probably as early as the seventh century BC. It was a decimal system similar to the earlier Egyptian (and even more similar to the later Roman system), with symbols for 1, 5, 10, 50, 100, 500, and 1000 repeated as many times as needed to represent the desired number. Addition was done by adding the symbols to the numbers being added separately, and multiplication was a laborious process based on successive doublings (division was based on the reverse of this process). Later,

Fig. 1.2 Attic/Herodian system

Ι	1
ΙΙ	2
ΙΙΙ	3
ΙΙΙΙ	4
Γ	5
ΓΙ	6
ΓΙΙ	7
ΓΙΙΙ	8
ΓΙΙΙΙ	9
Δ	10
ΔΓ	15
ΔΔ	20
Γ^{Δ}	50
Η	100
Χ	1000
Μ	10000

the alphanumeric system was adopted, which was also rather unwieldy (Figs. 1.2 and 1.3).

1.2.1 Thales and the Beginning of Geometry

Most of the Greek mathematics was based on geometry *of proof*. Thales of Miletus, one of the Seven Sages of Ancient Greece, who lived on the coast of Ionia in the first half of the sixth century BC, is usually considered the first to set guidelines for the abstract development of geometry, although what we know of his work (as on congruent and right triangles) now seems very rudimentary. We know very little about Thales, the first philosopher and mathematician in history. But what we do know is that he was a merchant and that he had traveled to Egypt. It is very likely that there he came into contact with people of the priesthood and became aware



Fig. 1.3 Thales, 626/623–548/545 BC



Fig. 1.4 Pythagoras, c. 570–c. 495 BC

of some of the mathematical achievements of the Egyptians. Many elementary theorems of geometry are attributed to Thales. Legend says that the first proven theorem was the one which states that *the opposite angles of an isosceles triangle are equal.* In his joy at this discovery, Thales offered a hectumbi (a hundred oxen) to Poseidon. Another theorem of Thales is the one which states that *if the side of an inscribed triangle is equal to the diametre of the circle then it faces a right angle.* But Thales is also credited with another great and by no means elementary theorem, the famous Thales's Intercept Theorem, about the ratios of line segments created if two intersecting lines are intersected by a pair of parallel lines (Fig. 1.4). (For a modern proof, see Exercise 33 in Chap. 5.)

1.2.2 Pythagoras

To a very large extent, the legend of Pythagoras of Samos became synonymous with the birth of Greek Mathematics. Indeed, he is believed to have coined both the words "philosophy" ("love of wisdom") and "mathematics" ("that which is learned"). Pythagoras was perhaps the first to realise that a complete system of mathematics

could be constructed, where geometric elements corresponded to numbers (in this case, positive expressions). The Pythagorean Theorem is one of the most well-known mathematical theorems. In any case, Greek mathematics was not limited to one man at all, it's just that Pythagoras is considered the first *true* mathematician and as many claim, we owe Pythagoras Pure Mathematics. Pythagoras himself was a rather controversial figure. Banished from his homeland of Samos because of his absolutist views, he founded a School in Croton in Lower Italy around 530 BC. He himself left no mathematical writings, and much of what we know about Pythagorean thought comes from the writings of Philolaus and other later Pythagorean scholars. It is not at all clear whether many (or indeed any) of the theorems attributed to him were actually proved by Pythagoras personally or by his followers. The members of his School were divided into the *mathematicians* (or *students*), who expanded and developed the more mathematical and scientific work begun by Pythagoras himself, and the *acousticians* (or *listeners*), who focused on the more religious and ritualistic aspects of his teachings. There was always a friction between the two groups and eventually the sect became embroiled in bitter feuds until it was finally disbanded. Resentment against Pythagorean mysticism escalated in 460 BC: all their meeting places were burned and destroyed, with at least 50 members of the School killed in Croton alone. The Pythagorean philosophy had the principle *Number is the beginning of everything*, or *God is number*. The number *one* was the generator of all numbers, two represented opinion, three harmony, four justice, five marriage, six creation, seven the seven planets, etc. The most sacred number of all was *Tetraktaeus* or ten, a *triangular* number consisting of the sum 1+2+3+4. This is a great intellectual achievement of the Pythagoreans: they deduced the special position of the number 10 from an abstract mathematical argument and not from something as simple as counting the fingers of two hands. Before Pythagoras, geometry was simply a collection of rules derived from empirical measurements; Pythagoras discovered that a complete system of Mathematics could be constructed, where geometrical elements corresponded to numbers, and where positive integers and their ratios were the only ones necessary to create a whole system of logic and truth. The name of Pythagoras remained in eternity mainly for the theorem that bears his name: *in a right-angled triangle, the square of the hypotenuse is equal to the sum of the squares of the vertical sides.* The Pythagorean Theorem and the properties of right triangles are the oldest and most widespread mathematical achievement after elementary arithmetic and geometry. While it has been traced to ancient mathematical texts of prior civilisations, it is Pythagoras who gave the theorem its definitive form, although it is not clear whether Pythagoras himself proved it definitively or merely described it. It soon became apparent that there can be a right triangle with vertical sides both equal to 1, as the Babylonians had discovered centuries earlier. However, it was Pythagoras' student Hippasus Metapontinus (c. 530–c. 450 BC) who, in trying to calculate the value of $\sqrt{2}$ (the hypotenuse), found that it was not possible to express it as a fraction. He thus demonstrated the possible existence of a completely new world of numbers, the *irrational* numbers. This very discovery destroyed the elegant mathematical world of Pythagorean fractions for good. The existence of a number that could

not be expressed as the ratio of two of God's creations (which was how the Pythagoreans thought of positive integers) jeopardised their entire cult system. The unfortunate Hippasus was apparently drowned by the mystical Pythagoreans, thus being punished for communicating this important discovery to the outside world. But the replacement of the idea of the deity of the integers by the richer concept of the continuum was a highly significant development in mathematics. It marked the actual birth of Greek Geometry, which deals with lines, planes and angles, which are continuous rather than discrete objects. Among others, Pythagoras and his students are believed to: proved that the sum of the angles of a triangle is equal to two right angles, were able to construct figures of given area and use simple geometric algebra, laid the foundations of number theory, and discovered that the intervals between harmonic musical notes always have positive integer ratios. Non-integer ratios tend to give unbearable sounds. Pythagoras described the first four tones of musical harmony: the octave (1:1), the perfect fifth (3:2), the perfect fourth (4:3) and the major third (5:4). The oldest way of tuning the 12-note chromatic scale is known as Pythagorean tuning.

1.2.3 Plato and Mathematical Rigour

Although usually referred to as a great philosopher, Plato was also one of the most important ancient Greek mathematicians. Inspired by Pythagoras, he founded his Academy in Athens in 387 BC, where he emphasised mathematics as a way of understanding reality (Fig. 1.5).

In particular, he was convinced that geometry was the key to unlocking the secrets of the Universe. Above the entrance of the Academy it was written: "No one ignorant of geometry may enter". Plato played an important role in encouraging and inspiring Greek intellectuals to study mathematics and philosophy. His Academy taught mathematics as a branch of philosophy, as Pythagoras had done, and the



Fig. 1.5 Plato, 428/427–348/3447 BC

first ten years of his fifteen-year course at the Academy were devoted to the study of geometry, astronomy, and music. Plato became known as the *creator of mathematics*, and his Academy boasted some of the most important mathematicians of the ancient world, including Eudoxus, Theaetetus of Athens (c. 417–c. 369 BC) and Archytas of Taras, 435/410–360/350 BC. He demanded of his students precise definitions, clearly stated assumptions, and logical inductive proof. He insisted that geometrical proofs should be demonstrated without any aids but the ruler and the compass. Among the many mathematical problems that Plato set for the investigation of his students were the so-called Three Classical Problems that we will see in the next section. To some extent, these problems have been identified with Plato, although he was not the first to raise them. The mathematician Plato is perhaps best known for identifying the five regular symmetrical three-dimensional figures, which he saw as the basis of the entire Universe, and which have become known as the Platonic Solids (see Sect. 6.4: these are the *tetrahedron* (made of four regular triangles and which for Plato represented fire), the *octahedron* (consisting of eight triangles, representing air), the *icosahedron* (consisting of twenty triangles, representing water), the *cube* (consisting of six squares, representing earth), and the *dodecahedron* consisting of twelve pentagons, which Plato vaguely described as the ones *God used to arrange the constellations in the whole sky*.

1.2.4 Three Problems and the First Elements

Three specific geometric problems, often referred to as the Three Classical Problems, all of which must be solved with a ruler and compass, date from the earliest days of Greek geometry: the *squaring of the circle* (to construct a square with area equal to the area of a given circle), the *Delian problem* (to construct a cube with twice the volume of a given cube) and the *angle trisection* (to construct an angle equal to the one third of a given angle). These unsolved problems greatly influenced future geometry and led to many fruitful discoveries, although their actual solutions (or, as it turned out, proofs of their insolvability) had to wait until the nineteenth century (Fig. 1.6).

Hippocrates of Chios (not to be confused with the great physician Hippocrates of Kos), was the mathematician who grappled with these problems during the fifth century BC, calculating the area of the famous menisci (lunes) that bear his name. It is known that he was born in Chios around 470 BC, but he was probably a ship owner and was involved in transporting wheat from Byzantium to Piraeus. Around

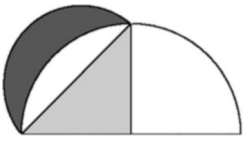


Fig. 1.6 Meniscus (lune)



Fig. 1.7 Democritus, c. 460–c. 370 BC

440 BC, during the war between the Athenians and the Samians, pirates from the wider Byzantium region seized his cargo ships and he traveled to Athens to file a claim for damages. Because the trial of the case took too long, Hippocrates, using his time, attended courses in the philosophical schools of the city, especially on geometry. His brilliant (and lost) book *The Elements*, dated around 440 BC, was the first collection of the elements of geometry, and his work was an important source for Euclid's later work. Hippocrates is considered to be the father of the method of *reductio ad absurdum-proof by contradiction* (Fig. 1.7).

1.2.5 Democritus and Eudoxus

Democritus of Abdera (c. 460–c. 370 BC), famous for his prophetic ideas about atoms, was also a pioneer of mathematics and geometry in the fifth–fourth century BC. His works though have all been lost. We know that he was one of the first to notice that a cone (resp. a pyramid) has a volume equal to one third of the volume of a cylindre (resp. a prism) with the same base and height. Democritus was perhaps the first to seriously consider the analysis of objects into an infinite number of cross-sections. Eudoxus of Cnidos, a student of Plato, is usually credited with the *method of exhaustion*, which was later ingeniously developed by Archimedes. This method is an early method of integration by successive approximations, which Eudoxus used to calculate the volumes of the pyramid and the cone. Eudoxus also developed a general theory of proportions, which was applicable to abstract quantities (which cannot be expressed as the ratio of two whole numbers), thus extending the incomplete ideas of Pythagoras.

1.2.6 Euclid and His Elements

Euclid of Alexandria, (c. 350–270 BC) the so-called *Father of Geometry* and *Elementor*, lived and flourished in Alexandria, Egypt about 300 BC, during the reign of Ptolemy I. Almost nothing is known about his life, and no description of his physical appearance has survived from antiquity. Thus, later depictions of him with a long beard and cloth hat in works of art are products of the Renaissance artist's imagination. He probably studied at Plato's Academy in Athens, but, by the time of Euclid, Alexandria under the patronage of the Ptolemies and with its famous and comprehensive Library, had already become a worthy rival to the great Academy.

Euclid wrote perhaps the most important and successful mathematical book of all time, *The Elements*. The Elements is the culmination of the mathematical revolution that had taken place in Greece up to that time. The Elements is a lucid and comprehensive collection and explanation of all the known mathematics of Euclid's time, including the works of Pythagoras, Hippocrates, Theaetetus, and Eudoxus of Cnidos (c. 390–c. 340 BC). Altogether, the Elements consists of 13 books, containing 465 theorems (propositions) and their proofs, described in a clean, concise, logical and elegant style. At the beginning of Book I, Euclid lists the framework of general assumptions within which he will move from now on. It lists 23 definitions, 5 postulates and then 5 general concepts to complete the previous ones. For the proofs only the ruler and the compass are allowed, in Plato's precepts. Among many other mathematical gems, the thirteen books of the Elements contain: formulae for calculating the volume of solids such as cones, pyramids and cylindres; algorithms for finding the greatest common divisor and least common multiple of two numbers; a proof of the Pythagorean Theorem and its converse; the definitive proof that there can be only five possible regular Platonic solids. The Elements also include a series of theorems about the properties of numbers, marking the first real principles of number theory. Euclid proved what has become known as the *Fundamental Theorem of Arithmetic*, he was the first to realise and prove that there are infinite prime numbers, and was the first to define the Golden Ratio (ϕ, roughly equal to 1.618), through the *mean ratio* ($AB{:}AC = AC{:}CB$) and showed its natural occurrence in many geometric shapes.

All in all, Euclid reworked and refined the mathematical concepts of his predecessors into a coherent whole, which became known as *Euclidean Geometry*. Euclidean Geometry is still as valid today as it was 2300 years ago, even in higher mathematics dealing with higher dimensional spaces. Only with the work of Bolyai, Lobachevsky and Riemann in the first half of the nineteenth century was any kind of non-Euclidean Geometry even considered. The Elements were the definitive book on geometry and mathematics for over two millennia, surviving in Europe during the Middle Ages through Arabic translations. Euclid set, forever, the model for mathematical writing, which uses logical deductions from initial assumptions in order to prove theorems (Fig. 1.8).



Fig. 1.8 Archimedes, c. 287–c. 212 BC

1.3 20 Centuries in One Breath

A little later than Euclid was Archimedes of Syracuse, the greatest Greek mathematician, physicist and engineer. Even a simple citation of Archimedes' achievements and the legends surrounding his person, is a rather lengthy process; we indicatively refer to [2]. We simply mention here that Archimedes, using the ingenious Eudoxus' *method of descent*, calculated the first integral (area of a parabolic section). Using methods beyond the compass and the ruler, he constructed complex curves, such as the Archimedean spiral. But Archimedes did not leave a School behind him, and Hellenistic Mathematics, for all its elegance, did not advance to a point beyond what the great geometer had reached. In the period approximately up to 400 AD, where the collapse of the ancient world was largely completed, we note the contribution of mathematicians such as Eratosthenes of Cyrene (c. 276 BC–c. 195/194 BC), Menaechmus of Alopeconnesus (c. 380–c. 320 BC), Claudius Ptolemy, Menelaus of Alexandria (c. 70–140 AD), Pappus of Alexandria and finally Theon (335–405 AD) and his daughter, the first recorded woman philosopher and mathematician Hypatia of Alexandria (c. 350/370–415 AD). Some of them already noticed some inconsistencies in Euclid's magnificent work; Pappus in particular, wrote commentaries on Book I.

An important observation that had already been made since then was that the Fifth Postulate (axiom of parallels) seemed independent of the other four Euclidean postulates, and there were attempts to prove it as a theorem. With the final ascendancy of Christianity, the closing of the philosophical schools and the entry into the Dark Ages, all mathematical activity ceased altogether. Until the Renaissance, the Arabs were the custodians of mathematics. Coming through their conquests into contact with Greek mathematics, they not only translated and elaborated it, but also made remarkable progress in their heyday by presenting an abundance of great mathematical achievements. It is known that the words *algebra* (al-gebr) and *algorithm* are of Arabic origin, the latter being a corruption of the surname of the great algebraist Al-Qwarismi (Muhammad ibn Musa Al-Qwarismi, c. 780–c. 850). The Arabs distinguished themselves, among other things, in the solution of equations and the transmission of the knowledge they had in Renaissance Italy, led e.g. the Italian mathematician and physician Cardano (Gerolamo Cardano,



Fig. 1.9 Carl Friedrich Gauss, 1777–1855

1501–1576) in finding formulae for the solution of the cubic equation in the fifteenth century AD. But the great leap in mathematics happened due to the works of Newton and Leibniz on calculus as well as Descartes with the introduction of his coordinates. After them, mathematics never looked back. At the beginning of the nineteenth century perhaps the most shocking event for Mathematics, since the time of Euclid occurred. Almost simultaneously, and independently of each other, Bolyai and Lobachevsky observed that simply by replacing the Fifth Postulate with another, a new, non-Euclidean Geometry, which would henceforth be known as *Hyperbolic Geometry*, results. It is thought that independently of these two, the great German mathematician Gauss (Carl Friedrich Gauss, 1777–1855) had also conceived the concept of a non-Euclidean geometry, but hesitated to publish it (Fig. 1.9).

A little later Riemann (Fig. 1.10), a student of Gauss, presents his revolutionary idea about the possibility of constructing various kinds of geometries, each time specifying the way we measure the distance on a geometric object. Klein, through the famous Erlangen Programme, considered Geometries as pairs consisting of an object and a group acting on the object through transformations (bijective mappings). Klein's idea method was fundamentally innovative in the following ways first, he emphasised Projective Geometry as the unifying frame for all geometries that he considered. Second, he introduced the utilisation of group theory to describe



Fig. 1.10 Bernhardt Riemann, 1826–1866

geometric features and in particular *symmetry*, and finally he made explicit the geometrical invariants as features of every geometry he considered. For instance, there are Euclidean invariants, e.g., lengths and angles which are not projective invariants. And Euclid? In all this modern consideration, Euclidean Geometry still plays a central role, at least locally. So the need for the Mathematicians of the nineteenth century, and in the context of the flourishing of Set Theory, to free Euclidean Geometry from any inconsistencies was unmistakable. This grandiose task was undertaken by many, but the famous German mathematician Hilbert (David Hilbert, 1862–1943) was the first to complete it.

1.4 Modern Foundation of Geometry

Standards of mathematical rigor have changed dramatically since Euclid wrote the Elements. There was a substantial need for a review of the Elements following the discovery of non-Euclidean geometries. There exists indeed a list of problematic points in Euclid's axiomatic foundation, including par example, the use of superposition in certain proofs without an axiomatic justification for it method; the lack of a continuity concept, the lack of clarity about whether a line is infinite or unbounded, etc. For the modern needs, it turned out that the list of axioms in the Elements was not exhaustive, but represented the principles that appeared to be the most important. Later, as we will see a little below, many Mathematicians incorporated new axioms into the official so-called axiom list of Euclid, greatly expanding it.

The foundation of Geometry is the study of Geometries as axiomatic systems and the first axiomatic system was that described by the demands, definitions, and common concepts of Euclid's Elements. The term *Axiomatic Geometry* can be applied to any Geometry developed by means of a system of axioms, but is often used to mean the Euclidean Geometry first studied from this point of view.

In an official system they play a special role *completeness* and *independence*. These are important mathematical properties, but there are also issues related to the teaching of Geometry that appear as part of the whole picture. To address the issues in the Euclidean axiomatic system, various authors have attempted either to fill in the gaps in Euclid's presentation, with the most notable effort being that of Hilbert, either to present an axiomatic system around different concepts, as Birkhoff (George David Birkhoff, 1884–1944) did. Of the pre-Hilbertian efforts, those of the German-Jewish mathematician Pasch (Moritz Pasch, 1843–1930), the Italian mathematician and glottologist Peano (Giuseppe Peano, 1858–1932) and the Italian mathematician Pieri (Mario Pieri, 1860–1913) stand out. In particular, Pasch laid the foundations of the modern axiomatic method; considered the meaning of primary terms and, along with the axioms he constructed a formalistic system relatively free from intuitive influences: he thought that the only place where intuition should play a role was in deciding which should be primary concepts and which axioms. Thus, for Pasch, the point is a primary concept but the line is not, since we have a good one intuition about points, but no one has ever seen or experienced an infinite straight line. The

primary concept Pasch uses in place of the line is the straight line segment. Pasch's work inspired both Hilbert and Peano. Peano's 1889 work on Geometry, which is largely a translation of Pasch's treatise on symbolism of logic (invented by Peano), uses the primary concepts of point and intermediate. Peano breaks for good the empirical link in the choice of primary terms and postulates that Pasch required. For Peano, the whole system is purely formalistic, devoid of any empirical information. Pieri took a different approach and considered a system in which there were only two primary concepts, those of point and motion. Pasch had used four primary concepts and Peano had reduced them to three, but both these approaches were based on the concept of the intermediate which Pieri replaced by the concept of movement. In 1905 Pieri gave the first axiomatic treatment in Complex Projective Geometry that did not rely on the construction of Real Projective Geometry.

1.4.1 Hilbert's Foundation

At the University of Göttingen, during the winter term of 1898–1899, the famous German mathematician Hilbert presented a series of lectures on Pieri's foundations. At Klein's request, Hilbert wrote the notes of these lectures in time for the summer of 1899, to be presented at a ceremony dedicated to the memory of Gauss and W. Weber held at the university. These notes were published in June 1899 under the title *Grundlagen der Geometrie* (Fundamentals of Geometry), [27]. His influence of this book was immediate. Eves states in his *History of Mathematics*: *The development of a set of axioms for Euclidean geometry does not depart too far from the spirit of Euclid. By using a minimum notation, Hilbert managed to convince mathematicians, in much larger area than Pasch and Peano, for the purely hypothetical-inductive nature of geometry. But, the influence of Hilbert's work went far beyond this: for, supported by the author's authority, he implanted the axiom as a key method not only in geometry, but also in virtually every other branch of mathematics. The motivation for the development of the foundations of mathematics provided in Hilbert's little book is hard to overestimate. Without the strange symbolism of the works of Pasch and Peano, the work of Hilbert can be read by any intelligent high school student.*

Hilbert's axiomatic system considers six primary concepts: point, line, plane, between, contains, and congruence. All points, lines, and planes in the following axioms are distinct unless otherwise noted.

I. Incidence.

1. For every two points A and B there is a line α that contains both.
2. For every two points there is no more than one line that contains both.
3. There are at least two points on a line. There are at least three points that are not on the same line.
4. For every three points A, B, C that are not on the same line there is a plane α that contains all the points. For every curve there is a point that lies on the plane.

5. For every three points A, B, C that are not on the same line, there is not more than one levels that contain them all.
6. If two points A, B of a line lie in plane α, then each point lies in α.
7. If two planes α, β have a common point A, then they have at least one second common point B.
8. There are at least four points that are not on the same plane.

II. **Ordering.**

1. If the point B is between the points A and C, B is also between C and A and there is a line containing the distinct points A, B, C.
2. If A and C are two points on a line, then there is at least one point B that lies between A and C.
3. Of the three points lying on a line, there is not more than one lying between the other two.
4. *Pasch's Axiom*: Let A, B, C be three points that are not on the same line and let α be a line that lies on the plane ABC and does not pass through any of the points A, B, C. Then, if the line α passes through point of the segment AB, will also pass through either a point of the segment BC or a point of section AC.

III. **Congruence.**

1. If A, B are two points on the line α, and if A' is a point on the same or another line α', then, on on a given half-line of α' starting at A', we can always find a point B' such that segment AB is equivalent to segment $A'B'$ ($AB \simeq A'B'$).
2. If the line segment AB is equivalent to the segment $A'B'$ and also to the segment $A''B''$, then the segment $A'B'$ is equivalent to the segment $A''B''$ ($AB \simeq A'B'$, $AB \simeq A'B'$, then $A'B' \simeq A'B'$.
3. Let AB and BC be two line segments α with only common point the B, and let $A'B'$ and $B'C'$ be two segments of the same or another line with only common point B'. Then, if $AB \simeq A'B'$ and $BC \simeq B'C'$, we have $AC \simeq A'C'$.
4. If $\angle ABC$ is an angle and if $B'C'$ is a segment, then there is exactly one segment $B'A'$ on each side of $B'C'$ such that $\angle A'B'C'$ be equivalent to $\angle ABC$. Furthermore, every angle is equivalent to itself.
5. The PGP criterion of the included angle.

IV. **Parallels.**

1. Euclidean Axiom: from a point outside a given line, passes at most one parallel to the line.

V. **Continuity.**

1. (Archimedean Axiom). If AB and CD are segments, then there exists a number n such that n copies of CD made continuously from A along the radius AB will pass the point B.



Fig. 1.11 David Hilbert, 1862–1943

2. (Completeness of lines). It is impossible to extend the set of points of a line with the arrangement and equivalence relations that would preserve the existing relations between the original elements as well as the fundamental properties of the arrangement of lines and equivalence (Axioms I–III and V-1).

Going beyond establishing a satisfactory set of axioms, Hilbert also proved the consistency of his system with respect to real number theory by constructing a model of the system from the real numbers. He proved the independence of some of his axioms by constructing models of geometries that satisfy all but one of the axioms under consideration. Thus, there are examples of Geometries that satisfy all except the Archimedean Axiom (non-Archimedean Geometry), except the Parallel Axiom IV.1 (non-Euclidean geometries), and so on (Fig. 1.11).

Using the same technique he also showed how certain important theorems depended on certain postulates and were independent of others. Some of his models were very complicated and other mathematicians tried to simplify them. For example, Hilbert's model which he used to show the independence of Desargues' Theorem (that we will see in Chap. 7) from certain axioms, led the American astronomer Moulton (Forest Ray Moulton, 1872–1952) to discover the non-Desarguesian Moulton plane. During the twentieth century there was a great debate about how Hilbert's view could pass into the teaching of school Geometry. This prompted the American mathematician MacLane (Leslie Saunders MacLane, 1909–2005) to introduce yet another foundation of Euclidean Geometry accessible to students. With the discovery of non-Euclidean geometries, the problem of their foundation arose among other things. Klein's genius idea to define as Geometry any pair consisting of a non-empty space and a group of its transformations, frees us from this problem. In such Geometries, there are no axioms but only theorems. The next section serves also as an introduction to the next chapter and describes in brief Klein's idea of ordering the geometric landscape. All notions and results referred below are described in detail into the next chapters of this book.

1.4.2 The Kleinian Hierarchy of Geometries

Felix Klein's Erlangen Programme revolutionised our understanding of geometry. Moving beyond the traditional focus on specific axioms or metric properties, Klein proposed a radical redefinition: *a geometry is fundamentally the study of the properties of a space that remain invariant under the action of a specific group of transformations.* This powerful idea not only unified diverse geometric systems but also naturally led to the concept of the *Kleinian hierarchy of geometries*.

The hierarchy arises from the relationship between different transformation groups. Larger, more general transformation groups preserve fewer properties, defining "broader" geometries. Smaller, more specialised subgroups preserve more properties, defining "narrower," more restrictive geometries. Crucially, if group H is a subgroup of group G, then the geometry defined by group H is a *subgeometry* of the geometry defined by group G. This means every invariant property of the Geometry of group G is automatically an invariant property of the Geometry of group H, but the latter possesses additional invariants not respected by group G. We shall see all this in detail in the next chapter. For the moment, we describe the structure of the hierarchy:

(1) *Projective Geometry*: occupying the broadest tier is projective geometry, defined by the *projective group*. This group includes all transformations that map straight lines to straight lines (collineations) and preserves the cross-ratio of four collinear points. It discards notions of distance, angle, and parallelism, focusing solely on incidence properties (points lying on lines, lines intersecting at points). Its invariants are projective properties like collinearity, concurrency, and cross-ratio. *All other classical geometries in this hierarchy are subgeometries of projective geometry.*
(2) *Affine Geometry:* restricting the projective group to transformations that map parallel lines to parallel lines (or preserve a designated "line at infinity") yields the *affine group*. Affine geometry preserves parallelism, ratios of lengths along parallel lines, and convexity, but still discards absolute distances and angles. Straight lines remain straight, but circles can become ellipses.
(3) *Metric Geometries:* Further restricting the affine group leads to geometries incorporating distance and angle:

- *Euclidean Geometry:* defined by the *group of Euclidean isometries* preserving absolute distances and angles. This includes rotations, translations, and reflections. Circles remain circles, triangles have fixed side lengths and angles.
- *Similarity Geometry:* defined by the *similarity group*, preserving angles and ratios of distances (shape), but not absolute size. Circles map to circles, angles are invariant, but lengths scale uniformly.
- *Elliptic/Hyperbolic (Non-Euclidean) Geometries:* defined by their respective isometry groups preserving the constant curvature metric on the sphere

(elliptic) or pseudo-sphere (hyperbolic). These preserve the intrinsic distances and angles within their curved spaces.

Klein's hierarchy provides a profound structural framework: it revealed deep connections between seemingly disparate geometries (e.g., Euclidean and non-Euclidean) by placing them within a single, group-theoretic framework relative to projective geometry. It offers a principled way to classify geometries based on their symmetry groups and the properties they preserve. It clarifies the relationships between geometries, showing how stricter geometries inherit invariants from broader ones while adding new ones.

The Erlangen Programme's group-theoretic perspective paved the way for the development of differential geometry (where É. Cartan generalised Klein's ideas using moving frames and Lie groups) and modern algebraic geometry. In essence, the Kleinian hierarchy organises the geometric universe by the *degree of symmetry* allowed. Starting from the highly symmetric yet property-poor projective plane, imposing progressively stricter conditions on the allowed transformations (smaller subgroups) carves out nested subgeometries, each preserving a richer set of specific properties like parallelism, similarity, or rigid congruence, culminating in the familiar, highly constrained world of Euclidean space. This elegant structure remains a cornerstone of geometric understanding.

Chapter 2
Kleinian Geometry



The presentation of mathematics in schools should be psychological and not systematic. The teacher, so to speak, should be a diplomat. He must take account of the psychic processes in the boy in order to grip his interest, and he will succeed only if he presents things in a form intuitively comprehensible. A more abstract presentation is only possible in the upper classes.

—Felix Klein

Groups play a central role in the discussion of this book and in geometry in particular. In 1872 Felix Klein (Fig. 2.1) published the Erlangen Program in which he proposed a system for classifying and comparing the geometries that had existed up to that time. Klein's idea was to use algebra to describe the transformations that preserve the invariants of the underlying space of the geometry, see [32]. The transformations form an algebraic group that acts on the space, making it *homogeneous.* We may imagine a homogeneous space as follows: standing at any point in this space, we see the same picture. It was Klein who first noticed that most (if not all) interesting geometric objects can be represented by quotients of groups. Furthermore, as we will see below, a Kleinian geometry is nothing more than a non-empty space equipped with a group of transformations acting on this space. The strict definition of a Kleinian Geometry due to Cartan, who also developed Klein's programme, uses Lie groups whose definition is beyond the scope of this book. We refer the interested reader to [47], but also to [4]. In this chapter, we will give the definition as well as the elements of Kleinian Geometry that we use, such as e.g., symmetries. We will close the chapter with the indicative example of the familiar Euclidean Planar Geometry viewed in Klein's way. Besides a section about group actions which can be found below, all the necessary elements from group theory are in Appendix B. There, we present the elements of group theory that are necessary for the better understanding of the text. In particular we stress on the general linear group and its subgroups, see Example B.1.3.

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I. D. Platis, *Geometries*, La Matematica per il 3+2 178,
https://doi.org/10.1007/978-3-032-08016-5_2



Fig. 2.1 Felix Klein, 1849–1925

There are a multitude of books on the subject, illustratively we mention here [19], [41], [26], [46], [48] for an exhaustive study of the subject as well as [4], [45] for a more geometric perspective of groups.

2.1 Geometry According to Klein

2.1.1 Group Actions

Groups can act either from the left or from the right on a set, assigning a permutation of the points of the set to an element of the group.

Definition 2.1.1 A group G is said to *act from the left* on a set $X \neq \emptyset$, if there exists a mapping

$$G \times X \to X, \quad (g, x) \mapsto gx, \ g \in G, \ x \in X,$$

such that:

1. $ex = x$ for each $x \in X$,
2. $g(hx) = (gh)x$, for each $g, h \in G$ and for each $x \in X$.

For each given $x \in X$, the set

$$\mathrm{Orb}(x) = \{gx \mid g \in G\},$$

is called the *orbit* of x. On each orbit $\mathrm{Orb}(x)$, we define an operation

$$(gx) \cdot (hx) = (gh)x$$

and with this operation the orbit acquires a group structure isomorphic to G. If G acts from the left on X, we define the relation

$$x \sim y \iff x \in \mathrm{Orb}(y).$$

This is an equivalence relation and the set of equivalence classes (orbits) is called *quotient space of X by G* or *orbit space* of the action of G on X, and is denoted by X/G (rarely by G/X).

Definition 2.1.2 A left action of G on X is called *transitive* if for every $y \in X$ there exists $x \in X$ such that $y = gx$ for some $g \in G$.

In the case of a transitive action, the quotient space X/G consists of a single element and then X is called *homogeneous*.

The *stabiliser* of an element $x \in X$ is the set

$$\mathrm{Stab}(x) = \{g \in G \mid gx = x\}.$$

Definition 2.1.3 A left action of G on X is called *free* if all stabilisers are trivial.

In the case of an action that is transitive and free, X is called *principally homogeneous*.

2.1.2 Definition and Properties of Kleinian Geometries

Let $X \neq \emptyset$. A *transformation g of X* is a bijective map $g : X \to X$. The set of all transformations of X is denoted by $\mathrm{Sym}(X)$ and is called the *symmetry group of X*.

Proposition 2.1.4 *The set* $\mathrm{Sym}(X)$ *is a group; its operation is composition of mappings.*

Proof It is clear that if $f, g \in \mathrm{Sym}(X)$ then the same holds for the composite transformation $f \circ g$. Associativity is a standard property of composition, the neutral element id_X is the identity map of X and finally, for each $f \in \mathrm{Sym}(X)$ we have that its inverse f^{-1} is also in $\mathrm{Sym}(X)$ and $f \circ f^{-1} = id_X$.

Definition 2.1.5 A *Kleinian Geometry* is a pair (X, G), where $X \neq \emptyset$ and $G < \mathrm{Sym}(X)$, that is, G is a subgroup of the symmetric group of X. The *points* of the geometry (X, G) are the elements of the set X and the *figures* of the geometry (X, G) are the subsets of X.

The transformation group G acts on X:

$$(g, x) \mapsto gx \,(= g(x)).$$

(The choice of the left action is arbitrary). Two figures P, Q are called *G-equivalent* if there exists a $g \in G$ such that $g(P) = Q$. A *geometric property* (G-property) is a property of X that remains invariant under the action of G.

Remark 2.1.6 If two figures P, Q are G-equivalent then they have identical G-properties, due to the invariance of the G-properties under the action of G.

The study of geometry (X, G) consists of:

1. the description of G-equivalence classes and in particular the orbit space X/G;
2. the determination of G-properties;
3. the theorems concerning G-equivalence classes and G-properties.

In Kleinian Geometry there are no axioms, only theorems. This has the advantage that such a "geometry" is always consistent. With the axiomatic approach, it is always possible that the axioms lead to paradoxes. In such a case, there can be no "model" for Geometry.

Another advantage of the Kleinian approach is that it allows us to relate different geometries. Suppose that G is the transformation group of the set X, and that H is a subgroup of G. Then, H is also a transformation group of X, and thus defines another geometry on X.

Theorem 2.1.7 (K1) *Suppose that G is a transformation group of a set X, and that H is a subgroup of G. If two figures are H-equivalent, then they are also G-equivalent.*

Proof Suppose that P and Q are H-equivalent. Then there exists a $h \in H$ with $h(P) = Q$. Since $H < G$, $h \in G$, so P and Q are G-equivalent. □

Remark 2.1.8 It follows that every G-equivalence class is the union of H-equivalence classes, so that in the geometry defined by G there are fewer (but larger) equivalence classes.

Theorem 2.1.9 (K2) *Suppose that G is a transformation group of a set X, and H is a subgroup of G. Then, if D is a G-property, then it is also an H-property.*

Proof Suppose that D is a G-property, so it is invariant under all g in G. If $h \in H$, $h \in G$ since $H < G$, so D is invariant from h. □

We conclude that the geometry defined by G may have fewer properties.

2.1.3 Symmetries

Usually, we introduce the concept of a group by mentioning some symmetries of Euclidean figures. The simplest example is that of the group of permutations of n objects, (Exercise 8, Sect. 2.2). In the same way, we can have the same view for transformations of any geometry (X, G). We have already said that a figure Σ in the geometry (X, G) is just a subset of X. It is reasonable to consider transformations

that preserve Σ: if $f \in G$ with $f(\Sigma) = \Sigma$ we say that the figure remains *invariant* under the transformation f. The transformation $f \in G$ is called a G-symmetry of the figure Σ.

Definition 2.1.10 Given a figure Σ of the geometry (X, G), the set of G-symmetries of Σ will be denoted by G_Σ.

Theorem 2.1.11 (G_Σ I) *If G_Σ is the set of G-symmetries of a figure Σ of the geometry (X, G), then $G_\Sigma < G$.*

Proof First, $g_1, g_2 \in G_\Sigma$. Then, since $g_i(\Sigma) = \Sigma$, $i = 1, 2$, we have

$$(g_1 \circ g_2)(\Sigma) = g_1(g_2(\Sigma)) = g_1(\Sigma) = \Sigma.$$

Next, if e is the neutral element of G, then

$$e(\Sigma) = \Sigma \implies e \in G_\Sigma.$$

Now, for $g \in G_\Sigma$,

$$g(\Sigma) = \Sigma \implies \Sigma = g^{-1}(\Sigma).$$

The proof is complete.

If $H < G$, the geometry (X, H) is defined and hence the symmetry group H_Σ of the shape Σ is also defined.

Theorem 2.1.12 (K3) *Let (X, G) be a geometry and $H < G$. Then if Σ is a figure of the geometry, then $H_\Sigma < G_\Sigma$.*

Proof From Theorem 2.1.11, it follows that it suffices to show that $H_\Sigma \subset G_\Sigma$. Let $h \in H_\Sigma$. Then $h(\Sigma) = \Sigma$ and since $H < G$, $h \in G$ and therefore $h \in G_\Sigma$.

From our intuition we can understand that in the above case the symmetry group H_Σ can be a subgroup of G_Σ and *different* from G_Σ. A simple example is the following:

Example 2.1.13 Consider the geometry $(\mathbb{R}, G)$, where the group G consists of translations T_a, $a \in \mathbb{R}$, $T_a(x) = a + x$, and dilations D_δ, $\delta > 0$, $D_\delta(x) = \delta(x)$. Let Σ be the half-line $[0, \infty]$. Then if $\mathrm{T}(1) < G$ is the translation group and $D(1)$ group of dilations, we have

$$\{e\} = \mathrm{T}_\Sigma \subsetneq G_\Sigma = D(1).$$

We observe that the definitions of G-equivalence and G-symmetry of a figure are similar. Those concepts are related, as the next theorem shows.

Theorem 2.1.14 (G_Σ II) *If Σ and Σ' are G-equivalent shapes of the geometry X, G, then G_Σ and $G_{\Sigma'}$ are conjugate subgroups of G.*

Proof Let $\Sigma' = g(\Sigma)$ for some $g \in G$. Then $\Sigma = g^{-1}(\Sigma')$. For $h \in G_\Sigma$, we have that $h(\Sigma) = \Sigma$, so that

$$(g \circ h \circ g^{-1})(\Sigma') = (g \circ h)(\Sigma) = g(\Sigma) = \Sigma'.$$

Therefore, $g \circ h \circ g^{-1} \in G'_\Sigma$. Similarly, we show that

$$k \in G'_\Sigma \implies h = g^{-1} \circ k \circ g \in G_\Sigma.$$

Since

$$h = g^{-1} \circ h \circ g, \quad k = g \circ h \circ g^{-1},$$

every $k \in G'_\Sigma$ can be written in the form $g \circ h \circ g^{-1}$ for some $h \in G_\Sigma$.

2.1.4 The Baby Examples of Low Dimensional Euclidean Geometries

The group $E(n)$ of *Euclidean isometries of* $\mathbb{R}^n$ will be the object of extensive study in Chap. 3. We will prove the Isometry Theorem 3.3.27, according to which every isometry of $\mathbb{R}^n$ is written in a unique manner as a composition of a translation and an element of the orthogonal group $\mathrm{O}(n)$. From this, we will conclude that $E(n) = \mathrm{T}(n) \rtimes \mathrm{O}(n)$, the semi-direct product of the translation group $\mathrm{T}(n)$ and the orthogonal group $\mathrm{O}(n)$. Since the proof does not differ in large dimensions, we will not present it here, but until the end of this section we will silently accept that $E(n) = \mathrm{T}(n) \rtimes \mathrm{O}(n)$.

For the cases $n = 1$ and $n = 2$, we consider the familiar Euclidean Line Geometry $(\mathbb{R}, E(1))$ and the Euclidean Planar Geometry $(\mathbb{R}^2, E(2))$, respectively. As an introduction to the general case, in what follows we will give a not-so-strict description of both these geometries; we refer to Chap. 3 for both the concepts as well as the strict proofs of what is mentioned below.

First, the real line is identified to the set $\mathbb{R}$ of real numbers. If $x \in \mathbb{R}$, then its absolute value $|x|$ (Euclidean norm) measures the (Euclidean) length of the vector with starting point 0 and endpoint x. Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be a surjective map; f will be called an *isometry* if

$$|f(x) - f(y)| = |x - y|,$$

for each $x, y \in \mathbb{R}$. In other words, *an isometry preserves lengths*. We note that an isometry is necessarily injective and thus a bijection. The set of isometries is the group $E(1)$(the multiplication is composition). It follows immediately that *length* is a geometric property in the Euclidean Geometry of the line.

Back to the isometry group $E(1)$, this is the group that comprises the following:

I. The *translation group* $\mathrm{T}(1)$ of $\mathbb{R}$,

$$\mathrm{T}(1) = \{T_a \mid T_a(x) = x + a,\ a \in \mathbb{R}\},$$

is a group, isomorphic to the additive group $(\mathbb{R}, +)$:

$$\mathrm{T}(1) \ni T_a \mapsto a \in (\mathbb{R}, +).$$

Clearly, each translation is an isometry and therefore $\mathrm{T}(1)$ is a subgroup of $E(1)$.

II. The orthogonal group $\mathrm{O}(1)$ comprises the 1×1 matrices (± 1) which correspond to the identity transformation and to the reflection in 0, respectively.

Now, the real plane is identified to the set $\mathbb{R}^2$ of ordered pairs (x, y), $x, y \in \mathbb{R}$. Recall that if $(x, y) \in \mathbb{R}^2$ then $\|(x, y)\| = \sqrt{x^2 + y^2}$ is the (Euclidean) length of the vector with starting point at $(0, 0)$ and endpoint (x, y) (Euclidean norm). Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be a surjective map; f here is an *isometry* if

$$\|f(x, y) - f(x', y')\| = \|(x, y) - (x', y')\|,$$

for every $(x, y), (x'y') \in \mathbb{R}^2$. Again, an isometry is a length preserving bijection. The set of isometries of $\mathbb{R}^2$ is the group $E(2)$. Besides length being a geometric property in Euclidean Planar Geometry, the same applies to *angle* and *area*. In this geometry, as we have already mentioned, all the Euclidean axioms may be proved as theorems.

The group $E(2)$ is the group that contains all the translations, rotations and reflections in lines of the plane. Explicitly, $E(2)$ comprises:

I. The *translation group* $\mathrm{T}(2)$ of $\mathbb{R}^2$,

$$\mathrm{T}(2) = \{T_{(a,b)} \mid T_{(a,b)}(x, y) = (x + a, y + b),\ (a, b) \in \mathbb{R}^2\},$$

is a group (multiplication is composition), isomorphic to the additive group $(\mathbb{R}^2, +)$:

$$\mathrm{T}(2) \ni T_{(a,b)} \mapsto (a, b) \in (\mathbb{R}^2, +).$$

If $T_{(a,b)} \in \mathrm{T}(2)$ and $(x_i, y_i) \in \mathbb{R}^2$, $i = 1, 2$, then

$$\begin{aligned} \|T_{(a,b)}(x_1, y_1) - T_{(a,b)}(x_2, y_2)\| &= \|(x_1 + a, y_1 + b) - (x_2 + a, y_2 + b)\| \\ &= \|(x_1 - x_2, y_1 - y_2)\| \\ &= \|(x_1, y_1) - (x_2, y_2)\|. \end{aligned}$$

Hence $\mathrm{T}(2) < E(2)$.

II. The orthogonal group O(2) contains 2×2 matrices A that satisfy the relations $AA^T = I$ and $A^T A = I$. We have that

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{O}(2)$$

is such that

$$\begin{aligned} a^2 + c^2 = b^2 + d^2 = 1, \quad ab + cd = 0, \\ a^2 + b^2 = c^2 + d^2 = 1, \quad ac + bd = 0. \end{aligned}$$

From the previous relations it follows that $A \in \mathrm{O}(2)$ if and only if

$$A = \begin{pmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{pmatrix}, \quad \text{or} \quad A = \begin{pmatrix} \cos\phi & \sin\phi \\ \sin\phi & -\cos\phi \end{pmatrix},$$

for some $\phi \in \mathbb{R}$. The elements on the left constitute the special orthogonal group SO(2). They represent counter-clockwise rotations $A_\phi : \mathbb{R}^2 \to \mathbb{R}^2$,

$$A_\phi(x, y) = (x, y)A = (x\cos\phi + y\sin\phi, \ -x\sin\phi + y\cos\phi)$$

by ϕ.

On the other hand, an element of the right may be written as

$$R(x, y) = (x, y) - 2((a, b) \cdot (x, y))(a, b), \tag{2.1}$$

where $(a, b) = (\sin(\phi/2), \ -\cos(\phi/2))$ and $\cdot$ represents the scalar product in $\mathbb{R}^2$:

$$(a_1, \ b_1) \cdot (a_2, \ b_2) = a_1 b_1 + a_2 b_2.$$

The above formula is the *reflection in a line from the origin in the direction of* (a, b).

Exercise Prove the relation (2.1).

The set of reflections *is not* a group. O(2) is a subgroup of $E(2)$: for $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{O}(2)$ and $(x, y), \ (x', y') \in \mathbb{R}^2$,

$$\begin{aligned} \|(x, y)A - (x', y')A\|^2 &= \big\|\big(a(x - x') + c(y - y'), \ b(x - x') + d(y - y')\big)\big\|^2 \\ &= (a^2 + c^2)(x - x')^2 + (b^2 + d^2)(y - y')^2 \\ &\quad + 2(ab + cd)(x - x')(y - y'). \end{aligned}$$

Since $a^2 + c^2 = b^2 + d^2 = 1$, we have that $ab + cd = 0$ and the desired result follows.

Example 2.1.15 Euclidean Geometry is a *metric* geometry: for instance, the space $\mathbb{R}^2$ is a metric space (endowed with the Euclidean metric, see also Sect. A.1.3). It is therefore natural to study the group of its isometries. In a metric space, the generalisation of an isometry is that of a *similarity*, see also Sect. 4.1 for the precise definition. The group of similarities of a metric space is wider than the group of isometries. However, we will see how there exist metric geometries, to be precise Spherical and Hyperbolic Geometry, in which the two groups are identical.

In the case of Euclidean Planar Geometry, the group $E(2)$ extends to the group $\mathrm{S}(2)$ which additionally contains *dilations*, that is, the isomorphisms

$$D_\delta(x, y) = (\delta x, \delta y),$$

$\delta > 0$, $\delta \neq 1$. Let C be the circle $x^2 + y^2 = 1$. Assuming that $E(2)$ comprises isometries, if $g \in E(2)$ then $g(C)$ will be a circle centred at $g(0, 0)$ with radius 1. Moreover, any circle centred at (x_0, y_0) and having radius 1 is the translation of C by (x_0, y_0). We conclude that the $E(2)$-class of C contains all circles of radius 1. Now, if $h \in \mathrm{S}(2)$, $h(C)$ is again a circle centred at some p, but its radius can be arbitrary. Again, if C' is a circle of arbitrary centre and radius, we can map it to C by applying a translation and an dilation. We conclude that the $\mathrm{S}(2)$-class of C comprises all circles of arbitrary radius, and thus the $\mathrm{S}(2)$-class of C contains the $E(2)$-class of C.

Example 2.1.16 Lengths and angles are preserved by the action of $E(2)$ and it is at least intuitively clear that a similarity of $\mathbb{R}^2$ preserves angles, but not lengths, since a dilation D_δ as above either stretches or shrinks and therefore changes lengths. We conclude that the invariance of lengths and angles are $E(2)$-properties, but only the invariance of angles is a $\mathrm{S}(2)$-property.

2.2 Exercises

1. Prove that in every left action $G \times X \to X$, the stabilisers of the elements of X are subgroups of G.
2. Prove that the set $\mathrm{Aut}(G)$ of automorphisms of a group G is a group (multiplication is composition). Show that $\mathrm{Aut}(G)$ acts from the left on G and determine the space of orbits $G/\mathrm{Aut}(G)$.
3. Show that the mapping $(\mathbb{R}_{>0}, \cdot) \times \mathbb{R}^n$, where

$$(\delta, \mathbf{x}) \mapsto \delta\mathbf{x},$$

is a left action. Find the orbit space. Is this action transitive? Free?

4. The orthogonal group $G = \mathrm{O}(2)$ acts on $X = \mathbb{R}^2$:

$$(A,\ (x, y)) \mapsto (x, y), \quad A \in \mathrm{O}(2),\ (x, y) \in \mathbb{R}^2.$$

Prove the following:

(a) The orbit space consists of all circles centred at the origin, along with the degenerate circle $(0, 0)$.
(b) The orbit space is in a bijective correspondence with $[0, +\infty)$.
(c) The action is transitive only on circles centred at the origin.
(d) For every $(x, y) \neq (0, 0)$, its stabiliser on $\mathrm{SO}(2)$ is trivial. What is its stabiliser on $\mathrm{O}(2)$?

5. Define the actions and orbit spaces of the geometries $(X, \{id_X\})$ and $(X, \mathrm{Sym}(X))$, describe the orbits and stabilisers, and examine whether each of these actions is a) transitive and b) free.
6. Let $X \neq \emptyset$ and let $\mathrm{Sym}(X)$ be its symmetric group. A geometry (X, G_1) is said to be contained in (X, G_2) if $G_1 < G_2$.

(a) Show that every geometry (X, G) contains $(X, \{id_X\})$ and is contained in $(X, \mathrm{Sym}(X))$.
(b) If $G_1 < G_2$ then show that $\mathrm{orb}_{G_1}(x) < \mathrm{orb}_{G_2}(x)$. It is true that then $X/G_1 \subseteq X/G_2$ also holds?

7. Let $X \neq \emptyset$. A collection τ_X of subsets of X is called a topology on X if the following properties are satisfied:

(a) $\emptyset,\ X \in \tau_X$.
(b) The union of any set of elements of τ_X belongs to τ_X.
(c) The intersection of any finite set of elements of τ_X belongs to τ_X.

A mapping $f : X \to X$ is called continuous if for every $B \in \tau_X$ the inverse image of $f^{-1}(B) \in \tau_X$. A bijection $f : X \to X$ is called a homeomorphism if f, f^{-1} are continuous. Show that the set $\mathrm{Hom}(X)$ of homeomorphisms of X is a group (multiplication is composition) and define the space of orbits of the geometry $(X, \mathrm{Hom}(X))$.
8. Let $X = \{1, 2, 3\}$. Here $\mathrm{Sym}(X)$ is the permutation group S_3 consisting of the elements

$$(123),\ (132),\ (213),\ (231),\ (312),\ (321).$$

To understand the geometry (X, S_3), we place 1,2, and 3 at the vertices of an equilateral triangle with the centre of gravity at the origin. By rotating in an angle $2\pi/3$ counterclockwise, vertex 1 will coincide with 2, 2 with 3 and 3 with 1 and we thus obtain the element (231). Next, by rotating in an angle $4\pi/3$ counterclockwise, vertex 1 will coincide with 3, 2 with 1 and 3 with 2 and we thus obtain the element (312). By reflecting on the line $x = 0$,

vertex 1 remains fixed, and vertices 2,3, are swapped; we thus obtain the element (132). By rotating in an angle $\pi/3$ followed by the above reflection, we obtain the element (213). Finally, by rotating in an angle $2\pi/3$ followed by the above reflection we obtain the element (321). Show that this geometry is the geometry of an equilateral triangle. (Here, the triangle is understood as a figure completely determined by its vertices, its centre of gravity is at the origin, and the transformation group is a finite subgroup of O(2).

9. (Symmetry with respect to a point in the plane.) Let $\mathbf{K} = (k_1, k_2) \in \mathbb{R}^2$ be a point. For each $(x, y) \in \mathbb{R}^2$ we define the symmetric of $\mathbb{S}_{\mathbf{K}}(x, y)$ with respect to $\mathbf{K}$ as follows: on the line

$$\mathbf{c}(t) = (1 - t)(k_1, k_2) + t(x, y), \quad t \in \mathbb{R}$$

joining $\mathbf{K}$ and (x, y), we consider the unique point $\mathbb{S}_{\mathbf{K}}(x, y) = \mathbf{c}(t') \neq (x, y) = \mathbf{c}(1)$ for which

$$\|\mathbf{c}(t') - \mathbf{K}\| = \|\mathbf{c}(1) - \mathbf{K}\|.$$

Prove the following:

(a) $\mathbb{S}_{\mathbf{K}}(x, y) = 2(k_1, k_2) - (x, y)$, for each $(x, y) \in \mathbb{R}^2$.
(b) $(\mathbb{S}_{\mathbf{K}} \circ \mathbb{S}_{\mathbf{K}})(x, y) = (x, y)$, for each $(x, y) \in \mathbb{R}^2$.
(c) $\mathbb{S}_{\mathbf{K}}(x, y) = (x, y)$, if and only if $(x, y) = (k_1, k_2) = \mathbf{K}$.
(d) Symmetry with respect to a point is an $E(2)$-property.

10. Prove that the translation group T(2) is a normal subgroup of $E(2)$.
11. Consider the mapping

$$(\mathbb{R}, +) \ni \phi \mapsto A_\phi \in \mathrm{SO}(2).$$

Prove the following:

(a) This mapping is an epimorphism; its kernel is $2\pi\mathbb{Z} = \{2k\pi \mid k \in \mathbb{Z}\}$.
(b)

$$\mathrm{SO}(2) \simeq \mathbb{R}/2\pi\mathbb{Z}.$$

(c) The group $\mathbb{R}/2\pi\mathbb{Z}$ contains elements $[x] = \{x + 2k\pi,\ k \in \mathbb{Z}\}$. If S^1 is the unit circle, considered as the set of complex numbers of modulus 1, then the mapping

$$\mathbb{R}/2\pi\mathbb{Z} \ni [x] \mapsto e^{ix} \in S^1,$$

is a continuous bijection.
(d) Conclude that there is a continuous bijection from SO(2) to S^1.

Chapter 3
Euclidean Geometry



... someone who had just started learning Geometry with Euclid, having just learned the first theorem, asked Euclid: "What will I gain by learning these things?" Euclid called his slave and said: "Give him three obols, because he must gain from what he learns".

—Euclid of Alexandria

We have already emphasised that Euclidean Geometry, named in honour of its founder Euclid, (Fig. 3.1) is the basis of classical geometry and has influenced in many and profound ways the overall development of the science of mathematics. In the context in which we examine Euclidean Geometry, it will be considered to be the Kleinian Geometry $(\mathbb{R}^n, E(n))$, with $\mathbb{R}^n$ being the space of ordered n-tuples of real numbers and $E(n)$ being the group of Euclidean isometries. As for $\mathbb{R}^n$, we will present it with all its properties that we will use: $\mathbb{R}^n$ is a vector, metric and normed space as well as a space with an inner product. We refer to any introductory Linear Algebra textbook for the details—although we assume that readers are familiar with these fundamental concepts and properties, we provide a somewhat extensive review at the Appendix A. Moreover, the isometry group $E(n)$ plays a central role and is examined in detail in this chapter. The role of $E(n)$ is also crucial from the following point of view: it is the group of transformations that preserve (Euclidean) distances, as well as angles and volumes. In this chapter, we will examine and analyse the various forms of isometries, and we will demonstrate the way in which they contribute to the understanding of space and figures. We will prove two central theorems, the Isometric Theorem that describes exactly the elements of the Euclidean isometry group and the Reflection Theorem that analyses the nature of these elements. For all of the above, we refer to [7, 8, 15] as well as to [18, 36]. It is worth noting at this point that Euclidean isometries, in addition to being able to be used to solve various geometric problems, find applications in various fields, from architecture to computer science.

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I. D. Platis, *Geometries*, La Matematica per il 3+2 178,
https://doi.org/10.1007/978-3-032-08016-5_3



Fig. 3.1 Euclid, c. 350–270 BC

3.1 The Real Space $\mathbb{R}^n$

The n-dimensional real space $\mathbb{R}^n$ is fundamental for the study of geometry. Its many and rich properties, briefly described in the Appendix A, are a point of reference for most of the geometric spaces and especially for those that we will study in the following chapters. $\mathbb{R}^n$ is defined as the set

$$\mathbb{R}^n = \{\mathbf{x} = (x_1, \ldots, x_n),\ x_i \in \mathbb{R},\ i = 1, \ldots, n\}.$$

Implicit to the above definition of $\mathbb{R}^n$ is the identification of *points* of $\mathbb{R}^n$ with *vectors*: any given point of $\mathbb{R}^n$ is identified to the vector having the origin $\mathbf{0} = (0, \ldots, 0)$ as an initial point and the given point as a terminal point. The space $\mathbb{R}^n$ is an *n-dimensional vector space*, see Sect. A.1 for details: The addition $+$ in $\mathbb{R}^n$ is defined by the relation

$$\mathbf{x} + \mathbf{y} = (x_1 + y_1, \ldots, x_n + y_n),$$

for each $\mathbf{x} = (x_1, \ldots, x_n)$ and $\mathbf{y} = (y_1, \ldots, y_n)$ vectors in $\mathbb{R}^n$, whereas the scalar multiplication $\cdot$ is given for each $\lambda \in \mathbb{R}$ and $\mathbf{x} \in \mathbb{R}^n$ by

$$\lambda \cdot \mathbf{x} = \lambda\mathbf{x} = (\lambda x_1, \ldots, \lambda x_n).$$

The *canonical basis* of the vector space $\mathbb{R}^n$ consists of the vectors $\mathbf{e}_1, \ldots, \mathbf{e}_n$, where

$$\mathbf{e}_i = (0, \ldots, 0, 1, 0, \ldots, 0)$$

with 1 being at the i-coordinate. If $\mathbf{x} = (x_1, \ldots, x_n)$ is any point, then we may write

$$\mathbf{x} = \sum_{i=1}^{n} x_i \mathbf{e}_i\,.$$

Linear mappings $f : \mathbb{R}^n \to \mathbb{R}^m$ are paramount in our study. Recall that such a map satisfies the condition

$$f(\lambda \mathbf{x} + \mu \mathbf{y}) = \lambda f(\mathbf{x}) + \mu f(\mathbf{y}),$$

for each $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and $\lambda, \mu \in \mathbb{R}$. We assume that the reader has a basic knowledge of linear mappings; we also refer to Sect. A.1.2.

The space $\mathbb{R}^n$ admits a natural topology which is inherited by its Euclidean metric structure. In this structure, the *Euclidean distance* $e_n(\mathbf{x}, \mathbf{y})$ between two points $\mathbf{x} = (x_1, \ldots, x_n)$ and $\mathbf{y} = (y_1, \ldots, y_m)$ in $\mathbb{R}^n$ is given by

$$e_n(\mathbf{x}, \mathbf{y}) = \left(\sum_{i=1}^{n} (x_i - y_i)^2 \right)^{1/2},$$

for details see also Sect. A.1.3. The space $\mathbb{R}^n$ is also a *normed space*; its norm is given by

$$\|\mathbf{x}\| = \left(\sum_{i=1}^{n} x_i^2 \right)^{1/2}, \quad \mathbf{x} = (x_1, \ldots, x_n) \in \mathbb{R}^n,$$

(see Sect. A.1.4 for details), and also an *inner product space*: the inner (dot) product is given by

$$\mathbf{x} \cdot \mathbf{y} = \left(\sum_{i=1}^{n} x_i y_i \right)^{1/2},$$

for each $\mathbf{x} = (x_1, \ldots, x_n)$ and $\mathbf{y} = (y_1, \ldots, y_n)$ in $\mathbb{R}^n$ (see Sect. A.1.5). The *angle* $\angle(\mathbf{x}, \mathbf{y})$ of two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ is defined to be zero if one of them is the zero vector; if both of them are nonzero then

$$\angle(\mathbf{x}, \mathbf{y}) = \arccos \left(\frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \cdot \|\mathbf{y}\|} \right) \in [0, \pi].$$

The angle is well defined due to the *Cauchy-Schwarz-Bunyakowski inequality*, see Sect. A.1.6 and in particular (A.1.13). The vectors $\mathbf{x}$ and $\mathbf{y}$ are said to be *perpendicular to each other (or orthogonal)* (we write $\mathbf{x} \perp \mathbf{y}$) if their angle is $\pi/2$. Equivalently, $\mathbf{x} \perp \mathbf{y} \iff \mathbf{x} \cdot \mathbf{y} = 0$. We assume that $\mathbf{0} \perp \mathbf{x}$ for all $\mathbf{u} \in \mathbb{R}^n$.

The vectors $\mathbf{x}$ and $\mathbf{y}$ are said to be *parallel* if their angle is 0 (i.e., when they point at the same direction), or π (i.e., when they point at the opposite direction).

3.2 Reflections

Reflections in hyperplanes in Euclidean space are linear transformations that invert a vector across a given hyperplane, acting as a mirror. We will see that reflections preserve distances and angles, making them isometries. Reflections are fundamental in geometry: they are used in constructing orthogonal groups and studying symmetries and evidently every Euclidean isometry is decomposed into reflections; this is the Reflection Theorem that we prove in Sect. 3.3.5. Besides geometry, they also play a key role in algorithms like the Householder transformation in numerical linear algebra. Reflections are defined via projections on hyperplanes; before we examine all this we recall what projection on a vector is, as well as the Gram-Schmidt orthonormalisation process.

3.2.1 Projection on Vectors

For $\mathbf{u} \in \mathbb{R}^n_*$, the *projection (on* $\mathbf{u}$*)* is defined by

$$\mathrm{pr}_{\mathbf{u}}(\mathbf{v}) = \frac{\mathbf{v} \cdot \mathbf{u}}{\|\mathbf{u}\|^2}\,\mathbf{u}, \tag{3.1}$$

for all $\mathbf{v} \in \mathbb{R}^n$. It will become clear in Sect. 3.2.3 that via $\mathrm{pr}_{\mathbf{u}}$ the vector $\mathbf{v}$ is projected orthogonally onto the straight line from $\mathbf{u}$ that has direction $\mathbf{u}$ (see also Exercise 1, Sect. 3.4). If $\mathbf{u} = \mathbf{0}$, we define $\mathrm{pr}_{\mathbf{0}}(\mathbf{v}) := \mathbf{0}$, that is, the projection map is the null map (it maps every vector to the null vector). Also, using (A.2) we can write

$$\mathrm{pr}_{\mathbf{u}}(\mathbf{v}) = \frac{\|\mathbf{v}\| \cos\theta}{\|\mathbf{u}\|}\,\mathbf{u},$$

where $\theta = \angle(\mathbf{u}, \mathbf{v})$ (Fig. 3.2).

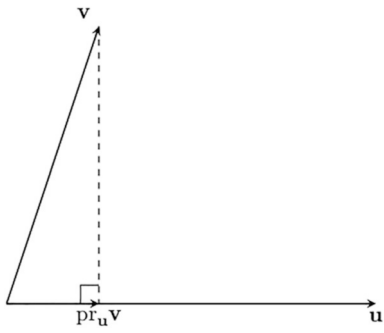


Fig. 3.2 Vector to vector projection

Proposition 3.2.1 *The projection* $\mathrm{pr}_{\mathbf{u}}$ *as defined in (3.1), is a linear map that satisfies the relation*

$$\mathrm{pr}_{\mathbf{u}} \circ \mathrm{pr}_{\mathbf{u}} = \mathrm{pr}_{\mathbf{u}}. \tag{3.2}$$

Proof We first show linearity. For $\mathbf{v}_i$, $i = 1, 2$ and $\lambda, \mu \in \mathbb{R}$:

$$\begin{aligned}
\mathrm{pr}_{\mathbf{u}}(\lambda\mathbf{v}_1 + \mu\mathbf{v}_2) &= \frac{(\lambda\mathbf{v}_1 + \mu\mathbf{v}_2)\cdot\mathbf{u}}{\|\mathbf{u}\|^2}\mathbf{u} \\
&= \frac{\lambda(\mathbf{v}_1\cdot\mathbf{u}) + \mu(\mathbf{v}_2\cdot\mathbf{u})}{\|\mathbf{u}\|^2}\mathbf{u} \\
&= \lambda\frac{\mathbf{v}_1\cdot\mathbf{u}}{\|\mathbf{u}\|^2} + \mu\frac{\mathbf{v}_2\cdot\mathbf{u}}{\|\mathbf{u}\|^2} \\
&= \lambda\mathrm{pr}_{\mathbf{u}}(\mathbf{v}_1) + \mu\mathrm{pr}_{\mathbf{u}}(\mathbf{v}_2).
\end{aligned}$$

Now, for each $\mathbf{v} \in \mathbb{R}^n$:

$$\begin{aligned}
\mathrm{pr}_{\mathbf{u}}(\mathrm{pr}_{\mathbf{u}}(\mathbf{v})) &= \mathrm{pr}_{\mathbf{u}}\left(\frac{\mathbf{v}\cdot\mathbf{u}}{\|\mathbf{u}\|^2}\mathbf{u}\right) \\
&= \frac{\mathbf{v}\cdot\mathbf{u}}{\|\mathbf{u}\|^2}\mathrm{pr}_{\mathbf{u}}(\mathbf{u}) \\
&= \frac{\mathbf{v}\cdot\mathbf{u}}{\|\mathbf{u}\|^2}\,\mathbf{u} \\
&= \mathrm{pr}_{\mathbf{u}}(\mathbf{v}),
\end{aligned}$$

where in the second line we used the linearity of $\mathrm{pr}_{\mathbf{u}}$ and in the third line we used the fact that $\mathrm{pr}_{\mathbf{u}}(\mathbf{u}) = \mathbf{u}$. □

Given a basis $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ of $\mathbb{R}^n$, it is possible to construct from it a new *orthonormal* basis $\{\mathbf{a}_1, \ldots, \mathbf{a}_n\}$, that is, a basis consisting of pairwise orthogonal unit vectors:

$$\|\mathbf{a}_i\| = 1, \quad \mathbf{a}_i \cdot \mathbf{a}_j = 0,\ i, j = 1, \ldots, n,\ i \neq j.$$

This process is called *Gram-Schmidt orthonormalisation process* and is given as follows:

$$\begin{aligned}
&\mathbf{u}_1 = \mathbf{v}_1, \quad \mathbf{a}_1 = \frac{\mathbf{u}_1}{\|\mathbf{u}_1\|}, \\
&\mathbf{u}_2 = \mathbf{v}_2 - \mathrm{pr}_{\mathbf{u}_1}(\mathbf{v}_2), \quad \mathbf{a}_2 = \frac{\mathbf{u}_2}{\|\mathbf{u}_2\|}, \\
&\vdots
\end{aligned}$$

$$\mathbf{u}_k = \mathbf{v}_k - \sum_{i=1}^{k-1} \mathrm{pr}_{\mathbf{u}_i}(\mathbf{v}_k), \quad \mathbf{a}_k = \frac{\mathbf{u}_k}{\|\mathbf{u}_k\|},\ 2 \le k \le n.$$

We note that the following useful lemma holds:

Lemma 3.2.2 *If $\mathbf{v}_1, \ldots, \mathbf{v}_n$ are unit vectors in $\mathbb{R}^n$, mutually orthogonal to each other, then they form an orthonormal basis of $\mathbb{R}^n$.*

Proof It suffices to show that $\mathbf{v}_i$ are linearly independent; to do so, let us assume that

$$\sum_{i=1}^{n} \lambda_i \mathbf{v}_i = \mathbf{0},$$

for some $\lambda_i \in \mathbb{R}$, $i = 1, \ldots, n$. Taking the dot product of each side of the equation with $\mathbf{v}_k$ we find $\lambda_k = 0$. Since this is true for every $k = 1, \ldots, n$, we obtain the desired result. □

We call the *orthocomplement* of a subspace V the set

$$V^{\perp} = \{\mathbf{w} \in \mathbb{R}^n \mid \mathbf{w} \cdot \mathbf{v} = 0,\ \mathbf{v} \in V\}.$$

Proposition 3.2.3 *Let V be a subspace of $\mathbb{R}^n$ of dimension $k \le n$ and let $V^{\perp}$ be the orthocomplement $V^{\perp}$ of V. The following hold:*

1. *$V \cap V^{\perp} = \{\mathbf{0}\}$.*
2. *$V^{\perp}$ is a subspace of $\mathbb{R}^n$ of dimension $n - k$.*
3. *For every $\mathbf{a} \in \mathbb{R}^n$ there exist $\mathbf{v} \in V$, $\mathbf{w} \in V^{\perp}$ such that $\mathbf{a} = \mathbf{v} + \mathbf{w}$.*

Proof If $\mathbf{a} \in V \cap V^{\perp}$, then $\mathbf{a} \in V$ and $\mathbf{a} \cdot \mathbf{v} = 0$ for every $\mathbf{v} \in V$. Therefore, $\mathbf{a} \cdot \mathbf{a} = 0 \iff \mathbf{a} = \mathbf{0}$.

Next, let $\mathbf{w}_1, \mathbf{w}_2 \in V^{\perp}$ and $\lambda, \mu \in \mathbb{R}$. For every $\mathbf{v} \in V$,

$$(\lambda \mathbf{w}_1 + \mu \mathbf{w}_2) \cdot \mathbf{v} = \lambda(\mathbf{w}_1 \cdot \mathbf{v}) + \mu(\mathbf{w}_2 \cdot \mathbf{v}) = 0.$$

Let $V = \langle \mathbf{v}_1, \ldots \mathbf{v}_k \rangle$, $\|\mathbf{v}_i\| = 1$, $\mathbf{v}_i \dot{\mathbf{v}}_j = 0$, $i, j = 1, \ldots, k$, $i \ne j$. We use the Gram-Schmidt process to complete V in an orthonormal basis

$$\{\mathbf{v}_1, \ldots, \mathbf{v}_k, \mathbf{v}'_{k+1}, \ldots, \mathbf{v}'_n\}$$

of $\mathbb{R}^n$. The subspace $V' = \langle \mathbf{v}'_{k+1}, \ldots, \mathbf{v}'_n \rangle$ is equal to $V^{\perp}$ and this completes the proof. □

3.2.2 Lines and Hyperplanes of $\mathbb{R}^n$

Every *straight line* of $\mathbb{R}^n$ passing through the point $\mathbf{a}$ in the direction of the vector $\mathbf{v}$ admits a parametric representation

$$\mathbf{c}(t) = \mathbf{a} + t\mathbf{v}, \quad t \in \mathbb{R}.$$

In particular, every straight line passing through the points $\mathbf{a}$ and $\mathbf{b}$ of $\mathbb{R}^n$ may be parameterised by

$$\mathbf{c}(t) = (1-t)\mathbf{a} + t\mathbf{b}, \quad t \in \mathbb{R}. \tag{3.3}$$

Even more specifically, the *line segment* with endpoints $\mathbf{a}$, $\mathbf{b}$, is parameterised by

$$\mathbf{c}(t) = (1-t)\mathbf{a} + t\mathbf{b}, \quad t \in [0,1].$$

The *Euclidean length* ℓ of the line segment with endpoints $\mathbf{a}$, $\mathbf{b}$ is

$$\ell = \|\mathbf{b} - \mathbf{a}\|.$$

In the case where $n = 2$, we know that the general equation of a straight line is

$$Ax + By = C, \quad |A| + |B| \neq 0.$$

Setting $\mathbf{x} = (x, y)$, $\mathbf{A} = (A, B)/\|(A, B)\|$ and $b = C/\|(A, B)\|$, we may equivalently write

$$\mathbf{x} \cdot \mathbf{A} = b.$$

A straight line as in (3.3) passes through the origin if and only if $b = 0$. This is generalised to many dimensions by the notion of hyperplane.

Definition 3.2.4 A *hyperplane of* $\mathbb{R}^n$ *is a set of the form*

$$W_{\mathbf{A},b} = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{x} \cdot \mathbf{A} = b,\ \mathbf{A} \in \mathbb{R}^n,\ \|\mathbf{A}\| = 1,\ b \in \mathbb{R}\}. \tag{3.4}$$

In the case where $b = 0$, that is, when the hyperplane passes through the origin, we have

$$W_{\mathbf{A}} = W_{\mathbf{A},0} = \{\mathbf{A}\}^{\perp}.$$

Therefore, the following holds:

Proposition 3.2.5 *If* $b = 0$*, then* $W_{\mathbf{A}} = W_{\mathbf{A},0}$ *is an* $(n-1)$*-dimensional vector subspace of* $\mathbb{R}^n$.

On the other hand, the converse also holds:

Corollary 3.2.6 *Every $(n-1)$-dimensional subspace of $\mathbb{R}^n$ is a hyperplane passing through the origin.*

Proof If W is a subspace of $\mathbb{R}^n$ of dimension $n-1$, then it is generated by some linearly independent vectors $\mathbf{v}_1, \ldots, \mathbf{v}_{n-1}$. The orthocomplement of W is a subspace of dimension 1; we may assume that is generated by a unit vector $\mathbf{A}$. Thus, for every $\mathbf{x} \in W$ we have $\mathbf{x} \cdot \mathbf{A} = 0$. □

3.2.3 Reflections in Hyperplanes

Let $\mathbf{A}$ be a unit vector and let $W = W_{\mathbf{A}} = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{x} \cdot \mathbf{A} = 0\}$ be a hyperplane. To construct reflection in the hyperplane W, we take an arbitrary $\mathbf{x} \in \mathbb{R}^n$ and consider the line $\mathbf{c}(t) = \mathbf{x} + t\mathbf{A}$ passing through $\mathbf{x}$ and such that is perpendicular to W. The line $\mathbf{c}(t)$ intersects W at the point where

$$\mathbf{c}(t) \cdot \mathbf{A} = 0 \iff t = -\mathbf{x} \cdot \mathbf{A}.$$

We denote the point $\mathbf{c}(-\mathbf{x} \cdot \mathbf{A})$ by

$$\mathrm{pr}_W(\mathbf{x}) = \mathbf{x} - (\mathbf{x} \cdot \mathbf{A})\mathbf{A}, \tag{3.5}$$

and call $\mathrm{pr}_W(\mathbf{x})$ the *projection of* $\mathbf{x}$ *on the hyperplane* W (Fig. 3.3).

Proposition 3.2.7 *Regarding the mapping* pr_W *as defined by (3.5):*

1. pr_W *is a linear mapping;*
2. *if* $\mathbf{x} \in W$, *then*

$$\mathrm{pr}_W(\mathbf{x}) = \mathbf{x};$$

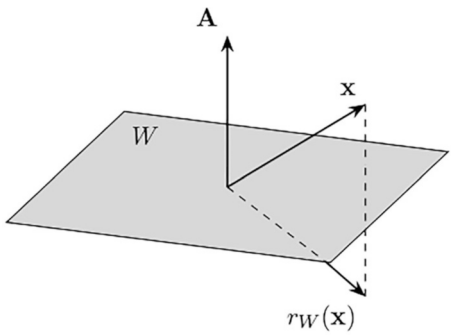


Fig. 3.3 Reflection in a hyperplane

3. for each $\mathbf{x} \in \mathbb{R}^n$ *it is*

$$\mathrm{pr}_W(\mathrm{pr}_W(\mathbf{x})) = \mathrm{pr}_W(\mathbf{x}).$$

Proof First, for each $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and for each $\lambda, \mu \in \mathbb{R}$, we have

$$\begin{aligned}\mathrm{pr}_W(\lambda\mathbf{x} + \mu\mathbf{y}) &= \lambda\mathbf{x} + \mu\mathbf{y} - ((\lambda\mathbf{x} + \mu\mathbf{y}) \cdot \mathbf{A})\,\mathbf{A} \\ &= \lambda\mathbf{x} + \mu\mathbf{y} - (\lambda\mathbf{x} \cdot \mathbf{A})\mathbf{A} - (\mu\mathbf{y} \cdot \mathbf{A})\mathbf{A} \\ &= \lambda(\mathbf{x} - (\mathbf{x} \cdot \mathbf{A})\mathbf{A}) + \mu(\mathbf{y} - (\mathbf{y} \cdot \mathbf{A})\mathbf{A} \\ &= \lambda\,\mathrm{pr}_W(\mathbf{x}) + \mu\,\mathrm{pr}_W(\mathbf{y}).\end{aligned}$$

Now, if $\mathbf{x} \in W$,

$$\mathrm{pr}_W(\mathbf{x}) = \mathbf{x} - (\mathbf{x} \cdot \mathbf{A})\mathbf{A} = \mathbf{x} - 0\,\mathbf{A} = \mathbf{x}.$$

Finally, for each $\mathbf{x} \in \mathbb{R}^n$,

$$\begin{aligned}\mathrm{pr}_W(\mathrm{pr}_W(\mathbf{x})) &= \mathrm{pr}_W(\mathbf{x} - (\mathbf{x} \cdot \mathbf{A})\mathbf{A}) \\ &= \mathrm{pr}_W(\mathbf{x}) - (\mathbf{x} \cdot \mathbf{A})\mathrm{pr}_W(\mathbf{A}) \\ &= \mathbf{x} - (\mathbf{x} \cdot \mathbf{A})\mathbf{A} - (\mathbf{x} \cdot \mathbf{A})(\mathbf{A} - (\mathbf{A} \cdot \mathbf{A})\mathbf{A}) \\ &= \mathrm{pr}_W(\mathbf{x}).\end{aligned}$$

□

If $\mathbf{x}$ is a given point, then the Euclidean distance of $\mathbf{x}$ from $\mathrm{pr}_W(\mathbf{x})$ is equal to

$$e_n(\mathbf{x},\ \mathbf{x} - (\mathbf{x} \cdot \mathbf{A})\mathbf{A}) = |\mathbf{x} \cdot \mathbf{A}|.$$

There are two points lying on the straight line $\mathbf{c}(t) = \mathbf{x} + t\mathbf{A}$ that are $2|\mathbf{x} \cdot \mathbf{A}|$ away from $\mathbf{x}$, those are $\mathbf{x}$ and $\mathbf{x} \pm 2(\mathbf{x} \cdot \mathbf{A})\mathbf{A}$. Indeed, $\mathbf{x}$ and $\mathbf{x} + 2(\mathbf{x} \cdot \mathbf{A})\mathbf{A}$ belong to the same half-space defined by W: if $\mathbf{x} \cdot \mathbf{A} > 0$ (resp. < 0) them also $(\mathbf{x} + 2(\mathbf{x} \cdot \mathbf{A})\mathbf{A}) \cdot \mathbf{A} = 3(\mathbf{x} \cdot \mathbf{A}) > 0$ (resp. < 0). On the other hand, if $\mathbf{x} \cdot \mathbf{A} > 0$ (resp. < 0) then $(\mathbf{x} - 2(\mathbf{x} \cdot \mathbf{A})\mathbf{A}) \cdot \mathbf{A} = -\mathbf{x} \cdot \mathbf{A} < 0$ (resp. > 0). It is therefore reasonable to give the following definition:

Definition 3.2.8 The *reflection* $r_W : \mathbb{R}^n \to \mathbb{R}^n$ *in the hyperplane* $W = W_{\mathbf{A}}$ is defined by the relation

$$r_W(\mathbf{x}) = \mathbf{x} - 2(\mathbf{x} \cdot \mathbf{A})\mathbf{A}, \quad \mathbf{x} \in \mathbb{R}^n. \tag{3.6}$$

Notice that

$$r_W(\mathbf{x}) = \mathrm{pr}_W(\mathbf{x}) - \mathrm{pr}_{\mathbf{A}}(\mathbf{x}), \tag{3.7}$$

where $\mathrm{pr}_{\mathbf{A}}$ is the projection on vector the $\mathbf{A}$, see Eq. (3.1).

Example 3.2.9 The simplest reflections are those in the coordinate hyperplanes $x_k = 0$ which we denote by r_k; they are given by the relations:

$$r_k(x_1, \ldots, x_{k-1}, x_k, x_{k+1} \ldots x_n) = (x_1, \ldots, x_{k-1}, -x_k, x_{k+1} \ldots x_n).$$

To see this, write the hyperplane $x_k = 0$ as $\mathbf{x} \cdot \mathbf{e}_k = 0$, where $\mathbf{e}_k$ is k-th element of the canonical basis of $\mathbb{R}^n$. Then for $\mathbf{x} = (x_1, \ldots, x_n)$,

$$\begin{aligned} r_k(\mathbf{x}) &= \mathbf{x} - 2x_k\, \mathbf{e}_k \\ &= (x_1, \ldots, x_{k-1}, x_k, x_{k+1}, \ldots, x_n) - (0, \ldots, 0, 2x_k, 0, \ldots, 0) \\ &= (x_1, \ldots, x_{k-1}, -x_k, x_{k+1} \ldots x_n). \end{aligned}$$

The following proposition describes the main properties of reflections in hyperplanes.

Proposition 3.2.10 *Let r_W be a reflection in a hyperplane as in (3.6). Then the following holds:*

1. *r_W is a linear mapping.*
2. *r_W satisfies: $r_W \circ r_W = id$.*
3. *r_W leaves the hyperplane W pointwise invariant.*
4. *If $\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ is the canonical basis of $\mathbb{R}^n$ then $\{r_W(\mathbf{e}_1), \ldots, r_W(\mathbf{e}_n)\}$ is an orthonormal basis of $\mathbb{R}^n$.*

Exercise Prove Proposition 3.2.10.

Remark 3.2.11 Recall that an *involution* is an invertible mapping that is its own inverse. It follows from the second statement of the above proposition that r_W is an involution.

If $\mathbf{A} = (A_1, \ldots, A_n)$, then for each $\mathbf{x} = (x_1, \ldots, x_n)$ we have

$$r_W(x_1, \ldots, x_n) = \left(x_1 - 2A_1 \sum_{i=1}^{n} A_i x_i, \ldots, x_n - 2A_n \sum_{i=1}^{n} A_i x_i \right).$$

Consequently, the matrix R_W of r_W with respect to the canonical basis is the symmetric matrix

$$R_W = \begin{pmatrix} 1-2A_1^2 & -2A_1A_2 & \dots & -2A_1A_n \\ -2A_2A_1 & 1-2A_2^2 & \dots & -2A_2A_n \\ \vdots & \vdots & \dots & \vdots \\ -2A_nA_1 & -2A_nA_2 & \dots & 1-2A_n^2 \end{pmatrix}.$$

Proposition 3.2.12 *Let R_W be the matrix of the reflection r_W with respect to the canonical basis of $\mathbb{R}^n$ as above. Then* $\det(R_W) = -1$.

Proof We observe first that the relation $r_W^2 = id.$ implies $R_W^2 = I$. Therefore, the eigenvalues of R_W may be $+1, -1$. If $\mathbf{v}$ is an eigenvector corresponding to the eigenvalue $+1$, then $r_W(\mathbf{v}) = \mathbf{v}$, which implies that $\mathbf{v} \perp \mathbf{A}$, i.e. that $\mathbf{v}$ belongs to W which is an $(n-1)$-dimensional subspace of $\mathbb{R}^n$. Now, if $\mathbf{w}$ is an eigenvector corresponding to the eigenvalue -1, then $r_W(\mathbf{w}) = -\mathbf{w}$, which implies that $\mathbf{w} = 2(\mathbf{A}\cdot\mathbf{w})\mathbf{A}$, i.e., that $\mathbf{w}$ belongs to a straight line (1-dimensional subspace), orthogonal to W. We conclude that R_W is diagonalisable, $R_W = QDQ^{-1}$, and that on the main diagonal of the diagonal matrix D, the value $+1$ appears $n-1$ times while the value -1 appears only once. The desired result follows immediately. □

Remark 3.2.13 If r_i, $i = 1, 2$ are reflections in hyperplanes W_i, then their composition is not a reflection on a hyperplane: $\det(R_1R_2) = 1$.

3.3 The Group $E(n)$

We begin by defining an isometry in an arbitrary metric space (see Definition A.1.5 in Sect. A.1.3).

Definition 3.3.1 Let (X, d) be a metric space. A surjective mapping $f : X \to X$ is called an *isometry of* X if for every $x, y \in X$ we have

$$d(f(x), f(y)) = d(x, y).$$

An isometry is automatically injective, hence a transformation of X. The set of isometries $\mathrm{Isom}(X, d)$ of X is a group; the group operation is composition of mappings.

Definition 3.3.2 We denote by

$$E(n) = \mathrm{Isom}(\mathbb{R}^n, e_n),$$

the group of isometries of Euclidean space and we call $E(n)$ the *Euclidean isometry group*.

Since for each $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$ we have that

$$e_n(\mathbf{a}, \mathbf{b}) = \|\mathbf{a} - \mathbf{b}\|,$$

we immediately have the following:

Proposition 3.3.3 *$f \in E(n)$ if and only if for each $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ we have*

$$\|f(\mathbf{x}) - f(\mathbf{y})\| = \|\mathbf{x} - \mathbf{y}\|.$$

We observe that if $f(\mathbf{0}) = \mathbf{0}$, then

$$\|f(\mathbf{x})\| = \|f(\mathbf{x}) - f(\mathbf{0})\| = \|\mathbf{x} - \mathbf{0}\| = \|\mathbf{x}\|.$$

Therefore every Euclidean isometry that fixes the origin is also norm preserving. Every such isometry also preserves the dot product: indeed, if $f : \mathbb{R}^n \to \mathbb{R}^n$ is an isometry such that $f(\mathbf{0}) = \mathbf{0}$, then for $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ we have

$$\begin{aligned} 2\, f(\mathbf{x}) \cdot f(\mathbf{y}) &= \|f(\mathbf{x})\|^2 + \|f(\mathbf{y})\|^2 - \|f(\mathbf{x}) - f(\mathbf{y})\|^2 \\ &= \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 - \|\mathbf{x} - \mathbf{y}\|^2 \\ &= 2\, \mathbf{x} \cdot \mathbf{y}. \end{aligned}$$

We have proved the following proposition which we will need in the proof of the Isometry Theorem.

Proposition 3.3.4 *Let $f \in E(n)$, $f(\mathbf{0}) = \mathbf{0}$. Then if $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$:*

1. $\|f(\mathbf{x})\| = \|\mathbf{x}\|$,
2. $f(\mathbf{x}) \cdot f(\mathbf{y}) = \mathbf{x} \cdot \mathbf{y}$.

Proposition 3.3.5 *Every reflection in a hyperplane from the origin is a Euclidean isometry.*

Proof Let $W = W_{\mathbf{A}} = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{x} \cdot \mathbf{A} = 0\}$ be a hyperplane and consider the reflection

$$r_W(\mathbf{x}) = \mathbf{x} - 2\, (\mathbf{x} \cdot \mathbf{A})\mathbf{A},$$

in W. For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$,

$$\begin{aligned} \|r_W(\mathbf{x}) - r_W(\mathbf{y})\|^2 &= \|\mathbf{x} - \mathbf{y} - 2\, ((\mathbf{x} - \mathbf{y}) \cdot \mathbf{A})\, \mathbf{A}\|^2 \\ &= \|\mathbf{x} - \mathbf{y}\|^2 + 4\, ((\mathbf{x} - \mathbf{y}) \cdot \mathbf{A})^2 - 4\, ((\mathbf{x} - \mathbf{y}) \cdot \mathbf{A})^2 \\ &= \|\mathbf{x} - \mathbf{y}\|^2. \end{aligned}$$

□

Since $r_W(\mathbf{0}) = \mathbf{0}$, we have from Proposition 3.3.4 the following

Corollary 3.3.6 *If r_W is a reflection in a hyperplane W, then for each $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ we have:*

1. $\|r_W(\mathbf{x})\| = \|\mathbf{x}\|$,
2. $r_W(\mathbf{x}) \cdot r_W(\mathbf{y}) = \mathbf{x} \cdot \mathbf{y}$.

The next Lemma will be particularly useful.

Lemma 3.3.7 *Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ with $\|\mathbf{x}\| = \|\mathbf{y}\|$. Then there exists a reflection r_W on a hyperplane W that maps $\mathbf{x}$ onto $\mathbf{y}$. If $\mathbf{x} \neq \mathbf{y}$, then this reflection is unique.*

Proof If $\mathbf{x} = \mathbf{y}$, then the reflection on every hyperplane containing $\mathbf{x}$ is the desired one. If $\mathbf{x} \neq \mathbf{y}$, we consider the unit vector $\mathbf{A} = (\mathbf{y} - \mathbf{x})/\|\mathbf{x} - \mathbf{y}\|$, the hyperplane $W = W_{\mathbf{A}}$ and the reflection r_W given by

$$r_W(\mathbf{u}) = \mathbf{u} - 2\,(\mathbf{u} \cdot \mathbf{A})\mathbf{A}.$$

Then we have:

$$\begin{aligned} r_W(\mathbf{x}) &= \mathbf{x} - 2\left(\mathbf{x} \cdot \frac{\mathbf{y} - \mathbf{x}}{\|\mathbf{y} - \mathbf{x}\|}\right)\frac{\mathbf{y} - \mathbf{x}}{\|\mathbf{y} - \mathbf{x}\|} \\ &= \mathbf{x} - 2\frac{\mathbf{x} \cdot \mathbf{y} - \|\mathbf{x}\|^2}{\|\mathbf{y} - \mathbf{x}\|^2}(\mathbf{y} - \mathbf{x}) \\ &= \mathbf{x} + \frac{2\|\mathbf{x}\|^2 - 2\mathbf{x} \cdot \mathbf{y}}{2\|\mathbf{x}\|^2 - 2\mathbf{x} \cdot \mathbf{y}}(\mathbf{y} - \mathbf{x}) \\ &= \mathbf{x} + (\mathbf{y} - \mathbf{x}) = \mathbf{y}. \end{aligned}$$

If there were another reflection $r_{W'}$ in a hyperplane W' with this property, then it should be $W' \perp \langle \mathbf{y} - \mathbf{x} \rangle$. Since W and W' are vector subspaces of the same dimension, this means that they are necessarily identical. □

3.3.1 The Translation Group $\mathrm{T}(n)$

The *translation group* $\mathrm{T}(n)$ consists of *translations*, that is, mappings $T_{\mathbf{a}} : \mathbb{R}^n \to \mathbb{R}^n$, $\mathbf{a} \in \mathbb{R}^n$, where

$$T_{\mathbf{a}}(\mathbf{x}) = \mathbf{x} + \mathbf{a}, \quad \mathbf{x} \in \mathbb{R}^n.$$

The next proposition describes the action of $\mathrm{T}(n)$ on $\mathbb{R}^n$.

Proposition 3.3.8 *The group* $\mathrm{T}(n)$ *is a subgroup of* $E(n)$ *isomorphic to the additive group* $(\mathbb{R}^n, +)$ *and acts transitively and freely on* $\mathbb{R}^n$. *The orbit space* $\mathbb{R}^n/\mathrm{T}(n)$ *comprises only one element, which is* $\mathbb{R}^n$.

Proof First, $\mathrm{T}(n)$ is a group; the operation is composition of mappings. The isomorphism of the additive group $(\mathbb{R}^n, +)$ and the translation group $\mathrm{T}(n)$ of $\mathbb{R}^n$ is given by the relation

$$(\mathbb{R}^n, +) \ni \mathbf{a} \mapsto T_{\mathbf{a}} \in \mathrm{T}(n).$$

We note that the side from which $T(n)$ acts on $\mathbb{R}^n$ is irrelevant here, since $\mathbf{x} + \mathbf{a} = \mathbf{a} + \mathbf{x}$, i.e., $\mathrm{T}(n)$ is abelian. The action is given by

$$(T_{\mathbf{a}}, \mathbf{x}) \mapsto \mathbf{a} + \mathbf{x}, \quad T_{\mathbf{a}} \in \mathrm{T}(n), \ \mathbf{x} \in \mathbb{R}^n.$$

The action is transitive: if $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, then

$$\mathbf{y} = (\mathbf{y} - \mathbf{x}) + \mathbf{x}.$$

Therefore the orbit space comprises a single point, that is, $\mathbb{R}^n$: $\mathbb{R}^n/\mathrm{T}(n) = \{\mathbb{R}^n\}$. Finally, the action is free: for every $\mathbf{x} \in X$, $\mathrm{Stab}(\mathbf{x}) = \{\mathbf{0}\}$. □

We now consider the action of $\mathrm{T}(n)$ on various geometric objects.

Proposition 3.3.9 *If* $\mathbf{c}(t) = \mathbf{u} + t\mathbf{v}$ *is a line from* $\mathbf{u}$ *in the direction of* $\mathbf{v}$, *then its* $\mathrm{T}(n)$*-orbit comprises all straight lines of* $\mathbb{R}^n$ *parallel to* $\mathbf{c}(t)$.

Proof If $T_{\mathbf{a}} \in \mathrm{T}(n)$, then

$$T_{\mathbf{a}}(\mathbf{c}(t)) = (\mathbf{a} + \mathbf{u}) + t\mathbf{v}$$

is a straight line parallel to $\mathbf{c}(t)$. On the other hand, all straight lines in the direction of $\mathbf{v}$ are translations of $t\mathbf{v}$, that is, translations of the straight line passing from the origin in the direction of $\mathbf{v}$. □

Corollary 3.3.10 *The action of* $T(n)$ *preserves parallelism.*

Remark 3.3.11 If two straight lines $\mathbf{c}_i$, $i = 1, 2$, intersect at $\mathbf{u}$, then

$$\mathbf{c}_1(t) = \mathbf{u} + t\mathbf{v}_1, \quad \mathbf{c}_2(t) = \mathbf{u} + t\mathbf{v}_2,$$

with the angle of intersection being $\angle(\mathbf{v}_1, \mathbf{v}_2) \neq 0, \pi$. If $T_{\mathbf{a}}$ is a translation, then their images are

$$T_{\mathbf{a}}(\mathbf{c}_i(t)) = (\mathbf{u} + \mathbf{a}) + t\mathbf{v}_i,$$

that is, straight lines intersecting in the image of the intersection point, at the same angle. Finally, if two straight lines are incompatible, their images under a translation will also be incompatible lines since all translations are bijective.

Proposition 3.3.12 *If $S(\mathbf{a}, \rho)$ is a sphere with centre $\mathbf{a}$ and radius ρ, then its $\mathrm{T}(n)$-orbit comprises all spheres with radius ρ.*

Exercise Prove Proposition 3.3.12.

Definition 3.3.13 A transformation $f : \mathbb{R}^n \to \mathbb{R}^n$ is said to be *volume preserving* if it preserves the volumes of figures of $\mathbb{R}^n$.

Recall that if $A \subset \mathbb{R}^n$ is a region, then its n-volume is given by

$$\mathrm{Vol}_n(A) = \int_A d\mathbf{x},$$

where the integral is the n-th integral and $d\mathbf{x} = dx_1 \dots dx_n$. The integral is a positive number if the region is bounded, while it is infinite otherwise.

Now let $T_{\mathbf{a}}$ be a translation: explicitly, if $\mathbf{a} = (a_1, \dots, a_n)$,

$$T_{\mathbf{a}}(x_1, \dots, x_n) = (x_1 + a_1, \dots, x_n + a_n).$$

It immediately follows that the Jacobian matrix $DT_{\mathbf{a}}$ is the identity matrix I_n. Therefore, if A is a region, by the Change of Variables Theorem we have that

$$\mathrm{Vol}_n(T_{\mathbf{a}}(A)) = \int_{T_{\mathbf{a}}(A)} d\mathbf{y} = \int_A |\det(DT_{\mathbf{a}}(\mathbf{x}))|\, d\mathbf{x} = \int_A d\mathbf{x} = \mathrm{Vol}_n(A).$$

We proved the following:

Proposition 3.3.14 *The translation group comprises volume-preserving mappings.*

3.3.2 The Orthogonal Group $\mathbf{O}(n)$

The orthogonal group is the matrix group comprising elements which are the associated matrices to orthogonal mappings:

Definition 3.3.15 A linear mapping $\alpha : \mathbb{R}^n \to \mathbb{R}^n$ is *orthogonal* if $\alpha(\mathbf{e}_1), \dots, \alpha(\mathbf{e}_n)$ form an orthonormal basis of $\mathbb{R}^n$.

From the definition of orthogonal mappings and Proposition 3.2.10 we immediately have the following:

Corollary 3.3.16 *Every reflection in a hyperplane from the origin is an orthogonal mapping.*

The next proposition gives equivalent orthogonality criteria for linear mappings.

Proposition 3.3.17 *Let $\alpha : \mathbb{R}^n \to \mathbb{R}^n$ be a linear mapping. Then the following are equivalent:*

1. *α is an orthogonal mapping.*
2. *α preserves dot products (for all $\mathbf{x}$ and $\mathbf{y}$, $\alpha(\mathbf{x}) \cdot \alpha(\mathbf{y}) = \mathbf{x} \cdot \mathbf{y}$).*
3. *α preserves vector lengths (for all $\mathbf{x}$, $\|\alpha(\mathbf{x})\| = \|\mathbf{x}\|$.)*
4. *If $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is an orthonormal basis, so is $\{\alpha(\mathbf{v}_1), \ldots, \alpha(\mathbf{v}_n)\}$).*

Proof (1) $\implies$ (2) Let $\mathbf{x} = (x_1, \ldots, x_n)$ and $\mathbf{y} = (y_1, \ldots, y_n)$. Due to linearity,

$$\alpha(\mathbf{x}) = \sum_{i=1}^{n} x_i \alpha(\mathbf{e}_i), \quad \alpha(\mathbf{y}) = \sum_{i=1}^{n} y_i \alpha(\mathbf{e}_i).$$

So,

$$\begin{aligned} \alpha(\mathbf{x}) \cdot \alpha(\mathbf{y}) &= \left(\sum_{i=1}^{n} x_i \alpha(\mathbf{e}_i) \right) \cdot \left(\sum_{j=1}^{n} y_j \alpha(\mathbf{e}_j) \right) \\ &= \sum_{i,j=1}^{n} x_i y_j \alpha(\mathbf{e}_i) \cdot \alpha(\mathbf{e}_j) \\ &= \sum_{i=1}^{n} x_i y_i = \mathbf{x} \cdot \mathbf{y}. \end{aligned}$$

(2) $\implies$ (3) We have:

$$\|\alpha(\mathbf{x})\|^2 = \alpha(\mathbf{x}) \cdot \alpha(\mathbf{x}) = \mathbf{x} \cdot \mathbf{x} = \|\mathbf{x}\|^2.$$

(3) $\implies$ (4) For $i, j = 1, \ldots, n$,

$$\alpha(\mathbf{v}_i) \cdot \alpha(\mathbf{v}_j) = \mathbf{v}_i \cdot \mathbf{v}_j.$$

(4) $\implies$ (1) This follows immediately. □

We now describe orthogonal mappings in terms of their associated matrices with respect to the canonical basis.

Proposition 3.3.18 *Let $\alpha : \mathbb{R}^n \to \mathbb{R}^n$ be a linear mapping and let A be the matrix of α with respect to the canonical basis $\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$. The following are equivalent:*

1. *α is an orthogonal mapping.*
2. *The columns of A form an orthonormal set of $(\mathbb{R}^n)^T$.*
3. *$A^T A = I$.*
4. *$AA^T = I$.*

5. The rows of A form an orthonormal set of $\mathbb{R}^n$.

Proof (1) $\iff$ (2) The columns $\mathbf{c}_j$ of A satisfy the relation $\mathbf{c}_j^T = \alpha(\mathbf{e}_j)$.
(2) $\iff$ (3) It follows directly from the definition of the matrix product.
(4) $\iff$ (5) It follows directly from the definition of the matrix product.
(3) $\iff$ (4) The proof is based on the argument: if $AB = I$ holds for two matrices A, B, then $BA = I$ also holds. Indeed, let α, β be the linear mappings corresponding to A, B. The relation $AB = I$ implies $\alpha \circ \beta = id$. But this means that $\alpha^{-1} = \beta$ and $\beta^{-1} = \alpha$, so $\beta \circ \alpha = id$. and therefore $BA = I$. □

Definition 3.3.19 The *orthogonal group* $\mathrm{O}(n)$ comprises matrices $A \in \mathrm{GL}(n, \mathbb{R})$ such that $A^{-1} = A^T$.

Proposition 3.3.20 *The group* $\mathrm{O}(n)$ *comprises Euclidean isometries:* $\mathrm{O}(n) < E(n)$.

Proof Indeed, if $A \in \mathrm{O}(n)$ and α is the corresponding linear transformation, then for each $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ we have:

$$\begin{aligned}\|\alpha(\mathbf{x}) - \alpha(\mathbf{y})\|^2 &= (\alpha(\mathbf{x}) - \alpha(\mathbf{y})) \cdot (\alpha(\mathbf{x}) - \alpha(\mathbf{y})) \\ &= \alpha(\mathbf{x}) \cdot \alpha(\mathbf{x}) + \alpha(\mathbf{y}) \cdot \alpha(\mathbf{y}) - 2\,\alpha(\mathbf{x}) \cdot \alpha(\mathbf{y}) \\ &= \mathbf{x} \cdot \mathbf{x} + \mathbf{y} \cdot \mathbf{y} - 2\,\mathbf{x} \cdot \mathbf{y} \\ &= \|\mathbf{x} - \mathbf{y}\|^2.\end{aligned}$$

□

In Comment B.1.4 we see that $\det(A) = \pm 1$ for every $A \in \mathrm{O}(n)$; also, in Example B.1.3 we show that the subset $\mathrm{SO}(n)$ of $\mathrm{O}(n)$ consisting of elements with determinant 1 is a subgroup of $\mathrm{O}(n)$. Considering the homomorphism

$$\mathrm{O}(n) \ni A \mapsto \det(A) \in \mathbb{R}_*,$$

we see that the kernel is a normal subgroup of $O(n)$: $\ker = \mathrm{SO}(n) \lhd \mathrm{O}(n)$. Furthermore, by the Isomorphism Theorem, see Sect. B.2.1, we have that the quotient group $\mathrm{O}(n)/\mathrm{SO}(n)$ is isomorphic to the set $\{-1, 1\}$:

$$\mathrm{O}(n)/\mathrm{SO}(n) \simeq \{1, -1\}.$$

It now follows immediately that $\mathrm{O}(n)$ is isomorphic to the semi-direct product (see Sect. B.2.2):

$$\mathrm{O}(n) = \mathrm{SO}(n) \rtimes \{\pm I_n\}.$$

Spectral Theorem for Orthogonal Matrices

We recall first the (real) spectral theorem, see Theorem 7.29 in [6]:

Theorem 3.3.21 *If A is a symmetric ($A = A^T$) matrix, then there exists a matrix $Q \in \mathrm{O}(n)$ and a diagonal matrix D such that*

$$A = QDQ^T. \tag{3.8}$$

Any $n \times n$ matrix admitting a representation of the form (3.8) is called orthogonally diagonalisable.

Theorem 3.3.22 *Let $A \in \mathrm{O}(n)$. Then:*

(a) All eigenvalues of A have absolute value 1 (i.e., they lie on the unit circle in the complex plane).
(b) A is orthogonally diagonalisable if and only if A is symmetric (i.e., $A = A^T$).
(c) In general, A can be orthogonally block-diagonalised as

$$A = Q \begin{pmatrix} R_{\theta_1} & & & & & \\ & \ddots & & & & \\ & & R_{\theta_k} & & & \\ & & & \pm 1 & & \\ & & & & \ddots & \\ & & & & & \pm 1 \end{pmatrix} Q^T \tag{3.9}$$

where $Q \in \mathrm{O}(n)$ and $R_{\theta_i} \in \mathrm{SO}(2)$, that is,

$$R_{\theta_i} = \begin{pmatrix} \cos\theta_i & -\sin\theta_i \\ \sin\theta_i & \cos\theta_i \end{pmatrix}, \quad \theta_i \in \mathbb{R}, \ i = 1, \ldots, k.$$

Proof

(a) We show that the eigenvalues of A have absolute value 1. Since A is a real matrix, if λ is an eigenvalue of A with eigenvector $\mathbf{v} = (v_1, \ldots, v_n)$, $A\mathbf{v}^T = \lambda\mathbf{v}^T$, then $\overline{\lambda}$ is also an eigenvalue of A with eigenvector $\overline{\mathbf{v}}$. Therefore,

$$|\lambda|^2 \overline{\mathbf{v}}\mathbf{v}^T = (\overline{\lambda}\overline{\mathbf{v}})(\lambda\mathbf{v}^T) = (\overline{\mathbf{v}}A^T)(A\mathbf{v}^T) = \overline{\mathbf{v}}(A^TA)\mathbf{v}^T = \overline{\mathbf{v}}\mathbf{v}^T.$$

Since $\overline{\mathbf{v}}\mathbf{v}^T = \sum_{i=1}^n |v_i|^2 > 0$ since $\mathbf{v}$ is an eigenvector, the desired result follows.

(b) If A is symmetric and orthogonal, then by the real spectral Theorem 3.3.21, A is orthogonally diagonalisable, $A = QDQ^T$, where Q is orthogonal and D is

diagonal. Since A is orthogonal we have

$$A^T A = (QDQ^T)^T(QDQ^T) = QD^T DQ^T = I,$$

and this implies $D^T D = I$, so the diagonal entries of D (that is, the eigenvalues) satisfy $\lambda_i^2 = 1$ and hence D has entries ± 1.

(c) Let $A \in \mathrm{O}(n)$ be a general orthogonal matrix and let λ be an eigenvalue; if $\lambda \in \mathbb{R}$, then $\lambda = \pm 1$ and the corresponding eigenspace is real and orthogonal. If $\lambda = e^{i\theta}$, then its complex conjugate $\overline{\lambda} = e^{-i\theta}$ is also an eigenvalue. The corresponding eigenvectors $\mathbf{v}$ and $\overline{\mathbf{v}}$ are complex. We can extract a real invariant subspace by taking the real and imaginary parts of $\mathbf{v}$. This leads to a 2×2 rotation block R_θ in the decomposition. By repeatedly finding invariant subspaces (for real eigenvalues) or 2×2 blocks (for complex eigenvalues), we conclude that any orthogonal matrix A can be written in the form (3.9).

□

Spectral Theorem 3.3.22 is fundamental in understanding the structure of orthogonal transformations, which include rotations and reflections in $\mathbb{R}^n$.

Corollary 3.3.23 *Let $A \in \mathrm{O}(n)$. The following holds.*

1. *If n is odd, then 1 or -1 is an eigenvalue of A.*
2. *If n is even, then 1 and -1 are not necessarily eigenvalues of A.*

Proof Our first claim follows from the fact that the characteristic polynomial of A is of odd degree with a constant term in the determinant (± 1) which is simultaneously the product of the eigenvalues.

For the second claim, let us observe that the matrix

$$\begin{pmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & \cos\phi & -\sin\phi \\ 0 & 0 & \sin\phi & \cos\phi \end{pmatrix} \in \mathrm{SO}(4),$$

has no real eigenvalues. □

The form (3.9) of an orthogonal matrix is called its *canonical form*. We observe that $A \in \mathrm{SO}(n)$ if there is an even number of -1's in its canonical form.

Corollary 3.3.24 (Euler's Rotation Theorem) *Every element of* SO(3) *is a rotation about a uniquely defined axis.*

Proof According to Theorem 3.3.22, rearranging if necessary we can assume that the canonical form of any element of SO(3) will be

$$A = \begin{pmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The corresponding mapping

$$(x, y, z) \mapsto (x, y, z)A = (x\cos\phi + y\sin\phi,\ -x\sin\phi + y\cos\phi,\ z),$$

is a rotation about the z axis. □

We now proceed to the study of the action of $\mathrm{O}(n)$ on $\mathbb{R}^n$; this is given by

$$(A, \mathbf{x}) \mapsto \alpha(\mathbf{x}),$$

where α is the orthogonal mapping corresponding to A. Since for each for each orthogonal transformation α we have

$$\|\alpha(\mathbf{x})\| = \|\mathbf{x}\|, \quad \mathbf{x} \in \mathbb{R}^n,$$

it follows that for each $\mathbf{x}$ the orbit $\mathrm{orb}(\mathbf{x})$ is contained in the sphere $S(\mathbf{0}, \|\mathbf{x}\|)$ of radius $\|\mathbf{x}\|$ centred at the origin. On the other hand, if $\mathbf{y}$ belongs to the sphere $S(\mathbf{0}, \|\mathbf{x}\|)$, then by Lemma 3.3.7 there is a reflection r_W on a hyperplane W that maps $\mathbf{x}$ to $\mathbf{y}$. We conclude that $\mathrm{orb}(\mathbf{x}) = S(\mathbf{0}, \|\mathbf{x}\|)$. We have proved:

Proposition 3.3.25 *The orbit space $\mathbb{R}^n/\mathrm{O}(n)$ comprises all spheres centred at the origin. $\mathrm{O}(n)$ acts transitively on every sphere centred at the origin.*

If $n = 1$, $\mathrm{O}(1) = \{\pm 1\}$ so the stabilisers are all trivial. For the case $n > 1$, we first find the stabiliser $\mathrm{Stab}(\mathbf{e}_n)$. If α is an orthogonal transformation in $\mathrm{Stab}(\mathbf{e}_n)$, $\alpha(\mathbf{e}_n) = \mathbf{e}_n$, then the matrix A of α will be of the form

$$A = \begin{pmatrix} A' & 0 \\ 0 & 1 \end{pmatrix}, \tag{3.10}$$

with the matrix A' being an element of $\mathrm{O}(n-1)$ corresponding to an orthogonal transformation α' of $\mathbb{R}^{n-1}$. We note at this point that the relation (3.10) is also the embedding of the group $\mathrm{O}(n-1)$ into a subgroup of $\mathrm{O}(n)$, that is, precisely the subgroup containing elements of the form (3.10):

$$\mathrm{O}(n-1) \ni A \mapsto \begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix} \in \mathrm{O}(n).$$

This embedding is a group monomorphism. Therefore, $\mathrm{Stab}(\mathbf{e}_n) \simeq \mathrm{O}(n-1)$.

Now, if $\mathbf{v} \neq \mathbf{0}$, $\|\mathbf{v}\| = 1$, by Lemma 3.3.7 there is a reflection r in a hyperplane orthogonal to

$$\mathbf{A} = \frac{\mathbf{v} - \mathbf{e}_n}{\|\mathbf{v} - \mathbf{e}_n\|} = \frac{(v_1, \cdots, v_{n-1}, v_n - 1)}{\sqrt{2(1 - v_n^2)}},$$

which maps $\mathbf{e}_n$ to $\mathbf{v}$:

$$r(\mathbf{x}) = \mathbf{x} + \frac{x_n - \mathbf{x}\cdot\mathbf{v}}{1 - v_n^2}(\mathbf{v} - \mathbf{e}_n), \quad \mathbf{x} \in \mathbb{R}^n.$$

We claim that $\mathrm{Stab}(\mathbf{v}) = \{r \circ \alpha \circ r, \mid \alpha \in \mathrm{Stab}(\mathbf{e}_n)\}$. Indeed, for every element $r \circ \alpha \circ r$ we have

$$r(\alpha(r(\mathbf{v}))) = r(\alpha(\mathbf{e}_n)) = r(\mathbf{e}_n) = \mathbf{v}.$$

Conversely, if $\alpha' \in \mathrm{Stab}(\mathbf{v})$,

$$\alpha'(\mathbf{v}) = \mathbf{v} \implies \alpha'(r(\mathbf{e}_n)) = r(\mathbf{e}_n),$$

and by composing from the left with r we get $r \circ \alpha' \circ r \in \mathrm{Stab}(\mathbf{e}_n)$. Therefore,

$$r \circ \alpha' \circ r = \alpha \in \mathrm{Stab}(\mathbf{e}_n) \implies \alpha' = r \circ \alpha \circ r.$$

It now follows that $\mathrm{Stab}(\mathbf{v}) \simeq \mathrm{Stab}(\mathbf{e}_n)$; the isomorphism is given by the relation

$$\mathrm{Stab}(\mathbf{v}) \ni \alpha' \mapsto r \circ \alpha' \circ r.$$

Therefore, $\mathrm{Stab}(\mathbf{v}) \simeq \mathrm{O}(n-1)$. Finally, let $\mathbf{w} \neq \mathbf{0}$ be an arbitrary point of $\mathbb{R}^n$. It is now straightforward that

$$\mathrm{Stab}(\mathbf{w}) = (\mathrm{Stab})(\mathbf{w}/\|\mathbf{w}\|) \simeq \mathrm{O}(n-1).$$

We have proved the following:

Proposition 3.3.26 *The action of* $\mathrm{O}(n)$ *is free for* $n = 1$. *When* $n > 1$, *the stabiliser of every nonzero element is a group isomorphic to* $\mathrm{O}(n-1)$.

3.3.3 The Isometry Theorem

The following theorem provides a complete description of the group of Euclidean isometries:

Theorem 3.3.27 (Isometry Theorem) *Every element of* $E(n)$ *is a composition of an element of the translation group* $\mathrm{T}(n)$ *and an element of the orthogonal group* $\mathrm{O}(n)$. *In fact,* $E(n) = \mathrm{T}(n) \rtimes \mathrm{O}(n)$.

Proof Let $f \in E(n)$. Composing by a translation if necessary, we may assume that $f(\mathbf{0}) = \mathbf{0}$. Therefore, it suffices to show that $f \in \mathrm{O}(n)$. Let $\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ be the

canonical basis of $\mathbb{R}^n$. By Proposition 3.3.4 we have

$$\|f(\mathbf{e}_i)\| = \|\mathbf{e}_i\| = 1, \quad f(\mathbf{e}_i) \cdot f(\mathbf{e}_j) = \mathbf{e}_i \cdot \mathbf{e}_j = 0,\ i, j = 1, \ldots, n,\ i \neq j.$$

Therefore, $f(\mathbf{e}_i)$, $i = 1, \ldots n$ form an orthonormal basis of $\mathbb{R}^n$. For $\mathbf{x} = (x_1, \ldots, x_n) \in \mathbb{R}^n$, we write

$$\mathbf{x} = \sum_{i=1}^{n} x_i \mathbf{e}_i, \quad f(\mathbf{x}) = \sum_{i=1}^{n} x_i' f(\mathbf{e}_i),$$

thus

$$x_i = \mathbf{x} \cdot \mathbf{e}_i = f(\mathbf{x}) \cdot f(\mathbf{e}_i) = x_i', \quad i = 1, \ldots n.$$

It follows that

$$f(\mathbf{x}) = f\left(\sum_{i=1}^{n} x_i \mathbf{e}_i\right) = \sum_{i=1}^{n} x_i f(\mathbf{e}_i).$$

Hence for $\mathbf{x} = (x_1, \ldots, x_n)$, $\mathbf{y} = (y_1, \ldots, y_n) \in \mathbb{R}^n$ and $\lambda, \mu \in \mathbb{R}$, we obtain

$$\begin{aligned} f(\lambda\mathbf{x} + \mu\mathbf{y}) &= f\left(\sum_{i=1}^{n} (\lambda x_i + \mu y_i)\mathbf{e}_i\right) \\ &= \sum_{i=1}^{n} (\lambda x_i + \mu y_i) f(\mathbf{e}_i) \\ &= \lambda \sum_{i=1}^{n} x_i f(\mathbf{e}_i) + \mu \sum_{i=1}^{n} y_i f(\mathbf{e}_i) \\ &= \lambda f(\mathbf{x}) + \mu f(\mathbf{y}). \end{aligned}$$

We deduce that f is a linear mapping. Its matrix with respect to the canonical basis has rows the orthonormal vectors $f(\mathbf{e}_i)$, $i = 1, \ldots, n$, so $f \in \mathrm{O}(n)$.

We now show that every element of $E(n)$ can be written as a composition of an element of $\mathrm{T}(n)$ and an element of $\mathrm{O}(n)$ in a unique way. Indeed, if there were $f \in E(n)$ with $f = T_{\mathbf{a}} \circ A = T_{\mathbf{b}} \circ B$, $T_{\mathbf{a}}, T_{\mathbf{b}} \in \mathrm{T}(n)$, $A, B \in \mathrm{O}(n)$, then

$$T_{\mathbf{a}-\mathbf{b}} = BA^{-1},$$

which is only possible if $T_{\mathbf{a}} = T_{\mathbf{b}}$ and $A = B$. □

3.3.4 Invariance of Curve Length

Let $\boldsymbol{\gamma}$ be a continuously differentiable curve of $\mathbb{R}^n$, $\boldsymbol{\gamma}(t) = (x_1(t), \ldots, x_n(t))$, $t \in I = [a, b]$. The Euclidean norm of the derivative is

$$\|\dot{\boldsymbol{\gamma}}(t)\| = \left(\sum_{i=1}^{n} \dot{x}_i^2(t)\right)$$

and the *Euclidean length* of $\boldsymbol{\gamma}$ is defined by the integral of its Euclidean norm:

$$\ell(\boldsymbol{\gamma}) = \int_a^b \|\dot{\boldsymbol{\gamma}}(t)\|\, dt.$$

Proposition 3.3.28 *The Euclidean length is invariant under the action of $E(n)$.*

Proof Let $\boldsymbol{\gamma} = \boldsymbol{\gamma}(t)$ be a continuously differentiable curve of $\mathbb{R}^n$ and also let $T_{\mathbf{a}}$ be a transformation and $A \in \mathrm{O}(n)$ be a corresponding orthogonal transformation α. Then the following hold:

(1) $\|(T_{\mathbf{a}} \dot{\circ}\, \boldsymbol{\gamma})(t)\| = \|(\boldsymbol{\gamma}(t)\dot{} + \mathbf{a})\| = \|\dot{\boldsymbol{\gamma}}(t)\|$ and
(2) $\|(\alpha\dot{(\boldsymbol{\gamma})})(t)\| = \|\dot{\boldsymbol{\gamma}}(t)\|$.

We only prove (2). Let $A = [a_{ij}]_{i,j=1,\ldots,n}$ be the matrix of α. Then $\alpha(\boldsymbol{\gamma}(t)) = \boldsymbol{\gamma}(t)A$ and taking derivatives on both sides using the chain rule[1] we have

$$\dot{\boldsymbol{\gamma}}(t) D\alpha(\boldsymbol{\gamma}(t)) = \dot{\boldsymbol{\gamma}}(t)A = \alpha(\dot{\boldsymbol{\gamma}}(t)),$$

with $D\alpha$ being the Jacobian matrix of α. Since for every $\mathbf{x} \in \mathbb{R}^n$,

$$\alpha(\mathbf{x}) = \left(\sum_{i=1}^{n} a_{i1}x_i, \ldots, \sum_{i=1}^{n} a_{in}x_i\right)$$

it immediately follows that $D\alpha(\mathbf{x}) = A^T$. It also holds that $\|\mathbf{x} \cdot A^T\| = \|\alpha^{-1}(\mathbf{x})\| = \|\mathbf{x}\|$. Therefore,

$$\|(\dot{\alpha(\boldsymbol{\gamma})}(t)\| = \|\dot{\boldsymbol{\gamma}}(t) D\alpha(\boldsymbol{\gamma}(t))\| = \|\alpha^{-1}(\dot{\boldsymbol{\gamma}}(t))\| = \|\dot{\boldsymbol{\gamma}}(t)\|.$$

□

The following proposition is important in many ways; it mainly opens the discussion about geodesics that we shall examine below in more detail.

[1] We write the chain rule in a way that preserves the vectors in the form of row matrices.

Proposition 3.3.29 *If* $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ *then the line segment with endpoints* $\mathbf{x}, \mathbf{y}$ *is the shortest path connecting them.*

Proof Applying a translation, we may assume $\mathbf{x} = \mathbf{0}$, and applying an orthogonal transformation we may also assume that $\mathbf{y} = (0, \ldots, 0, y_n)$, $y_n > 0$. Then the line segment joining $\mathbf{x}, \mathbf{y}$ is

$$\mathbf{c}(t) = t y_n \mathbf{e}_n, \quad t \in [0, 1],$$

and since $\|\dot{\mathbf{c}}(t)\| = y_n$, we get

$$\ell(\mathbf{c}) = \int_0^1 y_n \, dt = y_n.$$

Now, if $\boldsymbol{\gamma}$ is any continuously differentiable curve $\boldsymbol{\gamma} = \boldsymbol{\gamma}(t)$, $t \in [a, b]$, with $\boldsymbol{\gamma}(a) = \mathbf{0}$, $\boldsymbol{\gamma}(b) = \mathbf{y}$, then

$$\begin{aligned}
\ell(\boldsymbol{\gamma}) &= \int_a^b \left(\sum_{i=1}^n (\dot{x}_i(t))^2 \right)^{1/2} dt \\
&\geq \int_a^b |\dot{x}_n(t)| \, dt \\
&\geq \left| \int_a^b \dot{x}_n(t) \, dt \right| \\
&= |x_n(b) - x_n(a)| \\
&= |y_n - 0| = y_n = \ell(\mathbf{c}).
\end{aligned}$$

Taking infima over all $\boldsymbol{\gamma}$ that join $\mathbf{x}, \mathbf{y}$, we obtain the desired result. □

Geodesics

Let (X, d) be a metric space. We define a new metric d_{I} on X, the *induced intrinsic metric* as follows:

$$d_{\mathrm{I}}(x, y) = \inf\{\ell(\gamma)\},$$

where the infimum is taken over the lengths $\ell(\gamma)$ of all paths from x to y. Here, by a *path* we will mean a continuous curve

$$\gamma : [0, 1] \to X,$$

such that $\gamma(0) = x$ and $\gamma(1) = y$. (The lengths of such paths are defined in the same manner as of the lengths of rectifiable curves.) We set $d_1(x, y) = \infty$ if there is no finite-length path from x to y.

Definition 3.3.30 If

$$d_1(x, y) = d(x, y),$$

for all x and y we say that (X, d) is a *path metric space* and that d is *intrinsic*. If the infimum is realised for every $x, y \in X$, then the metric space (X, d) is called *geodesic* and the paths on which this occurs are called *geodesics*.

Corollary 3.3.31 *The Euclidean metric space $(\mathbb{R}^n, e_n)$ is a geodesic metric space. Its geodesics are straight lines.*

Chordal Metric in S^{n-1}

In what follows we give an example of a metric that is not intrinsic; we will encounter this metric again in Chap. 8. Let S^{n-1} be the unit sphere $S(\mathbf{0}, 1)$ and consider the restriction c_n of the Euclidean metric e_n on S^{n-1}:

$$c_n(\mathbf{x}, \mathbf{y}) = e_n(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|, \quad \mathbf{x}, \mathbf{y} \in S^{n-1}.$$

Let $C(n) = \mathrm{Isom}(S^{n-1}, c_n)$. The Spherical Geometry that we will examine in Chap. 6 is the geometry $(S^{n-1}, \mathrm{O}(n))$; here $\mathrm{O}(n)$ is the isometry group of the *spherical metric* s_{n-1} which is the intrinsic metric generated by the chordal metric c_n but is not equal to c_n. However, we will see that $C(n) = \mathrm{O}(n)$.

To be more specific, consider the points $\mathbf{x} = (1, 0)$ and $\mathbf{y} = (0, 1)$ on the circle S^1; we have that

$$c_2(\mathbf{x}, \mathbf{y}) = \sqrt{2}.$$

Let $\mathbf{c}$ be a curve of S^1, $\mathbf{c}(t) = (x(t), y(t))$, $\mathbf{c}(0) = \mathbf{x}$, $\mathbf{c}(1) = \mathbf{y}$. Since the curve lies in the circle, we may assume that $x(t) = \cos(\theta(t))$, $y(t) = \sin(\theta(t))$, for some (differentiable) function θ with $\theta(0) = 0$, $\theta(1) = \pi/2$. We have $\|\dot{\boldsymbol{\gamma}}(t)\| = |\dot{\theta}(t)|$ and

$$\ell(\boldsymbol{\gamma}) = \int_0^1 |\dot{\theta}(t)|\, dt \geq |\theta(1) - \theta(0)| = \pi/2.$$

The value $\pi/2$ is obtained for $\dot{\theta}(t) = \pm 1$ and thus

$$d_I(\mathbf{x}, \mathbf{y})(= s_1(\mathbf{x}, \mathbf{y})) = \pi/2 > \sqrt{2}.$$

3.3.5 *Reflection Theorem*

In 1937, Élie Cartan published a theorem stating that the orthogonal group $\mathrm{O}(n, F)$, $F = \mathbb{R}, \mathbb{C}$, is generated by reflections. This theorem was later generalised by Dieudonné. In its broad generalisation, the Reflection Theorem is proved for non-degenerate symmetric bilinear spaces over fields of characteristic $\neq 2$, [24]. In what follows, we prove the version of the Cartan-Dieudonné Theorem which is of our particular interest.

Theorem 3.3.32 (Cartan-Dieudonné) *Every element of the orthogonal group* $\mathrm{O}(n)$ *other than the identity element can be written as a composition of at most n reflections. When* $n \geq 2$*, the identity map is the composition of each reflection with itself.*

Proof The proof is by induction on the dimension. When $n = 1$, the only orthogonal transformations are the identity $id.$ and its inverse $-id.$ But $-id.$ is the reflection in $W = \{0\}$. When $n \geq 2$, we have seen that every reflection r_W is involutive, that is, $r_W \circ r_W = id.$ Now let $n \geq 2$ and also let α be an orthogonal transformation different from the identity mapping (Fig. 3.4). We distinguish two cases:

I. *1 is an eigenvalue of* α. In this case, let $\mathbf{v}$ be the corresponding eigenvector, $\alpha(\mathbf{v}) = \mathbf{v}$. If $\mathbf{A} = \mathbf{v}/\|\mathbf{v}\|$, let $W = W_{\mathbf{A}}$ be the hyperplane such that $\mathbb{R}^n = W \oplus \langle \mathbf{A} \rangle$. We will prove that $\alpha(W) \subset W$. Indeed, if $\mathbf{w} \cdot \mathbf{v} = 0$, for every $\mathbf{w} \in W$, then $\alpha(\mathbf{w}) \cdot \alpha(\mathbf{v}) = 0$, because α is an isometry with $\alpha(\mathbf{0}) = \mathbf{0}$. But since $\alpha(\mathbf{v}) = \mathbf{v}$, it follows that

$$\alpha(\mathbf{w}) \cdot \mathbf{v} = 0 \implies \alpha(W) \subset W.$$

Furthermore, since α is not the identity, the restriction α_W of α to W is not the identity of W. Since also $\dim(W) = n - 1$, by the inductive hypothesis we have that α_W can be written as a composition of at most $k \leq n - 1$ reflections in hyperplanes W_i, $i = 1, \ldots, k$ (here the hyperplanes are $n - 2$-dimensional):

$$\alpha_W = r'_k \circ \cdots \circ r'_1.$$



Fig. 3.4 Élie Joseph Cartan, 1869–1951

We now extend each reflection r_i' to a reflection r_i defined on all $\mathbb{R}^n$ in the following way: we first have that $W = W_i \oplus V_i$, $V_i = W_i^{\perp}$, $i = 1, \ldots, k$. If $V = \langle \mathbf{v} \rangle$, since $V \perp W$ it follows that $U_i = W_i \oplus V$, $i = 1, \ldots, k$, are hyperplanes. From the relation (3.7) we now obtain that for each $\mathbf{x} \in W$, we have:

$$r_i'(\mathbf{x}) = \mathrm{pr}_{W_i}(\mathbf{x}) - \mathrm{pr}_{V_i}(\mathbf{x}).$$

Now,

$$\mathbb{R}^n = U_i \oplus V_i = W_i \oplus V \oplus V_i,$$

so, we define the r_i in $\mathbb{R}^n$ by the relations:

$$r_i(\mathbf{x} + \lambda \mathbf{v}) = r_i'(\mathbf{x}) + \lambda \mathbf{v},$$

for each $\mathbf{x} + \lambda \mathbf{v} \in W \oplus V = \mathbb{R}^n$. Due to our assumptions,

$$r_i(\mathbf{x} + \lambda \mathbf{v}) = \mathrm{pr}_{U_i}(\mathbf{x} + \lambda \mathbf{v}) - \mathrm{pr}_{V_i}(\mathbf{x} + \lambda \mathbf{v}),$$

therefore r_i are reflections in the hyperplanes U_i, $i = 1, \ldots, k$. Since now α is the identity in V, it follows that

$$\alpha = r_k \circ \cdots \circ r_1.$$

II. *1 is not an eigenvalue of* α (Fig. 3.5).

Let $\mathbf{v} \neq \mathbf{0}$ be a hyperplane $W \perp (\alpha(\mathbf{v}) - \mathbf{v})$. Since α is an isometry that preserves the origin, it is $\|\alpha(\mathbf{v})\| = \|\mathbf{v}\|$ by Proposition 3.3.4. Also, by Lemma 3.3.7 we have

$$r_W(\mathbf{v}) = \alpha(\mathbf{v}),$$



Fig. 3.5 Jean Alexandre Eugène Dieudonné, 1906–1992

where r_W is the reflection in W. Now, $r_W \circ \alpha$ stabilises $\mathbf{v}$:

$$r_W(\alpha(\mathbf{v})) = r_W(r_W(\mathbf{v})) = \mathbf{v}.$$

Since $r_W^2 = id.$, if $r_W \circ \alpha = id.$, then

$$r_W \circ (r_W \circ \alpha) = r_W \implies \alpha = r_W,$$

which is absurd because $\alpha(\mathbf{v}) \neq \mathbf{v}$ for every $\mathbf{v}$. Hence we are in Case I: there are at most $k \leq n-1$ reflections r_k in hyperplanes W_k such that

$$r_W \circ \alpha = r_k \circ \cdots \circ r_1.$$

Therefore,

$$\alpha = r_W \circ r_k \circ \cdots \circ r_1.$$

□

The above theorem generalises to the case of any Euclidean isometry if we allow reflections on *affine* hyperplanes that is, hyperplanes that do not necessarily pass through the origin (Exercise 9, Sect. 3.4). If $f \in E(n)$, then it is written uniquely as

$$f = T_{\mathbf{a}} \circ \alpha, \quad T_{\mathbf{a}} \in \mathrm{T}(n),\ \alpha \in \mathrm{O}(n).$$

By Theorem 3.3.32 there are $k \leq n$ reflections r_i, $i = 1, \ldots, k$ on hyperplanes from the origin such that

$$f = T_{\mathbf{a}} \circ r_k \circ \ldots r_1.$$

However, the composition of translation and reflection in a hyperplane is a reflection in an affine hyperplane (Exercise 10, Sect. 3.4). We obtain the following:

Theorem 3.3.33 (Reflection Theorem) *Each element of $E(n)$ can be written either as a composition of at most n reflections or as a composition of a reflection on an affine hyperplane and at most n reflections.*

3.4 Exercises

1. Draw two vectors $\mathbf{u}$, $\mathbf{v}$ and with geometric arguments verify the type of the projection operator (3.1). Show that

$$\|\mathbf{v} - \mathrm{pr}_{\mathbf{u}}(\mathbf{v})\| = \|\mathbf{v}\| \sin\theta,$$

where $\theta = \angle(\mathbf{u}, \mathbf{v})$.

2. Verify that the Gram-Schmidt procedure indeed gives an orthonormal basis.
3. Let $\theta \in \mathbb{R}$ be and the transformation $\alpha : \mathbb{R}^2 \to \mathbb{R}^2$ be where

$$\alpha(x, y) = (x \cos\theta - y \sin\theta,\ x \sin\theta + y \cos\theta).$$

   A $\mathbf{e}_1 = (1, 0)$ and $\mathbf{e}_2 = (0, 1)$, show that $\alpha(\mathbf{e}_1), \alpha(\mathbf{e}_2)$ are an orthonormal basis of $\mathbb{R}^2$.
4. Prove that every line passing through $\mathbf{0}$ is a 1-dimensional vector subspace of $\mathbb{R}^n$, while a line not passing through the origin is not a vector subspace of $\mathbb{R}^n$.
5. Show that in the case of $\mathbb{R}^3$, two different 2-planes from the origin intersect in a line. Does this hold in dimensions higher than 3?
6. Let ℓ be a line of $\mathbb{R}^2$ with the formula $Ax + By = C$, $|A|^2 + |B|^2 \neq 0$. Prove that the reflection r in ℓ has the form

$$r(x, y) = \left(x - 2\frac{A(Ax + By)}{\|(A, B)\|^2},\ y - 2\frac{B(Ax + By)}{\|(A, B)\|^2}\right),$$

   for each $(x, y) \in \mathbb{R}^2$.
7. Prove that every isometry of a metric space is an injective mapping.
8. Prove that the set of isometries of a metric space (X, d) is a group with group operation the composition of mappings.
9. Working as in Sect. 3.2.3, show that the reflection in (any) hyperplane

$$W_{\mathbf{A},b} = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{x} \cdot \mathbf{A} = b,\ \|\mathbf{A}\| = 1,\ b \in \mathbb{R}\},$$

   is given by

$$r_W(\mathbf{x}) = \mathbf{x} + 2(b - (\mathbf{x} \cdot \mathbf{A}))\mathbf{A}, \quad \mathbf{x} \in \mathbb{R}^n.$$

   Show that r_W is a linear mapping if and only if $b = 0$. State and prove a proposition on reflections on affine hyperplanes, analogous to Proposition 3.2.10.
10. Prove the following.
    (a) The composition of two reflections in parallel hyperplanes is a translation.
    (b) The composition of translation and reflection is equal to the composition of a reflection on an affine hyperplane with a translation.
11. Show the converse of the previous exercise: each element of $\mathrm{T}(n)$ is the composition of two reflections in parallel hyperplanes.
12. Prove that each element of SO(2) is the product of two reflections.
13. Prove that each element of O(2) is the product of at most two reflections. What can you say about O(1)?
14. Prove that for every reflection r_W in a hyperplane $W = W_{\mathbf{A}}$ there exists an orthonormal basis in which the matrix of r_W is diagonal, such that the elements of the diagonal are all 1, except one which is -1.

15. The *symmetry operator* $s_n : \mathbb{R}^n \to \mathbb{R}^n$ *with respect to the origin* is given by $s_n(\mathbf{x}) = -\mathbf{x}$, $\mathbf{x} \in \mathbb{R}^n$ (see also Exercise 9, 2.2). The symmetry operator s_n is a linear mapping with matrix $-I_n$ with respect to the canonical basis. Using Theorem 3.3.22, prove that s_n cannot be the product of fewer than n reflections.
16. Let $\mathbf{x} = (x, y, z) \in \mathbb{R}^3$ be a point other than $\mathbf{0} = (0, 0, 0)$ and $\mathbf{x}' = (x, y, 0)$ its projection onto the plane $z = 0$. Let also

$$\theta = \angle(\mathbf{x}', \mathbf{e}_1), \quad \phi = \angle(\mathbf{x}, \mathbf{e}_3).$$

We apply the transformation $\alpha(\theta)$ with matrix

$$A(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

to $\|\mathbf{x}\|\mathbf{e}_1$ in order to obtain the point $\mathbf{x}'$. This transformation simultaneously defines a new orthonormal basis for $\mathbb{R}^3$, $\{\alpha(\theta)(\mathbf{e}_1), \alpha(\theta)(\mathbf{e}_2), \mathbf{e}_3\}$. Now, in the plane defined by $\alpha(\theta)(\mathbf{e}_1)$ and $\mathbf{e}_3$, we apply a rotation α' by $\pi/2-\theta$. The matrix of this rotation with respect to the basis $\{\alpha(\theta)(\mathbf{e}_1), \alpha(\theta)(\mathbf{e}_2), \mathbf{e}_3\}$ is

$$A'(\phi) = \begin{pmatrix} \sin\phi & 0 & -\cos\phi \\ 0 & 1 & 0 \\ \cos\phi & 0 & \sin\phi \end{pmatrix}$$

It follows that the transformation $\alpha' \circ \alpha$ with associated matrix with respect to the canonical basis

$$A(\theta) \cdot A'(\phi) = \begin{pmatrix} \cos\theta\sin\phi & -\sin\theta & -\cos\theta\cos\phi \\ \sin\theta\sin\phi & \cos\theta & -\cos\phi\sin\theta \\ \cos\phi & 0 & \sin\phi \end{pmatrix}, \tag{3.11}$$

maps $\|\mathbf{x}\|\mathbf{e}_1$ to $\mathbf{x}$.

(a) Complete the details of the above discussion.
(b) Prove that SO(3) acts transitively on spheres centred at the origin.
(c) Prove that every element of SO(3) is written in the form (3.11).

17. Prove that for every isometry $f \in E(n)$ there exists an isometry $\tilde{f} \in E(n+1)$ such that $\tilde{f}_{\mathbb{R}^n} = f$.
18. Let (X, G) be a geometry. G is said to act *doubly transitively* on X if for every pair (x_1, x_2), (y_1, y_2) of distinct points there exists $g \in G$ such that $g(x_i) = y_i$, $i = 1, 2$. Prove that in Euclidean Geometry if $(\mathbf{x}_1, \mathbf{x}_2)$, $(\mathbf{y}_1, \mathbf{y}_2)$ are pairs of

distinct points, then there exists a Euclidean isometry g with $g(\mathbf{x}_i) = \mathbf{y}_i$, $i = 1, 2$, if and only if

$$\|\mathbf{x}_1 - \mathbf{x}_2\| = \|\mathbf{y}_1 - \mathbf{y}_2\|.$$

19. Let $X \neq \emptyset$ be a set, and suppose that there is an injective mapping

$$\iota_X : X \to (\mathbb{R}^n, e_n).$$

For each $x, y \in X$ we define

$$d(x, y) = e_n(\iota_X(x),\ \iota_X(y)).$$

(a) Prove that (X, d) is a metric space.
(b) Identifying X through ι_X with its image in $\mathbb{R}^n$ which we will also denote by X, we write

$$d(\mathbf{x}, \mathbf{y}) = e_n(\mathbf{x}, \mathbf{y}), \quad \mathbf{x}, \mathbf{y} \in X.$$

Then d is called the *metric induced by the Euclidean metric on* X. Prove that if $f \in E(n)$, then the restriction of $f_X : X \to \mathbb{R}^n$ on X is an isometry of (X, d) if $f(X) = X$.
(c) The converse of the previous question is in principle very difficult to prove for arbitrary metric spaces. Here, we will consider the case where $X = S^{n-1}$, $d = c_n$. If $f : S^{n-1} \to S^{n-1}$ is an isometry, we define an extension of F to all $\mathbb{R}^n$ by the relation

$$F(\mathbf{0}) = \mathbf{0}, \quad F(\mathbf{x}) = f(x/\|\mathbf{x}\|) \cdot \|\mathbf{x}\|.$$

Prove in turn that

(i) $\|F(\mathbf{x})\| = \|\mathbf{x}\|$, for every $\mathbf{x} \in \mathbb{R}^n$.
(ii) $F(\mathbf{x}) \cdot F(\mathbf{y}) = \mathbf{x} \cdot \mathbf{y}$, for every $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and
(iii) F is a linear mapping (mimick the proof of the Isometry Theorem here).

Chapter 4
Similarity Geometry



Magnitudes are said to have a ratio to one another which can, when multiplied, exceed one another.

—Definition 4 in Book V of the Elements; this definition is due to Eudoxus of Cnidos.

Similarity Geometry is the geometry $(\mathbb{R}^n, S(n))$, where $S(n)$ is the group of similarities of $\mathbb{R}^n$. This geometry is broader than Euclidean Geometry, in the sense that it contains it, but the $S(n)$-properties are fewer than the $E(n)$-properties. In Euclidean Geometry, similarity refers to objects of the same shape but possibly different size. The principles of Similarity Geometry were already studied by the ancient Greeks; Thales' Intercept Theorem, for example, is a theorem of Similarity Geometry. More specifically, Euclid recorded in Book VI of the Elements the basic properties of similar figures and especially similar triangles; in the majority of them, all the relevant results are due to Eudoxus (Fig. 4.1). The group of similarities $S(n)$ contains the group of Euclidean isometries $E(n)$. An element of $S(n)$, a *similarity*, is a mapping that maps one object to another such that all distances are scaled (multiplied or divided) by the same constant. We emphasise that while similarities preserve angles, they generally do not preserve lengths or volumes. The concept of similarity is fundamental to various branches of mathematics besides geometry (and also algebra). Over time, the study of similarities has evolved, influencing modern mathematical fields such as Fractal Geometry and Complex Analysis. An important invariant of the Similarity Geometry is the metric simple ratio as well as the simple ratio of three points, whose fundamental properties we study in the last section of this chapter. Both of these invariants find their origins in Euclid. For this chapter, we refer mainly to [2, 10, 16].

4.1 Similarities in a Metric Space

We begin by defining what a similarity is in an abstract metric space.

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I. D. Platis, *Geometries*, La Matematica per il 3+2 178,
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Fig. 4.1 Eudoxus, c. 407–355 BC

Definition 4.1.1 Let (X, d) be a metric space and let $f : X \to X$ be a surjective mapping. f is called a *similarity* if there exists a $k = k(f) > 0$ such that

$$d(f(x), f(y)) = k \cdot d(x, y), \text{ for each } x, y \in X.$$

It immediately follows that a similarity f is also an injective mapping and we emphasise that the coefficient k which will also be called *coefficient (or ratio) of the similarity* does not depend on the choice of the points $x, y \in X$.

Proposition 4.1.2 *The set* $\mathrm{Sim}(X, d)$ *of similarities of a metric space is a group; the group multiplication is composition of mappings.*

Proof We straightforwardly verify that if f, g are similarities with similarity ratios k_f, k_g, respectively, then $f \circ g$ is also a similarity with similarity ratio $k_f k_g$; the neutral element is the identity mapping in X, and the inverse f^{-1} of a similarity f with similarity ratio k is a similarity with similarity ratio $1/k$. □

Since every isometry of X is automatically a similarity (with similarity ratio 1), we have that the group $\mathrm{Isom}(X, d)$ of isometries of X is a subgroup of $\mathrm{Sim}(X, d)$. But something more is also true:

Proposition 4.1.3 $\mathrm{Isom}(X, d) \lhd \mathrm{Sim}(X, d)$.

Proof For every $f \in \mathrm{Sim}(X, d)$ with similarity coefficient k, and for every $g \in \mathrm{Isom}(X, d)$, we have for every $x, y \in X$ that:

$$\begin{aligned} d(f(g(f^{-1}(x)),\ f(g(f^{-1}(y))) &= k\ d(g(f^{-1}(x)),\ g(f^{-1}(y))) \\ &= k\ d(f^{-1}(x),\ f^{-1}(y)) \\ &= k \cdot (1/k)\ d(x, y) \\ &= d(x, y). \end{aligned}$$

□

Corollary 4.1.4 *Let (X, d) be a metric space. Let $G = \mathrm{Isom}(X, d)$, $G' = \mathrm{Sim}(X, d)$ and consider the geometries (X, G) and (X, G'). The following hold:*

1. *For every $x \in X$, $\mathrm{orb}_G(x) \lhd \mathrm{orb}_{G'}(x)$.*
2. *If G acts transitively on X, then G' also acts transitively on X.*
3. *If G' acts freely on X, then G also acts freely on X.*

Exercise Prove Corollary 4.1.4.

4.2 The Similarity Group $S(n)$

We now come to the similarity group $S(n) = \mathrm{Sim}(\mathbb{R}^n, e_n)$ of $\mathbb{R}^n$. This group contains the group $E(n)$ as a normal subgroup (Proposition 4.1.3) as well as dilations which we define immediately below.

4.2.1 The Group $D(n)$ of Dilations

Definition 4.2.1 Let $\delta > 0, \delta \neq 1$. The mapping $D_\delta : \mathbb{R}^n \to \mathbb{R}^n$ defined by

$$D_\delta(\mathbf{x}) = \delta \cdot \mathbf{x}, \quad \mathbf{x} \in \mathbb{R}^n,$$

will be called *δ-dilation* or simply *dilation*.

Remark 4.2.2 When $\delta < 1$ we sometimes call D_δ a *contraction*.

It is clear that these mappings are linear mappings and since for every $\delta > 0$ and for every $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ we have

$$e_n(D_\delta(\mathbf{x}),\ D_\delta(\mathbf{y})) = \delta \cdot e_n(\mathbf{x}, \mathbf{y}),$$

it follows that dilations are Euclidean similarities with similarity ratio δ. Let us also observe that for every $D_\delta \in D(n)$,

$$\|D_\delta(\mathbf{x})\| = \delta \cdot \|\mathbf{x}\|, \quad \mathbf{x} \in \mathbb{R}^n. \tag{4.1}$$

Additionally, for each $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ we have

$$D_\delta(\mathbf{x}) \cdot D_\delta(\mathbf{y}) = (\delta\mathbf{x}) \cdot (\delta\mathbf{y}) = \delta^2 \cdot (\mathbf{x} \cdot \mathbf{y}). \tag{4.2}$$

The following proposition holds.

Proposition 4.2.3 *The set $D(n)$ of dilations of $\mathbb{R}^n$ together with the identity mapping, is an abelian group; the group multiplication is composition of mappings. The group $D(n)$ is isomorphic to the multiplicative group $(\mathbb{R}_{>0}, \cdot)$.*

Proof For each $\delta_1, \delta_2 > 0$,

$$D_{\delta_1} \circ D_{\delta_2} = D_{\delta_1 \delta_2} = D_{\delta_2 \delta_1} = D_{\delta_2} \circ D_{\delta_1}.$$

Also, for every $\delta > 0$, $(D_\delta)^{-1} = D_{1/\delta}$. Finally, we may verify straightforwardly that the map

$$D \ni D_\delta \mapsto \delta \in \mathbb{R}_{>0}$$

is a group isomorphism. □

The following holds.

Corollary 4.2.4 *The matrix of the linear mapping D_δ with respect to the canonical basis is δI_n. Its determinant is equal to δ^n.*

Exercise Prove Corollary 4.2.4.

The group $D(n)$ acts on $\mathbb{R}^n$ by

$$(D_\delta, \ \mathbf{x}) \mapsto \delta \mathbf{x}.$$

The orbit of the zero element consists only of this element, while for every $\mathbf{x} \neq \mathbf{0}$ we have

$$\mathrm{orb}(\mathbf{x}) = \{\delta \mathbf{x}, \ | \ \delta > 0\},$$

that is, the open half-line from $\mathbf{0}$ in the direction of $\mathbf{x}$. The orbit space $\mathbb{R}^n / D(n)$ consists of all these half-lines together with the origin. The action is free on $\mathbb{R}^n_*$ ($\mathrm{Stab}(\mathbf{0}) = D(n)$), but it is not transitive. This happens only on the open half-lines that constitute the orbits.

Proposition 4.2.5 *Let $\mathbb{R}^n_* / D(n)$ be the orbit space of the action of $D(n)$ on $\mathbb{R}^n_*$. Then, there is a bijection $\mathbb{R}^n_* / D(n) \to S^{n-1}$.*

Proof Every orbit is an open ray from the origin in the direction of some vector that we can assume to be unit. Conversely, every unit vector uniquely determines a ray from the origin. □

Proposition 4.2.6 *The action of $D(n)$ preserves angles.*

Proof If any of the vectors is zero, we have nothing to prove. If $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n_*$, and $\angle(\mathbf{x}, \mathbf{y}) = \theta$, let $D_\delta \in D(n)$. Using (4.1), (4.2), we have

$$\begin{aligned}\cos\left(\angle(D_\delta(\mathbf{x}),\ D_\delta(\mathbf{y}))\right) &= \frac{D_\delta(\mathbf{x}) \cdot D_\delta(\mathbf{y})}{\|D_\delta(\mathbf{x})\|\ \|D_\delta(\mathbf{y})\|} = \frac{\delta^2\ (\mathbf{x} \cdot \mathbf{y})}{(\delta\|\mathbf{x}\|)\ (\delta\|\mathbf{y}\|)} \\ &= \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\|\ \|\mathbf{y}\|} = \cos\theta.\end{aligned}$$

□

4.2.2 Similarity Theorem

Before proving the Similarity Theorem below, we will first describe the form of the elements of the similarity group $S(n)$.

Proposition 4.2.7 *Each element of the similarity group $S(n)$ can be written in as a composition of a Euclidean isometry and a dilation.*

Proof Let $f \in S(n)$ be a similarity with similarity ratio k. We define $g : \mathbb{R}^n \to \mathbb{R}^n$ by

$$g = f \circ D_{1/k}.$$

For each $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ we have:

$$\begin{aligned}\|g(\mathbf{x}) - g(\mathbf{y})\| &= \left\| f(D_{1/k}(\mathbf{x})) - f(D_{1/k}(\mathbf{y})) \right\| \\ &= k\ \|D_{1/k}(\mathbf{x}) - D_{1/k}(\mathbf{y})\| \\ &= k \cdot \frac{1}{k}\ \|\mathbf{x} - \mathbf{y}\| \\ &= \|\mathbf{x} - \mathbf{y}\|.\end{aligned}$$

Therefore $g \in E(n)$. □

From Proposition 4.2.7 and the Isometric Theorem, we obtain the following:

Theorem 4.2.8 (Similarity Theorem) *Every element of $S(n)$ can be written as a composition of an element of $E(n)$ and an element of $D(n)$: in fact,*

$$S(n) = E(n) \rtimes D(n) \simeq E(n) \rtimes (\mathbb{R}_{>0}, \cdot).$$

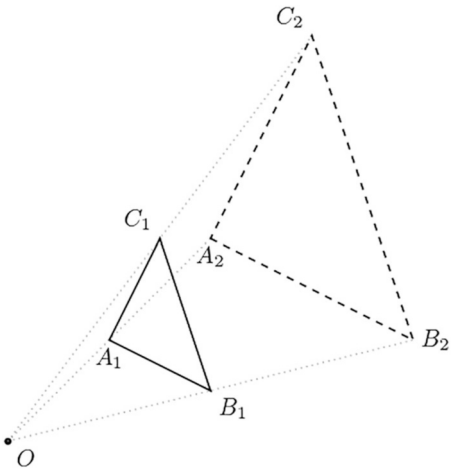


Fig. 4.2 A homothety is a similarity of a special form

Proof With the notation of the proof of Proposition 4.2.7, suppose now that there exist $g' \in E(n)$ and $D_{k'}D(n)$ such that

$$f = g \circ D_k = g' \circ D_{k'}.$$

Since the similarity ratio of f is k, we must have $k = k'$. But then $g' = g$. □

We observe that every element of $D(n)$ other than the identity cannot be written as a composition of two or more reflections, since the reflections are isometries.

Corollary 4.2.9 *Any element $f \in S(n)$ has the form*

$$f(\mathbf{x}) = \delta \mathbf{x} A + \mathbf{a},$$

where $\delta > 0$, $A \in \mathrm{O}(n)$ and $\mathbf{a} \in \mathbb{R}^n$.

Remark 4.2.10 Elements $f \in S(n)$ of the form $f(\mathbf{x}) = \delta \mathbf{x} + \mathbf{a}$ are called *homotheties*, see Fig. 4.2 as well as Exercise 4, Sect. 4.4.

An interesting application of the Similarity Theorem is the following:

Theorem 4.2.11 *In the Similarity (and therefore in the Euclidean) Geometry of $\mathbb{R}^n$, the sum of the angles of a triangle is π.*

Proof Let $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ be the vertices of a given triangle; since $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ are linearly independent, it follows that they lie in a plane. Therefore, we can restrict ourselves to the case $n = 2$. We first assume that the triangle is a right triangle. Applying appropriate similarities we may also assume that

$$\mathbf{x}_1 = (0, 0) \quad \mathbf{x}_2 = (1, 0), \quad \mathbf{x}_3 = (0, x),$$

with $x > 0$. For the acute angles α, β of the triangle, we have

$$\cos\alpha = \frac{x}{\sqrt{x^2+1}}, \quad \cos\beta = \frac{1}{\sqrt{x^2+1}}.$$

Therefore,

$$\alpha + \beta = \arccos(x/\sqrt{x^2+1}) + \arccos(1/\sqrt{x^2+1}).$$

Taking cosines on both sides and using the relation

$$\sin(\arccos y)) = \sqrt{1-y^2},$$

we end up with $\cos(\alpha+\beta) = 0$, so $\alpha + \beta = \pi/2$.

In the general case we can assume that

$$\mathbf{x}_1 = (0,0) \quad \mathbf{x}_2 = (1,0), \quad \mathbf{x}_3 = (x,y),$$

where $x, y > 0$. If $x = 1$, the triangle is a right triangle, thus we are interested in the cases $x < 1$, $x > 1$. If $x < 1$, we consider the two right triangles with vertices $\mathbf{x}_1$, $\mathbf{x}_3$, $(x,0)$ and $(x,0)$, $\mathbf{x}_3$, $\mathbf{x}_2$, respectively, and apply the result of the right triangle case. Similarly for $x > 1$. □

4.3 Metric Simple Ratios

Before going into details about metric simple ratios, we will examine the action of $S(n)$ on $\mathbb{R}^n$.

4.3.1 The Action of S(n)

We start with the following proposition:

Proposition 4.3.1 *$S(n)$ acts doubly transitively on $\mathbb{R}^n$.*

Proof We first observe that if $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^n$, then we may apply a translation followed by a rotation and a dilation to map them to the points $\mathbf{0}$, $\mathbf{e}_1$, respectively. We denote by f this similarity and by g the similarity that maps the points $\mathbf{y}_1, \mathbf{y}_2$ to $\mathbf{0}$, $\mathbf{e}_1$, respectively. Then $g \circ f^{-1}$ maps $\mathbf{x}_1, \mathbf{x}_2$ to $\mathbf{y}_1, \mathbf{y}_2$, respectively. □

We conclude that in Similarity Geometry we can assume that all line segments have length 1, since the $S(n)$-class of the segment with ends $\mathbf{0}$ and $\mathbf{e}_1$ contains all

line segments. In the same way, it follows that all spheres are in the $S(n)$-class of the unit sphere.

Proposition 4.3.1 has two interesting consequences. The first one has to do with those elements of $S(n)$ that stabilise two different points.

Proposition 4.3.2 *Let* $\mathbf{x}_1 \neq \mathbf{x}_2$ *be points of* $\mathbb{R}^n$. *Then*

1. $\mathrm{Stab}(\mathbf{x}_1, \mathbf{x}_2) = \{id.\}$ *if* $n = 1$.
2. $\mathrm{Stab}(\mathbf{x}_1, \mathbf{x}_2) \simeq \mathrm{O}(n-1)$ *if* $n \geq 2$.

Proof In the case where $n = 1$, we can assume by Proposition 4.3.1 that $\mathbf{x}_1 = 0$, $\mathbf{x}_2 = 1$ and the result follows immediately. In the case where $n \geq 2$, we can assume again by Proposition 4.3.1 that $\mathbf{x}_1 = \mathbf{0}$, $\mathbf{x}_2 = \mathbf{e}_n$. A similarity that leaves these two points unchanged can only be some orthogonal transformation α whose matrix will necessarily have the form

$$\begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix}, \quad A \in \mathrm{O}(n-1).$$

□

The second consequence of Proposition 4.3.1 is that the only element of $S(n)$ that stabilises $(n+1)$ pairwise distinct points such that any n of these points are linearly independent, is necessarily the identity transformation. The proof is left to Exercise 11, Sect. 4.4.

4.3.2 Fundamental Theorems of Metric and Simple Ratios

The metric simple ratio of three points arises naturally from the question of when $S(n)$ acts triply transitively on $\mathbb{R}^n$. In terms of classical Euclidean Geometry, and when the points are not collinear, this question is when two triangles are similar.

Definition 4.3.3 Let an ordered triple $\mathfrak{p} = (\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3)$ of pairwise distinct points of $\mathbb{R}^n$. The *metric simple ratio of* $(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3)$ is

$$s(\mathfrak{p}) = \|(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3)\| = \frac{\|\mathbf{x}_2 - \mathbf{x}_1\|}{\|\mathbf{x}_2 - \mathbf{x}_3\|}.$$

Proposition 4.3.4 *The metric simple ratio of three points remains invariant by the action of* $S(n)$.

Proof We will show that each element of $S(n)$ preserves the metric simple ratio of an ordered triple $\mathfrak{p} = (\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3)$ of pairwise distinct points. If $f \in S(n)$,

$$s(f(\mathfrak{p})) = \|(f(\mathbf{x}_1), f(\mathbf{x}_2), f(\mathbf{x}_3))\| = \frac{\|f(\mathbf{x}_2) - f(\mathbf{x}_3)\|}{\|f(\mathbf{x}_2) - f(\mathbf{x}_1)\|} = \frac{\|\mathbf{x}_2 - \mathbf{x}_3\|}{\|\mathbf{x}_2 - \mathbf{x}_1\|}$$
$$= \|(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3)\| = s(\mathfrak{p}).$$

□

For every triple of pairwise distinct points $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$, there are six permutations of them.

Proposition 4.3.5 *If $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ are pairwise distinct points in $\mathbb{R}^n$, then for $i, j, k = 1, 2, 3$, with $i \neq j \neq k$, the following hold:*

1. $\|(\mathbf{x}_i, \mathbf{x}_j, \mathbf{x}_k)\| \cdot \|(\mathbf{x}_k, \mathbf{x}_j, \mathbf{x}_i)\| = 1.$
2. $\|(\mathbf{x}_i, \mathbf{x}_j, \mathbf{x}_k)\| \cdot \|(\mathbf{x}_k, \mathbf{x}_i, \mathbf{x}_j)\| \cdot \|(\mathbf{x}_j, \mathbf{x}_k, \mathbf{x}_i)\| = 1.$
3. *All possible metric simple ratios are expressed in terms of the following two:*

$$s_1 = \|(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3)\|, \quad s_2 = \|(\mathbf{x}_1, \mathbf{x}_3, \mathbf{x}_2)\|.$$

4. *We have*

$$s_1 + s_2 \geq 1$$

where equality holds if and only if $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ lie on the same straight line.

Exercise Prove Proposition 4.3.5.

Theorem 4.3.6 (Fundamental Theorem of Metric Simple Ratios) *Let $\mathfrak{p} = (\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3)$ and $\mathfrak{q} = (\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3)$ be ordered triples of pairwise distinct points of $\mathbb{R}^n$, $n > 1$. There exists an $f \in S(n)$ with $f(\mathbf{x}_i) = \mathbf{y}_i$, $i = 1, 2, 3$, if and only if*

$$s_i(\mathfrak{p}) = s_i(\mathfrak{q}), \quad i = 1, 2.$$

Proof If there exists an $f \in S(n)$ such that $f(\mathbf{x}_i) = \mathbf{y}_i$, $i = 1, 2, 3$, then the equality $s_i(\mathfrak{p}) = s_i(\mathfrak{q})$, $i = 1, 2$ follows immediately from the invariance of metric simple ratios (Proposition 4.3.4).

For the converse, we can normalise so that

$$\mathbf{x}_1 = \mathbf{0}, \quad \mathbf{x}_2 = \mathbf{e}_n, \quad \mathbf{x}_3 = \mathbf{x}, \ \mathbf{x} \in \mathbb{R}^n \setminus \{\mathbf{0}, \mathbf{e}_n\}$$

and also that

$$\mathbf{y}_1 = \mathbf{0}, \quad \mathbf{y}_2 = \mathbf{e}_n, \quad \mathbf{y}_3 = \mathbf{y}, \ \mathbf{y} \in \mathbb{R}^n \setminus \{\mathbf{0}, \mathbf{e}_n\}.$$

If $s_i(\mathfrak{p}) = s_i(\mathfrak{q})$, $i = 1, 2$, then $\|\mathbf{x}\| = \|\mathbf{y}\|$ and $\|\mathbf{x} - \mathbf{e}_n\| = \|\mathbf{y} - \mathbf{e}_n\|$. From the last relation we get that $x_n = \mathbf{x} \cdot \mathbf{e}_n = \mathbf{y} \cdot \mathbf{e}_n = y_n$, therefore,

$$\|\mathbf{x}_{n-1}\| = \|(x_1, \ldots, x_{n-1})\| = \|(y_1, \ldots, y_{n-1})\| = \|\mathbf{y}_{n-1}\|.$$

If $\mathbf{x}_{n-1} = \mathbf{y}_{n-1}$, we are done. If not, then there exists a reflection r in $\mathbb{R}^{n-1}$ such that $r(\mathbf{x}_{n-1}) = \mathbf{y}_{n-1}$. The reflection with matrix

$$\begin{pmatrix} R & 0 \\ 0 & 1 \end{pmatrix} \in \mathrm{O}(n),$$

where R is the matrix of r, preserves $\mathbf{0}$, $\mathbf{e}_n$ and maps $\mathbf{x}$ to $\mathbf{y}$. □

Corollary 4.3.7 *For $n > 1$, the $S(n)$-class of a triangle T contains all triangles similar to T.*

We will now see what happens in the trivial case of ordered triples of pairwise distinct points x_1, x_2, x_3 and y_1, y_2, y_3 of the real line $\mathbb{R}$. As before, we can normalise so that

$$x_1 = y_1 = 0, \quad x_2 = y_2 = 1, \quad x_3 = x,\ y_3 = y,\ x, y \in \mathbb{R} \setminus \{0, 1\}.$$

The only element of $S(1)$ that stabilises $0, 1$, is the identity transformation; thus there exists an $f \in S(1)$ with $f(x_i) = y_i$, $i = 1, 2, 3$, if and only if $x = y$. This is equivalent to the relations of metric simple ratios, but it tells us something more: on the real line we have a relation between the points themselves and not just between their distances.

Simple Ratios

Definition 4.3.8 Let $\mathfrak{p} = (\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3)$ be an ordered triple of pairwise distinct collinear points in $\mathbb{R}^n$. Their simple ratio $\sigma(\mathfrak{p})$ is defined by

$$\sigma(\mathfrak{p}) = \sigma(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3) = \frac{\mathbf{x}_2 - \mathbf{x}_1}{\mathbf{x}_3 - \mathbf{x}_2}.$$

The simple ratio is well defined. Indeed, from our assumptions we have that the points of $\mathfrak{p}$ belong to a line $\mathbf{c}(t) = \mathbf{u} + t\mathbf{v}$, so that

$$\mathbf{x}_i = \mathbf{u} + t_i \mathbf{v}, \quad i = 1, 2, 3.$$

Therefore,

$$\sigma(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3) = \frac{t_2 - t_1}{t_3 - t_2}.$$

If $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ are three pairwise distinct collinear points, then 6 simple ratios are defined.

Proposition 4.3.9 *If $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ are pairwise distinct collinear points in $\mathbb{R}^n$, then for $i, j, k = 1, 2, 3$, with $i \neq j \neq k$, the following holds:*

1. $\sigma(\mathbf{x}_i, \mathbf{x}_j, \mathbf{x}_k) \cdot \sigma(\mathbf{x}_k, \mathbf{x}_j, \mathbf{x}_i) = 1$.
2. $\sigma(\mathbf{x}_i, \mathbf{x}_j, \mathbf{x}_k) \cdot \sigma(\mathbf{x}_k, \mathbf{x}_i, \mathbf{x}_j) \cdot \sigma(\mathbf{x}_j, \mathbf{x}_k, \mathbf{x}_i) = -1$.
3. *All simple ratios are expressed in terms of the simple ratio $\sigma = (\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3) \neq 0$.*

Exercise Prove Proposition 4.3.9.

Proposition 4.3.10 *The simple ratio of three collinear points remains invariant by the action of $S(n)$.*

Proof If $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ are collinear, then $\mathbf{x}_i = \mathbf{u} + t_i\mathbf{v}$ for some $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ and $t_i \in \mathbb{R}$, $i = 1, 2, 3$. If f is a similarity, then $f(\mathbf{x}_i) = f(\mathbf{u}) - t_i f(\mathbf{v})$ and therefore

$$\begin{aligned}\sigma(f(\mathbf{x}_i), f(\mathbf{x}_2), f(\mathbf{x}_3)) &= \frac{(f(\mathbf{u}) - t_2 f(\mathbf{v})) - (f(\mathbf{u}) - t_1 f(\mathbf{v}))}{(f(\mathbf{u}) - t_3 f(\mathbf{v})) - (f(\mathbf{u}) - t_2 f(\mathbf{v}))} \\ &= \frac{t_2 - t_1}{t_3 - t_2} = \sigma(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3).\end{aligned}$$

□

Theorem 4.3.11 (Fundamental Theorem of Simple Ratios) *Let $\mathfrak{p} = (\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3)$ and $\mathfrak{q} = (\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3)$ be ordered triples of pairwise distinct collinear points in $\mathbb{R}^n$. There exists an $f \in S(n)$ such that $f(\mathbf{x}_i) = \mathbf{y}_i$, $i = 1, 2, 3$, if and only if*

$$\sigma(\mathfrak{p}) = \sigma(\mathfrak{q}).$$

Proof Suppose there exists an $f \in S(n)$ such that $f(\mathbf{x}_i) = \mathbf{y}_i$, $i = 1, 2, 3$. Then, from the invariance of simple ratios (Proposition 4.3.10), we obtain the equality $\sigma(\mathfrak{p}) = \sigma(\mathfrak{q})$. For the converse, we normalise so that the straight line is $t\mathbf{e}_n$ and that

$$\mathbf{x}_1 = \mathbf{y}_1 = \mathbf{0}, \quad \mathbf{x}_2 = \mathbf{y}_2 = \mathbf{e}_n, \quad \mathbf{x}_2 = x\mathbf{e}_n, \ \mathbf{y}_3 = y\mathbf{e}_n, \quad x, y \in \mathbb{R} \setminus \{0, 1\}.$$

If $x/(x-1) = y/(y-1)$ then $x = y$ and $f = id$. □

4.4 Exercises

1. Prove that every similarity of a metric space is injective.
2. Let (X, d) be a metric space and let $f : X \to X$ be a self-mapping of X. f is called *continuous* at $x_0 \in X$ if

$$\forall \epsilon > 0, \ \exists \delta > 0 : \ d(f(x), f(x_0)) < \epsilon, \ \forall x \in B_{x_0}(\delta).$$

Prove that every similarity is continuous on X and the inverse f^{-1} of f is also continuous on X.

3. A self-mapping $f : X \to X$ of a metric space (X, d) is called *bi-Lipschitz*, if

$$\exists\, K = K(f) \geq 1 : \ \forall x, y \in X, \ (1/K)d(x, y) \leq d(f(x), f(y)) \leq Kd(x, y).$$

   (a) Prove that every bi-Lipschitz mapping with constant K is invertible and that its inverse is also a bi-Lipschitz mapping with constant K.
   (b) Prove that every bi-Lipschitz mapping with constant K is continuous and that its inverse is also continuous.
   (c) Prove that every similarity of (X, d) is a bi-Lipschitz mapping.
   (d) Give an example of a bi-Lipschitz mapping that is not an isometry.

4. Let $D_\delta \in D(n)$ and $T_{\mathbf{a}} \in \mathrm{T}(n)$.

   (a) Prove that

$$T_{\mathbf{a}} \circ D_\delta = D_\delta \circ T_{(1/\delta)\mathbf{a}}.$$

   (b) The mapping $D_{\delta,\mathbf{a}} = T_{\mathbf{a}} \circ D_\delta$, where

$$D_{\delta,\mathbf{a}}(\mathbf{x}) = \mathbf{a} + \delta\mathbf{x}, \quad \mathbf{x} \in \mathbb{R}^n,$$

   is called *a homothety, with homothetic center* $\mathbf{a}$, see also Remark 4.2.10. (Thus, a dilation is nothing more than a homothety centred at $\mathbf{0}$). Prove that the set of homotheties is a group with group multiplication the mapping composition, isomorphic to the group $G = (\mathbb{R}_{>0} \times \mathbb{R}^n, \star)$, with group multiplication $\star$ defined by

$$(\delta', \mathbf{a}') \star (\delta, \mathbf{a}) = (\delta'\delta, \ \mathbf{a}' + \delta'\mathbf{a}),$$

   for each $(\delta', \mathbf{a}')$ and $(\delta, \mathbf{a})$ in $\mathbb{R}_{>0} \times \mathbb{R}^n$.

5. (a) Let $D_\delta \in D(n)$ and $A \in \mathrm{O}(n)$ with corresponding orthogonal transformation α. Prove that

$$\alpha \circ D_\delta = D_\delta \circ \alpha.$$

   (b) Prove also that if $D_{\delta,\mathbf{a}}$ is a homothety, then

$$\alpha \circ D_{\delta,\mathbf{a}} = D_{\delta,\alpha(\mathbf{a})} \circ \alpha.$$

(c) Prove that mappings of the form $D_{\alpha,\delta,\mathbf{a}} = \alpha \circ D_{\delta,\mathbf{a}}$, where

$$D_{\alpha,\delta,\mathbf{a}}(\mathbf{x}) = \alpha(\mathbf{a}) + \delta\alpha(\mathbf{x}), \quad \mathbf{x} \in \mathbb{R}^n,$$

are similarities (with similarity ratio δ). The Similarity Theorem practically guarantees that all elements of the similarity group $S(n)$ are of the above form.

6. Let $\boldsymbol{\gamma} : [a, b] \to \mathbb{R}^n$, $\boldsymbol{\gamma}(t) = \mathbf{x}(t)$, be a continuously differentiable curve. Show the following:
   (a) $\|(D_\delta \circ \boldsymbol{\gamma}(t)\| = \delta\|\dot{\boldsymbol{\gamma}}(t)\|$.
   (b) $\ell(D_\delta \circ \boldsymbol{\gamma}) = \delta\ell(\boldsymbol{\gamma})$.
7. Let Ω be a closed and bounded region of $\mathbb{R}^n$ and $\Omega' = D_\delta(\Omega)$. Prove that

$$\mathrm{Vol}_n(\Omega') = \delta^n \mathrm{Vol}(\Omega).$$

8. Prove that the group of homotheties is a normal subgroup of $S(n)$.
9. Let $\triangle ABC$ and $\triangle A'B'C'$ be triangles in the plane. Use the Fundamental Theorem of Metric Simple Ratios to prove that these triangles are similar if and only if

$$\frac{\|\overrightarrow{AB}\|}{\|\overrightarrow{A'B'}\|} = \frac{\|\overrightarrow{AC}\|}{\|\overrightarrow{A'C'}\|} = \frac{\|\overrightarrow{BC}\|}{\|\overrightarrow{B'C'}\|}.$$

10. Prove that when $n = 2$, then the similarity mapping a pair of distinct points of $\mathbb{R}^2$ to another pair of distinct points is unique. This is known as the **Fundamental Theorem of Planar Similarity Geometry.** Due to Theorem 4.3.6, this result does not extend to dimension $n > 2$. Argue that in dimension 3 this can happen when the given triples are vertices of similar triangles. What happens in dimension $n > 3$?
11. Let $\mathbf{x}_0, \ldots, \mathbf{x}_n$ be $(n + 1)$ pairwise distinct points in $\mathbb{R}^n$. We also assume that any n-tuple of these points is an n-tuple of linearly independent points. Prove that if there exists an $f \in S(n)$ such that $f(\mathbf{x}_i) = \mathbf{x}_i$, $i = 0, \ldots, n$, then it is necessarily the identity transformation.

Chapter 5
Affine Geometry



... Although to penetrate into the intimate mysteries of nature and thence to learn the true causes of phenomena is not allowed to us, nevertheless it can happen that a certain fictive hypothesis may suffice for explaining many phenomena.

—Leonhard Euler

Affine Geometry is the geometry $(\mathbb{R}^n, A(n))$, with the affine group $A(n)$ being the semi-direct product $\mathbb{R}^n \rtimes \mathrm{GL}(n, \mathbb{R})$ of the translation group of $\mathbb{R}^n$ and the general linear group. In practice, Affine Geometry (which includes both Euclidean and Similarity Geometry) is what remains of Euclidean Geometry if we remove the ability of measuring lengths, angles and areas.

We may safely assume that the father of Affine Geometry is Euler (Leonhard Euler, 1707–1783) (Fig. 5.1). He is the one who also introduced the term *affine* (Latin affinis=related) in the book he wrote in 1748 entitled *Introductio in analysin infinitorum*, see [23]. In 1827 Möbius wrote about Affine Geometry in his book *Der barycentrische Calcul*, [37]. In the Erlangen Program, Klein recognises Affine Geometry as a generalisation of Euclidean Geometry. Since then, Affine Geometry has been used, e.g. in 1912, by Wilson and Lewis [58], to express the Special Theory of Relativity, by Weyl in 1918 [54], for his study on Mathematical Physics (Space, time, matter). Even more recently, in 1984, Birman and Nomizu used affine methods in a study of Lorentz Geometry, [12].

To get a first glimpse of Affine Geometry, we may imagine the sun shining through a window made of identical rectangular panes. The shadow on the floor will generally not be rectangular and as the Sun moves, the shadow will change shape. As the Sun sets, the shadows lengthen, so we see that the lengths are not constant. Also, experience shows that the shape of the shadow changes, so the angles are also not constant. On the other hand, we can observe that the shadows of rectangular windows are always parallelograms. This suggests that parallelism is preserved. Roughly speaking, the basic problem of Affine Geometry is to determine what properties all shadows have in common. Since the Sun is far from the Earth, a logical approach is to assume that the rays are parallel. This will lead us to the concept of parallel projection that we will examine in Sect. 5.2.3. What we will

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I. D. Platis, *Geometries*, La Matematica per il 3+2 178,
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Fig. 5.1 Leonhard Euler, 1707–1783

call affine transformations of $\mathbb{R}^n$ are composed of such parallel projections. The Fundamental Theorem of Affine Geometry is of particular importance; one of its many applications is the widespread use of various theorems of classical Euclidean Geometry, such as the theorems of Menelaus and Ceva (Giovanni Ceva, 1647–1734). For this chapter, we refer in particular to [5, 13, 16, 36, 50], but also to [24].

5.1 The Affine Space $\mathbb{R}^n$

Before focusing on the affine space $\mathbb{R}^n$, we first present the essentials of the general theory of affine spaces. As an affine space, $\mathbb{R}^n$ arises as an example of the general theory.

5.1.1 Affine Spaces

The elements of a vector space V are vectors (or translations, if viewed as elements of the additive group $(V, +)$), which we consider as objects of linear transformations. However, in many problems we are interested in geometric data concerning the relative position of shapes in the space under consideration, while linear transformations take a back seat. This is why we introduce the following definition.

Definition 5.1.1 An *affine space A with an associated vector space V*, is a geometric space (that is, its elements are points) related to the vector space V as follows: there is a transitive and free right action of V on A:

$$\begin{aligned} A \times V &\to A \\ (a, \mathbf{v}) &\mapsto a + \mathbf{v}. \end{aligned}$$

The action is denoted as addition, for reasons that will become apparent shortly. We observe that the following properties hold:

(1) *Right neutral element.* For each $a \in A$,

$$a + \mathbf{0} = a,$$

where $\mathbf{0}$ is the zero element of V.

(2) *Associativity.* For each $\mathbf{v}, \mathbf{w} \in V$, and for each $a \in A$,

$$(a + \mathbf{v}) + \mathbf{w} = a + (\mathbf{v} +_V \mathbf{w}),$$

where $+_V$ is the addition of V.

(3) *Transitive and free action.* For every $a \in A$, the mapping

$$V \to A : \mathbf{v} \mapsto a + \mathbf{v},$$

for every $\mathbf{v} \in V$, is bijective.

The first two properties are simply the properties of the definition of the right group action on a set (compare with the corresponding ones of the left group action on a set as described in Definition 2.1.1). The third property characterises free and transitive actions: surjectivity follows from transitivity and injectivity follows from freedom of the action.

Exercise After writing the definition of the *right action* of the group G on a set X, justify analytically the validity of properties (1)–(4) that follow from the Definition 5.1.1 of the affine space A with associated vector space V.

There is a fourth property that follows from the first two above.

(4) *Existence of injective transformations.* For every $\mathbf{v} \in V$, the mapping

$$A \to A : a \mapsto a + \mathbf{v},$$

is bijective.

The third property is very often used in its equivalent form below.

(3') *Subtraction.* For every ordered pair (b, a), $a, b \in A$, there exists a unique $\mathbf{v} \in V$, denoted by $\overrightarrow{ab}$ or $b - a$, such that

$$b = a + \mathbf{v}.$$

The vector $\overrightarrow{ab}$ is understood as the vector with initial point at a and terminal point at b.

Exercise Prove the equivalence of properties (3) and (3').

Remark 5.1.2 A very useful definition equivalent to Definition 5.1.1 of an affine space A with an associated vector space V is the following which is due to

the German mathematician, theoretical physicist, logician and philosopher Weyl (Hermann Klaus Hugo Weyl, 1885–1955). This definition uses two axioms that establish the correspondence between A and V through abstraction.

Definition 5.1.3 A set A with associated vector space V is called an affine space (of V) if the following are satisfied:

- **(A1)** *Existence of abstraction in* A: for every point $a \in A$ and every vector $\mathbf{v} \in V$ there is a unique point $b \in A$ such that $b - a = \overrightarrow{ab} = \mathbf{v}$.
- **(A2)** *Châsles' Law*: for every triple of points $a, b, c \in A$,

$$\overrightarrow{ac} = \overrightarrow{ab} + \overrightarrow{bc} \iff c - a = (c - b) + (b - a).$$

Property **(A2)** is is attributed to the French mathematician Michel Floréal Châsles, 1793–1880.

Exercise Show the equivalence of Definitions 5.1.1 and 5.1.3.

The point b in **(A1)** is understood as the point a that is displaced in the space A by the vector $\mathbf{v}$ (and therefore we also write $b - a = \overrightarrow{ab}$). Châsles' Law can be understood as the equivalence of the two displacements from a to b and then from b to c, to the single displacement from a to c. In Châsles' Law we also recognise the familiar parallelogramme rule for the addition of vectors.

Definition 5.1.4 The *dimension* of an affine space A with associated vector space V is the dimension of V. If $\dim(A) = n$, the set $\{x_0, x_1, \ldots, x_n\}$ is an affine basis of A if and only if the set $\{x_1 - x_0, \ldots, x_n - x_0\}$ is a basis of V.

Every vector space V can be considered as an affine space having itself as an associated vector space. That is, every element of V can be viewed either as a point or as a vector. The zero vector, when considered as a point, is usually denoted by O and is called *origin*. In particular, $\mathbb{R}^n$ is an affine space of dimension n and from now on we will only deal with this particular space. Up until now we have treated $\mathbb{R}^n$ as a vector space and denoted its elements as vectors (although we have agreed to identify points with vectors). To avoid overloading the notation, *we will continue to denote the elements of* $\mathbb{R}^n$ *by* $\mathbf{x}$ *in any case* (and not by x when it comes to points and by $\mathbf{x}$ when it comes to vectors).

Example 5.1.5 If $\{\mathbf{x}_0, \mathbf{x}_1, \ldots, \mathbf{x}_n\}$ is an affine basis of $\mathbb{R}^n$, we saw earlier that this is equivalent to saying that the set $\{\mathbf{x}_1 - \mathbf{x}_0, \ldots, \mathbf{x}_n - \mathbf{x}_0\}$ is a basis of $\mathbb{R}^n$. But then we know that the $n \times n$ determinant of the matrix

$$D(\mathbf{x}_1 - \mathbf{x}_0, \ldots, \mathbf{x}_n - \mathbf{x}_0) = \begin{pmatrix} \mathbf{x}_1 - \mathbf{x}_0 \\ \vdots \\ \mathbf{x}_n - \mathbf{x}_0 \end{pmatrix}, \tag{5.1}$$

is different from 0. Instead of the $n \times n$ matrix (5.1) we find useful to consider the $(n+1) \times (n+1)$ matrix

$$D(\mathbf{x}_0, \mathbf{x}_1, \ldots, \mathbf{x}_n) = \begin{pmatrix} \mathbf{x}_1 - \mathbf{x}_0 & 0 \\ \vdots & \vdots \\ \mathbf{x}_n - \mathbf{x}_0 & 0 \\ \mathbf{x}_0 & 1 \end{pmatrix} = \begin{pmatrix} \mathbf{x}_1 & 1 \\ \vdots & \vdots \\ \mathbf{x}_n & 1 \\ \mathbf{x}_0 & 1 \end{pmatrix}.$$

The set

$$B = \{\mathbf{0}, \mathbf{e}_1, \ldots, \mathbf{e}_n\}$$

is a basis of the affine space $\mathbb{R}^n$ which we will call *canonical basis* from now on.

Example 5.1.6 We will show that the points

$$\mathbf{x}_0 = (1, 1), \quad \mathbf{x}_1 = (2, 1), \quad \mathbf{x}_2 = (1, 2),$$

form a basis of the affine space $\mathbb{R}^2$ while this is not the case for the points

$$\mathbf{y}_0 = (-1, 3), \quad \mathbf{y}_1 = (0, 5), \quad \mathbf{y}_2 = (1, 7).$$

Indeed,

$$|D(\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_3)| = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 1 & 1 \\ 1 & 2 & 1 \end{vmatrix} = 1,$$

while,

$$|D(\mathbf{y}_0, \mathbf{y}_1, \mathbf{y}_3)| = \begin{vmatrix} -1 & 3 & 1 \\ 0 & 5 & 1 \\ 1 & 7 & 1 \end{vmatrix} = 0.$$

Let us examine the geometric significance of this fact. The vectors $\mathbf{y}_1 - \mathbf{y}_0, \mathbf{y}_2 - \mathbf{y}_0$ are linearly dependent, that is, there exists $\lambda \in \mathbb{R}_*$ such that

$$\mathbf{y}_2 - \mathbf{y}_0 = \lambda(\mathbf{y}_1 - \mathbf{y}_0) \iff \mathbf{y}_2 = \mathbf{y}_0 + \lambda(\mathbf{y}_1 - \mathbf{y}_0).$$

In other words, $\mathbf{y}_2$ belongs to the straight line passing through $\mathbf{y}_0$ in the direction of the vector $\mathbf{y}_1 - \mathbf{y}_0$. But this is equivalent to $\mathbf{y}_0, \mathbf{y}_1, \mathbf{y}_2$ being collinear.

Let $\{\mathbf{x}_0, \mathbf{x}_1, \ldots, \mathbf{x}_n\}$ be an affine basis of $\mathbb{R}^n$. Since the set

$$\{\mathbf{x}_1 - \mathbf{x}_0, \ldots, \mathbf{x}_n - \mathbf{x}_0\}$$

is a basis of $\mathbb{R}^n$, for a given $\mathbf{x} \in \mathbb{R}^n$ there exist λ_i, $i = 1, \ldots, n$, not all 0, such that

$$\mathbf{x} - \mathbf{x}_0 = \sum_{i=1}^{n} \lambda_i (\mathbf{x}_i - \mathbf{x}_0).$$

It follows that

$$\mathbf{x} = \left(1 - \sum_{i=1}^{n} \lambda_i\right) \mathbf{x}_0 + \sum_{i=1}^{n} \lambda_i \mathbf{x}_i.$$

We set $\lambda_0 = 1 - \sum_{i=1}^{n} \lambda_i$ and obtain the following proposition that defines the barycentric coordinates in the affine space $\mathbb{R}^n$.

Proposition 5.1.7 *If $\{\mathbf{x}_0, \mathbf{x}_1, \ldots, \mathbf{x}_n\}$ is an affine basis of $\mathbb{R}^n$, then for every $\mathbf{x} \in \mathbb{R}^n$ there exist λ_i, $i = 0, \ldots, n$, such that*

$$\mathbf{x} = \sum_{i=0}^{n} \lambda_i \mathbf{x}_i, \quad \sum_{i=0}^{n} \lambda_i = 1.$$

Definition 5.1.8 The numbers $\lambda_0, \lambda_1, \ldots, \lambda_n$ of Proposition 5.1.7 are called *barycentric coordinates corresponding to the basis* $\{x_0, x_1, \ldots, x_n\}$ with *barycentre* x.

Example 5.1.9 Taking the canonical basis $\{\mathbf{0}, \mathbf{e}_1, \ldots, \mathbf{e}_n\}$ of $\mathbb{R}^n$, the barycentric coordinates of a point $\mathbf{x} = (x_1, \ldots, x_n)$ are

$$\left(x_0 = 1 - \sum_{i=1}^{n} x_i,\ x_1, \ldots, x_n\right).$$

Actually,

$$\mathbf{x} = \sum_{i=1}^{n} x_i \mathbf{e}_i = (1 - \sum_{i=1}^{n} x_i)\mathbf{0} + \sum_{i=1}^{n} x_i \mathbf{e}_i.$$

More generally, if $\{\mathbf{x}_0, \mathbf{x}_1, \ldots \mathbf{x}_n\}$, $i = 0, 1, \ldots, n$ is an affine basis of $\mathbb{R}^n$, the barycentric coordinates $\boldsymbol{\lambda} = (\lambda_0, \lambda_1, \ldots \lambda_n)$ of any $\mathbf{x} = (x_1, \ldots, x_n) \in \mathbb{R}^n$ arise as the unique solution of the system

$$\boldsymbol{\lambda} D(\mathbf{x}_0, \ldots, \mathbf{x}_n) = \mathbf{x}. \tag{5.2}$$

The uniqueness of the solution follows from the fact that the set $\{\mathbf{x}_0, \mathbf{x}_1, \ldots \mathbf{x}_n\}$ is an affine basis, because then the determinant $|D(\mathbf{x}_0, \mathbf{x}_1, \ldots \mathbf{x}_n)|$ of the system (5.2) is different from 0. In practice, the $(n+1) \times (n+1)$ system (5.2) is not easily handled

for large n and many times the barycentric coordinates are obtained more easily by direct calculations, as we will see in the following example.

Example 5.1.10 The set

$$B = \{\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3\} = \{(0, 1, 0),\ (1, 1, 2),\ (-1, 0, 1),\ (2, 1, 1)\}$$

is an affine basis of $\mathbb{R}^3$. Indeed,

$$\begin{aligned} D(\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3) &= \begin{vmatrix} 1 & 0 & 2 & 0 \\ -1 & -1 & 1 & 0 \\ 2 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{vmatrix} \\ &= \begin{vmatrix} 1 & 0 & 2 \\ -1 & -1 & 1 \\ 2 & 0 & 1 \end{vmatrix} = -3 \neq 0. \end{aligned}$$

Instead of solving the system (5.2) we observe that the vectors

$$(1, 0, 2), \quad (-1, -1, 1), \quad (2, 0, 1),$$

form a basis of the *vector* space $\mathbb{R}^3$. Thus, for $\mathbf{x} = (x_1, x_2, x_3)$ there exist $\lambda_1, \lambda_2, \lambda_3 \in \mathbb{R}$ such that

$$(x_1, x_2 - 1, x_3) = \lambda_1(1, 0, 2) + \lambda_2(-1, -1, 1) + \lambda_3(2, 0, 1),$$

and we have the 3×3 system

$$\begin{aligned} \lambda_1 - \lambda_2 + 2\lambda_3 &= x_1, \\ -\lambda_2 &= x_2 - 1, \\ 2\lambda_1 + \lambda_2 + \lambda_3 &= x_3. \end{aligned}$$

It follows that

$$\begin{aligned} \lambda_1 &= -(1/3)x_1 + x_2 + (2/3)x_3 - 1, \\ \lambda_2 &= 1 - x_2, \\ \lambda_3 &= (2/3)x_1 - x_2 + x_3 - 1, \end{aligned}$$

and

$$\lambda_0 = 1 - \lambda_1 - \lambda_2 - \lambda_3 = 1 - (1/3)x_1 + x_2 - (5/3)x_3 + 2.$$

Definition 5.1.11 Let I be a set of indices and $\mathbf{x}_i$, $i \in I$, a family of points of $\mathbb{R}^n$. We will say that $\mathbf{x}_i$ are *affinely independent* if $\mathbf{x}_i - \mathbf{x}_\alpha$, $i \in I$, are linearly independent vectors for some $\alpha \in I$. On the other hand, $\mathbf{x}_i$, $i \in I$ are called *affinely dependent* if they are not affinely independent.

Exercise Prove that the definition of affine independence/dependence is independent of the choice of $\mathbf{x}_\alpha$. However, this relation is equivalent to

$$\mathbf{x}_j = \sum_{i \in I \setminus \{j\}} \lambda_i \mathbf{x}_i, \qquad \sum_{i \in I \setminus \{j\}} \lambda_i = 1.$$

This fact leads us to the following definition, which is the analogue of *linear combinations of vectors in a vector space.*

Definition 5.1.12 Let $\mathbf{x}_i$, $i \in I$, be a set of points in $\mathbb{R}^n$, I a set of indices. Every expression of the form

$$\sum_{i \in I} \lambda_i \mathbf{x}_i, \quad \sum_{i \in I} \lambda_i = 1,$$

will be called an *affine* (or *barycentric*) combination of $\mathbf{x}_i$, $i \in I$.

Remark 5.1.13 We have seen that affine independence of points is defined via linear dependence of associated vectors, Conversely, we may always pass from linear dependence of vectors to affine dependence of points. Indeed, let $\mathcal{B} = \{\mathbf{v}_i,\ i \in I\}$ be a set of linearly independent vectors. Then, the set $\{\mathbf{0}\} \cup \mathcal{B}$ is a set of affine independent points.

5.1.2 Affine Subspaces, k-Planes

Definition 5.1.14 An *affine subspace* of $\mathbb{R}^n$ is a subset B of $\mathbb{R}$ of the form:

$$B = \mathbf{a} + W = \{\mathbf{a} + \mathbf{w} : \mathbf{w} \in W\},$$

where $\mathbf{a}$ is a point of $\mathbb{R}^n$, and W is a vector subspace of V (Fig. 5.2).

If $\dim(W) = k \leq n$, then B is an affine space of dimension k with associated vector space W. The vector subspace W is also called the *direction* of the affine subspace B. Two affine subspaces with the same direction are called *parallel*. Given a direction W and a point $\mathbf{a} \in \mathbb{R}^n$, then the only affine subspace that passes from $\mathbf{a}$ in the direction of W is $a + W$. The following proposition follows immediately:

Proposition 5.1.15 *Every transformation* $\mathbb{R}^n \to \mathbb{R}^n : \mathbf{a} \mapsto \mathbf{a} + \mathbf{v}$, *maps every affine subspace to its parallel subspace.*

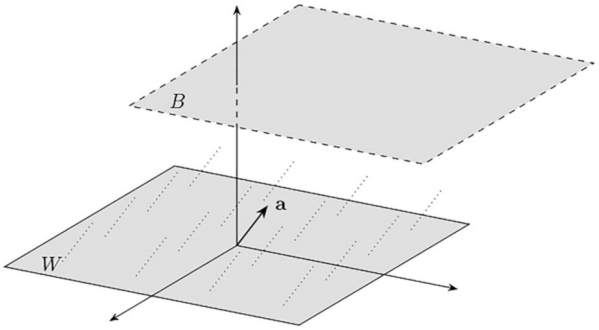


Fig. 5.2 Affine subspace $B = \mathbf{a} + W$ in $\mathbb{R}^3$

If $B = \mathbf{a} + W$ is a k-dimensional affine subspace of A and $W = \langle \mathbf{w}_1, \ldots, \mathbf{w}_k \rangle$, then the set

$$\{\mathbf{a}, \mathbf{w}_1 + \mathbf{a}, \ldots, \mathbf{w}_k + \mathbf{a}\},$$

is an affine basis of B.

Proposition 5.1.16 *The intersection of two affine subspaces of the same dimension is an affine subspace of smaller dimension.*

Proof Let $B = \mathbf{a} + W$ and $B' = \mathbf{a}' + W'$ be affine subspaces of the same dimension k. If the intersection $B \cap B'$ is empty, we have nothing to prove. If $B \cap B' \neq \emptyset$, then

$$B \cap B' = \mathbf{a} + \mathbf{a}' + W \cap W'$$

and the intersection $W \cap W'$ is a vector subspace of dimension $k - 1$. □

The affine k-dimensional subspaces of the affine space $\mathbb{R}^n$ are also called *affine k-planes* or *k-planes*. A k-plane W_k of $\mathbb{R}^n$, $1 \leq k \leq n - 1$ is the set of $\mathbf{x} \in \mathbb{R}^n$ determined by the parametric equation:

$$\mathbf{x} = \mathbf{x}(\lambda_1, \ldots, \lambda_k) = \mathbf{a} + \sum_{i=1}^{k} \lambda_i \mathbf{v}_i, \quad (\lambda_1, \ldots, \lambda_k) \in \mathbb{R}^k, \tag{5.3}$$

where $\mathbf{a} \in \mathbb{R}^n$ and $\mathbf{v}_i \in \mathbb{R}^n$, $i = 1, \ldots, k$, are linearly independent vectors of $\mathbb{R}^n$. Alternatively,

$$W_k = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{x} \cdot \mathbf{A}_i = b_i,\ i = 1, \ldots, n - k\}, \tag{5.4}$$

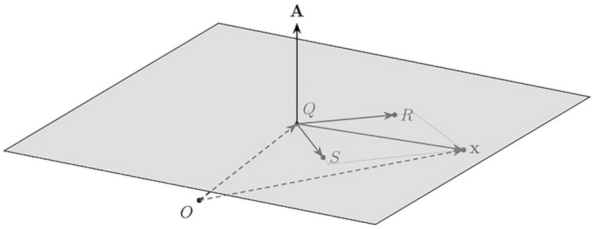


Fig. 5.3 2-Plane in $\mathbb{R}^3$: a point $\mathbf{x}$ in the plane is given by $\mathbf{x} = \overrightarrow{OQ} + \lambda\overrightarrow{QR} + \mu\overrightarrow{QS}$, or, alternatively, $(\mathbf{x} - \overrightarrow{OQ}) \cdot \mathbf{A} = 0$

where $\mathbf{A}_1, \ldots, \mathbf{A}_{n-k}$ is a set of $(n-k)$ orthonormal vectors of $\mathbb{R}^n$ and $b_i \in \mathbb{R}$, $i = 1, \ldots, n-k$. Finally, the barycentric coordinates for each $\mathbf{x}(\lambda_1, \ldots, \lambda_k) \in W_k$ are

$$1 - \sum_{i=1}^{k} \lambda_i, \lambda_1, \ldots, \lambda_k.$$

Example 5.1.17 Consider the 2-plane of $\mathbb{R}^3$ passing through $\mathbf{a} = \mathbf{e}_2$ and generated by $\mathbf{e}_1, \mathbf{e}_3$. Its parametric equation is

$$\begin{aligned} \mathbf{x}(\lambda, \mu) &= \mathbf{e}_2 + \lambda\mathbf{e}_1 + \mu\mathbf{e}_3 \\ &= (\lambda, 1, \mu), \quad \lambda, \mu \in \mathbb{R}. \end{aligned}$$

Therefore its Cartesian equation is $x_2 = 1$, or, alternatively, $(\mathbf{x} - \mathbf{e}_2) \cdot \mathbf{e}_2 = 0$ (Fig. 5.3).

Example 5.1.18 The 2-plane of $\mathbb{R}^4$ from the origin generated by $(1, 2, 0, 0)$ and $(0, 0, 1, 2)$ has a parametric equation

$$\begin{aligned} \mathbf{x}(\lambda, \mu) &= \lambda(1, 2, 0, 0) + \mu(0, 0, 1, 2) \\ &= (\lambda, 2\lambda, \mu, 2\mu), \quad \lambda, \mu \in \mathbb{R}. \end{aligned}$$

Its Cartesian equations are therefore $x_2 = 2x_1$, $x_4 = 2x_3$, or, alternatively, $\mathbf{x} \cdot (2, -1, 0, 0) = 0, \mathbf{x} \cdot (0, 0, 2, -1) = 0$.

Example 5.1.19 In $\mathbb{R}^3$ consider the parametric equation

$$\mathbf{x}(\lambda, \mu) = (2, 1, 0) + \lambda(1, -1, 2) + \mu(0, 2, 3), \quad \lambda, \mu \in \mathbb{R}.$$

The vectors $(1, -1, 2)$ and $(1, -1, 2)$ are linearly independent:

$$\lambda_1(1, -1, 2) + \lambda_2(1, -1, 2) = (0, 0, 0) \implies \lambda_1 = \lambda_2 = 0.$$

Therefore this parametric equation represents a two-dimensional affine subspace of $\mathbb{R}^3$. The affine basis is

$$\{(2, 1, 0),\ (3, 0, 2),\ (2, 3, 3)\},$$

and the barycentric coordinates of any $\mathbf{x}(\lambda, \mu)$ with respect to this basis are

$$(1 - \lambda - \mu,\ \lambda,\ \mu\}.$$

In the following example, we show how we can complete an affine basis of $\mathbb{R}^3$ from an affine basis of an affine subspace.

Example 5.1.20 In $\mathbb{R}^3$ consider the plane

$$P:\ x_1 - 2x_2 + x_3 = 1.$$

If $\mathbf{x} = (x_1, x_2, x_3)$, the equation of P is also written as $(1, -2, 1) \cdot \mathbf{x} = 1$. We set

$$x_1 = \lambda, \quad x_2 = \mu, \quad x_3 = 1 - \lambda + 2\mu,$$

so P is the set of points of the form

$$\begin{aligned}\mathbf{x} &= (\lambda, \mu, 1 - \lambda + 2\mu)\\ &= (0, 0, 1) + \lambda(1, 0, -1) + \mu(0, 1, 2)\\ &= (0, 0, 1) + \langle(1, 0, -1),\ (0, 1, 2)\rangle.\end{aligned}$$

Therefore, an affine basis for P is

$$B_P = \{(0, 0, 1),\ (1, 0, 0),\ (0, 1, 3)\}.$$

If $V = \langle(1, 0, -1),\ (0, 1, 2)\rangle$, then from the equation of P we get

$$V^{\perp} = \langle(1, -2, 1)\rangle.$$

It follows that the affine basis of $\mathbb{R}^3$ obtained by completing the affine basis of P is

$$B = B_P \cup \{(1, -2, 1) - (0, 0, 1)\} = \{(1, 0, 0),\ (2, 0, -1),\ (1, 1, 2),\ (1, -2, 0)\}.$$

Finally, if

$$\lambda_0, \lambda_1, \lambda_2, \quad \lambda_0 + \lambda_1 + \lambda_2 = 1,$$

are the barycentric coordinates for P with respect to the basis B_P, then the barycentric coordinates for $\mathbb{R}^3$ with respect to the basis B are

$$\lambda_0', \lambda_1, \lambda_2, \lambda_3, \quad \lambda_0' + \lambda_1 + \lambda_2 + \lambda_3 = 1.$$

It is $\lambda_0' = \lambda_0 - \lambda_3$, so we rewrite the barycentric coordinates of $\mathbb{R}^3$

$$\lambda_0 - \lambda_3, \lambda_1, \lambda_2, \lambda_3,$$

and with respect to these coordinates the equation of P is $\lambda_3 = 0$.

5.1.3 *Affine Mappings*

Definition 5.1.21 An *affine mapping from* $\mathbb{R}^n$ *to* $\mathbb{R}^m$ is a mapping $f : \mathbb{R}^n \to \mathbb{R}^m$ such that $g : \mathbb{R}^n \to \mathbb{R}^m$, defined by $g(\mathbf{b}-\mathbf{a}) = f(\mathbf{b}) - f(\mathbf{a})$ is a well-defined linear mapping.

Notice that g is well defined:

$$\mathbf{b} - \mathbf{a} = \mathbf{d} - \mathbf{c} \implies f(\mathbf{b}) - f(\mathbf{a}) = f(\mathbf{d}) - f(\mathbf{c}).$$

This has the following consequence: suppose we know the value $f(\mathbf{a})$ for some $\mathbf{a} \in \mathbb{R}^n$ and let $\mathbf{b} \in \mathbb{R}^n$. Then, there exists a unique $\mathbf{v} \in \mathbb{R}^n$ with $\mathbf{b} = \mathbf{a} + \mathbf{v}$ and

$$f(\mathbf{b}) - f(\mathbf{a}) = g(\mathbf{v}) \implies f(\mathbf{b}) = f(\mathbf{a}) + g(\mathbf{v}).$$

We conclude that f is completely determined by its value at a point and the associated linear mapping g.

Remark 5.1.22 If $f : \mathbb{R}^n \to \mathbb{R}^m$ is an affine mapping, we may always write it in the form

$$f(\mathbf{x}) = g(\mathbf{x}) + \mathbf{b},$$

where $g : \mathbb{R}^n \to \mathbb{R}^m$ is a linear map and $\mathbf{b} = f(\mathbf{0})$.

Example 5.1.23 The mapping $f : \mathbb{R}^3 \to \mathbb{R}^3$,

$$f(x_1, x_2, x_3) = (x_1 + x_2 + 1,\ x_2 + x_3,\ x_1 + x_3 - 1)$$

is affine: if $g(x_1, x_2, x_3) = (x_1 + x_2,\ x_2 + x_3,\ x_1 + x_3)$, then

$$f(x_1, x_2, x_3) = g(x_1, x_2, x_3) + (1, 0, -1) = g(x_1, x_2, x_3) + f(0, 0, 0).$$

Also, g is a linear mapping of $\mathbb{R}^3$ with matrix

$$A_g = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix} \in \mathrm{GL}(3, \mathbb{R}),$$

since $\det(A_g) = 2$.

On the other hand, the mapping $f : \mathbb{R} \to \mathbb{R}^2$, where

$$f(x) = (1 + x, x^2),$$

is not affine: if $x, y \in \mathbb{R}$, then

$$g(x - y) = f(x) - f(y) = (x - y, x^2 - y^2)$$

is not linear.

The following proposition tells us that a mapping between affine spaces is affine if and only if it preserves affine combinations.

Proposition 5.1.24 *The mapping $f : \mathbb{R}^n \to \mathbb{R}^m$ is an affine mapping if for every family of points $\mathbf{x}_i \in \mathbb{R}^n$, $i \in I$, and for every set of numbers λ_i, $i \in I$, we have that*

$$f\left(\sum_{i \in I} \lambda_i \mathbf{x}_i\right) = \sum_{i \in I} \lambda_i f(\mathbf{x}_i).$$

Proof Let $\mathbf{x}_i \in \mathbb{R}^n$, $i \in I$, and an affine combination $\sum_{i \in I} \lambda_i \mathbf{x}_i$. Then, if f is affine and g is its associated linear mapping, we choose a $\mathbf{x}_\alpha$ from $\mathbf{x}_i$, $i \in I$. We set $I' = I \setminus \{\alpha\}$ and we have:

$$\begin{aligned} f\left(\sum_{i \in I} \lambda_i \mathbf{x}_i\right) &= f(\mathbf{x}_0) + g\left(\sum_{i \in I} \lambda_i \mathbf{x}_i - \mathbf{x}_0\right) \\ &= f(\mathbf{x}_\alpha) + g\left(\sum_{i \in I'} \lambda_i (\mathbf{x}_i - \mathbf{x}_0)\right) \\ &= f(\mathbf{x}_\alpha) + \sum_{i \in I'} \lambda_i g(\mathbf{x}_i - \mathbf{x}_0) \\ &= f(\mathbf{x}_\alpha) + \sum_{i \in I'} \lambda_i \left(f(\mathbf{x}_i) - f(\mathbf{x}_0)\right) \\ &= \sum_{i \in I} \lambda_i f(\mathbf{x}_i), \quad \sum_{i \in I} \lambda_i = 1. \end{aligned}$$

Conversely, let $\{\mathbf{x}_0, \mathbf{x}_1, \ldots, \mathbf{x}_n\}$ be an affine basis of $\mathbb{R}^n$. We define for each $\mathbf{x} \in \mathbb{R}^n$ the mapping $g : \mathbb{R}^n \to \mathbb{R}^m$ by the relation

$$g(\mathbf{x} - \mathbf{x}_0) = f(\mathbf{x}) - f(\mathbf{x}_0), \quad x \in \mathbb{R}^n.$$

Then,

$$\begin{aligned} g(\mathbf{x} - \mathbf{x}_0) &= f\left(\sum_{i=0}^{n} \lambda_i \mathbf{x}_i\right) - f(\mathbf{x}_0) \\ &= \left(\sum_{i=0}^{n} \lambda_i f(\mathbf{x}_i)\right) - f(\mathbf{x}_0) \\ &= \sum_{i=1}^{n} \lambda_i \left(f(\mathbf{x}_i) - f(\mathbf{x}_0)\right) \\ &= \sum_{i=1}^{n} \lambda_i g(\mathbf{x}_i - \mathbf{x}_0), \end{aligned}$$

from which we conclude that g is a well-defined linear mapping, since it is completely determined by the above linear relation that contains its values in the elements of the basis of $\mathbb{R}^n$ resulting from the affine basis we chose. □

Definition 5.1.25 A bijective affine mapping $f : \mathbb{R}^n \to \mathbb{R}^n$ is called *affine automorphism.*

By definition, the associated linear mapping to an affine automorphism is a linear automorphism of vector spaces.

Corollary 5.1.26 *Let $f : \mathbb{R}^n \to \mathbb{R}^n$ be an affine automorphism. Then if $\{\mathbf{x}_0, \ldots, \mathbf{x}_n\}$ is an affine basis of $\mathbb{R}^n$, then $\{f(\mathbf{x}_0), \ldots, f(\mathbf{x}_n)\}$ is also an affine basis of $\mathbb{R}^n$.*

Proof Let $\{\mathbf{x}_0, \ldots, \mathbf{x}_n\}$ be a basis of $\mathbb{R}^n$ and $\mathbf{x} \in \mathbb{R}^n$:

$$\mathbf{x} = \sum_{i=0}^{n} \lambda_i \mathbf{x}_i, \quad \sum_{i=0}^{n} \lambda_i = 1.$$

From Proposition 5.1.24,

$$f(\mathbf{x}) = \sum_{i=0}^{n} \lambda_i f(\mathbf{x}_i), \quad \sum_{i=0}^{n} \lambda_i = 1,$$

so it remains to show the linear independence of the vectors $g(\mathbf{x}_i - \mathbf{x}_0) = f(\mathbf{x}_i) - f(\mathbf{x}_0)$, $i = 1, \ldots, n$, where g is the associated linear mapping to f. Since f is an

affine automorphism, g is a linear automorphism and the above vectors are linearly independent. □

Example 5.1.27 Let $\{(1,1),(2,1),(1,2)\}$ be an affine basis of $\mathbb{R}^2$ and let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be an affine mapping. To find the barycentric coordinates, we observe that since $(1,0)$ and $(0,1)$ are clearly linearly independent, there exist $\lambda_1, \lambda_2 \in \mathbb{R}$ such that for every $(x_1, x_2) \in \mathbb{R}^2$,

$$(x_1 - 1, x_2 - 1) = \lambda_1(1,0) + \lambda_2(0,1) = (\lambda_1, \lambda_2).$$

It follows that the barycentric coordinates of each $(x_1, x_2) \in \mathbb{R}^2$ are $(3 - x_1 - x_1, x_1 - 1, x_2 - 1)$. We now have from Proposition 5.1.24 that

$$f(x_1, x_2) = (3 - x_1 - x_2) f(1,1) + (x_1 - 1) f(2,1) + (x_2 - 1) f(1,2).$$

Let $f(1,1) = (a_1, a_2)$, $f(2,1) = (b_1, b_2)$, $f(1,2) = (c_1, c_2)$. Then,

$$\begin{aligned} f(x_1, x_2) = ((&3a_1 - b_1 - c_1,\ 3a_2 - b_2 - c - 2) \\ &+ (x_1, x_2) \begin{pmatrix} b_1 - a_1 & b_2 - a_2 \\ c_1 - a_1 & c_2 - a_2 \end{pmatrix}.\Big), \end{aligned}$$

and the matrix of the associated linear mapping has determinant

$$\begin{vmatrix} b_1 - a_1 & b_2 - a_2 \\ c_1 - a_1 & c_2 - a_2 \end{vmatrix} \neq 0,$$

since g is a linear automorphism.

Affine mappings have the following important property.

Proposition 5.1.28 *Parallelism is an affine invariant.*

Proof Let $\mathbf{a} + W$, $\mathbf{a}' + W$ be parallel subspaces of $\mathbb{R}^n$ and let $f : \mathbb{R}^n \to \mathbb{R}^m$ be an affine mapping,

$$f(\mathbf{x}) = g(\mathbf{x}) + \mathbf{b}, \quad \mathbf{x} \in \mathbb{R}^n,$$

where $g : \mathbb{R}^n \to \mathbb{R}^m$ is a linear mapping and $\mathbf{b} \in B$. Then,

$$\mathbf{a} + w \in \mathbf{a} + W \implies f(\mathbf{a} + \mathbf{w}) = f(\mathbf{a}) + g(\mathbf{w}) \in f(\mathbf{a}) + g(W).$$

Similarly,

$$\mathbf{a}' + \mathbf{w} \in \mathbf{a}' + W \implies f(\mathbf{a}' + \mathbf{w}) = f(\mathbf{a}') + g(\mathbf{w}) \in f(\mathbf{a}') + g(W)$$

and the desired result is obtained. □

Example 5.1.29 In $\mathbb{R}^2$ we consider the parallel lines

$$l_1 : \ x_1 + 3x_2 = 1,$$
$$l_2 : \ x_1 + 3x_2 = 2,$$

and the affine transformation

$$f(x_1, x_2) = (x_1 - 1, \ x_1 + x_2 - 2).$$

Notice that

$$f(x_1, x_2) = (x_1, \ x_1 + x_2) + (-1, -2) = g(x_1, x_2) + (-1, -2)$$

and the matrix of the linear transformation g is

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \in \mathrm{GL}(2, \mathbb{R}),$$

so indeed f is affine. One way to find the images $f(l_1)$, $f(l_2)$ is as follows: set

$$y_1 = x_1 - 1, \quad y_2 = x_1 + x_2 - 1.$$

Then,

$$x_1 = y_1 + 1, \quad x_2 = y_2 - y_1$$

and therefore:

$$f(l_1) : \ y_1 + 1 + 3(y_2 - y_1) = 1 \iff -2y_1 + 3y_2 = 0,$$
$$f(l_1) : \ y_1 + 1 + 3(y_2 - y_1) = 2 \iff -2y_1 + 3y_2 = 1,$$

which proves that the images are parallel lines. Alternatively, we could write l_1, l_2 in parametric (barycentric) form:

$$\mathbf{c}_1(t) = (1 - t)(1, 0) + t(0, 1/3),$$
$$\mathbf{c}_2(t) = (1 - t)(2, 0) + t(0, 2/3).$$

Then

$$f(\mathbf{c}_1(t)) = (1 - t) f(1, 0) + t f(0, 1/3) = (1 - t)(0, -1) + t(-1, -5/3)$$
$$= (0, -1) + t(-1, -2/3),$$

$$f(\mathbf{c}_2(t)) = (1-t)f(2,0) + tf(0,2/3) = (1-t)(1,0) + t(-1-4/3)$$
$$= (1,0) + t(-2,-4/3).$$

Again, we obtain that $f(l_1) \parallel f(l_2)$.

Example 5.1.30 Let $f : \mathbb{R}^3 \to \mathbb{R}^3$ be the affine mapping given by

$$f(x_1, x_2, x_3) = (x_1 + x_2 + 1,\ x_2 + x_3,\ x_1 + x_3 - 1),$$

for each $(x_1, x_2, x_3) \in \mathbb{R}^3$, and let also the plane

$$P :\ 2x_1 - x_2 + x_3 = 1.$$

A parametric equation of P is

$$\mathbf{x}(\lambda, \mu) = (\lambda, \mu, 1 - 2\lambda + \mu) = (1,0,0) + \lambda(1,0,-2) + \mu(0,1,1)$$
$$= (1-\lambda-\mu)(1,0,0) + \lambda(2,0,-2) + \mu(1,1,1). \tag{5.5}$$

So,

$$f(\mathbf{x}(\lambda,\mu)) = (1-\lambda-\mu)f(1,0,0) + \lambda f(2,0,-2) + \mu f(1,1,1)$$
$$= (1-\lambda-\mu)(2,0,0) + \lambda(3,0,-1) + \mu(3,2,1) \tag{5.6}$$
$$= (2,0,0) + \lambda(1,0,-1) + \mu(1,2,1)$$
$$= (2+\lambda+\mu,\ 2\mu\ -\lambda+\mu).$$

Eliminating λ, μ yields the Cartesian equation for $f(P) = P'$ as follows:

$$P' :\ x_1 - x_2 + x_3 = 2.$$

We now wish to write down a formula for the restriction $\tilde{f}$ of f as a mapping between the affine spaces P, P'. One way to do this is as follows. From the parametric equations (5.5) and (5.6) of P and P', respectively, we obtain the affine bases

$$B_P = \{\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2\} = \{(1,0,0),\ (2,0,-2),\ (1,1,1)\},$$
$$B_{P'} = \{\mathbf{x}'_0, \mathbf{x}'_1, \mathbf{x}'_2\} = \{(2,0,0),\ (3,0,-1),\ (3,2,1)\}.$$

Now $\tilde{f} : P \to P'$ is the mapping that satisfies the relations $\tilde{f}(\mathbf{x}_i) = \mathbf{x}'_i$, $i = 0, 1, 2$. The parametric equations $\mathbf{x}(\lambda, \mu)$ and $\mathbf{x}'(\lambda, \mu) = f(\mathbf{x}(\lambda, \mu))$ are bijections between

$\mathbb{R}^2$ and P and P', respectively, such that

$$\mathbf{x}(0,0) = \mathbf{x}_0, \quad \mathbf{x}(1,0) = \mathbf{x}_1, \quad \mathbf{x}(0,1) = \mathbf{x}_2,$$
$$\mathbf{x}'(0,0) = \mathbf{x}'_0, \quad \mathbf{x}'(1,0) = \mathbf{x}'_1, \quad \mathbf{x}'(0,1) = \mathbf{x}'_2.$$

We conclude that the parametric equations $\mathbf{x}$ and $\mathbf{x}'$ are essentially identifications of P, P' with $\mathbb{R}^2$, in such a way that the basis vectors are mapped to the vectors of the canonical basis of $\mathbb{R}^2$. In the next section we will see that the only affine mapping of $\mathbb{R}^2$ that stabilises three affinely independent points (in this case, $(0,0)$, $(1,0)$ and $(0,1)$) is the identity. Therefore, $\tilde{f} : P \to P'$ is given by the formula $\tilde{f} = (\mathbf{x}')^{-1} \circ \mathbf{x}$: $\tilde{f}(\lambda, \mu) = (\lambda, \mu)$.

5.2 The Affine Group $A(n)$

The following proposition follows directly from the definition of the affine mapping, see also Remark 5.1.22; in parallel, it constitutes the definition of the affine group of the affine space $\mathbb{R}^n$.

Proposition 5.2.1 *The set* $\mathrm{Aff}(\mathbb{R}^n)$ *of affine automorphisms of* $\mathbb{R}^n$ *is a group isomorphic to* $\mathbb{R}^n \rtimes \mathrm{GL}(n, \mathbb{R})$.

The affine group $\mathrm{Aff}(\mathbb{R}^n)$ will be denoted by $A(n)$.

5.2.1 $A(n)$ and its Action in $\mathbb{R}^n$

By definition, every element $f \in A(n)$, that is, every affine mapping $\mathbb{R}^n \to \mathbb{R}^n$, is written as

$$f(\mathbf{x}) = g(\mathbf{x}) + \mathbf{b}, \quad \mathbf{x} \in \mathbb{R}^n, \tag{5.7}$$

where g is an automorphism of the vector space $\mathbb{R}^n$ and $\mathbf{b} \in \mathbb{R}^n$.

If $A_g \in \mathrm{GL}(n, \mathbb{R})$ is the matrix of g, then it is clear that the inverse of the affine mapping f as in (5.7) is given by the formula

$$f^{-1}(\mathbf{x}) = g^{-1}(\mathbf{x}) - g^{-1}(\mathbf{b}), \quad \mathbf{x} \in \mathbb{R}^n, \tag{5.8}$$

and thus it is an affine automorphism itself.

Example 5.2.2 Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be the map defined by

$$f(x_1, x_2) = (2x_1 - x_2 + 1,\ x_1 + 2x_2 - 1), \quad (x_1, x_2) \in \mathbb{R}^2.$$

We write

$$f(x_1, x_2) = g(x_1, x_2)+(1, -1), \quad g(x_1, x_2) = (2x_1-x_2,\ x_1+2x_2) = (x_1, x_2)A_g,$$

where

$$A_g = \begin{pmatrix} 2 & 1 \\ -1 & 2 \end{pmatrix} \in \mathrm{GL}(2, \mathbb{R}).$$

It follows that $f \in A(2)$. To find the inverse, we first observe that

$$A_g^{-1} = (1/5)\begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix} \in \mathrm{GL}(2, \mathbb{R}),$$

so

$$g^{-1}(x_1, x_2) = (1/5)(2x_1 + x_2,\ -x_1 + 2x_2).$$

Furthermore,

$$g^{-1}(1, -1) = (1/5)(1, -3),$$

and therefore

$$f^{-1}(x_1, x_2) = (1/5)(2x_1 + x_2 - 1,\ -x_1 + 2x_2 + 3).$$

Example 5.2.3 Consider the affine mapping of the previous example and the canonical affine basis

$$B = \{(0, 0),\ (1, 0),\ (0, 1)\}$$

of $\mathbb{R}^2$. We have

$$f(0, 0) = (1, -1), \quad f(1, 0) = (3, 0), \quad f(0, 1) = (0, 1).$$

Since f is affine, the set

$$B' = \{(1, -1),\ (3, 0),\ (0, 1)\}$$

is in turn an affine basis of $\mathbb{R}^2$ (see Corollary 5.1.26). If, however, we only knew the set B' and that an affine mapping satisfied

$$(0, 0) \mapsto (1, -1), \quad (1, 0) \mapsto (3, 0), \quad (0, 1) \mapsto (0, 1),$$

then the formula of this linear mapping is found from

$$\begin{aligned}(x_1, x_2) &= (1 - x_1 - x_2)(0, 0) + x_1(1, 0) + x_2(0, 1)\\ &\mapsto (1 - x_1 - x_2)(1, -1) + x_1(3, 0) + x_2(0, 1)\\ &= (2x_1 - x_2 + 1,\ x_1 + 2x_2 - 1),\end{aligned}$$

that is, we get f from its values at its points of the canonical affine basis B. This can always be done, given that the set B' is an affine basis.

Example 5.2.4 Suppose we are given

$$A = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 2 & 0 \\ -3 & 0 & 2 \end{pmatrix} \in \mathrm{GL}(3, \mathbb{R}), \quad \mathbf{b} = (4, -2, 0).$$

Then, the affine mapping $f : \mathbb{R}^3 \to \mathbb{R}^3$ resulting from the linear mapping with matrix A and vector $\mathbf{b}$ is given for each $(x_1, x_2, x_3) \in \mathbb{R}^3$ by

$$\begin{aligned}f(x_1, x_2, x_3) &= (x_1, x_2, x_3) \begin{pmatrix} 1 & 0 & -1 \\ 0 & 2 & 0 \\ -3 & 0 & 2 \end{pmatrix} + (4, -2, 0)\\ &= (x_1 - 3x_3 + 4,\ 2x_2 - 2,\ -x_1 + 2x_3).\end{aligned}$$

At this point we make an important observation that will be useful in what follows. There is an embedding

$$A(n) = \mathbb{R}^n \rtimes \mathrm{GL}(n, \mathbb{R}) \hookrightarrow \mathrm{GL}(n + 1, \mathbb{R})$$

in which an affine transformation $f(\mathbf{x}) = g(\mathbf{x}) + \mathbf{b}$ of $\mathbb{R}^n$, with g having matrix $A_g \in \mathrm{GL}(n, \mathbb{R})$ and $\mathbf{b} \in \mathbb{R}^n$, is mapped to the matrix

$$A_f = \begin{pmatrix} A_g & 0 \\ \mathbf{b} & 1 \end{pmatrix} \in \mathrm{GL}(n + 1, \mathbb{R}). \tag{5.9}$$

The matrix A_f completely determines the affine transformation f and is called the *augmented matrix* f. The inverse of A_f is

$$(A_f)^{-1} = \begin{pmatrix} A_g^{-1} & 0 \\ -\mathbf{b}A_g^{-1} & 1 \end{pmatrix} = A_{f^{-1}}. \tag{5.10}$$

Example 5.2.5 Consider the matrix

$$A = \begin{pmatrix} 1 & -1 & 1 & 0 \\ 0 & 2 & 3 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix} \in \mathrm{GL}(4, \mathbb{R}).$$

This matrix is the augmented matrix of the affine mapping $f : \mathbb{R}^3 \to \mathbb{R}^3$ given by

$$f(x_1, x_2, x_3) = (x_1,\ -x_1 + x_2 + 1,\ x_1 + 3x_2 - 2x_3), \quad (x_1, x_2, x_3) \in \mathbb{R}^3.$$

Since

$$A^{-1} = \begin{pmatrix} 1 & 1 & 2 & 0 \\ 0 & 1 & 3/2 & 0 \\ 0 & 0 & -1/2 & 0 \\ 0 & -1 & -3/2 & 1 \end{pmatrix},$$

it follows that

$$f^{-1}(x_1, x_2, x_3) = (x_1,\ x_1 + x_2 - 1,\ 2x_1 + (3/2)x_2 - (1/2)x_3)\,.$$

The left action $A(n) \times \mathbb{R}^n \to \mathbb{R}^n$ of $A(n)$ on $\mathbb{R}^n$ is given by

$$(f,\ \mathbf{x}) \mapsto f(\mathbf{x}) = g(\mathbf{x}) + \mathbf{b},$$

for each $f \in A(n)$. This action is doubly transitive: indeed, consider two distinct pairs of ordered points $\mathfrak{p} = \{\mathbf{x}_0, \mathbf{x}_1\}$ and $\mathfrak{q} = \{\mathbf{y}_0, \mathbf{y}_1\}$. The sets

$$\mathbf{x}_0 + \langle \mathbf{x}_1 \rangle, \quad \mathbf{y}_0 + \langle \mathbf{y}_1 \rangle,$$

are both one-dimensional affine subspaces of $\mathbb{R}^n$, that is, straight lines. Let

$$\langle \mathbf{x}_1 \rangle^{\perp} = \langle \mathbf{x}_2, \ldots, \mathbf{x}_n \rangle$$

be the orthocomplement of $\mathbf{x}_1$ in $\mathbb{R}^n$. Then, $\{\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2 + \mathbf{x}_0, \ldots, \mathbf{x}_n + \mathbf{x}_0\}$ is an affine basis of $\mathbb{R}^n$ with augmented matrix

$$A_h = D(\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2 + \mathbf{x}_0, \ldots, \mathbf{x}_n + \mathbf{x}_0) = \begin{pmatrix} \mathbf{x}_1 - \mathbf{x}_0 & 0 \\ \mathbf{x}_2 & 0 \\ \vdots & \vdots \\ \mathbf{x}_n & 0 \\ \mathbf{x}_0 & 1 \end{pmatrix}.$$

This matrix is in $\mathrm{GL}(n+1,\mathbb{R})$ and if g is the linear mapping corresponding to

$$A_g = \begin{pmatrix} \mathbf{x}_1 - \mathbf{x}_0 \\ \mathbf{x}_2 \\ \vdots \\ \mathbf{x}_n \end{pmatrix},$$

then A_h is the augmented matrix of the affine mapping h defined $h(\mathbf{x}) = g(\mathbf{x}) + \mathbf{x}_0$ for each $\mathbf{x} \in \mathbb{R}^n$, which satisfies

$$h(\mathbf{0}) = \mathbf{x}_0, \quad h(\mathbf{e}_1) = \mathbf{x}_1.$$

In the same way we find that there exists an affine transformation h' that satisfies the relations

$$h'(\mathbf{0}) = \mathbf{y}_0, \quad h'(\mathbf{e}_1) = \mathbf{y}_1.$$

It turns out that the affine transformation $f = h \circ g^{-1}$ satisfies

$$f(\mathbf{x}_0) = \mathbf{y}_0, \quad f(\mathbf{x}_1) = \mathbf{y}_1,$$

which proves the doubly transitive action of $A(n)$. The following result is a direct consequence of the doubly transitive action in the dimension $n = 1$.

Corollary 5.2.6 *Every element of $f \in A(1)$ that stabilises two points is necessarily the identity transformation of $A(1)$.*

Proof We may assume that the points are 0, 1; then f is of the form $f(x) = ax + b$, and the relations $f(0) = 0$ and $f(1) = 1$ imply $a = 1, b = 0$. □

We will see a broad generalisation of this result in the next section.

Regarding the freedom of action of the affine group, we have that the stabiliser of $\mathbf{0}$ is

$$\mathrm{Stab}(\mathbf{0}) = \{f \in A(n) \mid f(\mathbf{0}) = \mathbf{0}\} \simeq \mathrm{GL}(n,\mathbb{R}).$$

Every $\mathbf{x} \neq \mathbf{0}$ is the translation of $\mathbf{0}$ by $\mathbf{x}$, therefore

$$f(\mathbf{x}) = \mathbf{x} \implies f(T_{\mathbf{x}}(\mathbf{0})) = T_{\mathbf{x}}(\mathbf{0}) \implies (T_{-\mathbf{x}} \circ f \circ T_{\mathbf{x}})(\mathbf{0}) = \mathbf{0}$$

and thus we obtain that the stabiliser of every $\mathbf{x} \in \mathbb{R}^n$ is isomorphic to the general linear group.

5.2.2 *Fundamental Theorem of Affine Geometry*

We have seen in the previous section that the action of the affine group is doubly transitive; we can also prove that it acts transitively on triples of affinely independent points ($n \geq 2$) (see Exercise 25, Sect. 5.4). Continuing this procedure we may obtain analogous results for the transitive action of $A(n)$ on $(k+1)$-sets of affinely independent points, $n \geq k$. The Fundamental Theorem of Affine Geometry that we prove below essentially tells us that the action of $A(n)$ is maximal, that is, it is transitive on $(n+1)$-sets of affinely independent points.

We first prove the following:

Lemma 5.2.7 *Every element $f \in A(n)$ that stabilises $n+1$ affinely independent points is the identity transformation.*

Proof If f stabilises the points $\mathbf{x}_0, \ldots, \mathbf{x}_n$, we can find an affine transformation h such that

$$h(\mathbf{0}) = \mathbf{x}_0, \quad h(\mathbf{e}_i) = \mathbf{x}_i,\ i = 1, \ldots, n.$$

Indeed, h is defined by the augmented matrix

$$A_h = \begin{pmatrix} \mathbf{x}_1 - \mathbf{x}_0 & 0 \\ \vdots & \vdots \\ \mathbf{x}_n - \mathbf{x}_0 & 0 \\ \mathbf{x}_0 & 1 \end{pmatrix}.$$

We can therefore assume by normalisation that f stabilises $\mathbf{0}$ and $\mathbf{e}_i$, $i = 1, \ldots, n$. Now let $f \in A(n)$ with $f(\mathbf{x}) = g(\mathbf{x}) + \mathbf{b}$.

First, since $f(\mathbf{0}) = \mathbf{0}$ it follows that $\mathbf{b} = \mathbf{0}$, so f is a linear mapping that stabilises the elements of the canonical basis. The desired result follows. □

Theorem 5.2.8 (Fundamental Theorem of Affine Geometry) *Given two affine bases $\{\mathbf{x}_0, \mathbf{x}_1, \ldots, \mathbf{x}_n\}$ and $\{\mathbf{y}_0, \mathbf{y}_1, \ldots, \mathbf{y}_n\}$ of $\mathbb{R}^n$, there exists a unique $f \in A(n)$ such that $f(\mathbf{x}_i) = \mathbf{y}_i$, $i = 0, 1, \ldots, n$.*

Proof From the proof of Lemma 5.2.7 we have that there exist $h, g \in A(n)$ that satisfy the relations:

$$h(\mathbf{0}) = \mathbf{x}_0, \quad h(\mathbf{e}_i) = \mathbf{x}_i,\ i = 1, \ldots, n$$

and

$$g(\mathbf{0}) = \mathbf{y}_0, \quad g(\mathbf{e}_i) = \mathbf{y}_i,\ i = 1, \ldots, n$$

The required element $f \in A(n)$ is $f = g \circ h^{-1}$. It is also unique; suppose to the contrary that there is another $f' \in A(n)$ such that $f'(\mathbf{x}_i) = \mathbf{y}_i, i = 0, 1, \ldots, n$. Then $f' \circ f^{-1}$ stabilises $n+1$ affinely independent points and therefore by Lemma 5.2.7 it cannot but none other than the identity element of $A(n)$. □

Example 5.2.9 To find a transformation $f \in A(2)$, $f(\mathbf{x}) = g(\mathbf{x}) + \mathbf{b}$ that maps $(0,0)$, $(1,0)$, $(0,1)$ to $(1,1)$ $(1,-2)$, $(2,1)$, respectively, we consider the determinant

$$\begin{vmatrix} 1 & 1 & 2 \\ 1 & -2 & 1 \\ 1 & 1 & 1 \end{vmatrix} = \begin{vmatrix} 0 & 1 & 1 \\ -3 & 0 & 1 \\ 0 & 0 & 1 \end{vmatrix} = 3 \neq 0.$$

The augmented matrix of the required $f \in A(2)$ is

$$A_f = \begin{pmatrix} 0 & -3 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}.$$

Hence

$$f(x_1, x_2) = (x_2 + 1,\ -3x_1 + 1), \quad (x_1, x_2) \in \mathbb{R}^2.$$

Example 5.2.10 We will find the affine transformation of $\mathbb{R}^2$ that maps

$$\mathbf{x}_0 = (1,2), \quad \mathbf{x}_1 = (2,-1), \quad \mathbf{x}_2 = (3,0),$$

to

$$\mathbf{y}_0 = (0,2), \quad \mathbf{y}_1 = (1,-4), \quad \mathbf{y}_2 = (-1,1),$$

respectively. To this end, we first identify the transformations h, g that map $\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2$ and $\mathbf{y}_0, \mathbf{y}_1, \mathbf{y}_2$ to $\mathbf{0}, \mathbf{e}_1, \mathbf{e}_2$ respectively, considering the corresponding generalized matrices

$$A_h = \begin{pmatrix} 1 & -3 & 0 \\ 2 & -2 & 0 \\ 1 & 2 & 1 \end{pmatrix} \quad \text{and} \quad A_g = \begin{pmatrix} 1 & -6 & 0 \\ -1 & -1 & 0 \\ 0 & 2 & 1 \end{pmatrix}.$$

Then, we verify that both are elements of $\mathrm{GL}(3, \mathbb{R})$. Indeed,

$$\det(A_h) = 4, \quad \det(A_g) = -7.$$

Now,

$$A_{h^{-1}} = \begin{pmatrix} -1/2 & 3/4 & 0 \\ -1/2 & 1/4 & 0 \\ 3/2 & -5/4 & 1 \end{pmatrix}.$$

Hence

$$A_g \cdot A_{h^{-1}} = \begin{pmatrix} 1 & -6 & 0 \\ -1 & -1 & 0 \\ 0 & 2 & 1 \end{pmatrix} \begin{pmatrix} -1/2 & 3/4 & 0 \\ -1/2 & 1/4 & 0 \\ 3/2 & -5/4 & 1 \end{pmatrix}$$
$$= \begin{pmatrix} -5/4 & 9/4 & 0 \\ -3/4 & 11/4 & 0 \\ 11/4 & -23/4 & 1 \end{pmatrix} = A_f.$$

The required transformation f is thus given by

$$f(x_1, x_2) = (1/4)(-5x_1 - 3x_2 + 11,\ 9x_1 + 11x_2 - 23), \quad (x_1, x_2) \in \mathbb{R}^2.$$

Example 5.2.11 To find the $f \in A(3)$ that maps

$$\mathbf{x}_0 = (0, 0, 0), \quad \mathbf{x}_1 = (1, 0, 0), \quad \mathbf{x}_2 = (0, 1, 0), \quad \mathbf{x}_3 = (0, 0, 1)$$

to

$$\mathbf{y}_0 = (1, 0, 1), \quad \mathbf{y}_1 = (1, 1, 0), \quad \mathbf{y}_2 = (0, 1, 1), \quad \mathbf{y}_3 = (1, -1, 1),$$

respectively, we examine whether

$$D = D(\mathbf{y}_0, \mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3) = \begin{pmatrix} 0 & 1 & -1 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 1 & 0 & 1 & 1 \end{pmatrix} \in \mathrm{GL}(4, \mathbb{R}).$$

Indeed, $|D| = 1$ and the desired transformation f has augmented matrix D. Therefore,

$$f(x_1, x_2, x_3) = (-x_2 + 1,\ x_1 + x_2 - x_3,\ -x_1 + 1), \quad (x_1, x_2, x_3) \in \mathbb{R}^3.$$

Remark 5.2.12 Three points $\mathbf{x}_0$, $\mathbf{x}_1$, $\mathbf{x}_2$ of $\mathbb{R}^n$ are affinely independent if they are not collinear, that is, if they do not belong to the same one-dimensional affine subspace of $\mathbb{R}^n$. Similarly, let $k + 1 \leq n + 1$ affinely independent points $\mathbf{x}_0, \ldots, \mathbf{x}_k$ of $\mathbb{R}^n$. This means that k-points $\mathbf{x}_1 - \mathbf{x}_0, \ldots, \mathbf{x}_k - \mathbf{x}_0$ are linearly independent points

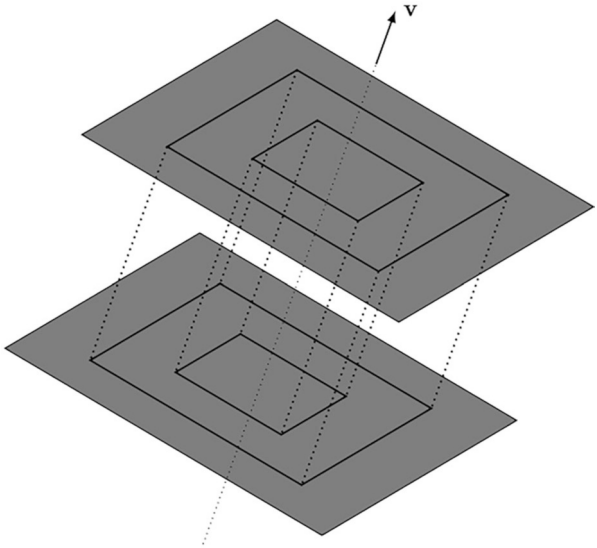


Fig. 5.4 Parallel projection preserves parallelism

of $\mathbb{R}^n$, in other words, they belong to a vector subspace of dimension k. But this is equivalent to saying that $\mathbf{x}_0, \ldots, \mathbf{x}_k$ belong to a k-plane, or else that cannot all belong to any l-plane when $l < k$.

5.2.3 *Parallel Projections*

Let $P = W_{\mathbf{A},b}$ be an n-plane of $\mathbb{R}^{n+1}$ defined by the equation $\mathbf{A} \cdot \mathbf{x} = b$, $\mathbf{x} \in \mathbb{R}^n$ (Fig. 5.4). If $\mathbf{v}$ is a direction ($\|\mathbf{v}\| = 1$) that is not orthogonal to $\mathbf{A}$, the *parallel projection* $\mathrm{pr}_{\mathbf{A}}^{\mathbf{v}}(\mathbf{x})$ *of* $\mathbf{x}$ *on* P *in the direction of* $\mathbf{v}$ is given by the formula

$$\mathrm{pr}_{\mathbf{A}}^{\mathbf{v}}(\mathbf{x}) = \mathbf{x} + \frac{b - \mathbf{A} \cdot \mathbf{x}}{\mathbf{A} \cdot \mathbf{v}}\mathbf{v}, \quad \mathbf{x} \in \mathbb{R}^{n+1}. \tag{5.11}$$

We observe that when $\mathbf{v} = \pm\mathbf{A}$, the above is exactly the orthogonal projection onto P. $\mathrm{pr}_{\mathbf{A}}^{\mathbf{v}}$ is an affine mapping $\mathbb{R}^{n+1} \to P$, with corresponding linear mapping

$$g(\mathbf{x}) = \mathbf{x} - \frac{\mathbf{A} \cdot \mathbf{x}}{\mathbf{A} \cdot \mathbf{v}}\mathbf{v}, \quad \mathbf{x} \in \mathbb{R}^{n+1}.$$

Indeed, for every $\mathbf{x} \in \mathbb{R}^n$,

$$\begin{aligned}\mathbf{A} \cdot \mathrm{pr}_{\mathbf{A}}(\mathbf{x}) &= \mathbf{A} \cdot \mathbf{x} + \frac{b - \mathbf{A} \cdot \mathbf{x}}{\mathbf{A} \cdot \mathbf{v}}(\mathbf{A} \cdot \mathbf{v}) \\ &= \mathbf{A} \cdot \mathbf{x} + b - \mathbf{A} \cdot \mathbf{x} = b.\end{aligned}$$

The proof of the next proposition is by straightforward computations.

Proposition 5.2.13 *The following holds for the parallel projection* $\mathrm{pr}_{\mathbf{A}}^{\mathbf{v}}$ *as defined in (5.11):*

(1) $\mathrm{pr}_{\mathbf{A}}^{\mathbf{v}} \circ \mathrm{pr}_{\mathbf{A}}^{\mathbf{v}} = \mathrm{pr}_{\mathbf{A}}^{\mathbf{v}}$,
(2) $\mathrm{pr}_{\mathbf{A}}^{\mathbf{v}}(\mathbf{x}) = \mathbf{x}$ *for each* $\mathbf{x} \in P$.

Exercise Prove Proposition 5.2.13.

An important property of the parallel projection is proved in the next proposition.

Proposition 5.2.14 *The parallel projection* $\mathrm{pr}_{\mathbf{A}}^{\mathbf{v}}$ *maps parallel lines of* $\mathbb{R}^{n+1}$ *(with a direction different from* $\mathbf{v}$*) to parallel lines of* P.

Proof Let $\mathbf{c}(t) = \mathbf{a} + t\mathbf{w}$, $t \in \mathbb{R}$ be a line. Then,

$$\begin{aligned}\mathrm{pr}_{\mathbf{A}}^{\mathbf{v}}(\mathbf{c}(t)) &= \mathbf{c}(t) + \frac{b - \mathbf{c}(t) \cdot \mathbf{A}}{\mathbf{A} \cdot \mathbf{v}}\mathbf{v} \\ &= \mathbf{a} + t\mathbf{w} + \frac{b - (\mathbf{a} + t\mathbf{w}) \cdot \mathbf{A}}{\mathbf{A} \cdot \mathbf{v}}\mathbf{v} \\ &= \mathbf{a} + \frac{b - \mathbf{a} \cdot \mathbf{A}}{\mathbf{A} \cdot \mathbf{v}} + t\left(\mathbf{w} - \frac{\mathbf{A} \cdot \mathbf{w}}{\mathbf{A} \cdot \mathbf{v}}\mathbf{v}\right).\end{aligned}$$

We observe that the direction of the projection of the line depends only on the direction of the line itself, which proves the desired result. Let us also observe that when $\mathbf{v} = \mathbf{w}$ then the parallel projection of the line is a point of P. □

If P' is another n-plane given by the equation $\mathbf{A}' \cdot \mathbf{x} = b'$, the restriction of $\mathrm{pr}_{\mathbf{A}}^{\mathbf{v}}$ to P' also defines an affine mapping $p : P' \to P$. We can also see this as follows: let $\mathbf{p}_0, \ldots, \mathbf{p}_n$ be an affine basis of P'. If

$$\mathbf{x} = \sum_{i=0}^{n} \lambda_i \mathbf{p}_i \in P, \quad \sum_{i=0}^{n} \lambda_i = 1,$$

then since every affine mapping preserves affine combinations, the restriction p of $\mathrm{pr}_{\mathbf{A}}^{\mathbf{v}}$ to P' is defined by the relation

$$p(\mathbf{x}) = \sum_{i=0}^{n} \lambda_i \mathrm{pr}_{\mathbf{A}}^{\mathbf{v}}(\mathbf{p}_i)$$

and is by construction an affine isomorphism.

Proposition 5.2.15 *Let P and P' be planes with equations $\mathbf{A}\cdot\mathbf{x} = b$ and $\mathbf{A}'\cdot\mathbf{x} = b'$, respectively. Let $\mathbf{v}$ also be a direction that is not orthogonal to any of $\mathbf{A}, \mathbf{A}'$. We denote by p the restriction of $\mathrm{pr}_{\mathbf{A}}^{\mathbf{v}}$ to P' and by p' the restriction of $\mathrm{pr}_{\mathbf{A}'}^{\mathbf{v}}$ to P. Then the affine isomorphisms p, p' satisfy the relations*

$$p \circ p' = id_P, \quad p' \circ p = id_{P'}.$$

Proof We will only show the left relation. For $\mathbf{x} \in P$, $\mathbf{A}\cdot\mathbf{x} = b$ we have:

$$\begin{aligned} p(p'(\mathbf{x})) &= p'(\mathbf{x}) + \frac{b - \mathbf{A}\cdot p'(\mathbf{x})}{\mathbf{A}\cdot\mathbf{v}}\mathbf{v} \\ &= \mathbf{x} + \frac{b' - \mathbf{A}'\cdot\mathbf{x}}{\mathbf{A}'\cdot\mathbf{v}}\mathbf{v} + \frac{b - \left(\mathbf{x} + \frac{b'-\mathbf{A}'\cdot\mathbf{x}}{\mathbf{A}'\cdot\mathbf{v}}\mathbf{v}\right)\cdot\mathbf{A}}{\mathbf{A}\cdot\mathbf{v}}\mathbf{v} \\ &= \mathbf{x} + \frac{b' - \mathbf{A}'\cdot\mathbf{x}}{\mathbf{A}'\cdot\mathbf{v}}\mathbf{v} + \frac{b - \mathbf{A}\cdot\mathbf{x}}{\mathbf{A}\cdot\mathbf{v}}\mathbf{v} - \frac{b' - \mathbf{A}'\cdot\mathbf{x}}{\mathbf{A}'\cdot\mathbf{v}}\,\frac{\mathbf{A}\cdot\mathbf{v}}{\mathbf{A}\cdot\mathbf{v}}\mathbf{v} \\ &= \mathbf{x}. \end{aligned}$$

□

Proposition 5.2.16 *Let $\mathbf{v}$, P and P' be as before and $p : P' \to P$ the parallel projection of P' onto P. Then the following holds.*

1. *If $P' \parallel P$ then p preserves Euclidean lengths.*
2. *If P' is not parallel to P, then p preserves lengths in at least one $(n-2)$-dimensional affine subspace of P'.*

Proof First suppose that P' is parallel to P. If $\mathbf{x}, \mathbf{y} \in P'$, then $\mathbf{A}\cdot\mathbf{x} = \mathbf{A}\cdot\mathbf{y} = b'$,

$$\|p(\mathbf{x})\|^2 = \|\mathbf{x}\|^2 + \frac{(b-b')^2}{(\mathbf{A}\cdot\mathbf{v})^2}\|\mathbf{v}\|^2 + 2\frac{b-b'}{\mathbf{A}\cdot\mathbf{v}}\,(\mathbf{x}\cdot\mathbf{v}),$$

$$\|p(\mathbf{y})\|^2 = \|\mathbf{y}\|^2 + \frac{(b-b')^2}{(\mathbf{A}\cdot\mathbf{v})^2}\|\mathbf{v}\|^2 + 2\frac{b-b'}{\mathbf{A}\cdot\mathbf{v}}\,(\mathbf{y}\cdot\mathbf{v})$$

and

$$p(\mathbf{x}) \cdot p(\mathbf{y}) = \mathbf{x} \cdot \mathbf{y} + \frac{b - b'}{\mathbf{A} \cdot \mathbf{v}} ((\mathbf{x} + \mathbf{y}) \cdot \mathbf{v}) + \frac{(b - b')^2}{(\mathbf{A} \cdot \mathbf{v})^2} \|\mathbf{v}\|^2.$$

So,

$$\begin{aligned} \|p(\mathbf{x}) - p(\mathbf{y})\|^2 &= \|p(\mathbf{x})\|^2 + \|p(\mathbf{y})\|^2 - 2p(\mathbf{x}) \cdot p(\mathbf{y}) \\ &= \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 - 2\,\mathbf{x} \cdot \mathbf{y} \\ &= \|\mathbf{x} - \mathbf{y}\|^2. \end{aligned}$$

If P' is not parallel to P, their intersection $P \cap P'$ is an affine subspace of P' (and P). The points of the intersection are fixed points of p, so the Euclidean lengths are preserved. □

A parallel projection $p : P' \to P$ as before, can be considered as an element of $A(n)$ provided that both P, P' are affinely isomorphic to $\mathbb{R}^n$. The question that arises naturally is whether every affine mapping of $\mathbb{R}^n$ is a parallel projection. The answer is generally negative, and this is because any dilation D_δ multiplies the length *each* interval by the factor δ. Nevertheless, we have the following theorem.

Theorem 5.2.17 (Parallel Projection Theorem) *The affine group $A(n)$ is generated by parallel projections.*

Proof We will use the Fundamental Theorem 5.2.8 which ensures that an element $f \in A(n)$ is uniquely defined by the values of the elements of an affine basis of $\mathbb{R}^n$. We will consider the case $n = 2$ and leave the general case as an exercise. Let $f \in A(2)$ and let also $B = \{\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2\}$ be an affine basis of $\mathbb{R}^2$; suppose also that

$$f(\mathbf{x}_i) = \mathbf{y}_i, \quad i = 0, 1, 2.$$

We will define two parallel projections that have the same effect on the elements of B. Let for this purpose be three affine copies P_0, P_1, P_2 of $\mathbb{R}^2$ on $\mathbb{R}^3$ (i.e., 2-planes of $\mathbb{R}^3$). At P_0 we mark the points $\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2$, and in the remaining planes we mark the points $\mathbf{y}_0, \mathbf{y}_1, \mathbf{y}_2$.

We first place the planes P_0 and P_2 so that $\mathbf{x}_0 \in P$ coincides with $\mathbf{y}_0 \in P_2$. By appropriate rotations of the planes about the point of coincidence, we can ensure that neither of $\mathbf{x}_1, \mathbf{y}_1$ lies on the intersection line. Then the vector $\mathbf{v} = \mathbf{y}_1 - \mathbf{x}_1$ is parallel to neither of the two planes, so we can define the parallel projection $pr^{\mathbf{v}} : P_0 \to P_2$ along $\mathbf{v}$. This projection maps $\mathbf{x}_0$ to $\mathbf{y}_0$ and $\mathbf{x}_1$ to $\mathbf{y}_1$.

Suppose that maps $\mathbf{x}_2$ to $\mathbf{y}_2'$. Since $\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2$ are not collinear, the same is true for their images $\mathbf{y}_0, \mathbf{y}_1, \mathbf{y}_2'$. We now place the planes P_1 and P_2 in such a way that the points $\mathbf{y}_0$ and $\mathbf{y}_1$ coincide. Since none of $\mathbf{y}_2, \mathbf{y}_2'$ is collinear with these points, they do not lie on the line of intersection. Therefore, $\mathbf{v}' = \mathbf{y}_2' - \mathbf{y}_2$ is not parallel to any

of these planes, and so we can define the projection $pr^{\mathbf{v}'} : P_2 \to P_1$ along $\mathbf{v}'$. This projection leaves $\mathbf{y}_0, \mathbf{y}_1$ fixed and maps $\mathbf{y}_2{}'$ onto $\mathbf{y}_2$.

It follows that $g = pr^{\mathbf{v}'} \circ pr^{\mathbf{v}}$ maps $\mathbf{x}_i$ to $\mathbf{y}_i$, $i = 0, 1, 2$. Since parallel projections are affine mappings (Proposition 5.2.15), the composition of is also an affine mapping. Since now f and g have the same values at three points, the required result follows from the Fundamental Theorem 5.2.8. □

Exercise Prove the general case of Theorem 5.2.17.

5.3 Three Euclidean Theorems

In this section we will prove by using affine methods, and in particular by using the Fundamental Theorem 5.2.8, three classical theorems of Euclidean Geometry. We begin with an observation that is a direct consequence of the Fundamental Theorem.

Corollary 5.3.1 *Any two triangles of* $\mathbb{R}^2$ *are affinely equivalent to the triangle with vertices* $\mathbf{0}$, $\mathbf{e}_1$, $\mathbf{e}_2$.

The second tool we will use is the simple ratio of collinear points, recall Definition 4.3.8.

Proposition 5.3.2 *The simple ratio of collinear points is an affine invariant.*

Proof Consider the straight line

$$\mathbf{c}(t) = \mathbf{a} + t\mathbf{v}, \quad t \in \mathbb{R}$$

and an ordered triple of pairwise disjoint points $\mathfrak{p} = (\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3)$, $\mathbf{x}_i = \mathbf{a} + t_i\mathbf{v}$, $i = 1, 2, 3$; their simple ratio is then

$$\sigma(\mathfrak{p}) = \frac{t_2 - t_1}{t_3 - t_2}.$$

If $f(\mathbf{x}) = \alpha(\mathbf{x}) + \mathbf{b}$ is an affine transformation, then $f(\mathbf{x}_i) = (\alpha(\mathbf{a}) + \mathbf{b}) + t_i\alpha(\mathbf{v})$ and therefore

$$\sigma(f(\mathfrak{p})) = \frac{t_2 - t_1}{t_3 - t_2}.$$

□

Remark 5.3.3 If O is the origin, then we can write

$$\mathbf{x}_i = \overrightarrow{OX_i}, \quad i = 1, 2, 3$$

and

$$\sigma(\mathfrak{p}) = \frac{\overrightarrow{OX_2} - \overrightarrow{OX_1}}{\overrightarrow{OX_2} - \overrightarrow{OX_3}} = \frac{\overrightarrow{X_1X_2}}{\overrightarrow{X_3X_2}}.$$

We now consider the following configuration for two ordered pairwise district triples of non-collinear points $\mathfrak{p} = (\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3)$ and $\mathfrak{q} = (\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3)$ of $\mathbb{R}^2$:

$$\begin{aligned}
\mathbf{y}_1 &= (1 - t_1)\mathbf{x}_2 + t_1\mathbf{x}_3, \quad t_1 \neq 0, 1,\\
\mathbf{y}_2 &= (1 - t_2)\mathbf{x}_3 + t_2\mathbf{x}_1, \quad t_2 \neq 0, 1,\\
\mathbf{y}_3 &= (1 - t_3)\mathbf{x}_1 + t_3\mathbf{x}_2, \quad t_3 \neq 0, 1.
\end{aligned}$$

In other words, $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ are vertices of a triangle while $\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3$ are points of the (extensions) of the sides of the triangle, other than their vertices. We consider the quantity

$$\Sigma = \sigma(\mathbf{x}_1, \mathbf{y}_3, \mathbf{x}_2) \cdot \sigma(\mathbf{x}_2, \mathbf{y}_1, \mathbf{x}_3) \cdot \sigma(\mathbf{x}_3, \mathbf{y}_2, \mathbf{x}_1) = \frac{t_1t_2t_3}{(1 - t_1)(1 - t_2)(1 - t_3)}.$$

From the Fundamental Theorem 5.2.8 it follows that we can assume that

$$\mathbf{x}_1 = (0, 0), \quad \mathbf{x}_2 = (1, 0), \quad \mathbf{x}_3 = (0, 1),$$

and then

$$\mathbf{y}_1 = (1 - t_1, t_1), \quad \mathbf{y}_2 = (0, 1 - t_2). \quad \mathbf{y}_3 = (t_3, 0).$$

We observe that the quantity Σ remains invariant by this normalisation.

Proposition 5.3.4 *Let one of the points $\mathbf{y}_i$, $i = 1, 2, 3$ belong to the extension of one of the sides of the normalised triangle. Then $\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3$ are collinear if and only if $\Sigma = -1$.*

Proof The points $\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3$ are collinear if and only if the determinant

$$|D(\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3)| = \begin{vmatrix} 0 & 1 - t_2 & 1 \\ t_3 & 0 & 1 \\ 1 - t_1 & t_1 & 1 \end{vmatrix}$$

vanishes. But,

$$|D(\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3)| = 0 \iff 1 - t_1 - t_2 - t_3 + t_1t_2 + t_2t_3 + t_3t_1 = 0 \iff \Sigma = -1.$$

□

Proposition 5.3.4 is the algebraic formulation of the following theorem due to Menelaus of Alexandria. In this version of the theorem, incorporated there is also its converse.

Theorem 5.3.5 (Menelaus' Theorem and its Converse) *Given a triangle $\triangle ABC$, and a transversal line intersecting the sides BC, AC, and AB at points D, E, and F respectively, with D, E, and F different from the vertices A, B, and C, then*

$$\frac{\overrightarrow{AF}}{\overrightarrow{FB}} \cdot \frac{\overrightarrow{BD}}{\overrightarrow{DC}} \cdot \frac{\overrightarrow{CE}}{\overrightarrow{EA}} = -1.$$

Conversely, if points $D, E,$ and F are taken on the sides $BC, AC,$ and AB respectively, so that the above relation is satisfied, then D, E, and F are collinear.

We see in Figs. 5.5 and 5.6 the two cases that can arise in the proof of Menelaus' Theorem. Due to our formulation, we did not have to examine these cases separately. In the first case, the transversal line intersects two sides of the triangle and the third in its extension.

In the second case, the transversal line does not intersect any side of the triangle (except only their extensions).

Continuing now with our configuration, we consider the lines connecting $\mathbf{x}_i$ $\mathbf{y}_i$, $i = 1, 2, 3$. The line from $\mathbf{x}_1$ and $\mathbf{y}_1$ is

$$l_1 : \ t_1 x_1 + (t_1 - 1)x_2 = 0,$$

the line from $\mathbf{x}_2$ and $\mathbf{y}_2$ is

$$l_2 : \ (t_2 - 1)x_1 - x_2 + 1 - t_2 = 0,$$

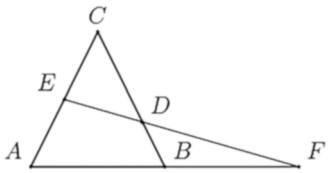


Fig. 5.5 Proof of Menelaus' Theorem, first case

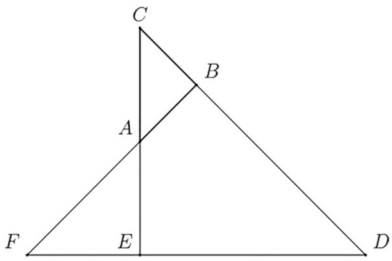


Fig. 5.6 Proof of Menelaus' Theorem, second case

and the line from $\mathbf{x}_3$ and $\mathbf{y}_3$ is

$$l_3 : \ x_1 + t_3 x_2 - t_3 = 0.$$

Proposition 5.3.6 *If the lines l_1, l_2, l_3 are concurrent, then $\Sigma = 1$. If $\Sigma = 1$, then l_1, l_2, l_3 are either parallel or concurrent.*

Proof Recall that if three (different) straight lines

$$\begin{aligned} l_1 : \ & a_1 x_1 + b_1 x_2 + c_1 = 0, \\ l_2 : \ & a_2 x_1 + b_2 x_2 + c_2 = 0, \\ l_3 : \ & a_3 x_1 + b_3 x_2 + c_3 = 0, \end{aligned}$$

are *concurrent* (i.e., they have a single common point) then the determinant

$$D(l_1, l_2, l_3) = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

vanishes. For the lines of our configuration we have:

$$\begin{aligned} D(l_1, l_2, l_3) &= \begin{vmatrix} t_1 & t_1 - 1 & 0 \\ t_2 - 1 & -1 & 1 - t_2 \\ 1 & t_3 & -t_3 \end{vmatrix} \\ &= t_1 \begin{vmatrix} -1 & 1 - t_2 \\ t_3 & -t_3 \end{vmatrix} - (t_1 - 1) \begin{vmatrix} t_2 - 1 & 1 - t_2 \\ 1 & -t_3 \end{vmatrix} \\ &= t_1 t_2 t_3 - (1 - t_1)(1 - t_2)(1 - t_3). \end{aligned}$$

So,

$$D(l_1, l_2, l_3) = 0 \iff \Sigma = 1.$$

If now conversely $\Sigma = 1$, then $D(l_1, l_2, l_3) = 0$. In the case where the minor determinants vanish

$$D(l_1, l_2) = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \quad \text{and} \quad D(l_2, l_3) = \begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix}.$$

then l_1, l_2, l_3 are parallel. If this is not the case, then they are concurrent. □

We observe that if the lines are concurrent, then their common point $C = (c_1, c_2)$ is the common point of l_1 and l_2, that is,

$$C = \left(\frac{(1-t_1)(1-t_2)}{t_1 + (1-t_1)(1-t_2)}, \frac{t_1(1-t_2)}{t_1 + (1-t_1)(1-t_2)} \right).$$

Since $t_1, t_2 \neq 0, 1$, this point does not lie on any of the perpendicular sides of the normalised triangle. Since

$$c_1 + c_2 = \frac{1-t_2}{1-t_2+t_1t_2},$$

the point C also does not lie on the hypotenuse of the triangle. We distinguish the following cases:

A. *The points* $\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3$ *all lie on the sides of the triangle (and not on their extensions).*
   Then, $c_1, c_2 > 0$, $c_1 + c_2 < 1$ and (c_1, c_2) belongs to the interior of the triangle.
B. *Two of the points* $\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3$ *all lie on the extensions of the sides of the triangle.* By direct calculation we can show that C is located on the exterior of the triangle.

The following theorems are direct consequences of Proposition 5.3.6; recall that a median of a triangle is the line segment joining a vertex of the triangle to the midpoint of the opposite side, thus bisecting that side:

Theorem 5.3.7 *The medians of a triangle intersect at a point (called the centroid) that is two-thirds of the distance from the vertices to the midpoint of the opposite sides.*

Proof In our formulation, we take $t_i = 1/2$, $i = 1, 2, 3$. Then $D(l_1, l_2, l_3) = 0$. The centroid is the common point of l_1, l_2, that is, $(1/3, 1/3)$ which belongs to the interior of the triangle. □

The following theorem is known as Ceva's Theorem, but it was proved much earlier than Ceva by the Arab king of Zaragoza (reigned from 1081 to 1085 AD) Yusuf Al-Mu'taman ibn Hud in the eleventh century AD.

Theorem 5.3.8 (Ceva's Theorem and Its Converse) *From vertices of a triangle* A, B, C *we draw lines* AE, BF *and* CD *to the corresponding opposite sides. These lines are concurrent if and only if*

$$\frac{\overrightarrow{AF}}{\overrightarrow{FB}} \cdot \frac{\overrightarrow{BD}}{\overrightarrow{DC}} \cdot \frac{\overrightarrow{CE}}{\overrightarrow{EA}} = 1.$$

Conversely, if the above relationship holds, then AE, BF *and* CD *are parallel or concurrent.*

5.4 Exercises

1. Determine whether the following sets are affine bases of the corresponding affine space $\mathbb{R}^n$:

   (a) $\{1, 2\}$ in $\mathbb{R}$.
   (b) $\{(-1, 0),\ (2, 2),\ (3, 5)\}$ in $\mathbb{R}^2$.
   (c) $\{(1, 0, 1),\ \ (2, 0, 7),\ (-3, 0, 12)\}$ in $\mathbb{R}^3$.

2. Find all $a \in \mathbb{R}$ for which the set

$$B(a) = \{(1, 2),\ (2, 4),\ (3, a)\},$$

   is an affine basis of $\mathbb{R}^2$. Then, find the barycentric coordinates corresponding to this basis.

3. After verifying that the set

$$B = \{(0, 1, 2),\ (-1, 1, 1),\ (-2, 1, 1),\ (1 - 2, 1)\}$$

   is a basis of the affine space $\mathbb{R}^3$, find the barycentric coordinates with respect to this basis.

4. Let $\mathbf{c}(t) = \mathbf{a} + t\mathbf{v}$, $t \in \mathbb{R}^n$, $\mathbf{a}, \mathbf{v} \in \mathbb{R}^n$, $\mathbf{v} \neq \mathbf{0}$, be a straight line in $\mathbb{R}^n$.

   (a) Prove that is a one-dimensional affine space with associated vector space $V = \langle \mathbf{v} \rangle$.
   (b) Prove that the set $\{\mathbf{a},\ \mathbf{v}\}$ is an affine basis of $\mathbf{c}$ and determine the barycentric coordinates with respect to this basis.

5. Prove the following.

   (a) Any two distinct points of the affine space $\mathbb{R}^n$ are affinely independent.
   (b) Any three collinear points of the affine space $\mathbb{R}^n$ are affinely dependent.

6. Write it out in detail in two ways:

   (a) the straight line of $\mathbb{R}^2$ that passes through $(-1, 2)$ in the direction of $(2, -1)$,
   (b) the plane of $\mathbb{R}^3$ from the origin produced by $(1, -1, 0)$ and $(0, 1, 1)$,
   (c) the 2-plane of $\mathbb{R}^4$ that passes through $\mathbf{e}_1$ and is produced by $\mathbf{e}_3, \mathbf{e}_4$.

7. Let the parametric equations

   (a) $\mathbf{x}(\lambda, \mu) = (1, -1, 2) + \lambda(2, 0, 1) + \mu(2, 1, 3)$,
   (b) $\mathbf{x}(\lambda, \mu) = (0, 2, 0) + \lambda(1, 1, 1) + \mu(1, 2, 0)$,

   where $\lambda, \mu \in \mathbb{R}$. Prove that both represent affine subspaces of $\mathbb{R}^3$ and for each subspace determine affine bases and the corresponding barycentric coordinates.

8. Let the parametric equation

$$\mathbf{x}(\lambda_1, \lambda_2) = (1, 2, 0, 0) + \lambda_1(1, -1, 2, 1) + \lambda_2(2, 0, 1, 0), \quad \lambda_i \in \mathbb{R},\ i = 1, 2.$$

Prove that it represents a 2-plane of $\mathbb{R}^4$. Determine an affine basis and the barycentric coordinates corresponding to this basis.

9. Prove that the parametric equation

$$\mathbf{x}(\lambda, \mu) = (1, 0, 1) + \lambda(1, 0, -1) + \mu(2, 0, -2), \quad \lambda, \mu \in \mathbb{R},$$

does not represent a 2-plane of $\mathbb{R}^2$. Show that the above equation represents a line.

10. Find an affine basis for the plane P of $\mathbb{R}^3$ with equation

$$2x_1 + 3x_2 - x_3 = 6,$$

and complete it in an affine basis of $\mathbb{R}^3$.

11. Find an affine basis for the 2-plane P of $\mathbb{R}^4$ with equations

$$x_1 - x_4 = 1, \quad x_2 + x_3 = 0,$$

and complete it in an affine basis of $\mathbb{R}^4$.

12. Let W_k be a k-plane as in (5.3) where we additionally assume that $\mathbf{v}_i$, $i = 1, \ldots, k$ are orthonormal. Prove that the orthogonal projection $\mathrm{pr}_{W_k}(\mathbf{x})$ of $\mathbf{x}$ onto W_k is given by

$$\mathrm{pr}_{W_k}(\mathbf{x}) = \mathbf{a} + \sum_{i=1}^{k}((\mathbf{x} - \mathbf{a}) \cdot \mathbf{v}_i))\mathbf{v}_i.$$

If W_k is given by (5.4), the orthogonal projection $\mathrm{pr}_{W_k}(\mathbf{x})$ of $\mathbf{x}$ onto W_k is given by

$$\mathrm{pr}_{W_k}(\mathbf{x}) = \mathbf{x} + \sum_{i=1}^{n-k}(b_i - (\mathbf{x} \cdot \mathbf{A}_i))\mathbf{A}_i.$$

13. Which of the following are affine mappings of the corresponding affine spaces? Justify your answer.

(i) $f_a : \mathbb{R} \to \mathbb{R}$, $f(x) = x + a^2$, $a \in \mathbb{R}$.
(ii) $f : \mathbb{R} \to \mathbb{R}^2$, $f(t) = (t + 3, t^2)$.
(iii) $f : \mathbb{R}^2 \to \mathbb{R}$, $f(x_1, x_2) = x_1 + x_2 - 3$.
(iv) $f : \mathbb{R}^2 \to \mathbb{R}^2$, $f(x_1, x_2) = (2,\ x_1 + x_2 - 3)$.
(v) $f : \mathbb{R}^n \to \mathbb{R}^n$, $f(\mathbf{x}) = g(\mathbf{x}) + \mathbf{b}$, where $g : \mathbb{R}^n \to \mathbb{R}^n$ is a mapping and $\mathbf{b}$ is a constant vector.

14. Let $f : \mathbb{R}^n \to \mathbb{R}^m$ and $g : \mathbb{R}^m \to \mathbb{R}^k$ be affine mappings and suppose that the composition $h = g \circ f$ is defined. Prove that h is an affine mapping.
15. Consider the points $(0, 1, 0)$, $(1, 0, 0)$ and $(1, 1, 1)$ of $\mathbb{R}^3$.

    (a) Show that they are affinely independent.
    (b) If $f : \mathbb{R}^3 \to \mathbb{R}^3$ be such that

    $$f(x_1, x_2, x_3) = (x_1 - 2x_2 + x_3 - 1,\ x_2 - 7,\ x_3 + 1), \quad (x_1, x_2, x_3) \in \mathbb{R}^3,$$

    prove that $f(0, 1, 0)$, $f(1, 0, 0)$ and $f(1, 1, 1)$ are affinely independent.
16. Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be such that

    $$f(x_1, x_2) = (2x_1 - x_2 + 1,\ x_1 + 2x_2 - 1) \quad (x_1, x_2) \in \mathbb{R}^2.$$

    (a) Prove that f is an affine map.
    (b) Find the f-image of the lines

    $$l_1 : x_1 - x_2 = 1,$$
    $$l_2 : x_1 - x_2 = 2.$$

17. Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be such that

    $$f(x_1, x_2) = (x_1 + x_2,\ x_1 - x_2 + 1), \quad (x_1, x_2) \in \mathbb{R}^2.$$

    (a) Prove that f is an affine map.
    (b) Consider the lines with parametric equations

    $$\mathbf{c}_1(t) = (1, 0) + t(0, 1), \quad \mathbf{c}_2(t) = (1, 0) + t(-1, 0),\ t \in \mathbb{R}$$

    After proving that the angle of these lines at their intersection point $(1, 0)$ is equal to $\pi/2$, then prove that the angle of their f-images at their intersection point is also equal to $\pi/2$. Why is this so? Show with an example that in general an affine mapping does not preserve angles.
18. Find an affine mapping of the plane that stabilises the line $x_1 = 0$.
19. Let $f : \mathbb{R}^3 \to \mathbb{R}^3$ be such that

    $$f(x_1, x_2, x_3) = (2x_1,\ x_2 - 1,\ x_1 + 2x_3) \quad (x_1, x_2, x_3) \in \mathbb{R}^3,$$

    and let P be the plane defined by the equation $x_1 - x_2 + x_3 = 2$. As in Example 5.1.30, find:

    (a) the image $f(P)$ and
    (b) a formula for the mapping $P \to f(P)$.

20. Identifying $\mathbb{R}^2$ with $\mathbb{C}$, prove that every element of $A(2)$ is of the form

$$f(z) = Az + B\overline{z} + C, \quad A, B, C \in \mathbb{C},\ |A| + |B| \neq 0.$$

21. Find the inverses of the following affine automorphisms:

(a) $f : \mathbb{R} \to \mathbb{R}$, $f(x) = -x + 1$,
(b) $f : \mathbb{R}^2 \to \mathbb{R}^2$, $f(x_1, x_2) = (x_1 - 2x_2 + 3,\ -2x_1 + x_2 + 1)$,
(c) $f : \mathbb{R}^3 \to \mathbb{R}^3$, $f(x_1, x_2, x_3) = (-x_1 + 1,\ x_2 + x_3,\ -x_3 + 3)$.

22. Find the augmented matrices of the affine mappings of Exercise 21.
23. Find the affine mappings resulting from the following:

(a)

$$A_g = \begin{pmatrix} 2 & 0 \\ 1 & -1 \end{pmatrix}, \quad \mathbf{b} = (0, 9),$$

(b)

$$A_g = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad \mathbf{b} = (-1, 1, -1).$$

24. After verifying that the following matrices are invertible, determine the affine mappings f that have them as augmented matrices:

(a)

$$A_f = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 2 & 3 & 1 \end{pmatrix},$$

(b)

$$A_f = \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{pmatrix}.$$

25. Given the affinely independent ordered triples of pairwise distinct points $\{\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2\}$ and $\{\mathbf{y}_0, \mathbf{y}_1, \mathbf{y}_2\}$, prove that for $n \geq 2$ there exists an $f \in A(n)$ such that $f(\mathbf{x}_i) = \mathbf{y}_i,\ = 0, 1, 2$. That is, $A(n)$ acts triply transitively on affinely independent points.

26. Determine whether there are affine transformations that map the points $(0, 0)$, $(1, 0)$, $(0, 1)$, respectively, to the points

    (a) $(1, 2)$, $(3, 4)$, $(5, 6)$.
    (b) $(-8, 0)$, $(0, 8)$ $(0, 0)$.
    (c) $(0, 1)$, $(0, 0)$, $(1, a)$, $a \in \mathbb{R}$.

27. Determine the number $a \in \mathbb{R}$ for which there exists an $f \in A(2)$ that maps

$$(1, a), \quad (1, 1), \quad (a, 1),$$

    respectively, to

$$(2, a), \quad (2, 2), \quad (a, 2).$$

28. Examine whether there exists a transformation $f \in A(3)$ that maps

$$\mathbf{e}_1 - \mathbf{e}_2, \quad \mathbf{e}_2 - \mathbf{e}_3, \quad \mathbf{e}_3 - \mathbf{e}_4, \quad \mathbf{e}_4 - \mathbf{e}_1,$$

    respectively, to

$$\mathbf{e}_2 - \mathbf{e}_3, \quad \mathbf{e}_3 - \mathbf{e}_4, \quad \mathbf{e}_3 - \mathbf{e}_1, \quad \mathbf{e}_1 - \mathbf{e}_2.$$

29. Show that there does not exist an $f \in A(2)$ that maps $(0, 0)$, $(1, 0)$, and $(0, 1)$, respectively, to $(1, 1)$, $(2, 2)$, and $(3, 3)$, in the following way: assume towards a contradiction that there exists such a transformation f, given by

$$f(x_1, x_2) = (x_1\, x_2) \begin{pmatrix} a & b \\ c & d \end{pmatrix} + (b_1, b_2),$$

    where $ad - bc \neq 0$. Then, the relations

$$f(0, 0) = (1, 1), \quad f(1, 0) = (1, 1), \quad f(0, 1) = (2, 2),$$

    lead to a contradiction.

30. Find $f \in A(3)$ that maps the 2-plane

$$P : \mathbf{x}(\lambda, \mu) = \lambda \mathbf{e}_1 + \mu \mathbf{e}_2$$

    in the plane

$$P' : \mathbf{x}'(\lambda, \mu) = \mathbf{e}_1 + \lambda \mathbf{e}_2 + \mu \mathbf{e}_3.$$

31. Let $\mathbf{v} = (1/\sqrt{3})(1, 1, 1)$ and the hyperplane $\mathbf{A} \cdot \mathbf{x} = 2$ of $\mathbb{R}^3$, $\mathbf{A} = (1/\sqrt{2})(-1, 1, 0)$. Write the formula of $\mathrm{pr}_{\mathbf{A}}^{\mathbf{v}}$ in detail.

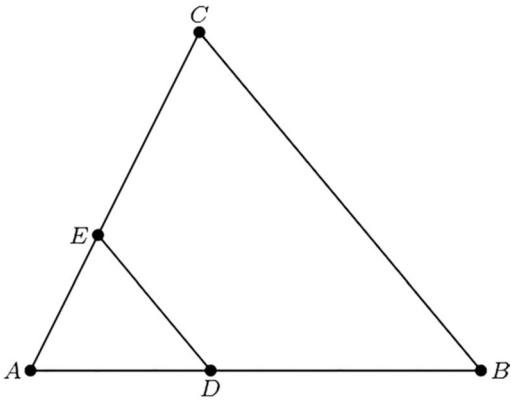


Fig. 5.7 Thales' Intercept Theorem

32. Let $\mathbf{v}$, P, P' and p be as in Proposition 5.2.16. Find conditions such that the parallel projection p preserves (a) the norms and (b) the inner products.
33. Prove Thales' Intercept Theorem in the following way: first use arguments from Affine Geometry to normalise so that the vertices of the triangle are

$$A = (0, 1), \quad B = (0, 0), \quad C = (1, 0),$$

and assume that

$$D = (0, t), \quad E = (s, 0), \ 0 < s, t < 1.$$

Due to the similarity of the triangles you can now use Theorem 4.3.6 (Fig. 5.7).

Chapter 6
Spherical Geometry



When I delightfully trace the orbits of heavenly bodies, I no longer touch the earth with my feet: I stand in the presence of Zeus himself and feast on ambrosia, food of the gods.

—Almagest I, Claudius Ptolemy

Spherical Geometry according to Klein is the pair $(S^{n-1}, \mathrm{O}(n))$. Spherical Geometry deals with the properties and relationships of points, lines, and shapes located on the surface of the unit sphere S^{n-1} of $\mathbb{R}^n$. In contrast to Euclidean Geometry, whose environment is flat, in Spherical Geometry we focus on figures like those encountered in geography and astronomy. One of the fundamental characteristics of Spherical Geometry is that its "straight lines" are *great circles*, meaning circles with radius 1 centered at the centre of S^{n-1}. These great circles play the role of straight lines in Euclidean Geometry. A tangible example is that of the meridians and parallels of the Earth: the former are great circles while the latter are *minor circles*, meaning their centre (except for the Equator) is not the centre of the Earth.

At the time when the Earth was discovered to be round and not flat, Spherical Geometry began to emerge to assist navigators in mapping the land and seas. However, even before Columbus, ancient Greeks and Phoenician navigators used the ideas of Spherical Geometry in their nautical explorations of the world they predominantly knew, mostly the Mediterranean, but not exclusively. We should really emphasise here that scientists such as Aristarchus of Samos, c. 310–c. 230 BC, Claudius Ptolemy, (Fig. 6.1) (Ptolemy's *Geography* applied spherical principles to Earth, defining latitude/longitude and projecting its curvature onto maps) and Eratosthenes of Cyrene believed that the Earth was round several centuries before Columbus. Therefore, Spherical Geometry began to be systematically studied already in ancient times due to its practical applications in navigation and astronomy.

The foundations were laid by ancient Greek mathematicians—we have the texts of the mathematician, astronomer and geographer Autolycus of Pitane, c. 360–c. 290 BC, and the astronomer and mathematician Theodosius of Bithynia (second–first century BC) from the second century BC, who even authored works on

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I. D. Platis, *Geometries*, La Matematica per il 3+2 178,
https://doi.org/10.1007/978-3-032-08016-5_6



Fig. 6.1 Claudius Ptolemy, c. 100–c. 170 AD

Spherics. Also, Menelaus's Theorem, which we saw in the previous chapter, was first proved by Menelaus as a spherical theorem. Moreover, Eratosthenes, known as the father of geography, measured the circumference of the Earth using elements of Spherical Geometry. Islamic scholars also studied and further developed the works of the Greeks, refined spherical trigonometry for calculating Earth's circumference and qibla directions.

Medieval navigators relied on spherical geometry for astrolabes and portolan charts, linking astronomy to terrestrial exploration. The Age of Discovery (fifteenth–seventeenth centuries) necessitated accurate spherical calculations for transoceanic voyages, driving innovations like Mercator's projection .

Euler's contributions are quite significant: he formalised spherical triangles, essential for celestial navigation and time keeping. Spherical Geometry's link with Geography is strong: Geography's core frameworks (e.g., great circles as shortest paths) derive from spherical geometry, enabling efficient air/sea routes. Gauss' nineteenth-century work on curved surfaces paved the way for non-Euclidean geometry, validating Earth as a "spheroid" model and modern geodesy uses spherical math for global positioning systems. Map projections (e.g., azimuthal, conic) reconcile spherical Earth with 2-dimensional maps, minimising distortions in area or distance. Climate science leverages spherical harmonics to model atmospheric/oceanic patterns across Earth's surface. From Eratosthenes' circumference estimate to digital globes, geography's evolution mirrors spherical geometry's precision. Today, the field remains vital for understanding plate tectonics, sea-level rise, and planetary-scale phenomena. All in all, spherical geometry's history is inseparable from geography—a dialogue between abstract math and Earth's tangible form.

There are many similarities but also characteristic differences between Spherical and Euclidean Geometry that will become apparent in this chapter. In the modern approach, the sphere S^2 has mostly been studied as part of three-dimensional Euclidean Geometry (stereometry), and there, the surface is considered as embedded in the surrounding space $\mathbb{R}^3$. However, in every dimension, the unit sphere can be examined with intrinsic methods that do not mention or require the existence of any surrounding space, either outside or within the sphere. Spherical Geometry is

a metric Geometry: we have already seen that the chordal metric (which refers to the surrounding space) has the orthogonal group as its group of isometries but is not intrinsic. For this purpose, we will consider S^{n-1} with its intrinsic metric, also called the spherical metric. Before we reach that point, we will examine some important properties of the sphere itself.

For this chapter, besides general geometry books, we refer specifically to [34] and [55]. Even more specifically, for a nice presentation of Spherical Trigonometry, we refer to [51], while for an extensive presentation of the version of Menelaus' Theorem by Arab Mathematicians, we refer to [43].

6.1 The Unit Sphere S^{n-1}

Let S^{n-1} be the unit sphere in $\mathbb{R}^n$ (Fig. 6.2),

$$S^{n-1} = \{\mathbf{x} \in \mathbb{R}^n \mid \|\mathbf{x}\| = 1\}.$$

We will initially examine the trivial cases $n = 1, 2$. When $n = 1$, S^0 consists simply of the points ± 1 in $\mathbb{R}$, which is a *compact* (i.e., closed and bounded) set. However, it is not connected subset of $\mathbb{R}$. We recall that a subset of $\mathbb{R}^n$ is *connected* if any two points can be joined by a continuous curve lying entirely within the set. In the case $n = 2$, we have the familiar unit circle S^1 of $\mathbb{R}^2$, which is a compact and connected subset of $\mathbb{R}^2$. Usually, to describe S^1 as a curve in $\mathbb{R}^2$, we consider the parameterisation of S^1 resulting from the polar coordinates of $\mathbb{R}^2$:

$$\gamma(\theta) = \mathbf{x}(\theta) = (\cos\theta, \sin\theta), \quad \theta \in \mathbb{R}.$$

We can also view θ as the angle of the vectors $\mathbf{e}_1$ and $\mathbf{x}$ and thus restrict θ to $[0, 2\pi)$ (or $[0, 2\pi]$, since in many cases we are interested in the injective embedding of a

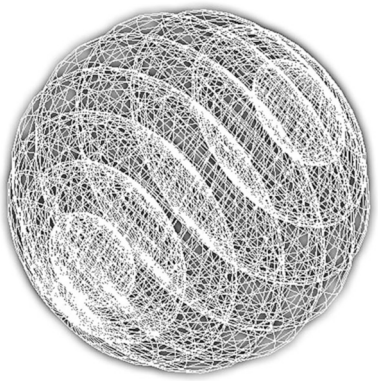


Fig. 6.2 The unit sphere S^{n-1}: embedded are shown spheres of smaller dimension

parameterised curve γ in the open interval of definition of the curve). We saw in Sect. 3.3.4 that the chordal metric is not intrinsic, and the proof of this implies that the intrinsic metric measures the distance between two points on the circle by the length of an arc. As we will see below, this is also what happens in the general case. In what follows, we will assume that $n \geq 3$. In these dimensions, as in the case $n = 2$, the sphere S^{n-1} is a compact and connected subset of $\mathbb{R}^n$. The sphere S^2 in particular will appear in a sufficient number of examples that we will give below.

6.1.1 Spherical Coordinates

The familiar spherical coordinate system of $\mathbb{R}^3$ generalises to any $n \geq 3$, according to the following theorem.

Theorem 6.1.1 *For each* $\mathbf{x} = (x_1, \ldots, x_n) \in \mathbb{R}^n$ *there is a radial coordinate* $r \geq 0$ *and a set of* $(n-1)$ *angular coordinates*

$$(\varphi_1, \ldots, \varphi_{n-1}) \in [0, \pi]^{n-2} \times [0, 2\pi),$$

such that $r = \|\mathbf{x}\|$ *and*

$$\begin{aligned}
x_1 &= r\cos(\varphi_1) \\
x_2 &= r\sin(\varphi_1)\cos(\varphi_2) \\
x_3 &= r\sin(\varphi_1)\sin(\varphi_2)\cos(\varphi_3) \\
&\vdots \\
x_{n-1} &= r\sin(\varphi_1)\cdots\sin(\varphi_{n-2})\cos(\varphi_{n-1}) \\
x_n &= r\sin(\varphi_1)\cdots\sin(\varphi_{n-2})\sin(\varphi_{n-1}).
\end{aligned}$$

Proof Let $\varphi_1 = \angle(\mathbf{x}, \mathbf{e}_1)$. Then

$$\cos(\varphi_1) = \frac{\mathbf{x} \cdot \mathbf{e}_1}{\|\mathbf{x}\|} = \frac{x_1}{r}.$$

We project $\mathbf{x}$ onto the coordinate hyperplane orthogonal to $\mathbf{e}_1$:

$$\mathrm{pr}_1(\mathbf{x}) = \mathbf{x} - (\mathbf{e}_1 \cdot \mathbf{x})\mathbf{e}_1 = \mathbf{x} - x_1\mathbf{e}_1 = (x_2, \ldots, x_n).$$

Let $\varphi_2 = \angle(\mathrm{pr}_1(\mathbf{x}), \mathbf{e}_2)$. Then

$$\cos(\varphi_2) = \frac{x_2}{\|(x_2, \ldots, x_n)\|}.$$

The triangle with vertices $\mathbf{0}$, $\mathbf{x}$, $\mathrm{pr}_1(\mathbf{x})$, is a right triangle with hypotenuse length r, so

$$\|(x_2, \ldots, x_n)\| = r\cos(\pi/2 - \varphi_1) = r\sin(\varphi_1),$$

and we get $x_2 = r\sin(\varphi_1)\cos(\varphi_2)$. We continue in this way, projecting $\mathrm{pr}_1(\mathbf{x})$ onto the coordinate hyperplane orthogonal to $\mathbf{e}_1, \mathbf{e}_2$ and then projecting $\mathrm{pr}_2(\mathrm{pr}_1(\mathbf{x}))$ onto the coordinate hyperplane orthogonal $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$, etc. In the last step, the corresponding projection belongs to a circle, so $\varphi_{n-1} \in [0, 2\pi)$.

The inverse transformation is uniquely defined except for some special cases. We have

$$r = \sqrt{{x_n}^2 + {x_{n-1}}^2 + \cdots + {x_2}^2 + {x_1}^2}$$

$$\varphi_1 = \operatorname{arccot}\frac{x_1}{\sqrt{{x_n}^2 + {x_{n-1}}^2 + \cdots + {x_2}^2}} = \arccos\frac{x_1}{\sqrt{{x_n}^2 + {x_{n-1}}^2 + \cdots + {x_1}^2}}$$

$$\varphi_2 = \operatorname{arccot}\frac{x_2}{\sqrt{{x_n}^2 + {x_{n-1}}^2 + \cdots + {x_3}^2}} = \arccos\frac{x_2}{\sqrt{{x_n}^2 + {x_{n-1}}^2 + \cdots + {x_2}^2}}$$

$$\vdots \qquad\qquad \vdots$$

$$\varphi_{n-2} = \operatorname{arccot}\frac{x_{n-2}}{\sqrt{{x_n}^2 + {x_{n-1}}^2}} = \arccos\frac{x_{n-2}}{\sqrt{{x_n}^2 + {x_{n-1}}^2 + {x_{n-2}}^2}}$$

$$\varphi_{n-1} = 2\operatorname{arccot}\frac{x_{n-1} + \sqrt{x_n^2 + x_{n-1}^2}}{x_n} = \begin{cases} \arccos\frac{x_{n-1}}{\sqrt{{x_n}^2 + {x_{n-1}}^2}} & x_n \geq 0 \\ 2\pi - \arccos\frac{x_{n-1}}{\sqrt{{x_n}^2 + {x_{n-1}}^2}} & x_n < 0 \end{cases}.$$

where if $x_k \neq 0$ for some k but all $x_{k+1}, \ldots, x_n$ are zero, then we can get $\varphi_k = 0$ when $x_k > 0$, and $\varphi_k = \pi$ when $x_k < 0$. Suppose that in the first step of the proof we implicitly assumed that $\mathbf{x}$ is not on the line $t\mathbf{e}_1$. According to the above, if this were the case and $x_1 > 0$ (resp. $x_1 < 0$), we would get $\varphi_1 = 0$ resp. $(\varphi_1 = \pi)$. □

Remark 6.1.2 In many occasions it is more useful to consider a simplified form of the coordinates of the Theorem 6.1.1 (Fig. 6.3). If $\mathbf{x} = x_1\mathbf{e}_1 + \cdots + x_n\mathbf{e}_n \in \mathbb{R}^n$, we write $\mathbf{x} = x_1\mathbf{e}_1 + \mathbf{x}_{n-1}$, where $\mathbf{x}_{n-1} = x_2\mathbf{e}_2 + \cdots + \mathbf{x}_n\mathbf{e}_n$. If $\mathbf{x}$ is not a multiple of $\mathbf{e}_1$, then since $x_1^2 + \|\mathbf{x}_{n-1}\|^2 = r^2 = \|\mathbf{x}\|^2$, there exists $\varphi \in (0, \pi)$, $\varphi = \angle(\mathbf{e}_1, \mathbf{x}_{n-1})$, such that

$$\mathbf{x} = r\cos\varphi\mathbf{e}_1 + r\sin\varphi\mathbf{z}, \quad \mathbf{z} = \frac{\mathbf{x}_{n-1}}{\|\mathbf{x}_{n-1}\|}. \tag{6.1}$$

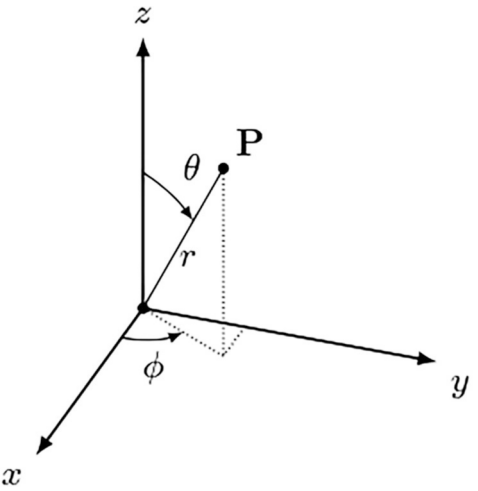


Fig. 6.3 Spherical coordinates in $\mathbb{R}^3$

Here $\mathbf{z}$ is a point of S^{n-2} which we consider as the image of the embedding

$$S^{n-2} \ni (\xi_1, \ldots, \xi_{n-1}) \mapsto (0, \xi_1, \ldots, \xi_{n-1}) \in S^{n-1}.$$

6.1.2 Intersections of k-Planes with S^{n-1}

Let W_k be a k-dimensional plane of $\mathbb{R}^n$, $1 \leq k \leq n-1$, with parametric equation:

$$\mathbf{x} = \mathbf{x}(\lambda_1, \ldots, \lambda_k) = \mathbf{a} + \sum_{i=1}^{k} \lambda_i \mathbf{v}_i, \quad (\lambda_1, \ldots, \lambda_k) \in \mathbb{R}^k, \tag{6.2}$$

where $\mathbf{a} \in \mathbb{R}^n$ and $\mathbf{v}_i \in \mathbb{R}^n$, $i = 1, \ldots, k$, are linearly independent vectors of $\mathbb{R}^n$. Alternatively,

$$W_k = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{x} \cdot \mathbf{A}_i = b_i,\ i = 1, \ldots, n-k\} \tag{6.3}$$

where $\mathbf{A}_1, \ldots, \mathbf{A}_{n-k}$ is a set of $(n-k)$ orthogonal unit vectors of $\mathbb{R}^n$ and $b_i \in \mathbb{R}$, $i = 1, \ldots, n-k$, see also Sect. 5.1.2.

It is clear that the orthogonal group $\mathrm{O}(n) < A(n)$ maps k-planes to k-planes. Indeed, if α is an orthogonal transformation then it is also an affine transformation; hence we have that

$$\alpha(W_k) = \left\{ \mathbf{x} \in \mathbb{R}^n \mid \mathbf{x} = \alpha(\mathbf{a}) + \sum_{i=1}^{k} \lambda_i \alpha(\mathbf{v}_k) \right\}.$$

We are interested in the relative position of a k-plane and S^{n-1}. Notice that if W_k is a k-plane, we can first replace $\mathbf{v}_i$ with an orthonormal set $\mathbf{v}'_i$, $i = 1, \ldots, k$, using the Gram-Schmidt orthonormalisation. Then, after completing this set in a basis

$$\{\mathbf{v}_1, \ldots, \mathbf{v}'_k, \mathbf{v}'_{k+1}, \ldots, \mathbf{v}'_n\}$$

of $\mathbb{R}^n$, we consider the orthogonal transformation α with matrix

$$A = \begin{pmatrix} \mathbf{v}'_1 \\ \vdots \\ \mathbf{v}'_n \end{pmatrix}.$$

Now, α satisfies the relations:

$$\alpha(\mathbf{e}_i) = \mathbf{v}'_i, \ i = 1, \ldots k, \quad \alpha(\mathbf{e}_j) = \mathbf{v}'_j, \ j = k+1, \ldots, n,$$

and in this way we can always normalise a k-plane to the form

$$\mathbf{x} = \mathbf{a} + \sum_{i=1}^{k} \lambda_i \mathbf{e}_i, \quad \lambda_i \in \mathbb{R}, \tag{6.4}$$

where also $\mathbf{a} \in \{\mathbf{e}_1, \ldots, \mathbf{e}_k\}^{\perp}$.

Proposition 6.1.3 *The intersection of the sphere with a k-plane is one of the following.*

1. *A single point.*
2. *The empty set.*
3. *A $(k-1)$-sphere of S^{n-1} with centre in the interior of S^{n-1} and radius < 1.*
4. *In particular, if the k-plane passes through the origin, this sphere has centre at the origin.*

Proof Consider a k-plane which we normalise to the form (6.4). Its intersection points with S^{n-1} are determined by the equation

$$\|\mathbf{x}\| = \left\| \mathbf{a} + \sum_{i=1}^{k} \lambda_i \mathbf{e}_i \right\| = 1.$$

Squaring, we obtain

$$\sum_{i=1}^{k} \lambda_i^2 = 1 - \|\mathbf{a}\|^2.$$

If $\|\mathbf{a}\| > 1$, the intersection is the empty set. If $\|\mathbf{a}\| = 1$, the intersection is the point $\mathbf{a}$. If $\|\mathbf{a}\| < 1$, we have the $(k-1)$-sphere

$$\|\mathbf{x} - \mathbf{a}\|^2 = 1 - \|\mathbf{a}\|^2.$$

In the special case where $\mathbf{a} = \mathbf{0}$, the above equation represents the embedding of S^{k-1} into S^{n-1}:

$$(\xi_1, \ldots, \xi_k) \mapsto (\xi_1, \ldots, \xi_k, \mathbf{0}_{n-k}).$$

□

There are two important conclusions that we deduce from Proposition 6.1.3. The first has to do with the intersection of the sphere S^{n-1} and a straight line (1-plane):

Corollary 6.1.4 *The intersection of S^{n-1} and a straight line is one of the following.*

1. *A single point.*
2. *The empty set.*
3. *Two points.*
4. *In particular, if the straight line passes through the origin, the intersection comprises two points $\pm\mathbf{v}$.*

As for its intersection with 2-planes, we have the following:

Corollary 6.1.5 *The intersection of S^{n-1} and a 2-plane is one of the following.*

1. *A single point.*
2. *The empty set.*
3. *A spherical circle.*
4. *In particular, if the 2-plane passes through the origin, the intersection is a circle centred at the origin.*

Example 6.1.6 The case $n = 3$ is of particular interest: the 2-planes are also hyperplanes. When the intersection is non-empty and different from a single point, we obtain from Corollary 6.1.5 that it is either a spherical circle centred at the origin (such a circle will be called a *great circle*, see next section), or a spherical circle centred inside the sphere S^2. Note that in this case a spherical circle centred at the origin is completely determined by the unit vector $\mathbf{A}$, which is perpendicular to the plane defined by this circle. The unit vectors $\pm\mathbf{A}$ are called *polar vectors* and the points of the sphere they determine are called *poles*. In particular, we consider the plane

$$x_3 = 0 \iff \mathbf{x} \cdot \mathbf{e}_3 = 0.$$

The intersection of this plane with S^2 is the set

$$x_1^2 + x_2^2 = 1, x_3 = 0,$$

that is, the equator of S^2 and $\pm\mathbf{e}_3 = (0, 0, \pm 1)$ are its polar vectors. Now, if we were given only one polar vector, say $\mathbf{e}_3$, then the equation $\mathbf{x} \cdot \mathbf{e}_3 = 0$ completely determines the equator.

In dimensions greater than 3, we do not have the ideal picture of Example 6.1.6, as the discussion that preceded this Section showed.

Example 6.1.7 In $\mathbb{R}^4$ we consider the 2-plane P_2 with parametric equation

$$\mathbf{x}(\lambda, \mu) = (0, 0, 0, -1) + \lambda(1, 1, 0, 0) + \mu(0, 0, 1, 1), \quad \lambda, \mu \in \mathbb{R},$$

and also the 3-plane (hyperplane) P_3 with Cartesian equation $x_1 = 0$. To find the intersection of P_2 with S^3, we first write $\mathbf{x} = (\lambda, \lambda, \mu, \mu - 1)$ and seek λ, μ such that

$$\|\mathbf{x}\|^2 = 2\lambda^2 + 2\mu^2 - 2\mu + 1 = 1 \iff \lambda^2 + \left(\mu - \frac{1}{2}\right)^2 = \frac{1}{4}.$$

We set $\lambda = (1/2)\cos\theta$, $\mu = 1/2 + (1/2)\sin\theta$, so the desired intersection is the curve

$$\mathbf{x}(\theta) = (1/2)(\cos\theta, \cos\theta, 1 + \sin\theta, -1 + \sin\theta),$$

which is also written equivalently as

$$\|\mathbf{x}(\theta) - (0, 0, 1/2, -1/2)\| = 1/2.$$

This equation represents a circle in S^3 centred at $(0, 0, 1/2, -1/2)$ and with radius 1/2.

Now, to find the intersection of P_3 with S^3, we set $x_1 = 0$ in the equation of S^3 and get

$$x_2^2 + x_3^2 + x_4^2 = 1.$$

This equation is a 2-sphere equation of S^3 centered at the origin and of radius 1. Let us note that it also represents the embedding of S^2 in S^3 given by the relation

$$S^2 \ni (\xi_1, \xi_2, \xi_3) \mapsto (0, \xi_1, \xi_2, \xi_3) \in S^3.$$

6.1.3 Great Circles

Definition 6.1.8 A *great circle* is a circle in S^{n-1} centred at the origin (see Corollary 6.1.5). A *minor circle* is a circle in S^{n-1} not centred at the origin.

The great circles are particularly important in Spherical Geometry, since, as we will see in Sect. 6.2 they constitute the geodesics of the spherical metric of S^{n-1}. This is the reason why they are also called *s-lines*.

Remark 6.1.9 We saw in Example 6.1.6 that when $n = 3$ a great circle is completely determined by one of its two poles. Let us observe, however, that if $n > 3$ a great circle does not have only two (opposite) poles. If a 2-plane is defined as the orthocomplement of two orthonormal vectors, then we have $n-2$ polar vectors along with their opposites. However, any choice of $n-2$ such vectors completely determines the great circle as long as do not chose a vector and its opposite as well.

Definition 6.1.10 Every pair of points $\mathbf{x}, -\mathbf{x} \in S^{n-1}$ is called a pair of *antipodal* points. The mapping $A : S^{n-1} \to S^{n-1}$ defined by the relation

$$S^{n-1} \ni \mathbf{x} \mapsto -\mathbf{x} = A(\mathbf{x}) \in S^{n-1}. \tag{6.5}$$

is called the *antipodal* mapping.

It is obvious that the line joining a pair of antipodal points passes through the origin, as it is represented by the parametric equation

$$\mathbf{c}(t) = (1-t)\mathbf{x} + t(-\mathbf{x}) = (1-2t)\mathbf{x}, \quad t \in \mathbb{R}.$$

The converse, that every straight line passing through the origin intersects S^{n-1} at a pair of antipodal points, follows from Corollary 6.1.4 (Fig. 6.4).

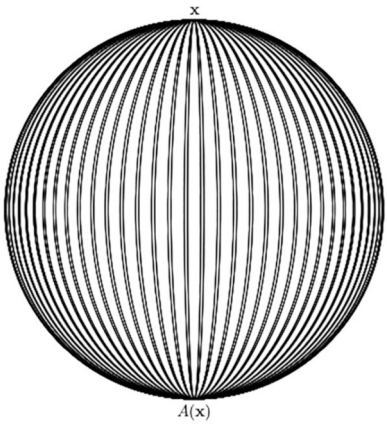


Fig. 6.4 An infinite number of great circles pass through two antipodal points

The great circles of S^{n-1} that pass through two antipodal points are called *meridians* (of the antipodal points). For example, we find all meridians of the antipodal points $\mathbf{e}_1, -\mathbf{e}_1$. The straight line joining these two points is $\mathbf{c}(t) = t\mathbf{e}_1$ and the orthonormal complement $\{\mathbf{e}_1\}^{\perp}$ is generated by the vectors $\mathbf{e}_2, \ldots, \mathbf{e}_n$. Therefore, all unit vectors perpendicular to $\mathbf{e}_1$ are of the form

$$\mathbf{v} = \mathbf{v}(\lambda_2, \ldots, \lambda_n) = \lambda_2\mathbf{e}_2 + \ldots \lambda_n\mathbf{e}_n, \quad \sum_{i=2}^{n} \lambda_i^2 = 1.$$

It turns out that all meridians passing through $\pm\mathbf{e}_1$ are the $(n-1)$-parametric family of great circles with equations

$$\mathbf{x}(\theta; \lambda_2, \ldots, \lambda_n) = \cos\theta\mathbf{e}_1 + \sin\theta\mathbf{v}(\lambda_2, \ldots, \lambda_n).$$

We note that when $n = 3$, setting $\lambda_2 = \cos\phi$ and $\lambda_3 = \sin\phi$, we have

$$\mathbf{x}(\theta; \phi) = \cos\theta\mathbf{e}_1 + \sin\theta\cos\phi\mathbf{e}_2 + \sin\theta\sin\phi\mathbf{e}_3.$$

That is,

$$\mathbf{x}(\theta; \phi) = \alpha_\phi(\mathbf{x}(\theta; \pi/2)),$$

where α_ϕ the orthogonal transformation with matrix

$$A_\phi = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos(\pi/2 - \phi) & -\sin(\pi/2 - \phi) \\ 0 & \sin(\pi/2 - \phi) & \cos(\pi/2 - \phi) \end{pmatrix}.$$

Proposition 6.1.11 *An infinite number of great circles pass through any point on the sphere S^{n-1}.*

Proof Let $\mathbf{v}$ be a given point on S^{n-1} and let $\mathbf{w} \in S^{n-1}$ be any point such that $\mathbf{w} \neq \mathbf{v}$. We distinguish two cases: first, if $\mathbf{v}$, $\mathbf{w}$ are linearly independent, then consider the 2-plane from the origin with parametric equation

$$\mathbf{x}(\lambda, \mu) = \lambda\mathbf{v} + \mu\mathbf{w}, \quad \lambda, \mu \in \mathbb{R}.$$

This plane intersects S^{n-1} at a great circle whose parametric equation we may write as

$$\mathbf{x}_\mathbf{v}(\theta) = \cos\theta\mathbf{v} + \sin\theta\mathbf{w}', \quad \theta \in \mathbb{R}.$$

Here, $\mathbf{w}'$ is a unit vector which belongs to the plane and is also perpendicular to $\mathbf{v}$. The curves $\mathbf{x}_\mathbf{v}$ determine an infinite number of great circles passing through $\mathbf{v}$.

Secondly, if $\mathbf{v}, \mathbf{w}$ are linearly dependent, then necessarily $\mathbf{v} = -\mathbf{w}$. Since there are infinitely many 2-planes from the origin that contain the line joining $\pm\,\mathbf{v}$, it follows that there are infinitely many great circles (meridians) passing through $\mathbf{v}$. □

We can always assume that the point $\mathbf{w} \in S^{n-1}$ as in the proof of Proposition 6.1.11 is perpendicular to the given point $\mathbf{v}$:

Example 6.1.12 The great circles passing through $(1/\sqrt{2})(1, -1, 0) \in S^2$ are all great circles of the form

$$\mathbf{x}(\theta) = (1/\sqrt{2})(\cos\theta, -\cos\theta, 0) + (\sin\theta v_1, \sin\theta v_2, \sin\theta v_3),$$

where $v_1^2 + v_2^2 + v_3^2 = 1$ and $v_1 - v_2 = 0$. These two relations are equivalent to $2v_1^2 + v_3^2 = 1$, therefore (v_1, v_2, v_3) belong to the one-parameter family consisting of the points

$$((1/\sqrt{2})\cos\phi, (1/\sqrt{2})\cos\phi, \sin\phi), \quad \phi \in \mathbb{R},$$

of S^2. Therefore, the equations of all the required great circles are given by

$$\mathbf{x}_\phi(\theta) = \left((1/\sqrt{2})(\cos\theta + \sin\theta\cos\phi),\ (1/\sqrt{2})(-\cos\theta + \sin\theta\cos\phi),\ \sin\phi)\right),$$

where each great circle also corresponds to a different choice of $\phi \in [0, 2\pi)$. Note at this point that for $\theta \in [0, \pi]$ the mapping $\mathbf{x}(\theta, \phi) = \mathbf{x}_\phi(\theta)$ gives a new coordinate system for S^2.

Proposition 6.1.13 *There exists a great circle that passes through any two given distinct points of S^{n-1}. This great circle is unique if the points are not antipodal. In the latter case, there is a family of great circles that passes through these points.*

Proof If $\mathbf{v}_1$ and $\mathbf{v}_2$ are points on the sphere, then we distinguish two cases:

- $\mathbf{v}_1, \mathbf{v}_2$ *are not antipodal.* Then, the desired great circle is

$$\mathbf{x}(\theta) = \cos\theta\mathbf{v}_1 + \sin\theta\mathbf{v}_2', \quad \theta \in \mathbb{R},$$

  with $\mathbf{v}_2'$ being a unit vector in the 2-plane from the origin defined by $\mathbf{v}_1$ and $\mathbf{v}_2$, orthogonal to $\mathbf{v}_1$. This great circle is the intersection of S^{n-1} with the 2-plane from the origin generated by $\mathbf{v}_1, \mathbf{v}_2$. This intersection is uniquely defined.
- $\mathbf{v}_1, \mathbf{v}_2$ *are antipodal.* Then, let $\mathbf{v}_2 = -\mathbf{v}_1$ and consider the line passing through $\mathbf{v} = \mathbf{v}_1$ and $-\,\mathbf{v}$:

$$c(t) = (1-t)\mathbf{v} + t(-\mathbf{v}) = (1-2t)\mathbf{v}, \quad t \in \mathbb{R}.$$

Then, there are infinitely many 2-dimensional planes passing through $\mathbf{v}$ and $-\mathbf{v}$: every 2-plane of the form

$$\mathbf{x} = \lambda\mathbf{v} + \mu\mathbf{w}, \quad \lambda, \mu \in \mathbb{R},$$

such that $\mathbf{w}$, $\mathbf{v}$ are linearly independent contains $\mathbf{v}$, $-\mathbf{v}$, and therefore the same is true for its intersection (the meridian of $\pm\mathbf{v}$) with S^{n-1}.

□

Remark 6.1.14 The proof of Proposition 6.1.13 in the case $n = 3$ is simpler. If $\mathbf{v}_1 \neq \pm\mathbf{v}_2$, then the unit vector

$$\mathbf{n} = \frac{\mathbf{v}_1 \times \mathbf{v}_2}{\|\mathbf{v}_1 \times \mathbf{v}_2\|},$$

is perpendicular to $\mathbf{v}_1$, $\mathbf{v}_2$ and uniquely defines a plane that passes through the origin and its intersection with S_2 contains $\mathbf{v}_1$, $\mathbf{v}_2$. The vector $\mathbf{n}$ is the polar vector of the great circle corresponding to $\mathbf{v}_1$, $\mathbf{v}_2$, and thus, in the case of linearly independent points, a great circle is uniquely defined by its polar vector. If $\mathbf{v}_1 = -\mathbf{v}_2 = \mathbf{v}$, the intersection of every plane that passes through the origin and contains the line $\mathbf{c}(t) = t\mathbf{v}$ is a great circle that contains $\mathbf{v}$ and $-\mathbf{v}$. The polar vectors of these great circles belong to the orthocomplement of $\mathbf{v}$.

Example 6.1.15 To find the great circle that passes through the points

$$\mathbf{v}_1 = (1/\sqrt{3})(1, -1, 1) \text{ and } \mathbf{v}_2 = (1/\sqrt{3})(1, 1, 1)$$

of S^2, we take their exterior product:

$$\begin{aligned}\mathbf{n} = \mathbf{v}_1 \times \mathbf{v}_2 &= (1/3)\begin{vmatrix}\mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3\\ 1 & -1 & 1\\ 1 & 1 & 1\end{vmatrix}\\ &= (1/3)(-2\mathbf{e}_1 + 2\mathbf{e}_3).\end{aligned}$$

So, the desired great circle is the intersection of the plane $x_1 - x_3 = 0$ with S^2. Thus the equations

$$2x_1^2 + x_2^2 = 1, \quad x_1 = x_3$$

from which we may obtain the parametric equation

$$\mathbf{x}(\theta) = \left(\frac{1}{\sqrt{2}}\cos\theta,\ \sin\theta,\ \frac{1}{\sqrt{2}}\cos\theta\right), \quad \theta \in \mathbb{R}.$$

Example 6.1.16 The great circle of S^3 that passes through $\mathbf{v}_1 = (1/\sqrt{5})(1, 2, 0, 0)$ and $\mathbf{v}_2 = (1/\sqrt{5})(0, 0, 1, 2)$, is given by

$$\mathbf{x}(\theta) = \cos\theta\mathbf{v}_1 + \sin\theta\mathbf{v}_2 = (1/\sqrt{5})(\cos\theta, 2\cos\theta, \sin\theta, 2\sin\theta), \quad \theta \in \mathbb{R}.$$

We observe that since $\mathbf{v}_1 \cdot \mathbf{v}_2 = 0$, we can take $\theta \in [0, 2\pi)$ and in this case we also have $\mathbf{x}(0) = \mathbf{v}_1$, $\mathbf{x}(\pi/2) = \mathbf{v}_2$.

Example 6.1.17 To find all the great circles of S^3 that pass through the antipodal points $\mathbf{v} = (1/2)(1, 1, 1, 1)$ and $A(\mathbf{v}) = -(1/2)(1, 1, 1, 1)$, we first consider the line that joins them:

$$\mathbf{c}(t) = t\mathbf{v} = (t/2)(1, 1, 1, 1), \quad t \in \mathbb{R}.$$

As in the proof of Proposition 6.1.13, let $\mathbf{w} \in S^{n-1}$, which we can assume without loss of generality to be orthogonal to $\mathbf{v}$. The 2-planes of the form

$$\mathbf{x}(\lambda, \mu) = \lambda\mathbf{v} + \mu\mathbf{w}, \quad \lambda, \mu \in \mathbb{R},$$

contain $\mathbf{v}$, $A(\mathbf{v})$, $\mathbf{w}$ and their intersection with S^{n-1} is a great circle:

$$\mathbf{x}_{\mathbf{w}}(\theta) = \cos\theta\mathbf{v} + \sin\theta\mathbf{w},$$

with $\mathbf{w}$ belonging to the orthocomplement of $\mathbf{v}$, that is, the 3-plane of $\mathbb{R}^4$ with equation

$$x_1 + x_2 + x_3 + x_4 = 0.$$

This plane is generated by the unit vectors

$$\mathbf{v}_1 = (1/\sqrt{2})(1, -1, 0, 0), \quad \mathbf{v}_2 = (1/\sqrt{2})(0, 0, 1, -1),$$
$$\mathbf{v}_3 = (1/2)(-1, 1, -1, 1).$$

(Notice that they are all perpendicular to each other.) Therefore, the vectors belonging to the unit sphere of this plane are of the form

$$\mathbf{w}(\varphi_1, \varphi_2) = \cos\varphi_1\mathbf{v}_1 + \sin\varphi_1\cos\varphi_2\mathbf{v}_2 + \sin\varphi_1\sin\varphi_2\mathbf{v}_3,$$

where $(\varphi_1, \varphi_2) \in [0, \pi] \times [0, 2\pi]$. It follows that the required family of great circles is given by the equation

$$\mathbf{x}_{\varphi_1,\varphi_2}(\theta) = \cos\theta\mathbf{v} + \sin\theta\mathbf{w}(\varphi_1, \varphi_2).$$

The next proposition concerns the relative position of two great circles (Fig. 6.5).

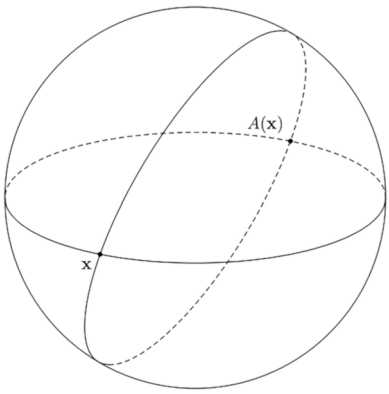


Fig. 6.5 Two great circles of S^2 intersect at antipodal points. In higher dimensions the intersection may be empty

Proposition 6.1.18 (Incidence) *Any two distinct great circles of S^2 intersect at exactly two antipodal points. For $n > 3$ this is not true in general: two great circles S^{n-1}, $n > 3$ may not have any intersection points. But if they do intersect, then they intersect at exactly two antipodal points.*

Proof The proof follows directly from the fact that any two different great circles lie in 2-planes from the origin. For $n = 3$, the intersection of two such planes is a straight line from the origin that intersects S^2 at two antipodal points. For $n = 3$, two 2-planes may not intersect. But if they do intersect, then their intersection is again a straight line from the origin that intersects S^{n-1} at two antipodal points. □

Remark 6.1.19 In the case of S^2, there is a neat way to find the intersection of two great circles. Let two great circles C_1, C_2 be in S^2 with polar vectors $\mathbf{n}_1$ and $\mathbf{n}_2$, respectively. The unit vector

$$\mathbf{n} = \frac{\mathbf{n}_1 \times \mathbf{n}_2}{\|\mathbf{n}_1 \times \mathbf{n}_2\|},$$

determines a set of points $L = \{t\mathbf{n}, \mid t \in \mathbb{R}\}$ that belongs to the intersection of the planes in which C_1, C_2 lie. On the other hand, every point that belongs to this intersection will necessarily be a point of the set L. Therefore, the intersection is the set L, that is, a line that intersects S^2 at the antipodal points $\pm\,\mathbf{n}$.

Corollary 6.1.20 *From a point outside a given great circle of S^{n-1} there pass no great circle which does not intersect the given great circle if $n = 3$; if $n > 3$ there is an infinite number of great circles that do not intersect the given great circle.*

Example 6.1.21 The great circles of S^3 with parametric equations

$$\mathbf{x}_1(\theta) = (\cos\theta, \sin\theta, 0, 0), \quad \mathbf{x}_2(\theta) = (0, 0, \cos\theta, \sin\theta),$$

have no common point. A famous result of Hopf tells us that we can locally represent S^3 as the union of S^2 and a great circle. More explicitly, S^3 is composed of fibers,

where each fiber is homeomorphic to S^1, one for each point of the sphere S^2. The interested reader should refer to [49], Section 20, for further details.

Example 6.1.22 Let in S^2 be the great circle C_1 that passes through

$$\mathbf{v}_1 = (1/\sqrt{2})(1, 1, 0) \quad \text{and} \quad \mathbf{v}_2 = (1/\sqrt{2})(1, 0, -1)$$

and let also C_2 be the great circle that passes through

$$\mathbf{v}_1' = (1, 0, 0) \quad \text{and} \quad \mathbf{v}_2' = (0, -1, 0).$$

We consider the exterior products

$$\mathbf{n} = \mathbf{v}_1 \times \mathbf{v}_2 = (1/2)(-1, 1, -1), \quad \mathbf{n}' = \mathbf{v}_1' \times \mathbf{v}_2' = (-1, 0, 0).$$

The points

$$\pm \frac{\mathbf{n} \times \mathbf{n}'}{\|\mathbf{n} \times \mathbf{n}'\|} = \pm(0, 0, 1),$$

are the intersection points of C_1 and C_2.

We close this section with a result concerning the action of the orthogonal group on great circles.

Proposition 6.1.23 *The orthogonal group* $\mathrm{O}(n)$ *acts transitively on the set of great circles of* S^{n-1}.

Proof It suffices to prove how every great circle can be mapped by an orthogonal transformation to the great circle

$$\mathbf{x}_e(\theta) = \cos\theta \mathbf{e}_1 + \sin\theta \mathbf{e}_2, \quad \theta \in \mathbb{R}. \tag{6.6}$$

Consider a great circle of the form

$$\mathbf{x}(\theta) = \cos\theta \mathbf{v}_1 + \sin\theta \mathbf{v}_2, \quad \theta \in \mathbb{R},$$

with $\|\mathbf{v}_i\| = 1$, $\mathbf{v}_1 \cdot \mathbf{v}_2 = 0$. There is an orthogonal transformation α that maps $\mathbf{v}_1$ to $\mathbf{e}_1$ and, furthermore, $\alpha(\mathbf{v}_2) \perp \mathbf{e}_1$. Since $\|\alpha(\mathbf{v}_2)\| = 1$, $\alpha(\mathbf{v}_2)$ will belong to the unit sphere of the hyperplane $x_1 = 0$, that is,

$$\alpha(\mathbf{v}_2) = (0, x_2, \ldots, x_n), \quad \sum_{i=2}^{n} x_i^2 = 1.$$

We now consider the orthogonal transformation that stabilises $\mathbf{e}_1$ and maps $\alpha(\mathbf{v}_2)$ to $\mathbf{e}_2$: it associated matrix is

$$\begin{pmatrix} 1 & 0 \\ 0 & B \end{pmatrix}, \quad B \in \mathrm{O}(n-1),$$

and B corresponds to the orthogonal transformation β that satisfies $\beta(x_2, \ldots, x_n) = (1, 0, \ldots, 0)$ on the unit sphere of $\mathbb{R}^{n-1}$; we have identified the latter with the hyperplane $x_1 = 0$. □

Remark 6.1.24 The normalisation of any great circle given by Eq. (6.6) is very useful. In the case $n = 3$, if C is a great circle with polar vector $\mathbf{n}$, there is an element of O(3) that maps it to $\mathbf{e}_3$; again, any great circle is normalised by the equator $x_3 = 0$ or $\mathbf{x}(\theta) = (\cos\theta, \sin\theta, 0)$.

6.2 Spherical Distance

The metric in Spherical Geometry follows from the spherical distance that we define immediately (Fig. 6.6).

Definition 6.2.1 The *spherical distance* $s_{n-1}(\mathbf{x}, \mathbf{y})$ between two points $\mathbf{x}, \mathbf{y} \in S^{n-1}$ is the angle of $\mathbf{x}, \mathbf{y}$:

$$s_{n-1}(\mathbf{x}, \mathbf{y}) = \angle(\mathbf{x}, \mathbf{y}) = \arccos(\mathbf{x} \cdot \mathbf{y}).$$

We note that the spherical distance $s_{n-1} : S^{n-1} \times S^{n-1} \to [0, \pi]$ is a bounded function: the maximum value π is obtained at antipodal points. We also observe that $\cos(s_{n-1}(\mathbf{x}, \mathbf{y})) = \mathbf{x} \cdot \mathbf{y}$ and that if $c_n(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$ is the chordal distance of $\mathbf{x}$

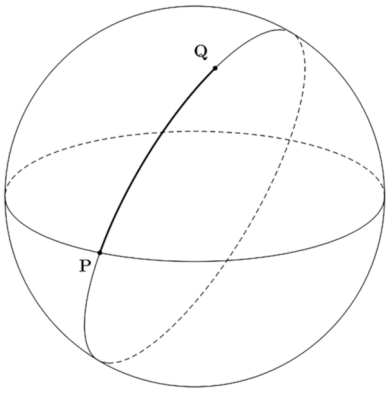


Fig. 6.6 The spherical distance between two points **P** and **Q** on the sphere is equal to the length of the shortest arc of the great circle that joins them

and $\mathbf{y}$, then

$$c_n^2(\mathbf{x},\mathbf{y}) = 2(1-(\mathbf{x}\cdot\mathbf{y})) = 2(1-\cos(s_{n-1}(\mathbf{x},\mathbf{y})) = 4\sin^2\left(\frac{s_{n-1}(\mathbf{x},\mathbf{y})}{2}\right).$$

We thus have the following relation between chordal and spherical distance:

$$c_n(\mathbf{x},\mathbf{y}) = 2\sin\left(\frac{s_{n-1}(\mathbf{x},\mathbf{y})}{2}\right), \quad \mathbf{x},\mathbf{y}\in S^{n-1}. \tag{6.7}$$

We will see in Proposition 6.2.6 that something much more is true: the spherical metric is the intrinsic metric that arises from the chordal metric. With that given, it immediately follows that the spherical distance defines a metric. However, for the sake of completeness, we give here an independent proof of this fact.

Theorem 6.2.2 *The distance s_{n-1} defines a metric on S^{n-1}: if $\mathbf{x},\mathbf{y},\mathbf{z}\in S^2$, then the following holds.*

1. $s_{n-1}(\mathbf{x},\mathbf{y})\geq 0$. *Also* $s_{n-1}(\mathbf{x},\mathbf{y}) = 0$ *if and only if* $\mathbf{x}=\mathbf{y}$.
2. $s_{n-1}(\mathbf{x},\mathbf{y}) = s_{n-1}(\mathbf{y},\mathbf{x})$.
3. *(Triangle Inequality.)*

$$s_{n-1}(\mathbf{x},\mathbf{z}) \leq s_{n-1}(\mathbf{x},\mathbf{y}) + s_{n-1}(\mathbf{y},\mathbf{z}).$$

The equality holds if and only if $\mathbf{x},\mathbf{y},\mathbf{z}$ belong to the same great circle.

Proof The proof of the first two statements is straightforward and is left to the reader as an exercise. For the triangle inequality, since the action of $\mathrm{O}(n)$ preserves spherical distances we can normalise so that

$$\mathbf{x} = \cos\theta\mathbf{e}_1 + \sin\theta\mathbf{e}_2, \quad \mathbf{y} = \mathbf{e}_1, \quad \mathbf{z} = \cos\varphi\mathbf{e}_1 + \sin\phi\mathbf{w},$$

where $\mathbf{w} = (0, w_2, \ldots, w_n)$, and $\theta,\varphi\in[0,\pi]$. We write

$$\theta_{\mathbf{xy}} = \angle(\mathbf{x},\mathbf{y}), \quad \theta_{\mathbf{xz}} = \angle(\mathbf{x},\mathbf{z}), \quad \theta_{\mathbf{yz}} = \angle(\mathbf{y},\mathbf{z}),$$

and we calculate

$$\begin{aligned}
\mathbf{x}\cdot\mathbf{y} &= \cos\theta_{\mathbf{xy}} = \cos\varphi,\\
\mathbf{x}\cdot\mathbf{z} &= \cos\theta_{\mathbf{xz}} = \cos\theta\cos\varphi + w_2\sin\theta\sin\varphi,\\
\mathbf{y}\cdot\mathbf{z} &= \cos\theta_{\mathbf{yz}} = \cos\theta.
\end{aligned}$$

Since the cosine function is strictly decreasing in $[0,\pi]$, it suffices to prove the relation

$$\cos\theta_{\mathbf{xz}} \geq \cos(\theta_{\mathbf{xy}} + \theta_{\mathbf{yz}}).$$

We have that

$$\cos\theta_{\mathbf{xy}}\cos\theta_{\mathbf{yz}} = \cos\theta_{\mathbf{xz}} - w_2\sin\theta_{\mathbf{xy}}\sin\theta_{\mathbf{yz}},$$

and since $|w_2| \leq 1$ we deduce

$$\cos\theta_{\mathbf{xz}} - \sin\theta_{\mathbf{xy}}\sin\theta_{\mathbf{yz}} \leq \cos\theta_{\mathbf{xy}}\cos\theta_{\mathbf{xz}} \leq \cos\theta_{\mathbf{xz}} + \sin\theta_{\mathbf{xy}}\sin\theta_{\mathbf{yz}}.$$

The Triangle Inequality follows from the right-hand side of the above inequality. Equality holds if and only if $w_2 = -1$, that is, equivalently when $\mathbf{z} = \cos\varphi\mathbf{e}_1 - \sin\varphi\mathbf{e}_2$. □

Example 6.2.3 The Triangle Inequality also gives us a way to examine when three points are s-collinear, that is, they belong to the same great circle. Take for example the points

$$\mathbf{x} = (1, 0, 0), \quad \mathbf{y} = (1/2)(\sqrt{3}, 1, 0), \quad \mathbf{z} = (0, 1, 0),$$

of S^2. Then

$$\begin{aligned}
\cos\theta_{\mathbf{xy}} = \sqrt{3}/2 &\implies \theta_{\mathbf{xy}} = \pi/6,\\
\cos\theta_{\mathbf{yz}} = 1/2 &\implies \theta_{\mathbf{yz}} = \pi/3,\\
\cos\theta_{\mathbf{xz}} = 0 &\implies \theta_{\mathbf{xy}} = \pi/2.
\end{aligned}$$

We observe that

$$\pi/2 = \theta_{\mathbf{xz}} = \theta_{\mathbf{xy}} + \theta_{\mathbf{yz}} = \pi/6 + \pi/3,$$

from which we conclude that $\mathbf{x}$, $\mathbf{y}$, $\mathbf{z}$ belong to the same great circle and furthermore, $\mathbf{y}$ is between $\mathbf{x}$ and $\mathbf{z}$. The general criterion is that one of the following relations should hold:

$$\begin{aligned}
\theta_{\mathbf{xz}} &= \theta_{\mathbf{xy}} + \theta_{\mathbf{yz}},\\
\theta_{\mathbf{xy}} &= \theta_{\mathbf{xz}} + \theta_{\mathbf{zy}},\\
\theta_{\mathbf{yz}} &= \theta_{\mathbf{yx}} + \theta_{\mathbf{xz}}.
\end{aligned}$$

Example 6.2.4 To find the points equidistant from the two different points $\mathbf{x}_1, \mathbf{x}_2$ of S^{n-1} joined by a great circle, we look for solutions $\mathbf{x}_m$ of the equation

$$s_{n-1}(\mathbf{x}_1, \mathbf{x}_m) = s_{n-1}(\mathbf{x}_2, \mathbf{x}_m),$$

that lie on the s-segment from $\mathbf{x}_1$ to $\mathbf{x}_2$. According to the previous example, we will have

$$\theta_{\mathbf{x}_1\mathbf{x}_2} = \theta_{\mathbf{x}_1\mathbf{x}_m} + \theta_{\mathbf{x}_m\mathbf{x}_2} = 2\theta_{\mathbf{x}_1\mathbf{x}_m}.$$

If we set

$$x_m = \frac{\mathbf{x}_1 + \mathbf{x}_2}{\|\mathbf{x}_1 + \mathbf{x}_2\|}, \tag{6.8}$$

we have:

$$\begin{aligned}\cos(s_{n-1}(\mathbf{x}_1, \mathbf{x}_m)) &= \frac{\mathbf{x}_1 \cdot (\mathbf{x}_1 + \mathbf{x}_2)}{\|\mathbf{x}_1 + \mathbf{x}_2\|} \\ &= \frac{1 + \mathbf{x}_1 \cdot \mathbf{x}_2}{\|\mathbf{x}_1 + \mathbf{x}_2\|} \\ &= \frac{\mathbf{x}_2 \cdot (\mathbf{x}_2 + \mathbf{x}_1)}{\|\mathbf{x}_1 + \mathbf{x}_2\|} \\ &= \cos(s_{n-1}(\mathbf{x}_2, \mathbf{x}_m)).\end{aligned}$$

Also,

$$\begin{aligned}\cos(s_{n-1}(\mathbf{x}_1, \mathbf{x}_m)) &= \frac{1 + \mathbf{x}_1 \cdot \mathbf{x}_2}{\sqrt{2 + 2\mathbf{x}_1 \cdot \mathbf{x}_2}} \\ &= \sqrt{\frac{1 + \mathbf{x}_1 \cdot \mathbf{x}_2}{2}}.\end{aligned}$$

So,

$$2\cos^2(s_{n-1}(\mathbf{x}_1, \mathbf{x}_m)) - 1 = \mathbf{x}_1 \cdot \mathbf{x}_2,$$

which is a relation equivalent to

$$\cos(2\theta_{\mathbf{x}_1\mathbf{x}_m}) = \cos(\theta_{\mathbf{x}_1\mathbf{x}_2}).$$

The median is therefore given by the relation (6.8).

We now prove the following important:

Proposition 6.2.5 *The isometry group* $\mathrm{Isom}(S^{n-1},\ s_{n-1})$ *of the spherical metric is the orthogonal group* $\mathrm{O}(n)$.

Proof First, from Eq. (6.7) it follows that

$$\mathrm{Isom}(S^{n-1},\ s_{n-1}) = \mathrm{Isom}(S^{n-1},\ c_n)$$

and furthermore we know that $\mathrm{O}(n) \subset \mathrm{Isom}(S^{n-1}, c_n)$. Therefore, it remains to prove that an isometry of S^{n-1} with respect to the chordal metric extends to an orthogonal transformation throughout $\mathbb{R}^n$. Let $f : S^{n-1} \to S^{n-1}$ be such an isometry and we set

$$\tilde{f}(\mathbf{0}) = \mathbf{0}, \quad \tilde{f}(\mathbf{x}) = \|\mathbf{x}\| \cdot f\left(\frac{\mathbf{x}}{\|\mathbf{x}\|}\right), \ \mathbf{x} \neq \mathbf{0}.$$

For $\mathbf{u}, \mathbf{v} \in S^{n-1}$ we have:

$$\begin{aligned} 2\, f(\mathbf{u}) \cdot f(\mathbf{v}) &= \|f(\mathbf{u})\|^2 + \|f(\mathbf{v})\|^2 - \|f(\mathbf{u}) - f(\mathbf{v})\|^2 \\ &= 1 + 1 - \|\mathbf{u} - \mathbf{v}\|^2 \\ &= 2\, \mathbf{u} \cdot \mathbf{v}. \end{aligned}$$

Now we take $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and we calculate:

$$\begin{aligned} \|\tilde{f}(\mathbf{x}) - \tilde{f}(\mathbf{y})\|^2 &= \|\tilde{f}(\mathbf{x})\|^2 + \|\tilde{f}(\mathbf{y})\|^2 - 2\, \tilde{f}(\mathbf{x}) \cdot \tilde{f}(\mathbf{y}) \\ &= \|\mathbf{x}\|^2 \cdot \|f(\mathbf{x}/\|\mathbf{x}\|)\|^2 + \|\mathbf{y}\|^2 \cdot \|f(\mathbf{y}/\|\mathbf{y}\|)\|^2 \\ &\quad -2\, \|\mathbf{x}\| \cdot \|\mathbf{y}\| f(\mathbf{x}/\|\mathbf{x}\|) \cdot f(\mathbf{y}/\|\mathbf{y}\|) \\ &= \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 - 2\, \mathbf{x} \cdot \mathbf{y} \\ &= \|\mathbf{x} - \mathbf{y}\|^2. \end{aligned}$$

Consequently, $\tilde{f}$ is a Euclidean isometry that preserves the origin, so it is an orthogonal transformation; the same holds for its restriction to the sphere which is f. □

6.2.1 Geodesics

We have already seen that the chordal metric in S^1 is not intrinsic. The same is true in the general case, but on the other hand, the intrinsic metric that arises in the metric space (S^{n-1}, c_n) is s_{n-1}. We prove this in Proposition 6.2.6, which also tells us that the geodesics of the sphere are the great circles: to go from one point on the sphere to another along the shortest path, we follow the smallest arc of the great circle that connects those points. But before proceeding with the proof, we will define the *spherical length* of a spherical curve. For this, let $\boldsymbol{\gamma} = \boldsymbol{\gamma}(t)$, $t \in [0, 1]$ be a differentiable curve of S^{n-1}:

$$\boldsymbol{\gamma}(t) = (x_1(t), \ldots, x_n(t)),$$

where x_i are all differentiable functions in $[0, 1]$ and $\sum_{i=1}^{n} x_i^2(t) = 1$. Then, using Theorem 6.1.1 we find differentiable functions

$$\varphi_i = \varphi_i(t), \quad i = 1, \ldots, n-1,\ t \in [0, 1]$$

such that

$$\begin{aligned}
x_1(t) &= \cos(\varphi_1(t)) \\
x_2(t) &= \sin(\varphi_1(t)) \cos(\varphi_2(t)) \\
x_3(t) &= \sin(\varphi_1(t)) \sin(\varphi_2(t)) \cos(\varphi_3(t)) \\
&\vdots \\
x_{n-1}(t) &= \sin(\varphi_1(t)) \cdots \sin(\varphi_{n-2}(t)) \cos(\varphi_{n-1}(t)) \\
x_n(t) &= \sin(\varphi_1(t)) \cdots \sin(\varphi_{n-2}(t)) \sin(\varphi_{n-1}(t)).
\end{aligned}$$

Calculating straightforwardly we find:

$$\sum_{i=1}^{n} \dot{x}_i(t) = \dot{\varphi}_1^2(t) + \sin^2 \varphi_1(t) \dot{\varphi}_2^2(t) + \cdots + \sin^2 \varphi_1(t) \ldots \sin^2 \varphi_{n-2}(t) \dot{\varphi}_{n-1}^2(t).$$

The square root of the quantity on the right hand-side is defined as the *spherical velocity* $\|\dot{\boldsymbol{\gamma}}(t)\|_s$. The *spherical length* $\ell_s(\boldsymbol{\gamma})$ of $\boldsymbol{\gamma}$ is then:

$$\ell_s(\boldsymbol{\gamma}) = \int_0^1 \|\dot{\boldsymbol{\gamma}}(t)\|_s \, dt.$$

Proposition 6.2.6 *If* $\mathbf{x}, \mathbf{y} \in S^{n-1}$*, then the smallest of arc of the great circle passing through* $\mathbf{x}, \mathbf{y}$ *is the shortest path connecting them.*

Proof Due to the transitivity of the action of $\mathrm{O}(n)$ as well as the invariance of the curve length by this action, we may normalise so that both points lie on the great circle passing through $\mathbf{e}_1, \mathbf{e}_2$ and in particular, we can assume that

$$\mathbf{x} = \mathbf{e}_1, \quad \mathbf{y} = (\cos\theta, \sin\theta, 0, \ldots, 0),$$

$\theta \in (0, \pi)$. In this way

$$s_{n-1}(\mathbf{x}, \mathbf{y}) = \theta.$$

Let now $\boldsymbol{\gamma} = \boldsymbol{\gamma}(t), t \in [0, 1]$ of S^{n-1} be a (differentiable) spherical curve $\boldsymbol{\gamma} = \boldsymbol{\gamma}(t)$, $t \in [0, 1]$ of S^{n-1} connecting $\mathbf{x}, \mathbf{y}$. According to the Comment 6.1.2 we can write:

$$\boldsymbol{\gamma}(t) = \cos(\varphi(t))\mathbf{e}_1 + \sin(\varphi(t))\mathbf{z}(t), \ t \in [0, 1].$$

Here, $\varphi : [0, 1] \to [0, \pi]$ is a differentiable function with $\varphi(0) = 0$, $\varphi(1) = \theta$ and $\mathbf{z}(t)$ is a differentiable curve of $\{0\} \times S^{n-2}$ embedded in S^{n-1} with $\mathbf{z}(0) = \mathbf{z}(1) = \mathbf{0}_{n-2}$. We have

$$\dot{\boldsymbol{\gamma}}(t) = -\sin(\varphi(t))\dot{\varphi}(t)\mathbf{e}_1 + \cos(\varphi(t))\dot{\varphi}(t)\mathbf{z}(t) + \sin(\varphi(t))\dot{\mathbf{z}}(t)$$

and so

$$\|\dot{\boldsymbol{\gamma}}(t)\|_s = \sqrt{(\dot{\varphi}(t))^2 + \sin^2(\varphi(t))\|\dot{\mathbf{z}}(t)\|^2}.$$

Now,

$$\begin{aligned}
\ell_s(\boldsymbol{\gamma}) &= \int_0^1 \|\dot{\boldsymbol{\gamma}}(t)\|_s \, dt \\
&= \int_0^1 \sqrt{(\dot{\varphi}(t))^2 + \sin^2(\varphi(t))\|\dot{\mathbf{z}}(t)\|^2} \, dt \\
&\geq \int_0^1 |\dot{\varphi}(t)| \, dt \\
&\geq \left| \int_0^1 \dot{\varphi}(t) \, dt \right| \\
&= |\varphi(1) - \varphi(0)| = \theta = s_{n-1}(\mathbf{x}, \mathbf{y}).
\end{aligned}$$

The equality holds on the s-segment of the greatest circle

$$(\cos\varphi, \sin\varphi, 0, \ldots, 0), \qquad \varphi \in [0, \pi].$$

Taking the infimum over the lengths of all the curves connecting $\mathbf{x}$, $\mathbf{y}$, we obtain the result. □

The above discussion justifies why we call the great circles s-lines: as geodesics they play the role that straight lines have in Euclidean Geometry. So the results that we proved in Sect. 6.1.3 can be reformulated as follows:

(i) There exists an infinite number of s-lines passing from any point of S^{n-1} (Proposition 6.1.11).
(ii) There exists an s-line passing through any two distinct points of S^{n-1}; it is unique if the points are not antipodal. If the points are antipodal, there is a family of s-lines that pass through these points (Proposition 6.1.13).
(iii) Two distinct s-lines always intersect at exactly two antipodal points if $n = 3$. For $n > 3$ the intersection of two lines can be either the empty set or two antipodal points (Proposition 6.1.18).

(iv) From a point outside an s-line of S^2 it is not possible to draw a line parallel to the given s-line. For $n > 3$, an infinite number of s-lines can be drawn, incompatible with the given s-line (Proposition 6.1.20).

6.3 Spherical Trigonometry

Consider three points in S^{n-1} and assume that they do not belong to the same great circle. From now on, we can also assume that three such points belong to the sphere S^2, due to the following proposition.

Proposition 6.3.1 *Any three points in S^{n-1} that do not lie on the same great circle belong to a 2-sphere.*

Proof Three points that do not belong to the same great circle define a 2-plane in $\mathbb{R}^n$ that does not pass through the origin. The intersection of this plane with S^{n-1} is a spherical circle of S^{n-1} of the form

$$\mathbf{x}(\theta) = \mathbf{a} + (r\cos\theta)\mathbf{v}_1 + (r\sin\theta)\mathbf{v}_2, \quad \theta \in \mathbb{R},$$

where $\mathbf{a} \neq 0$, $\mathbf{a} \cdot \mathbf{v}_i = 0$, $\|\mathbf{v}_1\| = \|\mathbf{v}_2\| = 1$, $\mathbf{v}_1 \cdot \mathbf{v}_2 = 0$ and $r = \sqrt{1 - \|\mathbf{a}\|^2} < 1$ (see Corollary 6.1.5). This circle belongs to the 2-sphere with centre $\mathbf{a}$ and radius r with parametric equation

$$\mathbf{x}(\theta, \phi) = \mathbf{a} + (r\cos\theta\cos\phi)\mathbf{v}_1 + (r\cos\theta\sin\phi)\mathbf{v} + (r\sin\theta)\mathbf{v}_2, \quad \theta, \phi \in \mathbb{R},$$

where $\mathbf{v}$ is a unit vector in the orthocomplement of $\{\mathbf{v}_1, \mathbf{v}_2\}$. □

In what follows, and until the end of the chapter, we will restrict ourselves to the sphere S^2.

6.3.1 Angles of Great Circles

Definition 6.3.2 The angle of two great circles C_1 and C_2 with polar vectors $\mathbf{n}_1$ and $\mathbf{n}_2$, respectively, is the angle $\angle(\mathbf{n}_1, \mathbf{n}_2)$. In other words, the angle of two great circles is the solid angle of the planes containing them:

$$\angle(C_1, C_2) = \arccos(\mathbf{n}_1 \cdot \mathbf{n}_2).$$

C_1 and C_2 will be called *orthogonal or perpendicular* if $\angle(C_1, C_2) = \pi/2$ (Fig. 6.7).

Proposition 6.3.3 *The following hold:*

(i) There is a unique great circle orthogonal to two given distinct great circles.

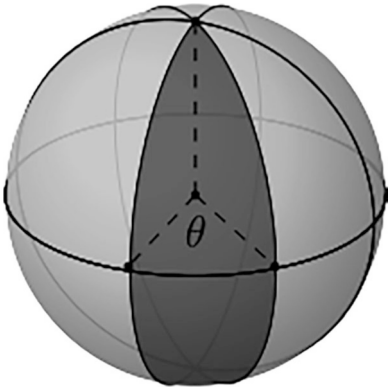


Fig. 6.7 The angle of two great circles and their meniscus

(ii) Let C be a great circle and a point lying in $S^2 \setminus C$ and is not a pole of C. Then there exists a unique great circle that passes through this point and is also orthogonal to C.

Proof

(i) Let C_1, C_2 be different great circles with polar vectors $\mathbf{n}_1$ and $\mathbf{n}_2$, respectively, which are not parallel. Then the unit vector

$$\mathbf{n} = \frac{\mathbf{n}_1 \times \mathbf{n}_2}{\|\mathbf{n}_1 \times \mathbf{n}_2\|}$$

is perpendicular to both $\mathbf{n}_1, \mathbf{n}_2$. Therefore, the great circle C with polar vector $\mathbf{n}$ is orthogonal to both C_1, C_2. It is unique, because the only other unit vector perpendicular to $\mathbf{n}_1, \mathbf{n}_2$ is $-\mathbf{n}$ which is also a polar vector of C.

(ii) Let $\mathbf{v} \in S^2$ and $\mathbf{n}$ be the polar vector of the great circle C with $\mathbf{v} \notin C$. Since by our assumptions $\mathbf{v}$ is not a pole of C, the vectors $\mathbf{v}$ and $\mathbf{n}$ are linearly independent. We consider the unit vector

$$\mathbf{n}' = \frac{\mathbf{v} \times \mathbf{n}}{\|\mathbf{v} \times \mathbf{n}\|} \perp \mathbf{n}.$$

Since $\mathbf{n}' \perp \mathbf{v}$, we get $\mathbf{v} \in C'$, where $C' \perp C$ is the great circle with polar vector $\mathbf{n}'$. The uniqueness follows as in i).

□

We close this section with the familiar *geographic coordinates* φ, ϑ (longitude and latitude, respectively). If $\mathbf{x} = (x_1, x_2, x_3) \in S^2$, then $\vartheta = \pi - \angle(\mathbf{x}, \mathbf{e}_3)$, $\vartheta \in [0, \pi]$, while the angle φ is the angle from $\mathbf{e}_1$ to $(x_1, x_2, 0)/\sqrt{x_1^2 + x_2^2}$ and belongs to $[0, 2\pi]$. Thus,

$$\mathbf{x} = \cos\vartheta \cos\varphi \mathbf{e}_1 + \cos\vartheta \sin\varphi \mathbf{e}_2 + \sin\vartheta \mathbf{e}_3.$$

If

$$\mathbf{x}' = \cos\vartheta'\cos\varphi'\mathbf{e}_1 + \cos\vartheta'\sin\varphi'\mathbf{e}_2 + \sin\vartheta'\mathbf{e}_3$$

is another point of S^2, then

$$\mathbf{x}\cdot\mathbf{x}' = \cos(\varphi-\varphi')\cos\vartheta\cos\vartheta' + \sin\vartheta\sin\vartheta'$$

and therefore

$$\cos(s_2(\mathbf{x},\mathbf{x}')) = \cos(\varphi-\varphi')\cos\vartheta\cos\vartheta' + \sin\vartheta\sin\vartheta'. \tag{6.9}$$

6.3.2 Spherical Triangles. Area of a Spherical Triangle

Let **A**, **B**, **C** be three points that do not lie on the same great circle and we draw the three great circles that pass through **B** and **C**, **C** and **A**, **A** and **B**, respectively. We will denote by:

- a, b, c the *smaller* arcs of the great circles that connect **B** and **C**, **C** and **A**, **A** and **B**, respectively.
- α, β, γ the angles of b and c, c and a, a and b, respectively, at the points **A**, **B**, **C**.

The angles are obtained as follows. Let $\mathcal{B}$ be the great circle passing through **A**, **C** and $\mathcal{C}$ be the great circle passing through **A**, **B**. The angle α is the angle between $\mathcal{B}$ and $\mathcal{C}$ at their common point **A**. Equivalently, the angles of the triangle can be defined as the angles of the tangents of the great circles at the corresponding vertices (this is the way we will use for *hyperbolic* triangles.

Definition 6.3.4 A *spherical triangle* $\triangle\mathbf{ABC}$ is the figure on the sphere resulting from three points **A**, **B**, **C** that do not belong to the same great circle and the arcs

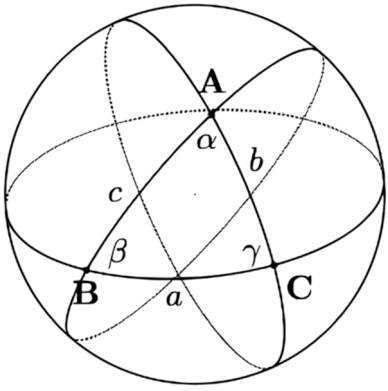


Fig. 6.8 Spherical triangle and its elements

a, b, c which are the smallest segments of the great circles connecting **B** and **C**, **C** and **A**, **A** and **B**, respectively, see Fig. 6.8. We call a, b, c the *sides* of the triangle, α, β, γ are the *angles* of the triangle and **A**, **B**, **C** are called *vertices* of the triangle.

Remark 6.3.5 The part of the sphere enclosed between a, b, c is called the *interior* of the triangle and many times, as in Euclidean Geometry, when we say triangle in Spherical Geometry, we will mean the figure together with its interior, unless we explicitly define it otherwise. Topologically, the existence of an interior spherical triangle is ensured by the Jordan Theorem for the sphere whose statement is due to the French topologist Camille Jordan, 1838–1922.

The first thing we notice about spherical triangles is that the well-known formula for the sum of the angles of a Euclidean triangle $\alpha + \beta + \gamma = \pi$ does not hold. For example, the spherical triangle with vertices

$$(1, 0, 0), \quad (0, 1, 0), \quad (0, 0, 1),$$

has an angle sum of $3\pi/2$ (Fig. 6.9).

Let $E = \alpha + \beta + \gamma - \pi$. Then $E = 0$ in Euclidean Geometry while in Spherical Geometry it is *always* $E > 0$. This is due to the following theorem of the French-Dutch mathematician Girard (Albert Girard, 1595–1632).

Theorem 6.3.6 (Girard $\sim$ 1620) *The number $E = \alpha + \beta + \gamma - \pi$ is equal to the area of the spherical triangle $T = \triangle\mathbf{ABC}$.*

This theorem had been proved earlier by the English mathematician and astronomer Harriot (Thomas Harriot, c. 1560–1621) but he had not published it. Let us note at this point that $E < 2\pi$ since $\alpha, \beta, \gamma \in (0, \pi)$. To prove Girard's Theorem, we first need to talk about *menisci* (lunes). The word is ancient Greek, a diminutive of *mene*=moon. Consider a great circle C and two antipodal points on C. These points divide C into two equal arcs which we call *semi-great circles*.

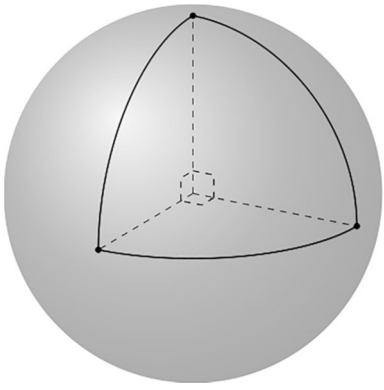


Fig. 6.9 Spherical triangle with three right angles

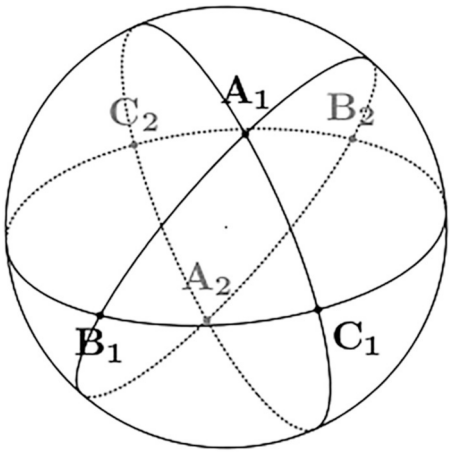


Fig. 6.10 Proof of Girard's Theorem

Definition 6.3.7 A meniscus or, lune is the region contained between two semi-great circles.

Two semi-great circles intersect at antipodal points and let θ be the solid angle of the planes containing them; this is equal to the angle of the meniscus defined by these semi-great circles at their intersection points. The spherical area E_θ of a meniscus with solid angle θ is equal to 2θ, since it is proportional to the angle θ and the total spherical area is 4π, see Fig. 6.7.

Proof of Theorem 6.3.6 The triangle T is formed by sides lying on the three great circles $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$. The great circle $\mathcal{A}$ divides the sphere into two hemispheres which we denote by $\mathcal{A}_1$ and $\mathcal{A}_2$. We similarly define hemispheres $\mathcal{B}_1, \mathcal{B}_2, \mathcal{C}_1, \mathcal{C}_2$, choosing them such that $T = \triangle\mathcal{A}_1 \cap \mathcal{B}_1 \cap \mathcal{C}_1$, with T having angles α, β and γ at the vertices at the intersections $\mathcal{B} \cap \mathcal{C}$, $\mathcal{C} \cap \mathcal{A}$ and $\mathcal{A} \cap \mathcal{B}$, respectively. $\mathcal{A}$, $\mathcal{B}$ and $\mathcal{C}$ divide the sphere into eight triangles $\mathcal{A}_i \cap \mathcal{B}_j \cap \mathcal{C}_k$, $i, j, k = 1, 2$, which we write for brevity as $\triangle\mathbf{A}_i\mathbf{B}_j\mathbf{C}_k$ (Fig. 6.10).

Now, the triangles $\triangle\mathbf{A}_1\mathbf{B}_1\mathbf{C}_1$ and $\triangle\mathbf{A}_2\mathbf{B}_1\mathbf{C}_1$ together form a meniscus of angle α defined by the great circles $\mathcal{B}$ and $\mathcal{C}$. In the same way, we find that

$$
\begin{aligned}
E(\triangle\mathbf{A}_1\mathbf{B}_1\mathbf{C}_1) + E(\triangle\mathbf{A}_2\mathbf{B}_1\mathbf{C}_1) &= 2\alpha,\\
E(\triangle\mathbf{A}_1\mathbf{B}_1\mathbf{C}_1) + E(\triangle\mathbf{A}_1\mathbf{B}_2\mathbf{C}_1) &= 2\beta,\\
E(\triangle\mathbf{A}_1\mathbf{B}_1\mathbf{C}_1) + E(\triangle\mathbf{A}_1\mathbf{B}_1\mathbf{C}_2) &= 2\gamma,\\
E(\triangle\mathbf{A}_2\mathbf{B}_2\mathbf{C}_2) + E(\triangle\mathbf{A}_1\mathbf{B}_2\mathbf{C}_2) &= 2\alpha,\\
E(\triangle\mathbf{A}_2\mathbf{B}_2\mathbf{C}_2) + E(\triangle\mathbf{A}_2\mathbf{B}_1\mathbf{C}_2) &= 2\beta,\\
E(\triangle\mathbf{A}_2\mathbf{B}_2\mathbf{C}_2) + E(\triangle\mathbf{A}_2\mathbf{B}_2\mathbf{C}_1) &= 2\gamma.
\end{aligned}
$$

Adding these six equations and noting that

$$E(\triangle \mathbf{A}_1\mathbf{B}_1\mathbf{C}_1) = E(\triangle \mathbf{A}_2\mathbf{B}_2\mathbf{C}_2), \qquad \sum_{i,j,k=1}^{2} E(\triangle \mathbf{A}_i\mathbf{B}_j\mathbf{C}_k) = 4\pi,$$

we end up with the relation $E(\triangle \mathbf{A}_1\mathbf{B}_1\mathbf{C}_1) + \pi = \alpha + \beta + \gamma$, which is what we wanted to prove. □

6.3.3 Spherical Trigonometry

Spherical trigonometry deals with the metric relations between the sides and the angles of spherical triangles. Its origin can be traced back to the Babylonians, but it was founded by the Ancient Greeks and was also studied extensively by Arab mathematicians. It was perfected in the modern era, around the end of the 19th century. In analogy to Euclidean trigonometry, there are also laws in the case of spherical trigonometry for solving spherical triangles. The following is fundamental:

Theorem 6.3.8 (First Cosine Law) *In a spherical triangle with sides a, b, c and angles α, β, γ we have:*

$$\cos a = \cos\alpha \sin b \sin c + \cos b \cos c. \tag{6.10}$$

Proof Due to the action of O(3), we may assume that the vertices of the triangle are the point $\mathbf{A} = \mathbf{e}_3$ with geographic coordinates $(\vartheta, \varphi) = (\pi/2 - c, 0)$, the point $\mathbf{B}$ with geographic coordinates $(c, 0)$ and the point $\mathbf{C}$ with geographic coordinates $(\pi/2 - b, A)$. Then, using Eq. (6.9) we get

$$\begin{aligned}\cos a &= \cos(s_2(\mathbf{B}, \mathbf{C})) \\ &= \cos\alpha \cos(\pi/2 - c)\cos(\pi/2 - b) + \sin(\pi/2 - c)\sin(\pi/2 - b) \\ &= \cos\alpha \sin b \sin c + \cos b \cos c.\end{aligned}$$

□

The spherical Pythagorean Theorem is a direct consequence of the First Cosine Law.

Corollary 6.3.9 (Pythagorean Theorem) *If the angle α is right, then*

$$\cos a = \cos b \cos c.$$

Theorem 6.3.10 (Sine Law)

$$\frac{\sin\alpha}{\sin a} = \frac{\sin\beta}{\sin b} = \frac{\sin\gamma}{\sin c}.$$

Proof From the identity $\sin^2 A = 1 - \cos^2 A$ and the First Law of Cosines for $\cos A$ as above, we get:

$$\begin{aligned}\sin^2\alpha &= 1 - \left(\frac{\cos a - \cos b\,\cos c}{\sin b\,\sin c}\right)^2 \\ &= \frac{(1-\cos^2 b)(1-\cos^2 c) - (\cos a - \cos b\,\cos c)^2}{\sin^2 b\,\sin^2 c} \\ \frac{\sin\alpha}{\sin a} &= \frac{[1-\cos^2 a - \cos^2 b - \cos^2 c + 2\cos a\cos b\cos c]^{1/2}}{\sin a\sin b\sin c}.\end{aligned}$$

Since the right-hand side remains unchanged by any permutation of $a,\ b,\ c$, the Sine Law follows immediately. □

In contrast to Euclidean Geometry, in Spherical Geometry we also have a Second Cosine Law. For its proof we need the concept of the *dual (or, polar)* triangle $T' = \triangle\mathbf{A}'\mathbf{B}'\mathbf{C}'$ which corresponds to a triangle $T = \triangle\mathbf{ABC}$ and is constructed as follows. We consider the great circle containing the side $a = \mathbf{BC}$. The perpendicular line from the origin to the plane of this circle intersects the sphere in two points and the point that is on the same side of the plane as the vertex $\mathbf{A}$ is denoted by $\mathbf{A}'$. Similarly, we take the points $\mathbf{B}'$ and $\mathbf{C}'$ and the triangle $\triangle\mathbf{A}'\mathbf{B}'\mathbf{C}'$ is the dual of the triangle $\mathbf{ABC}$ (Fig. 6.11). By construction,

$$\begin{aligned}&\alpha' = \pi - a, \qquad \beta' = \pi - b, \qquad \gamma' = \pi - c, \\ &a' = \pi - \alpha, \qquad b' = \pi - \beta, \qquad c' = \pi - \gamma.\end{aligned}$$

Theorem 6.3.11 (Second Cosine Law)

$$\cos\alpha = \frac{\cos a + \cos b\cos c}{\sin b\sin c}.$$

Proof The proof follows immediately by applying the First Cosine Law to the dual triangle $T' = \triangle\mathbf{A}'\mathbf{B}'\mathbf{C}'$. □

Corollary 6.3.12 *In Spherical Geometry, similar triangles are congruent and vice versa.*

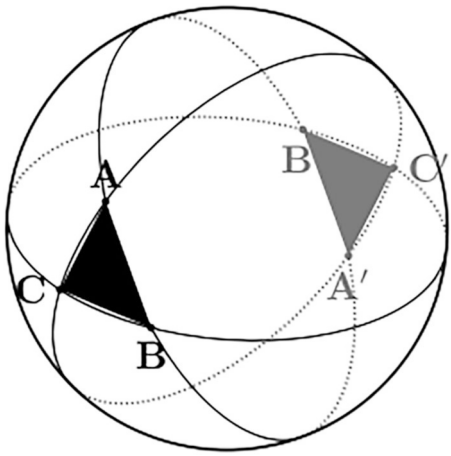


Fig. 6.11 Spherical triangle and its dual

6.4 Euler's Formula

Let $T = \triangle\mathbf{ABC}$ be a spherical triangle of S^2 with sides a, b, c, angles α, β, γ and vertices **A**, **B**, **C**. We will call a, b, c the *edges* of T. A *triangulation* of the sphere S^2 is a partition of it into a finite set of non-overlapping spherical triangles T_j such that the intersection of any two T_j is either empty, a common edge, or a common vertex. The *edges* of the triangulation are the edges of all T_j and the *vertices* of the triangulation are all the vertices of T_j (Fig. 6.12).

The simplest example of triangulation of the sphere (say, the surface of the Earth) is found by drawing the equator and n meridians. In this case, the triangulation contains $2n$ triangles, $n + 2$ vertices (n at the equator, and one at each pole), and $3n$ vertices. If we denote the number of triangles, edges, and vertices by F, E, and

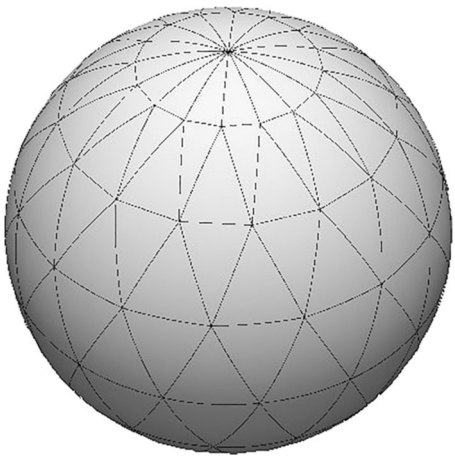


Fig. 6.12 Triangulation of the sphere

V, respectively, we get $F - E + V = 2$. That is, the expression $F - E + V$ does not depend on the choice of n. It is even more remarkable that the formula $F - E + V = 2$ holds for all triangulations of the sphere. This important topological result is due to Euler. The beautiful proof of the French mathematician Legendre (Adrien-Marie Legendre, 1752–1833) that we present here is based on the Girard's formula for the area of a spherical triangle.

Theorem 6.4.1 (Euler's Triangulation Theorem) *Suppose a triangulation of S^2 has F triangles, E edges and V vertices. Then*

$$F - E + V = 2.$$

Proof The area of a spherical triangle with angles θ_1, θ_2 and θ_3 is $\theta_1 + \theta_2 + \theta_2 - \pi$. Suppose there are F triangles, E edges and V vertices in the triangulation of the sphere. Summing all the angles in all the triangles, the total sum of angles is $2\pi V$ (because all the angles occur at a vertex without overlapping, and because the sum of the angles at any vertex is exactly 2π). Also, the sum of the areas of the triangles is the area of the sphere. Thus $2\pi V - F\pi = 4\pi$, or $2V = F + 4$. Now, enumerating the edges of each triangle and noting that in this enumeration each side counts twice, we obtain $3F = 2E$. Thus

$$F - E + V = F - 3F/2 + (F + 4)/2 = 2.$$

□

We close this section by stating that Euler's Formula also applies to regular polyhedra of $\mathbb{R}^3$. A regular polyhedron is one composed of p regular polygons, and exactly q such polygons intersect at each vertex of the polyhedron. Regular polyhedra are the famous Platonic Solids. Since the sum of the angles at each vertex is $< 2\pi$, we have

$$q(\pi - 2\pi/p) < 2\pi \iff (p-2)(q-2) < 4.$$

Therefore, since $p, q \geq 3$, the only possible pairs (p, q) are:

$$(3,3), \quad (3,4), \quad (3,5), \quad (4,3), \quad (5,3).$$

If F is the number of faces, E the number of edges, and V the number of vertices of the regular polyhedron, then since $2E = pF = qV$, Euler's formula gives

$$F = 2 + E - V = 2 + \frac{p}{2}F - \frac{p}{q}F \implies F = \frac{4q}{2p + 2q - pq}.$$

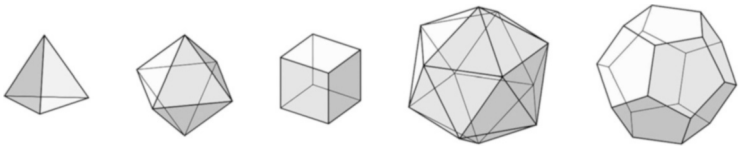


Fig. 6.13 The Platonic solids

In a similar manner we also obtain

$$E = \frac{2pq}{2p + 2q - pq}, \quad V = \frac{4p}{2p + 2q - pq}.$$

Substituting the above five pairs into the above relations we get the complete list of the Platonic Solids. Explicitly, as in Fig. 6.13 we have from left to right:

- *Tetrahedron*: 4 faces, 6 edges, 4 vertices.
- *Octahedron*: 8 faces, 12 edges, 6 vertices.
- *Hexahedron (cube)*: 6 faces, 12 edges, 8 vertices.
- *Icosahedron*: 20 faces, 30 edges, 12 vertices.
- *Dodecahedron*: 12 faces, 30 edges, 20 vertices.

6.5 Exercises

1. Determine 5 points in S^2 and 5 points in S^3.
2. Prove that the sphere S^{n-1}, $n \geq 3$, is compact.
3. Prove the following:
   (a) The restriction h on S^{n-1} of the mapping given by Eq. (6.1) is in S^{n-2}.
   (b) The inverse image $h^{-1}(\mathbf{z})$ of each point $\mathbf{z}$ of S^{n-2} is a circle of S^{n-1} centred at the origin.
4. Find the intersection
   (a) of the line $\mathbf{c}(t) = (2, 0, 0) + t(-1, 1, 1)$ and the sphere S^2,
   (b) of the line $\mathbf{c}(t) = (1, 0, 0, 1) + t(1, 1, 1, 0)$ and the sphere S^3,
   (c) of the 2-plane $\mathbf{x}(\lambda, \mu) = (0, 1, 1) + \lambda(0, 2, 0) + \mu(1, 0, 1)$ and the sphere S^2,
   (d) of the 2-plane $x_1 - x_4 = x_2 - x_3 = 0$ and the sphere S^3,
   (e) of the 3-plane $x_1 - x_4 = 0$ and the sphere S^3.
5. Find the one-parameter family $\mathbf{x}_\phi(\theta)$ of all great circles
   (a) of S^2 passing through $(1/\sqrt{3})(1, 1, 1)$,
   (b) of S^3 passing through $(1/\sqrt{2}(1, 1, 0, 0)$,
   (c) of S^4 passing through $(1/\sqrt{5})(1, 2, 0, 0)$.

6. As in Example 6.1.15, find the great circle of S^2 passing through the

   (a) $(1/\sqrt{5})(1, 0, 2)$ and $(1/\sqrt{5})(1, 2, 0)$ and
   (b) $(1/\sqrt{2})(1, 1, 0)$ and $(1/\sqrt{2})(1, 0, 1)$.

7. Argue the following: three (or more) intersecting circles of S^2 are meridians.
8. Find as in Example 6.1.17 all the great circles of S^3 passing through the points $\pm\,\mathbf{v} = (1/5)(1, 2, 2, -4)$.
9. Find the intersection points of the great circles L_1 and L_2 of S^2, if L_1 passes through

$$(1/\sqrt{3})(1, 1, -1). \quad (1/\sqrt{3})(1, -1, 1)$$

   and L_2 passes through

$$(1/\sqrt{3})(1, -1, -1), \quad (1/\sqrt{3})(-1, 1, -1).$$

10. Give an example of two non-intersecting great circles of S^3.
11. Let L_1 and L_2 be the great circles of S^2 with corresponding parametric equations

$$\mathbf{x}_1(\theta) = (1/\sqrt{7})(\cos\theta(1, -1, 2, 1) + \sin\theta(1, 1, 1, -2)),$$
$$\mathbf{x}_2(\theta) = (1/\sqrt{7})(\cos\theta(1, 2, -1, 1) + \sin\theta(2, -1, 1, 1)).$$

    Check whether these circles intersect or not using the method of proving the Proposition 6.1.18. You will have solve the homogeneous 4×4 linear system

$$\lambda_1(1, -1, 2, 1) + \mu_1(1, 1, 1, -2) = \lambda_2(1, 2, -1, 1) + \mu_2(2, -1, 1, 1).$$

12. Find the spherical distance and midpoint of the following pairs of points in S^2:

    (a) $(1, 0, 0)$ and $(-1, 0, 0)$,
    (b) $(0, 0, -1)$ and $(0, 1, 0)$,
    (c) $(1/2, -1/2, \sqrt{2}/2)$ and $(2/3, 1/3, -2/3)$,
    (d) $(0, 1/2, \sqrt{3}/2)$ and $(0, -1/2, -\sqrt{3}/2)$.

13. Show that the spherical distance of any two antipodal points in S^{n-1} is π.
14. Determine which of the following points belong to the same great circle:

    (a) $(0, 0, -1)$, $(0, 1, 0)$ and $(0, 0, 1)$,
    (b) $(1/2, -1/2, \sqrt{2}/2)$, $(2/3, 1/3, -2/3)$ and $(1, 0, 0)$.

15. Let $\mathbf{x}_0 \in S^{n-1}$ and $0 < \epsilon \leq 2\pi$. The open spherical neighbourhood $B^s_{\mathbf{x}_0}(\epsilon)$ with centre $\mathbf{x}_0$ and radius ϵ is the set

$$B^s_{\mathbf{x}_0}(\epsilon) = \{\mathbf{x} \in S^{n-1} \mid s_{n-1}(\mathbf{x}, \mathbf{x}_0) < \epsilon\}.$$

We denote by $B^c_{\mathbf{x}_0}(\delta)$ the open neighbourhood with centre $\mathbf{x}_0$ and radius $\delta > 0$ with respect to the chordal metric:

$$B^c_{\mathbf{x}_0}(\delta) = \{\mathbf{x} \in S^{n-1} \mid e_n(\mathbf{x}, \mathbf{x}_0) < \delta\}.$$

Prove that:

$$B^s_{\mathbf{x}_0}(\epsilon) = \begin{cases} B^c_{\mathbf{x}_0}(2\sin(\epsilon/2))) & \text{if } \epsilon \leq \pi, \\ \mathbb{R}^n \setminus B^c_{\mathbf{x}_0}(2\sin(\epsilon/2))) & \text{if } \pi < \epsilon \leq 2\pi. \end{cases}$$

16. Use the previous exercise to find the area of $B^s_{\mathbf{x}_0}(\epsilon)$ when $n = 3$.
17. In S^2, prove that the boundary of the open spherical neighbourhood $B^s_{\mathbf{x}_0}(\epsilon)$ is in general a minor circle and is a great circle if and only if $\epsilon = \pi/2, 3\pi/2$.
18. Determine the angles between the great circles with corresponding polar vectors:

    (a) $(1, 0, 0)$ and $(0, 0, 1)$,
    (b) $(0, 1, 0)$ and $(0, 0, 1)$,
    (c) $(1/3, 2/3, 2/3)$ and $(-3/5, 4/5, 0)$.

19. Let C be the great circle with parameteric equation

$$\mathbf{x}(\theta) = \cos\theta \mathbf{e}_1 + \sin\theta \mathbf{e}_2$$

and let a point

$$\mathbf{v} = \cos\varphi_1 \mathbf{e}_1 + \sin\varphi_1 \cos\varphi_2 \mathbf{e}_2 + \sin\varphi_1 \sin\varphi_2 \mathbf{e}_3,$$

which does not belong to C and is not a pole of C. Suppose that $\theta \in [0, \pi]$, $\varphi_1, \varphi_2 \in [0, \pi]$, $\varphi_1, \varphi_2 \neq \pi/2$.

    (a) Find a formula for the function $f : [0, \pi] \to [-1, 1]$,

$$f(\theta) = \cos(s_2(\mathbf{v}, \mathbf{x}(\theta))).$$

    (b) The distance $s_2(\mathbf{v}, C)$ is minimal at the point where f is maximised. Find this point and the corresponding distance.
    (c) Show that the great circle C' connecting its maximised point with $\mathbf{v}$ and C are orthogonal.

20. In Proposition 6.3.3 (ii), what happens when the point is a pole of the great circle?

21. Use the formula (6.9) to find the spherical distance between points with geographic coordinates (ϑ, φ) as follows:

    (a) $(0, 0)$ and $(\pi/3, \pi)$.
    (b) $(\pi/2, \pi/2)$ and $(\pi/6, 2\pi/3)$.

22. A spherical n-angle is the region enclosed by n intersecting great circles. Generalise Girard's formula for the area of a spherical n-angle.
23. Prove the Pythagorean theorem without using the First Cosine Law, by taking an appropriate normalisation for the vertices of the triangle.
24. By interchanging a, b, c and A, B, C, write all possible versions of the First and Second Cosine Laws.
25. Calculate the area of a spherical triangle whose angles are all $\pi/2$.
26. Prove the following celebrated theorem of Archimedes about the areas of a sphere and a cylindre: *The area of the part of the sphere S^2 cut by the planes $z = a$ and $z = b$, $-1 \leq a < b \leq 1$, is equal to the area of the circumscribed cylindre lying between these two planes.*
27. Let an equilateral spherical triangle of side a and interior angle A. Prove that

$$\cos(a/2)\sin(A/2) = 1/2.$$

    Conclude that in a spherical equilateral triangle, $A > \pi/3$.
28. Calculate the perimeter of a spherical triangle whose all angles are right angles.

Chapter 7
Projective Geometry



If one figure is derived from another by a continuous change and the latter is as general as the former, then any property of the first figure can be asserted at once for the second figure.

—Jean-Victor Poncelet

Projective Geometry according to Klein is the pair $(\mathbb{R}P^n, P(n))$, where $\mathbb{R}P^n$ is the n-dimensional *(real) projective space* and $P(n))$ is the *n-projective group*. These concepts will be defined precisely below, in Sects. 7.1 and 7.2, respectively. Projective Geometry deals with the properties of figures that remain invariant under *projections*. In contrast to Euclidean Geometry, which focuses primarily in measurements of length but also angles and volumes, Projective Geometry examines the properties that are preserved when figures are projected from one surface to another. One of the key features (and at the same time a key innovation) of Projective Geometry is the concept of the *point at infinity*. In this geometry, two parallel lines are considered to intersect at a point at infinity. This allows for the simplification of many geometric problems and the unification of the properties of shapes.

Figure 7.1 shows a *perspective position* of a circle located on a plane of three-dimensional space. From a point (here, this point is the origin, the *vertex* of the perspectivity) we draw lines to all points of the circle. Then, we consider the intersection of these lines with a plane at some other position. This intersection is an ellipse and is also called *perspective distortion* of the circle.

Projective Geometry has its roots in Euclid's Optics but it started developing fully at the times of the Renaissance, when artists and architects began to use *perspective* to create more realistic depictions of space (Fig. 7.2).

As early as the fifteenth century AD, it was understood that Euclidean Geometry was not sufficient to model the transformations related to the perspective of shapes. Perspective appeared frequently in painting and architecture, with painters and architects of the early Renaissance, in Italy and elsewhere, trying in various ways to handle it.

The method of perspective painting was then developed, initially in Italy by Brunelleschi (Filippo Brunelleschi, 1377–1446) and Alberti, (Leon-Battista Alberti,

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I. D. Platis, *Geometries*, La Matematica per il 3+2 178,
https://doi.org/10.1007/978-3-032-08016-5_7

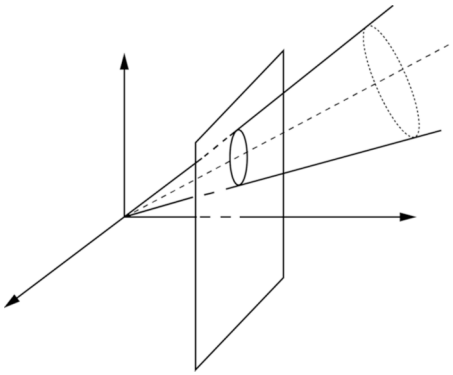


Fig. 7.1 The ellipse as a perspective distortion of a circle

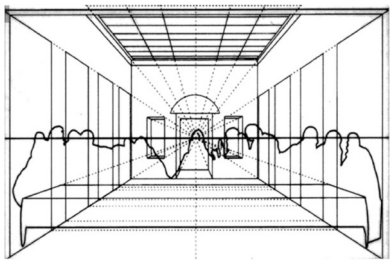


Fig. 7.2 Perspective lines in the Last Supper by Leonardo da Vinci

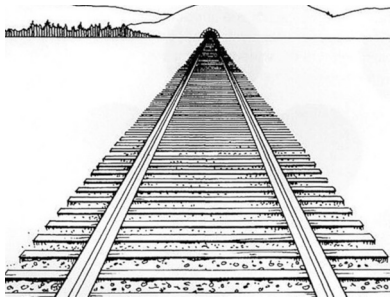


Fig. 7.3 A realisation of the point at infinity

1404–1472) and was widely used by the famous Italian painter and poly-scientist da Vinci (Leonardo da Vinci, 1452–1519). An open question in the Renaissance era was *what are the common properties of two perspective positions of the same figure?* During the seventeenth century, architect and engineer Desargues (Girard Desargues, 1591–1661) described all conics as perspective distortions of the circle. Desargues was the first to conceive the idea of the point at infinity as the *intersection of two parallel* lines. If the latter seems completely counter-intuitive to our mostly Euclidean intuition, we only have to look at Fig. 7.3 with the tracks of a railway line. Their intersection on the horizon can be thought of as the point at infinity.

In the eighteenth century, Descartes and Fermat (Pierre de Fermat, 1601–1665) had the revolutionary idea of describing geometry in an *analytic* way, in contrast to the synthetic way of the Greeks. This resulted in an unprecedented



Fig. 7.4 Jean-Victor Poncelet, 1788–1867

explosion of new geometric methods. Among many others, Desargues, based on ideas of the French mathematician, physicist and philosopher Pascal (Blaise Pascal, 1623–1662), focused on infinite approximations through Cartesian coordinates. The French mathematician Monge (Gaspard Monge, 1746–1818) introduced Descriptive Geometry and studied in particular the conservation of angles and lengths by projections. In 1822, Poncelet (Fig. 7.4), who was an officer in Napoleon's army, wrote - while imprisoned in a Russian prison- a thesis on the projective properties of figures and on the invariance of projections. This thesis is also considered the first in what we today call Projective Geometry: a projective property is a property that remains invariant under projections. The French mathematician Châsles that we encountered in Chap. 5 and the German mathematician Möbius (August-Ferdinand Möbius, 1790–1868) studied the most general projective transformations that map points to points and lines to lines and preserve the cross-ratio of four points.

Euclidean, Affine and Projective Geometry have the following hierarchy as Kleinian Geometries: Euclidean Geometry is part of Affine Geometry which in turn is contained in Projective Geometry. Indeed, Affine Geometry arises from Euclidean but with the only invariant being parallelism, while in Projective Geometry parallelism does not even exist.

In general terms we can say that Projective Geometry stands between Euclidean Geometry which is interested in lengths, angles and distances, and topology in which shape is not of particular interest. Projective Geometry is interested in properties invariant by projections, so straight lines remain straight under projective transformations. However, distances, lengths, and angles change. And of course, any two lines always intersect (again, see the train tracks).

One of the most important applications of Projective Geometry is in computer vision and graphic visualisation. Its principles are used to reconstruct 3-dimensional objects from 2-dimensional images, as well as to understand perspective in photographs and videos. In addition, Projective Geometry is fundamental to the understanding of curves and surfaces in Algebraic Geometry. Projective Geometry is also important in topology and differential geometry, where the properties of figures are studied under more general transformations.

Almost all the geometry books to which we have referred so far contain chapters dedicated to Projective Geometry, usually in the case of the projective plane,

such as e.g. [22]. More specifically, however, for treatises dedicated exclusively to Projective Geometry we refer indicatively to [11, 17, 25, 27], among the multitude of similar works.

7.1 The Projective Space $\mathbb{R}P^n$

As in Chap. 5, it is first appropriate to present some elements of the general theory of projective spaces which are defined for an arbitrary vector space. The real projective space emerges here as the main example of the general theory.

7.1.1 Projective Spaces

Let V be a vector space over a field $\mathbb{K}$. In the set $V_* = V \setminus \{\mathbf{0}\}$ we define the relation $\sim$ for each $\mathbf{x}, \mathbf{x}' \in V_*$ as follows:

$$\mathbf{x} \sim \mathbf{x}' \iff \mathbf{x}' = \lambda\mathbf{x} \text{ for some } \lambda \in \mathbb{K}_* = \mathbb{K} \setminus \{0\}.$$

The relation $\sim$ is an equivalence relation. Indeed, it is:

- reflexive because $\mathbf{x} \sim \mathbf{x}$ if $\mathbf{x} = 1\mathbf{x}$;
- symmetric because if $\mathbf{x} \sim \mathbf{x}' \iff \mathbf{x}' = \lambda\mathbf{x}$, $\lambda \in \mathbb{K}_*$, then $\mathbf{x} = (1/\lambda)\mathbf{x}' \iff \mathbf{x}' \sim \mathbf{x}$;
- transitive because if $\mathbf{x} \sim \mathbf{x}' \iff \mathbf{x}' = \lambda\mathbf{x}$, $\lambda \in \mathbb{K}_*$ and $\mathbf{x}' \sim \mathbf{x}'' \iff \mathbf{x}'' = \lambda'\mathbf{x}'$, $\lambda' \in \mathbb{K}_*$, then $\mathbf{x}'' = \lambda\lambda'\mathbf{x} \iff \mathbf{x} \sim \mathbf{x}''$.

Definition 7.1.1 The $\mathbb{K}$-*projective space of* V is defined as the set of equivalence classes:

$$\mathbb{K}P(V) = V_*/\sim = \{[\mathbf{x}] \mid \mathbf{x} \in V_*\}.$$

The mapping $\pi : V \to \mathbb{K}P(v)$,

$$\mathbf{x} \mapsto [\mathbf{x}], \ \mathbf{x} \in V$$

is called the *canonical projection.*

Remark 7.1.2 $\mathbb{K}P(V)$ is the set of straight lines passing through the origin of V where the point of the origin has been removed. Indeed, any such line $\mathbf{c}(t) = t\mathbf{v}$, $t \in \mathbb{K}_*$, $\mathbf{v} \neq \mathbf{0}$, corresponds to $[\mathbf{v}] \in \mathbb{K}P(V)$ and vice versa.

Every $(n+1)$-dimensional vector space V, $n > 1$, over a field $\mathbb{K}$ is isomorphic to the vector space $\mathbb{K}^{n+1}$, hence $\dim V = n+1$. In this case we will write $\mathbb{K}P^n$ instead

of $\mathbb{K}P(V)$ and we will say that $\mathbb{K}P^n$ has (projective) dimension n: $\dim(\mathbb{K}P^n) = n$. Hereafter we will assume $\mathbb{K} = \mathbb{R}$.

Definition 7.1.3 The set $\mathbb{R}P^n$ is the *real projective space of dimension n*.

We will call the elements of $\mathbb{R}P^n$ *points* and write them as $[\mathbf{x}]$. If $\mathbf{x} = (x_0, \ldots, x_n) \in \mathbb{R}^{n+1}$, we will write

$$[\mathbf{x}] = [x_0 : \cdots : x_n]. \tag{7.1}$$

The expression $[x_0 : \cdots : x_n]$ is called *homogeneous coordinates of* $[\mathbf{x}]$. We observe that if $[x_0 : \cdots : x_n]$ are homogeneous coordinates of some point $[\mathbf{x}]$, then $[\lambda x_0 : \cdots : \lambda x_n]$ are also homogeneous coordinates of the same point, since $[\mathbf{x}] = [\lambda \mathbf{x}]$ for each $\lambda \in \mathbb{R}_*$.

Now, for $i = 0, \ldots n$, we consider the sets

$$H_i = \{[x_0 : \cdots : x_n] \mid x_i \neq 0\}. \tag{7.2}$$

Since every point $[\mathbf{x}] \in \mathbb{R}P^n$ has at least one coordinate $x_i \neq 0$, it follows that

$$\mathbb{R}P^n = \bigcup_{i=0}^{n} H_i.$$

Let us observe that this is *is not* a disjoint union: for $i \neq j$ it is $H_i \cap H_j \neq \emptyset$. However, each of the sets H_i is in an injective correspondence with $\mathbb{R}^n$. For example, H_i comprises points of the form

$$\begin{aligned}[x_0 : x_1 : \cdots : x_{n-1} : x_n] &= [x_0/x_i : \cdots : x_{i-1}/x_i : 1 : x_{i+1}/x_i : \cdots : x_n/x_i] \\ &= [y_1 : \cdots : y_i : 1 : y_{i+1} : \cdots : y_n].\end{aligned}$$

The canonical projection $\mathbb{P} : \mathbb{R}P^n \to \mathbb{R}^n$ is therefore well defined: in the set H_i it is given by

$$H_i \ni [y_1 : \cdots : y_i : 1 : y_{i+1} : \cdots : y_n] \to (y_1, \ldots, y_n) \in \mathbb{R}^n.$$

Conversely, if $\mathbf{x} = (x_1, \ldots, x_n) \in \mathbb{R}^n$, its *standard lift* in $\mathbb{R}P^n$ is the point $[x_1 : \cdots : x_n : 1]$. Now, the set

$$\mathbb{R}P^n \setminus H_n = \{[\mathbf{x}] \in \mathbb{R}P^n \mid x_n = 0\},$$

is in an injective correspondence with the projective space $\mathbb{R}P^{n-1}$:

$$\mathbb{R}P^{n-1} \ni [x_0 : \cdots : x_{n-1}] \mapsto [x_0 : \cdots : x_{n-1} : 0] \in \mathbb{R}P^n \setminus H_n.$$

This correspondence represents the embedding of $\mathbb{R}P^{n-1}$ in $\mathbb{R}P^n$ (underlying is the embedding of $\mathbb{R}^n$ in $\mathbb{R}^{n+1}$ as the hyperplane $x_{n+1} = 0$). From the above discussion it follows that the projective space $\mathbb{R}P^n$ can be written as the disjoint union of the Euclidean space $\mathbb{R}^n$ and the projective space $\mathbb{R}P^{n-1}$.

Remark 7.1.4 The real projective space $\mathbb{R}P^n$ is a compact topological space, with the quotient topology: the open sets U of this topology are those for which $\pi^{-1}(U)$ is open in $\mathbb{R}^{n+1}_*$. The details of the proof of this fact are beyond the scope of this book; the interested reader can refer to Theorem 60.3, [39] for the two-dimensional case, and [3]. However, in Example 7.1.6 there is a relatively easy proof in the case of $\mathbb{R}P^1$.

Definition 7.1.5 Suppose that $B = \{\mathbf{v}_0, \ldots, \mathbf{v}_n\}$ is a basis of $\mathbb{R}^{n+1}$ and let also an element $\mathbf{v}_{n+1}$ such that its coordinates with respect to the basis B are all nonzero. Then the set

$$\{[\mathbf{v}_0], \ldots, [\mathbf{v}_n], [\mathbf{v}_{n+1}]\},$$

is called the *projective basis* of $\mathbb{R}P^n$.

We observe the following:

- if

$$\mathbf{v}_{n+1} = \sum_{k=0}^{n} \lambda_i \mathbf{v}_i, \quad \lambda_i \neq 0,\ i = 0, \ldots, n,$$

then $B' = \{\mathbf{v}'_i = \lambda_i \mathbf{v}_i,\ i = 0, \ldots, n\}$ is also a basis and

$$\mathbf{v}_{n+1} = \sum_{k=0}^{n} \mathbf{v}'_i.$$

From now on, we will assume that

$$\mathbf{v}_{n+1} = \sum_{i=0}^{n} \mathbf{v}_i,$$

for every basis $B\{\mathbf{v}_0, \ldots, \mathbf{v}_n\}$.

- The homogeneous coordinates as defined in 7.1 are the coordinates relative to the projective basis

$$[\mathbf{e}_0], \ldots, [\mathbf{e}_n], \left[\sum_{i=0}^{n} \mathbf{e}_i\right],$$

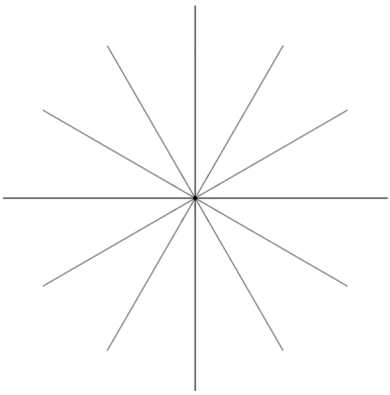


Fig. 7.5 The real projective space $\mathbb{R}P^1$ (projective line) is the set of all straight lines of $\mathbb{R}^2$ passing through the origin

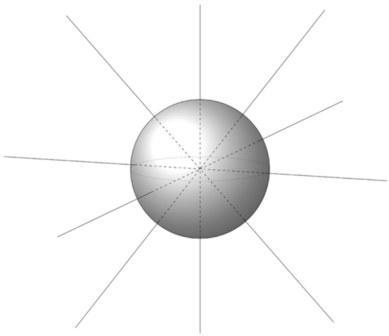


Fig. 7.6 Projective space $\mathbb{R}P^2$ (projective plane) and the unit sphere S^2

or, otherwise,

$$[1 : 0 : \cdots : 0], \ldots, [0 : \cdots : 0 : 1], \ [1 : 1 : \cdots : 1].$$

Another way to consider $\mathbb{R}P^n$ is via the identification of antipodal points of the sphere S^n. Indeed, we have already seen in Sect. 6.1.3 that every straight line of $\mathbb{R}^{n+1}$ passing through the origin is completely determined by two antipodal points of S^n and vice versa (Fig. 7.5). In other words, if in S^n we define the relation

$$\mathbf{x} \sim \mathbf{y} \iff \mathbf{x} = \pm\mathbf{y},$$

then we easily see that it is an equivalence relation and that

$$\mathbb{R}P^n = S^n / \sim .$$

The simplest real projective space is in dimension $n = 1$ (Fig. 7.6).

Example 7.1.6 When $n = 1$, the set $\mathbb{R}P^1$ is the *projective line* which is the set of all lines in the plane $\mathbb{R}^2$ that pass through the origin $(0, 0)$. The projective line is the

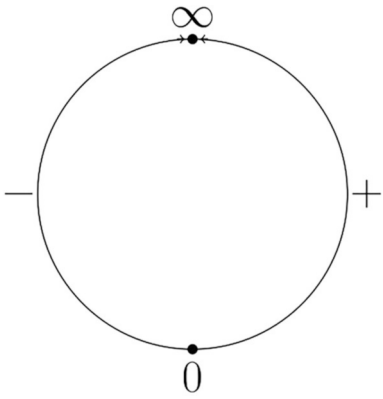


Fig. 7.7 Projective line and the unit circle

union of the subsets

$$H_0 = \{[x_0 : x_1] \in \mathbb{R}P^1 \mid x_0 \neq 0\} \quad \text{and} \quad H_1 = \{[x_0 : x_1] \in \mathbb{R}P^1 \mid x_1 \neq 0\}.$$

We can write

$$\mathbb{R}P^1 = H_0 \cup \{[0 : 1]\}, \quad \text{or} \quad \mathbb{R}P^1 = H_1 \cup \{[1 : 0]\},$$

that is, the projective line can be written as the (disjoint) union of the Euclidean line (in this case, H_0 or H_1, respectively) and a *point at infinity* (of $[0 : 1]$, or of $[1 : 0]$, respectively). The projective line $\mathbb{R}P^1$ is therefore in an injective correspondence with the circle S^1 itself and not simply with $S^1/\sim$. This is because S^1, through the stereographic projection (see Sect. 8.1.1), can be identified with the real line to which we have added a point at infinity. It therefore follows in a simple way that the projective line is a compact set and the same applies to the projective space of any dimension as pointed out in Comment 7.1.4 (Fig. 7.7).

We can pass from the projective to the Euclidean line via the *projection map* $\mathbb{P} : \mathbb{R}P^1 \to \mathbb{R}$ which is given in the set H_0 by $\mathbb{P}([x_0 : x_1]) = x_1/x_0$, while in the set H_1 it is given by $\mathbb{P}([x_0 : x_1]) = x_0/x_1$. Conversely, we can pass from the Euclidean line to the projective line via the standard lift $x \mapsto [x : 1]$. The canonical projective basis is

$$[1 : 0],\ [0 : 1],\ [1, 1].$$

7.1.2 Projective Subspaces

The definition of a projective subspace of the projective space $\mathbb{R}P^n$ is given via the canonical projection $\pi : \mathbb{R}^{n+1} \to \mathbb{R}P^n$.

Definition 7.1.7 Let V be a vector subspace of $\mathbb{R}^{n+1}$ of dimension $k+1 \leq n+1$. Then the set $\mathbf{V} = \pi(V)$ is called a *projective subspace* of $\mathbb{R}P^n$ of projective dimension k.

Since every $(k+1)$-dimensional subspace V of $\mathbb{R}^{n+1}$ is (isomorphic to a subspace) of the form $\mathbb{R}^{k+1}$, $k \leq n$, we can identify the image $\mathbf{V} = \pi(V)$ with the projective space $\mathbb{R}P^k$ according to the following:

Proposition 7.1.8 *If $\mathbf{V} = \pi(V)$ is a projective subspace of $\mathbb{R}P^n$ of projective dimension $k \leq n$, then there exists a bijective mapping $\mathbf{V} \to \mathbb{R}P^k$.*

Proof Every point $\mathbf{x} \in V$ is of the form

$$\mathbf{x} = \mathbf{x}(\lambda_0, \ldots, \lambda_k) = \sum_{i=0}^{k} \lambda_i \mathbf{v}_i,$$

where $\mathbf{v}_0, \ldots, \mathbf{v}_k$ are linearly independent vectors generating V. Then, if $\mathbf{v}_i = (v_i^0, \ldots, v_i^n)$, $\quad i = 0, \ldots, n$, we have that every point of $\mathbf{V} = \pi(V)$ is written as

$$[\mathbf{x}] = [\mathbf{x}(\lambda_0, \ldots, \lambda_k)] = \left[\sum_{j=0}^{k} \lambda_j v_j^0 : \cdots : \sum_{j=0}^{k} \lambda_j v_j^n \right],$$

with $\sum_{i=0}^{k} |\lambda_i| \neq 0$, that is, not all λ_i can vanish simultaneously. Due to the linear independence of $\mathbf{v}_i$, $i = 0, \ldots, n$, the mapping

$$\mathbb{R}P^k \ni [\lambda_0 : \cdots : \lambda_k] \mapsto [\mathbf{x}(\lambda_0, \ldots, \lambda_k)] \in \mathbf{V}, \tag{7.3}$$

is bijective. □

Remark 7.1.9 Through the mapping (7.3) a *projective basis for* $\mathbf{V}$ is also defined: this consists of the points

$$[\mathbf{x}(\mathbf{e}_0)], \quad [\mathbf{x}(\mathbf{e}_k)], \quad \left[\mathbf{x}\left(\sum_{i=0}^{k} \mathbf{e}_i \right) \right]. \tag{7.4}$$

Remark 7.1.10 Let V be a $(k+1)$-plane of $\mathbb{R}^{n+1}$, $V = \langle \mathbf{v}_0, \ldots, \mathbf{v}_k \rangle$. If

$$V^{\perp} = \langle \mathbf{A}_0, \ldots, \mathbf{A}_{n-k-1} \rangle,$$

then V is given by the equations

$$\mathbf{A}_i \cdot \mathbf{X} = 0, \ i = 0, \ldots, n-k-1, \ \mathbf{X} = (x_0, \ldots, x_n).$$

These *same* equations will be written in the form

$$[\mathbf{A}_i] \cdot \mathbf{X} = 0, \quad i = 0, \ldots, n-k-1,$$

as the equations of $\mathbf{V}$. The simplest case is that of a *projective hyperplane*. Such a hyperplane $\mathbf{V}$ is a projective subspace of dimension $n-1$ and is defined by the unique equation:

$$[\mathbf{A}] \cdot \mathbf{X} = 0 \iff \sum_{i=0}^{n} a_i x_i = 0.$$

The following proposition is equivalent to Proposition 5.1.16:

Proposition 7.1.11 *The intersection of two projective subspaces of the same dimension is a projective subspace of lower dimension.*

Projective Lines

Projective lines are the simplest projective subspaces. Let $\mathbf{L} = \pi(\Pi)$ be a projective line where Π is a 2-plane in $\mathbb{R}^{n+1}$. Then the parametric equation of Π is

$$\mathbf{x}(\lambda, \mu) = \lambda \mathbf{v} + \mu \mathbf{w}, \ \lambda, \mu \in \mathbb{R},$$

where $\mathbf{v}$ and $\mathbf{w}$ are linearly independent vectors. If $\mathbf{v} = (v_0, \ldots, v_n)$, $\mathbf{w} = (w_0, \ldots, w_n)$, then the projective line $\mathbf{L}$ is the set of points of $\mathbb{R}P^n$ with homogeneous coordinates

$$\mathbf{L} = \{[\lambda v_0 + \mu w_0 : \cdots : \lambda v_n + \mu w_n] \in \mathbb{R}P^n, \quad |\lambda| + |\mu| \neq 0\}.$$

Every such projective line $\mathbf{L}$ as above, can be identified with $\mathbb{R}P^1$ through the mapping (7.3):

$$\mathbb{R}P^1 \ni [\lambda_0 : \lambda_1] \mapsto [\mathbf{x}(\lambda_0, \ldots, \lambda_1)] \in \mathbf{L}.$$

A projective basis for $\mathbf{L}$ consists of the points

$$[\mathbf{x}(1,0)] = [\mathbf{v}], \quad [\mathbf{x}(0,1)] = [\mathbf{w}], \quad [\mathbf{x}(1,1)] = [\mathbf{v} + \mathbf{w}].$$

We conclude that a projective basis of a projective line consists of three pairwise distinct points.

Alternatively, if the 2-plane Π is given by the equations

$$\mathbf{A}_i \cdot \mathbf{X} = 0, \quad A_i \in \Pi^{\perp}, \ i = 0, \ldots, n-2,$$

then $\mathbf{L}$ is the set

$$\mathbf{L} = \{\mathbf{X} \in \mathbb{R}P^n : [\mathbf{A}_i] \cdot \mathbf{X} = 0, \quad \mathbf{A}_i \in \Pi^{\perp},\ i = 0, \ldots, n-2\}.$$

Example 7.1.12 The simplest case is that of projective lines in the projective plane $\mathbb{R}P^2$. Let Π be a 2-plane in $\mathbb{R}^3$ given by the equation

$$ax_0 + bx_1 + cx_2 = 0.$$

Then the projective line $\mathbf{L} = \pi(\Pi)$ of $\mathbb{R}P^2$ is given by the unique equation

$$[\mathbf{A}] \cdot \mathbf{X} = 0,$$

where $\mathbf{A} = (a, b, c) \perp \Pi$ and $\mathbf{X} = (x_0, x_1, x_2)$.

The proof of the following proposition follows directly from the fact that an infinite number of 2-planes pass through a line of $\mathbb{R}^{n+1}$ from the origin:

Proposition 7.1.13 *An infinite number of projective lines pass from any given point* $[\mathbf{x}] \in \mathbb{R}P^n$.

Example 7.1.14 Let $[1 : 0 : -1 : 1] \in \mathbb{R}P^3$. The set of lines of $\mathbb{R}P^3$ passing through this point is the set of points

$$[\lambda + \mu v_0 : \mu v_1 : -\lambda + \mu v_2 : \lambda + \mu v_3],\ |\lambda| + |\mu| \neq 0,$$

where (v_0, v_1, v_2, v_3) and $(1, 0, -1, 1)$ are linearly independent vectors. That is, the rank of the matrix

$$\begin{pmatrix} v_0 & v_1 & v_2 & v_3 \\ 1 & 0 & -1 & 1 \end{pmatrix},$$

is 2. The minor discriminants are equal to

$$v_1,\ v_0 + v_2,\ v_0 - v_3,\ v_2 + v_3.$$

Therefore, we must have that any of the above expressions must be different of zero; this defines five families of projective lines passing through $[1 : 0 : -1 : 1]$.

Similarly, since two linearly independent vectors of $\mathbb{R}^{n+1}$ define a unique 2-plane, we have the following:

Proposition 7.1.15 *Through every two distinct points of* $\mathbb{R}P^n$ *passes a unique projective line.*

The intersections of the projective lines are now directly related to the intersections of the projective planes. For any two 2-planes of $\mathbb{R}^{n+1}$ the following holds:

- They always intersect on a line passing through the origin when $n = 2$.
- They may not intersect except at the origin when $n > 2$. However, when their intersection is different from the origin, then it is a line.

We immediately conclude the following proposition.

Proposition 7.1.16 (Incidence) *Two projective lines of $\mathbb{R}P^n$ always intersect at a unique point if $n = 2$. When $n > 2$, their intersection may be empty. However, when they intersect, they intersect at a unique point.*

Example 7.1.17 In this example we show how to find the intersection of two projective lines of $\mathbb{R}P^2$. Two lines $\mathbf{L}_1 : ax_0 + bx_1 + cx_2 = 0$ and $\mathbf{L}_2 : a'x_0+b'x_1+c'x_2 = 0$ (i.e., π images of corresponding planes of $\mathbb{R}^3$ from the origin), intersect at a point (i.e., of the π-image of the line that constitutes the intersection of the planes) which is determined as follows: at least one of the minor determinants

$$D_0 = \begin{vmatrix} b & c \\ b' & c' \end{vmatrix}, \quad D_1 = \begin{vmatrix} a & c \\ a' & c' \end{vmatrix}, \quad D_2 = \begin{vmatrix} a & b \\ a' & b' \end{vmatrix},$$

is different from zero, since the system

$$\begin{aligned} ax_0 + bx_1 + cx_2 &= 0, \\ a'x_0 + b'x_1 + c'x_2 &= 0, \end{aligned}$$

has infinite solutions. For example, if $D_2 \neq 0$, then

$$x_0 = \frac{D_0}{D_2} \cdot x_2, \quad x_1 = -\frac{D_1}{D_2} \cdot x_2.$$

It turns out that the intersection of $\mathbf{L}_1 = \pi(\Pi_1)$ and $\mathbf{L}_2 = \pi(\Pi_2)$ is the (unique) projective point

$$[D_0 : -D_1 : D_2].$$

For further reference, observe that if $\mathbf{A} = (a, b, c)$ and $\mathbf{A}' = (a', b', c')$ then

$$[D_0 : -D_1 : D_2] = [\mathbf{A} \times \mathbf{A}'],$$

where $\mathbf{A} \times \mathbf{A}'$ is the exterior product of the vectors $\mathbf{A}$ and $\mathbf{A}'$.

7.1.3 Projective Mappings

By a *projective mapping* we mean a mapping from a projective space $\mathbb{R}P^n$ to a projective space $\mathbb{R}P^m$. In the Euclidean case the simplest mappings are linear; in

the projective case the simplest mappings arise from linear mappings and in this section we will mainly examine such projective mappings.

Let $f : \mathbb{R}^{n+1} \to \mathbb{R}^{m+1}$ be a monomorphism, that is, an injective linear mapping, with the formula

$$f(\mathbf{x}) = f(x_0, \ldots, x_n) = \mathbf{y} = (y_0, \ldots, y_n), \quad \mathbf{x} \in \mathbb{R}^n.$$

From f we obtain a mapping $[f] : \mathbb{R}P^n \to \mathbb{R}P^m$ defined by the relation

$$[f]([\mathbf{x}]) = [f(\mathbf{x})].$$

$[f]$ is well-defined: if $[\mathbf{x}_1], [\mathbf{x}_2] \in \mathbb{R}P^n$ then suppose $[f(\mathbf{x}_1)] \neq [f(\mathbf{x}_2)]$. Then for every $\lambda \in \mathbb{R}_*$ we have $f(\mathbf{x}_1) \neq \lambda f(\mathbf{x}_2)$, that is $f(\mathbf{x}_1 - \lambda\mathbf{x}_2) \neq \mathbf{0}$ due to linearity. Since f is injective, its kernel comprises of $\mathbf{0}$ only, therefore $\mathbf{x}_1 \neq \lambda\mathbf{x}_2$ for every $\lambda \in \mathbb{R}_*$, hence $[\mathbf{x}_1] \neq [\mathbf{x}_2]$.

Example 7.1.18 (Canonical Projections) The canonical projection $pr : \mathbb{R}^{n+1} \to \mathbb{R}^{k+1}$ is defined by the formula

$$pr(x_0, \ldots, x_n) = (x_0, \ldots, x_k), \quad (x_0, \ldots, x_n) \in \mathbb{R}^{n+1}.$$

Every projection is a monomorphism, so the corresponding projective mapping $[pr] : \mathbb{R}P^n \to \mathbb{R}P^k$ is defined by:

$$[pr]([x_0 : \cdots : x_n]) = [x_0 : \cdots : x_k], \quad [x_0 : \cdots : x_n] \in \mathbb{R}P^n.$$

Example 7.1.19 Let $f : \mathbb{R}^2 \to \mathbb{R}^3$ be the linear mapping with formula

$$f(x_0, x_1) = (3x_0,\ 3x_0, -6x_1).$$

f is injective and therefore the projective mapping

$$[f] : \mathbb{R}P^1 \to \mathbb{R}P^2, \quad [x_0 : x_1] \mapsto [x_0 : x_0 : -2x_1]$$

is well-defined. We observe that the image of the projective line $\mathbb{R}P^1$ is the projective line

$$[\lambda : \lambda : -2\mu] \in \mathbb{R}P^2, \quad \lambda, \mu \in \mathbb{R}.$$

Equivalently, the defining equation of this line may be written as

$$\begin{vmatrix} x_0 & x_1 & x_2 \\ 1 & 1 & 0 \\ 0 & 0 & -2 \end{vmatrix} = [1 : -1 : 0] \cdot \mathbf{X} = 0.$$

Example 7.1.20 Let $f : \mathbb{R}^3 \to \mathbb{R}^2$ be the linear mapping defined by the formula

$$f(x_0, x_1, x_2) = (x_0 - x_1,\ x_1 - x_2,\ x_2 - x_0).$$

f is not injective since $\ker(f) = \{(\lambda, \lambda, \lambda),\ | \lambda \in \mathbb{R}\}$. Then the relation

$$[x_0 : x_1 : x_2] \mapsto [x_0 - x_1 : x_1 - x_2 : x_2 - x_0],$$

is not a well-defined mapping: e.g., $[1 : 1 : 1] \mapsto [0 : 0 : 0]$ which makes no sense.

Example 7.1.21 The mapping $f : \mathbb{R}^2 \to \mathbb{R}^2$ with formula

$$f(x_0, x_1) = (x_0 - x_1\, x_0 + x_1)$$

is bijective (i.e., an automorphism). Therefore the transformation $[f] : \mathbb{R}P^1 \to \mathbb{R}P^1$ where

$$[f]([x_0 : x_1]) = [x_0 - x_1 : x_0 + x_1]$$

is well defined. We observe that the $[f]$-images of the elements of the canonical basis

$$f([1 : 0]) = [1 : 0], \quad f([0 : 1]) = [-1 : 1], \quad f([1 : 1]) = [0 : 1],$$

form another projective basis of $\mathbb{R}P^1$.

Example 7.1.22 Let

$$A_f = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 1 & 2 \\ 0 & -1 & 4 \end{pmatrix}.$$

Since $\det(A_f) = 12$, $A_f \in \mathrm{GL}(3, \mathbb{R})$ and the associated linear map $f : \mathbb{R}^3 \to \mathbb{R}^3$ with formula

$$f(x_0, x_1, x_2) = (2x_0 + x_1\, x_1 - x_2\, 2x_1 + 4x_2), \quad (x_0, x_1, x_2) \in \mathbb{R}^3,$$

is an automorphism and therefore the map $[f] : \mathbb{R}P^2 \to \mathbb{R}P^2$ with formula

$$[f]([x_0 :\ x_1 :\ x_2]) = [2x_0 + x_1 :\ x_1 - x_2 :\ 2x_1 + 4x_2], \quad [x_0 :\ x_1 :\ x_2] \in \mathbb{R}P^2,$$

is a well-defined projective mapping. We claim that it has fixed points. Indeed, the eigenvalues of f are 2 (with multiplicity 2) and 3. An eigenvector corresponding to

the eigenvalue 3 is $(1, 1, -2)$. We observe that

$$[f]([1 : 1 : -2]) = [3 : 3 : -6] = [1 : 1 : -2].$$

Also, an eigenvector corresponding to eigenvalue 2 is $(1, 0, 0)$. Again we observe that

$$[f]([1 : 0 : 0]) = [2 : 0 : 0] = [1 : 0 : 0].$$

The previous example is the reason for the next proposition:

Proposition 7.1.23 *Let $f : \mathbb{R}^{n+1} \to \mathbb{R}^{n+1}$ be an automorphism and $[f]$ be the corresponding projective mapping. The following hold:*

1. *If $n \geq 2$ is even, then $[f]$ has at least one fixed point in $\mathbb{R}P^n$.*
2. *If $n \geq 1$ is odd, then $[f]$ may or may not have a fixed point in $\mathbb{R}P^n$.*

Proof The characteristic polynomial p_f of f has at least one real root when $n \geq 2$ is even, but may or may not have real roots when $n \geq 1$ is odd. When there is a real root, say λ, it is an eigenvalue of f with a corresponding eigenvector $\mathbf{x}$. Then,

$$f(\mathbf{x}) = \lambda \mathbf{x} \iff [f]([\mathbf{x}]) = [\mathbf{x}].$$

□

Clearly, there are mappings between projective spaces that do not arise from any linear mapping of the corresponding vector spaces. We describe such a mapping in the following example.

Example 7.1.24 (Segre Embedding) Consider the mapping $\phi : \mathbb{R}P^1 \times \mathbb{R}P^1 \to \mathbb{R}P^3$ which is given by the formula

$$\phi([x_0 : x_1],\ [y_0 : y_1]) = [x_0y_0 :\ x_0y_1 :\ x_1y_0 :\ x_1y_1], \quad ([x_0 : x_1],\ [y_0 : y_1]) \in (\mathbb{R}P^1)^2.$$

The map ϕ is well defined: if $\lambda, \mu \in \mathbb{R}_*$, then

$$\begin{aligned}
&\phi([\lambda x_0 : \lambda x_1],\ [\mu y_0 : \mu y_1]) \\
&\quad = [(\lambda x_0)(\mu y_0) :\ (\lambda x_0)(\mu y_1) :\ (\lambda x_1)(\mu y_0) : (\lambda x_1)(\mu y_1)] \\
&\quad = [(\lambda\mu)(x_0y_0) :\ (\lambda\mu)(x_0y_1) :\ (\lambda\mu)(x_1y_0) :\ (\lambda\mu)(x_1y_1)] \\
&\quad = [x_0y_0 :\ x_0y_1 :\ x_1y_0 :\ x_1y_1].
\end{aligned}$$

We observe that

$$\phi(\mathbb{R}P^1 \times \mathbb{R}P^1) \subseteq \mathcal{S} = \{[v_0 : v_1 : v_2 : v_3] \in \mathbb{R}P^3 \mid v_0v_3 = v_1v_2\}.$$

On the other hand, let $[p] = [v_0 : v_1 : v_2 : v_3] \in \mathcal{S}$. Then, if par example $v_0 \neq 0$, we have:

$$\phi([v_0 : v_2],\ [v_0, v_1]) = [v_0v_0 :\ v_0v_1 :\ v_0v_2 :\ v_0v_3] = [p].$$

Therefore $\mathcal{S} = \phi(\mathbb{R}P^1 \times \mathbb{R}P^1)$. The map ϕ is injective: indeed, let $([x_0 : x_1],\ [y_0 : y_1])$ and $([x_0' : x_1'],\ [y_0' : y_1'])$ with

$$\phi([x_0 : x_1],\ [y_0 : y_1]) = \phi([x_0' : x_1'],\ [y_0' : y_1']) = [v_0 : v_1 : v_2 : v_3] \in \mathcal{S}.$$

This means that there exist $\lambda, \mu \in \mathbb{R}_*$ such that

$$(x_0y_0,\ x_0y_1,\ x_1y_0,\ x_1y_1) = \lambda(v_0, v_1, v_2, v_3),$$
$$(x_0'y_0',\ x_0'y_1',\ x_1'y_0',\ x_1'y_1') = \mu(v_0, v_1, v_2, v_3).$$

It follows that

$$\begin{vmatrix} x_0 & y_0 \\ x_1 & y_1 \end{vmatrix} = \begin{vmatrix} x_0' & y_0' \\ x_1' & y_1' \end{vmatrix} = \begin{vmatrix} v_0 & v_1 \\ v_2 & v_3 \end{vmatrix} = 0.$$

The set $\mathcal{S}$ is called the *Segre embedding* of $\mathbb{R}P^1 \times \mathbb{R}P^1$ in $\mathbb{R}P^3$. Through ϕ, $\mathbb{R}P^1 \times \mathbb{R}P^1$ is identified with the *quadratic surface* $\mathcal{S}$ of $\mathbb{R}P^3$: by setting $X_0 = x_0y_0$, $X_1 = x_0y_1$, $X_2 = x_1y_0$ and $X_3 = x_1y_1$ we have that

$$\mathcal{S} = \{[X_0 : X_1 : X_2 : X_3] \in \mathbb{R}P^3 \mid X_0X_3 = X_1X_2\}.$$

7.2 The Projective Group $P(n)$

The projective group is the set of projective mappings which are derived from linear *automorphisms* of $\mathbb{R}^{n+1}$. We study such mappings in the following section.

7.2.1 Homographies and the Group $P(n)$

Let $g : \mathbb{R}^{n+1} \to \mathbb{R}^{n+1}$ be an automorphism of $\mathbb{R}^{n+1}$ given for each $\mathbf{x} \in \mathbb{R}^{n+1}$ by the formula

$$\mathbf{x} = (x_0, \ldots, x_n) \mapsto (y_0, \ldots, y_n) = g(\mathbf{x}),$$

where

$$y_i = \sum_{j=0}^{n} a_{ij} x_j, \quad i = 0, 1, \ldots, n,$$

and $A_g = (a_{ij})_{i,j=0}^{n}$ is the matrix of g. From the transformation g there results a self-map $[g]$ of $\mathbb{R}P^n$ defined by the relation

$$[g]([\mathbf{x}]) = [g(\mathbf{x})]. \tag{7.5}$$

We prove that $[g]$ is a well-defined, bijective self-map of $\mathbb{R}P^n$.

1. $[g]$ is well-defined: if $[\mathbf{x}_1], [\mathbf{x}_2] \in \mathbb{R}P^n$, then

$$\begin{aligned} [\mathbf{x}_1] = [\mathbf{x}_2] &\iff \mathbf{x}_1 = \lambda \mathbf{x}_2 \implies g(\mathbf{x}_1) = g(\lambda \mathbf{x}_2) = \lambda g(\mathbf{x}_2) \\ &\iff [g(\mathbf{x}_1)] = [g(\mathbf{x}_2)]. \end{aligned}$$

2. $[g]$ is injective:

$$\begin{aligned} [g(\mathbf{x}_1)] = [g(\mathbf{x}_2)] &\iff g(\mathbf{x}_1) = \lambda g(\mathbf{x}_2) = g(\lambda \mathbf{x}_2) \implies \mathbf{x}_1 = \lambda \mathbf{x}_2 \\ &\iff [\mathbf{x}_1] = [\mathbf{x}_2]. \end{aligned}$$

3. $[g]$ is surjective: if $[\mathbf{y}] \in \mathbb{R}P^n$, then there exists $\mathbf{x} \in \mathbb{R}^{n+1}$ such that $g(\mathbf{x}) = \mathbf{y}$, hence,

$$[g(\mathbf{x})] = [g]([\mathbf{x}]) = [\mathbf{y}].$$

Definition 7.2.1 A mapping $[g] : \mathbb{R}P^n \to \mathbb{R}P^n$ defined by (7.5) will also be called *projective transformation* or *homography*.

Let $P(n)$ be the set of all homographies of $\mathbb{R}P^n$. The following are clear:

- The composition of two homographies $[g_1]$ and $[g_2]$ is defined by the relation

$$([g_1] \circ [g_2])([\mathbf{x}]) = [(g_1 \circ g_2)(\mathbf{x})].$$

- For the point $[id.]$ where $id.$ is the identity automorphism of $\mathbb{R}^{n+1}$,

$$([g] \circ [id.])([\mathbf{x}]) = [g]([\mathbf{x}]].$$

- The *inverse* $[g]^{-1}$ of the homography $[g]$ is also a homography defined by the relation

$$[g]^{-1}([x]) = [g^{-1}(x)],$$

where g^{-1} is the inverse of g.

Therefore $P(n)$ is a group.

Definition 7.2.2 The group $P(n)$ will be called the *projective group.*

7.2.2 $P(n)$ and its Action on $\mathbb{R}P^n$

The group of automorphisms of $\mathbb{R}^{n+1}$ is isomorphic to the general linear group $\mathrm{GL}(n, \mathbb{R})$. Consider the mapping

$$\mathrm{GL}(n+1, \mathbb{R}) \to P(n), \quad A_g \mapsto [g],$$

where

$$A_g = \begin{pmatrix} a_{00} & \dots & a_{0n} \\ \vdots & \dots & \vdots \\ a_{n0} & \dots & a_{nn} \end{pmatrix} \in \mathrm{GL}(n, \mathbb{R}^{n+1}),$$

is the matrix A_g of the automorphism g. This mapping is a group epimorphism. The kernel is the set of matrices of the form $A_g = \lambda I$ which is isomorphic to the multiplicative group $(\mathbb{R}_*, \cdot)$. By the First Group Isomorphism Theorem (Sect. B.2.1), we then have

$$P(n) \simeq \mathrm{GL}(n+1, \mathbb{R})/(\mathbb{R}_*, \cdot).$$

Definition 7.2.3 The *projective general linear group* is the set

$$\mathrm{PGL}(n+1, \mathbb{R}) = \mathrm{GL}(n+1, \mathbb{R})/(\mathbb{R}_*, \cdot), \tag{7.6}$$

which is a group isomorphic to the projective group $P(n)$.

The group $\mathrm{PGL}(n+1, \mathbb{R})$ comprises classes $[A_g]$, $A_g \in \mathrm{GL}(n+1, \mathbb{R})$,

$$[A_g] = \{\lambda A_g, \ \lambda \in \mathbb{R}_*\}.$$

The operation on $\mathrm{PGL}(n+1,\mathbb{R})$ is given by the relation

$$[A_{g_1}] * [A_{g_2}] = [A_{g_1} A_{g_2}].$$

If

$$[A_g] = \begin{bmatrix} a_{00} & \dots & a_{0n} \\ \vdots & \cdots & \vdots \\ a_{n0} & \dots & a_{nn} \end{bmatrix} \in \mathrm{PGL}(n+1,\mathbb{R}),$$

the corresponding homography is given by the relation

$$[g]([x_0 : \cdots : x_n]) = \left[\sum_{i=0}^{n} a_{i0} x_i : \cdots : \sum_{i=0}^{n} a_{in} x_i \right].$$

Example 7.2.4 The matrix of the automorphism $g : \mathbb{R}^2 \to \mathbb{R}^2$ with

$$g(x_0, x_1) = (2x_0 - x_1,\ -x_0 + 3x_1),$$

is

$$A_g = \begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix} \in \mathrm{GL}(2,\mathbb{R}).$$

Thus the map $[g] : \mathbb{R}P^1 \to \mathbb{R}P^1$ with formula

$$[g]([x_0 :\ x_1] = [2x_0 - x_1 :\ -x_0 + 3x_1],$$

is a homography. Its matrix is $[A_g] \in \mathrm{PGL}(2,\mathbb{R})$.

Example 7.2.5 The class

$$[A_g] = \begin{bmatrix} 0 & -1 & 1 \\ 2 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix} \in \mathrm{PGL}(2,\mathbb{R}),$$

since $\det(A_g) = -3$. The corresponding homography $[g] : \mathbb{R}P^2 \to \mathbb{R}P^2$ has formula

$$[g]([x_0 :\ x_1 :\ x_2] = [2x_1 - x_2 :\ -x_0 + x_1 :\ x_0 - x_2].$$

The left action of $P(n)$ on $\mathbb{R}P^n$ with projective transformations $[g]$ is given by

$$P(n) \times \mathbb{R}P^n \ni ([g], [\mathbf{x}]) \mapsto [g]([\mathbf{x}]) = [g(\mathbf{x})] \in \mathbb{R}P^n.$$

Proposition 7.2.6 *The action of $P(n)$ is doubly transitive and free.*

Proof If $[\mathbf{x}_0]$, $[\mathbf{x}_1]$ and $[\mathbf{y}_0]$, $[\mathbf{y}_1]$ are pairs of distinct points in the projective space, then since $\mathbf{x}_0, \mathbf{x}_1$ and $\mathbf{y}_0, \mathbf{y}_1$ are pairs of linearly independent points in $\mathbb{R}^{n+1}$, there exists an invertible transformation g of $\mathbb{R}^{n+1}$ that maps the subspace $\langle \mathbf{x}_0, \mathbf{x}_1 \rangle$ to the subspace $\langle \mathbf{y}_0, \mathbf{y}_1 \rangle$ with $g(\mathbf{x}_0) = \mathbf{y}_0$ and $g(\mathbf{x}_1) = \mathbf{y}_1$. Indeed, we consider the basis

$$B = \{\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n\}, \quad \mathbf{x}_i \in \langle \mathbf{x}_0, \mathbf{x}_1 \rangle^{\perp}, \ i = 2, \ldots, n,$$

and the mapping $h : \mathbb{R}^{n+1} \to \mathbb{R}^{n+1}$ with matrix

$$A_h = \begin{pmatrix} \mathbf{x}_0 \\ \mathbf{x}_1 \\ \vdots \\ \mathbf{x}_n \end{pmatrix}.$$

The projective transformation $[h]$ satisfies the relations

$$[h(\mathbf{e}_0)] = [\mathbf{x}_0], \quad [h(\mathbf{e}_1)] = [\mathbf{x}_1].$$

Similarly, we find a projective transformation $[h']$ that satisfies the relations

$$[h'(\mathbf{e}_0)] = [\mathbf{y}_0], \quad [h'(\mathbf{e}_1)] = [\mathbf{y}_1].$$

The desired transformation is $[g] = [h' \circ h^{-1}]$.

On the other hand, if $[\mathbf{x}] \in \mathbb{R}P^n$, its stabiliser

$$\mathrm{Stab}(\mathbf{x}) = \{[g] \in P(n) \mid [g]([\mathbf{x}]) = [\mathbf{x}]\},$$

is isomorphic to the set $(\mathbb{R}_*, \cdot)$, that is, the neutral element of $\mathrm{PGL}(n+1, \mathbb{R})$. □

A remarkable property of the group $P(n)$ is given in the following proposition:

Proposition 7.2.7 *Every element of $P(n)$ maps lines to lines.*

Proof The proof follows directly from the fact that every automorphism of $\mathbb{R}^{n+1}$ maps 2-planes to 2-planes. □

The next example shows us a way to find homographic images of lines in $\mathbb{R}P^2$.

Example 7.2.8 Let a line $\mathbf{L}$ of $\mathbb{R}P^2$ with equation $[\mathbf{A}] \cdot \mathbf{X} = 0$ and let $[g] \in P(2)$. For $[\mathbf{X}] \in \mathbf{L}$ we set $\mathbf{X}' = \mathbf{X}A_g$ and then

$$\mathbf{X}' = \mathbf{X}A_g \implies \mathbf{X}'A_{g^{-1}} = \mathbf{X}.$$

Thus,

$$0 = [\mathbf{A}] \cdot \mathbf{X} = [\mathbf{A}] \cdot (\mathbf{X}' A_{g^{-1}}) = [\mathbf{A}(A_{g^{-1}})^T] \cdot \mathbf{X}'.$$

In other words, the $[g]$-image of $\mathbf{L}: \ [\mathbf{A}] \cdot \mathbf{X} = 0$ is $\mathbf{L}': \ [\mathbf{A}(A_{g^{-1}})^T] \cdot \mathbf{X} = 0$.

As an application, let

$$\mathbf{L}: \ [\mathbf{A}] \cdot \mathbf{X} = 2x_0 + x_1 - 3x_2 = 0$$

and the transformation $[g]$ with

$$A_g = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 3 \\ -2 & 0 & 1 \end{pmatrix} \in \mathrm{GL}(3, \mathbb{R}).$$

The inverse of A_g is

$$A_{g^{-1}} = \begin{pmatrix} 1/3 & 0 & -1/3 \\ -7/3 & 1 & -2/3 \\ 2/3 & 0 & 1/3 \end{pmatrix}.$$

So, for $\mathbf{L}' = [g](\mathbf{L})$ we have

$$[\mathbf{A}(A_{g^{-1}})^T] = [5 : -5 : 1]$$

and thus $\mathbf{L}'$ has equation $5x_0 - 5x_1 + x_2 = 0$.

Corollary 7.2.9 *Every element of $P(n)$ preserves incidence: it maps the point of intersection of lines to the intersection of their images.*

Proof Let $\mathbf{L}$ and $\mathbf{L}'$ be lines of $\mathbb{R}P^n$ be and assume that they intersect at $[\mathbf{x}]$. If $[g] \in P(n)$, then $[g]([\mathbf{x}])$ belongs to the intersection $[g](\mathbf{L}) \cup [g](\mathbf{L}')$ consisting of a unique point. □

Remark 7.2.10 In the special case of lines $\mathbf{L}: \ [\mathbf{A}] \cdot \mathbf{X} = 0$ and $\mathbf{L}': \ [\mathbf{A}'] \cdot \mathbf{X} = 0$ of $\mathbb{R}P^2$, they intersect at $[\mathbf{A} \times \mathbf{A}']$. If $[g] \in P(2)$ then the point $[(\mathbf{A} \times \mathbf{A}')A_g])$ is the intersection point of the lines $[g](\mathbf{L}) \cup [g](\mathbf{L}')$.

Definition 7.2.11 A bijection of $\mathbb{R}P^n$ is called *collineation* if the images of collinear points are themselves collinear.

The set of collineations is a group (group operation is composition of mappings) which we will denote by $P_c(n)$. From Proposition 7.2.7 we immediately have the following:

Corollary 7.2.12 *Every element of $P(n)$ is a collineation.*

Remark 7.2.13 In principle, not all collineations are necessarily homographies. The projective group $P(n)$ is generally a normal subgroup of $P_c(n)$. In dimension 1, $P_c(1) = \mathrm{Sym}(\mathbb{R}P^1)$ since every bijective self-image of the projective line is trivially a collineation. In higher dimensions ($n \geq 2$), it is proved that every collineation is a homography, see [28], p. 30.

7.2.3 Fundamental Theorem of Projective Geometry

Following the same pattern as in the case of the affine group, from the double transitive action of $P(n)$ we can pass to the transitive action of the projective group on $k+1 < n+1$ points $[\mathbf{x}_0], \ldots, [\mathbf{x}_{k+1}]$, such that $\mathbf{x}_0, \ldots, \mathbf{x}_{k+1}$ are linearly independent vectors of $\mathbb{R}^{n+1}$. The Fundamental Theorem of Projective Geometry ensures the transitive action on $(n+1)$-sets of points that constitute a projective basis. As in the affine case, we need the following:

Lemma 7.2.14 *Every element $[g] \in P(n)$ that stabilises $n+2$ points of a projective basis of $\mathbb{R}P^n$ is the identity map.*

Proof Let the given points be $[\mathbf{v}_0], \ldots, [\mathbf{v}_n]$ and $[\mathbf{v}_{n+1}]$, where $\mathbf{v}_{n+1} = \sum_{i=0}^{n} \lambda_i \mathbf{v}_i$, $\lambda_0, \ldots, \lambda_n$ are all nonzero. We consider the linear map $h = h_{\lambda_0,\ldots,\lambda_n} : \mathbb{R}^{n+1} \to \mathbb{R}^{n+1}$ with matrix

$$A_h = \begin{pmatrix} \lambda_0 \mathbf{v}_0 \\ \vdots \\ \lambda_n \mathbf{v}_n \end{pmatrix}.$$

The matrix $A_h \in \mathrm{GL}(n, \mathbb{R})$: indeed

$$\det(A_h) = \left(\prod_{i=0}^{n} \lambda_i \right) \cdot \begin{vmatrix} \mathbf{v}_0 \\ \vdots \\ \mathbf{v}_n \end{vmatrix} \neq 0,$$

since all $\lambda_i \neq 0$ and $\mathbf{v}_i$ are linearly independent. Now,

$$[h]([1:0:\cdots:0]) = [\mathbf{v}_0], \quad [h]([0:\cdots:0:1]) = [\mathbf{v}_n]$$

and

$$[h]([1:1:\cdots:1]) = \left[\sum_{i=0}^{n} \lambda_i \mathbf{v}_i \right] = [\mathbf{v}_{n+1}].$$

Therefore, we can normalise and assume that there is a homography $[g]$ that stabilises the elements of the canonical projective basis. If

$$[A_g] = \begin{bmatrix} a_{00} & \dots & a_{0n} \\ \vdots & \dots & \vdots \\ a_{n0} & \dots & a_{nn} \end{bmatrix},$$

then for each $j = 0, \dots n$ we have

$$[g]([\mathbf{e}_j]) = [a_{j0} : \cdots : a_{jn}] = [\mathbf{e}_j],$$

from which we conclude that all non-diagonal elements of $[A_g]$ are zero. On the other hand,

$$[g]([1 : 1 : \cdots : 1]) = [g_{00} : g_{11} : \cdots : g_{nn}] = [1 : 1 : \cdots : 1],$$

which implies that all diagonal elements are equal. That is, $[g]$ is the identity mapping. □

Theorem 7.2.15 (Fundamental Theorem of Projective Geometry) *Given two projective bases*

$$[\mathbf{v}_0], \dots, [\mathbf{v}_n],\ [\mathbf{v}_{n+1}] \text{ and } [\mathbf{v}'_0], \dots, [\mathbf{v}'_n],\ [\mathbf{v}'_{n+1}],$$

of $\mathbb{R}P^n$*, there exists a unique* $[g] \in P(n)$ *such that*

$$[g]([\mathbf{v}_i]) = [\mathbf{v}'_i], \quad i = 0, 1, \dots, n+1.$$

Proof From the proof of Lemma 7.2.14 we have that there exist $[h], [h'] \in P(n)$ satisfying the relations:

$$[h]([\mathbf{e}_i]) = \mathbf{v}_i, \quad [h']([\mathbf{e}_i]) = \mathbf{v}'_i, \quad i = 0, \dots, n+1.$$

The homography $[g] = [h'] \circ [h^{-1}]$ is the desired one. It is also unique: if we assume that there exists a homography $[g']$ such that $[g']([\mathbf{v}_i]) = [\mathbf{v}'_i]$, $i = 0, 1, \dots, n+1$, then the element $[g'] \circ [g^{-1}]$ stabilises the $n+2$ points of the canonical projective basis. Therefore, by Lemma 7.2.14 it cannot be other but the identity element of $P(n)$. □

Example 7.2.16 We will find a homography $[g] \in P(1)$ that maps $[1 : 0]$, $[0 : 1]$, $[1 : 1]$ to $[1 : 2]$, $[-1 : 1]$ and $[2 : 1]$, respectively. Let $\lambda, \mu \in \mathbb{R}_*$ and let $h = h_{\lambda,\mu}$ be the automorphism of $\mathbb{R}^2$ with matrix

$$A_h = \begin{pmatrix} \lambda & 2\lambda \\ -\mu & \mu \end{pmatrix}.$$

Then

$$[h]([1:1]) = [\lambda - \mu \ : \ 2\lambda + \mu].$$

The system

$$\lambda - \mu = 2, \quad 2\lambda + \mu = 1$$

has solutions $\lambda = 1, \mu = -1$. Therefore, the desired transformation is $[h] = [h_{1,-1}]$ with matrix

$$\begin{bmatrix} 1 & 2 \\ 1 & -1 \end{bmatrix},$$

that is,

$$[h]([x_0 : x_1]) = [x_0 + x_1 \ : \ 2x_0 - x_1].$$

Example 7.2.17 Following the previous example, we now detect the element of $P(1)$ that maps $[1 : 2], [-1 : 1]$ and $[2 : 1]$ to $[2 : 1], [-1 : 1]$ and $[1 : 2]$, respectively. We have that $[h]$ maps $[1 : 0], [0 : 1]$ and $[1 : 1]$ to $[1 : 2], [-1 : 1]$ and $[2 : 1]$, respectively. So we only find the transformation $[h']$ that maps $[1 : 0], [0 : 1]$ and $[1 : 1]$ to $[2 : 1], [-1 : 1]$ and $[1 : 2]$, respectively. As before, we consider $h'_{\lambda,\mu}$ and its matrix

$$A_{h'} = \begin{pmatrix} 2\lambda & \lambda \\ -\mu & \mu \end{pmatrix}.$$

It is $[h']([1 : 1]) = [2\lambda - \mu \ : \ \lambda + \mu]$ and from the relations

$$2\lambda - \mu = 1, \quad \lambda + \mu = 2,$$

we get $\lambda = \mu = 1$. Therefore, $[h']$ has a matrix

$$\begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix},$$

that is,

$$[h']([x_0 : x_1]) = [2x_0 - x_1 \ : \ x_0 + x_1].$$

The desired transformation is $[g] = [h'] \circ [h^{-1}]$. Its matrix is

$$[A_{h'} A_{h^{-1}}] = \left[\begin{pmatrix} 2 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1/3 & 2/3 \\ 1/3 & -1/3 \end{pmatrix} \right] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Therefore,

$$[g]([x_0 : x_1]) = [x_1 : x_0].$$

Example 7.2.18 To find the homography $[g]$ of $\mathbb{R}P^2$ such that

$$[1 : 0 : 0] \mapsto [0 : 0 : 1],$$
$$[0 : 1 : 0] \mapsto [0 : 1 : 1],$$
$$[0 : 0 : 1] \mapsto [1 : 1 : 1],$$
$$[1 : 1 : 1] \mapsto [3 : 2 : 4],$$

let $\lambda, \mu, \nu \neq 0$ and

$$A_g = \begin{pmatrix} 0 & 0 & \lambda \\ 0 & \mu & \mu \\ \nu & \nu & \nu \end{pmatrix}.$$

Then from the relation

$$[g]([1 : 1 : 1]) = [\nu : \mu + \nu : \lambda + \mu + \nu] = [3 : 2 : 4],$$

we have $\lambda = 2$, $\mu = -1$, $\nu = 3$ and consequently

$$[g]([x_0 : x_1 : x_2]) = [3x_2 : -x_1 + 3x_2 : 2x_0 - x_1 + 3x_2].$$

7.2.4 Cross-Ratio

In this section we introduce the *cross-ratio of four collinear points*. The cross-ratio was known to Euclid and its fundamental property (Proposition 7.2.22 below) was proved by Pappus. Our motivation is the following: if we are given two ordered triples of pairwise distinct collinear points, there is a homography of $P(n)$ that maps the points of one triple to the corresponding points of the other. We can assume that the lines on which our triples lie are both the projective line $\mathbb{R}P^1$. Then by the Fundamental Theorem 7.2.15 the homography of $P(1)$ that maps the points of one triple to the corresponding points of the other is unique. The cross-ratio now

naturally appears when investigating the following question: if we are given two ordered quadruples of collinear points, when does there exist $[g] \in P(n)$ that maps the points of one quadruple to the corresponding points of the other?

We start by setting some requirements for the points that we will examine.

Definition 7.2.19 A quadruple of collinear points in the projective space has the *property* (T) if any three of its points are not the same. If the points of the quadruple are pairwise distinct, then we will say that the quadruple has the *property* (Π).

The cross-ratio is defined for quadruples of points with the property (T). We consider

$$[\mathfrak{p}] = \{[\mathbf{A}],\ [\mathbf{B}],\ [\mathbf{C}],\ [\mathbf{D}]\},$$

an ordered quadruple of points with the property (T) and we assume without any loss that $[\mathbf{A}] \neq [\mathbf{B}]$. Then, there exist $p, q, r, s \in \mathbb{R}$ such that

$$\mathbf{C} = p\,\mathbf{A} + q\,\mathbf{B}, \quad \mathbf{D} = r\,\mathbf{A} + s\,\mathbf{B}. \tag{7.7}$$

Definition 7.2.20 The *cross-ratio of the ordered quadruple* $[\mathfrak{p}]$ is defined as:

$$[\mathbf{A}, \mathbf{B}; \mathbf{C}, \mathbf{D}] = \frac{qr}{ps}.$$

The definition of the cross-ratio is independent of the choice of representatives: indeed, if $\mathbf{A}', \mathbf{B}', \mathbf{C}', \mathbf{D}'$ are some other representatives of $[\mathbf{A}], [\mathbf{B}], [\mathbf{C}], [\mathbf{D}]$, then there exist a, b, c, d all nonzero such that $\mathbf{A}' = a\,\mathbf{A}$, $\mathbf{B}' = b\,\mathbf{B}$, $\mathbf{C}' = c\,\mathbf{C}$ and $\mathbf{D}' = d\,\mathbf{D}$. Then,

$$\mathbf{C}' = c\,\mathbf{C} = cp\,\mathbf{A} + cq\,\mathbf{B} = (cp/a)\,\mathbf{A}' + (cq/b)\,\mathbf{B}',$$
$$\mathbf{D}' = d\,\mathbf{D} = dr\,\mathbf{A} + ds\,\mathbf{B} = (dr/a)\,\mathbf{A}' + (ds/b)\,\mathbf{B}',$$

therefore

$$[\mathbf{A}', \mathbf{B}'; \mathbf{C}', \mathbf{D}'] = \frac{(cq/b)(dr/a)}{(cp/a)(ds/b)} = \frac{qr}{ps} = [\mathbf{A}, \mathbf{B}; \mathbf{C}, \mathbf{D}].$$

We also observe that the cross-ratio of a quadruple of points with property (Π) takes values in $\mathbb{R} \setminus \{0, 1\}$. Indeed, first of all it cannot be 0: if $q = 0$ then $[\mathbf{C}] = [\mathbf{A}]$ while if $r = 0$ then $[\mathbf{D}] = [\mathbf{B}]$. Similarly, if the cross-ratio is equal to 1 we again end up with a contradiction. Indeed, from the relation $qr = ps$ we have:

$$\mathbf{C} = (qr/s)\,\mathbf{A} + q\,\mathbf{B} = (sq)\,\mathbf{D} \implies [\mathbf{C}] = [\mathbf{D}].$$

A natural and interesting question regarding the definition of the cross-ratio is the following: what happens to the cross-ratio of four points when we apply to it an

element of the permutation group S_4? There are known to be 24 permutations in S_4 but the cross-ratio resulting from any of them is a function of a predetermined cross-ratio. (See Exercise 22, Sect. 7.6.)

Remark 7.2.21 Let $\mathbf{L}$ be the line passing through the distinct points $[\mathbf{v}_1]$ and $[\mathbf{v}_2]$ of $\mathbb{R}P^n$. Each point of $[\mathbf{x}(\lambda, \mu)]$ is of the form

$$[\mathbf{x}(\lambda, \mu)] = [\lambda \mathbf{v}_1 + \mu \mathbf{v}_2], \quad |\lambda| + |\mu| \neq 0,$$

therefore $[\mathbf{x}(1, 0)] = [\mathbf{v}_1]$, $[\mathbf{x}(0, 1)] = [\mathbf{v}_2]$. Then if $[\mathbf{x}(\lambda_1, \mu_1)] = [\mathbf{v}_3]$ and $[\mathbf{x}(\lambda_2, \mu_2)] = [\mathbf{v}_4]$ are points on the line such that the quadruple $[\mathfrak{p}] = \{[\mathbf{v}_1], [\mathbf{v}_2], [\mathbf{v}_3], [\mathbf{v}_4]\}$ has the property (T), we get

$$[\mathbf{v}_1, \mathbf{v}_2; \mathbf{v}_3, \mathbf{v}_4] = \frac{\lambda_2 \mu_1}{\lambda_1 \mu_2}.$$

Proposition 7.2.22 *The cross-ratio is a projective invariant: it remains invariant under the action of $P(n)$.*

Proof Let $[\mathbf{A}]$, $[\mathbf{B}]$, $[\mathbf{C}]$, $[\mathbf{D}]$ be a quadruple of points with the property (T) and suppose without any loss that $[\mathbf{A}] \neq [\mathbf{B}]$, so that (7.7) holds. If $[g] \in P(n)$ with corresponding linear transformation g, then

$$g(\mathbf{C}) = p\, g(\mathbf{A}) + q\, g(\mathbf{B}), \quad g(\mathbf{D}) = r\, g(\mathbf{A}) + s\, g(\mathbf{B}).$$

Now, $[g(\mathbf{A}))] \neq [g(\mathbf{B})]$. This is because, if the opposite were true, there would be a $\lambda \in \mathbb{R}_*$ with $g(\mathbf{B}) = \lambda\, g(\mathbf{A})$. However, since g is invertible, this would imply that $\mathbf{B} = \lambda \mathbf{A}$ and therefore $[\mathbf{A}] = [\mathbf{B}]$, which contradicts the hypothesis. Therefore,

$$[g(\mathbf{A}), g(\mathbf{B}); g(\mathbf{C}), g(\mathbf{D})] = \frac{qr}{ps} = [\mathbf{A}, \mathbf{B}; \mathbf{C}, \mathbf{D}].$$

□

The following theorem is of particular importance.

Theorem 7.2.23 (Fundamental Theorem of Cross-Ratios) *Let*

$$[\mathfrak{p}] = \{[\mathbf{A}], [\mathbf{B}], [\mathbf{C}], [\mathbf{D}]\}, \quad [\mathfrak{p}'] = \{[\mathbf{A}'], [\mathbf{B}'], [\mathbf{C}'], [\mathbf{D}']\},$$

quadruples of collinear points with the property (Π). Then there exists a $[g] \in P(n)$ such that $[g]([\mathfrak{p}]) = [\mathfrak{p}']$, that is,

$$[g]([\mathbf{A}]) = [\mathbf{A}'], \ [g]([\mathbf{A}]) = [\mathbf{B}'], \ [g]([\mathbf{C}]) = [\mathbf{C}'], \ [g]([\mathbf{D}]) = [\mathbf{D}'],$$

if and only if

$$[\mathbf{A}, \mathbf{B}; \mathbf{C}, \mathbf{D}] = [\mathbf{A}', \mathbf{B}'; \mathbf{C}', \mathbf{D}'].$$

Proof If there exists a $[g] \in P(n)$ with

$$[g(\mathbf{A})] = [\mathbf{A}'], \quad [g(\mathbf{B})] = [\mathbf{B}'], \quad [g(\mathbf{C})] = [\mathbf{C}'], \quad [g(\mathbf{D})] = [\mathbf{D}'],$$

then the claim follows from Proposition 7.2.22. Now let

$$[\mathbf{A}, \mathbf{B}; \mathbf{C}, \mathbf{D}] = [\mathbf{A}', \mathbf{B}'; \mathbf{C}', \mathbf{D}'].$$

By applying appropriate homographies we can assume that both quadruples lie on the projective line. Then, we can also normalise so that

$$[\mathbf{A}] = [\mathbf{A}'] = [1 : 0], \quad [\mathbf{B}] = [\mathbf{B}'] = [0 : 1], \quad [\mathbf{C}] = [\mathbf{C}'] = [1 : 1].$$

Therefore, if $[\mathbf{D}] = [d_1 : d_2]$ and $[\mathbf{D}'] = [d_1' : d_2']$, we have

$$[\mathbf{A}, \mathbf{B}; \mathbf{C}, \mathbf{D}] = d_1/d_2, \quad [\mathbf{A}', \mathbf{B}'; \mathbf{C}', \mathbf{D}'] = d_1'/d_2'.$$

It follows that $[\mathbf{D}] = [\mathbf{D}']$ and this completes the proof. □

The following corollary of Theorem 7.2.23 is also called the *Fundamental Theorem of Planar Projective Geometry*.

Corollary 7.2.24 *Let* $\mathbf{L}$ *and* $\mathbf{L}'$ *be straight lines of* $\mathbb{R}P^2$. *A homography* $[g] \in P(2)$ *mapping* $\mathbf{L}$ *to* $\mathbf{L}'$ *is completely determined by its action on three pairwise distinct points of* $\mathbf{L}$.

Proof We can normalise so that $\mathbf{L} = \mathbf{L}' = \mathbb{R}P^1$. Let $[\mathbf{A}]$, $[\mathbf{B}]$ and $[\mathbf{C}]$ be three preselected pairwise distinct points of $\mathbb{R}P^1$ and let $[\mathbf{X}]$ be any point of $\mathbb{R}P^1$ different from the preselected points. Let

$$[\mathbf{A}, \mathbf{B}; \mathbf{C}, \mathbf{X}] = \lambda \in \mathbb{R} \setminus \{0, 1\}.$$

We observe that

$$[(1, 0), (0, 1); (1, 1), (\lambda, 1)] = \lambda = [\mathbf{A}, \mathbf{B}; \mathbf{C}, \mathbf{D}].$$

According to Theorem 7.2.15 and Lemma 7.2.14, there exists a *unique* $[g] \in P(1)$ such that

$$[g]([1 : 0]) = [\mathbf{A}], \quad [g]([0 : 1]) = [\mathbf{B}], \quad [g]([1 : 1]) = [\mathbf{C}], \quad [g]([\lambda : 1]) = [\mathbf{X}].$$

If we assume that $[\mathbf{X}]$ is identical to one of the preselected points, then we can also get the values $0, 1, \infty$ for λ. □

Remark 7.2.25 We conclude from the above that when we are given a quadruple points of $\mathbb{R}P^1$ with the property (Π), we may always assume that these points are

$$[1:0], \quad [0:1], \quad [1:1], \quad [\lambda:1],$$

where λ is the cross-ratio of the given points.

The next proposition follows immediately from Corollary 7.2.24, but we will also give another proof.

Proposition 7.2.26 (Four-Point Theorem) *Let* $[\mathbf{A}]$, $[\mathbf{B}]$, $[\mathbf{C}]$, $[\mathbf{X}]$, $[\mathbf{Y}]$ *be collinear points such that the quadruples*

$$[\mathfrak{p}] = \{[\mathbf{A}], [\mathbf{B}], [\mathbf{C}], [\mathbf{X}]\}, \quad [\mathfrak{p}'] = \{[\mathbf{A}], [\mathbf{B}], [\mathbf{C}], [\mathbf{Y}]\},$$

have the property (T). Then,

$$[\mathbf{A}, \mathbf{B}; \mathbf{C}, \mathbf{X}] = [\mathbf{A}, \mathbf{B}; \mathbf{C}, \mathbf{Y}] \implies [\mathbf{X}] = [\mathbf{Y}].$$

Proof Let

$$\mathbf{C} = p\mathbf{A} + q\mathbf{B}, \quad \mathbf{X} = r\mathbf{A} + s\mathbf{B}, \quad \mathbf{Y} = r'\mathbf{A} + s'\mathbf{B}.$$

Then,

$$[\mathbf{A}, \mathbf{B}; \mathbf{C}, \mathbf{X}] = [\mathbf{A}, \mathbf{B}; \mathbf{C}, \mathbf{Y}] \implies \frac{qr}{ps} = \frac{qr'}{ps'} \implies \frac{r}{s} = \frac{r'}{s'}.$$

So,

$$\begin{aligned}\mathbf{Y} &= r'\mathbf{A} + s'\mathbf{B} = (r'/r)r\mathbf{A} + (s'/s)s\mathbf{B} = (r'/r)(r\mathbf{A} + s\mathbf{B}) \\ &= (r'/r)\mathbf{X} \implies [\mathbf{X}] = [\mathbf{Y}].\end{aligned}$$

□

The Cross-Ratio on the Projective Line $\mathbb{R}P^1$

By identifying $\mathbb{R}P^1$ with $S^1 = \mathbb{R} \cup \{\infty\} = \overline{\mathbb{R}}$, the cross-ratio of a quadruple of points of $\overline{\mathbb{R}}$ with property (T) can be defined as follows: let $a, b, c, d \in \overline{\mathbb{R}}$ be such a quadruple. If none of the points is the ∞, we consider their lifts

$$[\mathbf{A}] = [a:1], \quad [\mathbf{B}] = [b:1], \quad [\mathbf{C}] = [c:1], \quad [\mathbf{D}] = [d:1].$$

If $a \neq b$, then since

$$c = \frac{c-b}{a-b}\,a + \frac{a-c}{a-b}\,b,$$
$$d = \frac{d-b}{a-b}\,a + \frac{a-d}{a-b}\,b,$$

we have

$$[\mathbf{A}, \mathbf{B}; \mathbf{C}, \mathbf{D}] = \frac{(d-b)(c-a)}{(d-a)(c-b)},$$

which we define as the *cross-ratio* $[a, b; c, d]$ of a, b, c, d. Of course, we need to make some assumptions in the case where two of the points are equal: for example, if $b = c$, then we say that $[a, b; c, d] = \infty$. Thus, the cross-ratio takes values in $\overline{\mathbb{R}}$. But again we have that if our quadruple of points has the property (Π), then the cross-ratio takes values in $\mathbb{R} \setminus \{0, 1\}$. Let us now observe that with the additional assumption $\infty : \infty = 1$, the cross-ratio

$$[a, b; c, d] = \frac{(d-b)(c-a)}{(d-a)(c-b)},$$

is also defined in the case where one of the points a, b, c, d is the point at infinity. We will return to this discussion of the cross-ratio in Sect. 8.6.3.

Example 7.2.27 We will find the cross-ratio of

$$[\mathbf{A}] = [1 : 2], \quad [\mathbf{B}] = [2 : 1], \quad [\mathbf{C}] = [1 : 0], \quad [\mathbf{D}] = [1 : 1]$$

in two ways. First, we have

$$\mathbf{C} = (1, 0) = (-1/3)\mathbf{A} + (2/3)\mathbf{B}, \quad \mathbf{D} = (1, 1) = (1/3)\mathbf{A} + (1/3)\mathbf{B}.$$

Then,

$$[\mathbf{A}, \mathbf{B}; \mathbf{C}, \mathbf{D}] = \frac{(2/3) \cdot (1/3)}{(-1/3) \cdot (1/3)} = -2.$$

Moreover, since $[\mathbf{A}]$, $[\mathbf{B}]$, $[\mathbf{D}]$ belong to the set H_0, we consider the projections

$$a = 1/2, \quad b = 2, \quad d = 1,$$

of $[\mathbf{A}]$, $[\mathbf{B}]$, $[\mathbf{D}]$, respectively, onto the real line, while taking $c = \infty$. Then,

$$[a, b; c, d] = \frac{(1-2)(\infty - 1/2)}{(1-1/2)(\infty - 2)} = \frac{-1}{1/2} \cdot \frac{\infty}{\infty} = (-2) \cdot 1 = -2.$$

Complete Quadrilaterals, Harmonic Quadrilaterals

Definition 7.2.28 Let $[\mathbf{A}_i]$, $i = 1, 2, 3, 4$ be coplanar, pairwise distinct points, such that there are no three collinear points. We will then say that they form a *complete quadrilateral*. The four points $[\mathbf{A}_i]$ will be called *vertices* of the quadrilateral and the six lines $\mathbf{L}_{i,j}$ from $[\mathbf{A}_i]$, $[\mathbf{A}_j]$, $i \neq j$, will be called *sides* of the quadrilateral. The intersection points of the opposite sides of the quadrilateral will be called *diagonal points*. The *opposite sides* of the quadrilateral are the pairs whose common point is not a vertex.

Without any loss, we consider four pairwise distinct points of $\mathbb{R}P^2$ such that no three of them are collinear. According to the Fundamental Theorem 7.2.15 we can assume that

$$[\mathbf{A}_1] = [1:0:0], \quad [\mathbf{A}_2] = [0:1:0], \quad [\mathbf{A}_3] = [0:0:1], \quad [\mathbf{A}_4] = [1:1:1].$$

We have the straight lines:

$$\mathbf{L}_{1,2}: \ [\mathbf{A}_3]\cdot\mathbf{X} = 0, \quad \mathbf{L}_{2,3}: \ [\mathbf{A}_1]\cdot\mathbf{X} = 0, \quad \mathbf{L}_{3,4}: \ [-1:1:0]\cdot\mathbf{X} = 0,$$
$$\mathbf{L}_{4,1}: \ [0:-1:1]\cdot\mathbf{X} = 0, \quad \mathbf{L}_{1,3}: \ [\mathbf{A}_2]\cdot\mathbf{X} = 0, \quad \mathbf{L}_{2,4}: \ [1:0:-1]\cdot\mathbf{X} = 0.$$

The *diagonal points* of the quadrilateral are:

- $[\mathbf{D}_1] = \mathbf{L}_{1,3} \cap \mathbf{L}_{2,4} = [1:0:1]$,
- $[\mathbf{D}_2] = \mathbf{L}_{1,2} \cap \mathbf{L}_{3,4} = [1:1:0]$,
- $[\mathbf{D}_3] = \mathbf{L}_{2,3} \cap \mathbf{L}_{4,1} = [0:1:1]$.

The line from $[\mathbf{D}_1]$ and $[\mathbf{D}_3]$ is

$$[-1:-1:1]\cdot\mathbf{X} = 0,$$

and let $[\mathbf{C}_1] = [-1:1:0]$ be its intersection with $\mathbf{L}_{1,2}$. Then, since

$$\mathbf{C}_1 = -\mathbf{A}_1 + \mathbf{A}_2, \quad \mathbf{D}_2 = \mathbf{A}_1 + \mathbf{A}_2 \implies [\mathbf{A}_1, \mathbf{A}_2; \mathbf{C}_1, \mathbf{D}_2] = -1.$$

When the cross-ratio of four points $[\mathbf{A}_1, \mathbf{A}_2; \mathbf{C}_1, \mathbf{D}_2]$ is equal to -1, we say that the points are in *harmonic position.* We also say that the point $[\mathbf{D}_2]$ is the *harmonic conjugate of* $[\mathbf{C}_1]$ *with respect to* $[\mathbf{A}_1]$, $[\mathbf{A}_2]$. We have proved the following:

Proposition 7.2.29 *A complete quadrilateral corresponds to four points in harmonic position.*

In Exercise 28, Sect. 7.6 we prove that the converse is also true.

7.3 Duality

Recall that if V is a vector space over the field $\mathbb{K}$, the *dual* V^* of V is the vector space consisting of all linear mappings $f : V \to \mathbb{K}$. The spaces V, V^* are isomorphic and also, there is a canonical isomorphism $(V^*)^* \to V$ which allows us to identify these two spaces. In the special case where $V = \mathbb{R}^{n+1}$ we have $(\mathbb{R}^{n+1})^* = \mathbb{R}^{n+1}$ since a linear mapping $f : \mathbb{R}^{n+1} \to \mathbb{R}$ is completely determined by its matrix with respect to the canonical basis; the latter is nothing but a vector of $\mathbb{R}^{n+1}$. The duality principle that we describe below is essentially an injective mapping of $\mathbb{R}P^n = \pi(\mathbb{R}^{n+1})$ to $\mathbb{R}P^n = \pi((\mathbb{R}^{n+1})^*)$. The idea of the duality principle is very old, it goes back to the French mathematician and logician Gergonne (Joseph Diez Gergonne, 1771–1859) and Poncelet.

7.3.1 Duality Principle

We denote by $\mathcal{H}$ the set of hyperplanes of $\mathbb{R}P^n$. We recall that if $\mathbf{W} \in \mathcal{H}$, then it is given by an equation of the form

$$[\mathbf{A}] \cdot \mathbf{X} = 0, \quad [\mathbf{A}] \in \mathbb{R}P^n, \ \mathbf{X} = (x_0, \ldots, x_n).$$

The point $[\mathbf{A}] = [a_0 : \cdots : a_n]$ is called *dual point of the hyperplane* $\mathbf{W}$. Conversely, if $[\mathbf{A}] \in \mathbb{R}P^n$, then the hyperplane $\mathbf{W}$ with equation $[\mathbf{A}] \cdot \mathbf{X} = 0$ is called *dual hyperplane of* $[\mathbf{A}]$.

We conclude that there exists a bijective correspondence

$$\mathfrak{D} : \mathbb{R}P^n \to \mathcal{H}$$

between points and hyperplanes, which comes from the 1–2 correspondence of hyperplanes of $\mathbb{R}^{n+1}$ from the origin and directions: to each hyperplane from the origin we correspond its two unit perpendicular vectors. In this way the set $\mathcal{H}$ acquires the structure of a projective space whose points are hyperplanes. The mapping $\mathfrak{D}$ is called *duality mapping*, or simply *duality*. There is no concept corresponding to duality in Euclidean Geometry.

It is interesting to examine the effect of duality on subspaces of $\mathcal{H}$. A k- subspace $\mathbf{V}$ of $\mathbb{R}P^n$ is the π-image of a $(k+1)$-dimensional subspace V of $\mathbb{R}^{n+1}$. Let $V^\perp$ be the orthocomplement of V: this is an $(n-k)$-dimensional subspace of V described as the intersection of $n-k$ hyperplanes of $\mathbb{R}^{n+1}$. Then

$$\mathfrak{D}(\mathbf{V}) = \mathbf{V}^\perp = \pi(\mathbf{V}^\perp).$$

Definition 7.3.1 A collection of points $\mathbf{\Sigma}$ of $\mathbb{R}P^n$ will be called a *range*. If $\mathbf{\Sigma}$ lies on a line $\mathbf{L}$, then $\mathbf{L}$ will be called an *axis* of the range $\mathbf{\Sigma}$. The dual set $\mathbb{P} = \mathfrak{D}(\mathbf{\Sigma})$ of the range $\mathbf{\Sigma}$ will be called a *pencil of hyperplanes homologous to* $\mathbf{\Sigma}$.

If $\mathbf{\Sigma}$ has an axis $\mathbf{L}: \ [\mathbf{O}] \cdot \mathbf{X} = 0$, then all hyperplanes of $\mathbb{P}$ contain a projective subspace of dimension $n-2$. In the case where $n = 2$, a pencil of hyperplanes is a pencil of *lines*, the lines of the pencil $\mathbb{P}$ passing through $[\mathbf{O}]$ of $\mathbf{L}$. In some cases the pencil $\mathbb{P}$ is called the *central pencil* and $[\mathbf{O}]$ is called the *vertex* of $\mathbb{P}$.

The simplest case of duality is of course when $n = 2$. There the set $\mathcal{H}$ is the set of lines and the mapping $\mathfrak{D}$ assures us that:

- A point $[\mathbf{A}]$ is mapped onto the line $[\mathbf{A}] \cdot \mathbf{X} = 0$ (the dual line of $[\mathbf{A}]$).
- A line $\mathbf{L}: \ [\mathbf{A}] \cdot \mathbf{X} = 0$ is mapped onto the point $[\mathbf{A}]$ (the dual point of $[\mathbf{L}]$).
- Two points $[\mathbf{A}]$, $[\mathbf{B}]$ are mapped onto their dual lines $\mathbf{L}_{[\mathbf{A}]}$, $\mathbf{L}_{[\mathbf{B}]}$, respectively, such that the intersection

$$L_{[\mathbf{A}]} \cup L_{[\mathbf{B}]} = [\mathbf{A} \times \mathbf{B}],$$

  that is the dual point of the line passing through $[\mathbf{A}]$, $[\mathbf{B}]$.
- A line $\mathbf{L}$ containing two points $[\mathbf{A}]$, $[\mathbf{B}]$ is mapped onto the intersection of the dual lines of $[\mathbf{A}]$, $[\mathbf{B}]$.
- If $[\mathbf{A}] \in \mathbf{L}$, then the dual line of $[\mathbf{A}]$ contains the dual point of $\mathbf{L}$.

The above is summarised in the following:

Duality Principle *The validity of a proposition of the geometry of* $\mathbb{R}P^2$ *referring to points and lines remains unchanged if we interchange the points with lines and the lines with points.*

Example 7.3.2 The vertices of the triangle defined by the two lines

$$\mathbf{L}_1: \ [\mathbf{A}] \cdot \mathbf{X} = 0, \quad \mathbf{L}_1: \ [\mathbf{B}] \cdot \mathbf{X} = 0, \quad \mathbf{L}_3: \ [\mathbf{C}] \cdot \mathbf{X} = 0,$$

are the points

$$[\mathbf{D}] = [\mathbf{A} \times \mathbf{B}], \quad [\mathbf{E}] = [\mathbf{B} \times \mathbf{C}], \quad [\mathbf{F}] = [\mathbf{C} \times \mathbf{A}].$$

On the other hand, given three pairwise disjoint and not collinear points $[\mathbf{A}]$, $[\mathbf{B}]$, $[\mathbf{C}]$, the sides of the triangle they form are

$$\mathbf{L}_1: \ [\mathbf{D}] \cdot \mathbf{X} = 0, \quad \mathbf{L}_2: \ [\mathbf{E}] \cdot \mathbf{X} = 0, \quad \mathbf{L}_3: \ [\mathbf{F}] \cdot \mathbf{X} = 0.$$

The triangles $\triangle\mathbf{ABC}$ and $\triangle\mathbf{DEF}$ as above are called *dual triangles*.

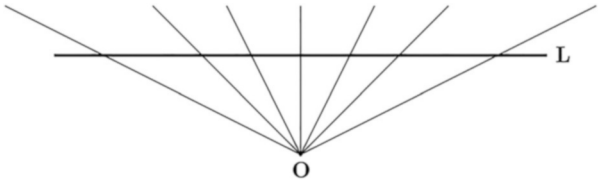


Fig. 7.8 Pencil with of seven lines with transverse line **L**

7.3.2 Cross-Ratios of Pencils

In a pencil $\mathbb{P}$ with axis **L** of the projective plane, it is possible to define the cross-ratio of any four lines of the pencil. This is because the corresponding dual points of each of them belong to the axis **L**. (In other words, four lines belong to a pencil, that is, they are concurrent, if and only if their corresponding dual points are collinear.) (Fig. 7.8).

Definition 7.3.3 Let $\mathbb{P}$ be a pencil of four lines

$$\mathbf{L}_i : \ [\mathbf{A}_i] \cdot \mathbf{X} = 0, \quad i = 1, \ldots, 4.$$

Then *the cross-ratio of* $\mathbb{P}$ is defined to be the cross-ratio $[\mathbf{A}_1, \mathbf{A}_1; \mathbf{A}_3, \mathbf{A}_4]$.

Example 7.3.4 Consider the lines:

$$[\mathbf{A}] \cdot \mathbf{X} = [1 : 1 : 2] \cdot \mathbf{X} = 0, \quad [\mathbf{B}] \cdot \mathbf{X} = [3 : -1 : 4] \cdot \mathbf{X} = 0,$$
$$[\mathbf{C}] \cdot \mathbf{X} = [5 : 1 : 8] \cdot \mathbf{X} = 0, \quad [\mathbf{D}] \cdot \mathbf{X} = [2 : 0 : 3] \cdot \mathbf{X} = 0,$$

We will show that they form a pencil of lines by proving that they are concurrent. We take any two dual points, let $[\mathbf{A}]$ and $[\mathbf{B}]$. Since $\mathbf{A} \times \mathbf{B} = (6, 2, -4)$, the line passing through them is

$$\mathbf{L} : \ [3 : 1 - 2] \cdot \mathbf{X} = 0.$$

We now observe that

$$[3 : 1 : -2] \cdot (5, 1, 8) = [3 : 1 : -2] \cdot (2, 0, 3) = 0,$$

and therefore all the points belong to **L**. We conclude that the given lines are concurrent. Their common point is $[\mathbf{O}] = [3 : 1 : -2]$ which is also the vertex of the pencil. We now calculate the cross-ratio of the pencil. We observe that

$$\mathbf{C} = 2 \cdot \mathbf{A} + 1 \cdot \mathbf{B}, \quad \mathbf{D} = (1/2) \cdot \mathbf{A} + (1/2) \cdot \mathbf{B},$$

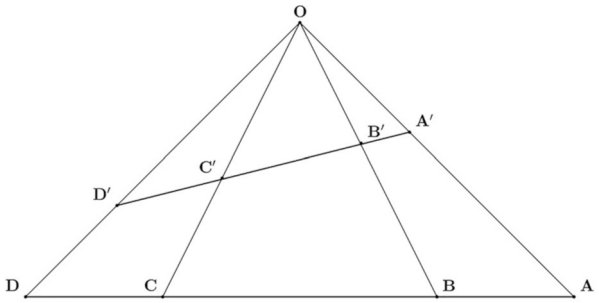


Fig. 7.9 Theorem 7.3.5

so

$$[\mathbf{A}, \mathbf{B}; \mathbf{C}, \mathbf{D}] = 1/2.$$

Every line that does not pass through the vertex of a central pencil $\mathbb{P}$ with axis $\mathbf{L}$ is called a *transverse line to* $\mathbb{P}$. If $\mathbb{P}$ is a pencil of *four* lines, then every transverse line meets $\mathbb{P}$ at four points of intersection. The next result ensures that the cross-ratio of a central pencil of four lines does not depend on any transverse line (not even on the axis) (Fig. 7.9).

Theorem 7.3.5 (Fundamental Theorem of Cross-Ratios of a Pencil of Four Lines.) *The four points of intersection determined by a line transverse to a pencil of four lines have a cross-ratio equal to the cross-ratio of the pencil.*

Proof Suppose that the lines of the pencil have dual points $[\mathbf{A}]$, $[\mathbf{B}]$, $[\mathbf{C}]$, $[\mathbf{D}]$ and let $\mathbf{C} = p\mathbf{A} + q\mathbf{B}$, $\mathbf{D} = r\mathbf{A} + s\mathbf{B}$. Let also $\mathbf{L} : \ [\mathbf{T}] \cdot \mathbf{X} = 0$ be the line transverse to the pencil. The intersection points of $\mathbf{L}$ with the lines of the pencil are

$$[\mathbf{T} \times \mathbf{A}], \quad [\mathbf{T} \times \mathbf{B}], \quad [\mathbf{T} \times \mathbf{C}], \quad [\mathbf{T} \times \mathbf{D}].$$

We thus have

$$\mathbf{T} \times \mathbf{C} = p(\mathbf{T} \times \mathbf{A}) + q(\mathbf{T} \times \mathbf{B}), \quad \mathbf{T} \times \mathbf{D} = r(\mathbf{T} \times \mathbf{A}) + s(\mathbf{T} \times \mathbf{B})$$

and therefore

$$[\mathbf{T} \times \mathbf{A}, \mathbf{T} \times \mathbf{B}; \mathbf{T} \times \mathbf{C}, \mathbf{T} \times \mathbf{D}] = [\mathbf{A}, \mathbf{B}; \mathbf{C}, \mathbf{D}].$$

□

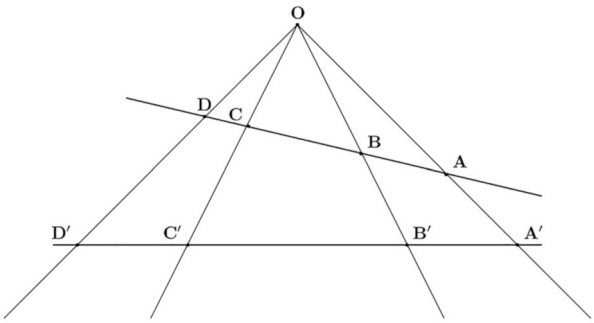


Fig. 7.10 Points in perspective position

Corollary 7.3.6 *Let* $\mathbf{L}$ *and* $\mathbf{L}'$ *be lines transverse to a pencil of four lines that intersect the pencil lines at* $[\mathbf{A}]$, $[\mathbf{B}]$, $[\mathbf{C}]$, $[\mathbf{D}]$ *and* $[\mathbf{A}']$, $[\mathbf{B}']$, $[\mathbf{C}']$, $[\mathbf{D}']$, *respectively. Then*

$$[\mathbf{A}, \mathbf{B}; \mathbf{C}, \mathbf{D}] = [\mathbf{A}', \mathbf{B}'; \mathbf{C}', \mathbf{D}'].$$

The following proposition is a partial converse of the Corollary 7.3.6 (Fig. 7.10).

Proposition 7.3.7 *Let* $\mathbf{L}$ *and* $\mathbf{L}'$ *be lines intersecting at* $[\mathbf{P}]$ *(Fig. 7.11). If* $[\mathbf{A}]$, $[\mathbf{B}]$, $[\mathbf{C}]$, *and* $[\mathbf{A}']$, $[\mathbf{B}']$, $[\mathbf{C}']$, *are triples of points in L and L′, respectively, with*

$$[\mathbf{P}, \mathbf{A}; \mathbf{B}, \mathbf{C}] = [\mathbf{P}, \mathbf{A}'; \mathbf{B}', \mathbf{C}'],$$

then the lines from $[\mathbf{A}]$ *and* $[\mathbf{A}']$, $[\mathbf{B}]$ *and* $[\mathbf{B}']$, $[\mathbf{C}]$ *and* $[\mathbf{C}']$ *are concurrent.*

Proof The lines from $[\mathbf{A}]$ and $[\mathbf{A}']$, $[\mathbf{B}]$ and $[\mathbf{B}']$ intersect at some point $[\mathbf{O}]$. The line from $[\mathbf{O}]$, $[\mathbf{C}]$ intersects $\mathbf{L}'$ at some $[\mathbf{C}'']$. Our goal is to show that $[\mathbf{C}'']$. Since $[\mathbf{P}]$, $[\mathbf{A}]$, $[\mathbf{B}]$, $[\mathbf{C}]$, and $[\mathbf{P}]$, $[\mathbf{A}']$, $[\mathbf{B}']$, $[\mathbf{C}'']$ belong to lines transversal to the pencil with vertex $[\mathbf{O}]$, we have that $[\mathbf{P}, \mathbf{A}; \mathbf{B}, \mathbf{C}] = [\mathbf{P}, \mathbf{A}'; \mathbf{B}', \mathbf{C}']$. But then from the

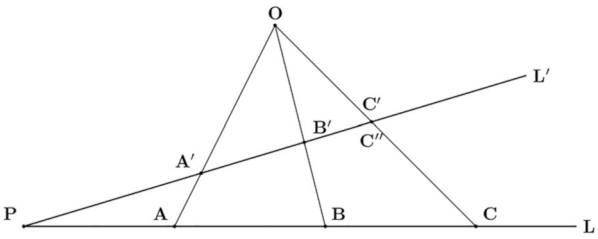


Fig. 7.11 Proposition 7.3.7

assumption we conclude that

$$[\mathbf{P}, \mathbf{A}'; \mathbf{B}', \mathbf{C}''] = [\mathbf{P}, \mathbf{A}'; \mathbf{B}', \mathbf{C}''] \implies [\mathbf{C}''] = [\mathbf{C}''],$$

as follows from the Four Points Theorem (Proposition 7.2.26). □

Remark 7.3.8 The conclusion of the above proposition could also be stated as follows: [**A**], [**B**], [**C**] *and* [**A**′], [**B**′], [**C**′] *are in a perspectivity from some point* [**O**].

7.4 Projectivities and Perspectivities

In this section we will analyse the projective transformations of $\mathbb{R}P^2$ a little more. We start with a general definition.

Definition 7.4.1 Consider the ranges $\mathbf{\Sigma} = ([\mathbf{A}_1], \ldots, [\mathbf{A}_m])$ and $\mathbf{\Sigma}' = ([\mathbf{B}_1], \ldots, [\mathbf{B}_m])$, on $\mathbb{R}P^n$ and suppose that there exists a homography $[g]$ of $P(n)$ such that $[g]([\mathbf{A}_i]) = [\mathbf{B}_i]$, $i = 1, \ldots, m$. We will say that $[g]$ is a *projectivity* between $\mathbf{\Sigma}$ and $\mathbf{\Sigma}'$ and we will write

$$\mathbf{\Sigma} \barwedge \mathbf{\Sigma}', \quad \text{or} \quad ([\mathbf{A}_1], \ldots, [\mathbf{A}_m]) \barwedge ([\mathbf{B}_1], \ldots, [\mathbf{B}_m]).$$

In general, the points of $\mathbf{\Sigma}$, $\mathbf{\Sigma}'$ can be in any position. The dual definition is:

Definition 7.4.2 Consider the pencils of hyperplanes

$$\mathbb{P} = (\mathbf{W}_1 : [\mathbf{A}_1] \cdot \mathbf{X} = 0, \ldots, \mathbf{W}_m : [\mathbf{A}_m] \cdot \mathbf{X} = 0) \quad \text{and}$$
$$\mathbb{P}' = (\mathbf{W}'_1 : [\mathbf{B}_1] \cdot \mathbf{X} = 0, \ldots, \mathbf{W}_m : [\mathbf{B}_m] \cdot \mathbf{X} = 0),$$

in $\mathbb{R}P^n$. We will say that $[g]$ is a *projectivity* between $\mathbb{P}$, $\mathbb{P}'$ and we will write

$$\mathbb{P} \barwedge \mathbb{P}', \quad \text{or} \quad (\mathbf{W}_1, \ldots, \mathbf{W}_m) \barwedge (\mathbf{W}'_1, \ldots, \mathbf{W}'_m),$$

if there is a homography $[g]$ of $P(n)$ such that $[g](\mathbf{W}_i) = \mathbf{W}'_i$, $i = 1, \ldots, m$.

The case $m = 4$ in dimension 2 is of particular interest; the following is a reformulation of the fundamental Theorem 7.2.23:

Proposition 7.4.3 *Consider the quadruples of collinear pairwise distinct points*

$$\mathbf{\Sigma} = ([\mathbf{A}_1], [\mathbf{A}_2], [\mathbf{A}_3], [\mathbf{A}_4]), \quad \mathbf{\Sigma}' = ([\mathbf{B}_1], [\mathbf{B}_2], [\mathbf{B}_3], [\mathbf{B}_4]).$$

Then

$$\mathbf{\Sigma} \barwedge \mathbf{\Sigma}', \quad \textit{or} \quad ([\mathbf{A}_1], [\mathbf{A}_2], [\mathbf{A}_3], [\mathbf{A}_4]) \barwedge ([\mathbf{B}_1], [\mathbf{B}_2], [\mathbf{B}_3], [\mathbf{B}_4]),$$

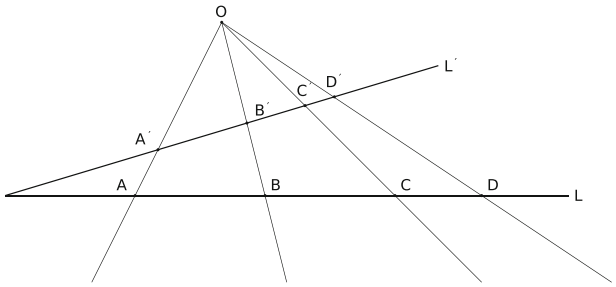


Fig. 7.12 Perspectivity

if and only if

$$[\mathbf{A}_1, \mathbf{A}_2; \mathbf{A}_3, \mathbf{A}_4] = [\mathbf{B}_1, \mathbf{B}_2; \mathbf{B}_3, \mathbf{B}_4].$$

In the special case that the quadruples of points lie on lines transverse to a pencil with vertex [**O**], then the projectivity is called a *perspectivity from* [**O**]. The point [**O**] is called the *centre of perspectivity* (Fig. 7.12). We then write

$$\mathbf{\Sigma} \overline{\wedge}^{[\mathbf{O}]} \mathbf{\Sigma}', \quad \text{or} \quad ([\mathbf{A}_1], [\mathbf{A}_2], [\mathbf{A}_3], [\mathbf{A}_4]) \overline{\wedge}^{[\mathbf{O}]} ([\mathbf{B}_1], [\mathbf{B}_2], [\mathbf{B}_3], [\mathbf{B}_4]).$$

A perspectivity (between two quadrilaterals) is always a projectivity. The converse is not always true; in Fig. 7.13, the projectivity

$$([\mathbf{A}], [\mathbf{B}], [\mathbf{C}], [\mathbf{D}]) \overline{\wedge} ([\mathbf{A}'], [\mathbf{B}'], [\mathbf{C}'], [\mathbf{D}']),$$

is not a perspectivity.

However, we will see below that there is a direct relationship between projectivities and perspectivities.

Theorem 7.4.4 *Every homography of $P(2)$ between two distinct ranges is either a perspectivity or a composition of two perspectivities.*

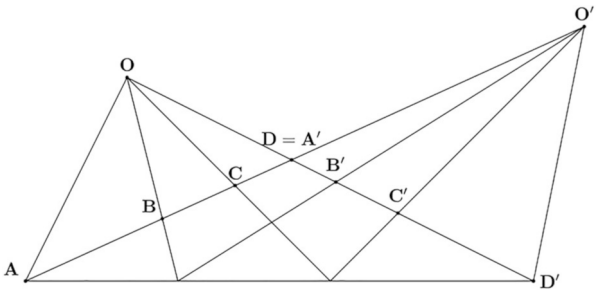


Fig. 7.13 A projectivity which is not a perspectivity

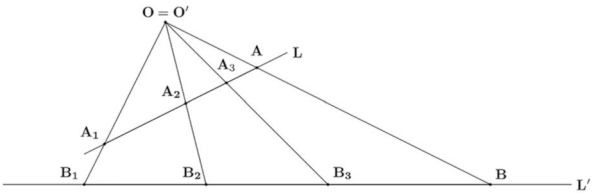


Fig. 7.14 Proof of Theorem 7.4.4: points in perspectivity

Proof It suffices to consider triples of points $[\mathbf{A}_1]$, $[\mathbf{A}_2]$, $[\mathbf{A}_3]$ and $[\mathbf{B}_1]$, $[\mathbf{B}_2]$, $[\mathbf{B}_3]$ lying in lines $\mathbf{L}$ and $\mathbf{L}'$, respectively. We distinguish two cases:

1. *The triples are in perspective position from point* $[\mathbf{O}]$ (Fig. 7.14).
   This means that the lines from $[\mathbf{A}_1]$, $[\mathbf{B}_1]$, from $[\mathbf{A}_2]$, $[\mathbf{B}_2]$ and from $[\mathbf{A}_3]$, $[\mathbf{B}_3]$, meet at $[\mathbf{O}]$. Suppose we are given a point $[\mathbf{A}]$ on the line $\mathbf{L}$. Then the line from $[\mathbf{O}]$ and $[\mathbf{A}]$ uniquely determines a point $[\mathbf{B}]$ on $\mathbf{L}'$ and also

$$([\mathbf{A}_1], [\mathbf{A}_2], [\mathbf{A}_3], [\mathbf{A}]) \overline{\overline{\wedge}}^{[\mathbf{O}]} ([\mathbf{B}_1], [\mathbf{B}_2], [\mathbf{B}_3], [\mathbf{B}]).$$

   We observe that conversely, if we were given a point $[\mathbf{B}]$ of $\mathbf{L}'$ then a point $[\mathbf{A}]$ on $\mathbf{L}$ is uniquely determined (Fig. 7.15).
2. *The triples are not in perspective position.* This means that the lines from $[\mathbf{A}_1]$, $[\mathbf{B}_1]$ and from $[\mathbf{A}_2]$, $[\mathbf{B}_2]$ meet at $[\mathbf{O}]$ but the lines from $[\mathbf{A}_1]$, $[\mathbf{B}_1]$ and from $[\mathbf{A}_3]$, $[\mathbf{B}_3]$ meet at a point $[\mathbf{O}'] \neq [\mathbf{O}]$. We assume again that we are given a point $[\mathbf{A}]$ on line $\mathbf{L}$. Then,

   - the line from $[\mathbf{O}]$, $[\mathbf{O}']$, intersects the line from $[\mathbf{A}_3]$, $[\mathbf{B}_2]$ at $[\mathbf{X}]$,
   - the line from $[\mathbf{O}]$, $[\mathbf{A}]$, intersects the line from $[\mathbf{A}_3]$, $[\mathbf{B}_2]$ at $[\mathbf{Y}]$,
   - the line from $[\mathbf{O}']$, $[\mathbf{Y}]$, intersects the line $\mathbf{L}'$ at $[\mathbf{B}]$.

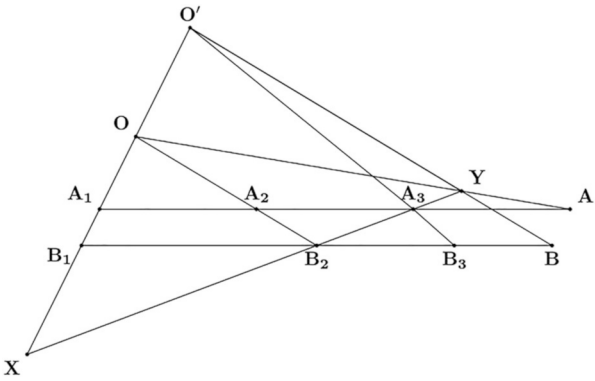


Fig. 7.15 Proof of Theorem 7.4.4: points in projectivity but not in perspectivity

Then

$$([\mathbf{A}_1], [\mathbf{A}_2], [\mathbf{A}_3], [\mathbf{A}]) \overline{\wedge}^{[\mathbf{O}]} ([\mathbf{X}], [\mathbf{B}_2], [\mathbf{A}_3], [\mathbf{Y}]) \overline{\wedge}^{[\mathbf{O}']} ([\mathbf{B}_1], [\mathbf{B}_2], [\mathbf{B}_3], [\mathbf{B}]).$$

□

The next proposition is a criterion of whether a projectivity is a perspectivity.

Proposition 7.4.5 *Let* $\mathbf{L}$ *and* $\mathbf{L}'$ *be lines intersecting at* $[\mathbf{P}]$. *Then a projectivity between two ranges of* $\mathbf{L}$ *and* $\mathbf{L}'$, *respectively, is a projectivity if and only if* $[\mathbf{P}]$ *is mapped onto itself.*

Proof It is obvious that $[\mathbf{P}]$ is mapped onto itself under a perspectivity. For the converse, we use the notation and the scheme in the proof of the second case of Theorem 7.4.4 in the construction of the two perspectivities. Suppose that the line from $[\mathbf{O}]$ and $[\mathbf{P}]$ intersects the line from $[\mathbf{A}_3]$ and $[\mathbf{B}_2]$ at a point $[\mathbf{X}']$. Since $[\mathbf{P}]$ is projected onto itself, $[\mathbf{O}']$ must lie on the line from $[\mathbf{O}]$ and $[\mathbf{X}']$. But $[\mathbf{O}]$, $[\mathbf{O}']$ also lie on the (different) line from $[\mathbf{A}_1]$ and $[\mathbf{B}_1]$. Consequently, $[\mathbf{O}]$ and $[\mathbf{O}']$ are identical and the projectivity is a perspectivity from $[\mathbf{O}]$. □

We finally state the duals of the propositions of this section.

- Two pencils $\mathbf{\Sigma}$ and $\mathbf{\Sigma}'$ are in perspectivity from a line $\mathbf{L}$ if each line of $\mathbf{\Sigma}$ intersects the corresponding line of $\mathbf{\Sigma}'$ on $\mathbf{L}$.
- A projectivity between two pencils is either a perspectivity or a composition of two perspectivities.
- A projectivity is a perspectivity if and only if the line joining the vertices of the two pencils is mapped onto itself.

7.5 Classic Theorems of Projective Geometry

In this section we present two classical theorems of Projective Plane Geometry. The first, due to Pappus of Alexandria, is essentially a theorem of Euclidean Geometry; its Euclidean proof is not trivial at all. The second, due to Desargues, has no analogue in Euclidean Plane Geometry.

7.5.1 Pappus' Theorem

Theorem 7.5.1 (Pappus) *Let* $[\mathbf{A}_1]$, $[\mathbf{A}_2]$, $[\mathbf{A}_3]$ *and* $[\mathbf{B}_1]$, $[\mathbf{B}_2]$, $[\mathbf{B}_3]$ *be triples of collinear points lying on different lines* $\mathbf{L}$ *and* $\mathbf{L}'$. *We assume that none of these points belongs to the intersection of the lines. For* $i, j = 1, 2, 3$ *let* $\mathbf{L}_{i,j}$ *be the line from* $[\mathbf{A}_i]$, $[\mathbf{B}_j]$ *and let also:*

- $[\mathbf{C}_1] = \mathbf{L}_{2,3} \cap \mathbf{L}_{3,2}$,

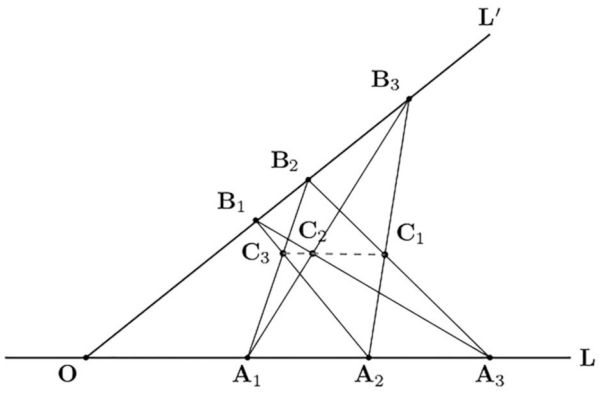


Fig. 7.16 Pappus' Theorem

- $[\mathbf{C}_2] = \mathbf{L}_{1,3} \cap \mathbf{L}_{3,1}$,
- $[\mathbf{C}_3] = \mathbf{L}_{1,2} \cap \mathbf{L}_{2,1}$.

Then $[\mathbf{C}_1]$, $[\mathbf{C}_2]$ *and* $[\mathbf{C}_3]$ *are collinear (Fig. 7.16).*

Proof We can normalise so that

$$[\mathbf{A}_3] = [1:0:0], \quad [\mathbf{B}_3] = [0:1:0], \quad [\mathbf{C}_3] = [0:0:1], \quad [\mathbf{A}_1] = [1:1:1].$$

Then since

- $\mathbf{L}: \ [0:1:-1] \cdot \mathbf{X} = 0$,
- $\mathbf{L}_{1,3}: \ [0:1:-1] \cdot \mathbf{X} = 0$ and
- the line passing through $[\mathbf{A}_1]$ and $[\mathbf{C}_1]$ is $[1:-1:0] \cdot \mathbf{X} = 0$,

we can assume that there exist a, b, c all different from 0 and 1 such that

$$[\mathbf{A}_2] = [a:1:1], \quad [\mathbf{C}_2] = [1:b:1], \quad [\mathbf{B}_2] = [1:1:c].$$

The point $[\mathbf{C}_3]$ is the intersection point of

$$\mathbf{L}_{2,3}: \ [-1:0:a] \cdot \mathbf{X} = 0 \text{ and } \mathbf{L}_{3,2}: \ [0:-c:1] \cdot \mathbf{X} = 0,$$

i.e.

$$[\mathbf{C}_3] = [(-1,0,a) \times (0,-c,1)] = [ac:1:c].$$

The points $[\mathbf{C}_i]$ are collinear if and only if the mixed product

$$\begin{vmatrix} 0 & 0 & 1 \\ 1 & b & 1 \\ ac & 1 & c \end{vmatrix} = 1 - abc = 0.$$

It remains to show that the relation $abc = 1$ holds. To do this, we find the point $[\mathbf{B}_1]$ which is the intersection of the lines $\mathbf{L}_{2,1}$ from $[\mathbf{A}_2]$ and $[\mathbf{C}_1]$ and $\mathbf{L}'$ from $[\mathbf{B}_2]$ and $\mathbf{B}_3]$:

$$\mathbf{L}_{2,1} : \ [1 : -a : 0] \cdot \mathbf{X} = 0, \quad \mathbf{L}' : \ [c : 0 : -1] \cdot \mathbf{X} = 0.$$

It follows that $[\mathbf{B}_1] = [a : 1 : ac]$. The relation $abc = 1$ now follows from the collinearity of the points $[\mathbf{C}_2]$, $[\mathbf{A}_3]$ and $[\mathbf{B}_1]$. □

Pappus' Theorem has interesting consequences. First, we have its dual version, the proof of which follows from the Duality Principle:

Theorem 7.5.2 (Pappus' Dual Theorem) *Consider lines with dual points* $[\mathbf{A}_1]$, $[\mathbf{A}_2]$, $[\mathbf{A}_3]$ *belonging to a pencil with vertex* $[\mathbf{O}]$ *and lines with dual points* $[\mathbf{B}_1]$, $[\mathbf{B}_2]$, $[\mathbf{B}_3]$ *belonging to a pencil with vertex* $[\mathbf{O}'] \neq [\mathbf{O}]$. *For* $i, j = 1, 2, 3$ *let* $\mathbf{L}_{i,j}$ *be the line from* $[\mathbf{A}_i]$, $[\mathbf{B}_j]$ *and let:*

- $[\mathbf{C}_1] = \mathbf{L}_{2,3} \cap \mathbf{L}_{3,2}$,
- $[\mathbf{C}_2] = \mathbf{L}_{1,3} \cap \mathbf{L}_{3,1}$,
- $[\mathbf{C}_3] = \mathbf{L}_{1,2} \cap \mathbf{L}_{2,1}$.

Then the lines with dual points $[\mathbf{C}_1]$, $[\mathbf{C}_2]$ *and* $[\mathbf{C}_3]$ *coincide.*

Definition 7.5.3 The line $\mathbf{C}$ from $[\mathbf{C}_i]$, $i = 1, 2, 3$ of Pappus' Theorem, is called *Pappus' line*.

An interesting application of the Pappus' line is the following:

Proposition 7.5.4 *Consider triples of points* $[\mathbf{A}_1]$, $[\mathbf{A}_2]$, $[\mathbf{A}_3]$ *and* $[\mathbf{B}_1]$, $[\mathbf{B}_2]$, $[\mathbf{B}_3]$, *which lie on lines* $\mathbf{L}$ *and* $\mathbf{L}'$, *respectively. If we are given* $[\mathbf{A}] \in \mathbf{L}$ *and the Pappus' line, then the point* $[\mathbf{B}] \in \mathbf{L}'$ *for which* $[\mathbf{A}_1, \mathbf{A}_2; \mathbf{A}_3, \mathbf{A}] = [\mathbf{B}_1, \mathbf{B}_2; \mathbf{B}_3, \mathbf{B}]$ *is completely determined.*

Proof A projectivity is completely determined by three pairs of corresponding points. In our case the correspondence is $[\mathbf{A}_i] \leftrightarrow [\mathbf{B}_i]$, $i = 1, 2, 3$. The points $[\mathbf{C}_1]$, $[\mathbf{C}_2]$, $[\mathbf{C}_3]$ lie on the Pappus line $\mathbf{C}$ of the projectivity. □

7.5.2 Desargues' Theorem

The theorem we study in this section was proved by the French mathematician G. Desargues in 1639 and essentially led to the substantial development of Projective



Fig. 7.17 Girard Desargues, 1591–1661

Geometry by Poncelet. We can safely claim that Desargues' theorem represents the geometric foundation of photography and perspective, which are used by painters and designers to represent objects in space on canvas or paper. The theorem was first proved as a theorem of the geometry of $\mathbb{R}^3$; we first give a definition for triangles lying either in real or projective space (Fig. 7.17).

Definition 7.5.5 Let $\triangle\mathbf{ABC}$ and $\triangle\mathbf{A}'\mathbf{B}'\mathbf{C}'$ be triangles. We will say that they are in *point perspective* if the lines passing through their respective vertices converge at a point (the *perspective centre*). On the other hand, we will say that the triangles are in *line perspective* if the intersection points of the lines of their respective sides belong to the same line (the *perspective axis*).

In Fig. 7.18 we see triangles of $\mathbb{R}^3$ in point perspective: the lines containing $\mathbf{AA}'$, $\mathbf{BB}'$, and $\mathbf{CC}'$, respectively, meet at $\mathbf{P}$.

Theorem 7.5.6 *Let the triangles* $\triangle\mathbf{ABC}$ *and* $\triangle\mathbf{A}'\mathbf{B}'\mathbf{C}'$ *of* $\mathbb{R}^3$ *be in point perspective, with the additional condition that no two corresponding sides are parallel. Then they are also in line perspective.*

If the latter condition does not hold, there will be only two points of intersection instead of three, and the theorem must be modified to include the result that these

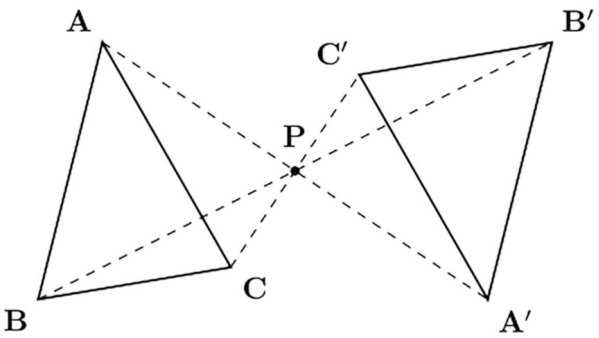


Fig. 7.18 Triangles in point perspective. Since perspective sides are parallel, Desargues' Theorem does not apply here

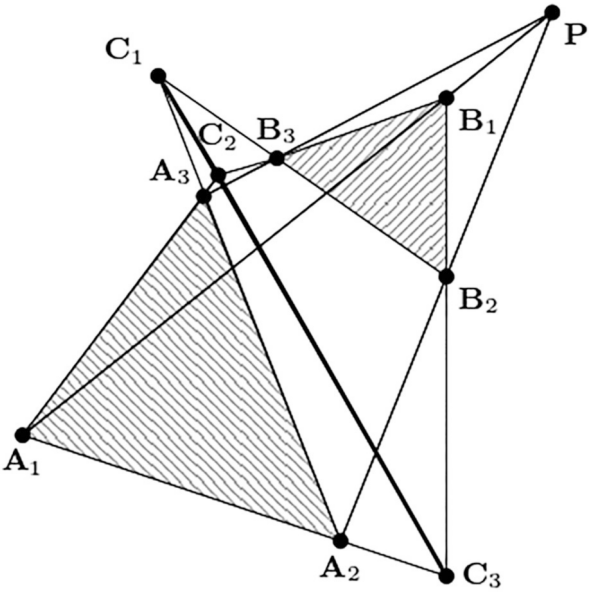


Fig. 7.19 Proof of Desargues' Theorem: the shaded triangles are in line perspective

two points will lie on a line parallel to the two parallel sides of the triangles. See also Exercise 37, Sect. 7.6. Instead of modifying the theorem to cover this special case, Poncelet modified Euclidean space itself. He did this by considering points at infinity; this consideration was key to the development of projective geometry: two "parallel lines" have a common point at infinity. Thus, Poncelet discovered that Desargues' theorem could be more simply stated as the projective plane theorem which we prove immediately in its stronger form.

Theorem 7.5.7 (Desargues) *Two triangles of* $\mathbb{R}P^2$ *are in point perspective if and only if they are also in line perspective (Fig. 7.19).*

Proof The triangles with vertices $[\mathbf{A}_1]$, $[\mathbf{A}_2]$, $[\mathbf{A}_3]$ and $[\mathbf{B}_1]$, $[\mathbf{B}_2]$, $[\mathbf{B}_3]$ are in point perspective. The intersection points of the corresponding sides are $[\mathbf{C}_1]$, $[\mathbf{C}_2]$, $[\mathbf{C}_3]$ which we will show are collinear. If $[\mathbf{P}]$ is the perspective centre, then since $[\mathbf{P}]$, $[\mathbf{A}_1]$ and $[\mathbf{B}_1]$ are collinear, there exist $\lambda, \lambda' \in \mathbb{R}_*$ with

$$\mathbf{P} = \lambda \mathbf{A}_1 + \lambda' \mathbf{B}_1.$$

Choosing representatives for $[\mathbf{A}_1]$, $[\mathbf{B}_1]$, $\lambda\mathbf{A}_1$ and $\lambda'\mathbf{B}_1$, respectively, we can write $\mathbf{P} = \mathbf{A}_1 + \mathbf{B}_1$, and similarly we will have $\mathbf{P} = \mathbf{A}_2 + \mathbf{B}_2\mathbf{A}_3 + \mathbf{B}_3$. The line from $[\mathbf{A}_1]$ and $[\mathbf{A}_2]$ is $[\mathbf{A}_1 \times \mathbf{A}_2] \cdot \mathbf{X} = 0$ and the line from $[\mathbf{B}_1]$ and $[\mathbf{B}_2]$ is $[\mathbf{B}_1 \times \mathbf{B}_2] \cdot \mathbf{X} = 0$. Therefore, their intersection point is $[(\mathbf{A}_1 \times \mathbf{A}_2) \times (\mathbf{B}_1 \times \mathbf{B}_2)]$. We set $\mathbf{C} = \mathbf{B}_1 \times \mathbf{B}_2$

and we have

$$(\mathbf{A}_1 \times \mathbf{A}_2) \times \mathbf{C} = (\mathbf{A}_1 \cdot \mathbf{C})\mathbf{A}_2 - (\mathbf{A}_2 \cdot \mathbf{C})\mathbf{A}_1.$$

But,

$$\begin{aligned}
\mathbf{A}_1 \cdot \mathbf{C} &= \mathbf{A}_1 \cdot (\mathbf{B}_1 \times \mathbf{B}_2) \\
&= \mathbf{A}_1 \cdot ((\mathbf{P} - \mathbf{A}_1) \times (\mathbf{P} - \mathbf{A}_2)) \\
&= (\mathbf{A}_1 - \mathbf{P}) \cdot (\mathbf{A}_1 \times (\mathbf{P} - \mathbf{A}_2)) \\
&= -\mathbf{P} \cdot (\mathbf{A}_1 \times (\mathbf{P} - \mathbf{A}_2)) \\
&= \mathbf{A}_1 \cdot (\mathbf{P} \times (\mathbf{P} - \mathbf{A}_2)) \\
&= -\mathbf{A}_1 \cdot (\mathbf{P} \times \mathbf{A}_2) \\
&= (\mathbf{A}_1, \mathbf{A}_2, \mathbf{P}),
\end{aligned}$$

i.e. the mixed product of $\mathbf{A}_1$, $\mathbf{A}_2$ and $\mathbf{P}$. Similarly we find:

$$\mathbf{A}_2 \cdot \mathbf{C} = (\mathbf{A}_1, \mathbf{A}_2, \mathbf{P})$$

and thus

$$[(\mathbf{A}_1 \times \mathbf{A}_2) \times (\mathbf{B}_1 \times \mathbf{B}_2)] = [\mathbf{A}_1 - \mathbf{A}_2].$$

In an analogous manner, we get that:

- the intersection point of the line through $[\mathbf{A}_1]$, $[\mathbf{A}_3]$ and the line through $[\mathbf{B}_1]$, $[\mathbf{B}_3]$ is $[\mathbf{A}_1 - \mathbf{A}_3]$ and
- the intersection point of the line through $[\mathbf{A}_2]$, $[\mathbf{A}_3]$ and the line through $[\mathbf{B}_2]$, $[\mathbf{B}_3]$ is $[\mathbf{A}_2 - \mathbf{A}_3]$.

However, since

$$(\mathbf{A}_1 - \mathbf{A}_2) + (\mathbf{A}_3 - \mathbf{A}_1) + (\mathbf{A}_2 - \mathbf{A}_3) = \mathbf{0},$$

these points are collinear.

Conversely, we assume that the points $\mathbf{C}_1$, $\mathbf{C}_2$ and $\mathbf{C}_3$ are collinear and apply the already proven part of the theorem to the triangles $\triangle\mathbf{C}_3\mathbf{B}_1\mathbf{A}_1$ and $\triangle\mathbf{C}_1\mathbf{B}_3\mathbf{A}_2$, which are in point perspective with perspective center $[\mathbf{C}_2]$. According to the proven part, the points

- $[\mathbf{P}]$, which is the intersection of the lines passing through $[\mathbf{B}_1]$, $[\mathbf{A}_1]$ and $[\mathbf{B}_3]$, $[\mathbf{A}_3]$, respectively,
- $[\mathbf{A}_2]$, which is the intersection of the lines passing through $[\mathbf{C}_1]$, $[\mathbf{A}_3]$ and $[\mathbf{A}_1]$, $[\mathbf{C}_3]$, respectively, and

- $[\mathbf{B}_2]$, which is the intersection of the lines passing through $[\mathbf{B}_1]$, $[\mathbf{C}_3]$ and $[\mathbf{B}_1]$, $[\mathbf{C}_3]$, respectively,

will be coincident. However, this is equivalent to the fact that the lines from $[\mathbf{A}_1]$, $[\mathbf{B}_1]$, $[\mathbf{A}_2]$, $[\mathbf{B}_2]$ and $[\mathbf{A}_3]$, $[\mathbf{B}_3]$ pass through the same point. In other words, the triangles $\triangle\mathbf{A}_1\mathbf{A}_2\mathbf{A}_3$ and $\triangle\mathbf{B}_1\mathbf{B}_2\mathbf{B}_3$ are in point perspective. □

7.6 Exercises

1. Let $\mathbb{R}P^2$ be the real projective plane.
   (a) Consider $\mathbb{R}P^2$ as the union of the following three sets:

$$\begin{aligned} H_0 &= \{[x_0 : x_1 : x_2] \in \mathbb{R}P^2 \mid x_0 \neq 0\}, \\ H_1 &= \{[x_0 : x_1 : x_2] \in \mathbb{R}P^2 \mid x_1 \neq 0\}, \\ H_2 &= \{[x_0 : x_1 : x_2] \in \mathbb{R}P^2 \mid x_2 \neq 0\}. \end{aligned} \tag{7.8}$$

   Write down the formulae for the projection of $\mathbb{R}P^1$ onto $\mathbb{R}$ and the standard lift map.
   (b) Let

$$L^2_\infty = \mathbb{R}P^2 \setminus H_2 = \{[x_0 : x_1 : 0] \in \mathbb{R}P^2\}.$$

   Prove that the mapping

$$\mathbb{R}P^1 \ni [x_0 : x_1] \mapsto [x_0 : x_1 : 0] \in \mathbb{R}P^2$$

   is injective and its image is L^2_∞. Conclude that the set L^2_∞ can be considered as a copy of the projective line $\mathbb{R}P^1$ embedded in $\mathbb{R}P^2$. Note that L^2_∞ is called the *line at infinity with respect to* H_2. This embedding implies the embedding of the Euclidean plane $\mathbb{R}^2$ into $\mathbb{R}^3$ as the plane $x_2 = 0$.
   (c) What are the lines at infinity of the sets H_0 and H_1?
2. (a) Let $F : \mathbb{R}^3 \to \mathbb{R}^3$ be such that

$$F(x, y, z) = (yz,\ xz,\ xy), \quad (x, y, z) \in \mathbb{R}^3.$$

   Show that the F-image of the unit sphere S^2 is the hypersurface $\mathcal{S}$ of $\mathbb{R}^3$ given by the equation

$$x^2y^2 + y^2z^2 + z^2x^2 - xyz = 0. \tag{7.9}$$

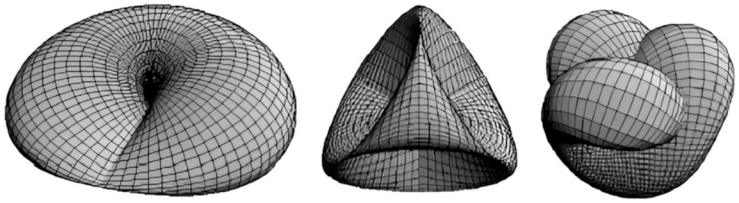


Fig. 7.20 From right to left: Cross-Cap, Steiner's Roman surface, Boy's surface

(b) Let $\mathbb{R}P^2 = S^2/\sim$ where $\sim$ is the equivalence relation identifying antipodal points in S^2. Show that F maps antipodal points of S^2 to the same point of $\mathcal{S}$ and hence we obtain a well defined map $\mathbb{R}P^2 \to \mathcal{S}$. The surface $\mathcal{S}$ is called *Steiner's Roman surface* and it was discovered in 1844 by the great Swiss geometer Jacob Steiner (1796–1863) when he was in Rome. Steiner's Roman surface is one of the three more famous representations of real projective plane, the other two being the Cross-Cap and Boy's surface, see Fig. 7.20.

3. Define the complex projective space $\mathbb{C}P^n$.

   (a) What kind of objects are its points?
   (b) Prove that when $n = 1$, the complex projective line $\mathbb{C}P^1$ is identical to the unit sphere S^2.

4. In the projective plane $\mathbb{R}P^2$ show that three points $[\mathbf{A}] = [a_0 : a_1 : a_2]$, $[\mathbf{B}] = [b_0 : b_1 : b_2]$ and $[\mathbf{C}] = [c_0 : c_1 : c_2]$ are collinear if and only if the mixed product

$$(\mathbf{A}, \mathbf{B}, \mathbf{C}) = \begin{vmatrix} a_0 & a_1 & a_2 \\ b_0 & b_1 & b_2 \\ c_0 & c_1 & c_2 \end{vmatrix} = 0.$$

5. As an application of Exercise 4:

   (a) Find the lines L_1 and L_2 that pass through $[\mathbf{A}]$ and $[\mathbf{B}]$ when these are respectively

      i. $[1 : 0 : 5]$, $[0 : -1 : 9]$,
      ii. $[2 : 3 : 2]$, $[-1 : 4 : 1]$.

   (b) Find the intersection point of the lines L_1 and L_2.
   (c) Consider which of the following triples of points are collinear:

      i. $[-2 : 8 : 1]$, $[3 : 4 : 0]$, $[-2 : 5 : 2]$.
      ii. $[1 : -1/2 : 2]$, $[-2 : 1 : 3]$, $[0 : 0 : 1]$.

6. Let the following hyperplanes of $\mathbb{R}P^3$:

$$\mathbf{P}: \ [\mathbf{A}] \cdot \mathbf{X} = ax_0 + bx_1 + cx_2 + dx_3 = 0,$$
$$\mathbf{P}': \ [\mathbf{A}'] \cdot \mathbf{X} = a'x_0 + b'x_1 + c'x_2 + d'x_3 = 0.$$

Argue for their intersection $\mathbf{P} \cap \mathbf{P}'$ by examining the solution space of the linear system

$$ax_0 + bx_1 + cx_2 + dx_3 = 0,$$
$$a'x_0 + b'x_1 + c'x_2 + d'x_3 = 0.$$

7. Examine whether the following linear maps result in projective maps. If so, write down the formula of the projective map.
   (a) $f : \mathbb{R}^2 \to \mathbb{R}^2$, $f(x_0, x_1) = (x_1, x_0)$.
   (b) $f : \mathbb{R}^2 \to \mathbb{R}^3$, $f(x_0, x_1) = (x_1 - 3x_0,\ x_0,\ x_0)$.
   (c) $f : \mathbb{R}^3 \to \mathbb{R}^2$, $f(x_0, x_1, x_2) = (x_2, x_0)$.

8. Determine, if they exist, the fixed points of the projective mappings $[m]$ whose matrix A_m of corresponding automorphisms is:

   (a)
   $$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

   (b)
   $$\begin{pmatrix} 1 & a & b \\ 0 & -1 & c \\ 0 & 0 & 1 \end{pmatrix}, \quad a, b, c \in \mathbb{R}.$$

9. Let the Segre embedding be as in Example 7.1.24. Prove the following:
   (a) For each $[p] = [x_0 : x_1] \in \mathbb{R}P^1$, the image $\phi(\{[p]\} \times \mathbb{R}P^1)$ is a line $\mathbf{L}_{[p]}$ on $\mathbb{R}P^3$.
   (b) If $[p] \neq [p']$, then $\mathbf{L}_{[p]} \cap \mathbf{L}_{[p']} = \emptyset$.
   (c) $\mathcal{S} = \bigcup_{[p] \in \mathbb{R}P^1} bL_{[p]}$.
   (d) Show that similar results also hold when we fix a $[q] = [y_0 : y_1]$.
   (e) For each $[p]$, $[q]$, the intersection $\mathbf{L}_{[p]} \cap \mathbf{L}_{[q]}$ is a point.

10. A mapping of projective spaces is projected onto a mapping of Euclidean spaces. If

$$\mathbb{R}P^n \ni [x_0 : \cdots : x_n] \mapsto [y_0 : \cdots : y_m] \in \mathbb{R}P^m,$$

then by projecting onto H_n and H_m respectively, we obtain the mapping

$$\mathbb{R}^n \ni (u_1, \ldots, u_n) \mapsto (v_1, \ldots, v_m),$$

where $u_i = x_{i-1}/x_n$ and $v_i = y_{i-1}/y_m$. Prove that this mapping is well-defined.

11. Let $f : \mathbb{R}^n \to \mathbb{R}^m$ be a mapping, $(u_1, \ldots, u_n) \mapsto (v_1, \ldots, v_m)$. Show that there is a lift of f to a mapping $[f] : \mathbb{R}P^n \to \mathbb{R}P^m$.
12. Find the $[g] \in P(n)$ inverses when

    (a) $g : \mathbb{R}^2 \to \mathbb{R}^2$, $g(x_0, x_1) = (x_1,\ x_0)$.
    (b) $g : \mathbb{R}^3 \to \mathbb{R}^3$, $g(x_0, x_1, x_2) = (x_0 + x_1 - x_2,\ x_1 - x_2,\ x_2))$.

13. Find the inverses of $[g] \in P(n)$ when the matrix $[A_g]$ is

    (a)

$$\begin{vmatrix} -1 & 2 \\ 2 & -1 \end{vmatrix}.$$

    (b)

$$\begin{vmatrix} 0 & 2 & 1 \\ 1 & 0 & 0 \\ 0 & -1 & 1 \end{vmatrix}.$$

14. Let the line $\mathbf{L} :\ [0 : 1 : -1] \cdot \mathbf{X} = 0$ be in $\mathbb{R}P^2$ and $[g] \in P(2)$ with

$$[A_g] = \begin{vmatrix} 1 & -1 & 1 \\ -1 & 0 & 1 \\ 0 & 1 & 0 \end{vmatrix}.$$

    Find $[g](\mathbf{L})$.
15. Let the line of $\mathbb{R}P^3$ be of the form

$$\mathbf{L} :\ [1 : 1 : -1 : 0] \cdot \mathbf{X} = 0, \quad [0 : 1 : 1 : 1] \cdot \mathbf{X} = 0,$$

    and the homography $[g] \in P(3)$ be of the matrix

$$[A_g] = \begin{vmatrix} 2 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & -1 \end{vmatrix}.$$

    Find $[g](\mathbf{L})$.

16. Argue as to why an element of the projective group maps a projective subspace into a projective subspace of the same projective dimension.
17. Find the homography $[g]$ of the projective line that maps

    (a) $[1:0]$, $[0:1]$, $[1:1]$ to $[-1,2]$, $[2,3]$, $[4:1]$, respectively and
    (b) $[-1:1]$, $[0:1]$, $[1:2]$ to $[0,1]$, $[3,-1]$, $[2:5]$, respectively.

18. Consider whether there is an element of $P(1)$ that maps $[1:0]$, $[0:1]$, $[1:1]$ and $[1:-1]$ to $[0:1]$, $[1:0]$, $[2:1]$, $[1:2]$, respectively.
19. Find all projective transformations that map $[1:0:0]$, $[0:1:0]$, $[0:0:1]$ to $[2:0:1]$, $[0:-1:2]$, $[0:0:1]$, respectively. Show that of all these, only one maps $[1:1:1]$ to $[3:-1:0]$.
20. Find the element of $P(2)$ that maps $[-1:2:0]$, $[2:-1:0]$, $[0:1:2]$, $[0:1:-2]$ to $[0:2:1]$, $[0:2:-1]$, $[1:2:0]$, $[2:1:0]$, respectively.
21. Let $\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_3$ and $\mathbf{B}_1, \mathbf{B}_2, \mathbf{B}_3$ be pairwise distinct triples of points in different projective lines $\mathbf{L}$ and $\mathbf{L}'$ respectively, of $\mathbb{R}P^n$. Show that there exists a $[g] \in P(n)$ such that $[g](\mathbf{A}_i) = [\mathbf{B}_i]$, $i = 1, 2, 3$
22. Show that the cross-ratio of four points of the projective line satisfies the properties:

$$[\mathbf{A}, \mathbf{B}; \mathbf{C}, \mathbf{D}] = [\mathbf{B}, \mathbf{A}; \mathbf{D}, \mathbf{C}] = [\mathbf{C}, \mathbf{D}; \mathbf{A}, \mathbf{B}] = [\mathbf{D}, \mathbf{C}; \mathbf{B}, \mathbf{A}] \tag{7.10}$$

    and

$$[\mathbf{A}, \mathbf{B}; \mathbf{C}, \mathbf{D}] = \lambda, \quad [\mathbf{A}, \mathbf{B}; \mathbf{D}, \mathbf{C}] = 1/\lambda,$$
$$[\mathbf{A}, \mathbf{C}; \mathbf{B}, \mathbf{D}] = 1 - \lambda, \quad [\mathbf{A}, \mathbf{C}; \mathbf{D}, \mathbf{B}] = 1/(1 - \lambda),$$
$$[\mathbf{A}, \mathbf{D}; \mathbf{B}, \mathbf{C}] = 1 - 1/\lambda, \quad [\mathbf{A}, \mathbf{D}; \mathbf{C}, \mathbf{B}] = \lambda/(\lambda - 1).$$

23. Find the cross-ratio of the following ordered quadruples of points:

    (a) $[0:-1:2]$, $[1:2:0]$, $[1:3:-2]$, $[1:1:2]$.
    (b) $[1:1:0:1]$, $[1:0:0:-2]$, $[1:2:0:4]$, $[2:1:0:-1]$.

24. Find in two ways the cross-ratio

$$[(1,3),\ (1,4);\ (1,2),\ (1,5)].$$

25. Given four pairwise distinct points of $\mathbb{R}P^2$, let $[\mathbf{A}_i]$, $i = 1, \ldots, 4$, we consider the lines $\mathbf{L}_i : [\mathbf{A}_i] \cdot \mathbf{X} = 0$. Prove that they meet at a point $[\mathbf{O}]$.
26. Continuing from the previous exercise, we assume that the points of the line are in $\mathbb{R}P^n$, $n > 2$. Here, we have hyperplanes $\mathbf{W}_i$; $[\mathbf{A}_i] \cdot \mathbf{X} = 0$. What can we say about their intersections?
27. Prove that the points (in the order given):

$$[1:2:1], \quad [3:4:0], \quad [4:6:7], \quad [-2:2:-5],$$

    are in harmonic position.

28. Construct a complete quadrilateral corresponding to a quadrilateral of points of the projective plane in harmonic position.
29. In the projective line $\mathbb{R}P^1$ we consider the space of four points with the property (Π): this can be described as the set

$$F_4 = \mathbb{R}P^1 \times \mathbb{R}P^1 \times \mathbb{R}P^1 \times \mathbb{R}P^1 \setminus \text{diagonals}.$$

The *four-point configuration space of the projective line* is the set $\mathfrak{F}_4 = F_4/\sim$, with $\sim$ being the equivalence relation on F_4 defined as follows: if $[\mathfrak{p}] = \{[\mathbf{A}_i],\ i = 1, \ldots, 4\}$ and $[\mathfrak{p}'] = \{[\mathbf{A}'_i],\ i = 1, \ldots, 4\}$ are elements of F_4 then

$$[\mathfrak{p}] \sim [\mathfrak{p}'] \iff \exists\, [g] \in P(1) :\ [g]([\mathbf{A}_i]) = [\mathbf{A}'_i].$$

Prove that the configuration space is in bijective correspondence with the set $\mathbb{R} \setminus \{0, 1\}$.

30. State the dual Fundamental Theorem of Projective Plane Geometry.
31. Find the vertices of the triangle defined by the lines

$$[1:1:1]\cdot\mathbf{X} = 0, \quad [-1:3:0]\cdot\mathbf{X} = 0 \quad \text{and} \quad [0:2:7]\cdot\mathbf{X} = 0.$$

Then find its dual triangle.

32. Argue why for every triangle $\mathbf{T}$ of the projective plane there exists $[g] \in P(2)$ such that $[g](\mathbf{T}) = \mathbf{T}^*$, where $\mathbf{T}^*$ is the dual triangle of $\mathbf{T}$. Is $[g]$ unique?
33. Consider the points be $[0:1:-2]$, $[1:0:-3]$, $[-2:3:0]$, $[1:1:-5]$.
    (a) Show that they are collinear.
    (b) Calculate their cross-ratio, in the order given above.
    (c) Calculate the vertex of the pencil defined by their dual lines.
34. Consider the lines

$$\mathbf{L}_1 :\ [1:1:2]\cdot\mathbf{X} = 0, \quad \mathbf{L}_2 :\ [1:0:0]\cdot\mathbf{X} = 0,$$
$$\mathbf{L}_3 :\ [1:1:0]\cdot\mathbf{X} = 0, \quad \mathbf{L}_4 :\ [-1:1:0]\cdot\mathbf{X} = 0.$$

Prove that the are concurrent and find the cross-ratio of their pencil in the given order.

35. Let $[\mathbf{A}]$, $[\mathbf{B}]$, $[\mathbf{C}]$, $[\mathbf{X}]$ be points of line $\mathbf{L}$ and $[\mathbf{A}]'$, $[\mathbf{B}']$, $[\mathbf{C}']$, $[\mathbf{X}']$ be points of line $\mathbf{L}'$ such that the intersection points
    (a) $[\mathbf{P}']$ of the lines from $[\mathbf{A}]$, $[\mathbf{B}']$ and from $[\mathbf{B}]$, $[\mathbf{A}']$,
    (b) $[\mathbf{P}'']$ of the lines from $[\mathbf{A}]$, $[\mathbf{C}']$ and from $[\mathbf{C}]$, $[\mathbf{A}']$,
    (c) $[\mathbf{P}''']$ of the lines from $[\mathbf{A}]$, $[\mathbf{X}']$ and from $[\mathbf{X}]$, $[\mathbf{A}']$,

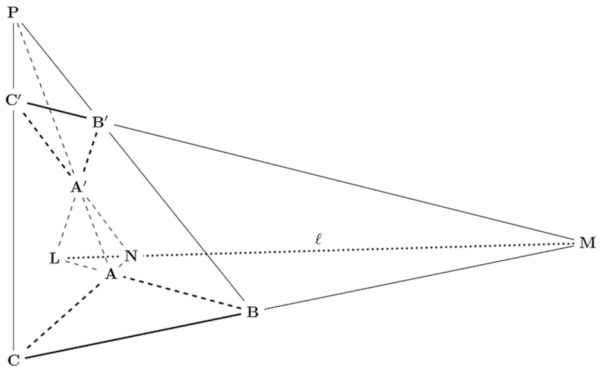


Fig. 7.21 Three-dimensional proof of the Desargues' Theorem

to be collinear. Prove that

$$[\mathbf{A}, \mathbf{B}; \mathbf{C}, \mathbf{X}] = [\mathbf{A}', \mathbf{B}'; \mathbf{C}', \mathbf{X}'].$$

36. Given the points

$$[\mathbf{A}_1] = [1 : 0 : -1], \quad [\mathbf{A}_2] = [0 : 1 : 0], \quad [\mathbf{A}_3] = [-1 : 1 : 1],$$

and the points

$$[\mathbf{B}_1] = [0 : 2 : 1], \quad [\mathbf{B}_2] = [1 : 0 : 0], \quad [\mathbf{B}_3] = [3 : 4 : 2]$$

(a) Show that $[\mathbf{A}_i]$ as well as $[\mathbf{B}_i]$, $i = 1, 2, 3$, are collinear and establish that they lie on different lines.
(b) Find the Pappus' line of $[\mathbf{A}_i]$, $[\mathbf{B}_i]$.

37. In this exercise we describe the proof of the Theorem 7.5.6 of Desargues in three-dimensional space, see Fig. 7.21.
Here the triangles with vertices $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$ and $\mathbf{A}'$, $\mathbf{B}'$, $\mathbf{C}'$ that are in perspective from $\mathbf{P}$ are assumed to lie in planes ϖ and ϖ', respectively. Suppose first that the planes are different and intersect in a line ℓ. The planes from $\mathbf{P}$, $\mathbf{A}$, $\mathbf{B}$ and from $\mathbf{P}$, $\mathbf{A}'$, $\mathbf{B}'$ coincide by hypothesis. This common plane intersects ϖ in the line from $\mathbf{A}$ and $\mathbf{B}$. It also intersects ϖ' in the line from $\mathbf{A}'$ and $\mathbf{B}'$. Therefore, these lines that intersect at $\mathbf{L}$ must also intersect at a point on ℓ. Similarly, the same happens with the other points $\mathbf{M}$, $\mathbf{N}$.
Now if $\varpi = \varpi'$ we cannot use the previous argument, since ℓ cannot be defined. Instead, we take a line ℓ from the perspective point $\mathbf{P}$, assume ℓ not to belong to ϖ and let points $\mathbf{P}_1, \mathbf{P}_2 \in \ell$ be different from $\mathbf{P}$. Then, the lines from $\mathbf{P}_1$, $\mathbf{A}$ and from $\mathbf{P}_2$, $\mathbf{A}'$ lie in the plane containing the lines from $\mathbf{P}_1$, $\mathbf{P}$, $\mathbf{P}_2$ and from $\mathbf{P}$, $\mathbf{A}'$, $\mathbf{A}$. Therefore, they intersect at a point, let $\mathbf{A}''$. Similarly, the

lines from $\mathbf{P}_1$, $\mathbf{B}$ and from $\mathbf{P}_2$, $\mathbf{B}'$ intersect at some $\mathbf{B}''$ and the lines from $\mathbf{P}_1$, $\mathbf{C}$ and from $\mathbf{P}_2$, $\mathbf{C}'$ intersect at some $\mathbf{C}''$. Then, the triangle with vertices $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$ is in perspective with the triangle with vertices $\mathbf{A}''$, $\mathbf{B}''$, $\mathbf{C}''$, with perspective vertex $\mathbf{P}_2$.

Let ϖ' be the plane of $\mathbf{A}''$, $\mathbf{B}''$, $\mathbf{C}''$. Then $\varpi \neq \varpi'$ and we apply Desargues's Theorem to this case; the difference lies in the fact that the perspective line of the sides is here the common line of ϖ and ϖ' containing $\mathbf{L}$, $\mathbf{M}$ and $\mathbf{N}$.

Chapter 8
Inversive Geometry



Persons, who have a decided mathematical talent, constitute, as it were, a favoured class. They bear the same relation to the rest of mankind that those who are academically trained bear to those who are not.

—August Möbius

Inversive Geometry according to Klein is the pair $(\overline{\mathbb{R}^n}, \mathrm{GM}(n))$: the set $\overline{\mathbb{R}^n}$ is the extended real space, that is, the real space $\mathbb{R}^n$ with the addition of a point at infinity, and $\mathrm{GM}(n)$ is the group of Möbius transformations in $\overline{\mathbb{R}^n}$. We will study this geometry in detail in what follows, but in rough lines we can say that Inversive Geometry may be considered as an extension of the Euclidean Geometry of $\mathbb{R}^n$ to the compact space $\overline{\mathbb{R}^n}$. The group $\mathrm{GM}(n)$ comprises transformations that are compositions of *inversions* and contain the similarity group $S(n)$ in a natural way. The novelty here is the addition of inversions in spheres and hyperplanes. The names *Möbius transformations* and *Möbius group* have prevailed for the finite products of inversions and $\mathrm{GM}(n)$, respectively, in honour of the great German mathematician A.F. Möbius (Fig. 8.1).

The extended real space $\overline{\mathbb{R}^n}$ is the compact space that naturally results from the *stereographic projection* that was known to Claudius Ptolemy. The inversion in a sphere, although it is based entirely on the geometric inversion that was known to the Ancient Greeks, was only defined around the beginning of the nineteenth century by the German-Jewish mathematician Magnus (Ludwig Immanuel Magnus, 1790–1861). In Inversive Geometry, a (metric) cross-ratio is naturally defined which is also the unique invariant of the Möbius transformations. Through Inversive Geometry we can naturally arrive to Hyperbolic Geometry and this is the path that we will follow here. Finally, it is worth emphasising that abstract Inversive Geometries can be defined in various metric (and semi-metric) spaces that are compactified pointwise and whose similarity group contains inversions on metric spheres. This is also the basis of modern *Möbius Geometry*.

In this chapter we basically follow the discussion given in [7]. In particular, we refer to Morley's classic book [38] on Inversive Geometry, as well as to [56]. For Möbius transformations of the plane we refer indicatively to [31] and for n-

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I. D. Platis, *Geometries*, La Matematica per il 3+2 178,
https://doi.org/10.1007/978-3-032-08016-5_8



Fig. 8.1 August Ferdinand Möbius, 1790–1868

dimensional transformations we refer to [57]. Finally, for the use of 2×2 matrices with elements from Clifford algebras to describe the Möbius transformations of $\mathbb{R}^n$ we refer to [53].

8.1 The Extended Real Space $\overline{\mathbb{R}^n}$

In this section we will construct the extended $\mathbb{R}^n$ via the stereographic projection and study its metric properties. The basic property is that the extended real space equipped with the metric topology inherited from the sphere via the stereographic projection and the usual Euclidean space with the standard metric topology are compatible: restricted to $\mathbb{R}^n$ the two topologies are identical (Fig. 8.2).

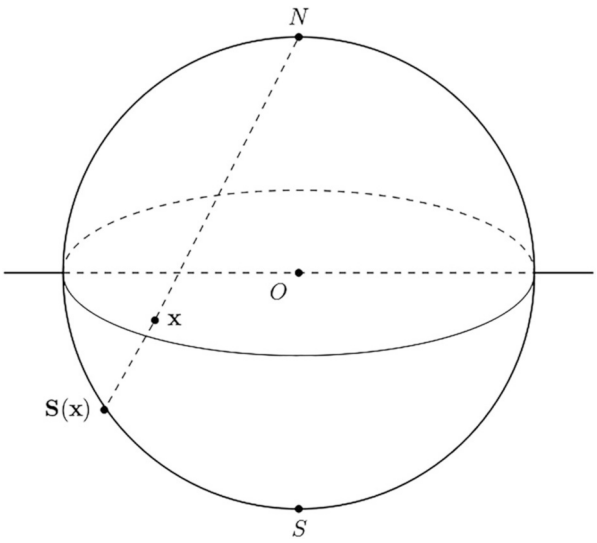


Fig. 8.2 Stereographic projection

8.1.1 Stereographic Projection

Let S^n be the unit sphere of $\mathbb{R}^{n+1}$:

$$S^n = \{\mathbf{x} \in \mathbb{R}^{n+1} \mid \|\mathbf{x}\| = 1\}.$$

We embed $\mathbb{R}^n$ in $\mathbb{R}^{n+1}$ as the hyperplane $x_{n+1} = 0$ and consider the mapping $\mathbf{S} : \mathbb{R}^n \to S^n$ that maps each point $(\mathbf{x}, 0) = (x_1, \ldots, x_n, 0)$ of $\mathbb{R}^n$ to the intersection point of the line segment with endpoints $(\mathbf{x}, 0)$ and $N = (\mathbf{0}, 1)$ and S^n. Explicitly, this line segment is the set of points

$$(1-t)(\mathbf{x},\ 0) + t(\mathbf{0},\ 1) = ((1-t)\mathbf{x}, t),\ t \in [0, 1]$$

and at the point where it intersects S^n we have

$$1 = \|((1-t)\mathbf{x},\ t)\|^2 = (1-t)^2\|\mathbf{x}\|^2 + t^2 \implies t = 1,\ \frac{\|\mathbf{x}\|^2 - 1}{\|\mathbf{x}\|^2 + 1}.$$

For $t = 1$ we obtain the point N. So we get for each $\mathbf{x} \in \mathbb{R}^n$,

$$\mathbf{S}(\mathbf{x}) = \frac{1}{\|\mathbf{x}\|^2 + 1} \cdot (2\mathbf{x},\ \|\mathbf{x}\|^2 - 1).$$

By construction, $\mathbf{S}$ is a bijective map of $\mathbb{R}^n$ to $S^n \setminus \{N\}$. Considering an additional *point at infinity* which we will denote by ∞, we consider the set $\overline{\mathbb{R}^n} = \mathbb{R}^n \cup \{\infty\}$ which we call *extended* $\mathbb{R}^n$. Then, by setting $\mathbf{S}(\infty) = N$ we extend the stereographic projection to a bijective map $\overline{\mathbb{R}^n} \to S^n$. The map $\mathbf{S}$ is called *stereographic projection* of the extended $\mathbb{R}^n$ onto the sphere S^n of $\mathbb{R}^{n+1}$.

To obtain the inverse map of $\mathbf{S}$, let $\mathbf{y} \in S^n$ be a point and consider the straight line from N that passes through $\mathbf{y}$. In parametric form:

$$(1-t)(\mathbf{0}, 1) + t(\mathbf{y}_n, y_{n+1}) = (t\mathbf{y}_n, 1 + t(y_{n+1} - 1)), \quad t \in \mathbb{R}.$$

This line meets $\mathbb{R}^n$ when

$$t = \frac{1}{1 - y_{n+1}}.$$

Thus

$$\mathbf{S}^{-1}(\mathbf{y}) = \begin{cases} \frac{\mathbf{y}_n}{1-y_{n+1}} & \mathbf{y} \in S^n \setminus \{N\}, \\ \\ \infty & \mathbf{y} = N. \end{cases}$$

Remark 8.1.1 The stereographic projection maps the unit sphere S^{n-1} of $\mathbb{R}^n$ pointwise to itself. Moreover, it maps the exterior of S^{n-1} to $S^n_+ = \{\mathbf{x} \in S^n \mid x_{n+1} > 0\}$ and the interior of S^{n-1} to $S^n_- = \{\mathbf{x} \in S^n \mid x_{n+1} < 0\}$.

Exercise Prove the claims of Remark 8.1.1.

8.1.2 Metric Topology of $\overline{\mathbb{R}^n}$

Through the stereographic projection we can equip $\overline{\mathbb{R}^n}$ with the metric topology of S^n which is induced from the Euclidean metric topology of $\mathbb{R}^{n+1}$ and make $\overline{\mathbb{R}^n}$ a *compact* metric space. The metric with which we equip $\overline{\mathbb{R}^n}$ is the *chordal* metric, which we will denote by d. The name is not accidental: we define

$$d(\mathbf{x}_1, \mathbf{x}_2) = \|\mathbf{S}(\mathbf{x}_1) - \mathbf{S}(\mathbf{x}_2)\| = c_{n+1}(\mathbf{S}(\mathbf{x}_1), \mathbf{S}(\mathbf{x}_2)), \quad \mathbf{x}_1, \mathbf{x}_2 \in \overline{\mathbb{R}^n},$$

with c_{n+1} being the chordal metric in S^n (Sect. 3.3.4). d is a metric; this follows from a general argument (see Exercise 2, Sect. 8.7), but we will give here a direct proof: first

$$d(\mathbf{x}_1, \mathbf{x}_2) = 0 \iff \|\mathbf{S}(\mathbf{x}_1) - \mathbf{S}(\mathbf{x}_2)\| = 0 \iff \|\mathbf{x}_1 - \mathbf{x}_2\| = 0 \iff \mathbf{x}_1 = \mathbf{x}_2,$$

where the second equivalence relation holds because $\mathbf{S}$ is bijective. Also,

$$d(\mathbf{x}_1, \mathbf{x}_2) = \|\mathbf{S}(\mathbf{x}_1) - \mathbf{S}(\mathbf{x}_2)\| = \|\mathbf{S}(\mathbf{x}_2) - \mathbf{S}(\mathbf{x}_1)\| = d(\mathbf{x}_2, \mathbf{x}_1),$$

and finally, for $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3 \in \overline{\mathbb{R}^n}$,

$$\begin{aligned} d(\mathbf{x}_1, \mathbf{x}_3) &= \|\mathbf{S}(\mathbf{x}_1) - \mathbf{S}(\mathbf{x}_3)\| \\ &\le \|\mathbf{S}(\mathbf{x}_1) - \mathbf{S}(\mathbf{x}_2)\| + \|\mathbf{S}(\mathbf{x}_2) - \mathbf{S}(\mathbf{x}_3)\| \\ &= d(\mathbf{x}_1, \mathbf{x}_2) + d(\mathbf{x}_2, \mathbf{x}_3). \end{aligned}$$

The inequality holds due to the triangle inequality for the Euclidean distance. The next proposition provides the explicit formula for d.

Proposition 8.1.2 *If $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, then*

$$d(\mathbf{x}, \mathbf{y}) = \frac{2\|\mathbf{x} - \mathbf{y}\|}{(\|\mathbf{x}\|^2 + 1)^{1/2}(\|\mathbf{y}\|^2 + 1)^{1/2}}.$$

If $\mathbf{y} = \infty$, then

$$d(\mathbf{x}, \infty) = \frac{2}{(\|\mathbf{x}\|^2 + 1)^{1/2}}.$$

Proof If $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, then

$$\begin{aligned} d^2(\mathbf{x}, \mathbf{y}) &= \|\mathbf{S}(\mathbf{x}) - \mathbf{S}(\mathbf{y})\|^2 \\ &= \|\mathbf{S}(\mathbf{x})\|^2 + \|\mathbf{S}(\mathbf{y})\|^2 - 2(\mathbf{S}(\mathbf{x}) \cdot \mathbf{S}(\mathbf{y})). \end{aligned}$$

But,

$$\|\mathbf{S}(\mathbf{x})\|^2 = \|\mathbf{S}(\mathbf{y})\|^2 = 1,$$

$$\mathbf{S}(\mathbf{x}) \cdot \mathbf{S}(\mathbf{y}) = \frac{\left(2\mathbf{x},\ \|\mathbf{x}\|^2 - 1\right) \cdot \left(2\mathbf{y},\ \|\mathbf{y}\|^2 - 1\right)}{(\|\mathbf{x}\|^2 + 1)(\|\mathbf{y}\|^2 + 1)} = \frac{4\mathbf{x} \cdot \mathbf{y} + (\|\mathbf{x}\|^2 - 1)(\|\mathbf{y}\|^2 - 1)}{(\|\mathbf{x}\|^2 + 1)(\|\mathbf{y}\|^2 + 1)}.$$

Consequently,

$$\begin{aligned} d^2(\mathbf{x}, \mathbf{y}) &= 2\left(1 - \frac{4\mathbf{x} \cdot \mathbf{y} + (\|\mathbf{x}\|^2 - 1)(\|\mathbf{y}\|^2 - 1)}{(\|\mathbf{x}\|^2 + 1)(\|\mathbf{y}\|^2 + 1)}\right) \\ &= \frac{4\|\mathbf{x} - \mathbf{y}\|^2}{(\|\mathbf{x}\|^2 + 1)(\|\mathbf{y}\|^2 + 1)}. \end{aligned}$$

Now,

$$\begin{aligned} d^2(\mathbf{x}, \infty) &= \|\mathbf{S}(\mathbf{x}) - \mathbf{S}(\infty)\|^2 \\ &= 1 + 1 - 2\frac{((2\mathbf{x},\ \|\mathbf{x}\|^2 - 1) \cdot (\mathbf{0}, 1))}{\|\mathbf{x}\|^2 + 1} \\ &= \frac{4}{\|\mathbf{x}\|^2 + 1}. \end{aligned}$$

□

The last proposition of this section shows us the relationship between the topologies of the chordal and Euclidean metrics on bounded sets of $\mathbb{R}^n$.

Proposition 8.1.3 *The chordal and Euclidean metrics are equivalent metrics on bounded sets of* $\mathbb{R}^n$.

Proof Let Ω be a bounded set of $\mathbb{R}^n$: there exists positive constant $M = M(\Omega)$ such that $\|\mathbf{x}\| \leq M$ for every $\mathbf{x} \in \Omega$. If $\mathbf{x}, \mathbf{y} \in \Omega$ then

$$d(\mathbf{x}, \mathbf{y}) = \frac{2\|\mathbf{x} - \mathbf{y}\|}{(\|\mathbf{x}\|^2 + 1)^{1/2}(\|\mathbf{y}\|^2 + 1)^{1/2}} \leq 2\|\mathbf{x} - \mathbf{y}\|.$$

(We note that this relation also holds for unbounded sets of $\mathbb{R}^n$.) Moreover,

$$\|\mathbf{x} - \mathbf{y}\| = (1/2)d(\mathbf{x}, \mathbf{y})(\|\mathbf{x}\|^2 + 1)^{1/2}(\|\mathbf{y}\|^2 + 1)^{1/2} < (1/2)(1 + M^2)d(\mathbf{x}, \mathbf{y}).$$

The inequalities

$$(1/2)d(\mathbf{x},\mathbf{y}) \leq \|\mathbf{x}-\mathbf{y}\| \leq (1/2)(1+M^2)d(\mathbf{x},\mathbf{y}), \quad \mathbf{x},\mathbf{y} \in \Omega,$$

prove our statement. □

8.2 Spherical and Hyperplanar Inversions in $\overline{\mathbb{R}}^n$

The concept of inversion in a sphere is the natural generalisation of inversion in a circle that was known since antiquity. In Fig. 8.3,

$$\|\mathbf{P}-\mathbf{O}\| \cdot \|\mathbf{P}'-\mathbf{O}\| = r^2,$$

where r is the radius of the circle. The points $\mathbf{P}, \mathbf{P}'$ are inverse points with respect to the circle. In Sect. 8.2.1 we will study the inversion in a sphere of $\mathbb{R}^n$, in Sect. 8.2.2 we will study inversions in hyperplanes that are nothing more than our familiar reflections, while finally, in Sect. 8.2.3 we will examine the action of inversions on $\overline{\mathbb{R}}^n$.

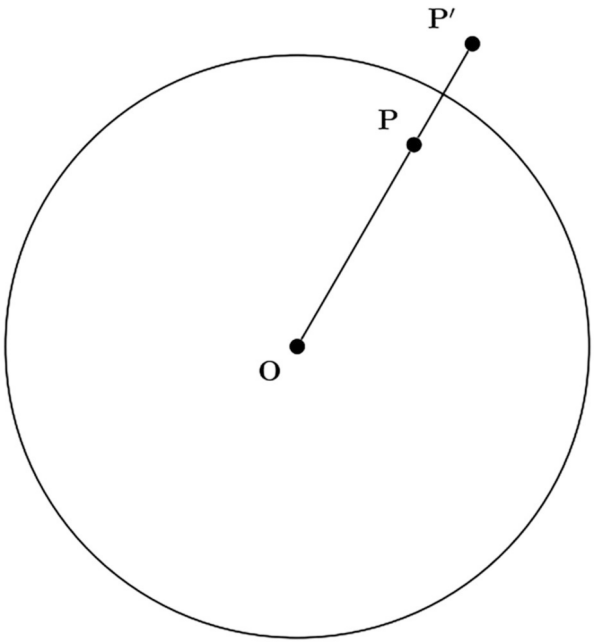


Fig. 8.3 Circular inversion

8.2.1 Inversions in Spheres of $\overline{\mathbb{R}^n}$

Let $S(\mathbf{a}, r)$ be a sphere in $\mathbb{R}^n$, $n \geq 2$, with centre $\mathbf{a}$ and radius $r > 0$:

$$S(\mathbf{a}, r) = \{\mathbf{x} \in \mathbb{R}^n \mid \|\mathbf{x} - \mathbf{a}\| = r\}.$$

We define the *inversion with respect to the sphere* $S(\mathbf{a}, r)$ to be the mapping $\iota_{\mathbf{a},r} : \overline{\mathbb{R}^n} \to \overline{\mathbb{R}^n}$ that corresponds to each $\mathbf{x} \in \mathbb{R}^n \setminus \{\mathbf{a}\}$ the point $\mathbf{y}$ on the line segment $[\mathbf{a}, \mathbf{x}]$ for which

$$\|\mathbf{x} - \mathbf{a}\| \cdot \|\mathbf{y} - \mathbf{a}\| = r^2. \tag{8.1}$$

In particular, $\mathbf{a}$ corresponds to ∞ and ∞ corresponds to $\mathbf{a}$. The point $\mathbf{a}$ is called the *centre of the inversion*, r the *power of the inversion*, and $S(\mathbf{a}, r)$ the *sphere of the inversion*. The points $\mathbf{x}$ and $\mathbf{x}' = \iota_{\mathbf{a},r}(\mathbf{x})$ are called *inverse points*. To calculate the exact formula of $\iota_{\mathbf{a},r}$, we consider the line segment $[\mathbf{a}, \mathbf{x}]$ with parametric equation

$$\mathbf{y} = (1 - t)\mathbf{a} + t\mathbf{x}, \quad t \in [0, 1].$$

Since

$$\mathbf{y} - \mathbf{a} = t(\mathbf{x} - \mathbf{a}) \implies \|\mathbf{y} - \mathbf{a}\| = t\|\mathbf{x} - \mathbf{a}\|,$$

we deduce from (8.1) that

$$r^2 = \|\mathbf{x} - \mathbf{a}\| \cdot \|\mathbf{y} - \mathbf{a}\| = t\|\mathbf{x} - \mathbf{a}\| \implies t = \frac{r^2}{\|\mathbf{x} - \mathbf{a}\|^2}.$$

We conclude:

$$\iota_{\mathbf{a},r}(\mathbf{x}) = \mathbf{a} + r^2 \frac{\mathbf{x} - \mathbf{a}}{\|\mathbf{x} - \mathbf{a}\|^2}, \quad \mathbf{x} \in \mathbb{R}^n \setminus \{\mathbf{a}\}. \tag{8.2}$$

Example 8.2.1 The inversion $\iota_{(0,1),1}$ in the sphere $S(0, 1), 1)$ of $\mathbb{R}^2$ is given by

$$\begin{aligned}\iota_{(0,1),1}(x_1, x_2) &= (0, 1) + 1^2 \frac{(x_1, x_2) - (0, 1)}{\|(x_1, x_2) - (0, 1)\|^2} \\ &= (0, 1) + \frac{(x_1, x_2 - 1)}{x_1^2 + (x_2 - 1)^2}, \quad (x_1, x_2) \in \mathbb{R}^2 \setminus \{(0, 1)\}.\end{aligned}$$

Example 8.2.2 By identifying $\mathbb{R}^2$ to the set of complex numbers $\mathbb{C}$, we may write the formula inversion $\iota_{a,r}$ in the sphere $S(a,r) = \{z \in \mathbb{C} \,|\, |z-a| = r\}$ as

$$\iota_{a,r}(z) = a + r^2 \frac{z-a}{|z-a|^2} = a + \frac{r^2}{\overline{z}-\overline{a}} = \frac{a\overline{z} + r^2 - |a|^2}{\overline{z}-\overline{a}}, \quad z \in \mathbb{C} \setminus \{a\}.$$

In this way, the formula in the previous example may be written equivalently as

$$\iota_{i,1}(z) = i + \frac{z-i}{|z-i|^2} = i + \frac{1}{\overline{z}+i} = \frac{i\overline{z}+2}{\overline{z}+i}.$$

Example 8.2.3 Let $S = S^{n-1} = S(\mathbf{0}, 1)$ and $\iota = \iota_{\mathbf{0},1}$ such that

$$\iota(\mathbf{x}) = \frac{\mathbf{x}}{\|\mathbf{x}\|^2}, \quad \mathbf{x} \in \mathbb{R}^n_* \tag{8.3}$$

defines inversion in the unit sphere S^{n-1} of $\mathbb{R}^n$. In the particular case of $\mathbb{R}^2 \equiv \mathbb{C}$, we have

$$\iota(z) = \frac{1}{\overline{z}}, \quad z \in \mathbb{C}_*.$$

The spherical inversion as defined in (8.2) may be extended into a bijection of $\overline{\mathbb{R}^n}$:

Definition 8.2.4 We define the *inversion* $\iota_{\mathbf{a},r} : \overline{\mathbb{R}^n} \to \overline{\mathbb{R}^n}$ by the formula

$$\iota_{\mathbf{a},r}(\mathbf{x}) = \begin{cases} \mathbf{a} + r^2 \frac{\mathbf{x}-\mathbf{a}}{\|\mathbf{x}-\mathbf{a}\|^2} & \mathbf{x} \in \mathbb{R}^n \setminus \{\mathbf{a}\}, \\ \infty & \mathbf{x} = \mathbf{a}, \\ \mathbf{a} & \mathbf{x} = \infty. \end{cases} \tag{8.4}$$

We can also extend similarities of $\mathbb{R}^n$ to $\overline{\mathbb{R}^n}$ in a natural way.

Definition 8.2.5 The *extended similarity group* $\overline{S(n)}$ of $\overline{\mathbb{R}^n}$ is defined as the group whose elements are compositions of extended translations, extended dilations, and extended elements of the orthogonal group $\mathrm{O}(n)$. Each such extended similarity is obtained by setting its value at ∞ to be ∞.

In other words, $\overline{S(n)}$ is $S(n)$ considered with its elements extended to leave the point at infinity fixed.

Remark 8.2.6 $\overline{S(n)}$ contains the extended group of Euclidean isometries $\overline{E(n)}$ as a proper subgroup. The extension of the Euclidean metric to $\overline{\mathbb{R}^n}$ is defined as follows:

$$\overline{e_n}(\mathbf{x}, \mathbf{y}) = \begin{cases} e_n(\mathbf{x}, \mathbf{y}) & \mathbf{x}, \mathbf{y} \in \mathbb{R}^n, \\ \\ +\infty & \mathbf{x} = \infty,\ \mathbf{y} \in \mathbb{R}^n \text{ or } \mathbf{x} \in \mathbb{R}^n,\ \mathbf{y} = \infty. \end{cases}$$

Hereafter, unless otherwise noted, we will talk about similarities and inversions and we will understand them as mappings of $\overline{\mathbb{R}^n}$.

Proposition 8.2.7 *The inversion of* $\iota_{\mathbf{a},r}$ *with respect to the sphere* $S(\mathbf{a}, r)$ *is written as*

$$\iota_{\mathbf{a},r} = T_{\mathbf{a}} \circ D_r \circ \iota \circ D_{1/r} \circ T_{-\mathbf{a}}, \tag{8.5}$$

where $T_{\mathbf{a}}$, $T_{-\mathbf{a}}$ *are the translations of* $\mathbb{R}^n$ *by* $\mathbf{a}$ *and* $-\mathbf{a}$, *respectively,* D_r, $D_{1/r}$ *are dilations by* r *and* $1/r$, *respectively, and* ι *is the inversion in* S^{n-1}.

Proof For $\mathbf{x} \in \overline{\mathbb{R}^n}$ we have:

$$\begin{aligned} (T_{\mathbf{a}} \circ D_r \circ \iota \circ D_{1/r} \circ T_{-\mathbf{a}})(\mathbf{x}) &= (T_{\mathbf{a}} \circ D_r \circ \iota \circ D_{1/r})(T_{-\mathbf{a}}(\mathbf{x})) \\ &= (T_{\mathbf{a}} \circ D_r \circ \iota \circ D_{1/r})(\mathbf{x} - \mathbf{a}) \\ &= (T_{\mathbf{a}} \circ D_r \circ \iota) \left(\frac{\mathbf{x} - \mathbf{a}}{r} \right) \\ &= (T_{\mathbf{a}} \circ D_r) \left(r \frac{\mathbf{x} - \mathbf{a}}{\|\mathbf{x} - \mathbf{a}\|^2} \right) \\ &= T_a \left(r^2 \frac{\mathbf{x} - \mathbf{a}}{\|\mathbf{x} - \mathbf{a}\|^2} \right) \\ &= \iota_{\mathbf{a},r}(\mathbf{x}). \end{aligned}$$

□

Remark 8.2.8 Let $Q_{\mathbf{a},r} = T_{\mathbf{a}} \circ D_r$. Then we can write

$$\iota_{\mathbf{a},r} = Q_{\mathbf{a},r} \circ \iota \circ Q_{\mathbf{a},r}^{-1}.$$

We will revisit this formula later.

Properties of Spherical Inversions

A direct consequence of the definition of inversion in a sphere (as well as of Eq. (8.5)) is the following:

Proposition 8.2.9 *Every inversion $\iota_{\mathbf{a},r}$ on a sphere $S(\mathbf{a}, r)$ is a bijective transformation of $\overline{\mathbb{R}^n}$. In particular, $\iota_{\mathbf{a},r}$ is an involution:*

$$\iota_{\mathbf{a},r}^2 = id.$$

Proof By Eq. (8.5) it suffices to show that ι is an involution. Indeed, if $\mathbf{x} \in \mathbb{R}^n_*$, then

$$(\iota \circ \iota)(\mathbf{x}) = \iota\left(\frac{\mathbf{x}}{\|\mathbf{x}\|^2}\right) = \mathbf{x}.$$

□

Remark 8.2.10 We observe that since an inversion is an involution we have $\iota_{\mathbf{a},r}^{-1} = \iota_{\mathbf{a},r}$. That is, for every $\mathbf{x} \in \overline{\mathbb{R}^n}$,

$$\iota_{\mathbf{a},r}(\mathbf{x}) = \mathbf{y} \iff \mathbf{x} = \iota_{\mathbf{a},r}(\mathbf{y}).$$

Proposition 8.2.11 *Every inversion $\iota_{\mathbf{a},r}$ in a sphere $S(\mathbf{a}, r)$ is a homeomorphism of $(\overline{\mathbb{R}^n}, d)$, where d is the chordal metric.*

Proof Due to Proposition 8.2.9 it suffices to show that $\iota_{\mathbf{a},r}$ is continuous under the chordal metric in $\overline{\mathbb{R}^n}$. This is true because $\iota_{\mathbf{a},r}$ is continuous under the Euclidean metric and hence under the chordal metric in every $\mathbf{x} \in \mathbb{R}^n \setminus \{\mathbf{a}\}$. Furthermore, we have

$$\lim_{\mathbf{x}\to\mathbf{a}} d(\iota_{\mathbf{a},r}(\mathbf{x}), \iota_{\mathbf{a},r}(\mathbf{a})) = \lim_{\mathbf{x}\to\mathbf{a}} d(\iota_{\mathbf{a},r}(\mathbf{x}), \infty) = \lim_{\mathbf{x}\to\mathbf{a}} \left(\frac{2}{(1 + \|\iota_{\mathbf{a},r}(\mathbf{x})\|^2)^{1/2}}\right) = 0,$$

since

$$\lim_{\mathbf{x}\to\mathbf{a}} \|\iota_{\mathbf{a},r}(\mathbf{x})\|^2 = +\infty.$$

Also, for the inverse ι we have

$$\begin{aligned}
\lim_{\mathbf{x}\to\infty} d(\iota(\mathbf{x}), \iota(\infty)) &= \lim_{\mathbf{x}\to\infty} d(\iota(\mathbf{x}), \mathbf{0}) \\
&= \lim_{\mathbf{x}\to\infty} \left(\frac{2\|\iota(\mathbf{x})\|}{(1 + \|\iota(\mathbf{x})\|^2)^{1/2}}\right) \\
&= \lim_{\mathbf{x}\to\infty} \left(\frac{2}{(1 + \|\mathbf{x}\|^2)^{1/2}}\right) \\
&= 0.
\end{aligned}$$

By Eq. (8.5) we then obtain

$$\lim_{\mathbf{x}\to\infty} d(\iota_{\mathbf{a},r}(\mathbf{x}), \iota_{\mathbf{a},r}(\infty)) = 0.$$

□

Proposition 8.2.12 *Every inversion $\iota_{\mathbf{a},r}$ in a sphere $S(\mathbf{a}, r)$ preserves the inversion sphere pointwise.*

Proof Let $\iota_{\mathbf{a},r}$ be the inversion in the sphere $S(\mathbf{a}, r)$ and let $\mathbf{x} \in S(\mathbf{a}, r)$, i.e., $\|\mathbf{x} - \mathbf{a}\| = r$. Then

$$\iota_{\mathbf{a},r}(\mathbf{x}) = \mathbf{a} + r^2 \frac{\mathbf{x} - \mathbf{a}}{r^2} = \mathbf{x}.$$

□

Remark 8.2.13 Notice that the converse of the above proposition is also true: the fixed points of $\iota_{\mathbf{a},r}$ are exactly the points of $S(\mathbf{a}, r)$.

Proposition 8.2.14 *Every inversion $\iota_{\mathbf{a},r}$ in a sphere $S(\mathbf{a}, r)$ maps the interior of $S(\mathbf{a}, r)$ to its exterior and vice versa.*

Proof Let the inversion $\iota_{\mathbf{a},r}$ in the sphere $S(\mathbf{a}, r)$ and let $\mathbf{x} \in \text{Int}(S(\mathbf{a}, r))$, that is, $\|\mathbf{x} - \mathbf{a}\| < r$. Then,

$$\|\iota_{\mathbf{a},r}(\mathbf{x}) - \mathbf{a}\| = \frac{r^2}{\|\mathbf{x} - \mathbf{a}\|} > r \implies \iota_{\mathbf{a},r}(\mathbf{x}) \in \text{Ext}(S(\mathbf{a}, r)).$$

Similarly, the converse is also proven. □

Proposition 8.2.15 *Let $\iota_{\mathbf{a},r}$ be an inversion in a sphere $S(\mathbf{a}, r)$. For any two points $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n \setminus \{\mathbf{a}\}$ and their corresponding inverses $\mathbf{x}', \mathbf{y}'$, we have:*

$$\|\mathbf{x}' - \mathbf{y}'\| = \frac{r^2 \|\mathbf{x} - \mathbf{y}\|}{\|\mathbf{x} - \mathbf{a}\| \cdot \|\mathbf{y} - \mathbf{a}\|}.$$

If $\mathbf{y} = \mathbf{a}$, then

$$\|\mathbf{x}' - \mathbf{a}\| = \frac{r^2}{\|\mathbf{x} - \mathbf{a}\|}.$$

Proof Since

$$\mathbf{x}' - \mathbf{y}' = r^2 \left(\frac{\mathbf{x} - \mathbf{a}}{\|\mathbf{x} - \mathbf{a}\|^2} - \frac{\mathbf{y} - \mathbf{a}}{\|\mathbf{y} - \mathbf{a}\|^2} \right),$$

by taking norms to the square we have

$$\begin{aligned}\|\mathbf{x}' - \mathbf{y}'\|^2 &= r^4 \left\| \frac{\mathbf{x}-\mathbf{a}}{\|\mathbf{x}-\mathbf{a}\|^2} - \frac{\mathbf{y}-\mathbf{a}}{\|\mathbf{y}-\mathbf{a}\|^2} \right\|^2 \\ &= r^4 \left(\frac{1}{\|\mathbf{x}-\mathbf{a}\|^2} + \frac{1}{\|\mathbf{y}-\mathbf{a}\|^2} - 2\left(\frac{\mathbf{x}-\mathbf{a}}{\|\mathbf{x}-\mathbf{a}\|^2} \right) \cdot \left(\frac{\mathbf{y}-\mathbf{a}}{\|\mathbf{y}-\mathbf{a}\|^2} \right) \right) \\ &= \frac{r^4}{\|\mathbf{x}-\mathbf{a}\|^2 \|\mathbf{y}-\mathbf{a}\|^2} \left(\|\mathbf{y}-\mathbf{a}\|^2 + \|\mathbf{x}-\mathbf{a}\|^2 - 2(\mathbf{x}-\mathbf{a}) \cdot (\mathbf{y}-\mathbf{a}) \right) \\ &= \frac{r^4 \|\mathbf{x}-\mathbf{y}\|^2}{\|\mathbf{x}-\mathbf{a}\|^2 \|\mathbf{y}-\mathbf{a}\|^2}.\end{aligned}$$

The second part of the proposition follows from the definition.

□

8.2.2 Inversions in Hyperplanes of $\overline{\mathbb{R}^n}$

Recall that an affine hyperplane $W_{\mathbf{A},b}$ of $\mathbb{R}^n$ is defined by the equation

$$\mathbf{x} \cdot \mathbf{A} = b, \quad \mathbf{x} \in \mathbb{R}^n,$$

where $\mathbf{A} \in \mathbb{R}^n$, $\|\mathbf{A}\| = 1$ and $b \in \mathbb{R}$. Such a hyperplane is extended to a hyperplane $\overline{W_{\mathbf{A},b}}$ of $\overline{\mathbb{R}^n}$ by adding the point at ∞: $\overline{W_{\mathbf{A},b}} = W_{\mathbf{A},b} \cup \{\infty\}$.

Definition 8.2.16 The *inversion in the hyperplane* $\overline{W_{\mathbf{A},b}}$ is the extended reflection $R_{\mathbf{A},b}$:

$$R_{\mathbf{A},b}(\mathbf{x}) = \begin{cases} \mathbf{x} + 2(b - (\mathbf{x} \cdot \mathbf{A}))\mathbf{A}, & \mathbf{x} \in \mathbb{R}^n, \\ \infty & \mathbf{x} = \infty. \end{cases} \tag{8.6}$$

Example 8.2.17 In $\mathbb{R}^2 = \mathbb{C}$ a straight line l can be written as $\Re(\overline{a}z) = b$, where $|a| = 1$ and $b \in \mathbb{R}$. Inversion in l is the extended reflection r defined by

$$r(z) = \begin{cases} z + 2(b - \Re(\overline{a}z))a, & z \in \mathbb{C}, \\ \infty & z = \infty. \end{cases} \tag{8.7}$$

In particular, for the straight line $\Re(-iz) = 0$, that is, the axis $x = 0$, the inversion is given for $z \in \mathbb{C}$ by

$$r(z) = \overline{z}.$$

Properties of Hyperplanar Inversions

The following proposition summarises some basic properties of inversions in hyperplanes.

Proposition 8.2.18 *Let $R_{\mathbf{A},b} : \overline{\mathbb{R}^n} \to \overline{\mathbb{R}^n}$ be the inversion in a hyperplane $\overline{W_{\mathbf{A},b}}$ of $\overline{\mathbb{R}^n}$. Then*

- *$R_{\mathbf{A},b}$ is bijective,*
- *$R_{\mathbf{A},b}$ is an involution: $R_{\mathbf{A},b} \circ R_{\mathbf{A},b} = id.$,*
- *$R_{\mathbf{A},b}$ is a homeomorphism of $\overline{\mathbb{R}^n}$,*
- *$R_{\mathbf{A},b}(\mathbf{x}) = \mathbf{x}$ if and only if $\mathbf{x} \in \overline{W_{\mathbf{A},b}}$.*

Exercise Prove Proposition 8.2.18.

8.2.3 Action of Inversions in $\overline{\mathbb{R}^n}$

In this section we study the manner that an inversion maps spheres and hyperplanes. We start with the following:

Proposition 8.2.19 *The inversion $\iota_{\mathbf{a},r}$ maps every hyperplane passing through the inversion centre to itself. Also, it maps every hyperplane not passing through the inversion centre to a sphere of $\mathbb{R}^n$ passing through the inversion centre.*

Proof Let $\iota_{\mathbf{a},r}$ be the inversion in the sphere $S(\mathbf{a}, r)$. A hyperplane $\overline{W}$ of $\overline{\mathbb{R}^n}$ passing through the centre $\mathbf{a}$ of the sphere has equation $\mathbf{x}\cdot\mathbf{A} = \mathbf{a}\cdot\mathbf{A}$, $\mathbf{x} \in \mathbb{R}^n$, where $\mathbf{A} \in \mathbb{R}^n$, $\|\mathbf{A}\| = 1$. If $\mathbf{x} \in W$, let $\mathbf{y} = \iota_{\mathbf{a},r}(\mathbf{x})$; then

$$\mathbf{x} = \mathbf{a} + r^2 \frac{\mathbf{y} - \mathbf{a}}{\|\mathbf{y} - \mathbf{a}\|^2}$$

and therefore

$$\mathbf{a} \cdot \mathbf{A} = \mathbf{x} \cdot \mathbf{A} = \mathbf{a} \cdot \mathbf{A} + r^2 \frac{\mathbf{y} - \mathbf{a}}{\|\mathbf{y} - \mathbf{a}\|^2} \cdot \mathbf{A},$$

from which we get $\mathbf{y} \cdot \mathbf{A} = \mathbf{a} \cdot \mathbf{A}$.

A hyperplane $\overline{W_{\mathbf{A},b}}$ that does not pass through the centre $\mathbf{a}$ of the inversion has equation $\mathbf{x} \cdot \mathbf{A} = b$, $b \neq \mathbf{a} \cdot \mathbf{A}$, where $\|\mathbf{A}\| = 1$. For $\mathbf{x} \in W_{\mathbf{A},b}$ and $\mathbf{y}$ as before, then

$$b = \mathbf{x} \cdot \mathbf{A} = \mathbf{a} \cdot \mathbf{A} + r^2 \frac{\mathbf{y} - \mathbf{a}}{\|\mathbf{y} - \mathbf{a}\|^2} \cdot \mathbf{A}.$$

We write equivalently

$$\|\mathbf{y}-\mathbf{a}\|^2+\frac{r^2}{\mathbf{a}\cdot\mathbf{A}-b}(\mathbf{y}-\mathbf{a})\cdot\mathbf{A}=0.$$

Completing the square yields

$$\left\|\mathbf{y}-\left(\mathbf{a}+\frac{r\mathbf{A}}{2(\mathbf{a}\cdot\mathbf{A}-b)}\right)\right\|^2=\frac{r^4}{4|\mathbf{a}\cdot\mathbf{A}-b|^2}.$$

This is an equation that represents a sphere passing through $\mathbf{a}$. □

Corollary 8.2.20 *The inversion $\iota_{\mathbf{a},r}$ maps every sphere of $\mathbb{R}^n$ that passes through the inversion centre to a hyperplane that does not pass through the inversion centre.*

Now let $S(\mathbf{a},r')$ be a sphere concentric with the inversion sphere. If $\mathbf{x}\in S(\mathbf{a},r')$ then $\|\mathbf{x}-\mathbf{a}\|=r'$. We set $\mathbf{y}=\iota_{\mathbf{a},r}(\mathbf{x})$. Then,

$$\|\mathbf{y}-\mathbf{a}\|=r^2/r',$$

that is, $\mathbf{y}\in S(\mathbf{a},r^2/r')$. Now if $S(\mathbf{a}',r')$ is any sphere, then $\|\mathbf{x}'-\mathbf{a}'\|=r'$ for every $\mathbf{x}'\in S(\mathbf{a}',r')$. We again set $\mathbf{y}=\iota_{\mathbf{a},r}(\mathbf{x})$ and then from the relation

$$\mathbf{y}=\mathbf{a}+r^2\frac{\mathbf{x}-\mathbf{a}}{\|\mathbf{x}-\mathbf{a}\|^2}\iff\mathbf{x}=\mathbf{a}+r^2\frac{\mathbf{y}-\mathbf{a}}{\|\mathbf{y}-\mathbf{a}\|^2},$$

it follows that

$$r'=\|\mathbf{x}-\mathbf{a}'\|=\left\|\mathbf{a}-\mathbf{a}'+r^2\frac{\mathbf{y}-\mathbf{a}}{\|\mathbf{y}-\mathbf{a}\|^2}\right\|.$$

Taking this to the square, the following relation follows after the calculations:

$$\left(\|\mathbf{a}-\mathbf{a}'\|^2-(r')^2\right)\|\mathbf{y}-\mathbf{a}\|^2+2r^2(\mathbf{a}-\mathbf{a}')\cdot(\mathbf{y}-\mathbf{a})+r^4=0.$$

We distinguish two cases:

1. $B=\|\mathbf{a}-\mathbf{a}'\|^2-(r')^2=0$: the sphere $S(\mathbf{a}',r')$ passes through the inversion centre. Then we get

$$2r^2(\mathbf{a}-\mathbf{a}')\cdot(\mathbf{y}-\mathbf{a})+r^4=0,$$

an equation representing a hyperplane that does not pass through the inversion centre.

2. $B \neq 0$: the sphere $S(\mathbf{a}', r')$ does not pass through the inversion centre. We write

$$\|\mathbf{y} - \mathbf{a}\|^2 + 2(r^2/B)(\mathbf{a} - \mathbf{a}') \cdot (\mathbf{y} - \mathbf{a}) + r^4/B = 0$$

and completing the square yields

$$\left\|\mathbf{y} - \Big(\mathbf{a} + (r^2/B)(\mathbf{a}' - \mathbf{a})\Big)\right\|^2 = r^4(r')^2/B^2,$$

which is an equation of a sphere that does not pass through the inversion centre.

All of the above is summarised as follows:

Proposition 8.2.21 *The inversion $\iota_{\mathbf{a},r}$ maps:*

1. *every sphere concentric with the inversion sphere to a sphere concentric with the inversion sphere,*
2. *every sphere passing through the inversion centre to a hyperplane that does not pass through the inversion centre, and*
3. *every sphere not passing through the inversion centre to a sphere that does not pass through the inversion centre.*

Proposition 8.2.22 *Let $R_{\mathbf{A},b}$ be an inversion in a hyperplane $\overline{W_{\mathbf{A},b}}$. Then $R_{\mathbf{A},b}$ maps spheres of $\mathbb{R}^n$ to spheres of $\mathbb{R}^n$ and hyperplanes of $\overline{\mathbb{R}^n}$ to hyperplanes of $\overline{\mathbb{R}^n}$.*

Exercise Prove Proposition 8.2.22.

Hereafter, by simply saying *inversion* of $\overline{\mathbb{R}^n}$ we will mean either an inversion in a sphere or an inversion in a hyperplane of $\overline{\mathbb{R}^n}$. Also, by *generalised sphere* we will call either a (Euclidean) sphere of $\mathbb{R}^n$ or a hyperplane of $\overline{\mathbb{R}^n}$. From the discussion of this section, we obtain:

Theorem 8.2.23 *Every inversion of $\overline{\mathbb{R}^n}$ maps generalised spheres onto generalised spheres of $\overline{\mathbb{R}^n}$.*

Example 8.2.24 In the complex plane $\mathbb{C}$ consider the inversion ι in the unit circle S^1 given by $\iota(z) = 1/\overline{z}$ and an arbitrary generalised circle S. If S is the straight line $\Re(\overline{a}z) = b$, then for $w = \iota(z) = 1/\overline{z}$, $z \in S$ we have

$$z = 1/\overline{w} \iff \Re(a/w) = b.$$

If $b = 0$ then $\iota(S) = S$. If $b \neq 0$ then straightforward calculations deduce the circle

$$\left|w - \frac{\overline{a}}{2b}\right| = \frac{|a|}{2b}.$$

Using the above, we calculate the image $\iota(S)$, where S is the *circle* $|z - 1| = 1$. Since $1 = 1/(2 \cdot (1/2))$ we set $a = 1$ and $b = 1/2$ to obtain that $\iota(S)$ is the straight line $\Re(z) = x = 1/2$.

Example 8.2.25 In the complex plane $\mathbb{C}$ consider the circle $S(r) = \{z \in \mathbb{C} \,|\, |z| = r\}$. If $z \in S(r)$ then

$$w = \iota(z) = 1/\overline{z} \implies |w| = 1/r.$$

Therefore $\iota(S(r)) \subset S(1/r)$ and similarly we show that $S(1/r) \subset \iota(S(r))$, which induces $\iota(S(r)) = S(1/r)$.

8.3 The General Möbius Group GM(n)

Definition 8.3.1 A *Möbius transformation of* $\overline{\mathbb{R}^n}$ is a composition of finitely many inversions of $\overline{\mathbb{R}^n}$. We denote by GM(n) the set of Möbius transformations of $\overline{\mathbb{R}}^n$ and call it the *general Möbius group*.

Exercise Prove that the set GM(n) is indeed a group.

From Theorem 8.2.23 we immediately have the following:

Theorem 8.3.2 *Every Möbius transformation of* $\overline{\mathbb{R}^n}$ *maps generalised spheres on generalised spheres of* $\overline{\mathbb{R}^n}$.

The Inversive Geometry (according to Klein) is the pair $(\overline{\mathbb{R}^n},\ \mathrm{GM}(n))$. In the following we will give a complete description of the general Möbius group GM(n) and its action on $\overline{\mathbb{R}^n}$.

8.3.1 Elementary Möbius Transformations

The elements of the extended similarity group $\overline{S(n)}$ are all Möbius transformations. Explicitly:

- Any translation $T_{\mathbf{a}}$ is written as the composition of the reflection in the hyperplane $\mathbf{x} \cdot \mathbf{a} = 0$ followed by the reflection in the hyperplane $\mathbf{x} \cdot \mathbf{a} = \|\mathbf{a}\|^2/2$.
- Due to the Reflection Theorem, every element of O(n) is a composition of reflections, that is, inversions in hyperplanes.
- On the other hand, dilations D_δ are also Möbius transformations: $D_\delta = \iota \circ \iota_{\mathbf{0},1/\sqrt{\delta}}$.

By Proposition 8.2.7 we now have the following:

Proposition 8.3.3 *Any two inversions in spheres of* $\mathbb{R}^n$ *are conjugate in the group* GM(n).

Proof Let ι_1, ι_2 be inversions in spheres S_1, S_2, respectively. There are elements Q_1, Q_2 in $\in \mathrm{GM}(n)$ such that

$$\iota_i = Q_i \circ \iota \circ Q_i^{-1}, \quad i = 1, 2,$$

where ι is the inversion in the sphere $S(\mathbf{0}, 1)$ (see Remark 8.2.8). Then

$$\iota = Q_1^{-1} \circ \iota_1 \circ Q_1 = Q_2^{-1} \circ \iota_2 \circ Q_2 \implies \iota_2 = Q \circ \iota_1 \circ Q^{-1},\ Q = Q_2 Q_1^{-1}.$$

□

The following theorem gives a complete description of the elements of $\in$ GM(n).

Theorem 8.3.4 *Every $g \in \mathrm{GM}(n)$ can be written as a composition of the following transformations of $\overline{\mathbb{R}^n}$:*

- *translations $T_{\mathbf{a}}$,*
- *orthogonal transformations $A \in \mathrm{O}(n)$,*
- *the inversion ι in the sphere $S(\mathbf{0}, 1)$,*
- *dilations D_δ.*

For the proof of Theorem 8.3.4 we need the following:

Lemma 8.3.5 *Let h be a Möbius transformation that has one of the forms of Theorem 8.3.4 and let j be an inversion of $\overline{\mathbb{R}^n}$. Then the composition $h \circ j$ can be written as a composition of transformations of the forms of Theorem 8.3.4.*

Proof The result follows immediately from Proposition 8.2.7 if j is an inversion in a sphere, while if it is an inversion of $R_{\mathbf{A},b}$ on a hyperplane $\overline{W_{\mathbf{A},b}}$, then it can be written as $R_{\mathbf{A},b} = T_{2b\mathbf{A}} \circ R_{\mathbf{A},0}$, that is, as a composition of a translation and an orthogonal transformation. □

Proof of Theorem 8.3.4 Let $g \in \mathrm{GM}(n)$, $g = j_1 \circ \cdots \circ j_n$, where j_i are inversions of $\overline{\mathbb{R}^n}$. The proof is by induction in n.

If $n = 1$ then g is either an inversion $\iota_{\mathbf{a},r}$ on a sphere $S(\mathbf{a}, r)$ or a reflection of $R_{\mathbf{A},b}$ in a hyperplane $\overline{W_{\mathbf{A},b}}$. In the first case, we have the result from Proposition 8.2.7. In the second case, we have from the proof of Lemma 8.3.5 that every inversion in a hyperplane can be written as a composition of a translation and an orthogonal transformation.

We now assume that the result holds for all numbers less than n and write $g = h \circ j_{n+1}$, where h is a composition of transformations as in the hypothesis of the theorem. The result follows immediately from Lemma 8.3.5. □

Comparison of Inversion Geometry with Other Geometries

From the above discussion, Definition 8.2.5 and Comment 8.2.6 we have the following relation:

$$(\overline{\mathbb{R}^n}, \overline{E(n)}) \subset (\overline{\mathbb{R}^n}, \overline{S(n)}) \subset (\overline{\mathbb{R}^n}, \overline{\mathrm{GM}(n)}).$$

We now examine the relation of Inversive Geometry with Affine Geometry. First, the elements of the affine group $A(n)$ are all extended to elements that act on $\overline{\mathbb{R}^n}$ and preserve the point at infinity; this procedure gives the extended affine group $\overline{A(n)}$. We will thus compare Inversive Geometry with $(\overline{\mathbb{R}^n}, \overline{A(n)})$. For convenience, we will limit ourselves to the case $n = 2$. Due to Exercise 20, Sect. 5.4 and the preceding discussion, it follows that every element of $\overline{A(2)}$ is of the form

$$f(z) = Az + B\overline{z} + C, \quad A, B, C \in \mathbb{C},\ |A| + |B| \neq 0,$$

if $z \in \mathbb{C}$ and $f(\infty) = \infty$. However, the inversions on circles of $\mathbb{C}$ cannot be written in the above form (justify!). It therefore follows that in general, a Möbius transformation of $\overline{\mathbb{C}}$ does not belong to $\overline{A(2)}$. On the other hand, the extension of an affine transformation on $\overline{\mathbb{C}}$ is not necessarily a Möbius transformation. For example, let

$$f(z) = ((1 + K)/2)z + ((1 - K)/2)\overline{z}, \quad 0 < K < 1,$$

and the unit circle $|z| = 1$. Then, since

$$w = f(z) \iff z = \Re(w) + i\frac{\Im(w)}{K},$$

it follows that

$$1 = |z| = \left|\Re(w) + i\frac{\Im(w)}{K}\right|,$$

which represents an ellipse. But this cannot be possible, since Möbius transformations map generalised circles to generalised circles.

8.4 Möbius Transformations are Conformal

Definition 8.4.1 A mapping $f : \mathbb{R}^n \to \mathbb{R}^n$ is said to be *conformal at* $\mathbf{x}_0 \in \mathbb{R}^n$ if it *preserves the angles at* $\mathbf{x}_0$.

If the conformal f is differentiable at $\mathbf{x}_0$ then this means that the Jacobian matrix $Df(\mathbf{x}_0)$ of f at $\mathbf{x}_0$ satisfies a relation of the form

$$(Df(\mathbf{x}_0))(Df(\mathbf{x}_0))^T = \lambda(\mathbf{x}_0)I_n,$$

where $\lambda(\mathbf{x}_0) \neq 0$ and I_n is the unitary $n \times n$ matrix. (Exercise 11, Sect. 8.7.)

Theorem 8.4.2 *Every $g \in \mathrm{GM}(n)$ is a conformal mapping of $\mathbb{R}^n$. If g contains an even number of inversions, then it preserves orientation. Otherwise, it reverses the orientation.*

Proof According to Theorem 8.3.4 it suffices to prove our theorem for the following cases:

1. *g is a translation.* If $g = T_{\mathbf{a}}$, then for every $\mathbf{x} \in \mathbb{R}^n$

$$T_{\mathbf{a}}(\mathbf{x}) = \mathbf{x} + \mathbf{a} \implies DT_{\mathbf{a}}(\mathbf{x}) = I_n,$$

the identity $n \times n$ matrix. But $\det(DT_{\mathbf{a}}(\mathbf{x})) = 1 > 0$ and therefore the transfers preserve orientation.

2. *g is an element of* $\mathrm{O}(n)$. If $g = A \in \mathrm{O}(n)$, then for every $\mathbf{x} \in \mathbb{R}^n$

$$g(\mathbf{x}) = \mathbf{x}A \implies Dg(\mathbf{x}) = A^T.$$

Now, $\det(Dg(\mathbf{x})) = 1$ if $A \in \mathrm{SO}(n)$ while $\det(Dg(\mathbf{x})) = -1$ when A is a reflection in hyperplanes from the origin.

3. *g is a dilation.* If $g = D_\delta$, $\delta > 0$, then for every $\mathbf{x} \in \mathbb{R}^n$

$$D_\delta(\mathbf{x}) = \delta\mathbf{x} \implies D_\delta(\mathbf{x}) = \delta I_n,$$

and also $\det(D_\delta(\mathbf{x})) = \delta^n > 0$.

4. *g is the inversion of ι.* If $g = \iota$, then for every $\mathbf{x} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$,

$$\iota(\mathbf{x}) = \frac{\mathbf{x}}{\|\mathbf{x}\|^2} \implies D\iota(\mathbf{x}) = D\left(\frac{1}{\|\mathbf{x}\|^2} \cdot \mathbf{x}\right).$$

Recall that if $f : U \to \mathbb{R}$ and $g : U \to \mathbb{R}^n$ are differentiable, where U is open in $\mathbb{R}^n$, then for every $\mathbf{x} \in U$,

$$D(f \cdot g)(\mathbf{x}) = g(\mathbf{x})^T(Df(\mathbf{x})) + f(\mathbf{x})Dg(\mathbf{x}).$$

In our case, $f(\mathbf{x}) = 1/\|\mathbf{x}\|^2$ and $g(\mathbf{x}) = \mathbf{x}$. So, $Dg(\mathbf{x}) = I_n$ and

$$Df(\mathbf{x}) = -\frac{1}{\|\mathbf{x}\|^4} D(\|\mathbf{x}\|^2) = -\frac{2\mathbf{x}}{\|\mathbf{x}\|^4}.$$

Therefore,

$$D\iota(\mathbf{x}) = -\frac{2\mathbf{x}^T\mathbf{x}}{\|\mathbf{x}\|^4} + \frac{1}{\|\mathbf{x}\|^2} I_n = \frac{1}{\|\mathbf{x}\|^2}(I_n - 2Q(\mathbf{x})), \quad Q(\mathbf{x}) = \frac{\mathbf{x}^T\mathbf{x}}{\|\mathbf{x}\|^2}.$$

We observe that $Q(x)$ is symmetric and

$$Q^2(\mathbf{x}) = \frac{(\mathbf{x}^T\mathbf{x})(\mathbf{x}^T\mathbf{x})}{\|\mathbf{x}\|^4} = \frac{\mathbf{x}^T(\mathbf{x}\mathbf{x}^T)\mathbf{x}}{\|\mathbf{x}\|^4} = \frac{\|\mathbf{x}\|^2(\mathbf{x}^T\mathbf{x})}{\|\mathbf{x}\|^4} = Q(\mathbf{x}).$$

So,

$$(D\iota(\mathbf{x}))(D\iota(\mathbf{x}))^T = \frac{1}{\|\mathbf{x}\|^4}(I_n - 2Q(\mathbf{x}))^2 = \frac{1}{\|\mathbf{x}\|^4}I_n$$

and ι is conformal. Now taking determinants we have that

$$(\det(D\iota(\mathbf{x})))^2 = \frac{1}{\|\mathbf{x}\|^{4n}} \implies \det(D\iota(\mathbf{x})) = \pm\frac{1}{\|\mathbf{x}\|^{2n}}.$$

Due to the continuity of $\mathbb{R}^n_*$ the determinant can be either only positive or only negative. To determine the sign, we examine the value of the determinant at the point $\mathbf{e}_1 = (1, 0, \ldots, 0)$:

$$D\iota(\mathbf{e}_1) = I_n - 2Q(\mathbf{e}_1) = J,$$

where J is a diagonal $n \times n$ matrix with $j_{11} = -1$ and $j_{ii} = 1$ for each $i > 1$. Therefore, its determinant is -1 and we conclude that

$$D\iota(\mathbf{x}) = -\frac{1}{\|\mathbf{x}\|^{2n}},$$

and therefore ι reverses the orientation.

□

Remark 8.4.3 In the complex plane $\mathbb{C}$, every holomorphic function $f : \mathbb{C} \supset D \to \mathbb{C}$ defined in an open domain D of $\mathbb{C}$ whose complex derivative is different from zero at a point $z_0 \in D$ is conformal (and orientation preserving) at z_0. Indeed, if $f(z) = f(x + iy) = u(x, y) + iv(x, y)$, then f satisfies the Cauchy-Riemann equations

$$u_x = v_y, \text{ and } u_y = -v_x,$$

everywhere in D. Therefore, at each $z_0 \in D$ we have

$$Df(z_0)(Df(z_0))^T = \begin{pmatrix} u_x & u_y \\ -u_y & u_x \end{pmatrix}_{z_0} \begin{pmatrix} u_x & -u_y \\ u_y & u_x \end{pmatrix}_{z_0} = (u_x^2(z_0) + u_y^2(z_0))I_2.$$

Following the same lines, one may show that a function $f : D \to \mathbb{C}$ which is antiholomorphic at a point $z_0 \in D$ (that is, the function $\overline{f}$ is holomorphic at z_0) is again conformal (and orientation reversing) at z_0.

Example 8.4.4 In this example we show that the condition $\lambda(x_0)$ in the definition 8.4.1 is necessary. Indeed consider the holomorphic map $f : \mathbb{C} \to \mathbb{C}$ given by $f(z) = z^2$ for each $z \in \mathbb{C}$. Since $f'(z) = 2z$ we have $f'(0) = 0$. We consider the straight lines $x = y$ and $x = -y$ which meet vertically at 0. However, their images are the positive and the negative half-lines of the y-axis, respectively, both emanating from zero. Clearly, their angle is $\pi \neq \pi/2$.

The abundance of conformal maps in the two dimensional case does not have a counterpart in higher dimensions. At this point, it is worth mentioning the following amazing (and profound) result of the French mathematician Liouville (Joseph Liouville, 1809–1882):

Theorem 8.4.5 (Liouville) *In dimensions $n > 2$, all conformal mappings of $\mathbb{R}^n$ are Möbius transformations.*

The proof of this theorem is beyond the scope of this book; for an elementary and relatively short proof we refer the interested reader to [33]. Theorem 8.4.2 leads us to the following definition:

Definition 8.4.6 The set of orientation-preserving Möbius transformations of $\overline{\mathbb{R}^n}$ will be denoted by $\mathrm{M}(n)$ and will be called the *Möbius group*.

Exercise Show that the set $\mathrm{M}(n)$ is a subgroup of $\mathrm{GM}(n)$.

8.5 Fundamental Theorem of Inversive Geometry

Regarding the transitivity of the action of $\mathrm{GM}(n)$ on $\overline{\mathbb{R}^n}$ we have the following theorem which is the Fundamental Theorem of Inversive Geometry:

Theorem 8.5.1 *The group $\mathrm{GM}(n)$ acts triply transitively on $\overline{\mathbb{R}^n}$: if $\mathbf{x}_i$ and $\mathbf{y}_i$, $i = 1, 2, 3$, are triplets of pairwise distinct points of $\overline{\mathbb{R}^n}$, then there exists $g \in \mathrm{GM}(n)$ such that*

$$g(\mathbf{x}_i) = \mathbf{y}_i, \quad i = 1, 2, 3.$$

If g' is another transformation that satisfies the above relations, then $g' \in gH$ where H is a subgroup of $\mathrm{GM}(n)$ conjugate to $\mathrm{O}(n-1)$. If $n = 2$, then either $g = g'$, or $g' = gh$ where h is conjugate with reflection in a line. If $n = 1$ then g is unique.

We will first prove two lemmas.

Lemma 8.5.2 *For any three pairwise distinct points $\mathbf{x}_i$, $i = 1, 2, 3$ of $\overline{\mathbb{R}^n}$, there exists an $h \in \mathrm{GM}(n)$ such that*

$$h(\mathbf{x}_1) = \infty, \quad h(\mathbf{x}_2) = \mathbf{0}, \quad h(\mathbf{x}_3) = \mathbf{e}_1 = (1, 0, \ldots, 0).$$

Proof Indeed, the translation $T_{-\mathbf{x}_1}$ maps $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ to

$$\mathbf{0}, \quad \mathbf{x}_2' = \mathbf{x}_2 - \mathbf{x}_2, \quad \mathbf{x}_3' = \mathbf{x}_3 - \mathbf{x}_1,$$

respectively. Accordingly, inversion ι maps $\mathbf{0}, \mathbf{x}_2', \mathbf{x}_3'$ to

$$\infty, \quad \mathbf{x}_2'' = \iota(\mathbf{x}_2'), \quad \mathbf{x}_3'' = \iota(\mathbf{x}_3'),$$

respectively. Then, translation $T_{-\mathbf{x}_2''}$ maps $\infty, \mathbf{x}_2'', \mathbf{x}_3''$ to

$$\infty, \quad \mathbf{0}, \quad \mathbf{x}_3''' = \iota(\mathbf{x}_3') - \mathbf{x}_2'',$$

respectively. Now, there an element of $\mathrm{O}(n)$ that maps $\mathbf{x}_3'''$ to a point $(a, \mathbf{0}_{n-1})$, $a = \|\mathbf{x}_3'''\| > 0$. Letting the similarity $D_{1/a}$ act on $\infty, \mathbf{0}, (a, \mathbf{0}_{n-1})$, we get the desired result. □

Lemma 8.5.3 *The stabiliser* $\mathrm{Stab}(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3)$ *of three pairwise distinct points* $\mathbf{x}_i$, $i = 1, 2, 3$, *of* $\overline{\mathbb{R}^n}$ *is a group conjugate to* $\mathrm{O}(n-1)$ *if* $n > 1$. *If* $n = 1$ *the stabiliser is trivial. If* $n = 2$, $\mathrm{Stab}(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3)$ *contains the identity transformation and an element conjugate with a reflection on a line.*

Proof Every Möbius transformation of $\overline{\mathbb{R}^n}$ that keeps the points $\infty, \mathbf{0}, \mathbf{1} = \mathbf{e}_1$ fixed is an element of $\mathrm{O}(n-1)$ as it is considered by its embedding

$$\mathrm{O}(n-1) \ni A \mapsto \begin{pmatrix} 1 & 0 \\ 0 & A \end{pmatrix} \in \mathrm{O}(n).$$

From Lemma 8.5.2 we now get

$$\mathrm{Stab}((\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3)) = h\mathrm{O}(n-1)h^{-1}, \quad h \in \mathrm{GM}(n).$$

□

Proof of Theorem 8.5.1 We conclude from Lemma 8.5.2 that there exist Möbius transformations h_1 and h_2 such that

$$h_1(\mathbf{x}_1) = h_2(\mathbf{y}_1) = \infty, \quad h_1(\mathbf{x}_2) = h_2(\mathbf{y}_2) = \mathbf{0}, \quad h_1(\mathbf{x}_3) = h_2(\mathbf{y}_3) = \mathbf{e}_1.$$

The transformation $g = h_2^{-1} \circ h_1 \in \mathrm{M}(\overline{\mathbb{R}^n})$ satisfies the requirements of the theorem. Suppose that there existed a g' such that $g'(\mathbf{x}_i) = \mathbf{y}_i$, $i = 1, 2, 3$. Then the transformation $f = h_1 g^{-1} g' h_1^{-1}$ stabilises the points $\infty, \mathbf{0}, \mathbf{1} = \mathbf{e}_1$ and therefore $f \in \mathrm{O}(n-1)$. Consequently, $g^{-1}g' = h_1^{-1} f h_1 = h$ which is an element that stabilises $\mathbf{x}_i$, $i = 1, 2, 3$. We thus obtain $g' = gh$.

If $n = 2$ then $h = id.$ if $f = id.$ and $h = h_1^{-1} r h_1$ if $f = r$, the reflection in $x_2 = 0$. It follows that either $g' = g$ or $g' = gh$ where h is conjugate by reflection. The case $n = 1$ is clear. □

Example 8.5.4 In order to find the Möbius transformation that maps three pairwise distinct points z_1, z_2, z_3 of $\mathbb{C} \equiv \mathbb{R}^2$ to the points $0, \infty, 1$, respectively, we proceed as in the proof of 8.5.2. Consider first the translation $f_1(z) = z - z_1$:

$$f_1(z_1) = 0, \quad f_1(z_2) = z_2 - z_1, \quad f_1(z_3) = z_3 - z_1.$$

Next, we apply inversion $\iota = f_2$ given by $f_2(z) = 1/\overline{z}$ to the points above:

$$(f_2 \circ f_1)(0) = \infty, \quad (f_2 \circ f_1)(z_2 - z_1) = 1/(\overline{z_2} - \overline{z_1}),$$
$$(f_2 \circ f_1)(z_3 - z_1) = 1/(\overline{z_3} - \overline{z_1}).$$

We now take the translation f_3 defined by $f_3(z) = z - 1/(\overline{z_2} - \overline{z_1})$:

$$(f_3 \circ f_2 \circ f_1)(\infty) = \infty, \quad (f_3 \circ f_2 \circ f_1)(1/(\overline{z_2} - \overline{z_1})) = 0,$$
$$(f_3 \circ f_2 \circ f_1)(1/(\overline{z_3} - \overline{z_1})) = \frac{\overline{z_2} - \overline{z_3}}{(\overline{z_3} - \overline{z_1})(\overline{z_2} - \overline{z_1})}.$$

We apply the reflection $f_4 = r$ given by $r(z) = \overline{z}$ to obtain:

$$(f_4 \circ f_3 \circ f_2 \circ f_1)(\infty) = \infty, \quad (f_4 \circ f_3 \circ f_2 \circ f_1)(0) = 0,$$
$$(f_4 \circ f_3 \circ f_2 \circ f_1)(\infty)\left(\frac{\overline{z_2} - \overline{z_3}}{(\overline{z_3} - \overline{z_1})(\overline{z_2} - \overline{z_1})}\right) = \frac{z_2 - z_3}{(z_3 - z_1)(z_2 - z_1)}.$$

We finally apply the similarity f_5 given by

$$f_5(z) = \frac{(z_3 - z_1)(z_2 - z_1)}{z_2 - z_3} z$$

to obtain the desired result. Notice that

$$(f_5 \circ f_4 \circ f_3 \circ f_2 \circ f_1)(z) = \frac{(z - z_3)(z_1 - z_3)}{(z - z_1)(z_2 - z_3)}.$$

We will revisit this formula in Sect. 8.6.3. Notice as well that $\mathrm{Stab}(\infty, 0, 1)$ comprises the identity and the conjugation j, $j(z) = \overline{z}$.

8.5.1 Metric Cross-Ratio

In analogy with the terminology we used in projective space, we will call *a quadruple with property (Π)* any ordered quadruple $\mathfrak{p} = \{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4\}$ of four *pairwise distinct points* of $\overline{\mathbb{R}^n}$.

Definition 8.5.5 Let d be the chordal metric in $\mathbf{R}^n$. For any quadruple $\mathfrak{p} = \{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4\}$ of points in $\overline{\mathbb{R}^n}$ with property (Π), the *metric cross-ratio* of $\mathfrak{p}$ is defined by the relation

$$\mathbb{X}(\mathfrak{p}) = (\mathbf{x}_1, \mathbf{x}_2; \mathbf{x}_3, \mathbf{x}_4) = \frac{d(\mathbf{x}_4, \mathbf{x}_2) \cdot d(\mathbf{x}_3, \mathbf{x}_1)}{d(\mathbf{x}_4, \mathbf{x}_1) \cdot d(\mathbf{x}_3, \mathbf{x}_2)}.$$

Exercise Use Proposition 8.1.3 to show that

$$\mathbb{X}(\mathfrak{p}) = (\mathbf{x}_1, \mathbf{x}_2; \mathbf{x}_3, \mathbf{x}_4) = \frac{\|\mathbf{x}_4 - \mathbf{x}_2\| \cdot \|\mathbf{x}_3 - \mathbf{x}_1\|}{\|\mathbf{x}_4 - \mathbf{x}_1\| \cdot \|\mathbf{x}_3 - \mathbf{x}_2\|},$$

with the assumption (which is now completely justified) $\infty : \infty = 1$.

The following holds:

Lemma 8.5.6 *If $\mathfrak{p} = \{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4\}$ is a quadruple of points in $\mathbf{R}^n$ with property (Π), then the metric cross-ratios $(\mathbf{x}_i, \mathbf{x}_j; \mathbf{x}_k, \mathbf{x}_l)$, $(i, j, k, l) \in S_4$, are functions of the metric cross-ratios*

$$\mathbb{X}_1(\mathfrak{p}) = (\mathbf{x}_1, \mathbf{x}_2; \mathbf{x}_3, \mathbf{x}_4) \quad \textit{and} \quad \mathbb{X}_2(\mathfrak{p}) = (\mathbf{x}_1, \mathbf{x}_3; \mathbf{x}_2, \mathbf{x}_4).$$

Exercise Prove Lemma 8.5.6.

Theorem 8.5.7 *A mapping $f : \overline{\mathbb{R}^n} \to \overline{\mathbb{R}^n}$ is a Möbius transformation if and only if it preserves the metric cross-ratio of every quadruple of points of $\overline{\mathbb{R}^n}$ with property (Π).*

Proof First, it suffices by Theorem 8.3.4 to show that translations, orthogonal transformations, dilations and the inversion in the sphere $S(\mathbf{0}, 1)$ all preserve the metric cross-ratio. We show here only the last one: if ι is the inversion in $S(\mathbf{0}, 1)$, then using Proposition 8.2.15 we have

$$\begin{aligned}
(\iota(\mathbf{x}_1), \iota(\mathbf{x}_2); \iota(\mathbf{x}_3), \iota(\mathbf{x}_4)) &= \frac{\|\iota(\mathbf{x}_4) - \iota(\mathbf{x}_2)\| \cdot \|\iota(\mathbf{x}_3) - \iota(\mathbf{x}_1)\|}{\|\iota(\mathbf{x}_4) - \iota(\mathbf{x}_1)\| \cdot \|\iota(\mathbf{x}_3) - \iota(\mathbf{x}_2)\|} \\
&= \frac{\frac{\|\mathbf{x}_4 - \mathbf{x}_2\|}{\|\mathbf{x}_4\| \cdot \|\mathbf{x}_2\|} \cdot \frac{\|\mathbf{x}_3 - \mathbf{x}_1\|}{\|\mathbf{x}_3\| \cdot \|\mathbf{x}_1\|}}{\frac{\|\mathbf{x}_4 - \mathbf{x}_1\|}{\|\mathbf{x}_4\| \cdot \|\mathbf{x}_1\|} \cdot \frac{\|\mathbf{x}_3\| \cdot \|\mathbf{x}_2\|}{\|\mathbf{x}_3\| \cdot \|\mathbf{x}_2\|}} \\
&= \frac{\|\mathbf{x}_4 - \mathbf{x}_2\| \cdot \|\mathbf{x}_3 - \mathbf{x}_1\|}{\|\mathbf{x}_4 - \mathbf{x}_1\| \cdot \|\mathbf{x}_3 - \mathbf{x}_2\|} \\
&= (\mathbf{x}_1, \mathbf{x}_2; \mathbf{x}_3, \mathbf{x}_4).
\end{aligned}$$

For the converse, composing f if necessary with a Möbius transformation, we can assume that $f(\infty) = \infty$. For $\mathbf{x}_i \in \mathbb{R}^n$, $i = 1, \ldots, 4$, the relation

$$\frac{(\infty, \mathbf{x}_2; \mathbf{x}_3, \mathbf{x}_4)}{(\mathbf{x}_1, \mathbf{x}_2; \infty, \mathbf{x}_4)} = \frac{(\infty, f(\mathbf{x}_2); f(\mathbf{x}_3), f(\mathbf{x}_4))}{(f(\mathbf{x}_1), f(\mathbf{x}_2); \infty, f(\mathbf{x}_4))}$$

is written equivalently as

$$\frac{\|f(\mathbf{x}_1) - f(\mathbf{x}_2)\|}{\|\mathbf{x}_1 - \mathbf{x}_2\|} = \frac{\|f(\mathbf{x}_3) - f(\mathbf{x}_4)\|}{\|\mathbf{x}_3 - \mathbf{x}_4\|}.$$

It follows that for any two points $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$,

$$\frac{\|f(\mathbf{x}) - f(\mathbf{y})\|}{\|\mathbf{x} - \mathbf{y}\|} = k > 0,$$

hence $f \in \overline{S(n)} \subset \mathrm{GM}(n)$. □

Exercise Complete the details in the proof of Theorem 8.5.7.

The next theorem is the *Fundamental Theorem of cross-ratios*.

Theorem 8.5.8 *Let* $\mathfrak{p} = \{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4\}$ *and* $\mathfrak{p}' = \{\mathbf{x}_1', \mathbf{x}_2', \mathbf{x}_3', \mathbf{x}_4'\}$ *be quadruples of points of* $\overline{\mathbb{R}^n}$ *with property* (Π)*. There exists an* $f \in \mathrm{GM}(n)$ *such that* $f(\mathbf{x}_i) = \mathbf{y}_i$, $i = 1, \ldots, 4$*, if and only if*

$$\mathbb{X}_1(\mathfrak{p}) = (\mathbf{x}_1, \mathbf{x}_2; \mathbf{x}_3, \mathbf{x}_4) = (\mathbf{y}_1, \mathbf{y}_2; \mathbf{y}_3, \mathbf{y}_4) = \mathbb{X}_1(\mathfrak{p}'),$$
$$\mathbb{X}_2(\mathfrak{p}) = (\mathbf{x}_1, \mathbf{x}_3; \mathbf{x}_2, \mathbf{x}_4) = (\mathbf{y}_1, \mathbf{y}_3; \mathbf{y}_2, \mathbf{y}_4) = \mathbb{X}_2(\mathfrak{p}').$$

Proof The sufficient direction is a consequence of Theorem 8.5.7. For the converse, we can normalise so that

$$\mathbf{x}_1 = \mathbf{y}_1 = \infty, \quad \mathbf{x}_2 = \mathbf{y}_2 = \mathbf{0}, \quad \mathbf{x}_3 = \mathbf{y}_3 = \mathbf{e}_1, \quad \mathbf{x}_4 = \mathbf{X}, \ \mathbf{y}_4 = \mathbf{Y},$$

with $\mathbf{X}, \mathbf{Y} \in \mathbb{R}^n_* \setminus \{\mathbf{e}_1\}$ (Lemma 8.5.2). From the equalities of the cross-ratios we then obtain, respectively,

$$\|\mathbf{X}\| = \|\mathbf{Y}\|, \quad \|\mathbf{X} - \mathbf{1}\| = \|\mathbf{Y} - \mathbf{1}\|.$$

If $\mathbf{X} = (X_1, \ldots, X_n)$, $\mathbf{Y} = (Y_1, \ldots, Y_n)$, then squaring the second relation and using the first equality yields $X_1 = Y_1$. Also, if $\mathbf{X}_{n-1} = (X_2, \ldots, X_n)$ and $\mathbf{Y}_{n-1} = (Y_2, \ldots, Y_n)$, then again from the first relation we get $\|\mathbf{X}_{n-1}\| = \|\mathbf{Y}_{n-1}\|$. Therefore, there exists an $A \in \mathrm{O}(n-1)$ such that $\mathbf{X}_{n-1}A = \mathbf{Y}_{n-1}$. It follows that the transformation $\phi : \overline{\mathbb{R}^n} \to \overline{\mathbb{R}^n}$ with

$$\phi(\mathbf{x}) = \mathbf{x}\begin{pmatrix} 1 & 0 \\ 0 & A \end{pmatrix}, \quad \mathbf{x} \in \overline{\mathbb{R}^n},$$

is a Möbius transformation with $\phi(\mathbf{X}) = \mathbf{Y}$. □

The Form of the Elements of GM(n)

From the proof of Theorem 8.5.7 it follows that every $f \in \mathrm{GM}(n)$ with $f(\infty) = \infty$ is of the form

$$f(\mathbf{x}) = \delta \mathbf{x} A + \mathbf{a}, \quad A \in \mathrm{O}(n),\ \delta > 0.$$

We observe that if it also preserves the unit ball B^n of $\mathbb{R}^n$, then

$$f(\mathbf{x}) = \mathbf{x}A.$$

If now $f(\infty) = \mathbf{b} \in \mathbb{R}^n$, $\iota_{\mathbf{b},r} \circ f$ stabilises ∞ for some $r > 0$. Therefore,

$$\begin{aligned} f(\mathbf{x}) &= \iota_{\mathbf{b},r}\,(\delta \mathbf{x} A + \mathbf{a}) \\ &= \mathbf{b} + r^2 \frac{\delta \mathbf{x} A + \mathbf{a} - \mathbf{b}}{\|\delta \mathbf{x} A + \mathbf{a} - \mathbf{b}\|^2} \\ (\mathbf{c} = (1/\delta)(\mathbf{b} - \mathbf{a})) \ &= \mathbf{b} + \frac{r^2}{\delta} \frac{\mathbf{x}A - \mathbf{c}}{\|\mathbf{x}A - \mathbf{c}\|^2}. \end{aligned}$$

We have proved the following:

Theorem 8.5.9 *Each element $f \in \mathrm{GM}(n)$ is written in the form*

$$f(\mathbf{x}) = \mathbf{b} + \rho \frac{\mathbf{x}A - \mathbf{c}}{\|\mathbf{x}A - \mathbf{c}\|^{\varepsilon}},$$

where $\mathbf{b}, \mathbf{c} \in \mathbb{R}^n$, $\rho > 0$, $A \in \mathrm{O}(n)$ and $\varepsilon = 0$ or 2.

8.6 Low Dimensional Inversive Geometry

In Sect. 8.3.1 we compared Inversive Geometry with Euclidean, Similarity and Affine Geometry. We will show in this section that in dimensions $n = 1, 2$, the geometries $(\overline{\mathbb{R}^n},\ \mathrm{GM}(n))$ and $(\mathbb{R}P^n,\ P(n))$ are identical.

8.6.1 Inversive Geometry of $\overline{\mathbb{R}}$

The group GM(1) has an interesting description:

Definition 8.6.1 In $\overline{\mathbb{R}}$ we consider mappings of the form

$$f(x) = \frac{ax+b}{cx+d}, \quad a, b, c, d \in \mathbb{R},\ ad - bc \neq 0, \tag{8.8}$$

where $f(\infty) = a/c$, $f(-d/c) = \infty$ (if $c = 0$ then we have only the relation $f(\infty) = \infty$). We call these mappings *linear rational transformations of the extended line*.

The next proposition summarises some structural properties of linear rational transformations of $\overline{\mathbb{R}}$.

Proposition 8.6.2 *Let G be the set of linear rational transformations of $\overline{\mathbb{R}}$. Then:*

(i) Every $f \in G$ is bijective: the inverse f^{-1} of f is given by

$$f^{-1}(x) = \frac{dx-b}{-cx+a},$$

and $f^{-1}(\infty) = -d/c$, $f^{-1}(a/c) = \infty$.

ii) Every $f \in G$ is a homeomorphism of $\overline{\mathbb{R}}$.

iii) The set G is a group with group operation is function composition.

Exercise Prove Proposition 8.6.2.

We now observe that all of the following transformations, considered as transformations of $\overline{\mathbb{R}}$, are elements of G:

- Translations: $T_a(x) = x + a$, $a \in \mathbb{R}$,
- Reflection in 0: $R(x) = -x$,
- Dilations $D_\delta(x) = \delta x$, $\delta > 0$,
- Inversion at 0: $\iota(x) = 1/x$.

In this way, GM(1) $\subset G$. Also, every $f \in G$ can be written as

$$f(x) = \begin{cases} (a/d)x + b/d & \text{when } c = 0, \\ \frac{a}{c} + \frac{bc-ad}{c^2} \cdot \frac{1}{x+(d/c)} & \text{when } c \neq 0, \end{cases}$$

which is a composition of all the above transformations; hence we also have $G \subset$ GM(1). We conclude the following:

Proposition 8.6.3 *The group* GM(1) *of Möbius transformations of the extended real line is the group G of linear rational transformations of $\overline{\mathbb{R}}$.*

We note at this point that the orientation preserving elementary Möbius transformations of $\overline{\mathbb{R}}$ are exactly the translations and the similarities; the reflection $x \mapsto -x$ and the inversion $x \mapsto 1/x$ are orientation reversing. Since each element of GM(1) is a composition of inversions, we have that M(1) is the set of elements that are compositions of an even number of reflections. Let us also observe that if $f \in \mathrm{GM}(1)$, then

$$f(x) = \frac{ax+b}{cx+d} \implies f'(x) = \frac{ad-bc}{(cx+d)^2},$$

and therefore f reverses or preserves the orientation depending on the sign of the determinant of its representative in $\mathrm{PGL}(2,\mathbb{R})$. We therefore have that

$$\mathrm{M}(1) = \{f \in \mathrm{GM}(1) \mid f'(x) > 0\}.$$

Let now

$$P(1) = \mathrm{PGL}(2,\mathbb{R}) = \left\{ [A] \mid A = \begin{pmatrix} a & c \\ b & d \end{pmatrix}, \quad ad - bc \neq 0 \right\}.$$

There exists a map $\Phi : \mathrm{PGL}(2,\mathbb{R}) \to \mathrm{GM}(1)$ given by

$$\Phi\left(\begin{bmatrix} a & c \\ b & d \end{bmatrix}\right) = f \mid f(x) = \frac{ax+b}{cx+d}. \tag{8.9}$$

Proposition 8.6.4 *The map Φ in (8.9) is a group isomorphism.*

Proof Let

$$[A] = \begin{bmatrix} a & c \\ b & d \end{bmatrix} \text{ and } [A'] = \begin{bmatrix} a' & c' \\ b' & d' \end{bmatrix}$$

be elements of $\mathrm{PGL}(2,\mathbb{R})$. Then

$$\begin{aligned} \Phi([A] \star [A']) = \Phi([AA']) &= \Phi\left(\begin{bmatrix} aa'+cb' & ac'+cd' \\ ba'+db' & bc'+dd' \end{bmatrix}\right) \\ &= f \circ f', \end{aligned}$$

where

$$f(x) = \frac{ax+b}{cx+d}, \quad f'(x) = \frac{a'x+b'}{c'x+d'}.$$

Now, if $\Phi([A]) = id$. we have $b = c = 0$ and $a = d$. Therefore Φ is a monomorphism. Finally, to every $f \in \mathrm{GM}(1)\mu\varepsilon f(x) = \frac{ax+b}{cx+d}$ is associated the element$[A] = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$. □

Consequently, the group GM(1) is identified to the projective group $P(1) = \mathrm{PGL}(2, \mathbb{R})$, that is, the transformation group of the projective line $\mathbb{R}P^1$. This is entirely natural since the circle $S^1 = \overline{\mathbb{R}}$ is identified with $\mathbb{R}P^1$. Hence we have the following:

Theorem 8.6.5 *Inversive Geometry* $(\overline{\mathbb{R}}, \mathrm{GM}(1))$ *and Projective Line Geometry* $(\mathbb{R}P^1, P(1))$ *are identical.*

There is also another description of GM(1). We consider the mapping

$$\Psi : \mathrm{SL}(2, \mathbb{R}) \to \mathrm{PGL}(2, \mathbb{R}), \quad A \mapsto [A].$$

The map Ψ is a group monomorphism and $\ker(\Psi) = \{I, -I\}$. Therefore from the First Group Isomorphism Theorem we obtain the following:

Proposition 8.6.6

$$\mathrm{GM}(1) = \mathrm{SL}(2, \mathbb{R})/\{I, -I\} := \mathrm{PSL}(2, \mathbb{R}).$$

Remark 8.6.7 We also denote M(1) by $\mathrm{PSL}^+(2, \mathbb{R})$.

8.6.2 Inversive Geometry of $\overline{\mathbb{R}^2} = \overline{\mathbb{C}}$

The study of the Inversive Geometry of the extended plane $(\overline{\mathbb{R}^2}, \mathrm{GM}(2))$, is done better when we identify the extended plane with the *extended complex line* $\overline{\mathbb{C}}$, or, alternatively, the *Riemann sphere*. We write

$$S^2 = \{(\zeta, t) \in \mathbb{C} \times \mathbb{R} \mid |\zeta|^2 + t^2 = 1\}.$$

If $z = x + iy \in \mathbb{C}$, then the stereographic projection $\mathbf{S} : \mathbb{C} \to S^2$ is given by the formula

$$\mathbf{S}(z) = \begin{cases} \left(\frac{2z}{1+|z|^2}, \frac{1-|z|^2}{1+|z|^2}\right) & z \in \mathbb{C} \\ (0, 1) & z = \infty. \end{cases},$$

whereas its inverse $\mathbf{S}^{-1}$ is given by the formula

$$\mathbf{S}^{-1}(\zeta, t) = \begin{cases} \frac{|\zeta|^2}{1-t}, & (\zeta, t)) \in S^2 \setminus \{(0,1)\} \\ \infty & (\zeta, t) = (0,1). \end{cases}$$

(See also Exercise 5c, Sect. 8.7). The following definition is equivalent to Definition 8.6.1:

Definition 8.6.8 A mapping of the form

$$f(x) = \frac{az+b}{cz+d}, \quad a, b, c, d \in \mathbb{C},\ ad - bc \neq 0, \tag{8.10}$$

where $f(\infty) = a/c$, $f(-d/c) = \infty$ (if $c = 0$ then we only have the relation $f(\infty) = \infty$), is called *linear rational transformation of the Riemann sphere*.

The following proposition is analogous to Proposition 8.6.2:

Proposition 8.6.9 *Let G_c be the set of linear rational transformations of the Riemann sphere. Then:*

(i) Every $f \in G_c$ is bijective: the inverse f^{-1} of f is given by

$$f^{-1}(z) = \frac{dz-b}{-cz+a},$$

and $f^{-1}(\infty) = -d/c$, $f^{-1}(a/c) = \infty$.

(ii) Every $f \in G_c$ is a homeomorphism of $\overline{\mathbb{C}}$.

(iii) The set G_c is a group; the group operation is function composition.

The following also holds.

Theorem 8.6.10 *The group* GM(2) *comprises transformations of $\overline{\mathbb{C}}$ with the following form:*

$$f(z) = \frac{az+b}{cz+d}, \quad \text{or} \quad f(z) = \frac{a\bar{z}+b}{c\bar{z}+d}, \tag{8.11}$$

where $ad - bc \neq 0$. In other words, every element in GM(2) *is either an element of G_c or an element of G_c followed by the conjugation map $j(z) = \bar{z}$.*

Proof We first write the elementary Möbius transformations in complex form:

- The translations $T_{(a,b)}(x, y) = (x + a, y + b)$ are written as

$$T_\alpha(z) = z + \alpha, \quad \alpha = a + ib.$$

- For the rotations R_θ, $\theta \in \mathbb{R}$ with

$$R_\theta(x, y) = (x, y)\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} = (x\cos\theta + y\sin\theta,\ -x\sin\theta + y\cos\theta)$$

we have

$$R_\theta(z) = e^{-i\theta}z.$$

- For the reflections r_θ, $\theta \in \mathbb{R}$ with

$$r_\theta(x, y) = (x, y)\begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix} = (x\cos\theta + y\sin\theta,\ x\sin\theta - y\cos\theta)$$

we have

$$r_\theta(z) = e^{i\theta}\overline{z}.$$

For the dilations D_δ, $\delta > 0$ with $D_\delta(x, y) = (\delta x, \delta y)$ we have

$$D_\delta(z) = \delta z.$$

- Finally, for the inversion ι with $\iota(x, y) = (x, y)/(x^2 + y^2)$ we have

$$\iota(z) = \frac{z}{|z|^2} = \frac{1}{\overline{z}}.$$

In this way, all elementary transformations of GM(2), and therefore all transformations of GM(2), are written in the form (8.11). To prove that each element of the form (8.11) is a composition of elementary Möbius transformations, we first observe that if $j(z) = \overline{z}$ is the reflection in the real line, then

$$f(z) = \frac{a\overline{z} + b}{c\overline{z} + d} = g(j(z)), \quad g(z) = \frac{az + b}{cz + d}.$$

On the other hand, for g as above, we have

$$g(z) = \begin{cases} (a/d)z + b/d & c = 0, \\ \frac{a}{c} + \frac{bc-ad}{c^2} \cdot \frac{1}{z+(d/c)} & c \neq 0. \end{cases}$$

The proof is complete. □

Corollary 8.6.11 *The group* M(2) *of orientation-preserving Möbius transformations consists of elements of the form*

$$f(z) = \frac{az+b}{cz+d}. \tag{8.12}$$

Every orientation-preserving Möbius transformation is a composition of an element of M(2) *with the reflection on the real line* $j(z) = \overline{z}$.

Proof Elements of the form (8.12) are holomorphic functions of $\mathbb{C} \setminus \{-d/c\}$ with complex derivative

$$f'(z) = \frac{ad-bc}{(cz+d)^2},$$

which is different from 0 in $\mathbb{C}\setminus\{-d/c\}$. Therefore they are conformal and orientation preserving. □

We now examine the relation of $GM(2)$ to the projective group $\mathrm{PGL}(2,\mathbb{C})$. In general, $\mathrm{PGL}(n+1,\mathbb{C})$ is defined in a manner completely analogous to the one we defined $\mathrm{PGL}(n+1,\mathbb{R})$ in Sect. 7.2.2. For the sake of completeness, we will deal here with the case where $n = 1$ in detail.

We start from the group $\mathrm{GL}(2,\mathbb{C})$ which is the set of invertible 2×2 matrices with complex coefficients. We define the following relation in $\mathrm{GL}(2,\mathbb{C})$: for each $A, B \in \mathrm{GL}(2,\mathbb{C})$,

$$A \sim B \iff \exists \lambda \in \mathbb{C}_* \ : A = \lambda B. \tag{8.13}$$

The relation $\sim$ is an equivalence relation and the set of trajectories

$$\mathrm{PGL}(2,\mathbb{C}) = \{[A] \mid A \in \mathrm{GL}(2,\mathbb{C})\},$$

is a group with operation

$$[A] \cdot [B] = [AB]. \tag{8.14}$$

The next proposition is the analogue of Proposition 8.6.4:

Proposition 8.6.12 *The mapping* $\Phi^{\mathbb{C}} : \mathrm{PGL}(2,\mathbb{C}) \to \mathrm{GM}(2)$ *given by*

$$\Phi^{\mathbb{C}}\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = f \mid f(z) = \frac{az+b}{cz+d}, \tag{8.15}$$

is a group isomorphism.

Thus GM(2) is identified to PGL(2, $\mathbb{C}$). The group PGL(2, $\mathbb{C}$) is the group $P^{\mathbb{C}}(1)$ of the transformations of the Complex Projective Line $\mathbb{C}P^1$ (Exercise 3, Sect. 7.6). This yields the following analogue of Theorem 8.6.5:

Theorem 8.6.13 *Inversive Geometry* ($\overline{\mathbb{C}}$, GM(2) = PGL(2, $\mathbb{C}$)) *is the geometry of the complex projective line* ($\mathbb{C}P^1$, $P^{\mathbb{C}}(1)$).

The alternative description of GM(2) results from the mapping

$$\Psi^{\mathbb{C}} : \mathrm{SL}(2, \mathbb{C}) \to \mathrm{PGL}(2, \mathbb{C}), \quad A \mapsto [A].$$

The map $\Psi^{\mathbb{C}}$ is a group monomorphism and $\ker(\Psi^{\mathbb{C}}) = \{I, -I\}$. Again, from the First Group Isomorphism Theorem we have:

Proposition 8.6.14

$$\mathrm{GM}(2) = \mathrm{SL}(2, \mathbb{C})/\{I, -I\} := \mathrm{PSL}(2, \mathbb{C}).$$

Remark 8.6.15 The group M(2) is also denoted by $\mathrm{PSL}^{+}(2, \mathbb{C})$.

8.6.3 Real and Complex Cross-Ratios

In addition to the metric cross-ratio defined in Sect. 8.5.1, in the cases $n = 1, 2$ there is also the real and the complex cross-ratio of four points. We have already seen the real cross-ratio in Sect. 7.2.4. We now rewrite the definition and main properties of the cross-ratio:

Definition 8.6.16 Let $\mathfrak{p} = \{x_1, x_2, x_3, x_4\}$ be an ordered quadruple of points of $\overline{\mathbb{R}}$ with property (Π). Then the *cross-ratio* $\mathbf{X}(\mathfrak{p})$ of $\mathfrak{p}$ is

$$\mathbf{X}(\mathfrak{p}) = [x_1, x_2; x_3, x_4] = \frac{(x_4 - x_2)(x_3 - x_1)}{(x_4 - x_1)(x_3 - x_2)},$$

with the necessary modifications if any of the points is ∞.

We notice that

$$|\mathbf{X}(\mathfrak{p})| = \mathbb{X}(\mathfrak{p}).$$

The following proposition summarises the main properties of the real cross-ratio.

Proposition 8.6.17 *The following hold:*

- *Let* $\mathfrak{p} = \{x_1, x_2, x_3, x_4\}$ *be a quadruple of points of* $\overline{\mathbb{R}}$ *with property* (Π) *and let* $\sigma \in S_4$ *be a permutation of the points of* $\mathfrak{p}$. *Then the cross-ratio* $\mathbf{X}(\sigma(\mathfrak{p})) = [x_{\sigma(1)}, x_{\sigma(2)}; x_{\sigma(3)}, x_{\sigma(4)}]$ *is a function of* $\mathbf{X}(\mathfrak{p}) = [x_1, x_2; x_3, x_4]$.

- *The cross-ratio remains invariant by the diagonal action of* GM(1).
- *There is a unique element of* GM(1) *that maps a quadruple* $\mathfrak{p} = \{x_1, x_2, x_3, x_4\}$ *of points of* $\overline{\mathbb{R}}$ *with property* (Π) *to a quadruple* $\mathfrak{p}' = \{x_1', x_2', x_3', x_4'\}$ *of points with property* (Π) *such that* $x_i \leftrightarrow x_i'$, $i = 1, \ldots, 4$, *if and only if*

$$\mathbf{X}(\mathfrak{p}) = [x_1, x_2; x_3, x_4] = [x_1', x_2'; x_3', x_4'] = \mathbf{X}(\mathfrak{p}').$$

Exercise Prove Proposition 8.6.17.

Recall that GM(1) acts triply transitively on $\overline{\mathbb{R}}$. We can prove this fact in a nice way via cross-ratios as follows. Let $\{x_1, x_2, x_3\}$ and $\{y_1, y_2, y_3\}$ be triples of pairwise distinct points in $\overline{\mathbb{R}}$ and let $x \in \overline{\mathbb{R}}$. The transformation g that sends x_1, x_2, x_3, x to $\infty, 0, 1, g(x)$, respectively, is given by the relation

$$[\infty, 0; 1, g(x)] = [x_1, x_2; x_3, x] \iff g(x) = \frac{(x - x_2)(x_3 - x_1)}{(x - x_1)(x_3 - x_2)}.$$

Similarly, the transformation h that sends y_1, y_2, y_3, x to $\infty, 0, 1, h(x)$ is given by the relation

$$h(x) = \frac{(x - y_2)(y_3 - y_1)}{(x - y_4)(y_3 - y_2)}.$$

We conclude that the desired transformation is $f = hg^{-1}$ and is given by

$$[y_1, y_2; y_3, f(x)] = [x_1, x_2; x_3, x], \quad x \in \overline{\mathbb{R}}. \tag{8.16}$$

This transformation is unique. Indeed, if there were f' such that $x_i \leftrightarrow y_i$ through f', then $F = hf^{-1}f'g^{-1}$ would stabilise $\infty, 0, 1$. Therefore

$$[\infty, 0; 1, z] = [\infty, 0; 1, F(z)] \iff z = F(z).$$

We also conclude that if a Möbius transformation of the extended line stabilises three points, then it is necessarily the identity. In fact, it suffices to assume that the three points are $\infty, 0, 1$ and then by (8.16) we have

$$[\infty, 0; 1, f(x)] = [\infty, 0; 1, x] \implies f(x) = x.$$

Remark 8.6.18 Notice that

$$g'(x) = \frac{(x_2 - x_1)(x_3 - x_1)}{x_3 - x_2} \cdot \frac{1}{(x - x_1)^2}, \quad h'(x) = \frac{(y_2 - y_1)(y_3 - y_1)}{y_3 - y_2} \cdot \frac{1}{(x - y_1)^2}.$$

Therefore f is an element of M(1) if the expressions $(x_2 - x_1)(x_3 - x_1)(x_3 - x_2)$ and $(y_2 - y_1)(y_3 - y_1)(y_3 - y_2)$ are identical. We conclude that whether f is an

element of M(1) or not depends on the *arrangement* of the points on the extended line.

Example 8.6.19 To find the Möbius transformation that maps $-1, 0, 1$ to $1, 0, -1$, respectively, we apply (8.16):

$$[1, 0; -1, f(x)] = [-1, 0; 1, x] \iff f(x) = -x.$$

The complex generalisations of the previous definitions and results follow. The complex cross-ratio comes to us from the geometry of the complex projective line:

Definition 8.6.20 Let $\mathfrak{p} = \{z_1, z_2, z_3, z_4\}$ be a quadruple of points of $\overline{\mathbb{C}}$ with property (Π). Then the *cross-ratio* of $\mathfrak{p}$ is

$$\mathbf{X}(\mathfrak{p}) = [z_1, z_2; z_3, z_4] = \frac{(z_4 - z_2)(z_3 - z_1)}{(z_4 - z_1)(z_3 - z_2)},$$

with the necessary modifications if any of the points is ∞.

The following proposition is the equivalent of Proposition 8.6.17 and summarises the properties of the complex cross-ratio.

Proposition 8.6.21 *The following hold:*

- *Let $\mathfrak{p} = \{z_1, z_2, z_3, z_4\}$ be a quadruple of points of $\overline{\mathbb{C}}$ with property (Π) and $\sigma \in S_4$ be a permutation of the points of $\mathfrak{p}$. Then the cross-ratio $\mathbf{X}(\sigma(\mathfrak{p})) = [z_{\sigma(1)}, z_{\sigma(2)}; z_{\sigma(3)}, z_{\sigma(4)}]$ is a function of $\mathbf{X}(\mathfrak{p}) = [z_1, z_2; z_3, z_4]$.*
- *The cross-ratio remains unchanged by the diagonal action of* M(2).
- *There is a unique element of* M(2) *that maps a quadruple $\mathfrak{p} = \{z_1, z_2, z_3, z_4\}$ of points of $\overline{\mathbb{C}}$ with property (Π) to a quadruple $\mathfrak{p}' = \{z_1', z_2', z_3', z_4'\}$ of points with property (Π) such that $z_i \leftrightarrow z_i'$, $i = 1, \ldots, 4$, if and only if*

$$\mathbf{X}(\mathfrak{p}) = [z_1, z_2; z_3, z_4] = [z_1', z_2'; z_3', z_4'] = \mathbf{X}(\mathfrak{p}').$$

Exercise Prove Proposition 8.6.21.

Remark 8.6.22 The complex cross-ratio *does not* remain invariant under the diagonal action of GM(2): for the reflection $j(z) = \overline{z}$ we have

$$[j(z_1), j(z_2); j(z_3), j(z_4)] = \overline{[z_1, z_2; z_3, z_4]}.$$

It is therefore natural to ask when the complex cross-ratio is real (and therefore remains invariant under the diagonal action of GM(2)). We answer this question in Proposition 8.6.25 below.

GM(2) acts triply transitively on $\overline{\mathbb{R}^2}$; by means of complex cross-ratios we can prove that M(2) acts triply transitively on $\overline{\mathbb{C}}$ following a method similar to the one we followed in the case of GM(1). For completeness we repeat the proof: let

(z_i), (w_i), $i = 1, 2, 3$, be triples of pairwise distinct points and let $z \in \overline{\mathbb{C}}$. The transformation g that sends z_1, z_2, z_3, z to $\infty, 0, 1, g(z)$, respectively, is given by the relation

$$[\infty, 0; 1, g(z)] = [z_1, z_2; z_3, z] \iff g(z) = \frac{(z - z_2)(z_3 - z_1)}{(z - z_4)(z_3 - z_2)}.$$

Similarly, the transformation h that sends w_1, w_2, w_3, z to $\infty, 0, 1, h(z)$, respectively, is given by the relation

$$h(z) = \frac{(z - w_2)(w_3 - w_1)}{(z - w_4)(w_3 - w_2)}.$$

We conclude that the desired transformation is $f = hg^{-1}$ and is given by

$$[w_1, w_2; w_3, f(z)] = [z_1, z_2; z_3, z], \quad z \in \overline{\mathbb{C}}. \tag{8.17}$$

It is proved in a manner similar to that of the case of GM(1) that it is unique. It also follows that every element of M(2) that leaves three different points unchanged is necessarily the identity transformation.

Example 8.6.23 Due to (8.17), the Möbius transformation that maps $-1, i, 1]$ to $\infty, 0, 1$ is given by

$$[\infty, 0; 1, f(z)] = [-1, i; 1, z] \implies f(z) = (1 + i)\frac{z - i}{z + 1}.$$

Example 8.6.24 To find $f \in \mathrm{M}(2)$ that maps the unit circle S^1 to the line $l : y = 0$, we choose arbitrary points $1, i, -1$ in S^1 and points $0, i, \infty$ in l and then f is defined by

$$[0, i; \infty, f(z)] = [1, i; -1, z] \implies f(z) = -(1 + i)\frac{z - i}{z + 1}.$$

Notice that this is *not* the unique element of M(2) that maps S^1 onto l; it is rather the unique element of M(2) that maps S^1 onto l such that 1 is mapped to 0, i is mapped to itself and -1 is mapped to ∞. By permuting the points of the triples we obtain different elements of M(2) that map S^1 to l.

Proposition 8.6.25 *The complex cross-ratio of an oriented quadruple* $\mathfrak{p}$ *of pairwise distinct points of* $\overline{\mathbb{C}}$ *is a real number if and only if the points of* $\mathfrak{p}$ *lie on the same generalised circle.*

Proof Let $\mathfrak{p} = \{z_1, z_2, z_3, z_4\}$ be an oriented quadruple of pairwise distinct points of $\overline{\mathbb{C}}$ and let $\mathbf{X} = \mathbf{X}(\mathfrak{p})$. There exists an element of M(2) that maps z_1, z_2, z_3, z_4 to $\infty, 0, 1, \mathbf{X}$ respectively. Then,

$$[z_1, z_2; z_3, z_4] = [\infty, 0; 1, \mathbf{X}] = \mathbf{X}.$$

If $\mathbf{X} \in \mathbb{R}$, then since $\infty, 0, 1, \mathbf{X}$ belong to the line $y = 0$ we have that z_1, z_2, z_3, z_4 also belong to a generalised circle. The converse is clear. □

Example 8.6.26 We consider rectangles R and R' with vertices $0, a, a + ib, b$ and $0, a', a' + ib', ib'$, respectively, $a, b > 0$. The cross-ratio $\mathbf{X}(R)$ of the ordered quadruple $\mathfrak{p} = \{0, a, a + ib, ib\}$ is $\mathbf{X}(R) = 1 + a^2/b^2$, and similarly the cross-ratio $\mathbf{X}(R')$ of the ordered quadruple $\mathfrak{p}' = \{0, a', a' + ib', ib'\}$ is $\mathbf{X}(R') = 1 + (a')^2/(b')^2$. Due to Proposition 8.6.21 there exists a Möbius transformation mapping the vertices of Q to the corresponding vertices of Q' (and thus mapping Q to Q') if and only if $a/b = a'/b'$, that is, if Q and Q' are *similar*.

8.7 Exercises

1. Prove that stereographic projection maps straight lines from the origin of $\mathbb{R}^n$ onto meridians of S^n.
2. Let (X, d) be a metric space and let Y be a set such that there exists a bijection $f : Y \to X$. Prove that the mapping

$$d' : Y \times Y \to \mathbb{R}_{\geq 0}, \quad d'(y_1, y_2) = d(f(y_1), f(y_2)), \ y_1, y_2 \in Y,$$

   is a well-defined metric on Y.
3. Prove that the stereographic projection $\mathbf{S} : (\overline{\mathbb{R}^n}, d) \to (S^n, c_{n+1})$ is a homeomorphism. Here c_{n+1} is the chordal metric on the sphere S^n.
4. This exercise provides a simple way to prove that the stereographic projection is a conformal mapping. Consider the curves $\boldsymbol{\gamma}_i : [a, b] \to \mathbb{R}^n$, $i = 1, 2$ and let $t_0 \in [a, b]$ with $\boldsymbol{\gamma}_1(t_0) = \boldsymbol{\gamma}_2(t_0) = \mathbf{x}_0$, that is, the curves intersect at $\mathbf{x}_0$. The angle θ of $\boldsymbol{\gamma}_1$ and $\boldsymbol{\gamma}_2$ at $\mathbf{x}_0$ is defined by the relation

$$\cos\theta = \frac{\dot{\boldsymbol{\gamma}}_1(t_0) \cdot \dot{\boldsymbol{\gamma}}_2(t_0)}{\|\dot{\boldsymbol{\gamma}}_1(t_0)\| \cdot \|\dot{\boldsymbol{\gamma}}_2(t_0)\|}.$$

   The curves $\boldsymbol{\gamma}_i' = \mathbf{S}(\boldsymbol{\gamma}_i)$, $i = 1, 2$ intersect at $\mathbf{S}(\mathbf{x}_0)$. If θ' is the angle of $\boldsymbol{\gamma}_1', \boldsymbol{\gamma}_2'$ at $\mathbf{S}(\mathbf{x}_0)$, then prove that $\theta = \theta'$.
5. Identifying the set of complex numbers $\mathbb{C}$ with $\mathbb{R}^2$, consider $\overline{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$. (The extended set of complex numbers is also called *Riemann* sphere). In a manner similar to that of Sect. 8.1:
   (a) define the stereographic projection $\mathbf{S} : \overline{\mathbb{C}} \to S^2$ and find its inverse;
   (b) define the chordal metric on $\overline{\mathbb{C}}$;
   (c) show that $\mathbf{S}$ maps generalised circles of $\overline{\mathbb{C}}$ to spherical circles;
   (d) define the inversion in a circle and the inversion in a line of $\mathbb{C}$ and formulate propositions analogous to those in Sect. 8.2 for this case.

6. In the plane $\mathbb{R}^2 \equiv \mathbb{C}$ consider two inverse points z and $z' = 1/\overline{z}$ with respect to the unit circle $S^1 = \{z \in \mathbb{C} \,|\, |z| = 1\}$. Show that the line connecting z and z' intersects S^1 in two antipodal points.
7. Generalise the result of the previous exercise to the arbitrary dimension.
8. Let $\iota_{\mathbf{a},r}$ be an inversion in a sphere $S(\mathbf{a}, r)$. For every two inverse points $\mathbf{x}, \mathbf{x}'$ with $\mathbf{x} \in \mathrm{Ext}(S(\mathbf{a}, r))$, consider the point $\mathbf{s}$ of the sphere, where

$$\mathbf{s} = S(\mathbf{a}, r) \cap [\mathbf{x}, \mathbf{x}'].$$

   (a) Prove that

$$\|\mathbf{x}' - \mathbf{s}\| = \frac{r\|\mathbf{x} - \mathbf{s}\|}{r + \|\mathbf{x} - \mathbf{s}\|}.$$

   (b) Express the point $\mathbf{s}$ as a function of $\mathbf{x}$ by solving for t the equation

$$\|(1-t)\mathbf{x} + t\,\iota(\mathbf{x})\| = 1, \quad t \in (0, 1).$$

9. Let $R_{\mathbf{A}}$ be the inversion in the hyperplane $\overline{W_{\mathbf{A},0}}$. If $\mathbf{x}$ and $\mathbf{x}'$ are inverse points, find the intersection point of the line segment $[\mathbf{x}, \mathbf{x}']$ and $\overline{W_{\mathbf{A},0}}$.
10. (a) Let three different points $\mathbf{x}_i$, $i = 1, 2, 3$ in $\overline{\mathbb{R}^2}$ (or $\overline{\mathbb{C}}$). Show that there is a unique generalised circle containing them.
    (b) Show that the result of the previous question also holds for every $\overline{\mathbb{R}^n}$, $n \geq 1$. Every three distinct points of $\overline{\mathbb{R}^n}$ are contained in a unique generalised circle of $\overline{\mathbb{R}^n}$.
11. Let $f : \mathbb{R}^n \to \mathbb{R}^n$ be a linear transformation, $\mathbf{v}, \mathbf{w}$ vectors of $\mathbb{R}^n_*$ and $\theta(\mathbf{v}, \mathbf{w})$ be the *angle* of $\mathbf{v}, \mathbf{w}$. Prove the following:

    (a) f preserves angles if for any nonzero $\mathbf{v}, \mathbf{w}$ it holds that

$$\frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\| \cdot \|\mathbf{w}\|} = \frac{f(\mathbf{v}) \cdot f(\mathbf{w})}{\|f(\mathbf{v})\| \cdot \|f(\mathbf{w})\|}. \tag{8.18}$$

    (b) If f is angle-preserving and $A = A_f$ is the matrix of f, then

$$AA^T = A^T A = \lambda I, \quad \text{for some } \lambda \neq 0.$$

12. (Normalisation of generalised spheres). Show the following:

    (a) If Σ is a sphere $S(\mathbf{a}, r)$ in $\overline{\mathbb{R}^n}$, then there exists $f \in \mathrm{M}(n)$ with $f(\Sigma) = S(\mathbf{0}, 1)$.
    (b) If P is a hyperplane $\overline{W(\mathbf{A}, b)}$ in $\overline{\mathbb{R}^n}$, then there exists $g \in \mathrm{M}(n)$ with $g(P) = \overline{W(\mathbf{e}_1, 0)}$.
    (c) Show that there exists $h \in \mathrm{GM}(n)$ with $h(S(\mathbf{0}, 1)) = \overline{W(\mathbf{e}_1, 0)}$.

(d) Using the above, conclude Theorem 8.2.23, to which the following proposition can now be added: *if* Σ *and* Σ' *are generalised spheres of* $\overline{\mathbb{R}^n}$*, then there exists* $\phi \in \mathrm{GM}(n)$ *with* $\phi(\Sigma) = \Sigma'$.

13. Show that translations, orthogonal transformations and similarities all preserve the metric cross-ratio.
14. **(Ptolemy's Theorem.)** Let $\mathbf{x}_i$, $i = 1, \ldots, 4$ be pairwise distinct points of $\overline{\mathbb{R}^n}$. Consider any permutation $(1, 2, 3, 4) \mapsto (\sigma(1), \sigma(2), \sigma(3), \sigma(4))$ of S_4 and the ordered quadruple

$$\mathfrak{p} = \{\mathbf{x}_{\sigma(1)}, \mathbf{x}_{\sigma(2)}, \mathbf{x}_{\sigma(3)}, \mathbf{x}_{\sigma(4)}\}$$

Consider also the metric cross-ratios of $\mathfrak{p}$:

$$\mathbb{X}_1(\mathfrak{p}) = (\mathbf{x}_{\sigma(1)}, \mathbf{x}_{\sigma(2)}; \mathbf{x}_{\sigma(3)}, \mathbf{x}_{\sigma(4)})$$
$$\text{and} \quad \mathbb{X}_2(\mathfrak{p}) = (\mathbf{x}_{\sigma(1)}, \mathbf{x}_{\sigma(3)}; \mathbf{x}_{\sigma(2)}, \mathbf{x}_{\sigma(4)}).$$

(a) Show that $\mathbb{X}_1(\mathfrak{p}) + \mathbb{X}_2(\mathfrak{p}) \geq 1$.
(b) Write the above inequality equivalently as

$$\|\mathbf{x}_{\sigma(4)} - \mathbf{x}_{\sigma(1)}\| \cdot \|\mathbf{x}_{\sigma(3)} - \mathbf{x}_{\sigma(2)}\| \leq \|\mathbf{x}_{\sigma(4)} - \mathbf{x}_{\sigma(2)}\| \cdot \|\mathbf{x}_{\sigma(3)} - \mathbf{x}_{\sigma(1)}\| + \|\mathbf{x}_{\sigma(4)} - \mathbf{x}_{\sigma(3)}\| \cdot \|\mathbf{x}_{\sigma(2)} - \mathbf{x}_{\sigma(1)}\|.$$

Assuming $\mathbf{x}_{\sigma(1)}$ vertices of a quadrilateral, interpret geometrically the above inequality known as *Ptolemy's Inequality*.
(c) Prove Ptolemy's Theorem: The equality in Ptolemy's Inequality holds if and only if $\mathbf{x}_i$ lie on a generalised circle of $\overline{\mathbb{R}}^n$.

15. **(Apollonian circles.)** For $\alpha, \beta \in \mathbb{C}$, $k > 0$, consider the equation

$$\left|\frac{z - \alpha}{z - \beta}\right| = k. \tag{8.19}$$

This equation defines the *Apollonian circles* in $\overline{\mathbb{C}}$.

(a) Prove that the equation of the Apollonian circles represents a line if $k = 1$ and a circle if $k \neq 1$. Therefore, Eq. (8.19) is an equation of a generalised circle in $\overline{\mathbb{C}}$.
(b) Prove that GM(2) maps Apollonian circles to Apollonian circles.
(c) Prove conversely that every generalised circle is Apollonian, that is, it is written in the form (8.19). Conclude directly that if $g \in \mathrm{M}(2)$, then g maps generalised circles to generalised circles.

16. Find the element of M(2) that maps $-i, 0, i$ to $-1, 0, 1$, respectively.

17. Let an oriented quadruple $\mathfrak{p} = \{z_1, z_2, z_3, z_4\}$ of pairwise distinct points and let

$$\mathbf{X}_1 = \mathbf{X}_1(\mathfrak{p}) = [z_1, z_2; z_3, z_4], \quad \mathbf{X}_2 = \mathbf{X}_2(\mathfrak{p}) = [z_1, z_3; z_2, z_4].$$

(a) Show that $\mathbf{X}_1 + \mathbf{X}_2 = 1$.
(b) If $\mathbb{X}_1 = |\mathbf{X}_1|$, $\mathbb{X}_2 = |\mathbf{X}_2|$ are the corresponding metric cross-ratios, then

$$\mathbb{X}_1 + \mathbb{X}_2 \geq 1 \quad \text{and} \quad |\mathbb{X}_1 - \mathbb{X}_2| \leq 1.$$

18. Let D be the unit disk $D = \{z \in \mathbb{C} \mid |z| < 1\}$ and let

$$f(z) = \frac{z - i}{z + i}, \quad z \in \overline{\mathbb{C}}.$$

(a) Show that f maps D on the domain $\mathcal{H}^2 = \{z \in \mathbb{C} \mid \Im(z) > 0\}$.
(b) Show that every element of M(2) of the form

$$f(z) = \frac{az + b}{cz + d}, \quad a, b, c, d \in \mathbb{R}, \quad ad - bc > 0.$$

maps $\mathcal{H}^2$ to itself. Show conversely that if any element $f \in \mathrm{M}(2)$ maps $\mathcal{H}^2$ to itself, then it is of the above form.

19. Show that every f of the form (8.8) is a homomorphism of $\overline{\mathbb{R}}$.
20. Show that the set G of transformations of the form (8.8) is a group with operation the composition of mappings.
21. Find the element of M(1) that maps $-1, 0, 1$ to $2, 3, 4$, respectively.
22. Find the element of M(2) that maps the straight line $l : x = 1$ to the unit circle S^1 by choosing three arbitrary pairwise distinct points in l and three arbitrary pairwise distinct points in S^1.
23. Find $r \in \mathbb{R}$, $|r| > 1$, such that there exists a transformation of GM(1) that maps $\infty, 0, 1, r$ to $-r, -1, 1, r,$, respectively. Then, find this transformation.
24. Determine the values of $\alpha \in \mathbb{C}$ for which there exists an element $f \in \mathrm{M}(2)$ at maps the oriented quadruple $\mathfrak{p} = \{-i, 1, i, -1\}$ to the oriented quadruple $\mathfrak{p}' = \{-1, 0, 1, a\}$ so that

$$f(-i) = -1, \quad f(1) = 0, \quad f(i) = 1, \quad f(-1) = \alpha.$$

25. Consider the rectangles of Example 8.6.26. Prove that there exists a linear map f defined in complex terms for each $z \in \mathbb{C}$ by $f(z) = Az + B\overline{z}$, where $A, B \in \mathbb{C}$, and is such that

$$f(0) = 0, \quad f(a) = a', \quad f(a + ib) = a' + ib', \quad f(ib) = ib'.$$

Chapter 9
Hyperbolic Geometry



Mathematicians do not study objects, but relations between objects. Thus, they are free to replace some objects by others so long as the relations remain unchanged. Content to them is irrelevant: they are interested in form only.

—Henri Poincaré

Hyperbolic Geometry according to Klein is the pair $(\mathcal{H}^{n+1}, \mathrm{GM}(\mathcal{H}^{n+1}))$, where

$$\mathcal{H}^{n+1} = \{\mathbf{x} = (x_1, \ldots, x_{n+1}) \in \mathbb{R}^{n+1} \mid x_{n+1} > 0\},$$

and $\mathrm{GM}(\mathcal{H}^{n+1})$ is the subgroup of $\mathrm{GM}(n+1)$ that leaves $\mathcal{H}^{n+1}$ invariant. The description of Hyperbolic Geometry in this way corresponds to the *upper half-space model* which is due to Poincaré (Fig. 9.1).

Hyperbolic Geometry in two dimensions appeared in the early nineteenth century in independent works by Lobachevsky and Bolyai, and was most probably known to Gauss. It is the first non-Euclidean Geometry, in the sense that it holds all the axioms of Euclidean Geometry but the Fifth Postulate is replaced by an axiom according to which from a point outside a given line an infinite number of parallel lines are drawn to it. It remains of course to clarify what exactly a line means in Hyperbolic Geometry. Hyperbolic Geometry is a metric geometry. The *hyperbolic distance* between two points is precisely that which remains unchanged by the action of $\mathrm{GM}(\mathcal{H}^{n+1})$. A line is therefore a geodesic curve with respect to this metric. We will examine all these issues, as well as others, in what follows; we emphasise here that there are other models of hyperbolic geometry besides that upper half-space model, which we will examine at the end of the chapter for the two-dimensional case.

Hyperbolic Geometry has important applications in Physics and Astronomy, especially in Einstein's Theory of Relativity, where space and time do not follow the rules of Euclidean Geometry.

In this chapter we basically follow [44] and [7]. There is an immense bibliography on Hyperbolic Geometry, indicatively we refer to Fenchel's classic book,

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I. D. Platis, *Geometries*, La Matematica per il 3+2 178,
https://doi.org/10.1007/978-3-032-08016-5_9



Fig. 9.1 Henri Poincaré, 1854–1912

[21], as well as to [1], [9], [29], [42] for modern readings of the subject in various dimensions.

9.1 The Group of Hyperbolic Transformations

9.1.1 Poincaré Extension of GM(n)

The Poincaré extension of a transformation $f \in \mathrm{GM}(n)$ to a transformation $\tilde{f} \in \mathrm{GM}(n+1)$ is based on the embedding of $\mathbb{R}^n$ into $\mathbb{R}^{n+1}$ as the hyperplane $x_{n+1} = 0$, that is, through the mapping

$$\mathbb{R}^n \ni \mathbf{x} \mapsto (\mathbf{x}, 0) \in \mathbb{R}^{n+1},$$

and is deduced from the extension of the elementary transformations of which it is composed, in the following way:

- The inversion in the sphere $S(\mathbf{a}, r)$ of $\mathbb{R}^n$ extends to the inversion in the sphere $S((\mathbf{a}, 0), r)$ of $\mathbb{R}^{n+1}$.
- The inversion in the hyperplane $\overline{W_{\mathbf{A},b}}$ of $\overline{\mathbb{R}^n}$ extends to the inversion in the hyperplane $\overline{W_{(\mathbf{A},0),b}}$ of $\overline{\mathbb{R}^{n+1}}$.

In this way, we always have $\tilde{f}(\mathbf{x}, 0) = (f(\mathbf{x}), 0)$, that is, $\tilde{f}$ preserves the hyperplane $x_{n+1} = 0$. Also, through the formulae of inversions in a sphere and a hyperplane, we see that $\tilde{f}$ preserves the upper half-space

$$\mathcal{H}^{n+1} = \{\mathbf{x} \in \mathbb{R}^{n+1} \mid x_{n+1} > 0\}.$$

The question here is whether the Poincaré extension $\tilde{f}$ of a transformation $f \in \mathrm{GM}(n)$ is unique. Suppose to the contrary that there exists a $g \in \mathrm{GM}(n+1)$ such that

$$g(\mathbf{x}, 0) = (f(\mathbf{x}), 0).$$

Then, $h = \tilde{f} \circ g^{-1}$ preserves the hyperplane $x_{n+1} = 0$ pointwise. Since h preserves $(\mathbf{0}_n, 0)$, $(\mathbf{e}_n, 0)$, ∞, we have

$$h(\mathbf{x}_n, x_{n+1}) = (\mathbf{x}_n, x_n + 1)A, \quad A \in \mathrm{O}(n+1).$$

Putting $\mathbf{x}_n = \mathbf{e}_i$, $i = 1, \ldots, n$, successively into the relation

$$(\mathbf{x}_n, 0)A = (\mathbf{x}_n, 0)$$

it follows due to the orthogonality of A that it is a diagonal matrix with $a_{ii} = 1$, for $i = 1, \ldots, n$ and $a_{n+1,n+1} = \pm 1$. Consequently, h is either the identity, or the reflection in the hyperplane $x_{n+1} = 0$. However, the latter is impossible since h preserves $\mathcal{H}^{n+1}$. We therefore conclude that the Poincaré extension $\tilde{f}$ of f is unique.

Definition 9.1.1 The set $\widetilde{\mathrm{GM}}(n)$ of Poincaré extensions of the elements of $\mathrm{GM}(n)$ is called *Poincaré extension of* $\mathrm{GM}(n)$.

Proposition 9.1.2 *The Poincaré extension* $\widetilde{\mathrm{GM}}(n)$ *of* $\mathrm{GM}(n)$ *is a subgroup of* $\mathrm{GM}(n+1)$ *isomorphic to* $\mathrm{GM}(n)$.

Proof We will only prove that $\widetilde{\mathrm{GM}}(n)$ is closed under group operation. Let $f, g \in \mathrm{GM}(n)$ and $\tilde{f}, \tilde{g}$ be the Poincaré extensions of f, g, respectively. Then

$$\widetilde{(f \circ g)}(\mathbf{x}, 0) = ((f \circ g)(\mathbf{x}), 0) = (f(g(\mathbf{x})), 0)$$

and

$$(\tilde{f} \circ \tilde{g})(\mathbf{x}, 0) = \tilde{f}(\tilde{g}(\mathbf{x}, 0)) = \tilde{f}(g(\mathbf{x}), 0) = (f(g(\mathbf{x})), 0).$$

□

Exercise Complete the proof of Proposition 9.1.2.

Proposition 9.1.3 *An element* $f \in \mathrm{GM}(n+1)$ *leaves the upper half-space* $\mathcal{H}^{n+1}$ *invariant if and only if* $f \in \widetilde{\mathrm{GM}}(n)$.

Proof We have already seen that every Poincaré extension leaves $\mathcal{H}^{n+1}$ invariant. Now let $f \in \mathrm{GM}(n+1)$ such that $f(\mathcal{H}^{n+1}) = \mathcal{H}^{n+1}$. Since f is a homeomorphism, we also have that it leaves the boundary of the upper half-space invariant and consequently is restricted to a homeomorphism $\overline{f}$ of $\overline{\mathbb{R}^n}$. Now, f preserves the

cross-ratios in $\overline{\mathbb{R}^{n+1}}$, hence $\overline{f}$ preserves the cross- ratios in $\overline{\mathbb{R}^n}$. But then it follows from Theorem 8.5.7 that $f \in \mathrm{GM}(n)$. If g is the Poincaré extension of $\overline{f}$, then $f \circ g^{-1}$ preserves pointwise the hyperplane $x_{n+1} = 0$. Arguing as in the definition of the Poincaré extension, the desired result is obtained. □

Definition 9.1.4 The *hyperbolic group* is the set of Möbius transformations that preserve the upper half-space $\mathcal{H}^{n+1}$ and will be denoted by $\mathrm{GM}(\mathcal{H}^{n+1})$. The subset of $\mathrm{GM}(\mathcal{H}^{n+1})$ consisting of elements that preserve the orientation will be denoted by $\mathrm{M}(\mathcal{H}^{n+1})$.

From Proposition 9.1.3 it is clear that

$$\widetilde{\mathrm{GM}}(n) = \mathrm{GM}(\mathcal{H}^{n+1}), \quad \widetilde{\mathrm{M}}(n) = \mathrm{M}(\mathcal{H}^{n+1}).$$

It is appropriate, before proceeding to the general properties of $\mathrm{GM}(\mathcal{H}^{n+1})$ (and $\mathrm{M}(\mathcal{H}^{n+1})$), to consider the example of the case $n = 1$.

9.1.2 *The Group* $\mathrm{GM}(\mathcal{H}^2)$

Let $\mathcal{H}^2 = \{z = x + iy \in \mathbb{C} \mid \Im(z) = y > 0\}$ be the upper half-plane. We immediately have from the discussion of the previous section that each element of $\mathrm{GM}(\mathcal{H}^2)$ maps the line $y = 0$ onto itself and the upper half-plane $\mathcal{H}^2$ onto itself. For the exact form of the elements of $\mathrm{GM}(\mathcal{H}^2)$ we will prove the following:

Theorem 9.1.5 *Let* $f \in \mathrm{GM}(1)$,

$$f(x) = \frac{ax + b}{cx + d}, \quad a, b, c, d \in \mathbb{R},\ ad - bc \neq 0.$$

The Poincaré extension $\tilde{f}$ *of* f *is given by*

$$\tilde{f}(z) = \frac{az + b}{cz + d}, \quad \text{if } ad - bc > 0, \tag{9.1}$$

while it is given by

$$\tilde{f}(z) = \frac{a\bar{z} + b}{c\bar{z} + d}, \quad \text{if } ad - bc < 0. \tag{9.2}$$

Proof We first identify the Poincaré extensions of the elementary transformations of GM(1). We have that:

- The extension of the translation $x \mapsto x + a$, $a \in \mathbb{R}$, is $(x, y) \mapsto (x + a, y)$. In complex form, $z \mapsto z + a$.

- The extension of the reflection $x \mapsto -x$ at the point 0 is the reflection $(x, y) \mapsto (-x, y)$, that is, the reflection on the line $x = 0$. In complex form, $z \mapsto -\overline{z}$.
- The extension of the dilation $x \mapsto \delta x$ is the dilation $(x, y) \mapsto (\delta x, \delta y)$. In complex form, $z \mapsto \delta z$.
- The extension of the inversion $x \mapsto 1/x$ is the inversion $(x, y) \mapsto (x, y)/\sqrt{x^2 + y^2}$. In complex form, $z \mapsto 1/\overline{z}$.

All the above extensions have one of the forms (9.1), (9.2). Now, f is written as

$$f(x) = \begin{cases} (a/d)x + (b/d) & c = 0, \\ \frac{a}{c} + \frac{bc-ad}{c^2} \cdot \frac{1}{x+(d/c)} & c \neq 0. \end{cases}$$

Let $ad - bc > 0$ first. If $c = 0$, then since $ad > 0$, f is the composition of the extensions of the dilation by a/d and the translation by b/d. If $c \neq 0$, then f is the composition of the extensions of the translation by d/c, the inversion at 0, the reflection at 0, the dilation by $(ad - bc)/c^2$, and the translation by a/c. Since the composition of extensions of Poincare transformations is the Poincare extension of their composition (Proposition 9.1.2), we get

$$\tilde{f}(z) = \begin{cases} (a/d)z + (b/d) & c = 0, \\ \frac{a}{c} + \frac{bc-ad}{c^2} \cdot \frac{1}{z+(d/c)} & c \neq 0. \end{cases}$$

Now suppose that $ad - bc < 0$. If $c = 0$, then since $ad < 0$, f is the composition of the extensions of the dilation by $-a/d$, the reflection at 0, and the translation by b/d. If $c \neq 0$, then f is the composition of the extensions of the translationd/c, the inversion at 0, the dilation by $(bc - ad)/c^2$, and the translation by a/c. Thus by the same argument as before, we get

$$\tilde{f}(z) = \begin{cases} (a/d)\overline{z} + (b/d) & c = 0, \\ \frac{a}{c} + \frac{bc-ad}{c^2} \cdot \frac{1}{\overline{z}+(d/c)} & c \neq 0. \end{cases}$$

□

Example 9.1.6 Let $f \in \mathrm{GM}(1)$,

$$f(x) = \frac{2x - 1}{x - 3}, \quad x \in \overline{\mathbb{R}}.$$

Since $2 \cdot (-3) - (-1) \cdot 1 = -5 < 0$, the Poincaré extension $\tilde{f}$ of f is given by

$$\tilde{f}(z) = \frac{2\overline{z} - 1}{\overline{z} - 3}.$$

On the other hand, for $g \in \mathrm{GM}(1)$,

$$g(x) = \frac{x-2}{x-1},$$

since $1 \cdot (-1) - (-2) \cdot 1 = 1 > 0$, the Poincaré extension $\tilde{g}$ of g is given by

$$\tilde{g}(z) = \frac{z-2}{z-1}.$$

Due to Proposition 8.6.6, we now have the relation

$$\mathrm{GM}(\mathcal{H}^2) \simeq \mathrm{GM}(1) \simeq \mathrm{PSL}(2,\mathbb{R}) = \mathrm{SL}(2,\mathbb{R})/\{I,-I\},$$

where I is the 2×2 identity matrix. The subgroup $\mathrm{M}(\mathcal{H}^2)$ of $\mathrm{GM}(\mathcal{H}^2)$ consisting of elements of the form

$$f(z) = \frac{az+b}{cz+d}, \quad ad - bc > 0,$$

is isomorphic to $\mathrm{SL}(2,\mathbb{R})$. Thus

$$\mathrm{M}(\mathcal{H}^2) = \mathrm{PSL}^+(2,\mathbb{R}) = \mathrm{SL}(2,\mathbb{R}).$$

Finally, we observe that the elements of $\mathrm{GM}(\mathcal{H}^2)$ of the form (9.1) are in a bijective correspondence with the elements of the form (9.2):

$$\mathrm{M}(\mathcal{H}^2) \ni f(z) = \frac{az+b}{cz+d} \mapsto g(z) = \frac{a(-\overline{z})+b}{c(-\overline{z})+d}.$$

Consequently, in the study of the Möbius transformations of $\mathcal{H}^2$ it is sufficient to deal with elements of $\mathrm{SL}(2,\mathbb{R})$. If $f \in \mathrm{M}(\mathcal{H}^2)$, then we can always write

$$f(z) = \frac{az+b}{cz+d}, \quad ad - bc = 1,$$

dividing if necessary by the root of $ad - bc$. The matrix

$$A_f = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2,\mathbb{R}),$$

is called the *associated matrix* to the transformation f.

Example 9.1.7 Consider the transformations of the extended real line given by

$$f(x) = \frac{2x-1}{x-1}, \quad g(x) = \frac{x-2}{x+1},$$

corresponding to the elements

$$[A] = \begin{bmatrix} 2 & -1 \\ 1 & -1 \end{bmatrix}, \quad [B] = \begin{bmatrix} 1 & -2 \\ 1 & 1 \end{bmatrix},$$

of $\mathrm{PGL}(2, \mathbb{R})$. Since $\det(A) = -1 < 0$, A is the matrix corresponding to the orientation reversing Möbius transformation $f \in \mathrm{GM}(\mathcal{H}^2)$ with

$$f(z) = \frac{2\overline{z} - 1}{\overline{z} - 1} = \frac{(-2)(-\overline{z}) - 1}{(-1)(-\overline{z}) - 1}.$$

This corresponds to the orientation preserving Möbius transformation

$$\tilde{f} : z \mapsto \frac{-2z - 1}{-z - 1} = \frac{2z + 1}{z + 1}$$

of $\mathrm{M}(\mathcal{H}^2)$. The matrix of this element has determinant 1, so it belongs to $\mathrm{SL}(2, \mathbb{R})$ and is the associated matrix of $\tilde{f}$.

On the other hand, $\det(B) = 3 > 0$, so g extends to the transformation $g \in \mathrm{M}(\mathcal{H}^2)$ with

$$g(z) = \frac{z - 2}{z + 1}.$$

The associated matrix of g is

$$(1/\sqrt{3})B = (1/\sqrt{3}) \begin{pmatrix} 1 & -2 \\ 1 & 1 \end{pmatrix} \in \mathrm{SL}(2, \mathbb{R}).$$

9.1.3 The Form of Hyperbolic Transformations

If $f \in \mathrm{GM}(\mathcal{H}^{n+1})$ then $f = \tilde{g}$, the Poincaré extension of a $g \in \mathrm{GM}(n)$. The transformation g is a composition of inversions on generalised spheres. The Poincaré extension of any such inversion is an inversion in $\overline{\mathbb{R}^{n+1}}$ in a sphere orthogonal to $\overline{\mathbb{R}^n}$. Since the extension $f = \tilde{g}$ is the composition of extensions of inversions, we have the following:

Proposition 9.1.8 *Each element of* $\mathrm{GM}(\mathcal{H}^{n+1})$ *is a composition of inversions of* $\overline{\mathbb{R}^{n+1}}$ *on spheres orthogonal to* $\overline{\mathbb{R}^n}$.

Poincaré extensions of all elementary transformations of $\mathrm{GM}(n)$ follow:

- The extension of the translation $\mathbf{x}_n \mapsto \mathbf{x}_n + \mathbf{a}$ is

$$(\mathbf{x}_n,\ x_{n+1}) \mapsto (\mathbf{x}_n + \mathbf{a},\ x_{n+1}).$$

  Here, $\mathbf{x}_n = (x_1, \ldots, x_n)$ and $\mathbf{a} \in \mathbb{R}^n$.
- The extension of the orthogonal transformation $\mathbf{x}_n \mapsto \mathbf{x}_n A$, $A \in \mathrm{O}(n)$, is

$$(\mathbf{x}_n,\ x_{n+1}) \mapsto (\mathbf{x}_n A,\ x_{n+1}) = (\mathbf{x}_n,\ x_{n+1}) \begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix}.$$

- The extension of the dilation $\mathbf{x}_n \mapsto \delta \mathbf{x}_n$, is

$$(\mathbf{x}_n,\ x_{n+1}) \mapsto (\delta \mathbf{x}_n,\ \delta x_{n+1}).$$

- The extension of the inversion $\mathbf{x}_n \mapsto \mathbf{x}_n / \|\mathbf{x}_n\|^2$, is

$$\mathbf{x} \mapsto \frac{\mathbf{x}}{\|\mathbf{x}\|^2}, \quad \mathbf{x} = (\mathbf{x}_n,\ x_{n+1}).$$

Theorem 9.1.9 *The general form of a hyperbolic transformation $f \in \mathrm{GM}(\mathcal{H}^{n+1})$ is given by the relation*

$$f(\mathbf{x}) = \mathbf{b} + \rho \frac{\mathbf{x}A - \mathbf{c}}{\|\mathbf{x}A - \mathbf{c}\|^{\varepsilon}}, \quad \mathbf{x} \in \mathcal{H}^{n+1},$$

where $\mathbf{b}, \mathbf{c} \in \mathbb{R}^n$, $\rho > 0$, $A \in \mathrm{O}(n+1)$ of the form

$$A = \begin{pmatrix} A' & 0 \\ 0 & 1 \end{pmatrix}, \quad A' \in \mathrm{O}(n)$$

and $\varepsilon = 0$ or 2.

Proof According to Theorem 8.5.9, each element $f \in \mathrm{GM}(n)$ is written in the form

$$f(\mathbf{x}) = \mathbf{b} + \rho \frac{\mathbf{x}A - \mathbf{c}}{\|\mathbf{x}A - \mathbf{c}\|^{\varepsilon}},$$

where $\mathbf{b}, \mathbf{c} \in \mathbb{R}^n$, $\rho > 0$, $A \in \mathrm{O}(n)$ and $\varepsilon = 0$ or 2. In other words, it is a composition of some of the above elementary transformations. In detail, if $\varepsilon = 0$, that is, if $f(\infty) = \infty$, then

$$f = T_{\mathbf{b} - \rho^2 \mathbf{c}} \circ D_{\rho^2} \circ \alpha,$$

where $\alpha \in \mathrm{O}(n)$ with matrix A. On the other hand, if $\varepsilon = 2$, then

$$f = T_{\mathbf{b}-\mathbf{c}} \circ \iota_{\mathbf{c},\rho} \circ \alpha.$$

We complete the proof by using Proposition 8.2.7 and the fact that the Poincaré extension of the composition of Möbius transformations is equal to the composition of the Poincaré extensions of the individual transformations. □

From the general form of Möbius transformations of Theorem 9.1.9, the chain rule and the properties of elementary Möbius transformations we have the following:

Corollary 9.1.10 *Let $f \in \mathrm{GM}(\mathcal{H}^{n+1})$. Then $f \in \mathrm{M}(\mathcal{H}^{n+1})$ if and only if:*

- $A' \in \mathrm{SO}(n)$ *when* $\varepsilon = 0$ *and*
- $\det(A') = -1$ *when* $\varepsilon = 2$.

9.1.4 *Classification of Elements of* $\mathrm{GM}(\mathcal{H}^{n+1})$

All elements of $\mathrm{GM}(\mathcal{H}^{n+1})$ are homeomorphisms and hence map $\overline{\mathbb{R}^n} = \partial\mathcal{H}^{n+1}$ onto itself. There is a classification of these transformations based on the number of points they stabilise; we recall that a Möbius transformation of $\overline{\mathbb{R}^{n+1}}$ that stabilises three points is conjugate to an element of $\mathrm{O}(n)$.

Definition 9.1.11 A transformation $f \in \mathrm{GM}(\mathcal{H}^{n+1})$ is called:

- *elliptic* if it stabilises one point of $\mathcal{H}^{n+1}$,
- *parabolic* if it stabilises a single point of $\overline{\mathbb{R}^n}$ and does not stabilise any point of $\mathcal{H}^{n+1}$,
- *loxodromic* if it stabilises two points of $\overline{\mathbb{R}^n}$ and does not stabilise any point of $\mathcal{H}^{n+1}$.

Example 9.1.12 For the elementary transformations of $\mathrm{GM}(\mathcal{H}^{n+1})$ we have the following:

- Translations

$$(\mathbf{x}_n,\ x_{n+1}) \mapsto (\mathbf{x}_n + \mathbf{a},\ x_{n+1}),$$

  where $\mathbf{x}_n = (x_1, \ldots, x_n)$ and $\mathbf{a} \in \mathbb{R}^n$ are parabolic: they stabilise ∞ and no other point in $\mathcal{H}^{n+1}$.
- Orthogonal transformations

$$(\mathbf{x}_n,\ x_{n+1}) \mapsto (\mathbf{x}_n A,\ x_{n+1}) = (\mathbf{x}_n,\ x_{n+1}) \begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix}, \quad A \in \mathrm{O}(n),$$

are elliptic: they stabilise $\infty, \mathbf{0}$ as well as the half-line $\mathbf{c}(t) = (\mathbf{y}_n, t)$, $t > 0$, where $\mathbf{y}_n$ is a fixed point of A.

- Dilations

$$(\mathbf{x}_n,\ x_{n+1}) \mapsto (\delta \mathbf{x}_n,\ \delta x_{n+1}), \quad \delta > 0,$$

stabilise only two points in the boundary, that is, $\mathbf{0}$ and ∞ and hence they are loxodromic.

- Inversion

$$\mathbf{x} \mapsto \frac{\mathbf{x}}{\|\mathbf{x}\|^2}, \quad \mathbf{x} = (\mathbf{x}_n,\ x_{n+1}),$$

stabilises pointwise the part of the unit sphere S^n where $x_{n+1} \geq 0$ and therefore is elliptic.

Our goal in this section is to establish criteria that will allow us to determine the type of hyperbolic transformations. We first consider the case where $f \in \mathrm{GM}(\mathcal{H}^{n+1})$ stabilises ∞, in other words, is the Poincaré extension of some element of $\overline{S(n)}$. Then,

$$f(\mathbf{x}) = r\mathbf{x}A + \mathbf{b},$$

where $r > 0$, $\mathbf{b} \in \mathbb{R}^n$ and A is an orthogonal matrix of the form

$$A = \begin{pmatrix} A' & 0 \\ 0 & 1 \end{pmatrix}, \quad A' \in \mathrm{O}(n). \tag{9.3}$$

The fixed point equation is then:

$$f(\mathbf{x}) = \mathbf{x} \iff \mathbf{x} \begin{pmatrix} A' - (1/r)I_n & 0 \\ 0 & 1 - 1/r \end{pmatrix} = -\frac{1}{r}\mathbf{b}. \tag{9.4}$$

Case 1 If f is elliptic, then there exists $\mathbf{x} \in \mathcal{H}^{n+1}$ which is a solution of (9.4). If $r \neq 1$, we end up with a contradiction since $\mathbf{b} \in \mathbb{R}^n$. Hence $r = 1$ and f is the Poincaré extension of

$$g(\mathbf{x}) = A'\mathbf{x} + \mathbf{b}.$$

g is a Euclidean isometry that stabilises the point

$$\mathbf{x}_n = -\mathbf{b}(A - I_n)^{-1} \in \mathbb{R}^n,$$

and therefore f stabilises all points of the line $(\mathbf{x}_n, x_{n+1})$ in $\mathcal{H}^{n+1}$.

Conversely, if f is the Poincaré extension of a Euclidean transformation that stabilises a point of $\mathbb{R}^n$, then it stabilises the line of $\mathcal{H}^{n+1}$ that is based at this point and is therefore elliptic.

Case 2 In the case of the parabolic f, and since it does not stabilise any point of $\mathbb{R}^{n+1}$, Equation (9.4) has no solution. Therefore,

$$\det\left(A - \frac{1}{r}I\right) = 0,$$

and thus $1/r$ is an eigenvalue of the orthogonal A. Since $r > 0$, then necessarily $r = 1$ and we conclude that f is the Poincaré extension of an element of $\overline{E(n)}$ that does not stabilise any point of $\mathbb{R}^n$.

Conversely, if f is the Poincaré extension of an element of $\overline{E(n)}$ that does not stabilise any point of $\mathbb{R}^n$, then since it cannot be loxodromic, nor elliptic from the first case, it is necessarily parabolic.

Case 3 Finally, in the case of loxodromic f, we have that $r \neq 1$ in Eq. (9.4). Then it has a solution only if $\mathbf{b} = \mathbf{0}$ and thus f is the Poincaré extension of the element $g(\mathbf{x}) = r\mathbf{x}A'$.

Conversely, if f is the Poincaré extension of an element of the form $g(\mathbf{x}) = r\mathbf{x}A'$, Eq. (9.4) has a unique solution belonging to $\overline{\mathbb{R}^n}$.

We have proved the following criterion:

Proposition 9.1.13 *Let $f \in \mathrm{GM}(\mathcal{H}^{n+1})$, $f(\infty) = \infty$. Then f is elliptic (resp. parabolic) if and only if it is the Poincaré extension of an element of $\overline{E(n)}$ which has (resp. does not have) a fixed point in $\mathbb{R}^n$. f is loxodromic if and only if it is the Poincaré extension of an element which is a composition of dilation and an orthogonal transformation of $\overline{\mathbb{R}^n}$.*

We continue with the case where $f(\infty) \neq \infty$. Let us assume that $f(\infty) = \mathbf{a}$. Then the inversion $g = \iota_{\mathbf{a},1}$ on the sphere $S(\mathbf{a}, 1)$ maps $\mathbf{a}$ to ∞ and consequently the transformation $g \circ f \circ g^{-1}$ then stabilises ∞. Applying Proposition 9.1.13 to this transformation, we obtain the following general criterion:

Proposition 9.1.14 *$f \in \mathrm{GM}(\mathcal{H}^{n+1})$ is elliptic (resp. parabolic) if and only if it is conjugate to $\mathrm{GM}(\mathcal{H}^{n+1})$ to the Poincaré extension an element of $\overline{E(n)}$ which has (resp. does not have) a fixed point in $\mathbb{R}^n$. It is loxodromic if and only if it is conjugate to an element in $\mathrm{GM}(\mathcal{H}^{n+1})$ which is the Poincaré extension of a composition of a dilation and an orthogonal transformation of $\overline{\mathbb{R}^n}$.*

From the proof of Propositions 9.1.13 and 9.1.14, we have the following:

Corollary 9.1.15 *A loxodromic transformation has exactly two fixed points in $\overline{\mathbb{R}^n}$.*

Example 9.1.16 Let $\theta \in \mathbb{R}$ and $A_\theta \in \mathrm{O}(3)$ be the matrix

$$A_\theta = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

and consider the map $f \in \mathrm{GM}(\mathcal{H}^3)$ given for each $(x_1, x_2, x_3) \in \mathcal{H}^3$ by

$$f(x_1, x_2, x_3) = (x_1, x_2, x_3)A_\theta + (0, 0, 1),$$

and $f(\infty) = \infty$. Explicitly,

$$f(x_1, x_2, x_3) = (x_1 \cos\theta + x_2 \sin\theta,\ -x_1 \sin\theta + x_2 \cos\theta,\ x_3 + 1).$$

It is clear that f does not stabilise any point in $\mathcal{H}^3$ and thus it is parabolic. On the other hand, the map $g \in \mathrm{GM}(\mathcal{H}^3)$ given for each $(x_1, x_2, x_3) \in \mathcal{H}^3$ by

$$g(x_1, x_2, x_3) = (x_1, x_2, x_3)A_\theta,$$

and $g(\infty) = \infty$, clearly stabilises the point $(0, 0, 0)$ and the axis of rotation $\mathbf{c}(t) = (0, 0, t)$, $t > 0$. Thus g is elliptic.

Classification of Transformations in $\mathrm{M}(\mathcal{H}^2)$

The case that is particularly accessible is of course when $n = 1$; then we can classify the elements of $\mathrm{M}(\mathcal{H}^2)$ via the identification of $\mathrm{M}(\mathcal{H}^2)$ with $\mathrm{SL}(2, \mathbb{R})$. Thus we will first present the classification of matrices in $\mathrm{SL}(2, \mathbb{R})$. Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2, \mathbb{R}),$$

let $\mathrm{tr}(A) = a + d$ be the trace of A and let also

$$p_A(\lambda) = \lambda^2 - \mathrm{tr}(A) \cdot \lambda + 1 = 0$$

be its characteristic equation. The discriminant of this quadratic equation is $D = \mathrm{tr}^2(A) - 4$. We distinguish the following cases:

- $D > 0$: we have two real eigenvalues

$$\lambda_{1,2} = \frac{\mathrm{tr}(A) \pm \sqrt{D}}{2}, \text{ and } \lambda_1 \cdot \lambda_2 = 1.$$

  Setting $\lambda = \lambda_1$, we obtain $\lambda + 1/\lambda = \mathrm{tr}(A)$.

- $D = 0$: we have a real double eigenvalue

$$\lambda = \frac{\operatorname{tr}(A)}{2} \implies \lambda^2 = 1.$$

- $D < 0$: we have two conjugate complex eigenvalues

$$\lambda_{1,2} = \frac{\operatorname{tr}(A) \pm i\sqrt{-D}}{2}, \text{ and } \lambda_1 \cdot \lambda_2 = 1$$

Setting $\lambda = \lambda_1$, we have $\lambda_2 = \overline{\lambda} = 1/\lambda$, so $\lambda = e^{i\theta}$ for some $\theta \in \mathbb{R}\backslash\{k\pi,\ k \in \mathbb{Z}\}$.

Proposition 9.1.17 *Every $A \in \mathrm{SL}(2, \mathbb{R})$ is conjugate in $\mathrm{SL}(2, \mathbb{R})$ to exactly one of the following matrices:*

(a)

$$\pm \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \quad \theta \in \mathbb{R} \setminus \{k\pi,\ k \in \mathbb{Z}\}.$$

(b)

$$\pm \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

(c)

$$\begin{pmatrix} \lambda & 0 \\ 0 & 1/\lambda \end{pmatrix},\ \lambda \in \mathbb{R},\ \lambda^2 \neq 1.$$

Proof We will only show the case of two different real eigenvalues: when $D > 0$, $\lambda_1 = \lambda, \lambda_2 = 1/\lambda$ and $\lambda + 1/\lambda = \operatorname{tr}(A)$, where $|\operatorname{tr}(A)| > 2$ and therefore $\lambda^2 \neq 1$. A is diagonalisable and thus there exists a matrix $P \in \mathrm{SL}(2, \mathbb{R})$ such that $A = PMP^{-1}$ with M being the matrix of the third form. □

Exercise Complete the proof of Proposition 9.1.17.

Example 9.1.18 Let

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \in \mathrm{SL}(2, \mathbb{R}).$$

Here $\mathrm{tr}(A) = 0$, $D = 1$ and thus we have two complex conjugate eigenvalues $\pm i$. Now observe that $i = e^{i\pi/2}$ and

$$A = \begin{pmatrix} \cos(\pi/2) & -\sin(\pi/2) \\ \sin(\pi/2) & \cos(\pi/2) \end{pmatrix}.$$

See also Example 9.1.20 below for further details.

We now give the following definition:

Definition 9.1.19 Let $f \in \mathrm{M}(\mathcal{H}^2)$ and suppose that

$$A_f = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2, \mathbb{R})$$

is the associated matrix of f.

- When A_f is conjugate to a matrix of the form (a) in $\mathrm{SL}(2, \mathbb{R})$, then f is called *elliptic* and is conjugate to a transformation of the form

$$z \mapsto \frac{\cos\theta z - \sin\theta}{\sin\theta z + \cos\theta}, \quad \theta \in \mathbb{R},$$

  in $\mathrm{M}(\mathcal{H}^2)$.
- When A_f is conjugate to a matrix of the form (b) in $\mathrm{SL}(2, \mathbb{R})$, then f is called *parabolic* and is conjugate to the transformation

$$z \mapsto z + 1,$$

  in $\mathrm{M}(\mathcal{H}^2)$
- When A_f is conjugate to a matrix of the form (c) in $\mathrm{SL}(2, \mathbb{R})$, then f is called *loxodromic* and is conjugate to the transformation

$$z \mapsto \lambda^2 z, \quad \lambda \in \mathbb{R},$$

  in $\mathrm{M}(\mathcal{H}^2)$.

Example 9.1.20 Let

$$f(z) = \frac{z-2}{z-1}$$

and

$$A = A_f = \begin{pmatrix} 1 & -2 \\ 1 & -1 \end{pmatrix} \in \mathrm{SL}(2, \mathbb{R}).$$

The characteristic polynomial

$$p_A(\lambda) = \lambda^2 + 1$$

has two conjugate complex roots $\pm i$. The eigenspace $V(i)$ is obtained by solving the system $(A - iI) \cdot (\xi_1\ \xi_2)^T = \mathbf{0}$, that is,

$$(1-i)\xi_1 - 2\xi_2 = 0,$$
$$\xi_1 - (1+i)\xi_2 = 0.$$

Hence,

$$V(i) = \left\{ \begin{pmatrix} (1+i)\mu \\ \mu \end{pmatrix}, \quad \mu \in \mathbb{C} \right\} \text{ and } \quad V(-i) = \left\{ \begin{pmatrix} (1-i)\mu \end{pmatrix}, \quad \nu \in \mathbb{C} \right\}.$$

Taking μ, ν with $2i\mu\nu = 1$ and the matrices

$$P = \begin{pmatrix} (1+i)\mu & (1-i)\nu \\ 1 & 1 \end{pmatrix}, \quad M = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix},$$

we have that $A = PMP^{-1}$ and hence A is conjugate in $\mathrm{SL}(2, \mathbb{C})$ to M. Note that M is the associated matrix of the transformation $z \mapsto -z$ in $\mathrm{SL}(2, \mathbb{C})$, representing a rotation by π in $\overline{\mathbb{C}}$. To see how the situation is in $\mathrm{SL}(2, \mathbb{R})$, we work as follows: we take the real and imaginary parts of the eigenvector $(1+i\ 1)^T$ and consider the matrix of $\mathrm{SL}(2, \mathbb{R})$ that has these vectors as columns:

$$Q = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

This is always possible and we can achieve it by choosing an appropriate $\mu \in \mathbb{C}$. We now write $i = \cos(\pi/2) + \sin(\pi/2) \cdot i = 0 + 1 \cdot i$ and consider the matrix

$$N = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

We can now see that $A = QNQ^{-1}$, f is an elliptic transformation conjugate to $z \mapsto -1/z$.

Example 9.1.21 Let

$$f(z) = \frac{-1}{z+2}$$

and

$$A = A_f = \begin{pmatrix} 0 & -1 \\ 1 & 2 \end{pmatrix} \in \mathrm{SL}(2, \mathbb{R}).$$

The characteristic polynomial

$$p_A(\lambda) = \lambda^2 - 2\lambda + 1$$

has a double root $\lambda = 1$. We are looking for a matrix

$$P = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \in \mathrm{SL}(2, \mathbb{R})$$

so that it satisfies

$$\begin{pmatrix} 0 & -1 \\ 1 & 2 \end{pmatrix} = \pm \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \delta & -\beta \\ -\gamma & \alpha \end{pmatrix} = \pm \begin{pmatrix} 1 - \alpha\gamma & \alpha^2 \\ -\gamma^2 & 1 + \alpha\gamma \end{pmatrix}.$$

We have solutions only for the case of the negative sign: $\alpha = \gamma = 1$, or $\alpha = \gamma = -1$. The values of β, δ are then determined by the equation $\beta\delta = \alpha\gamma - 1$. f is parabolic, conjugate to $z \mapsto z + 1$.

Example 9.1.22 Let

$$f(z) = \frac{2z + 1}{3z + 2}$$

and

$$A = A_f = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix} \in \mathrm{SL}(2, \mathbb{R}).$$

The characteristic polynomial

$$p_A(\lambda) = \lambda^2 - 4\lambda + 1$$

has roots $\lambda, 1/\lambda$, where $\lambda = 2 + \sqrt{3}$. The eigenspace $V(\lambda)$ results from the solution of the system $(A - \lambda I) \cdot (\xi_1 \ \xi_2)^T = \mathbf{0}$, that is,

$$-\sqrt{3}\xi_1 + \xi_2 = 0,$$
$$3\xi_1 - \sqrt{3}\xi_2 = 0.$$

Therefore,

$$V(\lambda) = \left\{ \begin{pmatrix} \mu \\ \sqrt{3}\mu \end{pmatrix}, \quad \mu \in \mathbb{R} \right\}.$$

Similarly we find

$$V(1/\lambda) = \left\{ \begin{pmatrix} \nu \\ -\sqrt{3}\nu \end{pmatrix}, \quad \nu \in \mathbb{R} \right\}.$$

Taking μ, ν satisfying $-2\sqrt{3}\mu\nu = 1$, we have that for

$$P = \begin{pmatrix} \mu & \nu \\ \sqrt{3}\mu & -\sqrt{3}\nu \end{pmatrix} \quad \text{and} \quad M = \begin{pmatrix} \lambda & 0 \\ 0 & 1/\lambda \end{pmatrix},$$

we obtain $A = PMP^{-1}$. Hence, f is conjugate in $\mathrm{SL}(2, \mathbb{R})$ to $z \mapsto \lambda^2 z = (5 + 4\sqrt{3})z$ and is loxodromic.

A point $z \in \overline{\mathcal{H}^2} = \mathcal{H}^2 \cup \overline{\mathbb{R}}$ is a fixed point of f if $f(z_0) = z_0$. The question of the set of fixed points of f is now answered by Proposition 9.1.17. The relation $f(z) = z$ leads:

- to the equation

$$(a - d)z + b = 0$$

  if $c = 0$ and
- to the quadratic equation

$$cz^2 + (d - a)z - b = 0,$$

  if $c \neq 0$.

When $c = 0$, we always have $f(\infty) = \infty$. Since $ad = 1$, $D \geq 0$ with the equality being true when f is parabolic and therefore does not stabilize another point. In the case of the inequality, it additionally stabilises $z = b/(d - a)$ and is hyperbolic.

When $c \neq 0$, the discriminant of the quadratic equation is equal to

$$(d - a)^2 + 4bc = \mathrm{tr}^2(A_f) - 4 = D.$$

Therefore, we arrive at the following:

Proposition 9.1.23 *$f \in \mathrm{M}(\mathcal{H}^2)$ has:*

- *one fixed point on $\overline{\mathcal{H}^2}$ if it is parabolic,*
- *one fixed point on $\mathcal{H}^2$ if it is elliptic,*
- *two fixed points on $\overline{\mathbb{R}}$ if it is hyperbolic.*

Remark 9.1.24 The elliptic transformation in Example 9.1.20 fixes the point $1+i$ in $\mathcal{H}^2$. Also, the parabolic transformation in Example 9.1.21 has the single point $z=1$ lying in $\partial\mathcal{H}^2$. Finally, the loxodromic transformation in Example 9.1.22 has two fixed points $\pm\sqrt{3}/3$ lying in $\partial\mathcal{H}^2$.

Example 9.1.25 Let

$$f(z)=\frac{2z}{z+2}.$$

The associated matrix to f is

$$A_f=\begin{pmatrix}1 & 0\\ 1/2 & 1\end{pmatrix}.$$

We have $\operatorname{tr}(A_f)=2$ and therefore f is parabolic. Its single fixed point is $z=0$.

Example 9.1.26 Let

$$f(z)=\frac{-z-1}{2z+1}.$$

The associated matrix to f is

$$A_f=\begin{pmatrix}-1 & -1\\ 2 & 1\end{pmatrix}.$$

We have $\operatorname{tr}(A_f)=0$ and therefore f is elliptic. Its fixed point in $\mathcal{H}^2$ is given by the equation

$$f(z)=z \iff 2z^2+2z+1=0 \implies z=\frac{-1+i}{2}.$$

Example 9.1.27 Let

$$f(z)=\frac{z+2}{z+3}.$$

The associated matrix to f is

$$A_f=\begin{pmatrix}1 & 2\\ 1 & 3\end{pmatrix}.$$

We have $\mathrm{tr}(A_f) = 3$ and therefore f is loxodromic. Its fixed points in $\overline{\mathbb{R}}$ are given by the equation

$$f(z) = z \iff z^2 + 2z - 2 = 0 \implies z = -1 \pm \sqrt{3}.$$

Example 9.1.28 Let

$$f(z) = \frac{\alpha z}{z + 1/\alpha}, \quad \alpha \in \mathbb{R}_*.$$

Then if $\alpha = 1$ f is parabolic and fixes 0; if $\alpha \neq 1$, then f is loxodromic and fixes 0 and $\alpha - 1/\alpha$.

9.2 Hyperbolic Geometry

9.2.1 Hyperbolic Lines

In Euclidean planar geometry, (straight) lines are the sets of fixed points of reflections (with respect to these lines) (Fig. 9.2). Every line can in other words be defined (in complex terms) as the set of solutions of an equation of the form

$$\Re(e^{-i\theta} z) = b.$$

In higher dimensions, reflections in (generally affine hyperplanes) pointwise preserve the hyperplanes and hence all the lines (the one-dimensional affine subspaces) they contain (Fig. 9.3).

We will use the same reasoning in the case of hyperbolic space. We have seen that the Poincaré extensions of the inversions on a sphere $S(\mathbf{a}, r)$ and on a hyperplane $\overline{W_{\mathbf{A},b}}$ of $\overline{\mathbb{R}^n}$ are respectively the inversions in a sphere $S((\mathbf{a}, 0), r)$ and



Fig. 9.2 János Bolyai, 1802–1860



Fig. 9.3 Nikolay Lobachevsky, 1792–1856

in a hyperplane $\overline{W_{(\mathbf{A},0),b}}$ of $\overline{\mathbb{R}^{n+1}}$, respectively, both of which are orthogonal on the boundary. When $n = 1$, a sphere of centre a and radius $r > 0$ consists simply of the points $a - r, a + r$. The corresponding sphere in the plane is the circle $S((a, 0), r)$ and its points lying in $\mathcal{H}^2$ are its open upper semicircle. On the other hand, a hyperplane in $\overline{\mathbb{R}}$ is simply a point x_0 of $\mathbb{R}$ and in the Poincaré extension we have the line $x = x_0$—the points (x_0, y) that lie in $\mathcal{H}^2$ are of course those with $y > 0$.

For $n > 1$, inversion in the sphere $S((\mathbf{a}, 0), r)$ preserve pointwise its great circles that are orthogonal to the boundary. The part of any such great circle that lies in $\mathcal{H}^{n+1}$ is the open upper semicircle. Similarly, the reflection in a hyperplane $\overline{W_{(\mathbf{A},0),b}}$ preserves pointwise the lines that are orthogonal to the boundary: such lines are of the form $x_i = c_i, i = 1, \ldots, n, x_{n+1} > 0$.

The following holds:

Proposition 9.2.1 *Let* $\mathrm{GM}(\mathcal{H}^{n+1})$ *be an inversion. Then:*

- *If* $n = 1$*, the set of its fixed points in* $\mathcal{H}^2$ *is either the points lying in the semicircle of the inversion circle with* $x_2 > 0$*, or the half-line of the inversion line with* $x_2 > 0$*.*
- *If* $n > 1$*, then the set of fixed points in* $\mathcal{H}^{n+1}$ *is either the upper part* ($x_{n+1} > 0$) *of the inversion sphere, or the upper part of the inversion hyperplane. In particular, the inversion preserves pointwise the semicircles and semi-lines of the corresponding inversion spheres and hyperplanes.*

Definition 9.2.2 A *hyperbolic line* (Fig. 9.4) is either:

- the part in the upper half-space of a great semicircle of a sphere of the form $\|\mathbf{x} - \mathbf{x}_0\| = r, \mathbf{x}_0 \in \mathbb{R}^n$, or
- a half line of the form $\mathbf{c}(t) = (\mathbf{x}_0, t), t > 0$.

If the hyperbolic line L has endpoints at infinity $\mathbf{z}, \mathbf{z}'$, we will write $L = (\mathbf{z}, \mathbf{z}')$ (one of the endpoints can of course also be ∞). The order in which we write the endpoints is not particularly important, except if we want to give an *orientation* to L. We note that if $\mathbf{z}, \mathbf{z}'$ are both in $\mathbb{R}^n$, then:

- $\|\mathbf{z} - \mathbf{z}'\| = 2r$, where r is the radius of the sphere containing the semicircle.

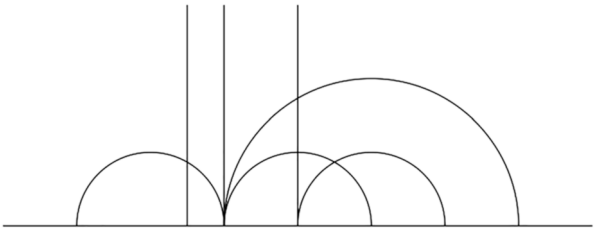


Fig. 9.4 Hyperbolic lines

- The centre of the semicircle (which is also the centre of the sphere containing it) is $\mathbf{a} = (\mathbf{z} + \mathbf{z}')/2$.
- The hyperbolic line $L = (\mathbf{z}, \mathbf{z}')$ lies in the 2-plane

$$\mathbf{x}(\lambda, \mu) = \mathbf{c}(\lambda) + \mu\, \mathbf{e}_{n+1}, \quad \lambda, \mu \in \mathbb{R}$$

  where $\mathbf{c}(\lambda) = \mathbf{a} + \lambda(\mathbf{z}' - \mathbf{z})/2$ is the line in $\mathbb{R}^n$ that passes through $\mathbf{z}$ and $\mathbf{z}'$.
- Substituting the above into the equation $\|\mathbf{x} - \mathbf{a}\| = r$ of the sphere, we obtain the parametric equation of the semicircle

$$\mathbf{x}(\theta) = \mathbf{a} + \cos\theta\, (\mathbf{z} - \mathbf{z}')/2 + r \sin\theta\, \mathbf{e}_{n+1}, \quad \theta \in (0, \pi). \tag{9.5}$$

Remark 9.2.3 Let the hyperbolic line $L = (z, z')$ in $\mathcal{H}^2$ $(z, z' \in \overline{\mathbb{R}})$. Then the formula of Eq. (9.5) is written equivalently

$$z = a + re^{i\theta},\ \theta \in (0, \pi), \quad a = (z + z')/2, \quad r = |z - z'|/2.$$

Example 9.2.4 In $\mathcal{H}^3$, the hyperbolic line with endpoints $\mathbf{z} = (1, 0, 1)$ and $\mathbf{z}' = (0, -1, 1)$ lies on a sphere with center $\mathbf{a} = (1/2, -1/2, 1)$ and radius $r = \sqrt{2}/2$. Its parameterisation is

$$\begin{aligned}\mathbf{x}(\theta) &= (1/2, -1/2, 1) + \cos\theta\, (1/2, 1/2, 0) + (\sqrt{2}/2) \sin\theta\, (0, 0, 1)\\ &= (1/2, -1/2, 1) + (\cos\theta/2, \cos\theta/2, \sin\theta/\sqrt{2}),\ \theta \in (0, \pi).\end{aligned}$$

Proposition 9.2.5 *From any point $\mathbf{x}_0 \in \mathcal{H}^{n+1}$ there pass infinitely many hyperbolic lines.*

Proof First, if $\mathbf{x}_0 = (x_1^0, \ldots, x_{n+1}^0) = (\mathbf{x}_n^0, x_{n+1}^0)$, then the line $x_i = x_i^0$, $i = 1, \ldots, n$, passes through $\mathbf{x}_0$. Then, if $\mathbf{a} \in \mathbb{R}^n$, we consider the sphere $S(\mathbf{a}, r)$, $r = \|\mathbf{x}_0 - \mathbf{a}\|$. If $\mathbf{x}_0$ is the north pole of the sphere (in this case, $\mathbf{a} = \mathbf{x}_n^0$), then we consider all meridians from $\mathbf{x}_0$. In the opposite case, we consider the unique great circle from the north pole $\mathbf{a} + r\mathbf{e}_{n+1}$ and $\mathbf{x}_0$. □

Example 9.2.6 In $\mathcal{H}^3$ we will find all hyperbolic lines passing through $\mathbf{x}_0 = (1, 1, 1)$. Its projection on the boundary is $(1, 1, 0)$, so we immediately take the vertical line $x_1 = x_2 = 1$. Now let $\mathbf{a} = (a_1, a_2, 0)$. If $\mathbf{a} = (1, 1, 0)$, we consider the sphere $S(\mathbf{a}, 1)$ and its meridians from $\mathbf{x}_0$. If $\mathbf{a} \neq (1, 1, 0)$, we consider $r = \|(1 - a_1, 1 - a_2, 1)\|$, the sphere $S(\mathbf{a}, r)$ and the unique great circle passing through $\mathbf{x}_0$ and $\mathbf{a} + r\mathbf{e}_3$.

Remark 9.2.7 When $n = 1$, if $z_0 \in \mathcal{H}^2$, the hyperbolic lines passing through z_0 are $x = \Re(z_0)$ and the semicircles

$$z(x_0; \theta) = a_0 + re^{i\theta}, \quad \theta \in (0, \pi),$$

where $a_0 \in \mathbb{R}$ is an arbitrary point in $\mathbb{R}$ and $r = |z_0 - a_0|$.

Proposition 9.2.8 *There exists a unique hyperbolic line passing through any two distinct points in* $\mathcal{H}^{n+1}$.

Proof Let $\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{H}^{n+1}$, $\mathbf{x}_1 \neq \mathbf{x}_2$. Let also π be a projection function on the plane $x_{n+1} = 0$ and set

$$\mathbf{z}_1 = \pi(\mathbf{x}_1), \quad \mathbf{z}_m = \pi(\mathbf{x}_m), \quad \mathbf{z}_2 = \pi(\mathbf{x}_2).$$

We distinguish two cases:

1. $\mathbf{x}_1 \cdot \mathbf{e}_i = \mathbf{x}_2 \cdot \mathbf{e}_i$, $i = 1, \ldots, n$. Then $\mathbf{x}_1, \mathbf{x}_2$ both belong to the same perpendicular line.
2. If the above is not the case, we consider the midpoint $\mathbf{x}_m$ of the line segment $[\mathbf{x}_1, \mathbf{x}_2]$ connecting $\mathbf{x}_1$ and $\mathbf{x}_2$:

$$\mathbf{x}_m = \frac{1}{2}(\mathbf{x}_1 + \mathbf{x}_2),$$

and the 2-plane W with parametric equation

$$\mathbf{x}(\lambda, \mu) = \mathbf{c}(\lambda) + \mu\, \mathbf{e}_{n+1}, \quad \lambda, \mu \in \mathbb{R},$$

where

$$\mathbf{c}(\lambda) = \mathbf{x}_m + \lambda(\mathbf{x}_2 - \mathbf{x}_1),$$

is the line joining $\mathbf{x}_1, \mathbf{x}_2$. Now, at the point $\mathbf{x}_m$ we draw the perpendicular line to $[\mathbf{x}_1,\ \mathbf{x}_2]$ which lies in W:

$$\mathbf{c}'(s) = \mathbf{x}_m + s\mathbf{v},$$

with $\mathbf{v} = \lambda(\mathbf{x}_2 - \mathbf{x}_1) + \mu\,\mathbf{e}_{n+1}$ such that

$$(\lambda(\mathbf{x}_2-\mathbf{x}_1)+\mu\,\mathbf{e}_{n+1})\cdot(\mathbf{x}_2-\mathbf{x}_1) = 0 \iff \lambda\,\|\mathbf{x}_1-\mathbf{x}_1\|^2+\mu\,(x^2_{n+1}-x^1_{n+1}) = 0,$$

where $x^i_{n+1} = \mathbf{x}_i \cdot \mathbf{e}_{n+1}$, $i = 1, 2$.

(2a) If $x^1_{n+1} = x^2_{n+1}$, then $\lambda = 0$ and we obtain $\mathbf{v} = \mathbf{e}_{n+1}$. Then the line $\mathbf{c}'$ intersects the boundary at $\mathbf{z}_m$ which is the centre of the semicircle. The radius is

$$r = \|\mathbf{z}_m - \mathbf{x}_i\|, \quad i = 1, 2.$$

Explicitly, a parametric equation of the semicircle is

$$\mathbf{x}(\theta) = \mathbf{z}_m + r\cos\theta\,\frac{\mathbf{z}_2 - \mathbf{z}_1}{\|\mathbf{z}_2 - \mathbf{z}_1\|} + r\sin\theta\,\mathbf{e}_{n+1}.$$

(2b) If $x^1_{n+1} \neq x^2_{n+1}$, we set

$$\lambda = x^1_{n+1} - x^2_{n+1}, \quad \mu = \|\mathbf{x}_1 - \mathbf{x}_2\|^2$$

and then

$$\mathbf{v} = (x^1_{n+1} - x^2_{n+1})\,(\mathbf{x}_2 - \mathbf{x}_1) + \|\mathbf{x}_1 - \mathbf{x}_2\|^2\,\mathbf{e}_{n+1}.$$

The line $\mathbf{c}'$ intersects the boundary at the point for which

$$\mathbf{c}'(s)\cdot\mathbf{e}_{n+1} = \mathbf{x}_m\cdot\mathbf{e}_{n+1} + s\|\mathbf{z}_1 - \mathbf{z}_2\|^2 = x^m_{n+1} + s\|\mathbf{z}_1 - \mathbf{z}_2\|^2 = 0.$$

That is, the centre of the sphere is

$$\mathbf{a} = \mathbf{c}'\left(-\frac{x^m_{n+1}}{\|\mathbf{z}_1 - \mathbf{z}_2\|^2}\right) = \mathbf{z}_m + \frac{(x^2_{n+1})^2 - (x^1_{n+1})^2}{2\|\mathbf{z}_1 - \mathbf{z}_2\|^2}\,(\mathbf{z}_2 - \mathbf{z}_1),$$

and its radius is $r = \|\mathbf{a} - \mathbf{x}_i\|$, $i = 1, 2$. The parametric equation of the hyperbolic line is

$$\|\mathbf{x}(\lambda, \mu) - \mathbf{a}\| = r.$$

□

Example 9.2.9 To find the hyperbolic line that passes through $z_1 = -1 + i$, $z_2 = 1 + i$ of $\mathcal{H}^2$, we observe first that the midpoint is $z_m = i$. The perpendicular from i to the line segment $[z_1, z_2]$ is the y axis that intersects the boundary at the origin 0.

The desired hyperbolic line is $L = (-\sqrt{2},\ \sqrt{2})$, that is,

$$z = \sqrt{2}e^{i\theta}, \quad \theta \in (0, \pi).$$

Example 9.2.10 We will find the hyperbolic line from the points $\mathbf{x}_1 = (0, 2, 1)$ and $\mathbf{x}_2 = (2, 0, 3)$ of $\mathcal{H}^3$. The midpoint of the line segment $[\mathbf{x}_1, \mathbf{x}_2]$ is $\mathbf{x}_m = (1, 1, 2)$ and its projection on the boundary is the point $\mathbf{z}_m = (1, 1, 0)$. We consider the line

$$\mathbf{c}(\lambda) = (1, 1, 2) + \lambda(2, -2, 2)$$

that joins $\mathbf{x}_1$ and $\mathbf{x}_2$, as well as the 2-plane

$$W:\ \mathbf{x}(\lambda, \mu) = \mathbf{c}(\lambda) + \mu(0, 0, 1), \quad \lambda, \mu \in \mathbb{R}.$$

The perpendicular vector $\mathbf{v} = (2\lambda, -2\lambda, 2\lambda + \mu)$ of $\mathbf{c}(\lambda)$ at $\mathbf{x}_m$ satisfies

$$(2\lambda, -2\lambda, 2\lambda + \mu) \cdot (2, -2, 2) = 0 \iff \mu = -6\lambda.$$

Thus we obtain $\mathbf{v} = (1, -1, -2)$ and $\mathbf{c}'(\lambda) = (1, 1, 2) + s(1, -1, -2)$ intersects the boundary at

$$\mathbf{a} = \mathbf{c}'(1) = (2, 0, 0).$$

The radius of the circle is $r = \|\mathbf{a} = \mathbf{x}_i\| = 3$ and therefore the equation of the hyperbolic line is obtained from the equations of the sphere

$$(x_1 - 2)^2 + x_2^2 + x_3^2 = 9,$$

and of the plane W:

$$x_1 + x_2 = 2,$$

when $x_3 > 0$. Observe that the endpoints are found by setting $x_3 = 0$: we obtain

$$L = \left(\frac{1}{2}(4 - 3\sqrt{2}, -3\sqrt{2}, 0),\ \frac{1}{2}(4 + 3\sqrt{2}, 3\sqrt{2}, 0)\right).$$

Proposition 9.2.11 *The group* $\mathrm{GM}(\mathcal{H}^{n+1})$ *acts transitively on the set of hyperbolic lines: for any two hyperbolic lines* $L = (\mathbf{z}_1, \mathbf{z}_2)$, $L' = (\mathbf{w}_1, \mathbf{w}_2)$ *and points* $\mathbf{z}_0 \in L$ *and* $\mathbf{w}_0 \in L'$, *there exists an element* $g \in \mathrm{GM}(\mathcal{H}^{n+1})$ *such that* $g(L) = L'$ *with* $g(\mathbf{z}_i) = \mathbf{w}_i$, $i = 0, 1, 2$. *When* $n = 1$, $g \in \mathrm{M}(\mathcal{H}^2)$ *is unique.*

Proof We claim that if $\mathbf{x}_1, \mathbf{x}_2 \in \overline{\mathbb{R}^n}$ and $\mathbf{x}_0 \in \mathcal{H}^{n+1}$, then there exists an $h \in \mathrm{GM}(\mathcal{H}^{n+1})$ such that

$$h(\mathbf{x}_1) = \mathbf{0}, \quad h(\mathbf{x}_2) = \infty, \quad h(\mathbf{x}_0) = \mathbf{e}_{n+1}.$$

Indeed, by inversion in a sphere with centre $\mathbf{x}_2$ we map $\mathbf{x}_2$ to ∞ and $\mathbf{x}_0, \mathbf{x}_1$ to points $\mathbf{x}_0' \in \mathcal{H}^{n+1}$ and $\mathbf{x}_1' \in \mathbb{R}^n$, respectively. With the translation $T_{-\mathbf{x}_1'}$ we now obtain the points $\mathbf{x}_0''$, $\mathbf{0}$ and ∞. Applying the dilation $D_{1/\|\mathbf{x}_0''\|}$ and then rotating on the unit sphere of $\mathbb{R}^{n+1}$, we have the desired result. From this, we obtain the first claim of the proposition since the elements of $\mathrm{GM}(\mathcal{H}^{n+1})$ map generalised spheres to generalised spheres. For the second claim, the transformation g is given by the equality of the complex cross-ratios

$$[z_0, z_1; z_2, z] = [w_0, w_1; w_2, g(z)].$$

This equation defines an element of $\mathrm{M}(\mathcal{H}^2)$ and is unique because if there were a g' with the same property, then the composition $g^{-1} \circ g'$ would stabilise three points, so it would have to be the identity transformation. □

Example 9.2.12 Let the line $L = (2, 4)$ and the point $3+i \in L$. The transformation g given by the equality of complex cross-ratios

$$[2, 4; 3 + i, z] = [\infty, 0; i, g(z)] \iff g(z) = \frac{z - 4}{z - 2},$$

satisfies the following: $g(2) = \infty$, $g(4) = 0$, $g(3 + i) = i$ and $g(L)$ is $(0, \infty)$. Notice that changing the orientation of the imaginary axis will give us another transformation:

$$[2, 4; 3 + i, z] = [0, \infty; i, h(z)] \iff h(z) = \frac{-z + 2}{z - 4}.$$

We have the following direct corollary of Proposition 9.2.11 which describes the transitive action of $\mathrm{GM}(\mathcal{H}^{n+1})$ on $\overline{\mathbb{R}^n}$ and $\mathcal{H}^{n+1}$:

Corollary 9.2.13 *The group* $\mathrm{GM}(\mathcal{H}^{n+1})$ *acts doubly transitively on* $\overline{\mathbb{R}^n}$*: for any points* $\mathbf{z}_1, \mathbf{z}_2$ *and* $\mathbf{w}_1, \mathbf{w}_2$ *there exists a* $g \in \mathrm{M}(\mathcal{H}^2)$ *such that* $g(\mathbf{z}_1) = \mathbf{w}_1$ *and* $g(\mathbf{z}_2) = \mathbf{w}_2$*. On the other hand,* $\mathrm{GM}(\mathcal{H}^{n+1})$ *acts transitively on* $\mathcal{H}^{n+1}$*: for any points* $\mathbf{z}_0, \mathbf{w}_0$ *there exists a* $g \in \mathrm{GM}(\mathcal{H}^{n+1})$ *such that* $g(\mathbf{z}_0) = \mathbf{w}_0$*.*

Proof For the first claim, it suffices to consider the hyperbolic lines $(\mathbf{z}_1, \mathbf{z}_2)$ and $(\mathbf{w}_1, \mathbf{w}_2)$ and to apply Proposition 9.2.11 by arbitrarily choosing points on these lines. For the second claim, let $L = (\mathbf{z}_0, \mathbf{z}_2)$ be a hyperbolic line containing $\mathbf{z}_0$ and let also $L' = (\mathbf{w}_1, \mathbf{w}_2)$ be a hyperbolic line containing $\mathbf{w}_0$. Again, the result follows immediately from Proposition 9.2.11. □

In the case $n = 1$, Corollary 9.2.13 is written as follows:

Corollary 9.2.14 *The group* $\mathrm{M}(\mathcal{H}^2)$ *acts doubly transitively on* $\overline{\mathbb{R}}$*: for any points* z_1, z_2 *and* w_1, w_2 *there exists a* $g \in \mathrm{M}(\mathcal{H}^2)$ *such that* $g(z_1) = w_1$ *and* $g(z_2) = w_2$*. Also,* $\mathrm{M}(\mathcal{H}^2)$ *acts transitively on* $\mathcal{H}^2$*: for any points* z_0, w_0 *there exists a* $g \in \mathrm{M}(\mathcal{H}^2)$ *such that* $g(z_0) = w_0$*.*

Proposition 9.2.15 (Incidence) *Two hyperbolic lines have at most one common point in* $\mathcal{H}^{n+1}$*.*

Proof A non-schematic proof is the following: let $L = (\mathbf{z}_1, \mathbf{z}_2)$, $L' = (\mathbf{w}_1, \mathbf{w}_2)$ be hyperbolic lines with two common points $\mathbf{z}_0, \mathbf{w}_0$. Through a transformation f we map L to $(\mathbf{0}, \infty)$ so that

$$f(\mathbf{z}_1) = \mathbf{0}, \quad f(\mathbf{z}_2) = \infty, \quad f(\mathbf{z}_0) = \mathbf{e}_{n+1}, \quad f(\mathbf{w}_0) = a\,\mathbf{e}_{n+1},$$

for some $a > 0$. Similarly, we map L to $(\mathbf{0}, \infty)$ again through a γ so that

$$g(\mathbf{z}_1) = \mathbf{0}, \quad g(\mathbf{z}_2) = \infty, \quad g(\mathbf{w}_0) = \mathbf{e}_{n+1}, \quad g(\mathbf{z}_0) = a'\,\mathbf{e}_{n+1},$$

for some $a' > 0$. Therefore, $h = g \circ f^{-1}$ satisfies the relations

$$h(\mathbf{0}) = \mathbf{0}, \quad h(\infty) = \infty, \quad h(\mathbf{e}_{n+1}) = a'\,\mathbf{e}_{n+1}.$$

On the other hand, $h^{-1} = f \circ g^{-1}$ satisfies the relations

$$h^{-1}(\mathbf{0}) = \mathbf{0}, \quad h^{-1}(\infty) = \infty, \quad h^{-1}(\mathbf{e}_{n+1}) = a\,\mathbf{e}_{n+1}.$$

It follows that $a = a' = 1$ and therefore $\mathbf{z}_0 = \mathbf{w}_0$. □

Example 9.2.16 Let $L_1 = (\mathbf{0}, \infty)$ and $L_2 = (-\mathbf{e}_1, \mathbf{b})$, $\mathbf{b} = (b_1, \ldots, b_n, 0)$, $b_1 > 0$ be hyperbolic lines in $\mathcal{H}^{n+1}$. L^2 lies on the sphere

$$\|\mathbf{x} - \mathbf{a}\| = r, \quad \mathbf{a} = \frac{\mathbf{b} - \mathbf{e}_1}{2}, \; r = \frac{\|\mathbf{b} + \mathbf{e}_1\|}{2}.$$

Any intersection points are obtained by setting $\mathbf{x} = t\,\mathbf{e}_{n+1}$, $t > 0$, in the above equation: after the calculations, we obtain that $t = \sqrt{b_1}$. The point $\sqrt{b_1}\,\mathbf{e}_{n+1}$ is the common point of $(\mathbf{0}, \infty)$ and the sphere to which $(-\mathbf{e}_1, \mathbf{b})$ belongs. When $n = 1$, this point is indeed the intersection point of the two hyperbolic lines. In higher dimensions, the points

$$-\mathbf{e}_1, \quad \sqrt{b_1}\,\mathbf{e}_{n+1}, \quad \mathbf{a} + r\,\mathbf{e}_{n+1}, \quad \mathbf{b},$$

must also belong to the same 2-plane (the $\mathbf{a} + r\,\mathbf{e}_{n+1}$ is the north pole of the sphere). If $\mathbf{a} + r\,\mathbf{e}_{n+1} = \sqrt{b_1}\,\mathbf{e}_{n+1}$, then $\mathbf{a} = \mathbf{0}$, $\mathbf{b} = \mathbf{e}_1$, $r = 1$ and the common point is $\mathbf{e}_{n+1}$. If this is not the case, then we consider the 2-plane

$$W : \; \mathbf{x}(\lambda, \mu) = \mathbf{a} + \lambda(\sqrt{b_1}\,\mathbf{e}_{n+1} - \mathbf{a}) + \mu(r\,\mathbf{e}_{n+1}), \quad \lambda, \mu \in \mathbb{R},$$

which contains $\mathbf{a} + r\ \mathbf{e}_{n+1},\ \sqrt{b_1}\,\mathbf{e}_{n+1}$. W must then contain $-\mathbf{e}_1$: there exist λ, μ such that $\mathbf{x}(\lambda, \mu) = -\mathbf{e}_1$ if and only if $\mathbf{a} = a_1\mathbf{e}_1$, $a_1 \neq 0$, i.e. $\mathbf{b} = b_1\,\mathbf{e}_1$, $b_1 \neq 1$. Then

$$\mathbf{x}\left(\frac{b_1+1}{b_1-1},\ \frac{2\sqrt{b_1}}{1-b_1}\right) = -\mathbf{e}_1.$$

It must also contain $\mathbf{b}$. Indeed,

$$\mathbf{x}\left(\frac{b_1+1}{1-b_1},\ \frac{2\sqrt{b_1}}{1-b_1}\right) = \mathbf{b}.$$

Remark 9.2.17 We can always normalise two intersecting hyperbolic lines L_1 and L_2 to be $L_1 = (\mathbf{0}, \infty)$ and $L_2 = (-be_1,\ b\ \mathbf{e}_1)$, $b > 0$. Indeed, let L_1 and L_2 be hyperbolic lines intersecting at $\mathbf{z}_0 \in \mathcal{H}^{n+1}$. We distinguish the following cases:

1. *L_i are semicircles.* Applying the transformation g that maps L_1 to $(\mathbf{0}, \infty)$ and $\mathbf{z}_0$ to $\mathbf{e}_{n+1}$, and then possibly rotating around the vertical x_{n+1} axis, we have that L_2 will be mapped through this transformation to a hyperbolic line of the form $(-a_1\mathbf{e}_1,\ a_2\mathbf{e}_1)$ with $a_i > 0\ i = 1, 2$. Applying the dilation D_{1/a_1} we get $(0, \infty)$, $(-\mathbf{e}_1,\ b\ \mathbf{e}_1)$, $b = (1/a_1)a_2$, which intersect at $\sqrt{b}\ \mathbf{e}_{n+1}$.
2. *One of the lines L_i, $i = 1, 2$, is a vertical half-line.* Suppose that L_1 is $(\mathbf{x}_0,\ \infty)$, $\mathbf{x}_0 \in \mathbb{R}^n$. Applying the translation $T_{-\mathbf{x}_0}$ and a dilation if necessary, we arrive in the previous picture.

9.2.2 Parallelism

Two hyperbolic lines of $\mathcal{H}^{n+1}$ lie in the same 2-plane if their endpoints all lie on a straight line of $\partial\mathcal{H}^{n+1}$. Two such hyperbolic lines are called *parallel* when they have no common point in $\mathcal{H}^{n+1}$. However, they may have some common point on $\overline{\mathbb{R}^n}$. Par example, in $\mathcal{H}^2$ the lines $(-1, 1)$ and $(1, 2)$ have a common end 1. Furthermore, all lines of the form $(\mathbf{x}, \infty)$, $\mathbf{x} \in \mathbb{R}^n$, have a common point ∞. In this case, the parallel lines are called *asymptotically parallel*, or *ultraparallel*. Finally, two lines that do not lie in the same plane and do not have a common point in $\mathcal{H}^{n+1}$ are called *incompatible*. In the case where two incompatible lines share a common endpoint, then the lines are called *asymptotically incompatible* (Fig. 9.5).

The next proposition is the *Axiom of Parallels of Hyperbolic Geometry*.

Proposition 9.2.18 *Let L be a hyperbolic line and let $\mathbf{z}_0$ a point, $z_0 \notin L$. Then infinitely many hyperbolic lines parallel to L pass from z_0.*

Proof Without loss we may assume that $n = 1$. Suppose first that $L = (0, \infty)$ and $z_0 \in \mathcal{H}^2$ with $\Re(z_0) = x_0 \neq 0$. An (asymptotically) parallel to L from z_0 is obviously $x = x_0$. Assume without that $x_0 > 0$. We draw the line segment joining 0

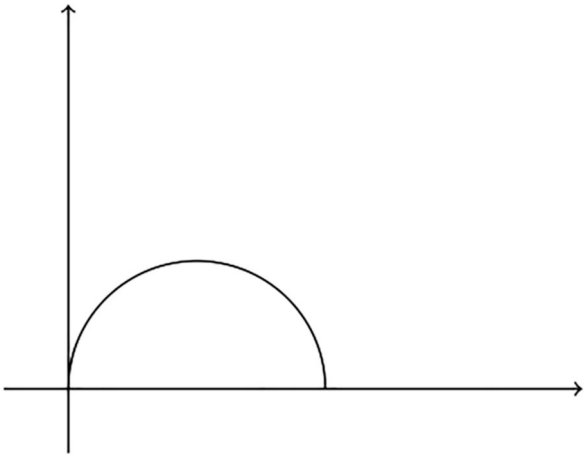


Fig. 9.5 Ultraparallel hyperbolic lines in $\mathcal{H}^2$

and z_0 and from z_0 we draw the perpendicular to this segment. This perpendicular intersects the boundary at some $2a$, $a > 0$. Then, the semicircle with centre a and equal radius is the hyperbolic line $(0, 2a)$ that passes through z_0 and is parallel to the hyperbolic line $(0, \infty)$. From there on, every semicircle with centre $a' > a$ and radius $|a' - z_0|$ is a hyperbolic line that passes through z_0, asymptotically parallel to L.

For the general case when z_0 and $L = (z_1, z_2)$ are arbitrary, we map z_0, z_1, z_2 to $z_0', 0, \infty$ through a g of $\mathrm{M}(\mathcal{H}^2)$, where z_0' is a point with $\Re(z_0') \neq 0$. □

9.2.3 Hyperbolic Subspaces

Definition 9.2.19 Let $1 < m \leq n$, $m, n \in \mathbb{N}$. The mapping $\iota_m : \mathcal{H}^{m+1} \to \mathcal{H}^{n+1}$ given by

$$(x_1, \ldots, x_{m+1}) \mapsto (0, \ldots, 0, x_1, \ldots, x_{m+1}), \tag{9.6}$$

is called the *canonical embedding* of $\mathcal{H}^{m+1}$ in $\mathcal{H}^{n+1}$.

From this definition follows the definition of hyperbolic $(m + 1)$-spaces:

Definition 9.2.20 A *hyperbolic subspace H of $\mathcal{H}^{n+1}$ of dimension $m + 1$* is a set of the form

$$H = g(\iota_m(\mathcal{H}^{m+1})), \quad g \in \mathrm{GM}(\mathcal{H}^{n+1}),$$

where ι_m is the canonical immersion (9.6).

We will agree that

Hyperbolic subspaces of dimension 1 are hyperbolic lines.

From Definition 9.2.20 we obtain the following:

Corollary 9.2.21 *The group* $\mathrm{GM}(\mathcal{H}^{n+1})$ *acts transitively on hyperbolic subspaces of the same dimension.*

9.3 Hyperbolic Metric

9.3.1 Hyperbolic Curve Length

Let $\boldsymbol{\gamma} : I = [a, b] \to \mathcal{H}^{n+1}$, $\boldsymbol{\gamma}(t) = \mathbf{x}(t) = (x_1(t), \ldots, x_{n+1}(t))$, be a continuously differentiable curve. The *hyperbolic norm* of the derivative $\dot{\boldsymbol{\gamma}}$ is defined by the relation

$$\|\dot{\boldsymbol{\gamma}}(t)\|_h = \frac{\|\dot{\mathbf{x}}(t)\|}{x_{n+1}(t)}.$$

We call the number

$$\ell_h(\boldsymbol{\gamma}) = \int_a^b \|\dot{\boldsymbol{\gamma}}(t)\|_h \, dt,$$

the *hyperbolic length* of $\boldsymbol{\gamma}$.

Proposition 9.3.1 *The hyperbolic length of a continuously differentiable curve* $\boldsymbol{\gamma}$ *of* $\mathcal{H}^{n+1}$*:*

- *does not depend on the parameterisation of* $\boldsymbol{\gamma}$ *and*
- *remains invariant under the action of* $\mathrm{GM}(\mathcal{H}^{n+1})$.

Exercise Prove Proposition 9.3.1.

Example 9.3.2 Let $\boldsymbol{\gamma}$ be the curve given by $\boldsymbol{\gamma}(t) = t\mathbf{e}_1 + b\mathbf{e}_{n+1}, b > 0, t \in (-a, a)$, $a > 0$. Then $\dot{\boldsymbol{\gamma}}(t) = \mathbf{e}_1$, $|\dot{\boldsymbol{\gamma}}(t)|_h = 1/b$ and $\ell_h(\boldsymbol{\gamma}) = 2a/b$.

Example 9.3.3 (Lengths of Hyperbolic Lines)

($\boldsymbol{\alpha}$) Consider the hyperbolic line segment with endpoints $\mathbf{z} = \mathbf{e}_{n+1}$ and $\mathbf{w} = b\,\mathbf{e}_{n+1}$, $b \neq 1$, on the hyperbolic line $(\mathbf{0}, \infty)$. A parameterisation of this segment is given by:

$$\boldsymbol{\gamma}(t) = (1-t)\mathbf{e}_{n+1} + tb\,\mathbf{e}_{n+1} = (1 + t(b-1))\,\mathbf{e}_{n+1}, \quad t \in [0, 1].$$

The derivative is thus $\dot{\boldsymbol{\gamma}}(t) = (b-1)\,\mathbf{e}_{n+1}$ and therefore $\|\dot{\boldsymbol{\gamma}}(t)\|_h = |(b-1)|/(1+t(b-1))$. Therefore, if $b > 1$,

$$\ell_h(\boldsymbol{\gamma}) = \int_0^1 \frac{(b-1)}{1+t(b-1)}\,dt = [\ln(1+t(s-1))]_{t=0}^1 = \ln b,$$

while for $b < 1$,

$$\ell_h(\boldsymbol{\gamma}) = \int_0^1 \frac{(1-b)}{1+t(b-1)}\,dt = -[\ln(1+t(b-1))]_{t=0}^1 = -\ln b.$$

We conclude that $\ell_h(\boldsymbol{\gamma}) = |\ln b|$. We set $\rho = \ell_h(\boldsymbol{\gamma})$. Then,

$$\begin{aligned}\cosh^2(\rho/2) &= \frac{(e^{\rho/2}+e^{-\rho/2})^2}{4} = \frac{e^{\rho}+e^{-\rho}+2}{4}\\ &= \frac{b+(1/b)+2}{4} = \frac{(b+1)^2}{4b}\\ &= \frac{\|\mathbf{z}-r(\mathbf{w})\|^2}{4\,z_{n+1}w_{n+1}},\end{aligned}$$

where r is the reflection on the hyperplane $x_{n+1} = 0$:

$$r(x_1,\ldots,x_n,x_{n+1}) = (x_1,\ldots,x_n,-x_{n+1}).$$

We obtain in this way the formula for the length ρ of the hyperbolic line segment with endpoints $\mathbf{z}$, $\mathbf{w}$ lying at $(\mathbf{0},\infty)$:

$$\cosh(\rho/2) = \frac{\|\mathbf{z}-r(\mathbf{w})\|}{2\sqrt{z_{n+1}}\sqrt{w_{n+1}}}. \tag{9.7}$$

($\boldsymbol{\beta}$) Consider now an arbitrary hyperbolic line segment $[\mathbf{z},\mathbf{w}]$ lying in a hyperbolic line $L = (\mathbf{z}_1,\mathbf{z}_2)$. We map L to $(\mathbf{0},\infty)$ via a $g \in \mathrm{GM}(\mathcal{H}^{n+1})$ in such a way that $\mathbf{z}$ is mapped to $\mathbf{e}_{n+1}$; in this way $\mathbf{w}$ is mapped to some $b\,\mathbf{e}_{n+1}$, $b \neq 1$. Since the hyperbolic length remains invariant under the action of $\mathrm{GM}(\mathcal{H}^{n+1})$, we have that the hyperbolic length ρ of $[\mathbf{z},\mathbf{w}]$ will be equal to the hyperbolic length ρ' of $[\mathbf{e}_{n+1},\ b\,\mathbf{e}_{n+1}]$.

Exercise Prove that the right-hand side of Eq. (9.7) also remains invariant under the action of $\mathrm{GM}(\mathcal{H}^{n+1})$.

We thus arrive again at the formula of Eq. (9.7). This formula is one of the expressions for the hyperbolic distance of any two points, as we will see in the next section.

Corollary 9.3.4 *Hyperbolic lines have infinite hyperbolic length.*

Exercise Prove the Corollary 9.3.4.

9.3.2 Hyperbolic Distance

The *hyperbolic distance* $d_h(\mathbf{z}, \mathbf{w})$ of two points $\mathbf{z}, \mathbf{w} \in \mathcal{H}^{n+1}$ is defined as follows: consider the set $\mathcal{C}(\mathbf{z}, \mathbf{w})$ comprising continuously differentiable curves $\boldsymbol{\gamma}$ of $\mathcal{H}^{n+1}$ with endpoints $\mathbf{z}$ and $\mathbf{w}$. Then,

$$d_h(\mathbf{z}, \mathbf{w}) = \inf_{\boldsymbol{\gamma} \in \mathcal{C}(\mathbf{z}, \mathbf{w})} \{\ell_h(\boldsymbol{\gamma})\}.$$

Proposition 9.3.5 *d_h as defined above is metric.*

Exercise Prove Proposition 9.3.5.

We will now show that the metric space $(\mathcal{H}^{n+1}, d_h)$ is a geodesic metric space.

Proposition 9.3.6 *The geodesics of $(\mathcal{H}^{n+1}, d_h)$ are the hyperbolic lines.*

Proof Consider two arbitrary and distinct points $\mathbf{z}, \mathbf{w} \in \mathcal{H}^{n+1}$. By normalisation, we can assume that $\mathbf{z} = \mathbf{e}_{n+1}$, $\mathbf{w} = b\,\mathbf{e}_{n+1}$, $b > 0$. Let $\boldsymbol{\gamma} : [0, 1] \to \mathcal{H}^{n+1}$ be any continuously differentiable curve such that

$$\boldsymbol{\gamma}(t) = \mathbf{x}(t) = (x_1(t), \ldots, x_{n+1}(t)), \quad \boldsymbol{\gamma}(0) = \mathbf{e}_{n+1},\ \boldsymbol{\gamma}(1) = b\,\mathbf{e}_{n+1},$$

Then,

$$\begin{aligned}
\ell_h(\boldsymbol{\gamma}) &= \int_0^1 \frac{\|\dot{\mathbf{x}}(t)\|}{x_{n+1}(t)}\, dt \\
&\geq \int_0^1 \frac{|\dot{x}_{n+1}(t)|}{x_{n+1}(t)}\, dt \\
&\geq \left| \int_0^1 \frac{\dot{x}_{n+1}(t)}{x_{n+1}(t)}\, dt \right| \\
&= \left| [\ln(x_{n+1}(t))]_{t=0}^1 \right| = |\ln b|.
\end{aligned}$$

The latter is exactly the length of the hyperbolic straight line segment connecting $\mathbf{e}_{n+1}, b\,\mathbf{e}_{n+1}$. Taking inf over all $\boldsymbol{\gamma} \in \mathcal{C}(\mathbf{e}_{n+1},\ b\mathbf{e}_{n+1})$ yields the desired result. □

The next proposition follows directly from formula (9.7).

Proposition 9.3.7 *Let r be the reflection in the plane $x_{n+1} = 0$. The hyperbolic distance $\rho = d_h(\mathbf{z}, \mathbf{w})$ of two points $\mathbf{z}, \mathbf{w}$ of $\mathcal{H}^{n+1}$ is given by the following*

equivalent formulae:

$$\cosh(\rho/2) = \frac{\|\mathbf{z} - r(\mathbf{w})\|}{2(z_{n+1})^{1/2}(w_{n+1})^{1/2}}, \tag{9.8}$$

$$\sinh(\rho/2) = \frac{\|\mathbf{z} - \mathbf{w}\|}{2\,(z_{n+1})^{1/2}(w_{n+1})^{1/2}}, \tag{9.9}$$

$$\cosh(\rho) = 1 + \frac{\|\mathbf{z} - \mathbf{w}|^2}{2\,z_{n+1}w_{n+1}}, \tag{9.10}$$

$$\rho = \frac{\|\mathbf{z} - r(\mathbf{w})\| + \|\mathbf{z} - \mathbf{w}\|}{\|\mathbf{z} - r(\mathbf{w})| - \|\mathbf{z} - \mathbf{w}\|}. \tag{9.11}$$

Exercise Prove Proposition 9.3.7.

The last proposition of this section essentially defines the concept of *between*:

Proposition 9.3.8 *Let* $\mathbf{z}, \mathbf{w} \in \mathcal{H}^{n+1}$ *be distinct points. Then*

$$d_h(\mathbf{z}, \mathbf{w}) = d_h(\mathbf{z}, \boldsymbol{\zeta}) + d_h(\boldsymbol{\zeta}, \mathbf{w}),$$

if and only if $\boldsymbol{\zeta}$ *belongs to the geodesic segment with endpoints* $\mathbf{z}, \mathbf{w}$.

Proof We may assume that $\mathbf{z}, \mathbf{w}$ are $\mathbf{e}_{n+1}, b\,\mathbf{e}_{n+1}$ for some $b > 0$. If $\boldsymbol{\zeta}$ belongs to the geodesic segment with endpoints $\mathbf{e}_{n+1}, b\,\mathbf{e}_{n+1}$, then $\boldsymbol{\zeta} = b'\,\mathbf{e}_{n+1}$, $1 \leq b' \leq b$ and the equality of the proposition holds. Conversely, again for $\mathbf{z}, \mathbf{w}$ as before, let us assume that equality is attained in formula of the proposition and we further assume that $\boldsymbol{\zeta}$ does not lie in $(\mathbf{0}, \infty)$. Then, however, the hyperbolic length of the curve comprising the hyperbolic line segments a) $s_1 = [\mathbf{e}_{n+1},\ \boldsymbol{\zeta}]$ and b) $s_2 = [\boldsymbol{\zeta},\ b\,\mathbf{e}_{n+1}]$, is strictly bigger than the length of the segment $[\mathbf{e}_{n+1},\ b\,\mathbf{e}_{n+1}]$. We end up with a contradiction. □

9.3.3 Isometries

We have already seen that $\mathrm{GM}(\mathcal{H}^{n+1})$ is a subgroup of the isometry group of $(\mathcal{H}^{n+1}, d_h)$. In this section we will show through a formula involving metric cross-ratios that $\mathrm{GM}(\mathcal{H}^{n+1})$ is *exactly* the isometry group (Fig. 9.6). We recall that the metric cross-ratio is invariant only under Möbius transformations (Theorem 8.5.7).

Let $\mathbf{z}, \mathbf{w} \in \mathcal{H}^{n+1}$ be any points lying in a hyperbolic line L. We assume that L has endpoints $\mathbf{z}^*, \mathbf{w}^*$ and that the four points appear in the order $\mathbf{z}^*$, $\mathbf{z}$, $\mathbf{w}$, $\mathbf{w}^*$, on L.

Proposition 9.3.9 *The hyperbolic distance* $d_h(\mathbf{z}, \mathbf{w})$ *is given by the formula:*

$$d_h(\mathbf{z}, \mathbf{w}) = \ln\left([\mathbf{w}^*, \mathbf{z}^*; \mathbf{z}, \mathbf{w}]\right). \tag{9.12}$$

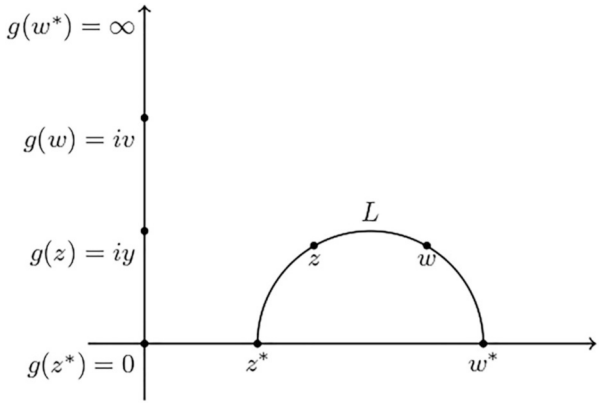


Fig. 9.6 Proposition 9.3.9 when $n = 1$

Proof We apply the element g of $\mathrm{GM}(\mathcal{H}^{n+1})$ that maps $\mathbf{w}^*, \mathbf{z}^*, \mathbf{z}, \mathbf{w}$, to $\infty, \mathbf{0}, \mathbf{e}_{n+1}, b\,\mathbf{e}_{n+1}i$ for some $s > 1$. Then

$$\begin{aligned} d_h(\mathbf{z}, \mathbf{w}) &= d_h(g(\mathbf{z}), g(\mathbf{w})) = \ln b \\ &= \ln([\infty, \mathbf{0}; \mathbf{e}_{n+1}, b\,\mathbf{e}_{n+1}i]) \\ &= \ln([\mathbf{w}^*, \mathbf{z}^*; \mathbf{z}, \mathbf{w}]). \end{aligned}$$

□

Remark 9.3.10 When $n = 1$, we can replace the metric cross-ratio with the complex cross-ratio.

Corollary 9.3.11 *The isometry group of* $(\mathcal{H}^{n+1}, d_h)$ *is* $\mathrm{GM}(\mathcal{H}^{n+1})$.

Corollary 9.3.12 *Let H be a hyperbolic subspace of dimension m: $H = (g \circ \iota_m)(\mathcal{H}^{m+1})$, $g \in \mathrm{GM}(\mathcal{H}^{n+1})$. Then the embedding of $\mathcal{H}^{m+1}$ into $\mathcal{H}^{n+1}$ via $g \circ \iota_m$ is isometric.*

9.3.4 Hyperbolic Balls and Spheres

The *open hyperbolic ball* of centre $\mathbf{z}_0 \in \mathcal{H}^{n+1}$ and radius $r > 0$ is the set

$$B_h(\mathbf{z}_0, r) = \{\mathbf{z} \in \mathcal{H}^{n+1} \mid d_h(\mathbf{z}, \mathbf{z}_0) < r\}.$$

The *closed hyperbolic ball* of center $\mathbf{z}_0 \in \mathcal{H}^{n+1}$ and radius $r > 0$ is the set

$$\overline{B_h(\mathbf{z}_0, r)} = \{\mathbf{z} \in \mathcal{H}^{n+1} \mid d_h(\mathbf{z}, \mathbf{z}_0) \leq r\},$$

and the *hyperbolic sphere* of center $\mathbf{z}_0 \in \mathcal{H}^{n+1}$ and radius $r > 0$ is the set

$$S_h(\mathbf{z}_0, r) = \{\mathbf{z} \in \mathcal{H}^{n+1} \mid d_h(\mathbf{z}, \mathbf{z}_0) = r\}.$$

We will show that every hyperbolic sphere is a Euclidean sphere whose centre and radius we will determine. Consider the sphere be $S_h(\mathbf{z}_0, r)$; then, from $d_h(\mathbf{z}, \mathbf{z}_0) = r$ and the formulae (9.8), (9.9) we obtain

$$k = \tanh(r/2) = \frac{\|\mathbf{z} - \mathbf{z}_0\|}{\|\mathbf{z} - r(\mathbf{z}_0)\|}.$$

This relation is an equation of an Apollonian sphere (see Exercise 15, Sect. 8.7 for the one-dimensional case) and it remains to determine its centre and radius. By squaring and then eliminating the denominator, we get

$$\|\mathbf{z}\|^2 - 2\left(\mathbf{z} \cdot \frac{r(\mathbf{z}_0) - k^2\mathbf{z}_0}{1 - k^2}\right) + |\mathbf{z}_0|^2 = |\mathbf{z}|^2 - 2\left(\mathbf{z} \cdot (\mathbf{z}_n^0 + z_{n+1}^0 \cosh r\, \mathbf{e}_{n+1})\right)$$
$$+|\mathbf{z}_0|^2 = 0.$$

Completing the square, we get

$$\|\mathbf{z} - (\mathbf{z}_n^0 + z_{n+1}^0 \cosh r\, \mathbf{e}_{n+1})\|^2 = (z_{n+1}^0)^2 \sinh^2 r.$$

We have proved:

Proposition 9.3.13 *The hyperbolic sphere* $S_h(\mathbf{z}_0, r)$, $\mathbf{z}_0 = \mathbf{z}_n^0 + z_{n+1}^0\, \mathbf{e}_{n+1}$, $r > 0$ *is the Euclidean sphere with centre* $\mathbf{z}_n^0 + z_{n+1}^0 \cosh r\, \mathbf{e}_{n+1}$ *and radius* $\sinh r\, z_{n+1}^0$.

9.3.5 Hyperbolic Volume

Let $\Omega \subset \mathcal{H}^{n+1}$ be a region. The *hyperbolic volume* $\mathrm{Vol_h}(\Omega)$ of Ω is defined by the $(n+1)$-integral

$$\mathrm{Vol_h}(\Omega) = \int_\Omega \frac{dx_1 \ldots dx_{n+1}}{x_{n+1}^2}. \tag{9.13}$$

Exercise Prove that the hyperbolic volume remains invariant under hyperbolic isometries.

Example 9.3.14 To calculate the hyperbolic area of the region $\Omega \subset \mathcal{H}^2$ bounded by the curves $\Re(z) = 0$, $|z| = 1$, $|z| = b$ and $\arg(z) = a$, where $0 < a < \pi/2$,

$b > 1$, we consider polar coordinates $z = re^{i\theta}$ so that

$$1 \le r \le b, \quad a \le \theta \le \pi/2.$$

Using the Change of Variables Theorem and formula (9.13) we calculate

$$\begin{aligned}\mathrm{Area_h}(\Omega) &= \int_1^b \int_a^{\pi/2} \frac{r\,dr d\theta}{r^2 \sin^2\theta} \\ &= \left(\int_1^b \frac{dr}{r}\right)\left(\int_a^{\pi/2} \frac{d\theta}{\sin^2\theta}\right) \\ &= \log b\, \cot a.\end{aligned}$$

9.4 Hyperbolic Trigonometry

9.4.1 Angles and Triangles

Let L_i, $i = 1, 2$ be hyperbolic lines in $\mathcal{H}^{n+1}$ that intersect at $\mathbf{z}_0$. If $\boldsymbol{\gamma}_i : [a, b] \to \mathcal{H}^{n+1}$ are continuously differentiable parametrisations of L_i, $\boldsymbol{\gamma}_i = \boldsymbol{\gamma}_i(t)$, $t \in [a, b]$, $i = 1, 2$, then there exists a $t_0 \in (a, b)$ such that $\boldsymbol{\gamma}_1(t_0) = \boldsymbol{\gamma}_2(t_0) = \mathbf{z}_0 \in \mathcal{H}^{n+1}$. The *hyperbolic scalar product* of the tangent vectors $\dot{\boldsymbol{\gamma}}_1(t_0)$ and $\dot{\boldsymbol{\gamma}}_2(t_0)$ at $\mathbf{z}_0$ is defined by the relation

$$(\dot{\boldsymbol{\gamma}}_1(t_0) \cdot \dot{\boldsymbol{\gamma}}_2(t_0))_h = \frac{\dot{\boldsymbol{\gamma}}_1(t_0) \cdot \dot{\boldsymbol{\gamma}}_2(t_0)}{(z^0_{n+1})^2},$$

where in the numerator of the right-hand fraction is the Euclidean scalar product. We observe that

$$|(\dot{\boldsymbol{\gamma}}_1(t_0) \cdot \dot{\boldsymbol{\gamma}}_2(t_0))_h| = \frac{|\dot{\boldsymbol{\gamma}}_1(t_0) \cdot \dot{\boldsymbol{\gamma}}_2(t_0)|}{(z^0_{n+1})^2} \le \frac{|\dot{\boldsymbol{\gamma}}_1(t_0)||\dot{\boldsymbol{\gamma}}_2(t_0)|}{(z^0_{n+1})^2} = |\dot{\boldsymbol{\gamma}}_1(t_0)|_h \cdot |\dot{\boldsymbol{\gamma}}_1(t_0)|_h,$$

with the inequality deduced from the Cauchy-Schwarz-Bunyakowski Inequality (A.1). We now call the number

$$\angle(L_1, L_2)_{\mathbf{z}_0} = \arccos\left(\frac{(\dot{\boldsymbol{\gamma}}_1(t_0) \cdot \dot{\boldsymbol{\gamma}}_2(t_0))_h}{|\dot{\boldsymbol{\gamma}}_1(t_0)|_h \cdot |\dot{\boldsymbol{\gamma}}_1(t_0)|_h}\right) \in [0, \pi], \tag{9.14}$$

the *angle* of L_1 and L_2 at z_0. Defined in this way, the angle $\angle(L_1, L_2)_{\mathbf{z}_0}$ *coincides with the angle defined in the Euclidean case* and therefore does not depend on the parametrisations of L_i, $i = 1, 2$.

Definition 9.4.1 Let A, B, C be three points of $\mathcal{H}^{n+1}$ not on the same hyperbolic line. Let also a, b, c be the segments of the hyperbolic lines connecting B and C, A and C, A and B, respectively. The resulting figure shall be called a *hyperbolic triangle* $\triangle ABC$. A, B, C are called the *vertices* of the triangle while a, b, c are the *sides* of the triangle. The *angles* α, β, γ of $\triangle ABC$ are the angle of b and c at A, of a and c at B, and of a and b at C, respectively.

In analogy with Spherical Geometry, it is sufficient to study the triangles of $\mathcal{H}^2$. This is ensured by the following:

Proposition 9.4.2 *Every three pairwise distinct points of $\mathcal{H}^{n+1}$ that do not lie in the same hyperbolic line, belong to some 2-hyperbolic subspace.*

Proof Let L be the hyperbolic line from A, C. There exists an $h \in \mathrm{GM}(\mathcal{H}^{n+1})$ such that

$$h(A) = \mathbf{e}_{n+1}, \quad h(B) = b\,\mathbf{e}_{n+1}, \quad h(C) = c_1\,\mathbf{e}_n + c_2\,\mathbf{e}_{n+1},$$

where $b \neq 1$, $c_1 \in \mathbb{R}_*$, $c_2 > 0$. Then the mapping $g : \mathcal{H}^2 \to \mathcal{H}^{n+1}$, $g = h \circ \iota_2$, where ι_2 is as in (9.6), maps $\mathcal{H}^2$ onto a hyperbolic subspace and also

$$g(i) = A, \quad g(bi) = B, \quad g(c_1 + ic_2) = C.$$

□

Hereafter we will refer exclusively to triangles of $\mathcal{H}^2$. The *interior* of a triangle is defined by Jordan's theorem for the plane when the vertices of the triangle all belong to $\mathcal{H}^2$ (Fig. 9.7).

However, there is also the case where we have triangles with one vertex at infinity. Let T be such a triangle. Then we may normalise by a transformation g such that $g(T) = T'$ is the triangle with sides

$$a = [ai, \infty], \quad b = [b_1 + ib_2, \infty], \quad c = [ai,\ b_1 + ib_2].$$

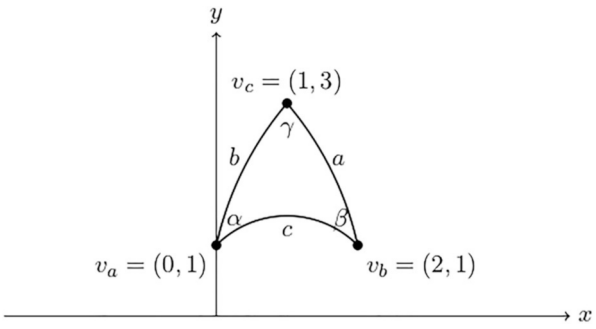


Fig. 9.7 Hyperbolic triangle in $\mathcal{H}^2$

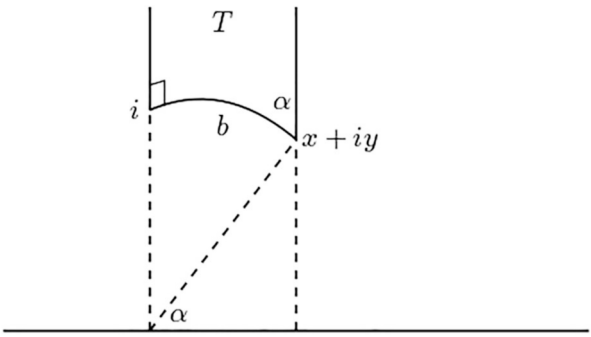


Fig. 9.8 Proposition 9.4.3: α is the angle of parallelism

Notice that this triangle has two sides of infinite length that meet at infinity at an angle 0. If $L: \ |z - x_0| = r$ is the hyperbolic line such that $c \in L$, we define

$$\text{int}(T') = \{z \ | 0 < \Re(z) < b_1, \ |z - x_0| > r\},$$

and consequently

$$\text{int}(T) = g(\text{int}(T')).$$

As usual, many times we will not distinguish between a triangle and its interior.

9.4.2 Angle of Parallelism

Consider a triangle with angles $\alpha, \pi/2, 0$, where $\alpha \neq 0$. The angle α is called the *angle of parallelism* (Fig. 9.8).

Proposition 9.4.3 *For T as above,*

$$\cosh b \sin \alpha = 1. \tag{9.15}$$

Proof We normalise so that $C = i$, $B = \infty$ and $A = z = x + iy$, $|z| = 1$. Using the distance formula (9.10) we have

$$\begin{aligned} \cosh b = \cosh(d_h(i, z)) &= 1 + \frac{|z - i|^2}{2\Im(z)} \\ &= 1 + \frac{2 - 2y}{2y} \\ &= 1/y = 1/\sin\alpha, \end{aligned}$$

with the latter relation resulting from the Tangent-Chord Theorem. □

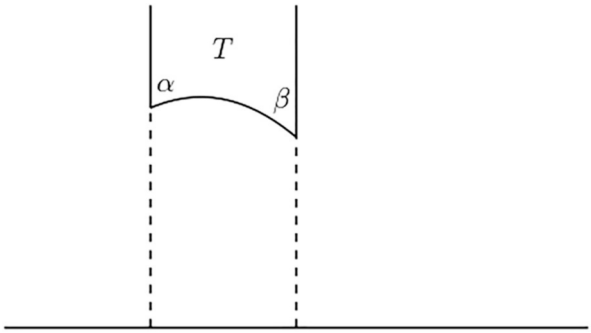


Fig. 9.9 Proposition 9.4.4: Law of Sines for triangles with a vertex at infinity

9.4.3 Triangles with a Vertex at Infinity

The case of the triangles of Sect. 9.4.2 is a special case of (right-angled) triangles with a vertex at infinity, that is, triangles with angles $\alpha, \beta, 0$. In this case, we have seen that the sides a, b are of infinite length (Fig. 9.9).

Proposition 9.4.4 (Law of Cosines for Triangles with a Vertex at Infinity) *For triangles with angles $\alpha, \beta, 0$, we have:*

$$\cosh c = \frac{\cos\alpha\cos\beta + 1}{\sin\alpha\sin\beta}. \tag{9.16}$$

Proof We can assume that $C = \infty$, $A = e^{i\theta}$, $B = e^{i\phi}$ with $0 < \theta < \phi < \pi$. Here, $\alpha = \theta$, $\beta = \pi - \phi$. Then (9.16) follows again from the distance formula (9.10). □

9.4.4 Hyperbolic Pythagorean Theorem

The Hyperbolic Pythagorean Theorem applies to hyperbolic right triangles, that is, triangles with angles $\alpha, \beta, \pi/2$ (Fig. 9.10).

Proposition 9.4.5 *In a hyperbolic right triangle with angles $\alpha, \beta, \pi/2$ the following holds:*

$$\cosh c = \cosh a \cos b. \tag{9.17}$$

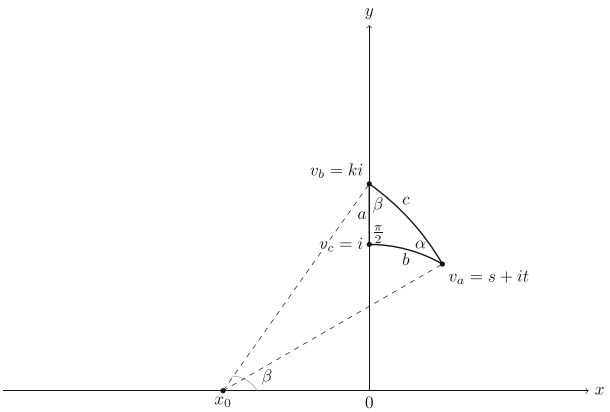


Fig. 9.10 Proposition 9.4.5: Hyperbolic Pythagorean Theorem

Proof We may assume that $A = z = s + it$, $|z| = 1$, $B = ki$, $s > 1$, and $C = i$. Then, applying the formula (9.10) again, we successively get

$$\cosh c = (1 + k^2)/(2kt),$$
$$\cosh b = 1/t,$$
$$\cosh a = (1 + k^2)(2k).$$

Formula (9.17) follows directly. □

9.4.5 Sine and Cosine Law

For an arbitrary hyperbolic triangle T with angles α, β, γ and sides a, b, c the following laws apply:

Law of Sines:

$$\frac{\sinh a}{\sin \alpha} = \frac{\sinh b}{\sin \beta} = \frac{\sinh c}{\sin \gamma}.$$

First Law of Cosines:

$$\cosh c = \cosh a \cosh b - \sinh a \sinh b \cos \gamma.$$

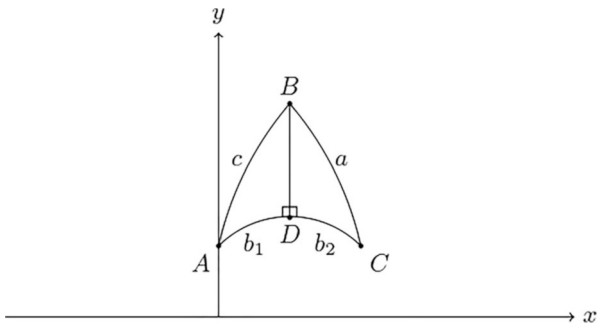


Fig. 9.11 Proof of the First Law of Cosines

Second Law of Cosines:

$$\cosh c = \frac{\cos\alpha\cos\beta + \cos\gamma}{\sin\alpha\sin\beta}.$$

We note that the Laws of Cosines have analogous symmetric expressions with respect to angles and sides.

Exercise Write all the symmetric expressions of the Laws of Cosines.

We also note that the Law of Sines and the Second Law of Cosines follow from the First Law of Cosines (see Exercises 18, Sect. 9.6 and 19, Sect. 9.6). Finally, in the same manner as in Spherical Geometry, we obtain from the Second Law of Cosines that similar hyperbolic triangles are also equal. That is, for every two triangles whose corresponding angles are equal, there is an isometry that maps one to the other. Therefore, as in Spherical Geometry, and in contrast to Euclidean Geometry, there is no similarity geometry that contains Hyperbolic Geometry; similarities are necessarily isometries (Fig. 9.11).

We will prove the First Law of Cosines. We need the following lemmas:

Lemma 9.4.6 *From the vertex of a hyperbolic triangle, a perpendicular is drawn to the opposite side. Of the three perpendiculars, at least one lies inside the triangle.*

Proof From a point outside the hyperbolic line, we can always draw a perpendicular to the line. To see this, it is enough to consider the line $L = (0, \infty)$ and the point z_0 with $\Re(z_0) \neq 0$. Then the line $|z| = |z_0|$ is perpendicular to L.

Now, we can show that the perpendicular to the longest side always lies inside the triangle. Indeed, we normalise again so that the largest side, let c be on $L = (0, \infty)$: $A = i$ and $B = e^c i$. The perpendicular from $C = z_1$ intersects L at $w = i|z_1|$ (Fig. 9.12).

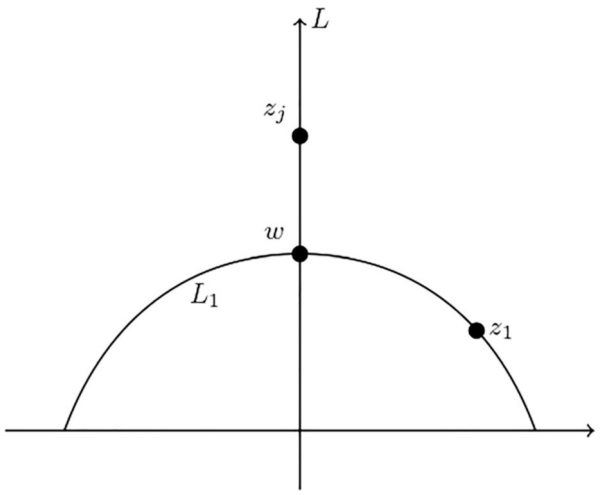


Fig. 9.12 Proof of Lemma 9.4.6

now,

$$d_h(z_1, i) \geq d_h(i|z_1|, i) = d_h(w, i),$$
$$d_h(z_1, e^c i) \geq d_h(i|z_1|, e^c i) = d_h(w, e^c i).$$

Since c is the longest side,

$$c = d_h(i, e^c i) \geq \max\{d_h(z_1, i),\ d_h(z_1, e^c i)\} \geq \max\{d_h(w, i),\ d_h(w, e^c i)\}.$$

The points i, w, $e^c i$ are collinear and one must lie between the other two. Our claim now follows from Proposition 9.3.8. □

Proof of the First Law of Cosines For the proof we use the Hyperbolic Pythagorean Theorem (Proposition 9.4.5) and its conclusions found in Exercises 16, Sect. 9.6 and 17, Sect. 9.6. Using also Lemma 9.4.6, and Fig. 9.12 as well, we have:

$$\begin{aligned}
\cosh c &= \cosh b_1 \cosh h \\
&= \cosh(b - b_2) \cosh h \\
&= (\cosh b \cosh b_2 - \sinh b \sinh b_2) \cosh h \\
&= \cosh b(\cosh b_2 \cosh h) - \sinh b \sinh b_2 \cosh h \\
&= \cosh b \cosh a - \sinh b \sinh b_2 \cdot \frac{\cosh a}{\cosh b_2} \\
&= \cosh b \cosh a - \sinh b \sinh a \cdot \frac{\cosh a}{\sinh a} \cdot \frac{\sinh b_2}{\cosh b_2}
\end{aligned}$$

$$= \cosh b \cosh a - \sinh b \sinh a \cdot \frac{\tanh b_2}{\tanh a}$$
$$= \cosh b \cosh a - \sinh b \sinh a \cos \gamma.$$

□

9.4.6 Area of a Hyperbolic Triangle

In this section we will prove the following important fact:

Theorem 9.4.7 *The sum of the angles of a hyperbolic triangle is a number greater than* π.

Theorem 9.4.7 arises as a direct consequence of the following proposition:

Proposition 9.4.8 *The hyperbolic area of a triangle* $T = T(\alpha, \beta, \gamma)$ *with angles* α, β, γ, *is equal to*

$$\text{Area}_h(T) = \pi - \alpha - \beta - \gamma.$$

Proof We first assume that $\gamma = 0$ and normalise so that the vertices A and B lie on the unit semicircle: $B = e^{i\theta}$, $A = e^{i\phi}$, $0 < \phi < \theta < \pi$. Then the interior of the triangle is:

$$T(\alpha, \beta, 0) = \{x + iy \in \mathcal{H}^2 \mid \cos(\pi - \alpha) < x < \cos\beta,\ \sqrt{1 - x^2} < y < +\infty\}.$$

Applying Fubini's Theorem, we calculate:

$$\begin{aligned}
\text{Area}_h(T(\alpha, \beta, 0)) &= \iint_{T(\alpha,\beta,0)} \frac{dxdy}{y^2} \\
&= \int_{\cos(\pi-\alpha)}^{\cos\beta} \left(\int_{\sqrt{1-x^2}}^{+\infty} \frac{dy}{y^2} \right) dx \\
&= \int_{\cos(\pi-\alpha)}^{\cos\beta ta} \left[-\frac{1}{y} \right]_{y=\sqrt{1-x^2}}^{+\infty} dx \\
&= \int_{\cos(\pi-\alpha)}^{\cos\beta} \frac{dx}{\sqrt{1-x^2}} \\
&= -\left[\arccos(x) \right]_{\cos(\pi-\alpha)}^{\cos\beta} \\
&= \pi - (\alpha + \beta).
\end{aligned}$$

If $T(\alpha, \beta, \gamma)$ is a triangle with non zero angles, we extend the side from A and C until it intersects infinity at point $x_0 \in \mathbb{R}$. Then, since

$$\text{Area}_h(T(\alpha, \beta, \gamma)) = \text{Area}_h(T(0, \beta', \gamma)) - \text{Area}_h(T(\pi - \alpha, \beta'', 0)),$$

where $\beta' - \beta'' = \beta$, we have

$$\text{Area}_h(T(\alpha, \beta, \gamma) = \pi - (\beta' + \gamma) - \pi + (\pi - a + \beta'') = \pi - (\alpha + \beta + \gamma).$$

□

Exercise Prove that the area of an hyperbolic n-angle P_n with angles α_i, $i = 1, \ldots, n$ is equal to

$$\text{Area}_h(P_n) = (n-2)\pi - \sum_{i=1}^{n} \alpha_i.$$

9.5 Models of Hyperbolic Planar Geometry

In this section we discuss the Poincaré and Beltrami-Klein disk models (Fig. 9.13). These models are generalised to the n-dimensional case, see e.g. [44]; here we will examine in Sects. 9.5.1 and 9.5.2 the direct connection of those models with the upper half-plane model. However, we should stress here the fundamental difference between the Poincaré models and the Beltrami-Klein model: the former are conformal models, that is, the angles are preserved and thus the hyperbolic disks are Euclidean disks. This does not happen in the Beltrami-Klein model: there is a distortion of the angles and the hyperbolic disks are Euclidean only when they are centred at the origin. On the other hand, this model has Euclidean straight lines as hyperbolic geodesics, which is in some way an advantage over the other two models.



Fig. 9.13 Eugenio Beltrami, 1836–1900

To introduce the Beltrami-Klein model we follow a method that highlights the relationship between Projective Geometry and Hyperbolic Geometry. We note that this method can be used to produce all the models of Hyperbolic Geometry. Interested readers should also see Exercise 26, Sect. 9.6 for a similar treatment of Poincaré models.

9.5.1 Poincaré's Disk Model

The transition from the upper half-plane model $\mathcal{H}^2$ to the unit disk model $\mathcal{B}^1_{\mathbb{C}}$ of $\mathbb{C}$ is done by the Cayley transformation $\mathcal{C} : \mathcal{H}^2 \to \mathcal{B}^1_{\mathbb{C}}$. This is a transformation $\mathcal{C} \in \mathrm{M}(2)$ given by

$$\mathcal{C}(z) = \frac{z-i}{z+i}, \tag{9.18}$$

restricted to $\mathcal{H}^2$. We observe that

$$\mathcal{C}(0) = -1, \quad \mathcal{C}(\infty) = 1, \quad \mathcal{C}(1) = -i,$$

and therefore $\mathcal{C}(\partial\overline{\mathcal{H}^2}) = S^1 = \partial(\mathcal{B}^1_{\mathbb{C}})$. On the other hand, for $z \in \mathcal{H}^2$ it is

$$w = \frac{z-i}{z+i} \iff z = -i \cdot \frac{(w+1)}{w-1},$$

and from the relation $\Im(z) > 0$ it follows that $|w| < 1$. All elements of the Hyperbolic Geometry of $\mathcal{H}^2$ are transferred to the disk $\mathcal{B}^1_{\mathbb{C}}$ via the Cayley transformation; below, we give some examples of this fact.

Proposition 9.5.1 *The hyperbolic length $\ell_h(\boldsymbol{\gamma})$ of a continuously differentiable curve $\boldsymbol{\gamma} : [a,b] \to \mathcal{B}^1_{\mathbb{C}}$ of $\mathcal{B}^1_{\mathbb{C}}$, $\boldsymbol{\gamma}(t) = z(t)$, is given by the relation*

$$\ell_h(\boldsymbol{\gamma}) = 2\int_a^b \frac{|\dot{z}(t)|\,dt}{1-|z(t)|^2}. \tag{9.19}$$

Proof Let $\mathbf{c} = \mathcal{C}^{-1} \circ \boldsymbol{\gamma}$ be the curve in $\mathcal{H}^2$. The hyperbolic length $\ell_h(\boldsymbol{\gamma})$ of $\boldsymbol{\gamma}$ is then equal to $\ell_h(\mathbf{c})$. Setting $f = \mathcal{C}^{-1}$ we have:

$$\mathbf{c}(t) = -i \cdot \frac{z(t)+1}{z(t)-1}, \quad \Im(\mathbf{c}(t)) = \frac{1-|z(t)|^2}{|z(t)-1|^2},$$

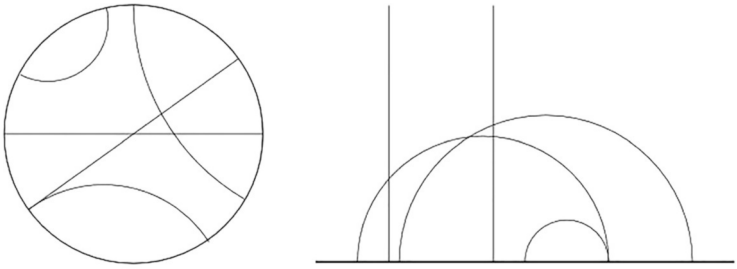


Fig. 9.14 Geodesics in Poincaré models

and

$$\dot{\mathbf{c}}(t)) = \frac{2i\ \dot{z}(t)}{(z(t)-1)^2}.$$

We conclude that the hyperbolic norm of $\dot{\boldsymbol{\gamma}}$ is:

$$|\dot{\boldsymbol{\gamma}}(t)|_h = \frac{2|\dot{z}(t)|}{1-|z(t)|^2},$$

Formula (9.19) follows. □

The hyperbolic distance ρ_h in the disk model is defined by

$$\rho_h(z,w) = d_h(\mathcal{C}^{-1}(z),\ \mathcal{C}^{-1}(w)). \tag{9.20}$$

Proposition 9.5.2 *The hyperbolic lines (and therefore, geodesics) in the disk model* $(\mathcal{B}^1_{\mathbb{C}}, \rho_h))$ *are either open diameters of* $\mathcal{B}^1_{\mathbb{C}}$*, or arcs of circles inside* $\mathcal{B}^1_{\mathbb{C}}$*, orthogonal to the boundary* S^1 *of* $\mathcal{B}^1_{\mathbb{C}}$ *(Fig. 9.14).*

Proof By definition, the geodesics of $(\mathcal{B}^1_{\mathbb{C}}, \rho_h))$ are the $\mathcal{C}$-images of the geodesics of $(\mathcal{H}^2, d_h))$. Let L be a hyperbolic line of the form (a, ∞), $a \in \mathbb{R}$. The image of $\mathcal{C}(L)$ is a segment of a generalised circle with endpoints 1 and $\frac{a-i}{a+i}$ on S^1. If $a = 0$, then $\mathcal{C}(L)$ is the open diameter $(-1, 1)$ that is orthogonal to S^1. If $a \neq 0$, then $\mathcal{C}(L)$ is given by:

$$w = u + iv = \mathcal{C}(a+iy) \iff a + iy = \mathcal{C}^{-1}(w) = -i \cdot \frac{w+1}{w-1}.$$

The relation $a = \Re(\mathcal{C}^{-1}(w))$ gives

$$a = \Im\left(\frac{w+1}{w-1}\right) = -\frac{2v}{|w-1|^2}.$$

If we assume with no loss that $a > 0$, then $v < 0$ and after the calculations we obtain the circular arc given by

$$\mathcal{C}(L): \ (u-1)^2 + \left(v + \frac{1}{a}\right)^2 = \frac{1}{a}, \quad |w| < 1, \ v < 0.$$

The vector normal to this circle is $(u-1, \ v+1/a)$. The intersection $\mathcal{C}(L) \cap S^1$ consists of the points 1 and $e^{i\theta}$, for some $\theta \in (\pi, 2\pi)$ which satisfies the relation

$$(\cos\theta - 1)^2 + \left(\sin\theta + \frac{1}{a}\right)^2 = \frac{1}{a^2} \iff \cos(\theta/2) + a\sin(\theta/2) = 0.$$

Setting $a = \tan(\pi b/2)$, $b \in (0, 1/2)$, we get

$$\cos\left(\frac{\theta - \pi b}{2}\right) = 0 \iff \theta = (b+1)\pi.$$

At the intersection points 1 and $e^{i\theta} = e^{i(b+1)\pi}$, the corresponding normal vectors to $\mathcal{C}(L)$ are

$$(0, \ 1/a), \quad (\cos\theta - 1, \ \sin\theta + 1/a),$$

while the corresponding normal vectors to S^1 are

$$(1, 0), \quad (\cos\theta, \sin\theta).$$

The desired result follows immediately. The case where $a < 0$ is similar. □

Exercise Complete the proof of Proposition 9.5.2 for the case where L is a semicircle with centre a and radius r.

Proposition 9.5.3 *The hyperbolic distance $\rho_h = \rho_h(z, w)$ of two points $z, w \in \mathcal{B}^1_{\mathbb{C}}T$ is given by the following equivalent formulae:*

$$\cosh^2(\rho_h/2) = \frac{|1 - z\overline{w}|^2}{(1-|z|^2)(1-|w|^2)}, \tag{9.21}$$

$$\tanh(\rho_h/2) = \left|\frac{z-w}{1-z\overline{w}}\right|,$$

$$\rho_h = \ln\left(\frac{|1-z\overline{w}| + |z-w|}{|1-z\overline{w}| - |z-w|}\right).$$

Exercise Prove Proposition 9.5.3.

Proposition 9.5.4 *The group of hyperbolic isometries of* $(B^1_{\mathbb{C}}, \rho_h)$ *comprises elements of the form*

$$z \to \frac{az + \overline{c}}{cz + \overline{a}}, \quad z \to \frac{a\overline{z} + \overline{c}}{c\overline{z} + \overline{a}}, \quad |a|^2 - |c|^2 = 1.$$

The elements of the form on the left side constitute the group $\mathrm{M}(\mathcal{B}^1_{\mathbb{C}})$ of conformal isometries also denoted by SU(1, 1). The general group of isometries $\mathrm{GM}(\mathcal{B}^1_{\mathbb{C}})$ comprises elements of SU(1, 1), or elements of SU(1, 1) followed by $z \mapsto \overline{z}$.

Exercise Prove Proposition 9.5.4.

The hyperbolic area of the region $\Omega \subset \mathcal{B}^1_{\mathbb{C}}$ is given in the disk model by the following formula:

$$\mathrm{Area}_h(\Omega) = 4 \iint_{\Omega} \frac{dxdy}{(1 - |z|^2)^2}. \tag{9.22}$$

We emphasise that the formulae that give the hyperbolic circumference and hyperbolic area of a hyperbolic disk $D(z_0, r)$, $z_0 \in \mathcal{B}^1_{\mathbb{C}}$, $r > 0$, are the same as those in the hyperbolic plane. This is because the Cayley transformation (9.18) is an isometry of the models $(\mathcal{H}^2, d_h)$ and $(\mathcal{B}^1_{\mathbb{C}}, \rho_h)$, and therefore an equiareal mapping.

Example 9.5.5 We will calculate the hyperbolic area of the region D_a bounded by the circle $|z| = a$ inside $\mathcal{B}^1_{\mathbb{C}}$. Using polar coordinates $z = re^{i\theta}$ and the Change of variables theorem, we have from formula (9.22) that

$$\begin{aligned}
\mathrm{Area}_h(D_a) &= 4 \int_0^a \int_0^{2\pi} \frac{r\, dr d\theta}{(1 - r^2)^2} \\
&= 4\pi \int_0^a \frac{2r\, dr}{(1 - r^2)^2} \\
&= 4\pi \left[\frac{1}{1 - r^2} \right]_0^a \\
&= \frac{4\pi a^2}{1 - a^2}.
\end{aligned}$$

9.5.2 Beltrami-Klein Model

The Beltrami-Klein model comes from Planar Projective Geometry (Fig. 9.15). Let $\mathbb{R}^{2,1}$ be the vector space $\mathbb{R}^3$ equipped with the inner product defined as follows: if

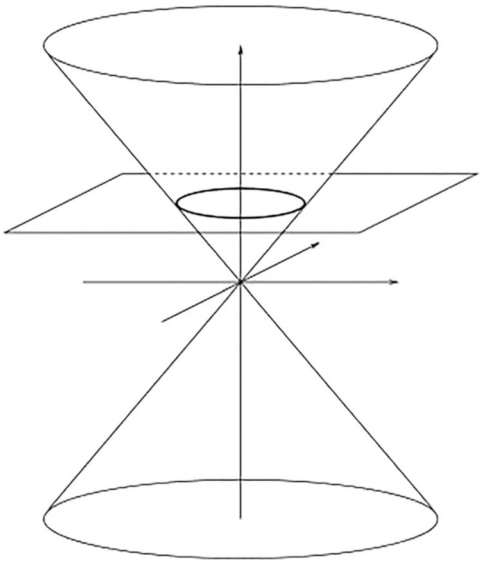


Fig. 9.15 The Beltrami-Klein model is the intersection of the plane $X_2 = 1$ with the light cone

$\mathbf{X} = (X_0, X_1, X_2)$ and $\mathbf{Y} = (Y_0, Y_1, Y_2)$, then

$$\langle \mathbf{X}, \mathbf{Y} \rangle = X_0 Y_0 + X_1 Y_1 - X_2 Y_2. \tag{9.23}$$

This product has a matrix

$$J = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \tag{9.24}$$

which has one double positive eigenvalue and one negative eigenvalue. Therefore, it defines what is called the (2, 1)-product: the product of a vector $\mathbf{X}$ with itself can be positive, but also negative or zero. We therefore consider the sets of *positive, zero* and *negative* vectors, respectively:

$$V_+ = \left\{ \mathbf{X} \in \mathbb{R}^{2,1} \mid \langle \mathbf{X}, \mathbf{X} \rangle > 0 \right\},$$
$$V_0 = \left\{ \mathbf{X} \in \mathbb{R}^{2,1} \setminus \{\mathbf{0}\} \mid \langle \mathbf{X}, \mathbf{X} \rangle = 0 \right\},$$
$$V_- = \left\{ \mathbf{X} \in \mathbb{R}^{2,1} \mid \langle \mathbf{X}, \mathbf{X} \rangle < 0 \right\}.$$

Remark 9.5.6 The space $\mathbb{R}^{2,1} \setminus \{\mathbf{0}\}$ can also be seen as the space of homogeneous coordinates of the projective plane $\mathbb{R}P^2$, that is, the union of the sets H_0, H_1, H_2 as

in (7.8). In other words, $\mathbb{R}^{2,1} \setminus \{\mathbf{0}\}$ is the projective plane equipped with the above $(2, 1)$-product.

Remark 9.5.7 The model is constructed for arbitrary n in the analogous manner. We stress here that for $n = 3$ the model is nothing but the model for spacetime in the Special Theory of Relativity.

Consider the subset H_2 of $\mathbb{R}^{2,1}$ defined by $X_2 \neq 0$ and consider also the projection $\mathbb{P} : \mathbb{R}^{2,1} \to \mathbb{R}^2$ given for points in H_2 by

$$\mathbf{X} = (X_0, X_1, X_2) \mapsto \left(\frac{X_0}{X_2}, \frac{X_1}{X_2}\right) = \mathbf{x} = (x_1, x_2).$$

We define the Beltrami-Klein disk $\mathcal{B}^2_{\mathbb{R}}$ to be the image $\mathbb{P}(V_-)$, that is,

$$\mathcal{B}^2_{\mathbb{R}} = \{(x_1, x_2) \in \mathbb{R}^2 \mid x_1^2 + x_2^2 < 1\}.$$

The boundary of $\mathcal{B}^2_{\mathbb{R}}$ is then the image $\mathbb{P}(V_0)$, that is,

$$S^1 = \{(x_1, x_2) \in \mathbb{R}^2 \mid x_1^2 + x_2^2 = 1\}.$$

We note here that if $(x_1, x_2) \in \mathcal{B}^2_{\mathbb{R}}$ then its standard lift in $\mathbb{R}^{2,1}$ is the vector $(x_1, x_2, 1)$. Topologically, $\mathcal{B}^1_{\mathbb{C}}$ and $\mathcal{B}^2_{\mathbb{R}}$ are clearly the same sets. It is important, however, to emphasise the *real structure* of the unit disk when talking about the Beltrami-Klein model: the model of the unit disk of Poincaré can be obtained by a similar procedure to the one we followed here, but instead of $\mathbb{R}^{2,1}$ and the projective plane $\mathbb{R}P^2$ we have the space $\mathbb{C}^{1,1}$ and the complex projective line $\mathbb{C}P^1$ (see Exercise 26, Sect. 9.6.). These sets differ, among other things, in that the complex projective line has an intrinsic complex structure, being in a bijective correspondence with the Riemann sphere. This is not the case for the real projective plane; it is considered to be without any complex structure. This is exactly the reason why we write $\mathcal{B}^1_{\mathbb{C}}$ instead of just $\mathcal{B}^2$; we do it in order to distinguish the complex model of the Poincaré disk from the real Beltrami-Klein model. As for the upper half-plane model, since we have also considered it with the complex structure inherited by $\mathbb{C}$, we will often encounter it with the notation $\mathcal{H}^1_{\mathbb{C}}$.

Proposition 9.5.8 *The isometries group of the* $(2, 1)$*-product (9.23) is the matrix group*

$$\mathrm{O}(2, 1) = \{A \in \mathrm{GL}(3, \mathbb{R}) \mid AJA^T = J\}, \tag{9.25}$$

where J *is the matrix of the product (9.24). If* $A \in \mathrm{O}(2, 1)$*, then*

$$A^{-1} = JA^TJ,$$

and therefore $\det^2(A) = 1$.

Proof If $A \in \mathrm{GL}(3, \mathbb{R})$ is an isometry, then for every $\mathbf{X}, \mathbf{Y} \in \mathbb{R}^{2,1}$ we have

$$\langle \mathbf{X}, \mathbf{Y} \rangle = \mathbf{X} J \mathbf{Y}^T = \langle \mathbf{X}A, \mathbf{Y}A \rangle = \mathbf{X} A J A^T \mathbf{Y}^T.$$

Since now $J^2 = I$, where I is the unit 3×3 matrix, we also have

$$AJA^T = J \iff A(JA^TJ) = I \implies A^{-1} = JA^TJ.$$

Finally, $\det(A^{-1}) = \det(J)\det(A)\det(J) = \det(A)$, so $\det^2(A) = 1$. □

The group $\mathrm{O}(2, 1)$ is called the *orthogonal* $(2, 1)$*-group* and by using the relation $AJA^T = J$ we can verify straightforwardly that the rows of each of its elements are vectors orthogonal to the product $\langle \cdot, \cdot \rangle$. Indeed, if

$$A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & j \end{pmatrix},$$

then $AJA^T = J$ gives

$$\begin{aligned} &a^2 + b^2 - c^2 = 1, \quad ad + be - cf = 0, \quad ag + bh - cj = 0, \\ &d^2 + e^2 - f^2 = 1, \quad dg + eh - fj = 0, \\ &g^2 + h^2 - j^2 = -1. \end{aligned}$$

Also, the relation $A^{-1} = JA^TJ$ means

$$A^{-1} = \begin{pmatrix} a & d & -g \\ b & e & -h \\ -c & -f & j \end{pmatrix}.$$

From $A^{-1}A = I$ we obtain the additional relations

$$\begin{aligned} &a^2 + d^2 - g^2 = 1, \quad ab + de - gh = 0, \quad ac + df - gj = 0, \\ &b^2 + e^2 - h^2 = 1, \quad bc + ef - hj = 0, \\ &c^2 + f^2 - j^2 = 1. \end{aligned}$$

We conclude that the columns of the matrix A are mutually orthogonal and that also:

- the first and second rows (i.e. the first and second columns) of A are positive unit vectors,
- the third row (i.e. the third column) of A is a negative unit vector.

The group O(2, 1) comprises elements A with $\det(A) = 1$ or $\det(A) = -1$. In the first case, we have the set SO(2, 1) which is a subgroup of O(2, 1). In the group O(2, 1) we define a relation $\sim$ as follows:

$$A \sim B \text{ if } A = \lambda B \text{ for some } \lambda \neq 0.$$

Note that since $\det(A) = \pm 1$ for every $A \in \mathrm{O}(2, 1)$, the above λ can take the values ± 1.

Exercise Show that $\sim$ is an equivalence relation.

Definition 9.5.9 The *projective orthogonal* (2, 1)*-group* is

$$\mathrm{PO}(2, 1) = \mathrm{O}(2, 1)/\sim .$$

Since the action of O(2, 1) preserves the spaces V_+, V_0 and V_-, it follows that the action of PO(2, 1) on $\mathcal{B}^2_{\mathbb{R}}$ leaves it invariant. Hyperbolic Geometry (according to Klein) is the pair $(\mathcal{B}^2_{\mathbb{R}}, \mathrm{PO}(2, 1))$. The group PSO(2, 1) comprises elements that preserve the orientation, that is, the group $\mathrm{PSO}^+(2, 1)$ that is isomorphic to SO(2, 1), and elements that do not preserve the orientation. The latter are compositions of elements of $\mathrm{PSO}^+(2, 1)$ followed by conjugation, that is, the projection of

$$(x_0, x_1, x_2) \mapsto (x_0, x_1, -x_2).$$

Finally, we mention that

$$\mathrm{PO}(2, 1) = \mathrm{SO}(2, 1)/\{I, -I\},$$

as follows from the embedding of SO(2, 1) in PO(2, 1) and the First Group Isomorphism Theorem.

The *hyperbolic distance* σ_h in the Beltrami-Klein model is now defined as follows: for any $\mathbf{x} = (x_1, x_2)$ and $\mathbf{y} = (y_1, y_2)$ points of $\mathcal{B}^2_{\mathbb{R}}$ we define $\sigma_h(\mathbf{x}, \mathbf{y})$ by the relation

$$\cosh^2(\sigma_h(\mathbf{x}, \mathbf{y})) = \frac{\langle \mathbf{X}, \mathbf{Y} \rangle^2}{\langle \mathbf{X}, \mathbf{X} \rangle \langle \mathbf{Y}, \mathbf{Y} \rangle}, \tag{9.26}$$

where $\mathbf{X}$, $\mathbf{Y}$ are lifts of $\mathbf{x}$, $\mathbf{y}$, respectively to $\mathbb{R}^{2,1}$. The definition of σ_h by (9.26) is invariant under the action of PO(2, 1) and does not depend on the lifts of $\mathbf{x}, \mathbf{y}$: indeed, if $\mathbf{X}'$, $\mathbf{Y}'$ are any lifts of $\mathbf{x}$, by, respectively, then there exist $\lambda, \mu \in \mathbb{R}_*$ with

$\mathbf{X}' = \lambda\mathbf{X}$ and $\mathbf{Y}' = \mu\mathbf{Y}$. then,

$$\begin{aligned}\frac{\langle \mathbf{X}', \mathbf{Y}'\rangle^2}{\langle \mathbf{X}', \mathbf{X}'\rangle\langle \mathbf{Y}', \mathbf{Y}'\rangle} &= \frac{\langle \lambda\mathbf{X}, \mu\mathbf{Y}\rangle^2}{\langle \lambda\mathbf{X}, \lambda\mathbf{X}\rangle\langle \mu\mathbf{Y}, \mu\mathbf{Y}\rangle} \\ &= \frac{\lambda^2\mu^2\langle \mathbf{X}, \mathbf{Y}\rangle^2}{\lambda^2\langle \mathbf{X}, \mathbf{X}\rangle \cdot \mu^2\langle \mathbf{Y}, \mathbf{Y}\rangle} \\ &= \frac{\langle \mathbf{X}, \mathbf{Y}\rangle^2}{\langle \mathbf{X}, \mathbf{X}\rangle\langle \mathbf{Y}, \mathbf{Y}\rangle}.\end{aligned}$$

For $\mathbf{X} = (x_1, x_2, 1)$ and $\mathbf{Y} = (y_1, y_2, 1)$ we get the formula

$$\cosh^2(\sigma_h(\mathbf{x}, \mathbf{y})) = \frac{(\mathbf{x}\cdot\mathbf{y} - 1)^2}{(\|\mathbf{x}\|^2 - 1)(\|\mathbf{y}\|^2 - 1)} = \frac{(x_1y_1 + x_2y_2 - 1)^2}{(x_1^2 + x_2^2 - 1)(y_1^2 + y_2^2 - 1)}. \tag{9.27}$$

It can be shown directly that σ_h is a well-defined metric. First, to show that is well-defined we first prove that if $\mathbf{X}, \mathbf{Y} \in V_-$, then

$$\langle \mathbf{X}, \mathbf{Y}\rangle^2 \geq \langle \mathbf{X}, \mathbf{X}\rangle\langle \mathbf{Y}, \mathbf{Y}\rangle, \tag{9.28}$$

where the equality holds if and only if $\mathbf{X}, \mathbf{Y}$ represent the same point in the projective plane (and hence, are lifts of the same point of $\mathcal{B}^2_{\mathbb{R}}$). The inequality (9.28) is also called *Inverse Cauchy-Schwarz-Bunyakowski Inequality* for obvious reasons. For its proof, see Exercise 22, Sect. 9.6. We can now show that $\sigma_h(\mathbf{x}, \mathbf{y})$ is positive and vanishes only when $\mathbf{x} = \mathbf{y}$. The symmetry is also immediately apparent: $\sigma_h(\mathbf{x}, \mathbf{y}) = \sigma_h(\mathbf{y}, \mathbf{x})$. We can also prove the Triangular Inequality; see Exercise 24, Sect. 9.6. However, we consider it appropriate here to highlight the relationship between the Beltrami-Klein model and the disk model and consequently the upper half-plane model. To this end, we consider the mapping $\mathfrak{C} : \mathcal{B}^2_{\mathbb{R}} \to \mathcal{B}^1_{\mathbb{C}}$ which is defined by the relation

$$\mathbf{x} = (x_1, x_2) \mapsto \frac{x_1 + ix_2}{1 + \sqrt{1 - \|\mathbf{x}\|^2}}. \tag{9.29}$$

Proposition 9.5.10 *The mapping $\mathfrak{C}$ given by (9.29) is a bijection and its $\mathfrak{C}^{-1}$ is given by*

$$z = x + iy \mapsto \left(\frac{2x}{1 + |z|^2}, \frac{2y}{1 + |z|^2}\right). \tag{9.30}$$

The map $\mathfrak{C}^{-1}$ is the composition of the stereographic projection from the disk to S^2:

$$z = x + iy \mapsto \left(\frac{2x}{1 + |z|^2}, \frac{2y}{1 + |z|^2}, \frac{1 - |z|^2}{1 + |z|^2}\right)$$

and the orthogonal projection from S^2 to the disk:

$$\left(\frac{2x}{1+|z|^2},\ \frac{2y}{1+|z|^2},\ \frac{1-|z|^2}{1+|z|^2}\right) \mapsto \left(\frac{2x}{1+|z|^2},\ \frac{2y}{1+|z|^2}\right).$$

Finally, $\mathfrak{C}$ leaves the boundary S^1 pointwise invariant.

Exercise Prove Proposition 9.5.10.

We now prove the following basic proposition:

Proposition 9.5.11 *Let $\mathbf{x}, \mathbf{y} \in \mathcal{B}^2_{\mathbb{R}}$. Then*

$$\sigma_h(\mathbf{x}, \mathbf{y}) = \rho_h(\mathfrak{C}(\mathbf{x}),\ \mathfrak{C}(\mathbf{y})),$$

where ρ_h is the hyperbolic metric on the disk $\mathcal{B}^1_{\mathbb{C}}$.

Proof We show equivalently that for every $z, w \in \mathcal{B}^1_{\mathbb{C}}$,

$$\rho_h(z, w) = \sigma_h(\mathfrak{C}^{-1}(z),\ \mathfrak{C}^{-1}(w)).$$

If $z = x + iy$ and $w = u + iv$, we set

$$x_1 = \frac{2x}{1+|z|^2}, \quad x_2 = \frac{2y}{1+|z|^2},$$
$$y_1 = \frac{2u}{1+|w|^2}, \quad y_2 = \frac{2v}{1+|w|^2}.$$

Then

$$\begin{aligned}
\cosh^2(\sigma_h(\mathfrak{C}^{-1}(z),\ \mathfrak{C}^{-1}(w))) &= \frac{(x_1y_1 + x_2y_2 - 1)^2}{(x_1^2 + x_2^2 - 1)(y_1^2 + y_2^2 - 1)} \\
&= \frac{\left(4\Re(z\overline{w}) - (1+|z|^2)\,(1+|w|^2)\right)^2}{(1-|z|^2)^2(1-|w|^2)^2} \\
&= \left(\frac{2|1-z\overline{w}|^2}{(1-|z|^2)(1-|w|^2)} - 1\right)^2 \\
&= \left(2\cosh^2(\rho_h(z,w)/2) - 1\right)^2 \\
&= \cosh^2(\rho_h(z,w)).
\end{aligned}$$

□

Proposition 9.5.12 *The hyperbolic lines (geodesics of $(\mathcal{B}^2_{\mathbb{R}}, \sigma_h)$) are Euclidean lines (Fig. 9.16).*

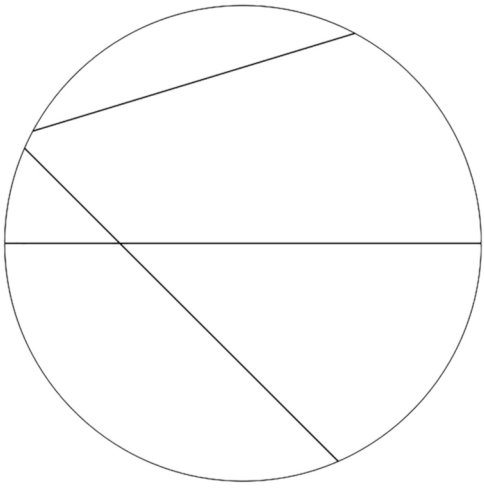


Fig. 9.16 Geodesics in the Beltrami-Klein model

Proof This can be seen either algebraically through the mapping $\mathfrak{C}$ and using the fact that $\mathfrak{C}$ leaves the unit circle pointwise invariant, or, in a manner similar to that of Exercise 26f, Sect. 9.6. However, this result is proved much more briefly and geometrically via the following simple observation: let L be a geodesic of $\mathcal{B}^1_{\mathbb{C}}$ and consider first the case where L is a Euclidean diameter. This is then lifted to a semi-meridian of S^2 which, through the inverse stereographic projection $\mathbf{S}^{-1}$, is again projected onto L. Finally, if L is a segment of a circle orthogonal to S^1, then this is lifted to the arc of the spherical circle of S^2 that passes through the endpoints of L and is orthogonal to S^1 at these points. However, the images of such circles through $\mathbf{S}^{-1}$ are again Euclidean lines. □

9.6 Exercises

1. Find the Poincaré extensions of the following elements of GM(1):

$$x \mapsto \frac{2x-3}{4x+5}, \quad x \mapsto \frac{-x+2}{x+4}.$$

   Specify which of them preserve orientation.
2. Consider the stabiliser $\mathrm{Stab}(\infty) = \{f \in \mathrm{GM}(1) \mid f(\infty) = \infty\}$.

   (a) Show that $\mathrm{Stab}(\infty)$ consists of elements of the form

$$x \mapsto ax + b, \quad a \neq 0.$$

   This is exactly the affine group $A(1)$ acting on $\mathbb{R}$.

(b) Find the Poincaré extension of Stab(∞) and show that its subgroup acting on $\mathcal{H}^2$ and consisting of orientation-preserving elements is exactly the set of elements

$$z \mapsto az + b, \quad a > 0,\ b \in \mathbb{R}.$$

This set is a group isomorphic to $A^+(1)$, that is, the subgroup of $A(1)$ that contains all orientation-preserving elements.

(c) Prove that there is a bijection

$$F : A^+(1) \to \mathcal{H}^2.$$

Through this mapping, provide $\mathcal{H}^2$ with a group structure.

3. Let A be an element of $\mathrm{SL}(2, \mathbb{R})$ of order 2. Show that then $A = \pm I$. If B is the adjoint matrix of $\mathrm{SL}(2, \mathbb{R})$ of some $f \in \mathrm{GM}(\mathcal{H}^2)$ of order 2, then the order of B is 4.
4. Find the associated matrix, the type, and the fixed points of the following hyperbolic transformations:

$$f(z) = \frac{-z}{z-4}, \quad g(z) = \frac{6z-5}{2z+1}, \quad h(z) = \frac{-z-5}{z+4}.$$

5. Find the point of intersection of $L = (1, 5)$ and $L' = (2, 6)$ in two ways: a) by writing the Cartesian equations of L, L' analytically and solving the system and b) using the transformation that maps $(1, 5)$ and its point $2 + i\sqrt{3}$ to $(0, \infty)$ and its point i, respectively.
6. Consider the hyperbolic lines $L = (-1, 1)$ and $L_b = (0, b)$, $b > 0$. Find conditions for b such that L and L' are: (a) intersecting and (b) parallel. In (a) determine the point of intersection, while in (b) determine the type of parallelism.
7. Prove that if $f \in \mathrm{M}(\mathcal{H}^2)$ is a loxodromic transformation, then there exists a hyperbolic line $\mathcal{A}_f$ that remains invariant under f: $f(\mathcal{A}_f) = \mathcal{A}_f$. The line $\mathcal{A}_f$ is called the *axis* of f.
8. Let the transformation f given by

$$f(z) = \frac{4z-1}{-3z+1}.$$

Show that f is loxodromic and find its axis $\mathcal{A}_f$.

9. Show that if $d_h(z, w) = \rho$, for some $z, w \in \mathcal{H}^2$, then

$$\rho = \tanh \left| \frac{z-w}{z-\overline{w}} \right|.$$

10. Consider the set $\Delta = \{z = x + iy \in \mathcal{H}^2 \mid x = y\}$. If $w \in \mathcal{H}^2$ then determine

$$d = \text{dist}(w, \Delta) = \inf\{d_h(z, w)]; \mid z \in \Delta\}.$$

11. Calculate the hyperbolic length of the hyperbolic circle $S_h(z_0, r)$.
12. Show directly that the group $\text{GM}(\mathcal{H}^2)$ preserves hyperbolic area.
13. Prove that the hyperbolic area $\text{Area}_h(D_h(z_0, r))$ of the hyperbolic disk $D_h(z_0, r)$ is equal to $4\pi \sinh^2(r/2)$.
14. Prove that the formula (9.15) of the Angle of Parallelism is equivalent to each of the following:

    (a) $\sinh b \tan \alpha = 1$.
    (b) $\tanh b \sec \alpha = 1$.

15. Prove that for triangles $T(\alpha, \beta, 0)$ we have

$$\sinh c = \frac{\cos \alpha + \cos \beta}{\sin \alpha \sin \beta}.$$

16. From the proof of Proposition 9.4.5 derive that $\tanh b = s$. Then show that

$$\tanh b = \sinh a \tan \beta.$$

17. With the assumptions of the previous exercise, show that the following hold:

    (a) $\sinh b = \sinh c \sin \beta$.
    (b) $\tanh a = \tanh c \cos \beta$.
    (c) $\cosh a \sin \beta = \cos \alpha$.
    (d) $\cosh c = \cot \alpha \cot \beta$.

18. Prove the Law of Sines.
19. Prove the Second Law of Cosines.
20. Prove that the angle of an equilateral hyperbolic triangle is less than $\pi/3$. Also, show that

$$2 \cosh(a/2) \sin(\alpha/2) = 1.$$

21. Show that the product (9.23) has all the properties of the inner product except $\langle \mathbf{x}, \mathbf{x} \rangle \geq 0$.
22. Prove the inverse Cauchy-Schwarz-Bunyakowski Inequality (9.28).
23. Prove that $\text{PO}(2, 1)$ acts transitively on $\mathcal{B}^2_{\mathbb{R}}$.
24. Prove directly the validity of the Triangular Inequality for σ_h.
25. Using the mapping $\mathfrak{C}$ given by (9.29), find the form of the hyperbolic isometries of $(\mathcal{B}^2_{\mathbb{R}}, \sigma_h)$.

26. Use the method we followed for the Beltrami-Klein model to define independently of the Inversive Geometry the Poincaré models of the upper half-plane and the disk as follows:

    (a) We denote by $\mathbb{C}^{1,1}$ the complex vector space $\mathbb{C}^2$ equipped with one of the following $(1, 1)$-Hermitian products $\langle \cdot, \cdot \rangle_1$ and $\langle \cdot, \cdot \rangle_2$ with matrices

    $$J_1 = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad J_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix},$$

    respectively. Explicitly, if $\mathbf{Z} = (Z_0, Z_1)$ and $\mathbf{W} = (W_0, W_1)$, then

    $$\langle \mathbf{Z}, \mathbf{W} \rangle_1 = \mathbf{Z} J_1 \mathbf{W}^* = -Z_0 \overline{W_0} + Z_1 \overline{W_1},$$
    $$\langle \mathbf{Z}, \mathbf{W} \rangle_2 = \mathbf{Z} J_2 \mathbf{W}^* = -i Z_0 \overline{W_1} + i Z_1 \overline{W_0},$$

    where $\mathbf{W}^* = (\overline{W_0}, \overline{W_1})^T$. Prove that all the properties of the Hermitian product hold, except positivity. Prove also that both matrices J_1 J_2 have one positive and one negative eigenvalue.

    (b) For both products, define the sets of positive, zero, and negative vectors, respectively. Considering the projection $\mathbb{P} : \mathbb{C}^{1,1} \to \mathbb{C}$ given by $\mathbb{P}(Z_0, Z_1) = Z_1/Z_0 = z \in \mathbb{C}$, show that the projection of the set of negative vectors with respect to the first product at the set where $Z_0 = 1$ is $\mathcal{B}^1_{\mathbb{C}}$, whereas with respect to the second product the projection is $\mathcal{H}^2$. Show also that the corresponding boundaries are the projections of the corresponding sets of zero vectors.

    (c) Find the isometry groups of $\langle \cdot, \cdot \rangle_1$ and $\langle \cdot, \cdot \rangle_2$ and from them define PSU(1, 1) and PSL(2, $\mathbb{R}$) respectively, as the transformation groups of $\mathcal{B}^1_{\mathbb{C}}$ and $\mathcal{H}^2$.

    (d) Define the hyperbolic distance ρ_i $i = 1, 2$ from the relation

    $$\cosh^2(\rho_i(z, w)/2) = \frac{|\langle \mathbf{Z}, \mathbf{W} \rangle_i|^2}{\langle \mathbf{Z}, \mathbf{Z} \rangle_i \langle \mathbf{W}, \mathbf{W} \rangle_i}, \quad i = 1, 2$$

    and show that it is well-defined, satisfies the axioms of the metric, and remains invariant under the action of PSU(1, 1) and PSL(2, $\mathbb{R}$), respectively.

    (e) Find the transformation that allows us to go from one model to the other.

    (f) Determine the geodesics of both models as follows: in the disk model, for example, take $z, w \in S^1 = \partial \mathcal{B}^1_{\mathbb{C}}$ and their elevations $\mathbf{Z}$ and $\mathbf{W}$, respectively, in the set of zero vectors. You can assume that $\langle \mathbf{Z}, \mathbf{W} \rangle_1 = -1$. The open

segment with ends $\mathbf{Z}$, $\mathbf{W}$ is

$$\gamma(t) = e^{t/2}\mathbf{Z} + e^{-t/2}\mathbf{W}, \quad t \in \mathbb{R}.$$

Now show that every point of this segment is a negative vector and using the above distance formula, also show that

$$\rho_1(\gamma(t),\ \gamma(s)) = |t - s|, \quad \text{for each } t, s \in \mathbb{R}.$$

Appendix A
The Real Space $\mathbb{R}^n$

A.1 The Vector Space $\mathbb{R}^n$

The n-dimensional real space is the set

$$\mathbb{R}^n = \{\mathbf{x} = (x_1, \ldots, x_n),\ x_i \in \mathbb{R},\ i = 1, \ldots, n\}.$$

From now on, we shall identify the points $\mathbf{x}$ with the vectors with initial point $\mathbf{0}$ and terminal point $\mathbf{x}$. The addition $+$ in $\mathbb{R}^n$ is defined by the relation

$$\mathbf{x} + \mathbf{y} = (x_1 + y_1, \ldots, x_n + y_n),$$

for each $\mathbf{x} = (x_1, \ldots, x_n)$ and $\mathbf{y} = (y_1, \ldots, y_n)$ vectors in $\mathbb{R}^n$. The following properties hold:

1. Commutative: $\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}$ for each $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$.
2. Associative: $\mathbf{x} + (\mathbf{y} + \mathbf{z}) = (\mathbf{x} + \mathbf{y}) + \mathbf{z}$ for each $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$.
3. Neutral element: $\mathbf{x} + \mathbf{0} = \mathbf{x}$ for each $\mathbf{x} \in \mathbb{R}^n$, where $\mathbf{0} = (0, \ldots, 0)$.
4. Opposite element: $\mathbf{x} + (-\mathbf{x}) = \mathbf{0}$ for each $\mathbf{x} \in \mathbb{R}^n$, where $-\mathbf{x} = (-x_1, \ldots, -x_n)$.

We conclude that the pair $(\mathbb{R}^n, +)$ is an abelian group (the *real additive group*).

We also define a scalar multiplication $\cdot$ as follows: if $\lambda \in \mathbb{R}$ and $\mathbf{x} \in \mathbb{R}^n$ then

$$\lambda \cdot \mathbf{x} = \lambda\mathbf{x} = (\lambda x_1, \ldots, \lambda x_n).$$

The scalar multiplication has the following properties:

1. $\lambda(\mathbf{x} + \mathbf{y}) = \lambda\mathbf{x} + \lambda\mathbf{y}$ for each $\lambda \in \mathbb{R}$ and $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$.
2. $(\lambda + \mu)\mathbf{x} = \lambda\mathbf{x} + \mu\mathbf{x}$ for every $\lambda, \mu \in \mathbb{R}$ and $\mathbf{x} \in \mathbb{R}^n$.
3. $\lambda(\mu\mathbf{x}) = (\lambda\mu)\mathbf{x}$ for every $\lambda, \mu \in \mathbb{R}$ and $\mathbf{x} \in \mathbb{R}^n$.
4. $1\mathbf{x} = \mathbf{x}$ for every $\mathbf{x} \in \mathbb{R}^n$.

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Therefore, the triple $(\mathbb{R}^n, +, \cdot)$ is a *vector space over* $\mathbb{R}$. The *dimension* of $\mathbb{R}^n$ is n.

Definition A.1.1 A set of vectors $\mathbf{v}_i$, $i \in I$ where I is a set of indices, is called *linearly independent* if for each $\lambda_i \in \mathbb{R}$, $i \in I$ such that

$$\sum_{i \in I} \lambda_i \mathbf{v}_i = \mathbf{0},$$

we have $\lambda_i = 0$, for each $i \in I$. If $I = \{1, \ldots, n\}$, and each element $\mathbf{x} \in \mathbb{R}^n$ is written as

$$\mathbf{x} = \sum_{i=1}^{n} \mu_i \mathbf{v}_i$$

for some $\mu_i \in \mathbb{R}$, $i = 1, \ldots, n$ (that is, $\mathbf{x}$ is written as *linear combination* of $\mathbf{v}_i$, $i = 1, \ldots, n$), then the set

$$\mathcal{B} = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$$

is called *basis* of $\mathbb{R}^n$. The numbers μ_i are called the *coordinates* of $\mathbf{x}$ with respect to the basis $\mathcal{B}$.

The *canonical basis* of $\mathbb{R}^n$ comprises $\mathbf{e}_1, \ldots, \mathbf{e}_n$, where

$$\mathbf{e}_i = (0, \ldots, 0, 1, 0, \ldots, 0)$$

with 1 being at the i-coordinate. If $\mathbf{x} = (x_1, \ldots, x_n)$ then

$$\mathbf{x} = \sum_{i=1}^{n} x_i \mathbf{e}_i .$$

Definition A.1.2 A set of vectors $\mathbf{v}_i$, $i \in I$ where I is a set of indices, is said to be *linearly dependent* if for some $j \in I$ there exist $\lambda_i \in \mathbb{R}$, $i \in I \setminus \{j\}$, such that

$$\mathbf{v}_j = \sum_{i \in I \setminus \{j\}} \lambda_i \mathbf{v}_i .$$

A useful criterion for the linear or non-linear independence of $k \leq n$ vectors in $\mathbb{R}^n$ is the following:

Criterion A.1.3 The vectors $\mathbf{v}_1, \ldots, \mathbf{v}_k$, $k \leq n$ of $\mathbb{R}^n$ are linearly independent (resp. dependent) if and only if the determinant

$$\begin{vmatrix} \mathbf{v}_1 \\ \vdots \\ \mathbf{v}_n \end{vmatrix},$$

which has the coordinates of $\mathbf{v}_i$ with respect to the canonical basis as rows, has a $k \times k$ minor which is different from 0 (resp., all $k \times k$ minors are equal to 0).

A.1.1 Subspaces

A subset $V \subset \mathbb{R}^n$ is called a *vector subspace of* $\mathbb{R}^n$ if

1. $\mathbf{0} \in V$,
2. $\lambda\mathbf{a} + \mu\mathbf{b} \in V$ for each $\lambda, \mu \in \mathbb{R}$, $\mathbf{a}, \mathbf{b} \in V$.

If $V = \langle \mathbf{v}_1, \ldots, \mathbf{v}_k \rangle$, that is, *is generated by k linearly independent vectors of* $\mathbb{R}^n$:

$$\mathbf{v} = \sum_{i=1}^{k} a_i \mathbf{v}_i, \ a_i \in \mathbb{R}, \ i = 1, \ldots, n,$$

for each $\mathbf{v} \in V$, then we say that it has *dimension k*, $\dim(V) = k$. Again, the linear independence of $\mathbf{v}_1, \ldots, \mathbf{v}_k$ ensures that any relation of the form

$$\lambda_1 \mathbf{v}_1 + \cdots + \lambda_k \mathbf{v}_k = \mathbf{0}, \ \lambda_i \in \mathbb{R}, \ i = 1, \ldots, k,$$

implies that $\lambda_i = 0$ for every $i = 1, \ldots, k$. We will assume that we can complete a basis of $\mathbb{R}^n$ from the vectors that generate V. That is, there exist linearly independent $\mathbf{v}'_{k+1}, \ldots, \mathbf{v}'_n$ of $\mathbb{R}^n$ such that $\mathbf{v}_1, \ldots, \mathbf{v}_k, \mathbf{v}'_{k+1}, \ldots, \mathbf{v}'_n$ are linearly independent and for every $\mathbf{a} \in \mathbb{R}^n$ we may write

$$\mathbf{a} = \sum_{i=1}^{k} a_i \mathbf{v}_i + \sum_{i=k+1}^{n} b_i \mathbf{v}'_i.$$

Finally, if V_1 and V_2 are subspaces of $\mathbb{R}^n$, such that $V_1 \cap V_2 = \{\mathbf{0}\}$ and for each $\mathbf{x} \in \mathbb{R}^n$ there exist $\mathbf{v}_i \in V_i i = 1, 2$, such that $\mathbf{x} = \mathbf{v}_1 + \mathbf{v}_2$, then we say that $\mathbb{R}^n$ is decomposed into the *direct sum* of V_1 and V_2 and we write

$$\mathbb{R}^n = V_1 \oplus V_2.$$

A.1.2 Linear Mappings

Definition A.1.4 A mapping $f : \mathbb{R}^n \to \mathbb{R}^m$ is called *linear*, if

$$f(\lambda \mathbf{x} + \mu \mathbf{y}) = \lambda\, f(\mathbf{x}) + \mu\, f(\mathbf{y}),$$

for each $\mathbf{x},\ \mathbf{y} \in \mathbb{R}^n$ and for each $\lambda,\ \mu \in \mathbb{R}$.

In particular, a linear mapping $f : \mathbb{R}^n \to \mathbb{R}^m$ will be called

- *monomorphism* if it is injective;
- *epimorphism* if it is surjective;
- *automorphism* (or, *isomorphism*) if it is bijective.

Let $f : \mathbb{R}^n \to \mathbb{R}^m$ be a linear mapping and suppose that

$$\mathcal{B} = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\} \text{ and } \mathcal{B}' = \{\mathbf{w}_1, \ldots, \mathbf{w}_m\},$$

are bases of $\mathbb{R}^n$ and $\mathbb{R}^m$, respectively. Then, for each $\mathbf{x} = \sum_{i=1}^n v_i \mathbf{v}_i$,

$$f(\mathbf{x}) = \sum_{i=1}^{n} v_i\, f(\mathbf{v}_i),$$

due to linearity. Moreover, for each $i = 1, \ldots, n$ it is:

$$f(\mathbf{v}_i) = \sum_{j=1}^{m} a_{ij}\, \mathbf{w}_j \implies f(\mathbf{x}) = \sum_{i=1}^{n} \sum_{j=1}^{m} a_{ij}\, \mathbf{w}_j.$$

The $m \times n$ matrix

$$A_f = \begin{pmatrix} a_{11} & \ldots & a_{1n} \\ & & \\ a_{m1} & \ldots & a_{mn} \end{pmatrix},$$

is called the *(associated) matrix of f (with respect to the bases $\mathcal{B}$ and $\mathcal{B}'$)*. We observe that in the case where the bases $\mathcal{B}$ and $\mathcal{B}'$ are the canonical bases of $\mathbb{R}^n$ and $\mathbb{R}^m$, respectively, then the images $f(\mathbf{e}_i)$ are the rows of A_f.

Conversely, given an $m \times n$ matrix $A = [a_{ij}]$ and given also the bases $\mathcal{B}$ and $\mathcal{B}'$ of $\mathbb{R}^n$ and $\mathbb{R}^m$ respectively, there exists a linear mapping $f : \mathbb{R}^n \to \mathbb{R}^m$ whose associated matrix is A. Due to linearity, this mapping is completely defined by the images of the elements $f(\mathbf{v}_i)$, $\mathbf{v}_i \in \mathcal{B}$, $i = 1, \ldots, n$ which are defined as follows:

$$f(\mathbf{v}_i) = \sum_{j=1}^{m} a_{ij}\, \mathbf{w}_j, \quad i = 1, \ldots, n.$$

Now an automorphism $f : \mathbb{R}^n \to \mathbb{R}^n$ has an associated matrix that is square and invertible, namely $A_f \in \mathrm{GL}(n, \mathbb{R})$ (Comment B.1.2). The set $\mathrm{Aut}(\mathbb{R}^n)$ of automorphisms of $\mathbb{R}^n$ is a group (the group operation is composition of mappings, (Definition B.2.3), and we can see that the mapping

$$\mathrm{Aut}(\mathbb{R}^n) \ni f \mapsto A_f \in \mathrm{GL}(n, \mathbb{R}),$$

where here, A_f is the associated matrix with respect to the canonical basis, is a group isomorphism.

A.1.3 Standard Topology of $\mathbb{R}^n$

The *Euclidean distance* $e_n(\mathbf{x}, \mathbf{y})$ of two points $\mathbf{x} = (x_1, \ldots, x_n)$ and $\mathbf{y} = (y_1, \ldots, y_n)$ of $\mathbb{R}^n$ is defined by the relation

$$e_n(\mathbf{x}, \mathbf{y}) = \left(\sum_{i=1}^{n} (x_i - y_i)^2 \right)^{1/2}.$$

The space $(\mathbb{R}^n, e_n)$ is a *metric space*. Recall the definition of a metric space:

Definition A.1.5 A non empty space X endowed with a function $d : X \times X \to \mathbb{R}$ is a *metric space* with *distance function* (or, *metric*) d if the following are satisfied:

1. d is a function $X \times X \to \mathbb{R}^+ = [0, +\infty)$.
2. $d(x, y) = 0 \iff x = y$ for each $x, y \in X$.
3. $d(x, y) = d(y, x)$ for each $x, y \in X$.
4. (Triangular inequality)

$$d(x, y) \leq d(x, z) + d(z, y)$$

for each $x, y, z \in X$. The equality holds if and only if $\mathbf{x}, \mathbf{y}, \mathbf{z}$, are linearly dependent.

For the case of the Euclidean metric in $\mathbb{R}^n$ we indeed have:

1. e_n is a function $\mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^+ = [0, +\infty)$.
2. $e_n(\mathbf{x}, \mathbf{y}) = 0 \iff \mathbf{x} = \mathbf{y}$ for each $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$.
3. $e_n(\mathbf{x}, \mathbf{y}) = e_n(\mathbf{y}, \mathbf{x})$ for each $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$.
4. (Triangle inequality)

$$e_n(\mathbf{x}, \mathbf{y}) \leq e_n(\mathbf{x}, \mathbf{z}) + e_n(\mathbf{z}, \mathbf{y})$$

for each $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$. The equality holds if and only if $\mathbf{x}, \mathbf{y}, \mathbf{z}$, are linearly dependent.

The set

$$B_{\mathbf{x}}(\epsilon) = \{\mathbf{y} \mid e_n(\mathbf{y}, \mathbf{x}) < \epsilon\}$$

is called the *open neighbourhood* of $\mathbf{x}$ of radius ϵ. The set

$$\overline{B_{\mathbf{x}}(\epsilon)} = \{\mathbf{y} \mid e_n(\mathbf{y}, \mathbf{x}) \leq \epsilon\}$$

is called the *closed neighbourhood* of $\mathbf{x}$ of radius ϵ.

Definition A.1.6 A set $A \subset \mathbb{R}^n$ is called *open*, if

$$\forall\, \mathbf{x} \in A,\ \exists\, \epsilon > 0 :\ B_{\mathbf{x}}(\epsilon) \subset A.$$

(Recall that the symbol $\forall$ stands for "for all" and the symbol $\exists$ stands for "there exists"). A set $A \subset \mathbb{R}^n$ is called *closed*, if $A^c = \mathbb{R}^n \setminus A$ is open.

Note that an open neighbourhood is an open set: if $\mathbf{y} \in B_{\mathbf{x}}(\epsilon)$, we consider the line segment from $\mathbf{x}$ that passes through $\mathbf{y}$. The neighbourhood $B_{\mathbf{y}}(\delta)$ where

$$\delta = \min\{e_n(\mathbf{x}, \mathbf{y}),\ \epsilon - e_n(\mathbf{x}, \mathbf{y})\}/2$$

is then entirely within $B_{\mathbf{x}}(\epsilon)$. On the other hand, a closed neighbourhood is a closed set. Indeed, if $\mathbf{y} \in \mathbb{R}^n \setminus \overline{B}_{\mathbf{x}}(\epsilon)$, we again take the line segment from $\mathbf{x}$ that passes through $\mathbf{y}$. The region $B_{\mathbf{y}}(\delta')$ where

$$\delta' = \min\{e_n(\mathbf{x}, \mathbf{y}) - \epsilon\}/2$$

then lies entirely within $\mathbb{R}^n \setminus \overline{B}_{\mathbf{x}}(\epsilon)$.

The following also hold:

1. $\emptyset$ (the null set) and $\mathbb{R}^n$ are the only sets that are simultaneously open and closed.
2. Any union of open sets is an open set (and any intersection of closed sets is a closed set).
3. Finite intersections of open sets give open sets (and finite unions of closed sets give closed sets).

Definition A.1.7 The collection of all open sets of $\mathbb{R}^n$ is called the *Euclidean (or standard) topology of* $\mathbb{R}^n$. The collection of all open domains is called the *basis of the topology*.

A.1.4 The Normed Space $\mathbb{R}^n$

The *Euclidean norm* $\|\cdot\|$ is defined for each $\mathbf{x} = (x_1, \ldots, x_n) \in \mathbb{R}^n$ by the relation

$$\|\mathbf{x}\| = \left(\sum_{i=1}^{n} x_i^2\right)^{1/2}.$$

We observe that the norm of $\mathbf{x}$ is equal to the distance $e_n(\mathbf{x}, \mathbf{0})$ of $\mathbf{x}$ from $\mathbf{0}$. Therefore, the Euclidean norm is defined in terms of the Euclidean distance. The space $(\mathbb{R}^n, \|\cdot\|)$ is thus a *normed space*.; the following axioms of norm are satisfied:

1. $\|\cdot\|$ is a function $\mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^+$.
2. $\|\mathbf{x}\| = 0 \iff \mathbf{x} = \mathbf{0}$ for each $\mathbf{x} \in \mathbb{R}^n$.
3. $\|\lambda\mathbf{x}\| = |\lambda|\|\mathbf{x}\|$ for each $\mathbf{x} \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$.
4. (Triangle inequality)

$$\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|,$$

for every $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. The equality holds if and only if $\mathbf{x}, \mathbf{y}$, are linearly dependent: there exists $\lambda \in \mathbb{R}$, $\lambda \neq 0$, such that $\mathbf{y} = \lambda\mathbf{x}$. Equivalently, $\mathbf{x}, \mathbf{y}$ and $\mathbf{0}$ are collinear.

Remark A.1.8 If $(X, \|\cdot\|_X)$ is a normed space, then we turn it into a metric space by defining the metric d_X on X as follows:

$$d_X(x, y) = \|x - y\|_X, \quad x, y \in X.$$

In the case of Euclidean space, we saw that the converse is also true; in general, this does not hold for every metric space.

Remark A.1.9 A vector $\mathbf{x} \in \mathbb{R}^n$ is *unit* if $\|\mathbf{x}\| = 1$.

A.1.5 Dot Product in $\mathbb{R}^n$

Definition A.1.10 Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, $\mathbf{x} = (x_1, \ldots, x_n)$ and $\mathbf{y} = (y_1, \ldots, y_n)$ be in $\mathbb{R}^n$. Their *dot product* $\mathbf{a} \cdot \mathbf{b}$ is defined by

$$\mathbf{x} \cdot \mathbf{y} = \sum_{i=1}^{n} x_i y_i.$$

Proposition A.1.11 (Polarisation Identity) *For each* $\mathbf{x} = (x_1, \ldots, x_n)$ *and* $\mathbf{y} = (y_1, \ldots, y_n)$ *in* $\mathbb{R}^n$*, we have*

$$4\,\mathbf{x} \cdot \mathbf{y} = \|\mathbf{x} + \mathbf{y}\|^2 - \|\mathbf{x} - \mathbf{y}\|^2.$$

Proof We compute straightforwardly:

$$\begin{aligned}\|\mathbf{x}+\mathbf{y}\|^2 - \|\mathbf{x}-\mathbf{y}\|^2 &= \sum_{i=1}^{n}(x_i+y_i)^2 - \sum_{i=1}^{n}(x_i-y_i)^2 \\ &= \sum_{i=1}^{n}\Big((x_i+y_i)^2 - (x_i-y_i)^2\Big) \\ &= 4\sum_{i=1}^{n} x_i y_i = 4\,\mathbf{x}\cdot\mathbf{y}.\end{aligned}$$

Therefore, the Euclidean dot product is defined by the Euclidean norm. The space $(\mathbb{R}^n, \cdot)$ is an *inner product space*. Indeed, the following properties hold for the dot product function $\cdot : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$:

1. $\mathbf{x}\cdot\mathbf{x} \geq 0$ and $\mathbf{x}\cdot\mathbf{x} = 0 \iff \mathbf{x} = \mathbf{0}$. (Positive definite).
2. $\mathbf{x}\cdot\mathbf{y} = \mathbf{y}\cdot\mathbf{x}$, for each $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. (Conjugation).
3. $(\mathbf{x}+\mathbf{y})\cdot\mathbf{z} = \mathbf{x}\cdot\mathbf{z} + \mathbf{y}\cdot\mathbf{z}$, for each $\mathbf{x}$, $\mathbf{y}$ and $\mathbf{z} \in \mathbb{R}^n$. (Linearity I).
4. $(\lambda\mathbf{x})\cdot\mathbf{y} = \lambda(\mathbf{x}\cdot\mathbf{y})$, for each $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$. (Linearity II).

Remark A.1.12 If we have a vector space endowed with an inner product, say $(X, \cdot)$, then we turn it into a normed space by defining the norm $\|\cdot\|_X$ on X as follows:

$$\|x\|_X = \sqrt{x\cdot x}, \quad x, \in X.$$

In the case of Euclidean space we saw that the converse is also true; in general, this is not true for any normed space.

A.1.6 CSB Inequality

Proposition A.1.13 *Let* $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. *Then the Cauchy-Schwarz-Bunyakowski inequality holds:*

$$|\mathbf{x}\cdot\mathbf{y}| \leq \|\mathbf{x}\|\cdot\|\mathbf{y}\|. \tag{A.1}$$

The equality holds if and only if $\mathbf{x}, \mathbf{y}$ *are linearly dependent.*

Proof Let $\mathbf{x} = (x_1, \ldots, x_n)$, $\mathbf{y} = (y_1, \ldots, y_n)$ and $\lambda \in \mathbb{R}$. If any of $\mathbf{x}, \mathbf{y}$ is the zero vector, the proposition holds trivially. So we assume that both vectors are nonzero. Then

$$\begin{aligned} 0 &\leq \sum_{i=1}^{n} (x_i \lambda + y_i)^2 \\ &= \|\mathbf{x}\|^2 \lambda^2 + 2(\mathbf{x} \cdot \mathbf{y})\ \lambda + \|\mathbf{y}\|^2. \end{aligned}$$

It follows that the discriminant of the quadratic in λ is less than or equal to zero:

$$(\mathbf{x} \cdot \mathbf{y})^2 \leq \|\mathbf{x}\|^2\ \|\mathbf{y}\|^2.$$

The discriminant is strictly less than zero if and only if the quadratic is greater than zero for all $\lambda \in \mathbb{R}$. Equivalently, if and only if there is no $\lambda \in \mathbb{R}$ such that $x_i \lambda + y_i = 0$ for each $i = 1, \ldots, n$, that is, if and only if the vectors are not linearly dependent.

By means of the inequality (A.1) we now define the *angle* $\angle(\mathbf{x}, \mathbf{y})$ of two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ as follows: if one of the two vectors is the zero vector, we define the angle to be 0. If both are nonzero then

$$\angle(\mathbf{x}, \mathbf{y}) = \arccos\left(\frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \cdot \|\mathbf{y}\|}\right) \in [0, \pi].$$

Consequently, if $\theta = \angle(\mathbf{x}, \mathbf{y})$, we have the formula

$$\mathbf{x} \cdot \mathbf{y} = \|\mathbf{x} \cdot \|\mathbf{y}\|\ \cos\theta. \tag{A.2}$$

A.2 Exercises

1. Let V be a subspace of $\mathbb{R}^n$, $V = \langle \mathbf{v}_1, \ldots, \mathbf{v}_k \rangle$, $k \leq n$.
   (a) Prove that any $l \leq k$ of the vectors $\mathbf{v}_i$, $i = 1, \ldots, k$ are linearly independent.
   (b) Let the sets $W = \langle \mathbf{v}_1, \ldots, \mathbf{v}_l \rangle$ and $W' = \langle \mathbf{v}_{l+1}, \ldots, \mathbf{v}_k \rangle$ be shown. Show the following.
      (i) W, W' are subspaces of V.
      (ii) $W \cap W' = \{\mathbf{0}\}$ and
      (iii) each $v \in V$ is written as $v = w + w'$, for some $w \in W$, $w' \in W'$.

      (The above defines the direct sum of subspaces: $V = W \oplus W'$).
2. Prove analytically that the Euclidean metric e_n satisfies Axioms 1–4 of the metric.

3. Prove that for each $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$:

$$|e_n(\mathbf{x}, \mathbf{z}) - e_n(\mathbf{z}, \mathbf{y})| \leq e_n(\mathbf{x}, \mathbf{y}).$$

   When does equality hold?
4. Prove that the Euclidean topology is indeed a topology, that is, it satisfies axioms 1–3 of topology.
5. Prove that the sphere $S_{\mathbf{x}}(\epsilon) = \overline{B_{\mathbf{x}}(\epsilon)} \setminus B_{\mathbf{x}}(\epsilon)$ is a closed set and has the following property: if $\mathbf{y} \in S_{\mathbf{x}}(\epsilon)$ and $B_{\mathbf{y}}(\delta)$ are open regions of it, then the sets $B_{\mathbf{y}}(\delta) \cap B_{\mathbf{x}}(\epsilon)$ and $B_{\mathbf{y}}(\delta) \cap (\mathbb{R}^n \setminus B_{\mathbf{x}}(\epsilon))$ are both nonempty.
6. Prove analytically that the Euclidean norm satisfies Axioms 1–4 of the norm. Also show that for each $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$ it is:

$$|\|\mathbf{x}\| - \|\mathbf{y}\|| \leq \|\mathbf{x}\| + \|\mathbf{y}\|.$$

   When does equality hold?
7. Prove that the Euclidean scalar product satisfies properties 1–4 of the inner product.
8. Prove the following for each $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$:

   (a) $\|\mathbf{x} \pm \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 \pm 2\, \mathbf{x} \cdot \mathbf{y}$.
   (b) $\|\mathbf{x} + \mathbf{y}\|^2 + \|\mathbf{x} - \mathbf{y}\|^2 = 2(\|\mathbf{x}\|^2 + \|\mathbf{y}\|^2)$ (Parallelogram Rule)
   (c) $\mathbf{x} \perp \mathbf{y}$ if and only if $\|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 = \|\mathbf{x} \pm \mathbf{y}\|^2$ (Pythagorean Theorem and its converse).
   (d) If $\theta = \angle(\mathbf{x}, \mathbf{y})$ then

$$\|\mathbf{x} \pm \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 \pm 2\, \|\mathbf{x}\| \cdot \|\mathbf{y}\|\, \cos\theta$$

   (Law of Cosines).

Appendix B
Elements of Group Theory

B.1 Groups

Let there be a set $G \neq \emptyset$ and an operation $\cdot$ defined on $G \times G$ with values in G that satisfies the following properties:

1. Associative: $(x \cdot y) \cdot z = x \cdot (y \cdot z)$, for each $x, y, z \in G$.
2. Neutral element: there exists an $e \in G$ such that $x \cdot e = e \cdot x = x$, for each $x \in G$.
3. Inverse element: for every $x \in G$ there exists an $x^{-1} \in G$ such that $x \cdot x^{-1} = x^{-1} \cdot x = e$.

If the above are satisfied then the pair $(G, \cdot)$ (or just G if we do not need to emphasise the group operation) is called *group with operation* $\cdot$. The operation $\cdot$ of the group is traditionally also called *multiplication*.

If in addition the property

4.

$$x \cdot y = y \cdot x \quad \text{for each } x, y \in G,$$

holds, then G is called *abelian*.

We list below some important examples of groups:

Example B.1.1 The *general linear group*

$$\mathrm{GL}(n, F) = \{A \in M^{n \times n}(F) \mid \det(A) \neq 0\},$$

where $F = \mathbb{R}$ or $\mathbb{C}$ and $M^{n \times n}(F)$, is the set of invertible square $n \times n$ matrices with elements from F. The operation is matrix multiplication. We show that $\mathrm{GL}(n, F)$ is a group:

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- for each $A, B \in \mathrm{GL}(n, F)$,

$$\det(AB) = \det(A) \cdot \det(B) \neq 0.$$

- Associativity follows from the properties of matrix multiplication: for each $A, B, C \in \mathrm{GL}(n, F)$,

$$(AB)C = A(BC).$$

- The neutral element is the identity square matrix I_n: for each $A \in \mathrm{GL}(n, F)$,

$$AI_n = I_n A = A.$$

- The inverse element of each $A \in \mathrm{GL}(n, F)$ its inverse A^{-1} since $\mathrm{GL}(n, F)$ comprises invertible matrices:

$$AA^{-1} = A^{-1}A = I_n.$$

Caution! $\mathrm{GL}(n, F)$ *is not* an abelian group since matrix multiplication is not a commutative operation in general.

Remark B.1.2 Every element A of $\mathrm{GL}(n, F)$ can be identified to an automorphism of $\mathbb{R}^n$ (Recall Sect. A.1.2).

Example B.1.3 All the following examples are subsets of the general linear group.

- The *special linear group* comprises matrices with determinant 1:

$$\mathrm{SL}(n, F) = \{A \in \mathrm{GL}(n, F) \mid \det(A) = 1\}.$$

- The *orthogonal group* is the set of matrices whose inverse is equal to its transpose:

$$\mathrm{O}(n) = \mathrm{O}(n, \mathbb{R}) = \{A \in \mathrm{GL}(n, \mathbb{R}) \mid A^{-1} = A^T\}.$$

- The *special orthogonal group* consists of orthogonal matrices with determinant 1:

$$\mathrm{SO}(n) = \mathrm{SO}(n, \mathbb{R}) = \{A \in \mathrm{O}(n) \mid \det(A) = 1\}.$$

- The *unitary group* is the set of complex orthogonal matrices:

$$\mathrm{U}(n) = \mathrm{O}(n, \mathbb{C}) = \{A \in \mathrm{GL}(n, \mathbb{C}) \mid A^{-1} = \overline{A^T}\}.$$

- The *special unitary group* comprises of complex orthogonal matrices with determinant 1:

$$\mathrm{SU}(n) = \mathrm{SO}(n, \mathbb{C}) = \{A \in \mathrm{U}(n) \mid \det(A) = 1\}.$$

Remark B.1.4 For each $A \in \mathrm{O}(n)$,

$$\det(A) = \det(A^T) = \det(A^{-1}) = 1/\det(A) \implies \det(A) = \pm 1.$$

Similarly, for each $A \in \mathrm{U}(n)$,

$$\begin{aligned} |\det(A)|^2 &= \det(A) \cdot \overline{\det(A)} \\ &= \det(A) \cdot \det(\overline{A^T}) \\ &= \det(A) \cdot \det(A^{-1}) = 1. \end{aligned}$$

Exercise Prove that all the sets in Example B.1.3 are groups.

So far, our examples have all been of matrix groups. Other, elementary groups, are the following.

- The additive group of integers $(\mathbb{Z}, +)$.
- The additive group of vectors in $\mathbb{R}^n$ (Sect. A.1).
- The multiplicative group $(\mathbb{R}_*, \cdot)$.

B.1.1 Subgroups

Let $(G, \cdot)$ be a group and $\emptyset \neq H \subset G$ be its subset for which the following holds:

1. $e \in H$.
2. If $h \in H$ then also $h^{-1} \in H$.
3. If $h_1, h_2 \in H$ then also $h_1 \cdot h_2 \in H$.

Then, H is called a *subgroup of* G and we write $H < G$ (the notation $H \leq G$ also applies).

The following criterion condenses the properties of the subgroup into one:

Criterion B.1.5 $H < G$ if and only if $e \in H$ and for every $h_1, h_2 \in H$ we have

$$h_1 \cdot h_2^{-1} \in H.$$

Example B.1.6 The following examples are elementary

- $\{e\}$ is a subgroup of every group that has e as a neutral element.
- $\mathrm{SL}(n, F) < \mathrm{GL}(n, F)$.

- $\mathrm{SO}(n) < \mathrm{O}(n)$.

Exercise Prove the relations in Example B.1.6.

Definition B.1.7 Let G be a group and $H < G$. The *left coset* gH *of* H *with respect to* $g \in G$ is

$$gH = \{gh \mid h \in H\}.$$

Remark B.1.8 In a similar way, we may define the right cosets Hg of H with respect to $g \in G$. The (left or right) cosets are equal-sized sets by which we can divide G through H.

Example B.1.9 Let $G = \mathbb{Z}$ be the additive group of integers and H be the subgroup of even numbers. If $g = 1$, then

$$1 + H = \{\ldots, -3, -1, 1, 3, \ldots\}.$$

We observe that $1 + H$ *is not* a subgroup of $\mathbb{Z}$.

If $g \in H$, then $gH = Hg = H$. In general, however, $gH = Hg$ does not hold for any $g \in G$.

Definition B.1.10 If H is a subgroup of G, then it is called *normal* if $gH = Hg$ for every $g \in G$. We write $H \lhd G$.

Example B.1.11 The following hold:

- $\{e\} \lhd G$, $G \lhd G$.
- Every subgroup of an abelian group is normal.

A criterion to determine whether a subgroup is normal is the following:

Criterion B.1.12 Let G be a group and $H < G$. Then $H \lhd G$ if and only if

$$ghg^{-1} \in H, \text{ for every } g \in G,\ h \in H.$$

Exercise Prove Criterion B.1.12.

Two subgroups H_1 and H_2 of a group G are called conjugate subgroups if there exists g of G such that $gH_1g^{-1} = H_2$. Subgroup conjugation is an equivalence relation.

B.1.2 Group Products

Definition B.1.13 Let G be a group, $H < G$ and $N \lhd G$. Then the set

$$HN = \{hn \mid h \in H,\ n \in N\}$$

is called the *product of* H and N.

Since $N \lhd G$, HN is a group with the operation being that of G. We note that in this case we have $HN = NH$, so the order in which we write the product does not matter.

Definition B.1.14 Let $(G_1, \cdot)$ and $(G_2, \star)$ be groups. The *direct product* of G_1, G_2 is the set $G_1 \times G_2$ equipped with the operation

$$(g_1, g_2)(h_1, h_2) = (g_1 \cdot h_1, g_2 \star h_2), \tag{B.1}$$

for each $(g_i, h_i) \in G_1 \times G_2$, $i = 1, 2$. This operation gives a group structure to $G_1 \times G_2$.

Example B.1.15 The additive group of $\mathbb{R}^n$ is an elementary example of a direct product of groups:

$$(\mathbb{R}^n, +) = \underbrace{(\mathbb{R}, +) \times \cdots \times (\mathbb{R}, +)}_{n \text{ times}}.$$

B.1.3 Group Quotient

Definition B.1.16 Let G be a group and $N \lhd G$. The set of all left cosets

$$G/N = \{gN \mid g \in G\},$$

is called the quotient group of G with N.

The quotient group G/N is indeed a group:

- the operation is defined by $(aN) \cdot (bN) = (a \cdot_G b)N$;
- its neutral element is eN, where e is the neutral element of G;
- for every $aN \in G/N$, $(aN)^{-1} = a^{-1}N$.

Exercise Fill in the missing details in the Definition B.1.16 of the quotient group.

B.2 Group Homomorphisms

Definition B.2.1 Let G_1, G_2 be groups and $\phi : G_1 \to G_2$ a mapping. ϕ is called a *homomorphism* if

$$\phi(x \cdot y) = \phi(x) \cdot \phi(y), \quad \text{for every } x, y \in G_1.$$

We emphasise that the operation on the left hand-side is the operation of G_1 while the operation on the right hand-side is the operation of G_2. ϕ is called:

- *monomorphism* if ϕ is injective;
- *epimorphism* if ϕ is surjective;
- *isomorphism* if ϕ is bijective.

B.2.1 Isomorphisms

Should we wish to check whether two groups have the same algebraic structure, we find it handy to examine if there is an isomorphism between those two groups. Let $\phi : G_1 \to G_2$ be a homomorphism.

- The *kernel* of the homomorphism ϕ is the set

$$\ker(\phi) = \{x \in G_1 \mid \phi(x) = e_2\}.$$

  We have $\ker(\phi) \lhd G_1$ and ϕ is a monomorphism if and only if $\ker(\phi) = \{e_1\}$.
- The *image* of the homomorphism ϕ is the set

$$\mathrm{Im}(\phi) = \{y \in G_2 \mid \exists\, x \in G_1 :\ y = \phi(x)\}.$$

  Here, $\mathrm{Im}(\phi) < G_2$ and ϕ is an epimorphism if and only if $\mathrm{Im}(\phi) = G_2$.
- $G_1/\ker(\phi)$ is isomorphic to $\mathrm{Im}(\phi)$ (First Group Isomorphism Theorem).

Example B.2.2 Using the above we will show now that $\mathrm{SL}(n, F) \lhd \mathrm{GL}(n, F)$. Let the multiplicative group $(F_*, \cdot)$ and consider the homomorphism

$$\mathrm{GL}(n, F) \ni A \mapsto \det(A) \in F_*.$$

The kernel is $\mathrm{SL}(n, F) \lhd \mathrm{GL}(n, F)$. Similarly, we may prove that

$$\mathrm{SO}(n) \lhd \mathrm{O}(n), \quad \mathrm{SU}(n) \lhd \mathrm{U}(n).$$

An isomorphism $\phi : G \to G$ is called an *automorphism*. The set of automorphisms of G is denoted by $\mathrm{Aut}(G)$ and is a group (the operation is composition of mappings).

Definition B.2.3 Aut(G) as above is called the *automorphism group of* G.

B.2.2 Semi-direct Product

The following proposition defines the semi-direct product of groups.

Proposition B.2.4 *Let G be a group, $H < G$, $N \lhd G$. The following are equivalent:*

1. $G = NH$, $N \cap H = \{e\}$.
2. *Every element $g \in G$ is uniquely written as $g = nh$, $n \in N$, $h \in H$.*
3. *The mapping $\phi : H \to G/N$, $h \mapsto hN$, is an isomorphism.*

Definition B.2.5 If one of the above holds, we say that G is the *semi-direct product of N and H* and we write

$$G = N \rtimes H.$$

We explain at this point that $\phi : H \to G/N$ is the composition $\pi \circ \iota$ of the epimorphism $\pi : G \to G/N$, $g \mapsto gN$, and the monomorphism $\iota : H \to G$, $h \mapsto h$. π is called the *normal projection of* G onto G/H and ι is called the inclusion of H into G.

The underlying set of the semi-direct product $G = N \rtimes H$ is the Cartesian product $N \times H$, but the group multiplication is different. Specifically, if $(n_i, h_i) \in N \times H$, $i = 1, 2$, then

$$(n_1, h_1)(n_2, h_2) = (n_1 h_1 n_2 h_1^{-1},\ h_1 h_2).$$

B.3 Exercises

1. Prove that $(\mathbb{R}^n, +)$, $(\mathbb{Z}, +)$, $(\mathbb{R}_*, \cdot)$ are abelian groups. Is $(\mathbb{Z}_*, \cdot)$ a group?
2. Let G be a group and $H_1 < H_2 < G$. Show that for every $g \in G$,

$$gH_1 \subseteq gH_2.$$

3. Prove that every abelian subgroup of G is a normal subgroup of G.
4. Let G be a group and $H \lhd G$. Also let $K < G$ with $H \subseteq K$. Prove that $H \lhd K$.
5. By using Criterion B.1.12, prove that $\mathrm{SL}(n, F) \lhd \mathrm{GL}(n, F)$ and $\mathrm{SO}(n) \lhd \mathrm{O}(n)$.
6. Prove that the Cartesian product $G_1 \times G_2$ of two groups $(G_1, \cdot)$ and $(G_2, \star)$ with the operation given by (B.1) is a group.
7. Prove that the direct product $G_1 \times G_2$ of two groups G_1, G_2 is an abelian group if both these groups are abelian.

8. Identifying G_1, G_2, with $G_1 \times \{e_2\}$ and $\{e_1\} \times G_2$, respectively, prove that they are normal subgroups of the direct product $G_1 \times G_2$.
9. Prove that if $H_i \lhd G_i$, $i = 1, 2$, then $H_1 \times H_2 \lhd G_1 \times G_2$.
10. Justify why in a homomorphism $\phi : G_1 \to G_2$ we always have

    (a) $\phi(e_1) = e_2$.
    (b) $\ker(\phi) \lhd G_1$, $\mathrm{Im}(\phi) < G_2$.

11. Let G be a group, $N \lhd G$ and $\pi : G \to G/N$ with $\pi(g) = gN$. Show that π is a group homomorphism (normal projection).
12. Let $\phi : G \to H$ be a homomorphism and $N \lhd G$. Prove that $\phi(N) \lhd H$.

Appendix C
Hints for Selected Exercises

Chapter 3, Section 3.4

2. For the Triangle Inequality, observe that

$$\begin{aligned} e_n^2(\mathbf{x},\mathbf{y}) &= \sum_{i=1}^{n}(x_i - y_i)^2 \\ &= \sum_{i=1}^{n}(x_i - z_i + z_i - y_i)^2 \\ &= \sum_{i=1}^{n}(x_i - z_i)^2 + \sum_{i=1}^{n}(y_i - z_i)^2 + 2\sum_{i=1}^{n}(x_i - y_i)(y_i - z_i) \\ &\leq e_n^2(\mathbf{x},\mathbf{y}) + e_n^2(\mathbf{y},\mathbf{z}) + 2\sum_{i=1}^{n}(x_i - y_i)(y_i - z_i). \end{aligned}$$

   To show that

$$\sum_{i=1}^{n}(x_i - y_i)(y_i - z_i) \leq e_n(\mathbf{x},\mathbf{y}) \cdot e_n(\mathbf{y},\mathbf{z}),$$

   mimic the proof of the CSB Inequality.

3. In this exercise, what we do is replace the usual coordinate system with a new system whose orthogonal axes have been obtained by rotating the original orthogonal axes by θ, counterclockwise.

5. Write the planes as

$$\begin{aligned} a_1x_1 + b_1x_2 + c_1x_3 &= 0, \\ a_2x_1 + b_2x_2 + c_2x_3 &= 0. \end{aligned}$$

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This system has nonzero solutions that are points on the line from the origin. For higher dimensions, take in $\mathbb{R}^4$ the planes

$$\mathbf{x} = (\lambda, \mu, 0, 0) \quad \text{and} \quad \mathbf{x}' = (0, 0, \lambda, \mu).$$

11. If $T_{\mathbf{a}} \in \mathrm{T}(n)$, then for each $\mathbf{x} \in \mathbb{R}^n$ the distance between $\mathbf{x}$ and $T_{\mathbf{a}}(\mathbf{x})$ is $\|\mathbf{a}\| = 2d$. Let the unit vector $\mathbf{A} = \mathbf{a}/\|\mathbf{a}\|$ and the hyperplanes $W_1 = W_{\mathbf{A},0}$ and $W_2 = W_{\mathbf{A},d}$. Then, if r_1, r_2 are the corresponding reflections, we have $r_2 \circ r_1 = T_{\mathbf{a}}$.
12. First show that the reflection r_θ on the line $y = \tan\theta$ has matrix

$$R_\theta = \begin{pmatrix} \cos(2\theta) & \sin(2\theta) \\ \sin(2\theta) & -\cos(2\theta) \end{pmatrix}.$$

You can also use Exercise 6, Sect. 3.4 for this, setting

$$(A, B) = (\sin\theta, \ -\cos\theta).$$

Then show using the above formula that the composition of two reflections $r_{\theta_1} \circ r_{\theta_2}$ is a rotation by $2(\theta_1 - \theta_2)$. Now if α_ϕ is a given element of SO(2), find the required reflections using the previous conclusion.
14. Since the eigenvalues of r_W are 1 and -1, and the elements of W are all eigenvectors of eigenvalue 1, the eigenspace of 1 is $(n-1)$-dimensional and its orthocomplement is $\{\mathbf{A}\}$. The desired orthonormal basis is an orthonormal basis of $\{\mathbf{A}\}^\perp = W$ together with $\mathbf{A}$.
17. Set

$$\tilde{f}(x_1, \ldots, x_n, x_{n+1}) = (f(x_1, \ldots, x_n), \ x_{n+1}),$$

and prove that $\tilde{f} \in E(n+1)$.
18. Consider the isometries mapping $\mathbf{x}_1, \mathbf{x}_2$ to $\mathbf{0}$, $\mathbf{x} = \mathbf{x}_1 - \mathbf{x}_2$, and $\mathbf{y}_1, \mathbf{y}_2$ to $\mathbf{0}$, $\mathbf{y} = \mathbf{y}_1 - \mathbf{y}_2$, respectively.

Chapter 4, Section 4.4

7. Use the Change of Variables Theorem.
8. If $D_{\delta,\mathbf{a}}$ is a homothety, then prove that the mapping

$$D_{\delta,\mathbf{a}} \circ D_{\alpha,\delta',\mathbf{b}} \circ (D_{\delta,\mathbf{a}})^{-1}$$

is a homothety, for every mapping of the form $D_{\alpha,\delta',\mathbf{b}}$. See also Exercise 5.
11. f cannot be a translation or a dilation, so it is an orthogonal transformation, which has n linearly independent vectors corresponding to the eigenvalue 1.

Chapter 5, Section 5.4

4. The right action of V

$$(\mathbf{c}(t), s\mathbf{v}) \mapsto \mathbf{c}(t) + s\mathbf{v}$$

is transitive and free.

11. The Cartesian equations are written equivalently as

$$(x_1, x_2, x_3, x_4) \cdot (1, 0, 0, -1) = 0, \quad (x_1, x_2, x_3, x_4) \cdot (0, 1, 1, 0) = 0.$$

A parametric equation for this plane can be found by setting $x_1 = \lambda_1, x_2 = \lambda_2$.

18. Such a mapping will have the form

$$f(x_1, x_2) = (ax_1 + bx_2 + c,\ a'x_1 + b'x_2 + c').$$

Since $f(0, x_2) = (bx_2 + c,\ b'x_2 + c')$, the desired affine mapping will be of the form

$$f(x_1, x_2) = (ax_1,\ a'x_1 + b'x_2 + c'), \quad ab' \neq 0.$$

25. We recall that affine independence of triples means that they are not triples of collinear points. The sets

$$\mathbf{x}_0 + \langle \mathbf{x}_1, \mathbf{x}_2 \rangle, \quad \mathbf{y}_0 + \langle \mathbf{y}_1, \mathbf{y}_2 \rangle,$$

are both 2-dimensional affine subspaces of $\mathbb{R}^n$. Working then according to the case of the doubly transitive action in the previous section, we can find transformations $h, g \in A(n)$ that map h the $\mathbf{0}, \mathbf{e}_1, \mathbf{e}_2$ to $\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2$ and g the $\mathbf{0}, \mathbf{e}_1, \mathbf{e}_2$ to $\mathbf{y}_0, \mathbf{y}_1, \mathbf{y}_2$. The composition $f = h \circ g^{-1}$ satisfies

$$f(\mathbf{x}_0) = \mathbf{y}_0, \quad f(\mathbf{x}_1) = \mathbf{y}_1, \quad f(\mathbf{x}_2) = \mathbf{y}_2,$$

which proves the claim.

30. Find affine bases of P, P' and complete them into affine bases of $\mathbb{R}^3$.

Chapter 6, Section 6.5

2a. S^{n-1} is a bounded set since $\|\mathbf{x}\| = 1$ for every $\mathbf{x} \in S^{n-1}$. To show that it is closed, it suffices to show that its complement is open. However, the complement consists of the open ball with center $\mathbf{0}$ and radius 1, and the set

$$B' = \{\mathbf{x} \in \mathbb{R}^n \mid \|\mathbf{x}\| > 1\}.$$

Show geometrically that B' is open.

16. Since the area of S^2 is 4π, the area of $B^s_{\mathbf{x}_0}(\epsilon)$ will be

$$\text{Area}(B^s_{\mathbf{x}_0}(\epsilon)) = \begin{cases} 4\sin^2(\epsilon/2) & \text{if } \epsilon \leq \pi, \\ 4(\pi - \sin^2(\epsilon/2)|)) & \text{if } \pi < \epsilon \leq 2\pi. \end{cases}$$

26. Parameterise the cut-off part with

$$\mathbf{x}(\theta_1, \theta_2) = (\sin\theta_1 \sin\theta_2,\ \sin\theta_1 \cos\theta_2,\ \cos\theta_1),$$

with $(\theta_1, \theta_2) \in D = [\arccos b, \arccos a] \times [0, 2\pi]$. If J is the Jacobian determinant of this transformation, then the desired area is

$$\text{Area} = \iiint_D |J|\, d\theta_1 d\theta_2 = 2\pi(b - a).$$

Another faster way is to consider the areas of specific spherical regions centred at $\mathbf{e}_3$.

Chapter 7, Section 7.6

25. Normalise the points appropriately.
28. Let $[\mathbf{A}_1], [\mathbf{A}_2], [\mathbf{C}_1]$ and $[\mathbf{D}_2]$ be such that $[\mathbf{A}_1, \mathbf{A}_2; \mathbf{C}_1, \mathbf{D}_2] = -1$. We normalise so that

$$[\mathbf{A}_1] = [1:0:0],\ [\mathbf{A}_2] = [0:1:0],\ \mathbf{C}_1 = [-1:1:0],\ \mathbf{D}_2 = [1:1:0].$$

Then the quadrilateral with sides $[\mathbf{A}_1], [\mathbf{A}_2], [\mathbf{A}_3] = [0:0:1]$ and $[\mathbf{A}_4] = [1:1:1]$ is the complete quadrilateral corresponding to the points of the harmonic position.

35. We first have that $[\mathbf{P}']$, $[\mathbf{P}''']$ and $[\mathbf{P}''']$ are collinear and let $\mathbf{L}''$ be the line containing them. Let $[\mathbf{A}'']$ be the intersection point of $\mathbf{L}''$ and the line from $[\mathbf{A}]$ and $[\mathbf{A}'']$. Then $[\mathbf{A}]$, $[\mathbf{B}]$, $[\mathbf{C}]$, $[\mathbf{X}]$ and $[\mathbf{A}'']$, $[\mathbf{P}'']$, $[\mathbf{P}'']$ and $[\mathbf{P}''']$ are in perspective position from $[\mathbf{A}'']$. Therefore,

$$[\mathbf{A}, \mathbf{B}; \mathbf{C}, \mathbf{X}] = [\mathbf{A}'', \mathbf{P}'; \mathbf{P}'', \mathbf{P}'''].$$

On the other hand, $[\mathbf{A}'']$, $[\mathbf{B}'']$, $[\mathbf{C}'']$, $[\mathbf{X}'']$ and $[\mathbf{A}'']$, $[\mathbf{P}'']$, $[\mathbf{P}'']$ and $[\mathbf{P}''']$ are in perspective position from $[\mathbf{A}]$.

Chapter 8, Section 8.7

5a. You can use the description of S^2 as the set $\{(\zeta, t) \in \mathbb{C} \times \mathbb{R} \mid |\zeta|^2 + t^2 = 1\}$. This will give $\mathbf{S}^{-1}(\zeta, t) = \zeta/(1 - t)$.

5c. You may need the following: every Euclidean line in $\mathbb{C}$ has an equation of the form $\Re(Az) = b$, where A is complex with $|A| = 1$ and $b \in \mathbb{R}$, while every circle $S(a, r)$ consists of elements of the form $z = a + re^{i\theta}$, $\theta \in \mathbb{R}$.

10a. If any of $\mathbf{x}_i$ is ∞, then $\mathbf{x}_i$, $i = 1, 2, 3$ are contained in a Euclidean line. If all $\mathbf{x}_i$, $i = 1, 2, 3$ are different from ∞, then they are either contained in a Euclidean line or in a circle of your choice.

11b. Let $\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ be the canonical basis of $\mathbb{R}^n$. For $\mathbf{v} = \mathbf{e}_i$, $\mathbf{w} = \mathbf{e}_j$, $i \neq j$ we get from Eq. (8.18) that $(\mathbf{e}_i A) \cdot (\mathbf{e}_j A) = 0$ and therefore the rows of A are mutually orthogonal vectors. This means, however, that the (ij)-elements of AA^T are all 0 for $i \neq j$. Consequently, AA^T is a diagonal matrix. Now again for $i \neq j$ we have

$$0 = (\mathbf{e}_i - \mathbf{e}_j) \cdot (\mathbf{e}_i + \mathbf{e}_j) = ((\mathbf{e}_i - \mathbf{e}_j)A) \cdot ((\mathbf{e}_i + \mathbf{e}_j)A) = (A^T A)_{ii}(A^T A)_{jj}$$

and we conclude that all diagonal elements of AA^T are equal to a number $\lambda \neq 0$.

12a. Use the translation $T_{-\mathbf{a}}$ to map Σ to $S(\mathbf{0}, r)$. Then apply the dilation $D_{1/r}$.

12b. Write the equation of the hyperplane as $(\mathbf{x} - \mathbf{a}) \cdot \mathbf{A} = 0$. Use the translation $T_{-\mathbf{a}}$ to map the hyperplane to $\mathbf{x} \cdot \mathbf{A} = 0$. Then take that element of $\mathrm{O}(n)$ that maps $\mathbf{A}$ to a vector $\delta\mathbf{e}_1$, $\delta > 0$.

12c. First, use a suitable translation, then use Proposition 8.2.19.

14a. Normalise so that

$$\mathbf{x}_{\sigma(1)} = \infty, \quad \mathbf{x}_{\sigma(2)} = \mathbf{0}, \quad \mathbf{x}_{\sigma(3)} = \mathbf{1}, \quad \mathbf{x}_{\sigma(4)} = \mathbf{x} \in \mathbb{R}^n.$$

Then

$$\begin{aligned} \mathbb{X}_1(\mathfrak{p}) + \mathbb{X}_2(\mathfrak{p}) &= \|\mathbf{x}\| + \|\mathbf{x} - \mathbf{1}\| \\ &\geq \|\mathbf{x} - (\mathbf{x} - \mathbf{1})\| = 1. \end{aligned}$$

Justify each step.

14c. In the inequality of the first question, the equality holds if and only if $\mathbf{x}$, $\mathbf{x} - \mathbf{1}$ are collinear.

15a. If $k = 1$, we have the line

$$2\Re(\overline{\alpha}z) = |\alpha|^2 - |\beta|^2,$$

If $k \neq 1$ we have the circle

$$\left| z - \frac{\alpha - k^2\beta}{1 - k^2} \right| = \frac{k|\alpha - \beta|}{|1 - k^2|}.$$

15b. The proof follows directly from Theorem 8.3.2, but you can also consider the cases of elementary transformations separately.

15c. Let C be a generalised circle. If C is the line $\Re(e^{-i\theta}z) = 0$, then it is written

$$|z - e^{-i\theta/2}| = |z + e^{-i\theta/2}|.$$

If C is the line $\Re(e^{-i\theta}z) = b$, $b \in \mathbb{R}_*$, then it is the transpose of $\Re(e^{-i\theta}z) = 0$ by a (which one?) and therefore an Apollonian circle. If C is the circle $|z| = r$, consider the transformation

$$f(z) = \frac{z-r}{z+r},$$

and show that $f(C)$ is an Apollonian circle, so $|z| = r$ is also an Apollonian circle. If C is the circle $|z - z_0| = r$, follow a procedure analogous to that for a line.

18a. Write

$$w = \frac{z-i}{z+i} \implies z = -i\frac{w+1}{w-1}.)$$

18b. For the converse, note that f will preserve the line $y = 0$, so it will be an element of $\mathrm{PGL}(2, \mathbb{R})$.

19. First, f is obviously a homeomorphism on $\mathbb{R}\setminus\{-d/c\}$. Now, if d is the chordal metric on $\overline{\mathbb{R}}$, it is

$$d(x, y) = \frac{2|x-y|}{(1+|x|^2)^{1/2}(1+|y|^2)^{1/2}}, \quad x, y \in \mathbb{R},$$
$$d(x, \infty) = \frac{2}{(1+|x|^2)^{1/2}}.$$

Show then that

$$d(f(x), f(\infty)) \to 0 \quad \text{when } d(x, \infty) \to 0,$$
$$d(f(x), f(-d/c)) \to 0 \quad \text{when } d(x, -d/c) \to 0.$$

23. It is $r = -1/3$ and if f is the desired transformation, then

$$f^{-1}(x) = \frac{x+1}{3x+1}.$$

Notice that

$$(f^{-1})'(x) = -\frac{2}{(3x+1)^2} < 0,$$

so f^{-1} and therefore f reverses orientation.

Chapter 9, Section 9.6

2c. F is defined by

$$A^+(1) \ni (x \mapsto ax + b) \mapsto b + ia \in \mathcal{H}^2.$$

Now if $f(x) = ax + b$ and $f'(x) = a'x + b'$ are elements of $A^+(1)$, then since

$$(f \circ f')(x) = aa'x + ab' + b,$$

we define a group operation $\star$ on $\mathcal{H}^2$ as

$$\begin{aligned} z \star z' &= (u + iv) \star (u' + iv') \\ &= F(vx + u) \star F(v'x + u') := F(vv'x + vu' + u) \\ &= vu' + u + ivv'. \end{aligned}$$

You can also observe that $A^+(1)$ is isomorphic to the subgroup of $\mathrm{SL}(2, \mathbb{R})$ containing elements of the form

$$\left\{ \begin{pmatrix} a^{1/2} & b/a^{1/2} \\ 0 & 1/a^{1/2} \end{pmatrix} = \begin{pmatrix} a^{1/2} & 0 \\ 0 & 1/a^{1/2} \end{pmatrix} \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}, \quad a > 0,\ b \in \mathbb{R} \right\}.$$

3. Consider the characteristic polynomial of A:

$$p_A(\lambda) = \lambda^2 - \mathrm{tr}(A) \cdot \lambda + 1 = 0.$$

From the Cayley-Hamilton Theorem we obtain:

$$A^2 - \mathrm{tr}(A) \cdot A + I = 0.$$

Since $A^2 = I$, we get

$$\mathrm{tr}(A) \cdot A = 2I \implies A = \frac{2}{\mathrm{tr}(A)} I.$$

Since $A \in \mathrm{SL}(2, \mathbb{R})$, we get $\mathrm{tr}^2(A) = 4$.

5. *Answer:* The Cartesian equations:

$$L : \ |z - 3| = 2, \quad L' : \ |z - 4| = 2.$$

The transformation is given by the equality of the cross-ratios

$$[1, 5; 2 + i\sqrt{3}, z] = [\infty, 0; i, g(z)].$$

Without writing the exact formula of g, find $g(2)$, $g(6)$.

7. After normalisation, we can assume that $f(z) = \delta z$, $\delta > 0$, $\delta \neq 1$. The fixed points of f are $0, \infty$. For each $si \in (0, \infty)$, $f(si) = \delta si \in (0, \infty)$.
8. *Answer:* The fixed points are $(-1 \pm \sqrt{7/3})/2$. Axis:

$$\mathcal{A}_f = \{z \in \mathcal{H}^2 \mid |z + 1/2| = \sqrt{7/3}/2\}.$$

10. If $w = u + iv$ $u > 0$, then $d = 0$ (why?). If $u \leq 0$, then from the previous exercise it is

$$(\tanh^{-1} \rho)^2 = \frac{(x-u)^2 + (x-v)^2}{(x-u)^2 + (x+v)^2} = f(x), \ x > 0.$$

Now show that f has a global minimum.

11. Let $\ell = \ell(S_h(z_0, r))$ be the hyperbolic length of $S_h(z_0, r)$: we consider the parametrisation of the hyperbolic circle

$$\boldsymbol{\gamma}(\theta) = \Re(z_0) + i\Im(z_0)\cosh r + \sinh r \Im(z_0) e^{i\theta}, \quad \theta \in [0, 2\pi].$$

We have $\dot{\boldsymbol{\gamma}}(\theta) = i \sinh r \Im(z_0) e^{i\theta}$ and

$$|\dot{\boldsymbol{\gamma}}(\theta|_h = \frac{\sinh r}{\cosh r + \sinh r \sin\theta}.$$

Therefore,

$$\ell = \int_0^{2\pi} \frac{\sinh r \ d\theta}{\cosh r + \sinh r \sin\theta}.$$

We set $\zeta = e^{i\theta}$, thus $d\zeta = ie^{i\theta} d\theta = i\zeta d\theta \kappa\alpha\iota \sin\theta = (\zeta - \overline{\zeta})/(2i) = (\zeta - (1/\zeta))/(2i)$. After the calculations, the integral is then written in complex form as

$$\ell = 2 \oint_{|\zeta|=1} \frac{d\zeta}{\zeta^2 + 2i \coth r \ \zeta - 1}.$$

The denominator has roots $i(1 \pm \cosh r)/\sinh r$, namely

$$\zeta_1 = -i \coth(r/2), \quad \zeta_2 = -i \tanh(r/2),$$

of which, because they are inverse, only one lies inside the unit disk. Since $\cosh x > \sinh x$ for every $x > 0$, this root is ζ_2. Then the integral residue is equal to

$$\mathrm{Res}(\zeta_2) = \lim_{\zeta \to \zeta_2} \frac{2(\zeta - \zeta_2)}{(\zeta - \zeta_1)(\zeta - \zeta_2)} = \frac{2}{\zeta_2 - \zeta_1} = -i \sinh r.$$

From the Integral Residue Theorem, it then follows

$$\ell = \ell(S_h(z_0, r)) = 2\pi i \text{Res}(\zeta_2) = 2\pi \sinh r.$$

12. Write the hyperbolic area element as

$$\frac{dxdy}{y^2} = \frac{dzd\overline{z}}{2i\Im^2(z)}.$$

There is an explanation of why this is possible, but it is beyond the scope of this book. We write for $\Omega \subset \mathcal{H}^2$ and $g \in \text{GM}(\mathcal{H}^2)$,

$$\text{Area}(g(\Omega)) = \frac{1}{2i} \iint_{g(\Omega)} \frac{dzd\overline{z}}{\Im^2(z)}.$$

By the Change of Variable Theorem, we have for $g \in \text{M}(\mathcal{H}^2)$ that

$$\text{Area}(g(\Omega)) = \frac{1}{2i} \iint_{\Omega} |\det(g(\zeta))| \frac{d\zeta d\overline{\zeta}}{\Im^2(g(\zeta))}.$$

If $g(\zeta) = (a\zeta + b)(c\zeta + d)$, $ad - bc = 1$, then

$$\det(g(\zeta)) = |g'(\zeta)|^2 = \frac{1}{|cz + d|^4}$$

and also $\Im(g(\zeta)) = \Im(\zeta)/|c\zeta + d|^2|$. Therefore,

$$\text{Area}(g(\Omega)) = \frac{1}{2i} \iint_{\omega} \frac{d\zeta d\overline{\zeta}}{\Im^2(\zeta)} = \text{Area}_h(\Omega).$$

Similarly in the case where $g(\zeta) = (a(-\overline{\zeta}) + b)/(c(-\overline{\zeta}) + d)$, $ad - bc = 1$.

13. The desired hyperbolic area is equal to the hyperbolic area of the hyperbolic disk $D = D(i, R)$, where $\sinh R = \tanh r$ (why?). This disk consists of the points

$$z(\rho, \theta) = i + \sinh \rho e^{i\theta}, \quad (\rho, \theta) \in [0, R] \times [0, 2\pi].$$

So, from the Change of Variables Theorem we have

$$\begin{aligned} \text{Area}_{\text{h}}(D) &= \int_0^R \int_0^{2\pi} \frac{\cosh \rho \sinh \rho \; dr d\theta}{(1 + \sinh \rho \sin \theta)^2} \\ &= \int_0^R \left(\int_0^{2\pi} \frac{\cosh \rho \sinh \rho \; d\theta}{(1 + \sinh \rho \cos \theta)^2} \right) d\rho \end{aligned}$$

Now calculate the inner integral using the method of Exercise 11.

16. If the geodesic passing through $x + iy$ and i has centre x_0, then

$$|x_0 - z|^2 = (x_0 - x)^2 + y^2 = 1 - 2x_0 x = s^2 = |x_0 - si|^2.$$

So, if we have assumed $x > 0$, then $x_0 < 0$. The Euclidean triangle with vertices $x_0, 0, si$ has angle β at x_0 (why?). Then, $\tan\beta = 2xs/(s^2 - 1)$.

18. Using the First Law of Sines, we write

$$\left(\frac{\sinh c}{\sin\gamma}\right)^2 = \frac{1}{1 - \left(\frac{\cosh a \cosh b - \cosh c}{\sinh a \sinh b}\right)^2}.$$

The Law of Sines will hold if the right-hand side is symmetric with respect to a, b, c, that is, if the expression

$$\sinh^2 a \sinh^2 b - (\cosh a \cosh b - \cosh c)^2$$

is symmetric. This is proven with the help of the relation $\cosh^2 x - \sinh^2 x = 1$.

19. We write for brevity

$$A = \cosh a, \quad B = \cosh B, \quad c = \cosh C.$$

The First Law of Cosines gives

$$\cos\gamma = \frac{AB - C}{(A^2 - 1)^{1/2}(B^2 - 1)^{1/2}}.$$

Squaring, we get

$$\sin^2\gamma = \frac{1 + 2ABC - (A^2 + B^2 + C^2)}{(A^2 - 1)(B^2 - 1)}.$$

The numerator D is positive and symmetric with respect to A, B, C. Now prove that

$$\frac{\cos\alpha\cos\beta + \cos\gamma}{\sin\alpha\sin\beta} = \frac{\cos\alpha\cos\beta + \cos\gamma}{\sin\alpha\sin\beta} \cdot \frac{(A^2 - 1)^{1/2}(B^2 - 1)^{1/2}(C^2 - 1)}{(A^2 - 1)^{1/2}(B^2 - 1)^{1/2}(C^2 - 1)} = C,$$

which proves the point.

22. Since the inequality does not depend on the choice of representatives, we may consider $\mathbf{X}, \mathbf{Y} \in V_-$ to be lifts of $\mathbf{x}, \mathbf{y} \in \mathcal{B}^2_{\mathbb{R}}$, respectively. Thus,

$$\langle \mathbf{X}, \mathbf{Y}\rangle^2 = (\mathbf{x} \cdot \mathbf{y} - 1)^2,$$
$$\langle \mathbf{X}, \mathbf{X}\rangle\langle \mathbf{Y}, \mathbf{Y}\rangle = (\|\mathbf{x}\|^2 - 1)(\|\mathbf{y}\|^2 - 1),$$

where $\cdot$ is the Euclidean scalar product and $\|\cdot\|$ is the Euclidean norm. The inequality (9.28) is equivalently written as

$$\|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 - 2\mathbf{x}\cdot\mathbf{y} \geq \|\mathbf{x}\|^2 \cdot \|\mathbf{y}\|^2 - (\mathbf{x}\cdot\mathbf{y})^2.$$

If either of $\mathbf{x}, \mathbf{y}$ is $(0,0)$, the inequality holds. Suppose that neither is $(0,0)$ and let $\theta \in [0,\pi]$ be the angle between $\mathbf{x}$ and $\mathbf{y}$. Then, since

$$\mathbf{x}\cdot\mathbf{y} = \|\mathbf{x}\| \cdot \|\mathbf{y}\| \cos\theta,$$

we write equivalently

$$\|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 - 2\|\mathbf{x}\| \cdot \|\mathbf{y}\| \cos\theta \geq \|\mathbf{x}\|^2 \cdot \|\mathbf{y}\|^2 \sin^2\theta.$$

Dividing by $\|\mathbf{x}\|^2 \cdot \|\mathbf{y}\|^2$ we get the equivalent expression

$$\frac{1}{\|\mathbf{x}\|^2} + \frac{1}{\|\mathbf{y}\|^2} \geq \frac{2\cos\theta}{\|\mathbf{x}\| \cdot \|\mathbf{y}\|} + \sin^2\theta.$$

The function

$$f(\theta) = \frac{2\cos\theta}{\|\mathbf{x}\| \cdot \|\mathbf{y}\|} + \sin^2\theta, \quad \theta \in [0,\pi],$$

is strictly decreasing, therefore $f(\theta) \leq f(0) = 2/(\|\mathbf{x}\| \cdot \|\mathbf{y}\|)$. On the other hand,

$$\left(\frac{1}{\|\mathbf{x}\|} - \frac{1}{\|\mathbf{y}\|}\right)^2 \geq 0 \implies \frac{1}{\|\mathbf{x}\|^2} + \frac{1}{\|\mathbf{y}\|^2} \geq \frac{2}{\|\mathbf{x}\| \cdot \|\mathbf{y}\|},$$

which proves the inequality. We observe that equality holds only if $\theta = 0$, equivalently if $\|\mathbf{x}\| = \|\mathbf{y}\|$, consequently $\mathbf{x} = \mathbf{y}$.

23. It suffices to show that for every $\mathbf{x} = (x_1, x_2) \in \mathcal{B}^2_{\mathbb{R}}$ there exists $A \in \mathrm{SO}(2,1)$ with $\mathbf{X}A = [1:0:0]$, where you can assume that $\mathbf{X} = [1 : x_1 : x_2]$.
24. Using Exercise 23, take the points $\mathbf{0} = (0,0)$, $\mathbf{x} = (x_1, x_2)$, $\mathbf{y} = (y_1, y_2)$ and prove that

$$\sigma_h(\mathbf{x},\mathbf{y}) \leq \sigma_h(\mathbf{0},\mathbf{x}) + \sigma_h(\mathbf{0},\mathbf{y}).$$

Bibliography

1. Anderson, J.W.: Hyperbolic Geometry. Springer Undergrad. Math. Ser., 2nd edn. Springer, London (2005)
2. Archimedes: The works of Archimedes. Edited in modern notation, with introductory chapters by Thomas L. Heath. Camb. Libr. Collect., Math. Cambridge University Press, Cambridge. Reprint of the 1897 original edition (2009)
3. Armstrong, M.A.: Basic topology. Corr. repr. Undergraduate Texts Math. Springer, Cham (1983)
4. Armstrong, M.A.: Groups and Symmetry. Undergraduate Texts Math. Springer, New York (1988)
5. Artin, E.: Geometric Algebra. Wiley/Interscience Publishers, New York. repr. of the orig., publ. by Interscience Publ., Inc. in the series 'Interscience Tracts in Pure and Applied Mathematics (1957) no. 3. Paperbk. ed. edition (1988)
6. Axler, S.: Linear Algebra Done Right. Undergraduate Texts Math, 4th edn. Springer, Cham (2024)
7. Beardon, A.F.: The Geometry of Discrete Groups, volume 91 of Grad. Texts Math. Springer, Cham (1983)
8. Beardon, A.F.: Algebra and Geometry. Cambridge University Press, Cambridge (2005)
9. Benedetti, R., Petronio, C.: Lectures on Hyperbolic Geometry. Universitext. Springer, Berlin (1992)
10. Berger, M.: Geometry. I, II. Transl. from the French by M. Cole and S. Levy. Universitext, corrected 4th printing edn. Springer, Berlin (2009)
11. Beutelspacher, A., Rosenbaum, U.: Projektive Geometrie. Von den Grundlagen bis zu den Anwendungen, volume 41 of Vieweg Stud., 2nd revised and expanded edn. Vieweg, Braunschweig (2004)
12. Birman, G.S., Nomizu, K.: The Gauss-Bonnet theorem for 2-dimensional spacetimes. Mich. Math. J. **31**, 77–81 (1984)
13. Brannan, D.A., Esplen, M.F., Gray, J.J.: Geometry, 2nd edn. Cambridge University Press, Cambridge (2012)
14. Bunt, L.N.H., Jones, P.S., Bedient, J.D.: The Historical Roots of Elementary Mathematics, reprint edition. Dover Publications, New York (1988)
15. Cederberg, J.N.: A Course in Modern Geometries, corr. reprint of the 1989 orig. edn. Undergraduate Texts Math. Springer, New York (1995)
16. Coxeter, H.S.M.: Introduction to Geometry, XIV, 443 p. John Wiley & Sons, New York (1961)
17. Coxeter, H.S.M.: Projective Geometry, 1st paperback reprint of the 1987 2nd edn. Springer, New York (2003)

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https://doi.org/10.1007/978-3-032-08016-5

18. Coxeter, H.S.M., Greitzer, S.L.: Geometry Revisited, volume 19 of New Math. Libr. The Mathematical Association of America, Washington, DC (1967)
19. Dillon, M.I.: Geometry Through History. Euclidean, Hyperbolic, and Projective Geometries. Springer, Cham (2018)
20. Eves, H.: An Introduction to the History of Mathematics. With Cultural Connections by Jamie H. Eves, 6th edn. Saunders College Publishing, Philadelphia (1990)
21. Fenchel, W.: Elementary Geometry in Hyperbolic Space, volume 11 of De Gruyter Stud. Math. Walter de Gruyter & Co., Berlin (1989)
22. Fenn, R.: Geometry. Springer Undergrad. Math. Ser. Springer, London (2001)
23. Ferraro, G.: Euler's treatises on infinitesimal analysis: Introductio in analysin, infinitorum, Institutiones calculi differentialis, Institutionum calculie integralis. In: Euler Reconsidered. Tercentenary Essays, pp. 39–101. Kendrick Press, Heber City, UT (2007)
24. Gallier, J.: Geometric Methods and Applications. For Computer Science and Engineering, volume 38 of Texts Appl. Math., 2nd edn. Springer, Dordrecht (2011)
25. Hartshorne, R.: Foundations of Projective Geometry, vol. VII, 167 p. W.A. Benjamin, New York (1967)
26. Henle, M.: Modern Geometries: The Analytic Approach. Prentice Hall, Upper Saddle River, NJ (1997)
27. Hilbert, D.: Grundlagen der Geometrie (Festschrift 1899). Klass. Texte Wiss. Springer Spektrum, Heidelberg (2015)
28. Hirschfeld, J.W.P.: Projective Geometries over Finite Fields, 2nd edn. Oxford Math. Monogr. Clarendon Press, Oxford (1998)
29. Iversen, B.: Hyperbolic Geometry, volume 25 of Lond. Math. Soc. Stud. Texts. Cambridge University Press, Cambridge (1992)
30. Katz, V.J.: A History of Mathematics. An Introduction, 2nd edn. Addison-Wesley, Reading, MA (1998)
31. Kisil, V.V.: Geometry of Möbius Transformations. Elliptic, Parabolic and Hyperbolic Actions of SL $_2(\mathbb{R})$. With DVD-ROM. World Scientific, Hackensack, NJ (2012)
32. Klein, F.: Elementary Mathematics from a Higher Standpoint. Volume II: Geometry. Translated from the 4th German edition by Gert Schubring. Springer, Berlin (2016)
33. Kushelman, M., McGrath, P.: On Liouville's theorem for conformal maps. Am. Math. Mon. **131**(7), 619–623 (2024)
34. Maehara, H., Martini, H.: Circles, Spheres and Spherical Geometry. Birkhäuser, Cham (2024)
35. Martzloff, J.C.: A History of Chinese Mathematics. Transl. from the French by S. Wilson. With forewords by Jaques Gernet and Jean Dhombres, corrected 2nd printing edition. Springer, Berlin (2006)
36. Meserve, B.E.: Fundamental Concepts of Geometry. Preliminary edn., 296 p. Addison-Wesley Publishing Company, Cambridge (1953)
37. Möbius, A.F.: Der barycentrische Calcul. Ein neues Hülfsmittel zur analytischen Behandlung der Geometrie. Nachdruck der Ausgabe Leipzig 1827, XXIV, 454 S. DM 74.00. Georg Olms Verlag, Hildesheim (1976)
38. Morley, F., Morley, F.V.: Inversive Geometry, vol. ix, 273 S., 67 Fig. G. Bell & Sons, London (1933)
39. Munkres, J.R.: Topology, 2nd edn. Prentice Hall, Upper Saddle River, NJ (2000)
40. Plofker, K.: Mathematics in India. Princeton University Press, Princeton, NJ (2009)
41. Ramírez Galarza, A.I., Seade, J.: Introduction to Classical Geometries. Birkhäuser, Basel (2007)
42. Ramsay, A., Richtmyer, R.D.: Introduction to Hyperbolic Geometry. Universitext. Springer, New York, NY (1995)
43. Rashed, R., Papadopoulos, A.: Menelaus' "Spherics". Early translation and al-Māhānī/al-Harawī's version, volume 21 of Sci. Graeco-Arabica. De Gruyter, Berlin (2017)
44. Ratcliffe, J.G.: Foundations of Hyperbolic Manifolds, volume 149 of Grad. Texts Math., 3rd expanded edition. Springer, Cham (2019)

45. Reid, M., Szendrői, B.: Geometry and Topology. Cambridge University Press, Cambridge (2005)
46. Rotman, J.J.: An Introduction to the Theory of Groups., volume 148 of Grad. Texts Math., 4th edn. Springer, New York, NY (1995)
47. Sharpe, R.W.: Differential Geometry: Cartan's Generalization of Klein's Erlangen Program. Foreword by S. S. Chern, volume 166 of Grad. Texts Math. Springer, Berlin (1997)
48. Sossinsky, A.B.: Geometries, volume 64 of Stud. Math. Libr. American Mathematical Society (AMS), Providence, RI (2012)
49. Steenrod, N.: The Topology of Fibre Bundles, 1st paperback edn. Princeton University Press, Princeton, NJ (1999)
50. Szmielew, W.: From Affine to Euclidean Geometry. An Axiomatic Approach. Ed., prep. for publ. and transl. from the Polish by Maria Moszynska, , XIII, 194 p. Dfl. 80.00; $ 34.95. D. Reidel Publishing Company/Warszawa/PWN- Polish Scientific Publishers, Dordrecht/Boston/London (1983)
51. Van Brummelen, G.: Heavenly Mathematics. The Forgotten Art of Spherical Trigonometry, reprint of the 2013 hardback edition. Princeton University Press, Princeton, NJ (2017)
52. van der Waerden, B.L.: Science Awakening. Translated from the Dutch text by Arnold Dresden with additions of the author, 360 p. P. Noordhoff Ltd, Groningen (1954)
53. Waterman, P.L.: Möbius transformations in several dimensions. Adv. Math. **101**(1), 87–113 (1993)
54. Weyl, H.: Raum, Zeit, Materie. Vorlesungen über allgemeine Relativitätstheorie. J, VIII u. 234 S. 8°. Springer, Berlin (1918)
55. Whittlesey, M.A.: Spherical Geometry and Its Applications. Textb. Math. CRC Press, Boca Raton, FL (2020)
56. Wilker, J.B.: Inversive Geometry. The Geometric Vein, The Coxeter Festschr, pp. 379–442 (1982)
57. Wilker, J.B.: Moebius transformations in dimension n. Period. Math. Hung. **14**, 93–99 (1983)
58. Wilson, E.B., Lewis, N.: The space-time manifold of relativity. The non-euclidean geometry of mechanics and electromagnetics. Amer. Acad. Proc. **48**, 387–507 (1912)

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